



**Swiss Centre
for Life Cycle
Inventories**

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Life Cycle Inventories of Agricultural Production Systems

Data v2.0 (2007)



Thomas Nemecek and Thomas Kägi

Agroscope Reckenholz-Tänikon Research Station ART

ecoinvent report No. 15

Zürich and Dübendorf, December 2007

ecoinvent report No. 15 consists of two parts:

**ecoinvent report No. 15a: Life cycle inventories of Swiss and
European agricultural production systems**

**ecoinvent report No. 15b: Life cycle inventories of U.S. agricultural
production systems**

Project "ecoinvent data v2.0"

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by Thomas Nemecek and Thomas Kägi, ART

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Summary

The agricultural sector has various relevant impacts on the environment. The study of agricultural systems and their environmental impacts are of high importance.

More than 330 datasets have been defined for agricultural production systems in ecoinvent data. The aim was both to provide data for the modelling of agricultural production systems and to provide data on a number of agricultural products. Datasets for the following categories were defined (number of modules in brackets):

- infrastructure:
 - o buildings: infrastructure (21) and operation (8)
 - o machinery: infrastructure (6) and work processes (35)
 - o drying processes (4)
- inputs:
 - o fertilisers (mineral (24) and organic (6))
 - o pesticides (68)
 - o seed (22)
 - o animal feed (10)
- outputs:
 - o arable crop products (115)
 - o dried roughage (hay) (3)
 - o starch (2)
 - o animal products (3)

The relevant input data for modelling agricultural systems are included for arable and fodder crops and for cattle and pig production. They are also partly available for special crops and for poultry production. Products at farm level are included for arable crops and three types of hay.

Most datasets refer to Switzerland, except the inventories for the 12 European crop products, fertilisers and pesticides, which were defined for a European context. Datasets for U.S. conditions are described in ecoinvent report no. 15b. Further datasets for the conditions of Asia, Brazil and the USA are described in ecoinvent reports no. 17 and 21.

The direct field emissions of NH_3 , NO_3^- , N_2O , P, heavy metals and the tractor exhaust gases NMVOC, NO_x and CO have been calculated using emission models.

ecoinvent data provides datasets for calculating LCAs in the most important production branches in Swiss conditions and also a variety of arable and fodder crop products.

Abbreviations

4WD	4 wheel drive
ACW	Agroscope Changins-Wädenswil Research Station, Nyon, Switzerland
ART	Agroscope Reckenholz-Tänikon Research Station ART
Ba	bale
BOD	biological oxygen demand
CED	cumulative energy demand
CH	Switzerland
CH ₄	methane
DM	dry matter
EFMA	European Fertilizer Manufacturers Association
FADN	farm accountancy data network
FAL	Swiss Federal Research Station for Agroecology and Agriculture, Zurich-Reckenholz (today part of ART)
FAT	Swiss Federal Research Station for Agricultural Economics and Engineering, Tänikon (today part of ART)
FiBL	Research Institute of Organic Agriculture, Frick, Switzerland
FU	functional unit
hp	horse-power
IP	integrated production
kg	kilogram (measurement of weight)
LCI	life cycle inventory
LCIA	life cycle impact assessment
LU	livestock unit
m ²	square metre (measurement of area)
m ³	cubic metre (measurement of volume)
mFC	mean fuel consumption
MU	milking unit, milking cluster
N ₂ O	nitrous oxide, dinitrogen monoxide
n.a.	not available
NH ₃	ammonia
ÖLN	“ökologischer Leistungsnachweis” (ecological requirements for agricultural production)
PM	particulate matter
PTO	power take-off

RAC	Swiss Federal Research Station for Plant Production (Agroscope RAC Changins, today belonging to ACW)
RER	Europe
SALCA	Swiss Agricultural Life Cycle Assessment
Ta	tanker
Th	tractor hour
TL	trailer load
vkm	vehicle kilometre
WU	working unit

Changes to ecoinvent Data v1.01 in v2.0

Correction of inventories

Carbon dioxide uptake through plants and the energy in biomass was corrected in the crop production inventories avoiding double counting of the CO₂- and energy content in the seeds. The carbon and energy content of the seed is now subtracted from the carbon and energy content of the exported products in the unit process inventories.

Update of the datasets for crop production

The calculation of the direct emissions in the crop inventories has been completely revised based on the SALCA methodology (Swiss Agricultural Life Cycle Assessment). The following models were used:

- Nitrate: Richner *et al.* (2006),
- Phosphorus: Prasuhn (2006), including erosion according to Oberholzer *et al.* (2006),
- Heavy metals: (2006).

Furthermore, we updated yield data by using more up to date statistics and a longer times series. The related processes like amount of fertilisers, use of machinery for harvest and transport were adapted accordingly. Due to these changes, also the other direct emissions had to be adapted, even if the emission models remained unchanged.

New inventories

For Saxony-Anhalt (Germany), Barrois (France) and Castilla-y-Leon (Spain) four inventories for agricultural production were added. Furthermore, 11 inventories considering US agricultural production were added (see ecoinvent report no. 15b).

- protein peas conventional, Saxony-Anhalt, at farm, DE
- barley grains conventional, Saxony-Anhalt, at farm, DE
- rape seed conventional, Saxony-Anhalt, at farm, DE
- wheat grains conventional, Saxony-Anhalt, at farm, DE
- protein peas conventional, Castilla-y-Leon, at farm, ES
- barley grains conventional, Castilla-y-Leon, at farm, ES
- sunflower conventional, Castilla-y-Leon, at farm, ES
- wheat grains conventional, Castilla-y-Leon, at farm, ES
- protein peas conventional, Barrois, at farm, FR
- barley grains conventional, Barrois, at farm, FR
- rape seed conventional, Barrois, at farm, FR
- wheat grains conventional, Barrois, at farm, FR
- irrigating, US
- potatoes, at farm, US
- rape seed, at farm, US
- rice, at farm, US

- rice seed, at regional storehouse, US
- wheat grains, at farm, US
- wool, sheep, at farm, US
- cotton fibres, at farm, US
- cotton seed, at farm, US
- cotton seed, at regional storehouse, US
- sheep for slaughtering, live weight, at farm, US

Part I: Introduction

1 Introduction and Overview

The impacts of agricultural production on the environment are manifold. Although its share of the gross domestic product has declined steadily over a number of decades, farming still exerts a significant impact on the environment, and has even exacerbated some environmental problems. Agricultural productivity increased significantly during the 20th century, with mechanisation leading to a dramatic rise in labour productivity. Improved production techniques, intensive use of fertilisers and pesticides and progress in animal husbandry helped to increase yields. However, excessive use of these inputs has resulted in a variety of problems, such as e.g. eutrophication or toxicity. Agriculture (together with forestry) is responsible for a large part of land use. Agricultural production is the main source of several major emissions. Examples are ammonia (NH₃), 93% of which comes from agricultural sources (Thöni *et al.* 2007), methane (CH₄) (Minonzio *et al.* 1998) and nitrate (NO₃⁻). For these reasons, the study of agricultural production systems is a major priority.

This report documents the life cycle inventories for agricultural production systems contained in ecoinvent data. The documentation of further inventories of agricultural products can be found in reports no. 15b, 17 and 21.

The aims of the report and the corresponding data are twofold:

- to provide datasets for infrastructure and inputs used in agricultural production necessary for calculating agricultural production systems,
- to provide datasets on several agricultural products that are typical for Switzerland, Europe and the USA, with a focus on plant production.

Users of these data should bear in mind that they are intended for use within life cycle studies of agricultural systems, and not for other purposes, like e.g. comparison of the pesticides with other chemicals.

The most relevant modules for agricultural infrastructure (buildings and machines), work processes and inputs are available for arable and fodder crops as well as for cattle and pig production (Tab. 1.1). They are also available in part for special crops and for poultry production. Datasets on products are included for arable crops, and a few are included for fodder crops. Other products can be calculated using the ecoinvent modules presented in this report.

Tab. 1.2 shows an overview of the modules defined in the different categories.

Tab. 1.1 Overview of the agricultural production branches covered by ecoinvent data.

Production branches	Buildings	Machinery	Work processes	Inputs	Products CH	Products EU	Product USA
Arable crops							
Fodder crops							
Horticulture (Field)							
Horticulture (Greenhouse)							
Fruit growing							
Vineyards							
Cattle production							
Pig production							
Poultry production							
Sheep & Wool production							

relevant datasets available
 partly available
 not available

Tab. 1.2 Overview of the available modules of agricultural production systems included in ecoinvent data.

	Subcategory	Number of modules	Example of inventories for the subcategories		
			Name	Location	Unit
Infra-structure	Buildings	21	dried roughage store, air dried, solar	CH	kg
	Machinery	6	label housing system, pig	CH	pig place
Operation of infrastructure	Building usage	8	agricultural machinery, tillage, production	CH	kg
	Machinery and equipment usage	35	loose housing system, cattle, operation	CH	LU
			dried roughage store, air dried, solar, operation	CH	kg
			hayage, by rotary tedder	CH	ha
Agricultural inputs	Drying	4	tillage, ploughing	CH	ha
	Mineral fertilisers	24	milking	CH	kg
			grain drying, high temperature	CH	kg
			lime, from carbonation, at regional storehouse	CH	kg
	Organic fertilisers	6	ammonium nitrate, as N, at regional storehouse	RER	kg
			urea, as N, at regional storehouse	RER	kg
			horn meal, at regional storehouse	CH	kg
	Pesticides	68	cyclic N-compounds, at regional storehouse	RER	kg
			[Sulfonyl]urea-compounds, at regional storehouse	CH	kg
			pesticide unspecified, at regional storehouse	CH	kg
Agricultural outputs	Seed	24	sugar beet seed IP, at regional storehouse	CH	kg
	Feed	10	wheat organic, at fodder mill	CH	kg
			wheat IP, at fodder mill	CH	kg
			potatoes organic, at farm	CH	kg
	Plant production	120	rape seed extensive, at farm	CH	kg
			wheat grains conventional, Barrois, at farm	FR	kg
			wool, sheep, at farm	US	kg
	Animal production	4			
	Total	330			

The documentation is structured as follows:

- Part I (Introduction) provides some general information on agriculture with special reference to Switzerland, and on the calculation of life cycle inventories in agriculture. General information relating to more than one chapter can also be found in this part.
- Part II (Agricultural infrastructure and its operation) describes the infrastructure in agricultural production (machinery, buildings). The use of buildings and machinery (mainly in field work processes) is also described in this part.
- Part III (Agricultural inputs) describes the inputs in agricultural production, such as fertilisers, pesticides, seed and feedstuffs. As the agricultural sector relies heavily on product cycles, several outputs from agriculture are also used as inputs (e.g. seed). The separation of inputs and outputs therefore depends on point of view, and can never be absolute.
- Part IV (Agricultural outputs) describes the products from agriculture (plant production only was included in the version 1.1), such as food raw materials, hay and starch. For feedstuffs, agricultural field production is included in Part IV, whereas their processing is described in Part III.
- Part V refers to some selected publications.
- Part VI contains the appendices (supplementary information and the unit process inventories) as well as the literature.

2 Raw Material Reserves and Resources

The main resources for agricultural production are

- agricultural land
- water
- solar radiation
- nutrients (major elements N, P, K, oligo-elements and micro-nutrients)
- fossil energy
- ores and minerals, used for the production of machines, construction of buildings, etc.

Agricultural land is a limiting resource for agricultural production in Switzerland. Competition for “land” resources is high in the densely populated areas of Switzerland, where the main areas of agricultural production are located. Some figures on the usage of agricultural areas are given in chapter 3 (Tab. 3.1).

In most areas of Switzerland, *rainfall* is usually not a limiting factor for agricultural production. The average rainfall on the Swiss plateau was 1067 mm/year in the years 1961-1990 (SBV 2000b), which is sufficient for most crops. Nevertheless, vegetables are regularly irrigated in most regions. In some regions with less rainfall, several other crops are also irrigated.

Solar radiation is often limiting for agricultural production, but as it is a renewable, non-depletable resource, it is not conceived of as being an environmental problem.

The nutrient N is taken from the air, where its availability is not a limiting factor. Conversion of nitrogen from the air (N_2) to a form available to plants (NH_4^+ or NO_3^-) is performed either by biological nitrogen fixation (by *Rhizobium* bacteria associated with the roots of leguminous plants) or synthetically in industrial processes (chapter 8). In the latter case, the use of fossil energy is the limiting factor, not the availability of nitrogen.

Phosphorus reserves are documented in Althaus *et al.* (2007).

Known reserves of potassium amount to between $8.4 \cdot 10^9$ and $17 \cdot 10^9$ tonnes of K_2O (US Geological Survey 2002). Most of the reserves are located in North America and the former Soviet Union. The amount of potassium salts mined was $26.5 \cdot 10^6$ tonnes of K_2O in 2000, which is between 0.16% and 0.32% of known reserves. If annual consumption remains constant, known reserves will be sufficient for the next 320 to 640 years.

Ore and mineral availability and reserves are documented in the respective ecoinvent reports.

3 Agricultural Production in Switzerland

Of Switzerland's total area (4.13 million ha), 1.60 million ha are used for agricultural production (36.9% of the total area, BFS 2002). Permanent pastures and meadows comprise by far the largest part of the agricultural surface (Tab. 3.1). About half of this surface is alpine pastures of only low productivity. Of the arable land, about half is devoted to cereals, and one quarter to temporary leys.

In 2004, Swiss agriculture employed about 190,000 workers - 70% full-time and 30% part-time - on approx. 64,000 farms (BLW 2006). The average farm size was 19.37 ha that year. In a comparison between farm sizes in Switzerland on the one hand and Austria, Germany, France and Great Britain on the other, Baur (2000) showed that Swiss farms are substantially smaller than those of the other four countries, being namely between 14% and 60% of their size. Likewise, the number of agricultural workers per 100 ha of agricultural land in 1995/1996 was much higher in Switzerland (14.1 workers/100 ha), than in the other countries (5.4 in Austria, 4.1 in Germany, 3.3 in France and only 2.2 in Great Britain). These figures make it clear that Swiss farms are comparatively small with a high number of agricultural workers per area unit. This must be borne in mind if the data, defined for Swiss agricultural production, should be used for other countries.

The value of total Swiss agricultural production in 2005 amounted to CHF 10.3 billion. Milk accounted for 22% of this figure, meat and egg production for 25%, and plant production for 43%. In many parts of Switzerland, e.g. in the mountains and in regions with too-high precipitation levels for arable crops, climatic or soil conditions imply that grazing is the only agricultural use to which the land may be put. Consequently, animal production with an emphasis on cattle is of great importance in Switzerland.

Since 1993, Swiss agricultural policy has turned from the promotion of agricultural production first and foremost, towards more environmentally friendly production practices (Anwander Phan-Huy 2000). Special direct payments were introduced to reduce the use of pesticides and mineral fertilisers and promote biodiversity (see BLW 2002). Other measures were geared towards animal welfare. This change in agricultural policy led to a marked increase in ecological compensation areas and a decrease of about one third in pesticide use in the period between 1989 and 1998. The surplus in the national N-balance decreased from 133,000 t in 1990-1992 to 115,000 t in 1998, while the surplus in the national P-balance has decreased by more than 10,000 t (BLW 2001). The change in agricultural policy has also led to an extensification of agricultural production.

In 2004, 10.8% of agricultural land was cultivated according to the rules of organic production, while less than 6% was cultivated conventionally. The rest was farmed according to "integrated production" standards (BLW 2006).

Tab. 3.1 Agricultural land usage in the year 2004. Source: BLW (2006), completed from SBV (2006) for alpine pastures.

	Area in ha in 2004
Cereals (including grain maize)	161753
Wheat	85735
Spelt	2249
Rye	1680
Barley	37401
Oats	3028
Triticale	12400
Grain maize	18816
Cereal mixes	281
Silage maize	42433
Potatoes	13335
Sugar beets	18622
Fodder beets	1652
Grain legumes	4925
Protein peas	4600
Faba beans (broad beans)	249
Lupins	76
Oil seeds	23227
Rape seed	15751
Sunflower	4981
Soybean	2495
Renewable resources	1239
Vegetables	8813
Fallow	3592
Other arable land	1712
Open arable land	281303
Temporary leys	124474
Other surfaces	3069
Total arable land	408846
Fruit production	6733
Vineyards	14937
China reed	238
Permanent meadows and pastures	624337
Other usage	9483
Total agricultural land (except alpine pastures)	1064574
Alpine pastures	537801
Total agricultural land (including alpine pastures)	1602375

4 System Characterisation

This chapter documents the principal assumptions made for the establishment of the life cycle inventories for agricultural production (chapters 5 to 15). Points concerning specific inventories only can be found in the relevant chapters.

4.1 LCAs of Agricultural Systems

The life cycle assessment method was first developed to assess the environmental impacts of industrial processes. In terms of their impact on the environment, agriculture and industrial processes differ in some important respects:

- Agriculture is very intensive in terms of land use.
- Agricultural production relies heavily on natural resources.
- Agricultural production is dependent on soil, water availability, the weather, and the presence or absence of weeds, insect pests and pathogens. For these reasons, yields can vary greatly from year to year.
- There is a strong seasonality of agricultural production in most regions, which depends on temperature and the availability of water.

Various adaptations were necessary in order to apply the LCA method to agricultural systems. These questions, which concern the aspects of system boundaries, allocation and environmental impacts, were addressed by Sleeswijk *et al.* (1996), Audsley *et al.* (1997), FAL (2002) and others.

Below, we describe the system boundaries, basic assumptions, organisation of the infrastructure modules and models for the calculation of direct field emissions.

4.2 System Boundaries

4.2.1 Temporal System Boundaries

The temporal system boundaries of products from plant production were fixed as follows: the inventory starts after the harvest of the preceding crop and ends at the harvest of the crop in question. In Switzerland a large variety of crop rotations are practised. There is no “typical” crop rotation; in fact any crop can be preceded by a variety of other crops. The start of the inventory was therefore set at the time of soil cultivation. Post-harvest treatments directly related to the crop, such as stubble cultivation, were included in the inventories of the harvested crop.

Spring-sown crops have a considerably shorter vegetation period than autumn-sown crops. In integrated and organic production in Switzerland (see *Direktzahlungsverordnung*¹), the soil must be covered to a large extent during winter. A green manure (catch crop) reduces the risk of soil erosion and nitrate leaching during winter and helps to preserve the nutrients in the soil on the one hand, but on the other hand requires the use of machinery and energy. To take this into account, green manure has been included in the inventories of spring-sown crops. This means that green manure during the winter is part of the system of a spring-sown crop. For autumn-sown crops, not only is green manuring unnecessary, it is not possible in most cases, as the soil is already covered by the crop itself.

For permanent crops (permanent meadows in chapter 16), the inventories were calculated for a period of 12 months, from January to December.

¹ Verordnung über die Direktzahlungen in der Landwirtschaft (Direktzahlungsverordnung, DZV), 7.12.1998.

4.2.2 Process-Related System Boundaries

The diagrams in Fig. 4.1 and Fig. 4.2 show how the agricultural system was modelled for the inventories in ecoinvent, as well as the recommended way to use the ecoinvent modules for other studies of arable and roughage crops.

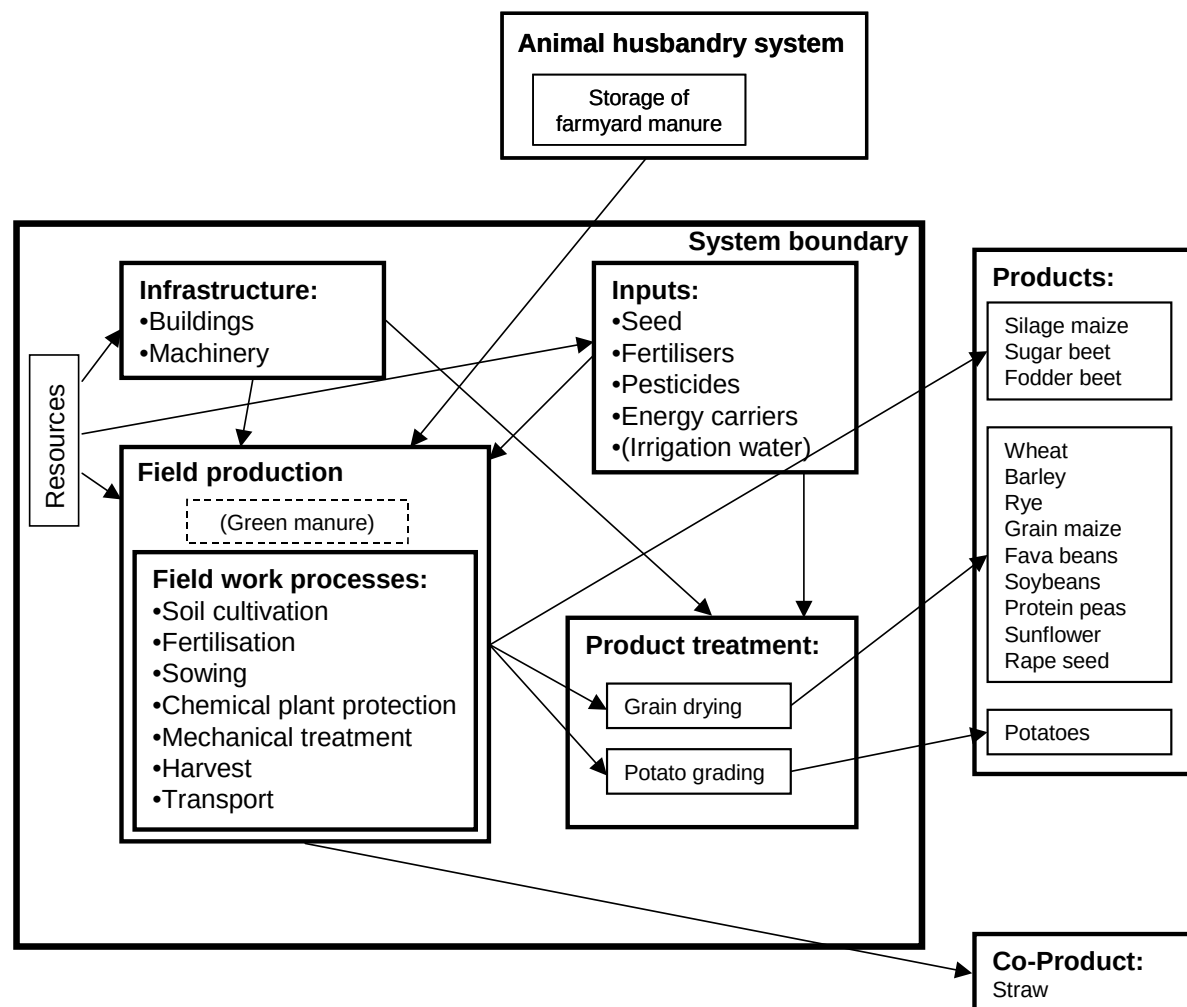


Fig. 4.1 Schematic representation of the processes considered in a life cycle inventory of an arable cropping system (products “at farm”, as considered in chapters 11, 14 and 15). Items in parentheses are included in some inventories only (green manure) or not at all (irrigation).

Fig. 4.1 shows the process-related system boundaries for the modelling of arable cropping systems (cultivation of arable crops, seed, etc.).

Green manure was included for spring-sown crops only (see chapter 4.2.1).

Irrigation was not included in the ecoinvent data inventories, according to the basic assumptions (see chapter 4.3). Irrigation should, however, be considered in applications where the land is irrigated. Chapter 7 describes modules that can be used for this purpose.

Agricultural land use was included directly in field production, since it is an integral part of the production process.

The production and storage of farmyard manure was wholly allocated to the animal husbandry system. For this reason, no emissions from animal husbandry or from manure storage were included in the plant production inventories, where farmyard manure is used. Likewise the infrastructure and

feedstuffs used for animal husbandry and manure storage were not included in the inventories for plant production, but in the animal husbandry system. On the other hand, emissions from manure spreading on fields and meadows, e.g. ammonia, heavy metals, etc., were included.

Fig. 4.2 shows how roughage production systems were modelled in ecoinvent. Soil cultivation, seed and sowing are used only if a temporary ley on arable land is being considered. The three inventories described in chapter 16 deal with permanent grassland and thus do not include these processes or the input of seed. Unlike the inventories for arable crops, those for roughage (hay) also include conservation and storage of the dry roughage.

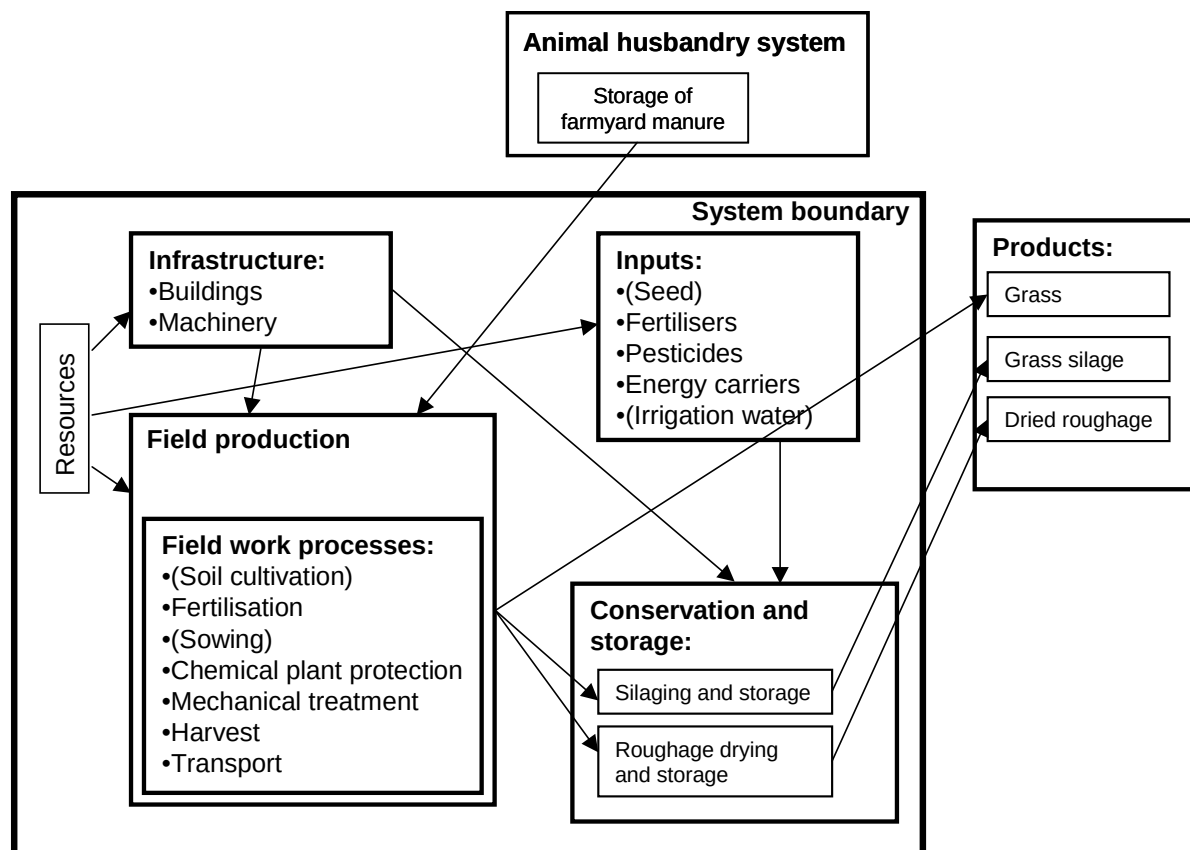


Fig. 4.2 Schematic representation of the processes considered in a life cycle inventory of a roughage production system (products “at farm”, as considered in chapter 15). Items in parentheses are not included in the inventories in ecoinvent.

Fig. 4.3 shows the recommended use of an animal production system. The only datasets of this category available in ecoinvent data are the sheep husbandry datasets for the USA (ecoinvent report no. 15b). The system is described here as an example of this kind of usage. Most of the elements (except the animals) are available as modules in ecoinvent data.

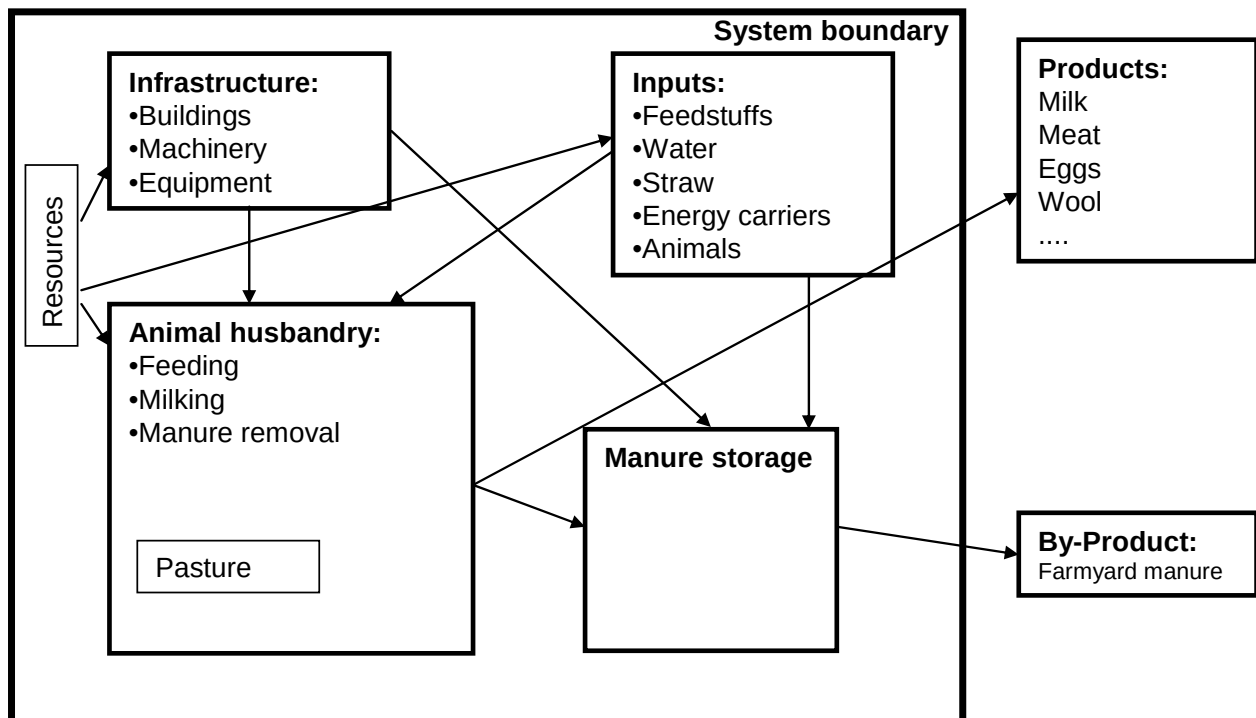


Fig. 4.3 Schematic representation of the processes considered in a life cycle inventory of an animal production system "at farm". No such system was included in the ecoinvent data.

The following exchanges (resources and emissions) were not considered in the inventories:

- toxic emissions to food and feedstuffs (e.g. heavy metals and pesticides), since the 'food' and 'feedstuffs' categories are not taken account of in ecoinvent
- impacts on the soil other than inputs of toxic substances: physical impacts (e.g. soil compaction), supply of organic material influencing soil biology, etc.
- effects on landscape structure and image
- effects on biodiversity
- odours
- noise.

These aspects should be borne in mind when studying the environmental impacts of agricultural systems.

Use of sewage sludge as an agricultural fertiliser is not included in the inventories in this report. This subject is treated in Doka (2007). For the life cycle inventories of wastewater treatment services, the spreading of sewage sludge on agricultural land and the emissions related to this process are assigned to the process of wastewater treatment. Emissions from the spreading of sewage sludge can be calculated according to the models described in chapter 4.4.

4.2.3 Infrastructure and its Operation

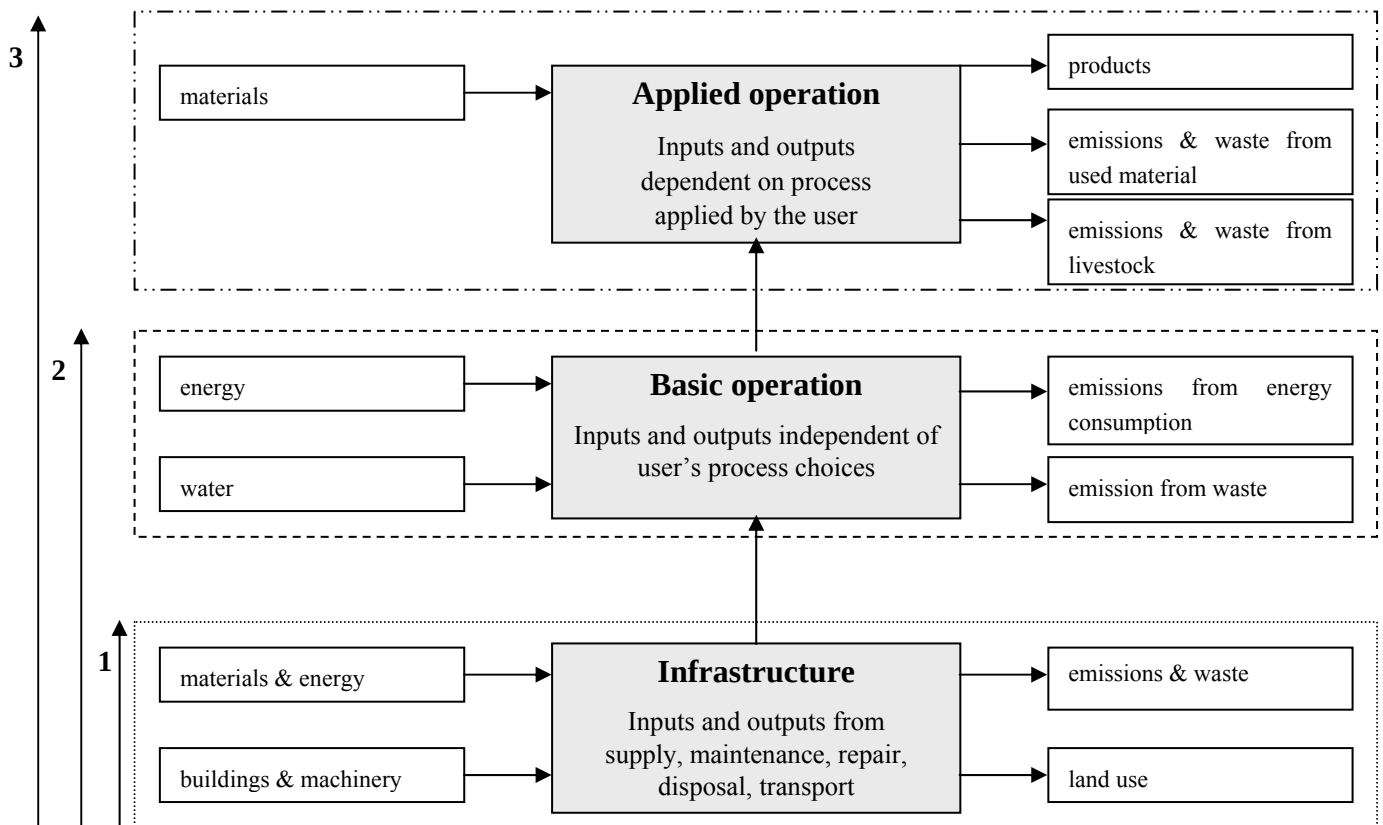


Fig. 4.4 Modelling of infrastructure and its operation in ecoinvent data.

Infrastructure and its operation were modelled on three levels (Fig. 4.4). Because of the various ways of using the infrastructure, a modular representation offers maximum flexibility of use of these modules.

Level 1: Infrastructure

The infrastructure modules (buildings and machinery, chapters 5 and 6) include, respectively, the construction and production of the infrastructure unit, as well as the transport (raw materials, machinery), maintenance, repair and final disposal. Examples of such modules are “tied housing system, cattle” or “tractor, production”.

Level 2: Basic operation

The basic operation modules in chapters 5 and 7 include the inputs and outputs independent of user choices, as well as the infrastructure used (level 1). The module “fertilising, by broadcaster”, for example, includes the amount of infrastructure used (tractor, fertiliser spreader and shed, as well as the input of diesel and the emissions related to the use of the tractor (exhaust gases, tyre wear). Because the module can be used for a variety of different fertilisers, the fertiliser itself and all emissions related to the spreading of fertiliser on the field (e.g. N- and P-containing emissions, heavy metals contained in the fertilisers) are not included in this module.

Examples of basic operation modules are “tied housing system, cattle, operation” or “slurry spreading, by vacuum tanker”.

Level 3: Applied operation

The applied operation includes all inputs and outputs related to operation which are dependent on user choices as well as on the infrastructure used (level 1) and the basic operation (level 2). Examples of such inputs are fertilisers, seed, pesticides, farmyard manure, feed for animals, etc. Examples of emissions are field emissions from fertilisers and pesticides, emissions from livestock, and emissions from the storage of farmyard manure.

No modules for applied operation in the narrower sense were included in ecoinvent data; these processes have to be modelled by the user of the datasets. The applied operation was considered in the modules in chapters 11, 14, 15 and 16, where inputs and emissions caused by applied operation were included.

Users of the infrastructure- and basic-operation modules (levels 1 and 2, respectively) must ensure that all inputs and outputs stemming from applied operation are properly taken account of.

4.3 Basic Assumptions for the Inventories of Plant Production Systems

The following general assumptions are valid for the Swiss plant-production system modules in this report:

- Fields and meadows are *not irrigated*. This is the most frequent practice in Switzerland and Europe for the crops considered in ecoinvent, and tallies with the data source used (LBL *et al.* 2000).
- The field was assumed to have a *slight slope of 5%* (Nemecek *et al.* 2005, Appendix 3.1.3; value valid for the lowlands). The field slope mainly affects soil erosion and P-emissions to the water. For the European datasets the values given by local experts were used.
- *Humus content* was assumed to be 2%, *clay content* 20% and *potential rooting depth* 80 cm (Nemecek *et al.* 2005, Appendix 3.1.3). These factors affect the quantity of nitrate leached.
- The field is situated in the *lowlands*. The majority of arable crops are cultivated in the lowlands, and most seed production takes place there as well. Nevertheless, a large proportion of grassland is located in the hills and mountains, and although studies (Nemecek & Huguenin 2002, Nemecek *et al.* 2005) have shown that the differences between the lowlands and mountainous regions in terms of environmental impacts were found to be relatively small, this fact must be borne in mind.
- The soil was assumed to be of *average erodibility*.
- The field plot was assumed to have *no drainage*. The majority of the fields and meadows in Switzerland are not drained². For the canton of Zurich, for instance, the percentage of drained agricultural area lies between 7 and 38%, depending on the region (Schmid & Prasuhn 2000). For the other regions in Europe the same assumption was made.
- *Fertilisation* follows *current recommendations* (Walther *et al.* 2001). In order to obtain direct payments³, the farmer must have a balanced nutrient balance. The fertilising recommendations (Walther *et al.* 2001) form the basis for calculating the nutrient balance. Consequently, it is likely

² Personal communication from V. Prasuhn, ART, September 2002.

³ Verordnung über die Direktzahlungen in der Landwirtschaft (Direktzahlungsverordnung, DZV), 7.12.1998.

that farmers generally follow these recommendations. Nevertheless, it is possible to deviate from these recommendations to a certain extent: there is a tolerance of up to 10% for a positive nutrient balance. Furthermore, a farmer may apply more fertilisers than recommended to one crop, and less to another.

- No special measures are taken to prevent *ammonia losses*. This corresponds to the most frequent practice in Switzerland (Reidy & Menzi 2005) and Europe and is in accordance with the data source chosen for the use of machinery (LBL *et al.* 2000).
- No special measures are taken to prevent *soil erosion*, except the application of green manure for spring-sown crops. This is in accordance with the data source chosen for the use of machinery (LBL *et al.* 2000).
- The average density of livestock units (LU) per hectare was set at 1.2 LU/ha (BLW 2001). This value was used to calculate the potential N-mineralisation of the soil, except for the extensive meadow, where no fertiliser is applied at all. No distinction has been made between integrated and organic farming, even if fertilising practise is different. Organic farms apply more manure to arable crops than do integrated farms. If the entire crop rotation is considered, however, this difference almost disappears (FAT 2000a), since the farmyard manure is applied to a larger extent to the meadows in the integrated farm.

The Swiss plant production inventories in ecoinvent refer to this “standard situation”. In conditions differing from this situation, the emissions may differ substantially from the values in ecoinvent data.

4.4 Direct Field Emissions

4.4.1 Emissions of Ammonia to the Air

Ammonium (NH_4^+) contained in fertilisers can easily be converted into ammonia (NH_3) and released to the air. Agriculture is the biggest source of ammonia emissions in Switzerland. For 2000, Thöni *et al.* (2007) estimated the total emissions of NH_3 to be 53,000 tonnes, thereof 93% from agriculture. Animal husbandry (emissions in the stable, during manure storage and spreading) is the largest source. About 30% of the excretions of N are lost in the form of ammonia. By taking appropriate measures, these emissions could be reduced by about 20-40% (Menzi *et al.* 1997).

Ammonia contributes to acidification and the eutrophication of sensitive ecosystems. Its impact is mainly local and regional.

A comparison of different emission factors for ammonia can be found in Menzi *et al.* (1997).

Slurry and Liquid Manure

The losses of NH_3 during the spreading of farmyard manure were calculated with the models given by Katz (1996) and Menzi *et al.* (1997).

The NH_3 -emissions during the spreading of slurry and liquid manure are:

$$\text{NH}_{3s} = 17/14 * (-9.5 + 19.4 \text{ TAN} + 1.1 \text{ SD}_m) * (0.0214 \text{ S} + 0.358) * A_s$$

NH_{3s} = emission of NH_3 from slurry or liquid manure (kg NH_3 /ha)

TAN = total content of ammonium-N in the slurry or liquid manure (kg $\text{NH}_4\text{-N}/\text{m}^3$)

S = quantity of slurry spread (m^3/ha of fertilised surface), including the dilution water

SD_m = saturation deficit of the air in month *m*

A_s = fraction of the total area, where slurry is spread (%/100)

The specific weight of slurry was taken as 1t/m^3 (Menzi *et al.* 1997).

The saturation deficit of the air was calculated from the average monthly values for temperature (T_m) and relative humidity (rH_m) of 25 stations in the Swiss lowlands for the years 1994-2000 (see Tab. 4.1). The saturation deficit is given by:

$$SD_m = (1 - rH_m) * 6.112 * e^{((17.67T_m)/(243.5 + T_m))}$$

rH_m = average relative humidity in month m (%/100)

T_m = average temperature in month m ($^{\circ}\text{C}$)

In the calculations for ecoinvent data, it was assumed that no additional measures are taken to reduce the ammonia emissions.

Tab. 4.1 Saturation deficit (SD) in the different months. Values refer to the average values for the Swiss lowlands.

Month	Jan.	Feb.	March	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
SD	1.3	1.8	3.1	3.7	5.0	5.5	6.7	6.3	3.7	2.4	1.6	1.3

The factor A_s is used to adjust the emissions from average fertilisation values. The problem of the NH_3 -emission equations is that they result in too high emissions, if the rate of slurry or liquid manure application is unrealistically low (the same applies for solid manure below). For very low values of S , more N would be released than is present in the slurry or liquid manure. As the values are averages of many farms, where some farmers apply manure and others do not, this situation occurs quite often. To correct this we calculate with the average manure application rate of the farmers using slurry and liquid manure ($40\text{ m}^3/\text{ha}$ and application) and adjust A_s accordingly. This is explained by the following example: in case the average slurry application rate of all farmers is $10\text{ m}^3/\text{ha}$, we set $S=40\text{ m}^3/\text{ha}$ and $A_s=0.25$, in case it is $20\text{ m}^3/\text{ha}$, we set $S=40\text{ m}^3/\text{ha}$ and $A_s=0.5$. Should the average application rate be above $40\text{ m}^3/\text{ha}$, then no correction is applied and $A_s=1$.

For liquid sewage sludge (not used in this ecoinvent report, but in Doka 2007), the NH_3 -emission was estimated as follows:

60% of the ammonium is emitted as ammonia (Menzi *et al.* 1997). 43% of the total N is in the form of NH_4^+ (Külling *et al.* 2002, data for 1999). This means that 26% of the total N is emitted in the form of ammonia.

Solid Manure

The emissions during the application of solid manure (from cattle and pigs) were calculated as follows:

$$\text{NH}_{3M} = 17/14 * (0.787 \text{ TAN} * M + 0.757) * 0.75 * A_M$$

NH_{3M} = emission of NH_3 from solid manure ($\text{kg NH}_3/\text{ha}$)

TAN = total content of ammonium-N in the manure ($\text{kg NH}_4\text{-N/t}$)

M = quantity of solid manure spread (t/ha of fertilised surface)

A_M = fraction of the total area, where solid manure is spread (%/100)

For poultry manure, the following emission factors were used (N-losses in the form of ammonia referring to the total N-content): 20% (litter from deep pits from laying hens), 25% (litter from belts from laying hens) and 15% (litter from broilers, see Menzi *et al.* 1997).

The contents of total N and soluble N were taken from Walther *et al.* (2001).

The average application rate of farmers using solid manure is 16 t/ha and application. Similarly to slurry and liquid manure, this value was used in the formula and A_m was adjusted accordingly. Example: in case the average of farmers is 2 t/ha, then $M=16$ t/ha and $A_m=0.125$.

Mineral Fertilisers

The emission factors for mineral fertilisers given by Asman (1992) were used (Tab. 4.2). The emissions were calculated as % of N emitted and subsequently converted to NH_3 .

Tab. 4.2 NH_3 -emissions from mineral fertilisers (% N emitted in form of NH_3).

Type of fertiliser	Emission factor for NH_3 -N
ammonium nitrate, calcium ammonium nitrate	2 %
ammonium sulphate	8 %
urea	15 %
multinutrient fertilisers (NPK-, NP-, NK-fertilisers)	4 %
urea ammonium nitrate	8.5 % ^{*)}

^{*)} The average of ammonium nitrate and urea was taken, since no emission factor is given by Asman (1992).

4.4.2 Nitrate Leaching to Ground Water

Nitrate (NO_3^-) is either supplied to the soil by fertilisers or produced by micro-organisms in the soil via the mineralisation of organic matter. Nitrate in the soil can be absorbed as a nutrient by the plants. In periods of heavy rainfall, however, precipitation exceeds soil evaporation and transpiration of the plants, which leads initially to saturation of the soil with water, and afterwards to percolation to the ground water. As nitrate is easily dissolved in water, the risk of leaching is high. This situation is quite frequent in Switzerland.

The risk of nitrate leaching is highest in autumn and winter, when precipitation often or always exceeds uptake by the plants. Moreover, nitrogen mineralisation is highest in late summer, when the nitrogen often cannot be taken up by the plants (Stauffer *et al.* 2001).

Experiments have shown that it is not the choice of crops but rather the succession of crops in a crop rotation that is decisive for the amount of nitrate leached (Stauffer *et al.* 2001). Since the modules in the ecoinvent database are life cycle inventories of products taking into account one single crop only, the succession of crops cannot be accounted for properly. This fact should be borne in mind when interpreting the nitrate leaching values.

Nitrate losses are undesirable for several reasons:

- From the agricultural point of view, valuable nutrients are lost from the soil, increasing the need for fertilisers.
- Nitrate in ground water used as drinking water may have a toxic impact to humans. Although the acute toxicity of nitrate is low, nitrate is easily converted into nitrite, which has a higher acute toxicity and is supposed to be indirectly carcinogenic (Surbeck & Leu 1998).
- Once ground water becomes surface water, nitrate contributes to eutrophication and also induces emissions of nitrous oxide, a major greenhouse gas (Schmid *et al.* 2000).

The tolerance level for nitrate in drinking water is 40 mg/l in Switzerland and 50 mg/l in the EU, while the Swiss quality goal is 25 mg/l maximum. Results from the Swiss monitoring network NAQUA (Greber *et al.* 2002) show that these levels are exceeded only in areas with arable crops, or in fruit- and wine-growing areas. In areas with forests or permanent grassland, these levels are never exceeded. This shows the importance of arable crops and soil cultivation in nitrate leaching.

Nitrate emissions to ground water can be estimated by simulation models, although this method is very complex and time-consuming and does not always lead to very satisfactory results (Oberholzer *et al.* 2001). A comparison of different methods for estimating nitrate leaching is given in Audsley *et al.* (1997).

Potential nitrate leaching was calculated by a model comprising the following elements (Richner *et al.* 2006):

- Nitrogen mineralisation from the soil organic matter per month,
- Nitrogen uptake by vegetation (if any) per month,
- Nitrogen input from the spreading of fertiliser and
- soil depth.
- Not considered factors:
- amount of seepage,
- denitrification.
- The following description is an extract of the full description of the model SALCA-nitrate by Richner *et al.* (2006). The reader is referred to this report for further information.

The model of Richner *et al.* (2006) calculates the potential nitrate leaching of arable crops, meadows and pasture land considering not only crop rotation, soil cultivation, nitrate fertilising but also nitrate mineralisation from the soil organic matter, nitrate uptake by the plants and various soil conditions. The model is valid for the Swiss lowland and adjoining regions. The calculation bases on the monthly difference between the amount of mineralized nitrate in the soil and the nitrate uptake potential of the plants. Furthermore, the nitrate leaching risk from fertiliser application during inappropriate time periods is taken into account. The nitrate leaching potential of pastures rises because of locally high nitrate concentrations. Therefore the total amount of nitrate on pastures is calculated against the amount of animals and the grazing duration.

The total potential nitrate leaching of an arable crop is assessed by the sum of the monthly values within the assessment period starting one month after the harvest of the former crop and ending in the month of harvesting of the given crop.

Tab. 4.3 Nitrogen mineralisation potential ($N_{\min}$, kg N per ha and month, from Richer *et al.* 2006) in soils with 15% clay and 2% humus Intensive soil cultivation means treatment by a rotary cultivator or a rotary harrow in the respective month. In months where there is no intensive soil cultivation, the value "Without intensive soil cultivation" is used.

	Jan.	Feb.	March	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Without intensive soil cultivation	0	0	10	15	20	25	30	35	40	20	10	0
With intensive soil cultivation	0	0	15	20	30	35	50	60	70	40	20	0

In addition, nitrogen mineralisation was further corrected for clay and humus content of the soil (Tab. 4.4) as well as for green manuring and tillaging of pastures (see Richner *et al.* 2006).

Tab. 4.4 Correction of nitrate mineralization against the clay and humus content of the soil.

		Humus content (%)			
		<3	3-5	5-8	8-15
Clay content (%)	0-20	0	+10	+20	+40
	20-30	-10	-5	+5	+25
	30-40	-20	-20	-10	+5
	>40	-30	-30	-25	-15

Tab. 4.5 Nitrogen uptake by vegetation ($N_{\text{upt m}}$, kg N per ha and month, from Richer *et al.* 2006). Int = with intensive disease and insect control, Ext = with extensive disease and insect control. For the inventories for integrated production (IP), the columns denoted by "Int" were applied, whilst the columns denoted by "Ext" were applied for the inventories for "extensive" and "organic" production.

Months	Winter wheat		Winter barley		Winter rye		Grain maize	Silage maize	Winter rape seed		Potatoes	
	Int	Ext	Int	Ext	Int	Ext	Int/Ext	Int/Ext	Int	Ext	Int	Ext
January	0	0	0	0	0	0	-	-	5	5	-	-
February	15	15	15	15	15	15	-	-	10	10	-	-
March	30	30	30	30	25	25	-	-	30	30	-	-
April	40	40	45	45	35	35	-	-	40	35	15	15
May	60	50	50	40	45	40	10	10	50	40	30	30
June	50	40	20	10	40	35	40	40	40	30	120	80
July	30	20	0	0	20	15	50	50	20	10	30	15
August	-	-	-	-	-	-	40	40	-	-	20	10
September	-	-	5	-	-	-	30	30	30	30	10	0
October	10	10	15	15	10	10	20	10	60	60	-	-
November	10	10	10	10	10	10	10	-	20	20	-	-
December	5	5	5	5	5	5	-	-	10	10	-	-

Months	Potato seed		Sugar beets	Fodder beets	Sun-flower	Faba beans (spring sown)	Protein peas (spring sown)	Soy beans	Permanent meadow	Permanent meadow
	Int	Ext	Int/Ext	Int/Ext	Int/Ext	Int/Ext	Int/Ext	Int/Ext	Int	Ext
January	-	-	-	-	-	-	-	-	0	0
February	-	-	-	-	-	-	-	-	15	5
March	10	10	-	-	-	5	5	-	30	10
April	30	30	15	20	10	15	15	-	40	20
May	50	45	35	40	35	20	20	10	40	25
June	90	60	70	75	50	25	25	25	40	25
July	-	-	60	65	50	30	30	30	40	25
August	-	-	45	60	35	30	20	35	40	25
September	-	-	40	50	10	20	-	20	40	20
October	-	-	20	20	-	-	-	-	30	15
November	-	-	-	-	-	-	-	-	10	5
December	-	-	-	-	-	-	-	-	5	0

Based on the average number of livestock units of farms in the Swiss lowlands (BLW 2001), St was set at 1.2 LU/ha for all calculations, except for the extensive meadow, where St=0, since no fertiliser is applied (see chapter 4.3). For the European datasets, a farm without livestock was assumed.

The risk of nitrogen leaching due to fertiliser application is dependent on the crop and the month in which fertiliser was applied (Tab. 4.6.; Richner *et al.* 2006).

Tab. 4.6 Risk of nitrogen leaching (fraction of potentially leachable nitrogen of the N applied through fertilisers in %, from Richner *et al.* 2006).

Months	Winter cereals		Maize, soya beans	Winter rape seed and green manure		Potato, sugar and fodder beets	Faba beans, protein peas (spring sown)	Sun-flowers	Perma- nent meadow Int	Perma- nent meadow Ext
	sowing year	harvest- year	harvest- year	sowing year	harvest- year	harvest- year	harvest- year	harvest- year		
January	100	50	100	100	20	100	100	20	100	20
February	100	30	100	100	10	100	100	10	100	20
March	100	10	100	100	0	50	50	0	50	0
April	100	0	80	100	0	30	30	0	30	0
May	100	0	70	100	0	10	0	0	0	0
June	100	0	0	100	0	0	0	0	0	0
July	100	-	0	100	-	0	0	0	0	0
August	100	-	0	80	-	0	-	0	0	0
September	90	-	0	0	-	0	-	0	-	0
October	90	-	-	0	-	-	-	0	-	0
November	90	-	-	20	-	-	-	10	-	20
December	90	-	-	20	-	-	-	20	-	20

The correction of the potential nitrate leaching due to fertiliser application against the depth of the soil is listed in Tab. 4.7.

Tab. 4.7 The correction of the potential nitrate leaching due to fertiliser application against the depth of the soil (Richner *et al.* 2006)

Soil depth (cm)	Correction (%)
>100	0
90-100	+5
80-90	+10
70-80	+15
60-70	+20
50-60	+25
40-50	+30
<40	+35

There is no leaching water during the intensive vegetation period because the evapotranspiration is similar or higher than the precipitation. Therefore no nitrate leaching occurs during these periods. For various crops fertilising is only possible shortly before the growing period due to agronomic or technical reasons. The model accumulates the monthly values of nitrate mineralisation, nitrate uptake by the plants and the nitrate from fertilising during this period (Tab. 4.8).

Tab. 4.8 Accumulation of the monthly values of nitrate mineralisation, nitrate uptake by the plants and the nitrate from fertilising for various crops (Richner *et al.* 2006).

	Month											
crops	J	F	M	A	M	J	J	A	S	O	N	D
winter cereals												
spring cereals												
maize, soy beans												
potatoes												
sugar beets, fodder beets												
sunflowers												
faba beans, protein peas (spring sown)												
permanent meadow												

A transformation factor of 0.8 is applied to the calculated potential emissions, in order to estimate the effective leaching rate. The amount of NO₃ leached calculated by the method above was multiplied with this factor (see Richner *et al.* 2006).

4.4.3 Emissions of Phosphorus to the Water

Phosphorus (P) is an important plant nutrient and must be supplied to the plants in sufficient quantities. A part of the phosphorus is lost to water due to leaching, run-off and soil erosion through water, causing eutrophication, P is a limiting element. Soil erosion by wind is not of importance in Switzerland.

Phosphorus can cause eutrophication of water (Prasuhn & Grünig 2001).

We distinguish between three different kinds of phosphorus emissions to water:

- leaching of soluble phosphate to ground water (inventoried as “phosphate, to ground water”),
- run-off of soluble phosphate to surface water (inventoried as “phosphate, to river”),
- erosion of soil particles containing phosphorus (inventoried as “phosphorus, to river”).

The emission models SALCA-P (Prasuhn 2006) developed by ART are applied in ecoinvent data. A comparison of different emission coefficients is given by Audsley *et al.* (1997), Schmid & Prasuhn (2000) and Prasuhn & Grünig (2001).

The following factors are considered for the calculation of P emissions in ecoinvent data inventories:

- type of land use
- type of fertiliser
- quantity of P in fertilisers
- type and duration of soil cover for the calculation of the soil erosion (C-factor).

For other factors, considered in the model SALCA-P, default values are used:

- distance to next river or lake
- topography
- chemical and physical soil properties
- drainage. As the field was assumed to have no drainage (see chapter 4.3), the emissions to surface water through drainage were not taken in account.

The model takes soil erosion, surface run off and drainage losses to surface water and leaching to ground water into account.

It should be borne in mind that the values are valid for the soil and site parameters chosen. Changes in soil conditions or in cropping practice could lead to emissions substantially different from the ones calculated in ecoinvent data.

The key factors of the model are listed below. Please see Prasuhn (2006) for detailed calculations.

Phosphorus Leaching to Ground Water

P leaching to the ground water was estimated as an average leaching, corrected by P-fertilisation:

$$P_{gw} = P_{gwl} * F_{gw}$$

$$P_{gw} = \text{quantity of P leached to ground water (kg/(ha*a))}$$

$$P_{gwl} = \text{average quantity of P leached to ground water for a land use category (kg/(ha*a)), which is } \begin{matrix} 0.07 \text{ kg P/(ha*a) for arable land and} \\ 0.06 \text{ kg P/(ha*a) for permanent pastures and meadows.} \end{matrix}$$

$$F_{gw} = \text{correction factor for fertilisation by slurry (-)}$$

$$F_{gw} = 1 + 0.2/80 * P_{2O_{5sl}}$$

$$P_{2O_{5sl}} = \text{quantity of } P_2O_5 \text{ contained in the slurry or liquid sewage sludge (kg/ha). The values of } P_2O_5\text{-content were taken from Walther } et al. (2001).$$

P Run-Off to Surface Waters

Run-off to surface waters was calculated in a similar way to leaching to ground water:

$$P_{ro} = P_{rol} * F_{ro}$$

$$P_{ro} = \text{quantity of P lost through run-off to rivers (kg/(ha*a))}$$

$$P_{rol} = \text{average quantity of P lost through run-off for a land use category (kg/(ha*a)), which is}$$

- 0.175 kg P/(ha*a) for open arable land,
- 0.25 kg P/(ha*a) for intensive permanent pastures and meadows and
- 0.15 kg P/(ha*a) for extensive permanent pastures and meadows

$$F_{ro} = \text{correction factor for fertilisation with P (-), calculated as:}$$

$$F_{ro} = 1 + 0.2/80 * P_{2O_{5min}} + 0.7/80 * P_{2O_{5sl}} + 0.4/80 * P_{2O_{5man}}$$

- $P_{2O_{5min}}$ = quantity of P_2O_5 contained in mineral fertilisers (kg/ha)
- $P_{2O_{5sl}}$ = quantity of P_2O_5 contained in slurry or liquid sewage sludge (kg/ha)
- $P_{2O_{5man}}$ = quantity of P_2O_5 contained in solid manure (kg/ha)

The values of P_2O_5 -content for slurry and manure were taken from Walther *et al.* (2001).

P Emissions Through Erosion by Water to Surface Waters

P emissions through erosion to surface waters were calculated as follows:

$$P_{er} = 10000 * S_{er} * P_{cs} * F_r * F_{erw}$$

$$P_{er} = \text{quantity of P emitted through erosion to rivers (kg P/(ha*a))}$$

$$S_{er} = \text{quantity of soil eroded (kg/(ha*a))}$$

$$P_{cs} = \text{P content in the top soil (kg P/kg soil). The average value of 0.00095 kg/kg was used.}$$

$$F_r = \text{enrichment factor for P (-). The average value of 1.86 was used (Wilke & Schaub 1996).}$$

This factor takes account of the fact that the eroded soil particles contain more P than the average soil.

$$F_{erw} = \text{fraction of the eroded soil that reaches the river (-). The average value of 0.2 was used.}$$

The amount of eroded soil S_{er} is calculated according to Oberholzer *et al.* (2006, Appendix A4.1).

4.4.4 Emissions of Nitrous Oxide (N₂O) to the Air

Nitrous oxide is produced as an intermediate product in the denitrification process (conversion of NO_3^- into N_2) by soil micro-organisms. It can also be produced as a by-product in the nitrification process (conversion of NH_4^+ into NO_3^- , Schmid *et al.* 2000). The total emissions of N_2O caused by the Swiss agricultural sector in 1996 were estimated at 8,600 tonnes. N losses in the form of N_2O are closely linked to the nitrogen cycle in agriculture; intensive agriculture with a high input of nitrogen fertiliser contributes to the increase in N_2O -emissions. N_2O is a greenhouse gas with a high impact.

Calculations of N_2O emissions are based on the IULIA model described by Schmid *et al.* (2000), which in turn is an adaptation of the IPCC method for calculating N_2O emissions (IPCC *et al.* 1996).

Direct emissions of N₂O and indirect or induced emissions are included. In the case of indirect N₂O emission, nitrogen is first emitted as NH₃ or NO₃⁻ and subsequently converted to N₂O.

N₂O emissions [kg N₂O] from mineral and organic fertilisers were calculated on the basis of the available nitrogen (N_{av} [kg N]). The factor of 1.25% N lost as N₂O was used. For mineral fertilisers, it was assumed that 100% of the nitrogen was available. The quantity of available nitrogen was reduced by losses in the form of ammonia. On the other hand, N₂O emissions induced by ammonia (NH₃ [kg NH₃]) are included. A part of the nitrogen that is leached in the form of nitrate is converted to N₂O. The emission factor of 2.5% (%N emitted in the form of N₂O) from Schmid *et al.* (2000) was used. No correction for nitrogen lost through nitrate leaching was applied in the equation. Nitrate leaching occurs partly due to N-fertilisation, but also partly due to nitrogen mineralised from the organic matter in the soil.

The content of available nitrogen in farmyard manure was taken from Walther *et al.* (2001).

$$\text{N}_2\text{O} = 44/28 * (0.0125 (\text{N}_{\text{av}} - 14/17 * \text{NH}_3 + \text{N}_{\text{cr}} + 0.6\text{N}_{\text{bf}}) + 0.01 * 14/17 * \text{NH}_3 + 0.025 * 14/62 * \text{NO}_3^-)$$

$$\text{N}_2\text{O} = \text{emission of N}_2\text{O (kg N}_2\text{O/ha)}$$

$$\text{N}_{\text{av}} = \text{available nitrogen (kg N/ha)}$$

$$\text{N}_{\text{cr}} = \text{nitrogen contained in the crop residues (kg N/ha)}$$

$$\text{N}_{\text{bf}} = \text{nitrogen from biological N fixation (kg N/ha), estimated by the quantity of N contained in the shoots of legumes.}$$

$$\text{NH}_3 = \text{losses of nitrogen in the form of ammonia (kg NH}_3\text{/ha)}$$

$$\text{NO}_3^- = \text{losses of nitrogen in the form of nitrate (kg NO}_3^-\text{/ha).}$$

4.4.5 Emissions of NO_x to the Air

During denitrification processes in soils, NO_x may also be produced. These emissions were estimated from the emissions of N₂O ⁴:

$$\text{NO}_x = 0.21 * \text{N}_2\text{O}$$

Since this process is not one of conversion from N₂O to NO_x, but a parallel process, no correction of the N₂O emissions is required.

This equation includes the direct NO_x emissions from fertilisers and the soil only. Other sources such as tractor exhaust gases are included in the respective inventories (see chapter 7).

4.4.6 Nutrient Inputs in Agricultural Soils

The input of nutrients (N, P, K, Ca, etc.) into the agricultural soil was not inventoried as emissions to the soil for the following reason: The inventories of agricultural products in ecoinvent data are based on the fertilising recommendations (Walther *et al.* 2001). These recommendations in turn are based on the assumption that the fertiliser should cover the needs of the plants. In a first step, the export of nutrients through the products (main- and co-products) was calculated. In a second step, the recommended fertiliser dose was calculated by accounting for various other aspects. The nutrients supplied to the soil will therefore either be exported in the products or lost to the air or water. The quantity of nutrients in the soil should not be changed on average in the long term.

⁴ personal communication from Grub, 1996

4.4.7 Release of Fossil CO₂ after Urea Applications

During the urea production process, CO₂ is used, which is chemically bound in the molecule. After application and transformation processes in the soil, this CO₂ is released to the atmosphere. Per kg of applied urea-N, 1570 g of fossil CO₂ are released that are inventoried as “Carbon dioxide, fossil” to “air, low population density” in ecoinvent data.

4.4.8 Emissions of Heavy Metals to Agricultural Soil, Surface Water and Ground Water

According to an analysis of the heavy metals that are causing problems in Swiss agriculture (Kühnholz 2001), the following seven were selected for the inventories in ecoinvent data:

Cadmium (Cd), Chromium (Cr), Copper (Cu), Lead (Pb), Mercury (Hg), Nickel (Ni) and Zinc (Zn).

Typical heavy-metal content of agricultural and non-agricultural soils is given by Desaulles & Dahinden (2000).

Kühnholz (2001) gives a comparison of different emission factors and methods for calculating heavy metal balances.

The heavy metal emissions were calculated by SALCA-heavy metal (Freiermuth 2006). Inputs into farm land and outputs to surface water and groundwater are calculated on the basis of heavy metal input from seed, fertilisers, plant protection products and deposition. Residues left on the field are not considered because they do not leave the system. For erosion of soil average heavy metal contents for arable land, pastures, meadows and intensive crops are used. The amount of eroded soil is calculated as for P-emissions with the method described in Oberholzer *et al.* (2006). An allocation factor is used to distinguish between diffuse and agriculture-related introduction (Freiermuth 2006). We give only a summary description of the method here. For a full description, the reader is referred to Freiermuth (2006).

Three types of emissions are considered in ecoinvent data:

- Leaching of heavy metals to the ground water (always positive values)
- Emissions of heavy metals into surface waters through erosion of soil particles (always positive values)
- Emissions of heavy metals to agricultural soil (positive or negative values).

The following sources were used to calculate heavy-metal contents:

- Mineral fertilisers: Desaulles & Studer (1993, p. 153), see Tab. A. 2 in the Appendix,
- Farmyard manure: Menzi & Kessler (1998) and Desaulles & Studer (1993, p. 152), see Tab. A. 3 in the Appendix,
- pesticides: FAW & BLW (2000),
- biomass (seed and products from plant production): Houba & Uittenbogaard (1994, 1995, 1996 & 1997), von Steiger & Baccini (1990) and Wolfensberger & Dinkel (1997), see Tab. A. 1 in the Appendix. For grass seed, the values of wheat grains were used; for clover seed, the values of protein peas.

Heavy metal emissions into ground and surface water are calculated with constant leaching rates as:

$$M_{\text{leach } i} = m_{\text{leach } i} * A_i$$

$M_{\text{leach } i}$ agricultural related heavy metal i emission

$m_{\text{leach } i}$ average amount of heavy metal emission (Tab. 4.9)

A_i allocation factor for the share of agricultural inputs in the total inputs for heavy metal i

Tab. 4.9 Heavy metal leaching to groundwater according to Wolfensberger & Dinkel (1997).

	Cd	Cu	Zn	Pb	Ni	Cr	Hg
mg/ha/year	50	3600	33000	600	n.a.	21200	1.3

Heavy metal emissions through erosion are calculated as follows:

$$M_{\text{erosion } i} = c_{\text{tot } i} * B * a * f_{\text{erosion}} * A_i$$

M_{erosion} agricultural related heavy metal emissions through erosion

$c_{\text{tot } i}$ total heavy metal content in the soil (Keller & Desaulles 2001, see Tab. 4.10)

B amount of soil erosion according to Oberholzer *et al.* (2006)

a accumulation factor 1.86 (according to Prasuhn 2006 for P)

f_{erosion} erosion factor considering the distance to river or lakes with an average value of 0.2 (considers only the fraction of the soil that reaches the water body, the rest is deposited in the field)

A_i allocation factor for the share of agricultural inputs in the total inputs for heavy metal i

Tab. 4.10 Heavy metal contents in mg per kg soil (from Keller & Desaulles 2001).

Land use	Cd [mg/kg]	Cu [mg/kg]	Zn [mg/kg]	Pb [mg/kg]	Ni [mg/kg]	Cr [mg/kg]	Hg [mg/kg]
Permanent grassland	0.309	18.3	64.6	24.6	22.3	24.0	0.088
Arable land	0.24	20.1	49.6	19.5	23.0	24.1	0.073
Intensive crops	0.307	39.2	70.1	24.9	24.8	27.0	0.077

The balance of all inputs into the soil (fertilisers, pesticides, seed and deposition) and outputs from the soil (exported biomass, leaching and erosion), multiplied by the allocation factor is calculated as an emission to agricultural soil.

$$M_{\text{soil } i} = (\sum \text{inputs}_i - \sum \text{outputs}_i) * A_i$$

Some of the values for emissions of heavy metals to the soil are negative. This means that more heavy metals are exported than imported. It must, however, be borne in mind that these heavy metals are transferred either to the water bodies or to the products harvested from the field (food, feed and straw).

A certain fraction of the heavy metal input into the soil stems from atmospheric deposition. The deposition would occur even without any agricultural production and is therefore not charged to the latter. An allocation factor accounts for this. The farmer is therefore responsible for a part of the inputs only (the rest stems mainly from other economic sectors), therefore only a part of the emissions is calculated in the inventory.

$$A_i = M_{\text{agro } i} / (M_{\text{agro } i} + M_{\text{deposition}})$$

A_i allocation factor for the share of agricultural inputs in the total inputs for heavy metal i

$M_{\text{agro } i}$ total input of heavy metal from agricultural production in mg/(ha*year) (fertilisers + seeds + pesticides)

$M_{\text{deposition}}$ total input of heavy metal from atmospheric deposition in mg/(ha*year) (Tab. 4.11)

Tab. 4.11 Heavy metal deposition (see Freiermuth 2006).

	Cd	Cu	Zn	Pb	Ni	Cr	Hg
Deposition [mg/ha/year]	700	2400	90400	18700	5475	3650	50

4.4.9 CO₂-Binding and Solar Energy in Biomass

The energy content and CO₂-binding in according to Tab. 4.12. Crop residues remaining in the field are not considered, since these usually decompose within a few years. The change in soil organic-matter content was not considered, i.e. the organic C-content of the soil was assumed to be constant.

Attention must be paid during application of these datasets in life cycle studies (e.g. if straw is combusted to produce heat or food products are consumed): the CO₂ and energy released from the agricultural products must be considered.

Tab. 4.12 CO₂-binding and gross calorific value of the exported biomass (referring to the dry matter, DM). These flows were inventoried as resources.

Product	A) Resource "Carbon dioxide, in air" (kg CO ₂ /kg DM):	B) Resource "Energy, gross calorific value, in biomass" (MJ/kg DM):	Sources:
Grain maize	1.60	18.52	A) Leifeld: Personal communication*), B) Diepenbrock (1995)
Silage maize	1.72	19.00	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Potatoes and potato seed	1.55	17.59	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Rape seed	2.86	26.48	A) Leifeld: Personal communication, B) Maier <i>et al.</i> (1998)
Sugar beets	1.40	16.43	A) RAP (1999), B) Diepenbrock (1995)
Fodder beets	1.40	16.51	A) RAP (1999), B) Diepenbrock (1995)
Sunflower grains	2.34	30.25	A) Leifeld: Personal communication, B) Maier <i>et al.</i> (1998)
Faba beans	1.46	18.52	A) RAP (1999), B) Diepenbrock (1995)
Soya beans	1.60	22.98	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Protein peas	1.47	18.48	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Wheat grains	1.55	18.11	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Rye grains	1.55	17.87	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Barley grains	1.56	18.12	A) Leifeld: Personal communication, B) Diepenbrock (1995)
Wheat straw, rye straw, barley straw	1.61	20.54	A) Leifeld: Personal communication, B) Maier <i>et al.</i> (1998)

Product	A) Resource "Carbon dioxide, in air" (kg CO ₂ /kg DM):	B) Resource "Energy, gross calorific value, in biomass" (MJ/kg DM):	Sources:
Hay intensive	1.65	17.9	A) Leifeld: Personal communication, B) S. Erzinger, FAT: pers. comm. 28.8.02
Hay extensive	1.65	18.9	A) Leifeld: Personal communication, B) S. Erzinger, FAT: pers. comm. 28.8.02

*) Personal communication from Jens Leifeld, FAL, 23 April 2002.

The binding of CO₂ from the atmosphere was estimated by the C-content in the dry matter multiplied by the stoichiometric factor 44/12, based on the assumption that the carbon in the biomass is completely taken from the air. The carbon content was derived from measurements, where available or calculated from the composition of the biomass, by taking the following C-contents: carbohydrates 44%, proteins 40%, fat 75%, fibres 44% and ash 0%⁵.

The energy content (gross calorific value) corresponds to the upper heating value of the dry biomass.

The CO₂ and energy content of the planted seeds were subtracted from the CO₂ and energy content of the products in order to avoid double counting.

For grass seed, the values of wheat grains were used; for clover seed, the values of protein peas.

⁵ Personal communication from Jens Leifeld, ART, 23 April 2002.

Part II: Agricultural Infrastructure and its Operation

5 Agricultural Buildings

5.1 Characteristics

Buildings create considerable environmental impacts throughout their life cycles, from the exploitation of resources, through manufacture of the construction materials, construction, and use, to dismantling (Lalivé d'Epinay 2000). In a life cycle assessment of 50 farms, the buildings contributed on average one third of the “use of non-renewable energy resources” (Rossier & Gaillard 2001). Before LCIs for agricultural building became available, LCIs of industrial buildings have had to be used for agricultural life cycle assessments. However, agricultural buildings differ considerably from industrial buildings, as they are frequently timber, not metal structures.

The type of construction, and hence the use of resources, strongly depends on building use. Energy consumption in the “utilisation” phase is determined by the machinery, ventilation and heating system installed in the buildings, and differs according to animal category and technology installed.

5.1.1 Important Agricultural Buildings

Tab. 5.1 lists the different farm types in Swiss agriculture in the year 2000 (Hausheer & Meier 2001). These data show that more than a third of Swiss farms are dairy farms, and one quarter are combined farms with pig, poultry and cattle (farm types “combined pigs/poultry” and “combined others”). This means that the most important and widespread agricultural building types in Switzerland are:

- housing for dairy cattle (tied and loose systems),
- housing for fattening pigs (conventional and label systems) and
- sheds and garages.

Tab. 5.1 Share of selected farm types in Swiss agriculture 2000.

Farm type	Share
Arable crops	6.4%
Special crops	6.5%
Dairying	36.0%
Other cattle	7.1%
Pigs/poultry	2.2%
Combined dairying/arable crops	10.9%
Combined pigs/poultry (min. 25% of the LU are pigs and poultry)	11.3%
Combined others (highly varied farms)	13.8%
Other farm types	5.8%
Total of farms	100.0%

Statistical data on animal husbandry for 1999 and 2000 are available from the farm census carried out by the Swiss Federal Statistical Office (SBV 2001b). Tab. 5.2 gives these data and the calculated average number of animals per farm. Since we can assume that all the animals of a given category on a farm are housed in one building, these numbers correspond roughly to average Swiss housing sizes. These data cover all Swiss agricultural areas including the mountain regions, where farm size is smaller than in the plain regions.

Tab. 5.2 Extract from the statistical data on animal husbandry (SBV 2001b) and calculated number of animals per farm.

Animal category	Number of farms	Number of animals CH	Average number of animals per farm	Reference year
Cows	50,274	724,778	14.4	1999
Fattening pigs	8,618	1,356,775	157.4	2000

Compared to those of other countries, agricultural farm buildings in Switzerland are of expensive, solid construction. To a certain extent, this building style is justified by the Swiss climate, e.g. low temperatures and heavy snow in winter, but other factors play a role as well (economic situation, tradition, legislation).

5.2 Life Cycle Inventories of Agricultural Buildings

The life cycle of a building can be divided into three phases (Tab. 5.3). In the construction phase, the building is built from various construction materials. This phase includes construction work and transport. The utilisation phase extends throughout the service life of the building. Repairs must be carried out and individual building parts or fittings replaced because their service life is shorter than that of the building as a whole. Energy, water (for cleaning and drinking) and other auxiliary materials (e.g. cleaning agents and lubricating oil) are also required during use. The waste materials accruing during repair and replacement must be appropriately disposed of, as well as the materials when the building is dismantled at the end of its service life.

Tab. 5.3 The three phases of a building life cycle.

Construction phase	Construction materials, construction work, transport
Utilisation phase	Building maintenance, energy, water, auxiliary materials
Waste disposal	Dismantling, transport, recycling, incineration, dumping

5.2.1 The Buildings Selected

The construction solutions on which the building module calculations are based are selected examples of the types of housing and buildings commonly found in Switzerland in the year 2000. They do not conform to an – inexistent – “average Swiss building style”. This procedure corresponds to other building modules within ecoinvent (Kellenberger *et al.* 2007).

The following life cycle inventories were established:

- two different housing systems for dairy cattle (tied and loose housing),
- two different housing systems for fattening pigs (fully slatted floor and multi-surface system),
- milking parlour,
- shed with a garage section,
- four types of dried roughage and silage stores,
- tower silo,
- slurry store.

Tab. A. 4 and Tab. A. 5 in the Appendix give the sizes, description and service lives of the farm buildings and building parts selected for the ecoinvent database, together with the functional units of the modules. Fig. A. 1 to Fig. A. 6 show the construction plans of the buildings.

5.2.2 Method

The material quantity calculations for the construction of an agricultural building are based on the FAT Modular Construction Standards for Farm Building Costs (Hilty & Herzog 2000). This database is normally used to estimate investment demand of agricultural buildings.

The *element method*, described in Lalive d'Épinay (2000) and Ogip (2002), is the way chosen for calculating the material demand for buildings (Fig. 5.1).

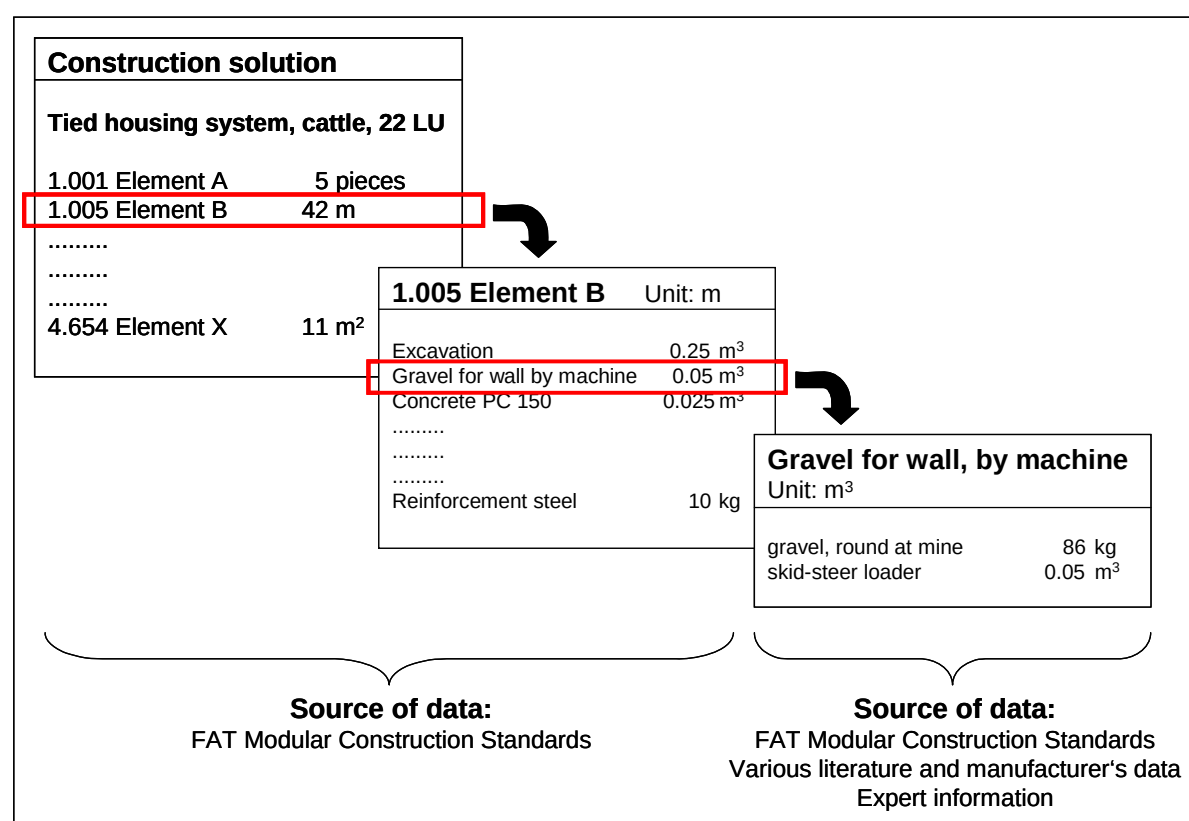


Fig. 5.1 The “element method” as used to calculate the material demand of buildings.

Construction Phase

The FAT Modular Construction Standards divide the buildings into individual elements (e.g. wall structure, slurry store, pen fittings, etc.). The elements may be individually combined to give a construction solution for a specific farm building. To calculate the material input for a building, the material quantities of the individual elements were determined (Fig. 5.2: columns E and F) by using the information in the FAT Modular Construction Standards and by compiling expert information. The construction solution, defined by building experts, shows how many of which elements are required. The total material quantity of a building was calculated by multiplying the material quantity per element by the number determined for the element. The material-type and quantity data were linked to the relevant life cycle inventories of construction materials from the ecoinvent database.

This method could also be used to determine partial areas of a farm building. For farm installations such as the dried roughage store and the milking parlour, the corresponding parts of the building shell were included in addition to the installation.

Tied housing system, cattle, 22 LU									
service life of building 50 years									
A Element (ELE)	B Quantity ELE	C REP [% per year]	D Deprecia- tion [%]	E Construction material, construction process	F Quantity/ELE	G Material constr. phase	H Material REP/ service life	I Material RPL/ service life	J Material overall service life
hay drop 1.00/1.00 m with insulated cover	3 St	1.5	4	cast iron	kg 27.2	81.6	61.2	81.6	224.4
				PUR foam slab	kg 10.8	32.4	24.3	32.4	89.1
				sawn timber, for sheeting	kg 62.0	186.0	139.5	186.0	511.5
tied housing, two rows with slurry channel	11 St	1	4	reinforcing steel	kg 134.0	1474.0	737.0	1474.0	3685.0
				excavation skid-steer-loader	m3 3.4	37.2	0.0	0.0	37.2
				excavation hydraulic digger	m3 10.5	115.8	0.0	0.0	115.8
				concrete pc150	kg 1669.8	18367.8	9183.9	18367.8	45919.5
				concrete pc 300	kg 5112.8	56240.8	28120.4	56240.8	140602.0
				iron enamel	kg 6.5	71.5	35.8	71.5	178.8
									
				chromium steel	kg 1.0	10.6	5.3	10.6	26.4
				sheet steel, galvanised	kg 0.5	5.5	2.8	5.5	13.8
				cement PC-CH	kg 780.0	8580.0	4290.0	8580.0	21450.0
.....									

Fig. 5.2 Extract from the material calculations for the cattle tied housing system, 22 LU.

ELE = element, REP = repair, RPL = replacement.

To calculate the area for the land-use, the used building area was taken from the construction plan (Appendix Fig. A. 1 to Fig. A. 6). In addition to the building area, four metres of concrete court on each side of the building were included (area only, material not included⁶). For building parts (e.g. milking parlour, slurry store), no additional area was included.

Diesel consumption for construction processes was calculated by excavation and gravel volume. Electricity consumption for construction processes was estimated on the supposition that diesel accounts for 70 % of the overall energy consumption of construction processes, and electricity, 30 % ⁷.

Utilisation Phase

The material input for repair throughout the full service life of the building was calculated using the percentage of repair costs defined for each individual element in the FAT Modular Construction Standards⁸ (Fig. 5.2: columns C and H).

⁶ Personal communication from H.J. Althaus, EMPA Dübendorf, 27 January 2003.

⁷ Personal communication from D. Kellenberger, EMPA Dübendorf, 5 December 2002.

⁸ Exception: For the construction materials “gravel, crushed” and “gravel, round” as well as for the construction works “excavation, skid-steer loader” and “excavation, hydraulic digger” the values for repair and replacement were set to zero, given that they are carried out only once during the entire service life of the building.

The material input for replacement throughout the full service life of the building was calculated using the reciprocal of the depreciation period defined for each individual element in the FAT Modular Construction Standards⁹ (Fig. 5.2: columns D and I).

Electricity consumption for replacement and repair processes was estimated by the ratio of the material input during utilisation phase over construction phase. Diesel consumption for replacement and repair was estimated by the volume of concrete used.

The energy, water and auxiliary materials used annually on the farm were also compiled from literature data and expert information on the basis of the elements from the FAT Modular Construction Standards (Hilty & Herzog 2000). However, they were also supplemented by other areas important to resource consumption but not incurred in direct conjunction with building parts and installations (e.g. pen cleaning, drinking water, etc). The calculated data were linked to the corresponding ecoinvent modules.

For the module “dried roughage store, air dried, solar, operation”, the use of solar energy was included according to Jungbluth (2007) with a mean insolation of 1093 kWh/(m² * year) over the whole year and on the whole collector surface.

Waste Disposal

The disposal of the construction materials (from repair, replacement and dismantling) was derived from material consumption in the building and utilisation phase. Simplification was necessary when assigning the construction materials to the ecoinvent modules for disposal systems, because for some construction materials there is no corresponding module for disposal. Tab. A. 6 (Appendix) shows how the construction materials are assigned to the modules for disposal.

5.2.3 System Boundaries

The building modules in the database are grouped into infrastructure and basic operation (see Fig. 4.4). The infrastructure modules contain everything connected with the building shell and fittings: building construction, repair and replacement of building elements and the disposal of all waste materials, including dismantling (see Fig. 5.3). They cover the whole service life of the building, which is 50 years for most modules (Tab. A. 4 and Tab. A. 5).

The (basic) operation modules relate to the functional units given in Tab. A. 5 and contain the input required to operate the housing or building parts: energy, water and other auxiliary materials (see Fig. 5.3). The operation module was linked to the corresponding infrastructure module by the inclusion of a fraction of the infrastructure (see 5.2.4). Because an infrastructure module includes the repair and replacement of elements with a shorter service life, the life cycle inventory always represents a “new” – well-maintained – building. In the operation module, therefore, the material quantity per year remains the same for the use of the infrastructure in differently inserted service lives.

No operation module was defined for building parts incurring no operational expenditure apart from use of the infrastructure (e.g. shed).

⁹ Exception: For the construction materials “gravel, crushed” and “gravel, round” as well as for the construction works “excavation, skid-steer loader” and “excavation, hydraulic digger” the values for repair and replacement were set to zero because they are carried out only once during the entire service life of the building.

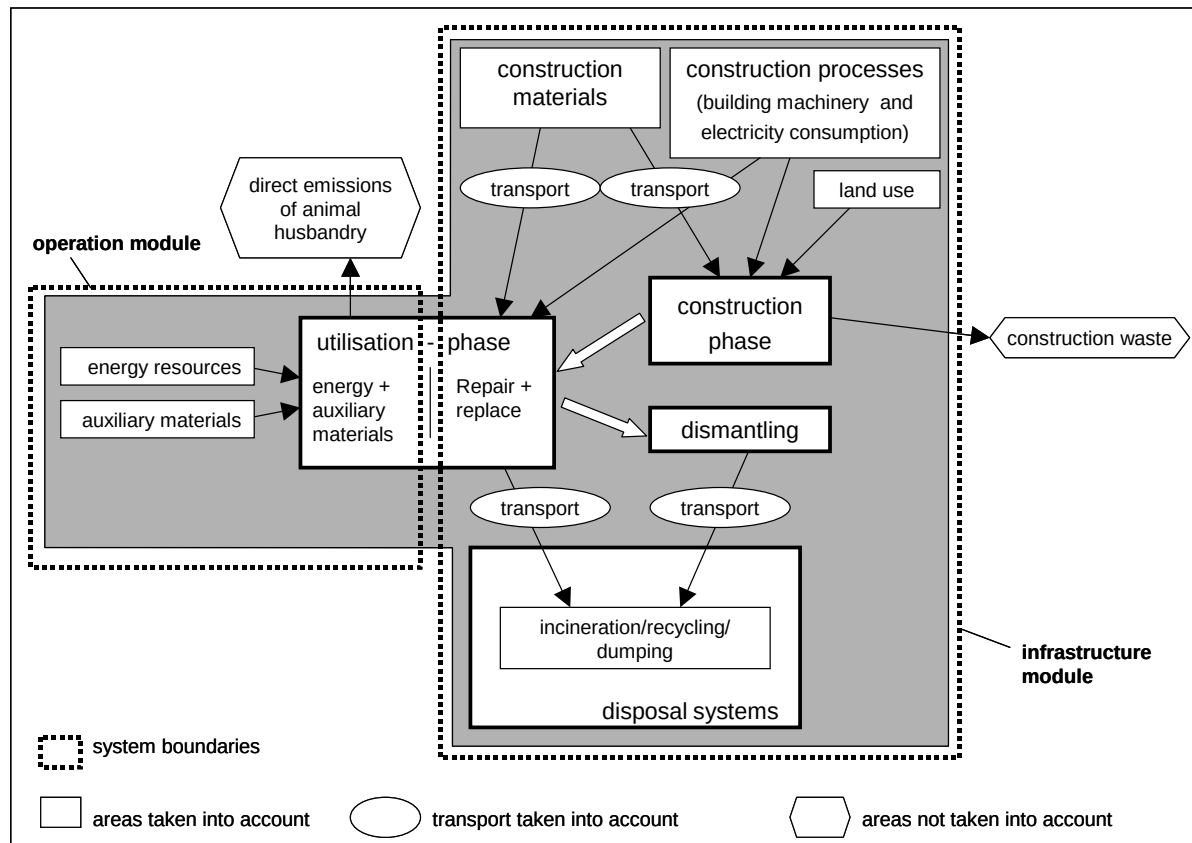


Fig. 5.3 System boundaries of the life cycle inventories of agricultural buildings.

In addition to the material used in the construction phase, these life cycle inventories also include the use of construction machinery (hydraulic diggers, skid-steer loaders). The utilisation phase includes the materials and energy used for repairs and replacements to the building shell and fittings, as well as the consumption of energy and auxiliary materials (water, lubricating oil, cleaning agents, etc.). The infrastructure modules of agricultural buildings take account of both the inputs associated with construction, repair and replacement as well as of the disposal of waste during repair, replacement and dismantling via links with ecoinvent modules for construction materials, building processes and disposal systems.

According to the principle explained in Fig. 4.4 in chapter 4.2.3, the modules investigated do not take account of the following areas:

- Direct emissions of animal farming (CH_4 , NH_3 , N_2O , etc.), since these are dependent on animal feeding and farm management (e.g. pasture). The literature data on this subject are sparse.
- Fodder production, fodder and straw consumption, farmyard manure (direct emissions, spreading), wastewater.
- Disposal of construction waste (construction phase) such as packing materials and remnants of construction materials, because it represents only a small proportion of the overall disposal quantity.

Virtually no data are available for determining the transport distances for construction materials from manufacturers or regional warehouses to building sites. Concrete, gravel, sand and timber are frequently bought locally, so transportation distances are short¹⁰. There are only a few manufacturers of other building materials in Switzerland. Certain pen fittings are also bought in neighbouring countries. Since an average 85% of the total weight of the buildings in this study comprised concrete, gravel and sand, the average distance travelled by all construction materials was estimated to be 50 km.

5.2.4 Functional Units, Service Life and Application

The functional unit of the module depends on the building type. Tab. A. 4 shows the functional units of the infrastructure building modules investigated, whilst Tab. A. 5 shows those of the operation modules. The service life given in Tab. A. 4 and Tab. A. 5 was used to link the infrastructure to the relevant operation module (see also section 5.2.3).

A user wishing to inventory a product from animal farming, e.g. milk, should use, among others, the modules "loose housing system, cattle, operation" for the use of the building, modules for feedstuffs from chapter 12 and 15, as well as own data for air emissions from animals, building surfaces, slurry handling or pasture (see Fig. 4.4). The module "loose housing system, cattle" for the building infrastructure is already included in the operation module and must not be added again.

5.3 Data Quality Considerations

Determination of the construction materials, their quantities, and the building processes for erection and replacement are based on a solid data foundation. Repair input calculations may differ from reality to a certain extent, as they were calculated using cost-based annual repair factors for individual construction elements.

The construction solutions on which the building module calculations are based are selected examples of the types of housing and buildings commonly found in Switzerland in the year 2000. They do not conform to an average Swiss building style. This procedure corresponds to that of other building modules in ecoinvent (see Kellenberger *et al.* 2007). However, since the inventories for the agricultural buildings are more detailed than those of the building hall calculated by Kellenberger *et al.* (2007), a comparison of agricultural buildings and building halls is not meaningful.

The following limitation on the use of individual modules should be noted: to obtain realistic values when applying the modules, building sizes should not differ too widely from the sizes on which the module calculations were based (see Tab. A. 4 and Tab. A. 5). An acceptable approximation can be achieved for building sizes within a range of about half to twice the value of the calculated size of the modules. A wider difference would not be reasonable, since the type of construction would be too different from the calculated module.

¹⁰ Ready-mix concrete travels an average 8 km, according to the Ready-Mix Concrete Association (Weibel & Stritz 1995)

6 Agricultural Machinery

6.1 Characteristics

The term "agricultural machinery" is defined as machines designed for and used in agricultural production. The applications for which this machinery is employed range from road transport to field work, such as soil management, seeding, fertilising and harvesting. Machinery not specifically designed for agricultural purposes (e.g. lawn-mowers) or which is mainly employed in livestock husbandry (e.g. milking equipment, see chapter 5) has not been considered in this chapter.

6.1.1 Classes of Agricultural Machinery

Based on the estimates of experts from ART, agricultural machinery was categorised into six classes as shown in Tab. 6.1. The first two classes contain automotive vehicles. Endless-track vehicles were not considered in the present inventories. The representatives of the other four classes are all non-automotive, and are pulled by tractors during operation.

Tab. 6.1 Classes of agricultural machinery (Zimmermann 2002).

Class	Tractors	Harvesters	Trailers	Agricultural machinery, general	Agricultural machinery, tillage	Slurry tankers
Examples	Tractor Transporter Two-axle mower	Combine-harvester	Tyre trailer	Seeder Hoe Self-loading trailer	Plough Harrow Roller	Vacuum tanker Pump tanker

Tab. 6.2 provides the approximate figures for the different representatives of agricultural machinery in operation in 1996 (for some representatives, in 1990) in Switzerland. The largest representative of general agricultural machinery is hay-harvesting machines (about 30 % of the total general machinery).

More than three quarters of the approx. 150,000 machines attributed to the "tractors" class were tractors in the narrower sense (around 125,000 units in the year 2000). The number of tractors recently put into operation was between 2,500 and 3,000 per year (SBV 2000b and SBV 2001b).

The degree of mechanisation can be estimated by comparing the number of machines with the number of farms (about 73,500 in 1999) or the utilized agricultural area (about 1.08 million ha); data from SBV (2001b).

Tab. 6.2 1996 Federal farm census stock data on the different classes of agricultural machinery (SBV 2001b; rounded values).

Class	Tractors	Harvesters	Trailers ^a	Agricultural machinery, general ^b	Agricultural machinery, tillage	Slurry tankers
Stock 1996	147,000	4,000	141,000	362,000	116,000	35,000

^a Data from federal farm census 1990.

^b Where no data were available for 1996, they were taken from the 1990 federal farm census.

6.2 Life Cycle Inventories of Agricultural Machinery

6.2.1 Scope of the Life Cycle Inventories

The inventories of the six classes of agricultural machinery are based on different data sources and expert estimates.

The inventories take into account resource use and the emission levels during the production, maintenance and disposal of agricultural machinery. Impacts incurred during operation of the machinery (e.g. emissions from fuel consumption and from tyre abrasion) were not taken into account. These impacts were included in the modules of agricultural field-work processes (chapter 7). Nor were infrastructure data for production buildings, for which no data were available, taken into account. Inventory system boundaries are illustrated in Fig. 6.1 below.

The inventories refer to the supply of machinery intended for use in Switzerland. Transport from production in Western Europe to Switzerland was included.

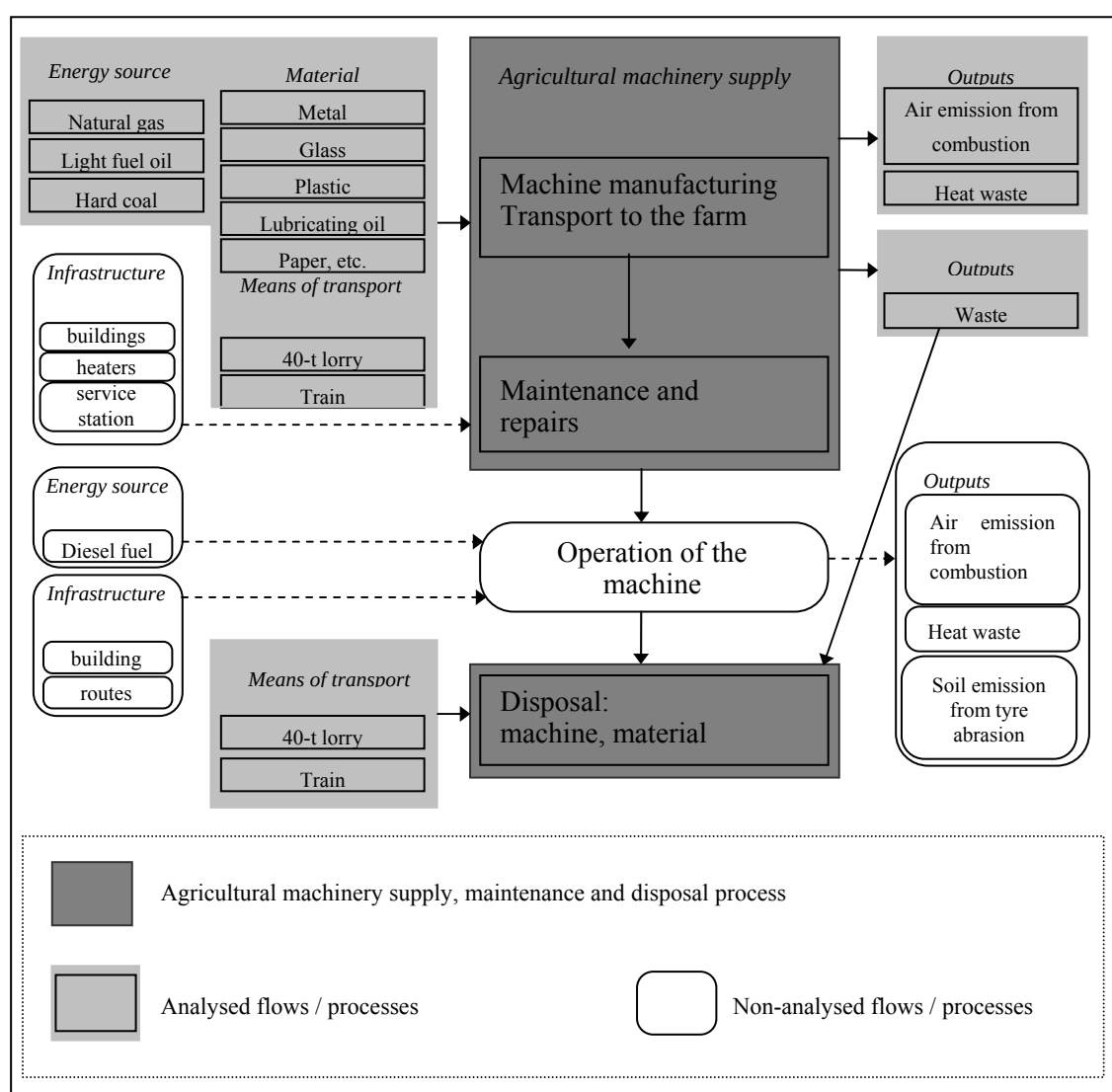


Fig. 6.1 System boundaries of the life cycle inventories of agricultural machinery.

6.2.2 Composition of Agricultural Machinery

Neither manufacturer's nor bibliography data contained details on the composition of the agricultural machinery at our knowledge. More-detailed information is available for the composition of lorries, which can be used as support for an approximation.

According to expert information, the material composition of agricultural machines and vehicles not only varies between different classes of machines, but also within the same class between different models of machines. Nevertheless, some main components of agricultural machinery can be identified, the most important of which is steel (Fig. 6.2). The synthetic-rubber content is determined mainly by the tyres¹¹.

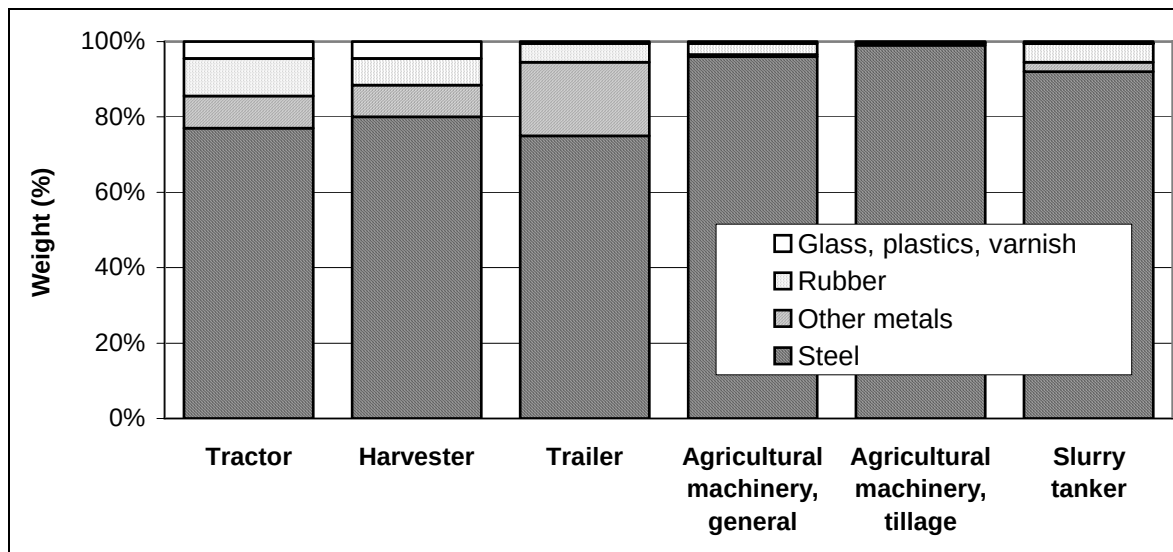


Fig. 6.2 Principal components of the different machinery classes (% of weight).

6.2.3 Manufacture of Agricultural Machinery

In Frischknecht *et al.* (1996) the process energy needed for the manufacture of a 16-t lorry is 12 MJ/kg (of which 45 % electricity, 41 % natural gas, 7 % light fuel oil and 7 % coal). Since the construction processes of lorries and agricultural vehicles are comparable, this value was taken for the supply of tractors and harvesters. The other classes of agricultural machinery undergo a simpler construction process, for which reason the assumed energy consumption was reduced to 10 MJ/kg.

No literature data were available on direct emissions during the manufacturing process. The assumptions made to obtain the inventories are described in chapter 6.2.6. Transport of the machinery from place of manufacture to the farm was calculated as 400 km by lorry- and 100 km by rail¹².

¹¹ Personal communication from M. Rinaldi, Swiss Federal Research Station for Agricultural Economics and Engineering of Taenikon (FAT), November 2001.

¹² Personal communication from E. Stadler, Swiss Federal Research Station for Agricultural Economics and Engineering of Taenikon (FAT), November 2001.

6.2.4 Maintenance and Repair of Agricultural Machinery

Materials and energy consumption and direct emissions resulting from the maintenance and repair of agricultural machinery were calculated based on Ammann (2001), Frischknecht *et al.* (1996) and Maibach *et al.* (1999). Waste resulting from maintenance and repair of the machinery is summarised with the waste of end-of-life disposal (chapter 6.2.5)

Maintenance

The number of tyre sets needed for replacement is obtained by dividing the average lifetime of the machinery class by the lifetime of one tyre set and subtracting the set fitted during manufacture. Multiplying this value by the weight of one set of tyres yielded the amount of synthetic rubber for tyres in the inventory (for the tractor class: $[7000 \text{ h} / 2500 \text{ h} - 1] * 0.098 \text{ kg}_{\text{tyre}}/\text{kg}_{\text{tractor}} = 0.176 \text{ kg}_{\text{tyre}}/\text{kg}_{\text{tractor}}$).

The amount of engine- and hydraulic oil needed during the lifetime of the agricultural vehicles was taken from Ammann & Stadler (1998). The value is expressed in litres per hour ($0.047 \text{ l}_{\text{engine oil}}/\text{h} + 0.050 \text{ l}_{\text{hydraulic oil}}/\text{h}$). This value was transformed by multiplying it by the specific weight (0.88 kg/l) and average lifetime of the vehicles, and dividing by the average weight of the vehicles (for the tractor class: $0.097 \text{ l}_{\text{oil}}/\text{h} * 0.88 \text{ kg/l} * 7000 \text{ h} / 3000 \text{ kg tractor} = 0.199 \text{ kg}_{\text{oil}}/\text{kg tractor}$).

The amount of paper (0.068 g/vkm^{13}), polypropylene (0.034 g/vkm) and lead (0.34 g/vkm) for filters, brake blocks and batteries for 16-t lorries was taken from Maibach *et al.* (1999). For transformation, a lorry weight of 6800 kg and a travelling speed of 80 km/h were assumed (paper for the tractor class: $0.068 \text{ g}_{\text{paper}}/\text{km} * 80 \text{ km/h} / 6800 \text{ kg}_{\text{machinery}} * 7000 \text{ h} = 5.6 \text{ g/kg}_{\text{machinery}}$).

Repair

No literature data were found on the amount of materials needed to repair the machinery. An estimate was therefore made based on cost-related repair factors given in Ammann (2001). The repair factor is defined as the repair costs during life divided by the price of the new machinery (Tab. 6.3). The types of repair materials were assumed to be appropriate for the composition of the new machinery.

Tab. 6.3 Repair factors for different classes of agricultural machinery.

Class	Tractors	Harvesters	Trailers	Agricultural machinery, general	Agricultural machinery, tillage	Slurry tankers
Repair factor ^a	0.20	0.32	0.11	0.34	0.45	0.21

^a A repair factor of 0.20 means, that 20% of the initial materials of a new tractor are replaced during its lifetime. This in addition to the materials used for maintenance.

In Frischknecht *et al.* (1996), the energy consumption for maintenance of a 16-t lorry is given as 0.134 MJ/vkm (of which 22 % is electricity and 78 % heating oil), which corresponds to an energy consumption of $27.2 \text{ MJ per kg maintenance materials}$. This energy consumption factor was applied to the need of the agricultural machinery for maintenance- and repair materials.

For direct air emissions from maintenance and repair, see chapter 6.2.6.

¹³ vkm = vehicle kilometre

6.2.5 Waste Disposal of Agricultural Machinery

The waste to be disposed of includes the waste owing to maintenance and repair on the one hand and the disposal of the machinery at the end of its service life on the other hand. The percentage of waste produced by maintenance and repair ranges between 12 % (trailers) and 38 % (tractors) of total waste incurred.

For the different components, the following disposal paths were chosen:

- Used oil: incineration in a hazardous waste incineration plant.
- Tyres: About 50 % of tyres are burned as fuel in cement-producing plants. Of the remaining 50 %, a small proportion are recycled and a larger proportion are exported abroad, where the tyres continue to be used. For incineration in cement-producing plants, the former owner must pay a fee¹⁴. Impacts for transport are therefore attributed to the disposal of tyres, but a cut-off was applied for emissions from combustion, as the cement plant also has an interest in burning the tyres for energy gain¹⁵. No impacts were calculated for recycling. For export, a cut-off for re-use was made and no impacts of disposal were calculated for first-user in Switzerland.
- Metals: it was estimated that all metals are recycled and no impacts need be allocated to their disposal.
- Resh: Glass, paper, plastics, rubber and varnish are shredded and treated as resh. For ecoinvent, no module such as "disposal, resh to municipal incineration" or "disposal, shredded car waste to municipal incineration" was available, so the materials were considered to be disposed of separately in municipal incineration. The following modules were chosen: "disposal, building, glass, sheet, to final disposal" for glass, "disposal, paper, 11.2% water, to municipal incineration" for paper and "disposal, plastics, mixture, 15.3% water, to municipal incineration" for plastics, rubber and varnish¹⁶.
- Transport: a distance of 40 km by lorry was assumed for all components.

Energy consumption for disassembly was estimated at 0.5 MJ electricity per kg machinery (Frischknecht *et al.* 1996, Maibach *et al.* 1999).

Air emissions resulting from disposal of agricultural machinery were calculated in the utilised ecoinvent modules "... at municipal incineration". For additional air emissions, see chapter 6.2.6 below.

6.2.6 Direct Air Emissions from Manufacture, Maintenance, Repair and Disposal of Agricultural Machinery

According to Maibach *et al.* (1999), NMVOCs are emitted during the maintenance of vehicles, with a 16-t lorry emitting 0.053 g/vkm. Dividing this value by the amount of energy required for maintenance (0.134 MJ/vkm) gives a figure of 0.4 g of NMVOCs per MJ of energy used. This value was also assumed to be correct for maintenance and repair of agricultural machinery.

As mentioned above, no literature data concerning direct emissions during the manufacturing process were available. It was therefore assumed that the same quantity of NMVOCs (0.4 g per MJ of energy used) is emitted during manufacture as during maintenance and repair.

¹⁴ Personal communication from M. Spielmann, Natural and Social Science Interface (ETH-UNS), Swiss Federal Institute of Technology, Zurich, Switzerland, 16 December 2002.

¹⁵ Personal communication from G. Doka, Doka Ökobilanzen. Zurich, Switzerland, 16 December 2002.

¹⁶ Personal communication from G. Doka, Doka Ökobilanzen. Zurich, Switzerland, 10 December 2002.

Data on additional carbon dioxide emission from corrosion of the varnish (1.089 kg carbon dioxide/kg varnish) were taken from SAEFL (1995b). These constitute an insignificant proportion (0.1%) of the carbon dioxide emissions of the entire inventory.

6.2.7 Functional Unit and Application of the Modules

The functional unit of all six classes of machinery is **one kilogram machine** during its entire lifetime.

The machinery is allocated between the process considered and other usages using information on weight, operation time and lifetime of the machinery. Operation time and lifetime must be declared in the same unit, either in years or in a working unit such as hours or hectares. The operation time must be known by the module user. If no data for weight or lifetime are available, Tab. 6.4 gives an overview of mean values for some agricultural machines (the lifetime is listed twice in different units).

The amount of machinery (AM) needed for a specific process was calculated (formula 6.1) by multiplying the weight of the machinery by the operation time (how long the machinery is used for the process) and dividing the result by the lifetime of the machinery.

$$AM [kg/WU] = Weight [kg] * \frac{Operation\ time [h/WU]}{Lifetime [h]} \quad (6.1)$$

For example, a 3600-kg tractor used for 10 hours' field work was calculated thus: $3600 [kg] * 10 [h] / 7000 [h] = 5.14 [kg]$ of module "tractors, production".

The assumed lifetime in the calculated inventories has no effect on emissions during manufacture and disposal of the machinery, but does determine emissions during maintenance and repair. If the lifetime of a particular piece of machinery is lower or higher than a mean value, the emissions during maintenance and repair change proportionally. We estimate that a 10% change in lifetime results in a change in the emissions of the entire inventory of between 2% for trailers and 4% for a tractor, but these changes vary for the different emissions. The inventories should therefore not be applied if the lifetime of a piece of machinery varies too greatly from the values given in Tab. 6.4.

For further application examples see chapter 7, where the present modules were used to balance several agricultural fieldwork processes, creating ecoinvent modules for typical agricultural work.

Tab. 6.4 Typical power, working capacity, weight and life-time of selected agricultural machinery.

Machinery	Weight [kg]^a	Life-time [years]^b	WU^d	Utilisation [WU/year]^b	Life-time [WU]
Tractors					
Tractor, 4WD, -29 kW (-40 hp) ^c	1900	12	h	500	6000
Tractor, 4WD, 30 -64 kW (41 -87 hp)	3300	12	h	600	7200
Tractor, 4WD, 65 -94 kW (88-128 hp)	5300	12	h	600	7200
Tractor, 4WD, 95-128 kW (129-163 hp)	6450	12	h	600	7200
Transporter	1900	12	h	550	6600
Two-axle mower	1500	12	h	500	6000
Harvesters					
Combine harvester, 3-4.2 m, 80-130 hp (60 -95 kW)	7000	12	ha	75	900
Combine harvester, 4.2-6 m, 170-240 hp (125-175 kW)	9500	12	ha	125	1500
Trailers					
Tyre trailer, 2 axes, 8 t	1500	15	h	80	1200
Tyre trailer, 2 axes, 10 t, tilting hydraulically	2400	15	h	80	1200
Agricultural machinery, general					
Seeder	500	15	ha	35	525
Hoe	500	15	ha	35	525
Fertiliser spreader	200	10	ha	100	1000
Maize chopper	1000	10	ha	11	110
Complete harvester, potatoes	3200	12	ha	10	120
Rotary mower	600	12	ha	50	600
Self-loading-trailer	2600	12	TL ^e	300	3600
Agricultural machinery, tillage					
Plough, 2 furrow	500	12	ha	20	240
Plough, 4 furrow	1050	12	ha	40	480
Disc harrow	1400	15	ha	50	750
Spring tooth harrow	600	12	ha	45	540
Rotary harrow	900	12	ha	30	360
Roller, 2.5-3 m	750	20	ha	25	500
Roller, 6 m	2700	20	ha	50	1000
Field cultivator	550	12	ha	40	480
Slurry tankers					
Pump and vacuum tanker, 2000-4000 l	1050	12	Ta ^f	500	6000
Pump and vacuum tanker, 5000-8000 l	2150	12	Ta ^f	500	6000

^a Personal communication from H. Ammann, FAT, Ettenhausen, Switzerland 7 August, 2002.^b Ammann (2001)^c 4WD = 4 Wheel Drive, hp = horse power^d WU = Working Unit^e TL = Trailer Load^f Ta = Tanker

6.3 Data Quality Considerations

Data on material composition of the agricultural machinery are recent estimates from experts at ART, based on machines and vehicles whose use is permitted in Switzerland.

Data on energy consumption, waste heat and air emissions refer to the values given for lorries in Western Europe in Frischknecht *et al.* (1996) and Maibach *et al.* (1999). Their application to the agricultural machinery is based on expert statements. Data are considered to be a good estimate for tractors and harvesters, but only a rough estimate for the other four classes of agricultural machinery.

Data on transport from the place of manufacture to the agricultural farm and on transport for waste disposal are recent estimates from experts at ART, based on European data for agricultural machinery.

Data on waste disposal are recent estimates from experts at ART, based on literature data on waste disposal for similar machinery in Western Europe.

Data on infrastructure required for the manufacture, maintenance and repair of the machinery were not included in the present inventories, since no adequate values were available.

The application of the module "tractor, production" for big tractors leads to a slight overestimate of the energy and material flows for the maintenance of the vehicle.

The impacts arising from the operation time of the machinery were not included since they are included in other ecoinvent modules, balancing the operation of the machinery for different field work processes (chapter 7).

7 Agricultural Field Work Processes

7.1 Characteristics

Environmental protection, and in particular the prevention of air pollution, is an important issue in most European countries. Studies have shown that a significant part of the total air pollution can be attributed to off-road vehicles (SAEFL 1995a). Agricultural field work processes are typical activities where off-road vehicles are employed. In Switzerland, agricultural and forestry machinery is responsible for about half of the total diesel exhaust gas produced by off-road vehicles (Rinaldi & Stadler 2002).

This chapter investigates typical arable-farming and forage-crop-production processes in Switzerland. A total of 32 processes have been modelled:

- Tillage (9 modules)
- Fertilisation (3)
- Sowing (3)
- Plant protection (2)
- Irrigation (2)
- Harvesting (13)
- Transport (1)

Tab. A. 9 in the Appendix gives an overview and short description of the field work processes included in ecoinvent data. Work processes for the cultivation of special crops were not considered. For animal farming processes, see chapter 5.

7.2 Life Cycle Inventories of Agricultural Field Work Processes

7.2.1 Scope of the Life Cycle Inventories

According to the principle explained in Fig. 4.4 in chapter 4.2.3, the inventories are designed as basic operation modules. The following inputs were included in the inventories:

- the infrastructure (machinery and shed attributable to the process, see modules described in chapters 5 and 6)
- the energy input (diesel fuel burned during the fieldwork) and
- the water consumption (for irrigation).
-
- The following emissions were considered:
- emissions to air from combustion of the fuel and
- emissions to soil from tyre abrasion during the work process.

Inputs and outputs related to applied operation (see Fig. 4.4) were not included: materials such as seed, pesticides, transported material, etc.; land use of the cultivated field; products such as maize, potatoes or milk; and emissions originating from the materials used and the livestock in question. chapters 14 and 15 show examples, how applied operation can be implemented. An exception was made for the work process "baling", for which the standard working material "PE film" was taken into account.

Emissions to soil (except from tyre abrasion), soil compaction, emissions to water, and noise and dust were not considered, either owing to their negligibility, or to the unavailability of data, or to their dependence on materials or livestock. Aspects that might be important to consider for further analysis are soil compaction, erosion, biodiversity, dust and noise.

The road between farm and field was not included for different reasons. In the first place, the available ecoinvent modules on roads describe bigger roads than the country lane connecting field and farm. In addition, it would have required considerable effort to allocate the environmental impacts between agricultural- and other vehicles driving on the road, with a great likelihood that the resulting impacts (land use, emissions, road repair) caused by the agricultural work processes would be negligible.

In general terms, the following activities were considered part of the work process:

- Preliminary work at the farm, such as attaching the appropriate machine to the tractor.
- Driving to field (with an assumed distance of 1 km).
- Field work, for a field plot of 1 ha surface area.
- Driving to farm.
- Concluding work, such as uncoupling the machine.

A short description of the activities considered and the inputs and outputs for each work process is given in Tab. A. 9 in the Appendix. The system boundaries of the life cycle inventories are shown in Fig. 7.1.

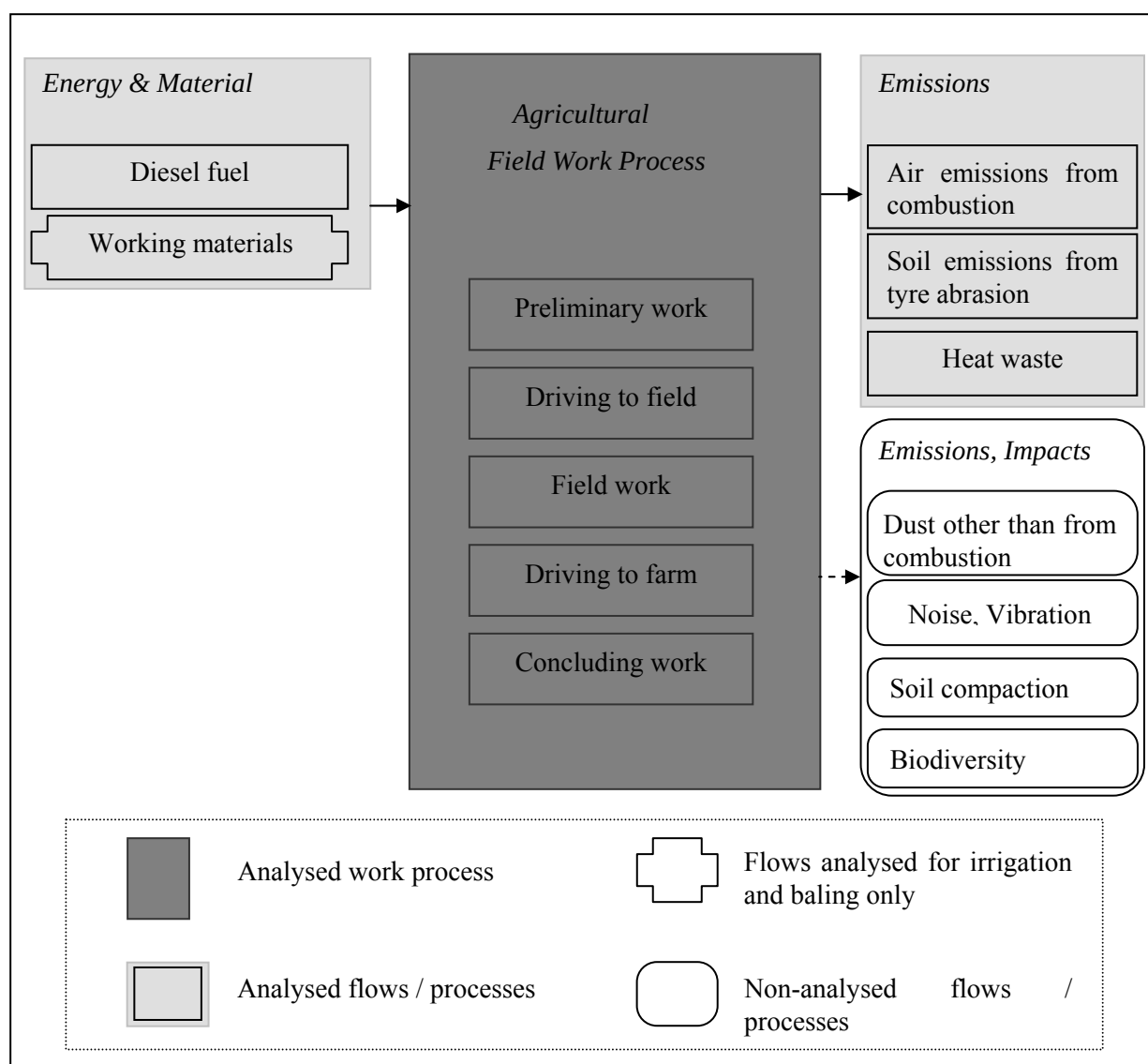


Fig. 7.1 System boundaries of the inventories for field work processes.

7.2.2 Application and Functional Unit of the Field Work Process Modules

Users wishing to inventory a product, e.g. round-bale silage, must choose and combine different ecoinvent modules described in this and other chapters of this section, *inter alia* the modules "tillage, ploughing", "clover seed, IP, at farm", "grass seed, IP, at farm", "sowing", "mowing, by rotary mower", "baling", "loading bales" and "transport, tractor and trailer".

The functional units of the modules are listed in Tab. A. 9 in the Appendix. The impacts are given relative to a working unit such as hectare (ha), tonnes kilometre (tkm), cubic metres (m³), kilogram (kg) or units (bales).

7.2.3 Infrastructure

The amount of agricultural machinery (AM) required for a specific work process (hours per functional unit (FU)) was obtained by multiplying the weight of the utilised machinery by the time required for the field work (operation time) over the lifetime of the machinery (LT_m).

$$AM [kg/WU] = \text{Weight [kg]} * \frac{\text{operation time [h/WU]}}{LT_m [h]} \quad (7.1)$$

Data from the Swiss Federal Research Station for Agricultural Economics and Engineering of Taenikon (FAT) were used for the weight of the machinery¹⁷ and the operation time¹⁸. Weight and operation time are listed in Tab. A. 9 and Tab. A. 10. For the correct use of the agricultural machinery modules, the lifetime (or useful life, Tab. 6.4) and repair factors (Tab. 6.3) of the balanced machinery must be close to those assumed in chapter 6.

For the motor mower, a model weighing 370 kg with a cutter bar of 1.9 m and an 8-kW petrol engine was chosen. The material composition can be assumed to be almost the same as for a tractor, and is described by the ecoinvent-module “tractor, production”.

The amount of shed space (AS) to be attributed to a work process was calculated by multiplying the surface occupied by the machinery by the field work time and dividing this by the lifetime of the shed (LT_{shed}) and the annual employment of the machinery (AE_M).

$$AS [m^2/WU] = \text{Surface [m}^2] * \frac{\text{operation time [h/WU]}}{LT_{shed} [a] * AE_M [h/a]} \quad (7.2)$$

The surface occupied by each individual machine was taken from Ammann (2002). The lifetime of the shed was defined in chapter 5 as 50 years. The annual employment of the machines was taken from Ammann (2001). The surface occupied, the operation time and the annual employment are listed in Tab. A. 9 and Tab. A. 10.

The country lane connecting the farm with the field was not included (see chapter 7.2.1).

7.2.4 Fuel Consumption

Data on mean fuel consumption (mFC) by an agricultural work process, taken from Rinaldi & Stadler (2002) and expert information¹⁹, were given in litres per hour of running time of the vehicle in question. This fuel consumption was measured on-site for various agricultural work processes, including the work steps described in chapter 7.2.1 and Fig. 7.1. For work processes where no measurements were available, an estimate was made based on workload (see Fig. 7.3 and Fig. 7.4 for examples of load spectra). By multiplying the values given in litres per hour by the duration per working unit of the field work process (= operation time of the machinery) and the specific weight of diesel (0.84 kg/l), the fuel consumption (FC) for the field work in question was obtained.

$$FC [kg/WU] = mFC [l/h] * \text{operation time [h/WU]} * \delta_{\text{Diesel}} [kg/l] \quad (7.3)$$

See Tab. A. 10 for mFC values, and Tab. A. 9 for operation time.

The fuel consumption of harvesting with the combine harvester was calculated with formula (7.4) from Rinaldi & Gaillard (1999).

¹⁷ Personal communication from H. Ammann, Swiss Federal Research Station for Agricultural Economics and Engineering (FAT), Taenikon, Ettenhausen, Switzerland, 7 August 2002.

¹⁸ Personal communication from R. Stark, Swiss Federal Research Station for Agricultural Economics and Engineering (FAT), Taenikon, Ettenhausen, Switzerland, July 2002.

¹⁹ Personal communication from M. Rinaldi, Swiss Federal Research Station for Agricultural Economics and Engineering (FAT), Taenikon, Ettenhausen, Switzerland, July 2002.

$$FC [kg/WU] = (1.5 [kg/h] + 0.23 [kg/kWh] * PTO \text{ power [kW]}) * \text{operation time [h/WU]} \quad (7.4)$$

PTO = power take-off

For the motor mower, data were taken from Stadler *et al.* (1999). The fuel consumption for the different processes is summarised in Tab. A. 10 in the Appendix.

7.2.5 HC-, NO_x- and CO emissions from Combustion

Engine emissions of HC, NO_x and CO were taken from Rinaldi & Stadler (2002), as well as from unpublished data. Data are based on emission models (Fig. 7.2) combined with measurements of load spectra (Fig. 7.3 and Fig. 7.4).

The emission models were calculated from test-bed measurements on 112 different tractors of different sizes, types and manufacturers. Two test cycles (ISO 8178 C1 test and a specific 6-level test created by the ART) were measured for each tractor. Fig. 7.2 shows a typical representation of the statistical function obtained during test-bed measurements.

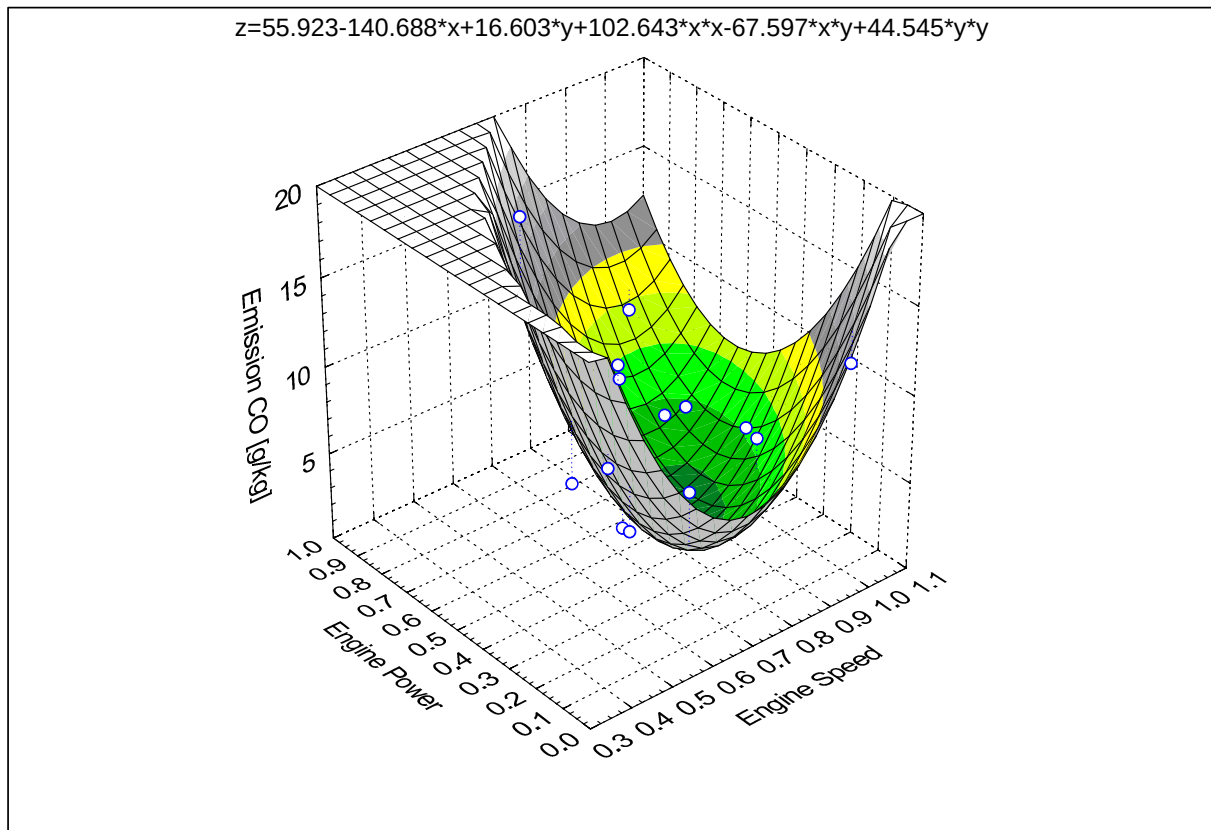


Fig. 7.2 Graphical representation of the statistical function for engine-power- and engine-speed-dependent CO emission (Rinaldi & Stadler 2002).

The amount of exhaust gas produced by a diesel engine depends on engine speed and power. These variables change constantly during practical application, which is why a specific work process must be characterised by a load spectrum. A load spectrum is a mathematical matrix showing how long the engine works at a determined speed and a given power demand (Rinaldi & Stadler 2002) during the

work process. Fig. 7.3 and Fig. 7.4 show two load spectra illustrating a big difference in the demands made by a work process on the engine of a tractor.

The amount of the three waste gases (WGs) HC²⁰, NO_x and CO is given in grams per hour, so the amount per working unit was calculated very similarly to the fuel consumption. The value from reference (WG_{reference}) was multiplied by the duration of the field work.

$$\text{WG [g/WU]} = \text{WG}_{\text{reference}} [\text{g/h}] * \text{operation time [h/WU]} \quad (7.5)$$

The values for HC-, NO_x- and CO emissions per hour are reported in Tab. A. 10 in the Appendix.

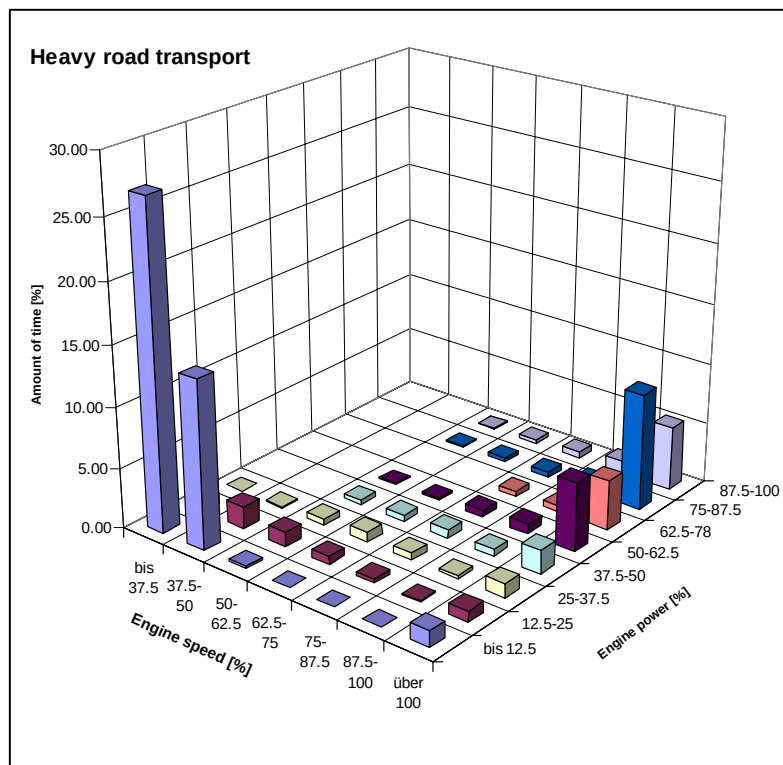


Fig. 7.3 Load spectrum obtained by measuring a heavy road transport with tractor (for other examples see Rinaldi & Stadler 2002). Relative time that the engine runs at a specific speed and power.

²⁰ HC was inventoried as NMVOC; the amounts of methane, benzene, B(a)P and PAH were therefore deduced.

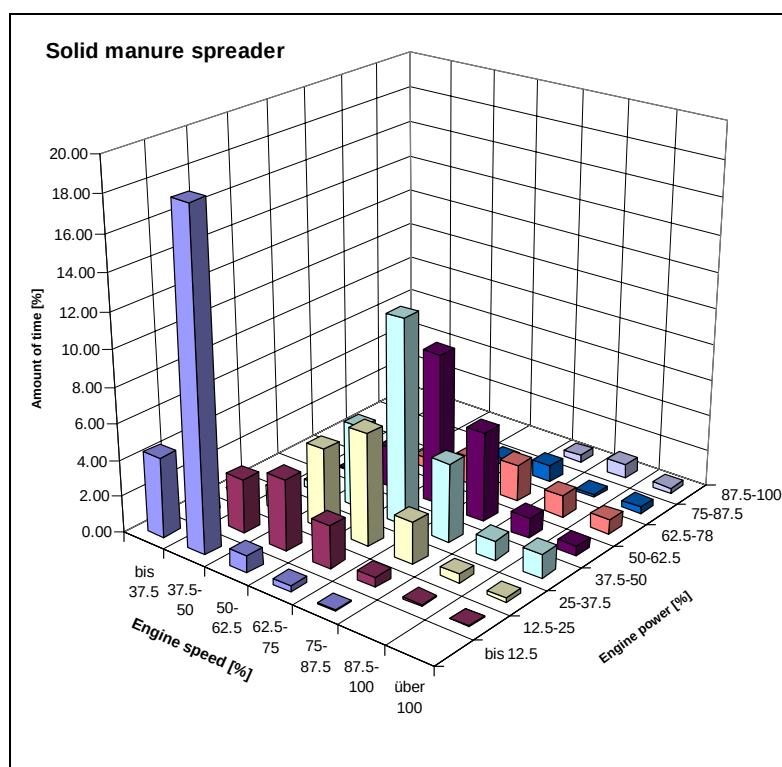


Fig. 7.4 Load spectrum obtained by measuring the work with a solid-manure spreader (for other examples see Rinaldi & Stadler 2002). Relative time that the engine runs at a specific speed and power.

Emission data for the motor mower were taken from Stadler *et al.* (1999).

7.2.6 Other Air Emissions from Combustion

Emission factors for other air emissions than HC, CO and NO_x were taken from SAEFL (2000), except for benzene (Wörgetter, 1991: 53). Substances and emission factors listed in Tab. 7.1 were used; the values for petrol are given for the motor mower.

Tab. 7.1 Emission factors for air emissions from fuel combustion (SAEFL 2000).

Substance	Formula	Emission factor [g / kg fuel consumption]		Significance / remarks
		Diesel	Petrol	
Carbon dioxide	CO ₂	3.12E+03	3.00E+03	Global warming
Sulphur dioxide	SO ₂	1.01E+00	7.20E-02	Photochemical oxidation, acidification
Lead	Pb	0	1.46E-01	Terrestrial ecotoxicity
Methane	CH ₄	1.29E-01	2.92E+00	Global warming
Benzene	C ₆ H ₆	7.30E-03	9.48E+00	Human toxicity
Particulate Matter	PM _{2.5}	EF _{PM2.5}	EF _{PM2.5}	See formula 7.8
Cadmium	Cd	1.00E-05	1.00E-05	Terrestrial ecotoxicity
Chromium	Cr	5.00E-05	5.00E-05	Terrestrial ecotoxicity
Copper	Cu	1.70E-03	1.70E-03	Human toxicity, freshwater aquatic ecotoxicity, terrestrial ecotoxicity
Dinitrogen monoxide	N ₂ O	1.20E-01	1.30E-01	Global warming
Nickel	Ni	7.00E-05	7.00E-05	Human toxicity, freshwater aquatic ecotoxicity, terrestrial ecotoxicity

Substance	Formula	Emission factor [g / kg fuel consumption]		Significance / remarks
		Diesel	Petrol	
Zinc	Zn	1.00E-03	1.00E-03	Terrestrial ecotoxicity
Benzo(a)pyrene	C ₂₀ H ₁₂	3.00E-05	4.00E-05	ecoinvent quality guidelines requires to separate B(a)p from other PAHs
Ammonia	NH ₃	2.00E-02	4.00E-02	Acidification, eutrophication
Selenium	Se	1.00E-05	1.00E-05	Human toxicity, freshwater aquatic ecotoxicity, terrestrial ecotoxicity
Polycyclic aromatic hydrocarbons		Sum of different PAHs listed below		Human toxicity, freshwater aquatic ecotoxicity, terrestrial ecotoxicity
Benz(a)-Anthracene		8.00E-05	7.50E-05	
Benzo(b)-Fluor-anthracene		5.00E-05	4.00E-05	
Chrysene		2.00E-04	1.00E-05	
Dibenzo(a,h)-Anthracene		1.00E-05	1.50E-04	
Fluoranthene		4.50E-04	4.50E-04	
Phenanthrene		2.50E-03	1.20E-03	

The output of these other waste gases (WG)²¹ was calculated by multiplying the fuel consumption (formula 7.3 and Tab. A. 10) of the work process by the emission factor (EF) listed in Tab. 7.1.

$$WG [g/WU] = FC [kg_{fuel}/WU] * EF [g_{WG}/kg_{fuel}] \quad (7.6)$$

For particulate matter, only those particles with a diameter of less than 2.5 µm (PM_{2.5}) were inventoried, since measurements showed that no particles of greater diameter are detectable in flue gas from diesel fuel combustion (Stein *et al.* 2002: 99 and Mayer²²).

The emission factor for PM_{2.5} was calculated with formula 7.7 from SAEFL (2000). For diesel, the values are A = 7.25 and B = 3.62, whereas for petrol the values are A = 2.23 and B = -3.9.

$$EF_{PM_{2.5}} [g/kg_{fuel}] = A [g_{PM}/kg_{fuel}] - B [g_{PM}/(kg_{fuel} * kW)] * nominal\ power^{0.1} [kW] \quad (7.7)$$

The output of PM_{2.5} was obtained by multiplying the emission factor from formula 7.7 by a correction factor (CF), by the mean power (MP) of the tractor during the field work, by the time taken by the field work.

$$PM_{2.5} [g/WU] = EF_{PM_{2.5}} [g_{PM}/kg_{fuel}] * CF [kg_{fuel}/kWh] * MP [kW] * operation\ time [h/WU] \quad (7.8)$$

The correction factor is needed in order to adapt the data given for the reference year 1990 to the year for which the calculation is made. The value of the correction factor for the year 2002 is 0.854 for diesel and 0.75 for petrol (SAEFL 2000). For operation time and nominal and mean power, see Tab. A. 9 and Tab. A. 10.

²¹ For CO₂ output the amount of CO output was deduced.

²² Personal communication from A. Mayer, TTM Andreas Mayer, Niederrohrdorf, Switzerland, July 2002.

7.2.7 Soil emissions from tyre abrasion

The heavy-metal emissions (HM) from tyre abrasion were calculated bearing in mind the number of tyre sets used during the lifetime of the machinery. The number of tyre sets needed is obtained by dividing the lifetime of the machinery (LT_m) by the lifetime of one tyre set (LT_t ; Tab. 7.2). Multiplying this value by the weight of one set of tyres (per one kilogram of machinery; Tab. 7.2), and the amount of tyre rubbed off (RO; Tab. 7.2) gives the amount of synthetic rubber from tyre abrasion for one kilogram of agricultural machinery during the machinery's entire lifetime. Hence, in order to calculate the emission from one field-work process, we must multiply the amount of synthetic rubber by the heavy-metal content ($HM_{content}$) and by the calculated amount of agricultural machinery (AM; formula 7.1). The content of heavy metals in the synthetic rubber are 1.6 % (or 16 g/kg_{rubber}) for zinc, 0.0026 % (or 2.6 g/kg_{rubber}) for lead and 0.0006 % (or 0.6 g/kg_{rubber}) for cadmium (Frischknecht *et al.* 1996).

$$HM [g/WU] = (LT_m [h]/LT_t [h]) * Weight [kg_{tyres}/kg_{machinery}] * RO [kg_{rubber}/kg_{tyres}] * HM_{content} [g_{HM}/kg_{rubber}] * AM [kg_{machinery}/WU] \quad (7.9)$$

Tab. 7.2 Basic values for the calculation of heavy-metal emission from tyre abrasion; Frischknecht *et al.* (1996) and Zimmermann (2002).

	Tractors	Harvesters	Trailers	Agricultural machinery, general	Slurry tankers
Lifetime of tyres [h]	2,500	1,080	750	2,500	2,500
Weight of one set of tyres [kg _{tyres} /kg _{machinery}]	0.0975	0.0675	0.0475	0.0275	0.0475
Synthetic rubber, rubbed off [kg _{rubber} /kg _{tyres}]	0.29	0.29	0.56	0.56	0.56

The class "agricultural machinery, tillage" has no tyres.

7.2.8 Irrigation

In some regions of Switzerland, during certain seasons, rainfall alone does not provide agricultural surfaces with enough water for plant cultivation, and crops must be supplied with additional water from irrigation. Regions where irrigation is typically practised are the canton of Wallis, where little rain falls over the year, and arable crops, meadows and pastures are irrigated; and the southern part of Switzerland, where annual rainfall is very high, but irregularly distributed over the year, and mainly arable crop and vegetable plantations are irrigated. The Swiss Central Plateau receives an average amount of rainfall per year which is well distributed over the growing period, and it is mainly the vegetable plantations that are irrigated (see also chapter 2).

The irrigation of vegetables differs from the irrigation of arable crops or meadows in terms of technique and the amount of water sprayed. Since the vegetable plantations are not considered in the ecoinvent modules for agricultural field work and the irrigation of green land is less common, the overhead irrigation system with high precipitation sprinkler (or rainer) (Strasser 1990) was chosen as a characteristic irrigation technique for arable crops.

An average amount of water sprayed on the fields of 120 mm per year was assumed. With a field size of one hectare, this means that 1200 m³ water per hectare is sprayed. This amount is distributed over four irrigation units per year. It was assumed that the irrigation water was taken from surface water (such as a river, creek or channel, or perhaps a lake).

There are two irrigation modules available, per hectare (“irrigating, CH, ha”) and per cubic meter irrigated water (“irrigating, CH, m³”). Both base on exactly the same data inventory (the second datasets equals 1/1200 of the first dataset).

Infrastructure

The 300-kg pump is equipped with a 22-kW engine with a volumetric capacity of 30 m³/h (hence, the operation time is 10 hours per irrigation unit) and a delivery pressure of 7 to 8 bar (Tsurumi 2002). A 100-m-long high-density polyethylene water pipe with a weight of 4.1 kg per metre (values are based on manufacturer’s information) is fitted to carry the water from the pump to the hydrant near the field. Installing the water pipe in the soil requires the excavation of 100 m³ of soil per hectare. Both the pump and the hydrant are made of cast iron. Since the hydrant is designed for three hectares, a third of a unit (20 kg) was used for one-hectare calculations. Fifty metres of polyvinyl chloride (PVC) water hose (0.9 kg/m) are needed to connect the hydrant to the irrigation automaton (DLG 2000 and manufacturer’s information).

The mobile, turbine-driven, surface-irrigation automaton is equipped with 300 m of water hose with an outer diameter of 75 mm. The hose is made of polyethylene, and weighs 400 kg (based on Strasser 1990). The irrigation automaton was inventoried with 1300 kg (Eggers *et al.* 1992) of general agricultural machinery (ecoinvent module “agricultural machinery, general, production”).

The lifetimes of the different irrigation components were taken from Ammann (2001) and personal communications²³.

Energy use

Electricity consumption for the water pump is 880 kWh/ha of sprinkled surface ($22 \text{ kW} \times 1200 \text{ m}^3 / 30 \text{ m}^3/\text{h}$). A tractor is used to install the equipment. Diesel consumption is 3.78 kg/ha ($4 \times 0.25 \text{ h} \times 4.5 \text{ l/ha} \times 0.83 \text{ kg/l}$) (see also Tab. A. 9 and Tab. A. 10).

7.3 Data Quality Considerations

Data on infrastructure (machinery and shed), irrigation and heavy-metal emissions from tyre abrasion were based on recent literature – mainly from ART, but also from international sources – and on experts’ estimates or unpublished data.

The values for fuel consumption and HC-, NO_x- and CO emissions are recent published and unpublished ART test-bed and field-work measurements valid for Swiss conditions.

The other air-emission values were taken from a current Swiss publication on off-road vehicles.

Emissions to soil (except from tyre abrasion), soil compaction, erosion, biodiversity, emissions to water, noise and dust were not considered.

Not considering the working materials (except for the PE film for baling) increases accuracy, since no assumptions need to be made, and the actual materials employed, which are known to the user, will be used in the calculations.

²³ Personal communication from H. Ammann, Swiss Federal Research Station for Agricultural Economics and Engineering (FAT), Taenikon, Ettenhausen, Switzerland, September 2002.

Part III: Agricultural Inputs

8 Mineral Fertilisers

8.1 Characteristics

Categories of Fertilisers

Fertilisers are substances that furnish essential nutrients for plant growth, thereby improving the yield and in some cases the quality of plant products. In agricultural production, the major elements limiting growth are nitrogen (N), phosphorus (P) and potassium (K) (Tab. 8.1).

On the basis of these primary nutrients, fertilisers are classified into three main categories of agricultural relevance:

- N fertilisers,
- P fertilisers,
- K fertilisers.

Moreover, there are multinutrient fertilisers composed of two or more nutrients, namely:

- NP fertilisers containing nitrogen and phosphorus,
- NK fertilisers containing nitrogen and potassium,
- PK fertilisers containing phosphorus and potassium, and
- NPK fertilisers containing nitrogen, phosphorus and potassium.

Apart from these three primary nutrients needed for growth, there are secondary nutrients that are also needed, albeit generally in lower quantities than the primary nutrients.

The secondary nutrient calcium (Ca) is not only applied to the soil in order to provide this nutrient; in general, calcium-containing products such as lime are applied to increase soil pH. A balanced pH is essential for the fertility of agricultural soils.

Despite its twofold use, lime is generally categorised as a fertiliser. The same applies to magnesium (Mg), though the application of magnesium to adjust soil pH is less relevant in agricultural use. In order to provide both these elements simultaneously, dolomite $\text{CaMg}(\text{CO}_3)_2$ containing 50 - 55% CaCO_3 and 40 - 44% MgCO_3 is widely applied.

Plants also require micronutrients, though in much lower quantities than primary and secondary nutrients. The application of micronutrients as fertilisers is less frequent and the quantities used are relatively small. No inventories for fertilisers containing micronutrients are included in ecoinvent data.

Tab. 8.1 Essential elements required for plant growth (Ohio Agronomy Guide 2002).

Nutrient category	Element
Primary nutrients	Nitrogen (N), Phosphorus (P), Potassium (K)
Secondary nutrients^{*)}	Sulphur (S), Magnesium (Mg), Calcium (Ca)
Micronutrients (usually < 100 mg/kg)	Iron (Fe), Manganese (Mn), Boron (B), Chlorine (Cl), Zinc (Zn), Copper (Cu), Molybdenum (Mo)

^{*)} Besides the essential elements Hydrogen (H), Oxygen (O) and Carbon (C).

Based on their chemical nature, fertilisers can be divided into mineral and organic fertilisers. Mineral fertilisers consist mainly of inorganic substances such as ammonium nitrate. By contrast, organic fertilisers are composed of organic compounds and materials, such as compost. Most organic fertilisers are by-products of industrial processes or outputs from waste recycling. Although urea-containing products are organic compounds, these are generally categorised under mineral fertilisers, since their manufacturing process and their application in agriculture is very similar to that of mineral N fertilisers. Note that the urea-containing fertilisers are categorised as “chemicals” in ecoinvent data, subcategory “organic” for urea and “inorganic” for urea ammonium nitrate.

Importance of Fertiliser Production and Use

370 million tonnes of fertilisers were produced worldwide in 2000 (Tab. 8.2, FAOSTAT Database, FAO 2002). 13% of this quantity was consumed in Western Europe, of which nitrogen fertilisers comprised the largest share (64% related to the weight of the product), followed by phosphate fertilisers (22%) and potassium fertilisers (14%).

Roughly 250,000 tonnes of mineral fertilisers were consumed in Switzerland in 2000, representing less than 0.1% of worldwide fertiliser production. Total consumption of fertilisers in 2000 amounted to 66% in the form of nitrogen fertilisers, 14% in the form of phosphate fertilisers, and 20% in the form of potassium fertilisers (on the basis of the weight).

No mineral fertilisers have been produced in Switzerland since 1997, except for calcium ammonium nitrate (CAN). Production of this fertiliser covers approx. 45% of the domestic consumption, and roughly 27% of the total consumption of N fertilisers in Switzerland (FAOSTAT Database, reference year 2000, FAO 2002).

Tab. 8.2 Key statistics related to fertilisers (Source: FAOSTAT Database, reference year 2000, FAO 2002). Values given refer to thousands of tonnes of fertiliser as well as to thousands of tonnes of plant nutrients per annum. Mean nutrient content according to Patyk & Reinhardt (1997).

Values in 1000 tons for the year 2000		Switzerland				Western Europe		Europe		World
		Consumption	Production	Imports	Exports	Consumption	Production	Consumption	Production	Production
Nitrogenous fertilisers (mean N- content: 32.5%)	as N	55	15	46	6	10,158	7,419	13,355	20,294	84,616
	as product	169	46	142	18	31,254	22,829	41,094	62,444	260,356
Phosphate fertilisers (mean P ₂ O ₅ - content 46.7%)	as P ₂ O ₅	17	0	18	1	3,105	1,725	4,115	5,320	31,704
	as product	36	0	39	3	6,648	3,694	8,812	11,392	67,890
Potash fertilisers (mean K ₂ O - content 60%)	as K ₂ O	30	0	30	0	3,938	4,489	4,911	11,596	25,541
	as product	50	0	50	0	6,564	7,482	8,186	19,328	42,569
Total fertilisers	as product	256	46	230	22	48,597	34,004	58,092	93,164	370,814

The use of fertilisers has helped increase agricultural crop yields over the last few decades (Maene 2000), and has thus contributed to the nutrition of a growing world population. On the other hand,

intensive fertiliser use can lead to numerous environmental problems, e.g. eutrophication through loss of nitrogen and phosphorus to water, acidification caused by loss of ammonia from N fertilisers, pollution of drinking water by nitrates, increased greenhouse effect from emissions of N₂O during the denitrification of N fertilisers, and from the use of energy in the production and transport of fertilisers. Over the past decade, measures have been taken in Switzerland to reduce the negative environmental impacts of agriculture. Among these are measures to reduce the use of fertilisers (BLW 2000 & 2001, see also chapter 3).

Emissions caused by the use of fertilisers in plant production are described in chapter 4.4.

8.2 Life Cycle Inventories of Mineral Fertilisers

Life cycle inventories were established for 26 mineral fertilisers listed in Tab. 8.3. These represent a selection of fertilisers that are widely applied in Swiss agriculture.

Since the procedures followed to compile the life cycle inventories of mineral and organic fertilisers differed significantly, organic fertilisers will be treated separately in chapter 9.

The mineral fertilisers belong to the following categories:

- Nitrogen- or N fertilisers (inventories 1 to 6),
- Phosphate- or P fertilisers (inventories 7 to 9),
- Potash- or K fertilisers (inventories 10 to 11),
- Multinutrient fertilisers (NP- and, NK fertilisers, inventories 12 to 23),
- Lime and stone meal (inventories 24 to 26).

The inventory data for mineral fertilisers were compiled from several sources, requiring different procedures and assumptions. For this reason, the documentation was divided into Subchapters 8.2.1 to 8.2.4, according to the basic references. Tab. 8.3 gives an overview of the different modules, the production processes, and the chapters in which the documentation can be found.

Tab. 8.3 Overview of the mineral fertilisers included in the ecoinvent database. The reference function is 1 kg of the nutrient content in the final product.

No.	Inventory name	Location	Abbr.	Short description of the production processes and sources of further information	Chapter
1	ammonium nitrate, as N, at regional storehouse	RER	AN	Ammonium nitrate is produced by the neutralisation of ammonia with nitric acid. The resulting solution is evaporated and then granulated (Davis & Haglund 1999). Supply materials: Ammonia and nitric acid.	8.2.1
2	calcium ammonium nitrate, as N, at regional storehouse	RER	CAN	The production process starts with the production of the ammonium nitrate. The final product is obtained by adding dolomite or limestone to the solution before drying and granulating it (Davis & Haglund 1999). Supply materials: Ammonia, nitric acid and dolomite.	8.2.1

No.	Inventory name	Location	Abbr.	Short description of the production processes and sources of further information	Chapter
3	urea, as N, at regional storehouse	RER		The first step for the synthesis of urea is the production of ammonium carbonate by combining ammonia and carbon dioxide at high pressure. During the next step, carbonate is dehydrated to form urea and water, then urea is concentrated to form a solid product (Davis & Haglund 1999). Supply materials: Ammonia and carbon dioxide.	8.2.1
4	urea ammonium nitrate, as N, at regional storehouse	RER	UAN	The production of urea ammonium nitrate consists in mixing solutions of urea and ammonium nitrate; the resulting solution is then cooled (Davis & Haglund 1999). Supply materials: liquid ammonium nitrate and liquid urea.	8.2.1
5	calcium nitrate, as N, at regional storehouse	RER	CN	Most of the calcium nitrate is a by-product of the NPK production process by the nitrophosphate route. The precipitated crystals of calcium nitrate are melted and neutralised by gaseous ammonia. The resulting melt is granulated (Davis & Haglund 1999). Supply materials: Nitric acid and limestone or rock phosphate.	8.2.3
6	ammonium sulphate, as N, at regional storehouse	RER	AS	"Most of the ammonium sulphate used in fertilisers is a by-product of caprolactam production or gas scrubbing. A small amount is produced synthetically or produced captive in fertiliser processes" (Davis & Haglund 1999). Supply materials: Sulphuric acid and ammonia.	8.2.3
7	single superphosphate, as P ₂ O ₅ , at regional storehouse	RER	SSP	SSP is the product of the reaction between rock phosphate and sulphuric acid. The phosphate must be converted to water-soluble monocalcium, as this process allows retention of the calcium phosphate in the product (Davis & Haglund 1999). Supply materials: Sulphuric acid and rock phosphate.	8.2.1
8	triple superphosphate, as P ₂ O ₅ , at regional storehouse	RER	TSP	TSP results from the reaction of rock phosphate with phosphoric acid. In the final product, the phosphorus must be supplied as 70% acid and 30% rock (Davis & Haglund 1999). Supply materials: Phosphoric acid and rock phosphate.	8.2.1
9	thomas meal, as P ₂ O ₅ , at regional storehouse	RER		Thomas meal is a by-product in the form of slag, obtained during steel production from low-grade ore (Thomasdünge GmbH 2003).	8.2.4

No.	Inventory name	Location	Abbr.	Short description of the production processes and sources of further information	Chapter
10	potassium sulphate, as K ₂ O, at regional storehouse	RER		There are two ways to produce potassium sulphate: The first consists in causing potassium chloride to react with sulphate-bearing materials, e.g. kieserite. The second involves the reaction of potassium chloride with sulphuric acid (Davis & Haglund 1999).	8.2.1
11	potassium chloride, as K ₂ O, at regional storehouse	RER		Potassium chloride is mainly produced by mining and beneficiation of potash salt (Kongshaug 1998, K+S Aktiengesellschaft 2001). A part of potassium chloride is produced from sea water. Raw material: potash salt.	8.2.2
12	ammonium nitrate phosphate, at regional storehouse	RER	ANP	Ammonium nitrate phosphate or nitro AP is produced by the nitrophosphate route (David & Haglund 1999). Supply materials: Ammonia and rock phosphate.	8.2.1
13	ammonium nitrate phosphate, as N, at regional storehouse	RER	ANP		8.2.1
14	ammonium nitrate phosphate, as P ₂ O ₅ , at regional storehouse	RER	ANP		8.2.1
15	potassium nitrate, at regional storehouse	RER	KN	Potassium nitrate is mainly produced "by ion exchange process based on nitric acid and muriate of potash" (Kongshaug 1998). Supply materials: potassium chloride and nitric acid.	8.2.1
16	potassium nitrate, as N, at regional storehouse	RER	KN		8.2.1
17	potassium nitrate, as K ₂ O, at regional storehouse	RER	KN		8.2.1
18	monoammonium phosphate, at regional storehouse	RER	MAP	"Production of ammonium phosphates, diammonium phosphates and monoammonium phosphates, has developed through phases from atmospheric neutralisation in tanks, to pressure neutralisation and finally to pipe reactors in the granulator and/ or dryer. Today the most energy-efficient processes utilise pipe reactors" (Davis & Haglund 1999). Supply materials: Phosphoric acid and ammonia.	8.2.1
19	monoammonium phosphate, as N, at regional storehouse	RER	MAP		8.2.1
20	monoammonium phosphate, as P ₂ O ₅ , at regional storehouse	RER	MAP		8.2.1

No.	Inventory name	Location	Abbr.	Short description of the production processes and sources of further information	Chapter
21	diammonium phosphate, at regional storehouse	RER	DAP	"Production of ammonium phosphates, diammonium phosphates and monoammonium phosphates, has developed through phases from atmospheric neutralisation in tanks, to pressure neutralisation and finally to pipe reactors in the granulator and/ or dryer. Today the most energy-efficient processes utilise pipe reactors" (Davis & Haglund 1999). Supply materials: Phosphoric acid and ammonia.	8.2.1
22	diammonium phosphate, as N, at regional storehouse	RER	DAP		8.2.1
23	diammonium phosphate, as P ₂ O ₅ , at regional storehouse	RER	DAP		8.2.1
24	lime, algae, at regional storehouse	CH		Lime from algae is extracted from the sea bed, dried, milled and used as fertiliser (Lithofertil 2003).	8.2.4
25	lime, from carbonation, at regional storehouse	CH		Lime from carbonation is a by-product of sugar manufacture, used to precipitate the sugar water. The remaining lime content is used as fertiliser (Ricoter 2003).	8.2.4
26	stone meal, at regional storehouse	CH		Stone meal is used as a fertiliser owing to its content in silicic acid (between 34 – 56%), lime (between 30 – 56%) and other inorganic components (Ulrich & Partner 2003).	8.2.4

For applications of agricultural lime (CaCO₃), use of the module "limestone, milled, packed, at plant CH" documented in Kellenberger *et al.* (2007), is recommended.

In order to give an overview of the production routes of fertilisers, we adapted and completed the block model by Davis & Haglund (1999) represented in Fig. 8.1. This flow chart shows the relationships between raw materials, intermediates and final products, as well as the interrelationships of the different production processes.

Multinutrient Fertilisers

The fertiliser industry also produces compound fertilisers with several nutrients (NPK fertiliser). These products contain nitrogen (N), phosphorus (P) and potassium (K). The total nutrient content (N+P₂O₅+K₂O) is normally between 40 and 50%. This category of products can also contain magnesium, boron, sulphur and trace elements (EFMA 1995).

Basically, there are two ways to manufacture these products: the mixed acid route and the nitrophosphate route.

- Production of NPK fertilisers via the *mixed acid route* (EFMA 2000c):
This production method allows the creation of a large variety of multinutrient fertilisers by combining phosphoric, sulphuric and nitric acid as well as ammonium nitrate solution in some cases. The manufacture of these products begins with the production of phosphoric acid, a step which creates a large quantity of gypsum. The mixing of the acids, with ammonium nitrate in some cases, is followed by a neutralization step in which gaseous ammonia is added. Other materials may be added at the end of or during this production step in order to enlarge the variety of the final products. The last step consists in the granulation of the final product.
- Production of NPK fertilisers via the *nitrophosphate route*:
The aim of this production method is to create a final product containing ammonium nitrate, phosphate and potassium salts "starting from rock phosphate and using all the nutrient components in an integrated process without solid wastes and with minimal gaseous and liquid emissions" (EFMA 2000b).
At the beginning of this process, the rock phosphate is digested with nitric acid to produce nitrophosphoric acid and calcium nitrate. The solution is cooled to form crystals of calcium

nitrate. The nitrophosphoric acid is neutralized with ammonia, after which reaction other substances can be added to the solution. Finally, the resulting mixture is granulated or prilled to obtain a solid final product.

The flow chart in Fig. 8.1 also represents NPK fertilisers. Some examples of multinutrient fertilisers with two major fertilising elements (NP- and NK fertilisers) were included in ecoinvent data. No NPK fertilisers were inventoried, however, because there are many different ways to mix N-, P- and K fertilisers to create various multinutrient products. Davis & Haglund (1999) give some examples for this group of products. For practical application, such multinutrient fertilisers can be approximated by combining the inventories given in ecoinvent data.

The following information must be known:

- the N-, P- and K-nutrient content,
- the form of the nutrient (e.g. ammonium, nitrate or urea for N).

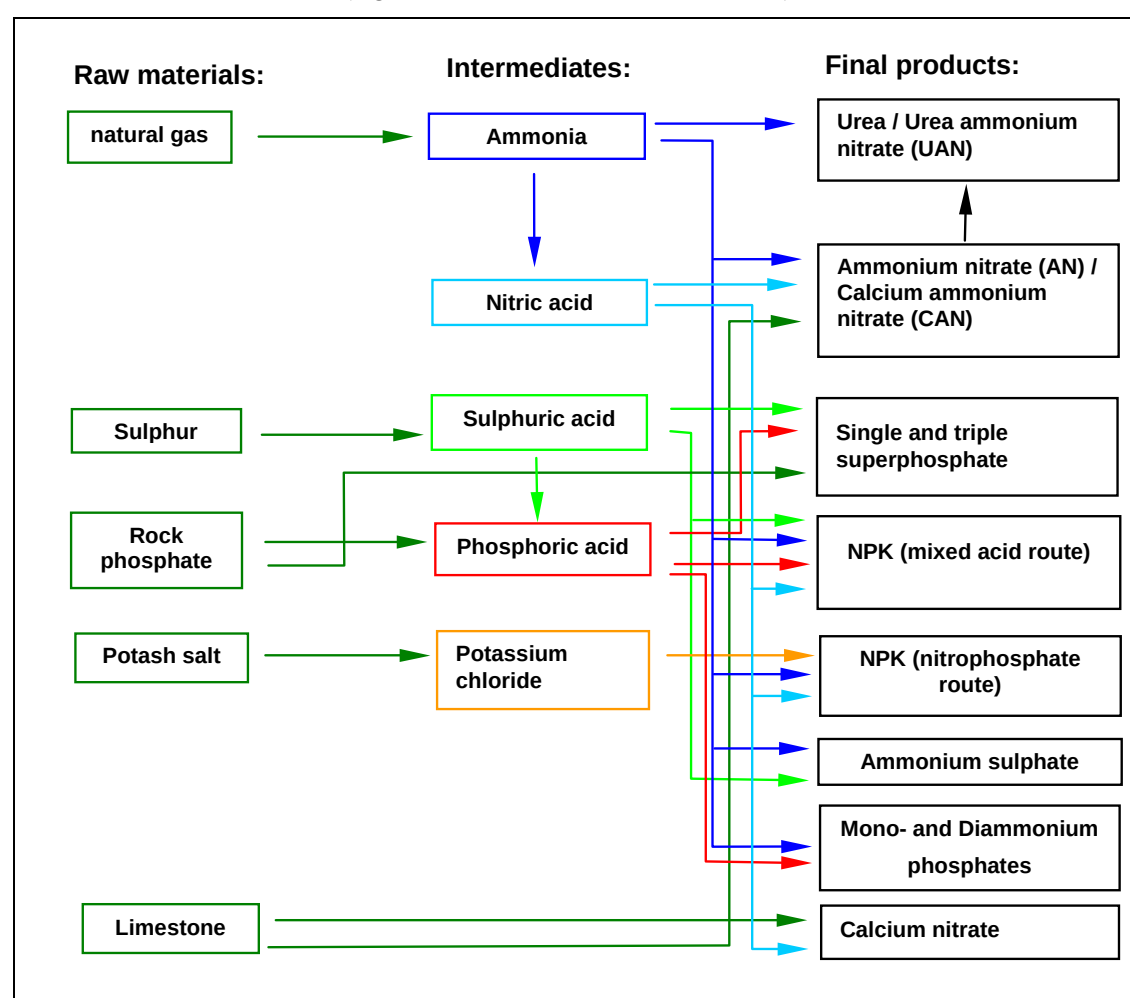


Fig. 8.1 Schematic representation of the production routes of fertilisers elaborated from Davis & Haglund (1999), completed from the EFMA (1997) and UNEP, UNIDO & IFA (1998). Ammonium sulphate is normally produced as by-product of the nylon-manufacturing process. Calcium nitrate mainly emerges as a by-product of the production of NPK fertilisers via the nitrophosphate route.

Evaluation of Data Sources for Fertiliser Inventories:

Because there is no single source containing all the required data, various data sources were used to compile the fertiliser inventories.

The unit process inventories for most of the mineral fertilisers were taken from Davis & Haglund (1999). This publication, which is near to the target year 2000, incorporates older publications by Patyk (1996), UNEP (1996), Kongshaug (1998) and the EFMA (1995). These inventories consider the different steps of the production processes, such as the use of raw materials and semi-finished products, the process energy, the transport of raw materials and intermediate products, and the relevant emissions. The inventories for the products used in Sweden were calculated from values measured in local production plants. The inventories created for the fertilisers used in Western Europe are composed of the average values of the European Fertilizer Manufacturers Association (EFMA). A comparison of the two types of inventories reveals minimal differences between the resulting data. It is therefore possible to affirm that the technological differences between the production plants in Sweden and Western Europe are insignificant, and that the data used for the Western Europe inventories can be considered to approximate the values measured directly in the production plants.

The data used for the ammonium sulphate and calcium nitrate inventories were taken from Kongshaug (1998). This publication considers only the process energy required for the production steps from the intermediates to the final product. Three different scenarios are illustrated for the inventories studied in this report: production using modern technology, production using old technology (in 30-year-old plants), and the average Europe scenario. Because of the significant differences between the technologies used, the energy values vary widely. This publication does not furnish a detailed description of the processes considered for fertiliser production, so it is difficult to make comparisons with values from other sources.

The aim of the European Fertilizer Manufacturers Association (EFMA) publications is to describe the best available technique for fertiliser production in Europe, and consequently, the relevant ways to prevent and control emissions. The different reports describe the product manufacturing processes and the emissions data in detail, but give no information on energy use. Inventories not found in the above publications were compiled from Garcia & Nemecek (2000) and Audsley *et al.* (1997).

System Boundaries and Included Processes:

The following production steps for fertilisers are included in the ecoinvent unit process inventories:

1. *The transport of raw materials and intermediates to the production plant:* The unit-process inventory starts with the transport of the different raw materials from the extraction plant, or from the plants where the intermediates are produced, to the final processing plant. The mining process is included in the unit-process inventory for KCl only; the other mining processes are already included in the modules for the intermediate products such as phosphoric acid or phosphate rock.
2. *The synthesis of the different components to produce the final product:* The inventories include energy use in the production processes, the use of supply materials, and process emissions at the production plant. The only waste listed in this group of inventories is gypsum, produced during the manufacture of phosphoric acid (Davis & Haglund 1999). This waste is already included in the inventory for phosphoric acid (Althaus *et al.* 2007). Salt residues and wastes arising during potash mining were included in the inventory for KCl.
3. *The transport of the final product to the regional storehouse in Europe* (Central Europe was taken as the reference).
4. *Infrastructure usage* (buildings, equipment and land use) was approximated in these inventories by means of the generic infrastructure module “chemical plant, organics, RER”.

The following processes and inputs are not part of the systems considered:

- *Transport of the final products from the regional storehouse in Europe to the user in Switzerland* was not included in these inventories, since this depends largely on the country and region under consideration. Users of the modules must include these distances in their own calculations. Tab. 8.6 gives typical transport distances from the regional storehouse in RER to the final user in Switzerland.
- In the inventories based on data from Davis & Haglund (1999), the following elements are not included: “coating of the final product”, “small amounts of salts and other additives”, and “micronutrients”.
- The inventories of fertilisers listed in Tab. 8.3 do not include the packaging of the final product. The environmental impact of packaging is considered to be negligible.

8.2.1 Inventories of Mineral Fertilisers²⁴ Based on the Unit Process Inventories Specified in Davis & Haglund (1999)

The unit-process inventory data for 20 mineral fertilisers were taken from inventories in Davis & Haglund (1999) (see “Basic reference” column in Tab. 8.12), which were in turn composed of averaged data from the EFMA (1995), Kongshaug (1998) and Patyk (1996).

Only those inventories in Davis & Haglund (1999) referring to the Western European average were selected. According to Davis & Haglund (1999, p.37), the European average, created from the averaged data of several fertiliser plants in Europe and general information from the European Fertilizer Manufacturers Association (EFMA), can be considered representative for the production of fertilisers in Western Europe. These data include a certain amount of imports to Western Europe.

The inventories given in Davis & Haglund (1999) take into account the use of resources for all production steps ranging from the extraction of the raw materials to the production of the intermediates and the final product. Catalyst production and waste treatment as well as production of capital goods (machinery and buildings) have been left out. For the modules in the ecoinvent database, infrastructure use was approximated by the “chemical plant, organics” proxy module. Moreover, coating and packaging of the final fertiliser products were not included in the original reference.

Please refer to Davis & Haglund (1999) for a detailed description of the data sources and the methodology and assumptions applied.

For the calculation in ecoinvent data, the following assumptions and modifications were applied:

Inputs of Materials and Energy

Inputs specified in Davis & Haglund (1999) were interpreted and assigned to ecoinvent inventories as shown in Tab. 8.4.

²⁴ including urea-containing fertilisers.

Tab. 8.4 Inventories in ecoinvent data assigned to input specifications given in Davis & Haglund (1999).

Input specification in Davis & Haglund (1999)	Inventory assigned in ecoinvent data		
	Name	Location/Cate gory	Unit
Natural gas	heat, natural gas, at industrial furnace >100kW	RER	MJ
Heavy oil	heat, heavy fuel oil, at industrial furnace	RER	MJ
Steam	Inventories using HNO ₃ as input: energy categorised as 'steam' which is required for fertiliser production is considered to result from the heat released during HNO ₃ -production. The energy contained in the steam was quantified with a negative sign using the exchange ^{*)} : Heat, waste.	air/high population density	MJ
Steam	Inventories requiring no HNO ₃ as input: we assumed that steam is produced by the combustion of natural gas using the inventory: heat, natural gas, at industrial furnace >100kW	RER	MJ
Electricity, European average	electricity, medium voltage, production UCTE, at grid	UCTE	kWh
Ammonia	ammonia, steam reforming, liquid, at plant	RER	kg
Nitric acid	nitric acid, 50% in H ₂ O, at plant	RER	kg
H ₃ PO ₄	phosphoric acid, fertiliser grade, 70% in H ₂ O, at plant	MA	kg
Commercial rock phosphate	phosphate rock, as P ₂ O ₅ , beneficiated, dry, at plant	MA	kg
Dolomite	limestone, powdered, at plant	CH	kg
KCl	potassium chloride, as K ₂ O, at regional storehouse	RER	kg
H ₂ SO ₄	sulphuric acid, liquid, at plant	RER	kg
Carbon dioxide	We refrained from considering CO ₂ -consumption during the production of urea and urea ammonium nitrate (UAN) as quantified in Davis & Haglund (1999). In accordance with Davis & Haglund (1999, p. 85), we assumed that urea synthesis took place at the same plant as NH ₃ -production. CO ₂ is produced as a by-product during NH ₃ -production, and is consequently used as input for the synthesis of urea (see also Althaus <i>et al.</i> 2007). To produce 1 t of urea, 0.733 t of CO ₂ is required (Davis & Haglund 1999). This CO ₂ is released after the application of urea in the field and has to be considered in the inventory for the agricultural production. 1.57 kg CO ₂ is released per kg of "urea, as N, at regional storehouse".	-	-

^{*)} Quoting Davis & Haglund (1999, p.69): "In reality, steam consumed during production of ammonium nitrate is in most cases taken from steam produced in a nearby nitric acid plant."

Transports

Standard transport distances for basic chemicals listed in Frischknecht *et al.* (2007) were used as reference for N-, P- and K fertilisers, distances for the transport of lime were taken from Patyk & Reinhardt (1997) and those for Thomas meal from Gaillard *et al.* (1997) (Tab. 8.5). These distances refer to the transport from the manufacturer in RER to the regional storehouse in RER.

Tab. 8.5 Transport distances used to calculate fertiliser transport from the manufacturer in RER to the regional storehouse (RER), according to specifications given in Frischknecht *et al.* (2007), Patyk & Reinhardt (1997) and Gaillard *et al.* (1997).

Transport carrier (name of inventory used in ecoinvent data)	transport, lorry >16t, fleet average; RER in km	transport, lorry 3.5-16t, fleet average; RER in km	transport, freight, rail; RER in km	transport, barge; RER in km
N-fertiliser	100		600	
P-fertiliser	100		600	
K-fertiliser	100		600	
Lime fertiliser		79	291	141
Thomas meal	100	100	50	

Generally speaking, except for the inventories of phosphoric acid, phosphate rock and limestone, the ecoinvent inventories used to calculate the inputs required for fertiliser production already refer to the situation in Europe (RER). Consequently, transport specifications for basic raw materials and intermediate products in Davis & Haglund (1999) were not included. Transport to the fertiliser plant was supplemented for the inputs phosphoric acid, phosphate rock and limestone only:

- The inventories phosphoric acid and phosphate rock refer to the location “at plant, Morocco (MA)”. For the transport of these materials from the plant in Morocco to a fertiliser plant in central Europe, transport distances of 2500 km by transoceanic freight and 400 km by lorry (own estimates) were taken into account.
- The inventory limestone refers to the situation “at plant, Switzerland (CH)”. To convert the inventory “limestone, at plant, CH” into an inventory for RER, transport distances were adopted from Patyk & Reinhardt (1997), who estimated that 291 km by train, 79 km by lorry and 141 km by barge are typical transport distances connected with the production of lime fertilisers and their transport to a regional storehouse in Europe. As no indication is given in the source, we assumed that half of this transport is connected with the transport of the raw material lime to a fertiliser plant in Europe.

For including the transport associated with the transfer of the fertilisers from the regional storehouse (RER) to the user in Switzerland, the transport distances in Tab. 8.6 may be used; this information was provided by a major fertiliser retailer in Switzerland²⁵. These distances are not included in the inventories described in this chapter, but should be included in applications of these modules for Switzerland.

Tab. 8.6 Typical transport distances of fertilisers from the regional storehouse (RER) to the user in CH.

Transport carrier/ Fertiliser category	Train in km	Road (lorry) in km	Barge in km
N- fertiliser	100	100	900
P- fertiliser	100	100	400
K- fertiliser	100	100	100
Lime	120	120	-
NPK-fertiliser	100	100	600
PK-fertiliser	100	100	500

²⁵ Personal communication from C. Kopp, Landor GmbH, 6 February 2001.

Infrastructure and Land Use

No data were available on the use of infrastructure and land for the production of mineral fertilisers. The infrastructure in all modules of mineral fertilisers described in chapters 8.2.1, 8.2.2, 8.2.3 and 8.2.4 (except for lime from carbonation, for which no production process is included) was therefore approximated by the inventory “chemical plant, organics, RER” (assuming a production of 50,000 tonnes/year and a plant lifetime of 50 years (Althaus *et al.* 2007)). As all modules refer to the nutrient content, the quantity of the plant used was corrected for the respective nutrient content of each fertiliser.

Emissions

The data on the process emissions produced during the fertiliser manufacturing were taken from Davis & Haglund (1999). If data in this source were absent or incomplete, the process emissions were supplemented by specifications given in Audsley *et al.* (1997), Gaillard *et al.* (1997), Patyk & Reinhardt (1997) and the EFMA (2000a) (see column “Complemented or modified emission factors” in Tab. 8.12). The following general approach was used:

- N-containing emissions to water stated as the total amount of nitrogen (N_{tot}), i.e. those not differentiated into $\text{NH}_3/\text{NH}_4^+$ or NO_3^- species, were converted into NH_3 emissions to rivers on the basis of N content.
- The data on P-containing emissions to water were inventoried in the sources we consulted as nutrient in the form of P_{tot} . In these cases, the amount of P_{tot} was converted in ecoinvent into PO_4^{3-} emissions on the basis of P content. We assumed that these substances are emitted into rivers.
- Process emissions of heavy metals to water during production of phosphorus-containing fertilisers were taken from Audsley *et al.* (1997) (Tab. 8.7). No data on process emissions of heavy metals were available for any other category of fertilisers; however, it is known that heavy metals cause problems mainly in the case of P fertilisers (see Desaulles & Studer 1993, p. 152).

Tab. 8.7 Process emissions during manufacture of mineral P fertilisers (including P-containing multinutrient fertilisers) according to Audsley *et al.* (1997).

Heavy metal emissions during production of P-containing fertilisers into rivers, in mg								
per kg P_2O_5	As	Cd	Cr (III)	Cu	Hg	Ni	Pb	Zn
P-fertiliser	4.4	4.4	22	22	4.2	17	19	26

- Particulate emissions released during the production of N-containing fertilisers result on the one hand from gas-particle conversion of gaseous precursor compounds (e.g. NH_3) (EFMA 1995), producing nucleation-mode particles (particle diameter mostly below $2.5 \mu\text{m}$ ($\text{PM}_{2.5}$)), and on the other hand from resuspension of the fertiliser product during subsequent manufacturing processes, which produces mainly mechanically derived particles in the coarse mode ($\text{PM} > 2.5 \mu\text{m}$). For all N-containing mineral fertilisers, including multinutrient fertilisers, we assume that $\text{PM}_{2.5}$ and PM_{10} contribute 40% and 60%, respectively, to the total particulate matter (TPM) emitted. Particles emitted into the air during the manufacturing process of P-containing fertilisers containing no nitrogen, i.e. where no gas-particle processes are involved, are considered to consist predominantly of coarse-mode particles ($\text{PM} > 2.5 \mu\text{m}$) emerging from mechanical processes. For the P-containing mineral fertilisers SSP and TSP, we assume that $\text{PM}_{2.5}$ and PM_{10} contribute 20% and 60%, respectively, to the total particle mass (TPM) emitted.
- Heat released due to electricity use was taken into account, assuming that heat would be released into the air.

It was assumed that fertiliser production takes place in an urban/industrial area. As a consequence, all aerial emissions were categorised as emanating in an area of high population density.

Allocations

Multinutrient fertilisers were inventoried as referring to the fertiliser product, i.e. as these were specified in Davis & Haglund (1999); in addition, they are presented as referring to the respective nutrient. This nutrient-based allocation is required for practical considerations; when applying N-, P-, and K fertilisers, farmers calculate the quantities to be applied based on the respective nutrient. Furthermore, this procedure increases the comparability of the fertilisers.

IMPORTANT: Both allocated modules belonging to a multinutrient fertiliser must always be used together. The user of these allocated inventories must ensure that both components of the multinutrient fertilisers are taken in the correct quantities, i.e. that the ratio of the nutrients corresponds to the one given in Tab. 8.12.

The inputs and outputs of the multinutrient fertiliser (NP- and NK fertilisers) were allocated to each nutrient as follows:

- Energy requirements of the fertiliser product manufacturing process as given in Davis & Haglund (1999) were allocated to each nutrient using the averages of the total process energy for the production of each nutrient as stated by Patyk & Reinhardt (1997) (see Tab. 8.8). Differences in nutrient content between both references were adjusted. For ammonium nitrate phosphate, the same allocation factor as for MAP and DAP was applied.
- N-containing resources as inputs, such as ammonia and nitric acid, required for the multinutrient fertiliser product were allocated in their entirety to the fertiliser referring to the nutrient nitrogen. Similarly, P-containing inputs, such as phosphoric acid and phosphate rock, were attributed in their entirety to the fertiliser referring to the nutrient phosphorus. K-containing inputs, such as potassium chloride, were in turn assigned in their entirety to the fertiliser referring to the nutrient K.
- N-containing emissions as outputs required for the multinutrient fertiliser product were allocated in their entirety to the fertiliser referring to the nutrient nitrogen. Similarly, P-containing emissions were attributed in their entirety to the fertiliser referring to the nutrient phosphorus.
- Toxic metal emissions to the river evolving during production of P-containing fertilisers were attributed in their entirety to the fertiliser referring to the nutrient phosphorus.
- Allocation of the particle fractions of airborne emissions to the inventory referring to the nutrient nitrogen: PM_{2.5} 100%, PM_{2.5-10} 60% and TPM-PM₁₀ 40%. Allocation to the inventory referring to the nutrient phosphorus: PM_{2.5-10} 40% and TPM-PM₁₀ 60%.

Tab. 8.8 Allocation of energy requirements for the mineral fertiliser production process based on data in Patyk & Reinhardt (1997, pp. 88, 113 and 122) and Davis & Haglund (1999).

	MAP		DAP		ANP		KN	
	MJ/t N	MJ/t P ₂ O ₅	MJ/t N	MJ/t P ₂ O ₅	MJ/t N	MJ/t P ₂ O ₅	MJ/t N	MJ/t K ₂ O
Specific process energy input MJ/t nutrient	45070	11825	45070	11825	45431	6805	41981	8409
Nutrient content	11%	52%	18%	46%	8%	52%	14%	44%
Specific process energy input MJ/t	4958	6149	8113	5440	3816	3538	5877	3700

fertiliser product ^{*)}	MAP		DAP		ANP		KN	
	MJ/t N	MJ/t P ₂ O ₅	MJ/t N	MJ/t P ₂ O ₅	MJ/t N	MJ/t P ₂ O ₅	MJ/t N	MJ/t K ₂ O
Total process energy/kg product	11107		13552		7355		9577	
Allocation factor	45%	55%	60%	40%	52%	48%	61%	39%

^{*)} Calculated from the product of the nutrient content as given in Davis & Haglund (1999) and the total process energy specifications for each nutrient stated in Patyk & Reinhardt (1997).

- For multinutrient fertilisers, notably PK- and NP fertilisers, transport distances from the manufacturer in RER to the regional storehouse in RER were allocated on the basis of the ratio of the nutrient to the total nutrient content of the fertiliser (Tab. 8.9).

Tab. 8.9 Allocation of transport distances for multinutrient fertilisers based on the respective nutrient.

Mean nutrient content of multinutrient fertilisers (Davis & Haglund 1999)	Total nutrient content of the fertiliser product	Ratio of the nutrient to the total nutrient content of the multinutrient fertiliser (Allocation factors)		
In weight percent	%N + %P ₂ O ₅ resp. %N + %K ₂ O	N	P ₂ O ₅	K ₂ O
MAP (11%N, 52% P₂O₅)	63%	17%	83%	0%
DAP (18%N, 46% P₂O₅)	64%	28%	72%	0%
ANP (8.4%N, 52% P₂O₅)	60%	14%	86%	0%
KNO₃ (14%N, 44% K₂O)	58%	24%	0%	76%

8.2.2 Life Cycle Inventory of Potassium Chloride

Germany is the major producer of potassium fertilisers in Western Europe. Large quantities are also produced in Eastern Europe, mainly in the Russian Federation and Belarus.

Davis & Haglund (1999) did not give any data for potassium chloride. Patyk & Reinhardt (1997) give an overview of literature data applying to Germany. Overcash (2000) also provides data on KCl production. More-recent data for potassium fertiliser production than Patyk & Reinhardt's (1997) were found in the environmental reports of K+S Aktiengesellschaft (1999 & 2001), the only producers of potassium chloride in Germany and the largest one in Europe.

Patyk & Reinhardt (1997) and K+S Aktiengesellschaft (1999 & 2001) describe the production process. The salt is mined from underground mines, then crushed and milled. Three different processes are applied to concentrated KCl to produce a usable fertiliser: solution in hot water, flotation and electrostatic separation. The choice of method depends on the composition of the raw salts and the requirements for the final product. The data refer to a mixture of these three methods.

The unit-process inventory contains the processes of mining, concentration and drying of the salts as well as transport to the regional storehouse. The final product is solid and has a K₂O content of 60%, which corresponds to a purity of 95% KCl in the final product.

The data for the inventory were taken from K+S Aktiengesellschaft (2001, p.36), using the year 2000 (for the "Geschäftsbereich Kali- und Magnesiumprodukte" [potassium and magnesium products sector]).

K+S Aktiengesellschaft's (2001) total energy-consumption value (2.25 MJ/kg KCl and 3.75 MJ/kg K₂O) is only half Patyk & Reinhardt's (1997) older values (7.2 MJ fossil energy/kg K₂O and 100 kWh electricity/kg K₂O). Overcash (2000) gives the process energy input as 1.5 MJ/kg KCl and 2.5 MJ/kg K₂O. Energy requirements decreased over the past few years owing to technological progress (K+S Aktiengesellschaft 2001), which might explain why the figures are substantially lower than older values, but similar to Overcash's (2000). K+S Aktiengesellschaft's figures (2001) therefore seem reasonable.

Natural gas is the only source of fossil energy for the production of heat and electricity. Electricity and heat are produced in the factory's own cogeneration units. The percentages of energy used in cogeneration units (86%) and for direct heat production (14%) were taken from K+S Aktiengesellschaft (1999, p.23). The values were considered in the inventories by means of the modules "natural gas, burned in industrial furnace >100kW, RER" and "natural gas, burned in cogeneration unit 1MWe lean burn, CH".

Cooling- and process-water quantities were taken from K+S Aktiengesellschaft (2001, p.36) and inventoried as water from river. The report gives the use of fuels, which was inventoried as "diesel, burned in building machine, GLO", since the report states that 95% diesel is used (mainly in machines for the mining process). The auxiliary materials cited in the environmental report were inventoried as "chemicals inorganic, at plant, GLO".

The quantities of the process emissions HCl and dust were inventoried as emissions to "air/high population density". The figures given by K+S Aktiengesellschaft (2001) for HCl and dust emissions are 90 times and 3 times lower, respectively, than the values given by Patyk & Reinhardt (1997). The latter authors, however, had no specific information on process emissions. Particle size distribution was taken from EPA (1995) as PM_{2.5}/TPM = 5%, PM₁₀/TPM = 9%.

The environmental report gives information on the quantity of waste and hazardous waste, but not on their composition. Therefore the generic waste modules "disposal, municipal solid waste, 22.9% water, to municipal incineration, CH" and "disposal, hazardous waste, 25% water, to hazardous waste incineration, CH" respectively, were used.

4.5 kg of raw salt is used per kg of product. The composition of the different salts is given by Patyk & Reinhardt (1997, Tab. 6-30). Sylvite (KCl) content in the raw salt is 15-35% (K+S Aktiengesellschaft 1999, p.18). The use of the resource was inventoried as "sylvite, 25 % in sylvinit, in ground" (1.13 kg/kg KCl resp. 1.89 kg/kg K₂O).

Only 22% of the raw salt can be used as fertiliser; the rest must be disposed of. The largest portion, namely 2.75 kg/kg KCl resp. 4.58 kg/kg K₂O is stored in huge heaps. This quantity was inventoried as "disposal, salt tailings, potash mining, 0% water, to residual material landfill". The composition of this residue was derived from Patyk & Reinhardt (1997, Tab. 6-30), after removal of the 95% KCl component²⁶. Another portion of the material (about 10%) is used as *Versatz*, i.e. to fill the cavities created by mining. This portion was not inventoried in ecoinvent, since the material is returned to the cavities from which it was excavated. Part of the residue is contained in brine, 50% of which is injected into rock (*Plattendolomit*) in a process called *Versenkung* ("sinking"). This part is not inventoried, since there appear to be no emissions. The other 50% of the brine is released to rivers. The corresponding emissions (Na⁺, K⁺, Ca⁺⁺, Mg⁺⁺, S⁻) were estimated from the composition of the residues and inventoried as emissions to rivers (Tab. 8.10). This composition was used to calculate emissions to rivers, as well as for the disposal of the salt residues in the module "disposal, salt tailings, potash mining, 0% water, to residual material landfill".

²⁶ According to Titkov *et al.* (1998) the average recovery rate of KCl from sylvite ores is about 95%.

Tab. 8.10 Calculated composition of the salt residues from potash salt mining.

Na	K	Ca	Mg	Cl	S	O
29.1%	0.6%	0.8%	4.8%	49.9%	5.0%	9.9%

Transport to the regional storage was supplemented according to Tab. 8.5.

8.2.3 Life Cycle Inventories of Mineral Fertilisers Approximated from Specifications for the Process Energy in Kongshaug (1998)

No unit-process inventories are given in Davis & Haglund (1999) for ammonium sulphate and calcium nitrate; however, Kongshaug (1998) quotes a process energy of 28.7 MJ/kg N and 41.4 MJ/kg N, respectively, for the production of these two products.

Ammonium sulphate can be produced by different paths, the most important of which is nylon (caprolactam) manufacture, in which ammonium sulphate emerges as a by-product from the caprolactam oxidation stream and the rearrangement reaction stream (Kongshaug 1998; EPA 1995). The synthetic path consists in combining anhydrous ammonia and sulphuric acid in a reactor. Yet another way of producing ammonium sulphate is to capture ammonia from coke-oven exhaust gases and to cause a reaction with sulphuric acid (EPA 1995). The ammonium sulphate solution is then processed in a water evaporator to obtain a thicker fluid. Ammonium sulphate crystals are obtained by centrifugation of the solution. After centrifugation, the solution is transferred to a water evaporator until the crystals contain about 1% to 2.5% water by weight. Lastly, the crystals are dehydrated in a rotary drum dryer or a fluidised-bed dryer (EPA 1995). Since the most important production path is caprolactam manufacture, which produces ammonium sulphate as a by-product, only the further processing was considered.

Calcium nitrate is a by-product of the manufacture of nitrophosphate fertilisers by the nitrophosphate route (EFMA 1995). The production of NPK fertiliser begins with the dissolving of the rock phosphate in nitric acid. The resultant solution is cooled to obtain calcium nitrate tetrahydrate crystals and nitrophosphoric acid. The following production steps consist in different reactions involving nitrophosphoric acid and other materials, and which yield a nitrate-bearing fertiliser. These reactions produce a calcium nitrate solution as by-product which can be neutralised and evaporated to obtain a solid fertiliser (EFMA 2000b).

The data on the process energy used in the life cycle inventories of ammonium sulphate and calcium nitrate were taken from Kongshaug (1998). From the latter's explanations, it was concluded that the energy use refers to the further treatment of ammonium sulphate and calcium nitrate as by-products. These values must be considered as uncertain, since the source contains no detailed description of the production processes.

The values from Kongshaug (1998) represent the situation in average European plants, and are based on the averaged data of several fertiliser plants in Europe and general information from the European Fertilizer Manufacturers Association (EFMA). This information on process energy was used and split into the different energy carriers according to the shares of the energy carriers given in Patyk & Reinhardt (1997) for an average N fertiliser (Tab. 8.11). An adjustment was made for the differences between the N content given in Patyk & Reinhardt (1997) and Kongshaug (1998).

The process emissions for ammonium sulphate were supplemented by EPA (1995). The main emissions were 0.38 g particles and 0.52 g VOC per kg N (taking the average of the different dryer types and assuming that emission control is performed). A particle-size distribution of 40% PM_{2.5} and 60% PM₁₀ was assumed, as for the other N fertilisers.

Process emissions for calcium nitrate were taken from EFMA (1995). They were calculated as 0.12 g nitrate/kg N and 2.7 g particles/kg N as emissions to air, and 8.6 g nitrate/kg N as emissions to water. The same particle-size distribution as for the other N fertilisers was assumed.

Transport from an average manufacturer in RER to a regional storehouse in RER was added on (Tab. 8.5).

Tab. 8.11 Shares of different process-energy carriers for an average N fertiliser (28.6% N), as quoted in Patyk & Reinhardt (1997, p. 94).

Energy Carrier	Inventory in ecoinvent data	Unit	Location	in% (total 100%)
Natural gas	heat, natural gas, at industrial furnace >100kW	MJ	RER	77.8
Fuel oil	heat, heavy fuel oil, at industrial furnace	MJ	RER	15.3
Hard coal	heat, at hard coal industrial furnace 1-10MW	MJ	RER	5
Diesel	diesel, burned in building machine	MJ	GLO	0.04
Electricity	electricity, medium voltage, production UCTE, at grid	kWh	UCTE	1.8

8.2.4 Inventories of Mineral Fertilisers Based on Garcia & Nemecek (2000) and Audsley *et al.* (1997)

The unit process inventory data for lime (algae and carbonation) and stone meal (Nos. 24 - 26 in Tab. 8.12) were based on a survey conducted by Garcia & Nemecek (2000), including several factories in Switzerland and France. The unit-process inventory data for Thomas meal (No. 9 in Tab. 8.12), for their part, were taken from Audsley *et al.* (1997).

For applications of lime obtained directly from the ground, we recommend the use of the inventory “limestone, milled, packed, at plant CH”, described in ecoinvent report No. 7 (Althaus *et al.* 2007).

With the exception of lime from algae, these mineral fertilisers emerge as by-products during the manufacture of other products. Hence, only energy consumption for the further processing of these by-products to obtain a usable fertiliser product, as well as their transport, were accounted for in these inventories. In the inventory “lime from algae”, the resource “calcite” contained in the algae was also included. The derivation of these inventories is explained below.

Lime from Algae

The unit-process inventory data come from a producer of lime from algae in France²⁷. Lime from algae is obtained from the seabed only for the purpose of producing this lime fertiliser, i.e. is not a by-product of any production process (Lithofertil 2003). The original water content of algae brought up from the seabed is 25% by weight. After being transported to the lime plant by ship and lorry (over distances of 60 km and 15 km, respectively), the algae are dried to a water content of 2.5%.

According to the manufacturer, natural-gas combustion is used as an energy source for drying. We assumed that the energy demand for drying algae comes to 5 MJ per kg water removed, which is the value given in chapter 13 for high-temperature grain drying. Using electricity as the energy source (electricity demand 20 kWh/t final product), the dried algae are then ground. Transport of the product to a regional storehouse in Switzerland was estimated as 1000 km by train.

According to LBL *et al.* (2001, p. 493), lime from algae in a commercial form contains 89% CaCO₃. Demand for the resource calcite contained in the algae was quantified as “calcite, in ground”.

²⁷ Personal communication from P. Letertre, Lithofertil, Landaul, F-56690, France, 22 March 2002.

Lime from Carbonation

Lime from carbonation is a by-product of sugar manufacture, in which lime is used to precipitate the sugar water (Ricoter 2003). The remaining lime-containing filter residue may be disposed of at the manufacturer's expense²⁸ or alternatively be used as liming in agriculture.

According to a provider of lime from carbonation²⁹ in Switzerland, no further processing is required to produce a commercial fertiliser.

Consequently, only transport from the manufacturer to the regional storehouse (which, according to the manufacturer, averages 60 km by lorry) was taken into consideration.

Stone Meal

Depending on the quality of the stone used to produce it, stone meal generally contains between 34 – 56% silicic acid and 30 – 56% lime, as well as other inorganic components (Ulrich & Partner 2003). Used in agriculture, stone meal improves pH-balance in soils and thus biological activity, in addition to supplying a certain amount of nutrients. Furthermore, it is used in organic farming to increase plant vigour and defence mechanisms against pathogens (Ulrich & Partner 2003).

Stone meal generally accrues as filter residue in stone mines and can thus be considered a by-product. According to a Swiss manufacturer³⁰, the following further processing is required to obtain a usable fertiliser: mixing, belt conveying and weighing with energy requirement (electricity) averaging 5 kWh per tonne stone meal.

In addition to the energy requirement for further processing, transport to the regional storehouse of 50 km by train and 50 km by lorry was included in this inventory.

Thomas Meal

Thomas meal (or Thomas slag) is obtained as a by-product in the form of slag during steel production from low-grade ore (Thomasdünge GmbH 2003). Thomas meal is applied in agriculture as a P fertiliser, but also improves soil pH due to its lime component. According to Audsley *et al.* (1997), the energy required to convert this waste product into a useful agricultural product is 9.6 MJ/kg P or 4.2 MJ/kg P₂O₅. Audsley *et al.* (1997) assumed that the energy was provided by the combustion of fuel oil.

To compile the Thomas meal inventory in ecoinvent data, the specifications given in Audsley *et al.* (1997) were adopted, supplemented by transport from the plant to the regional storehouse (see Tab. 8.5). The values were related to a P₂O₅ content of 17% by weight (LBL, 2001, p. 493).

²⁸ Personal communication from Mr. Fankhauser, Zuckerfabrik Aarberg, Switzerland, 23 March 2002.

²⁹ Personal communication from Mr. Würsch, Ricoter AG, Aarberg, Switzerland, 10 May 2000.

³⁰ Personal communication from Mr. Schildknecht, Bernasconi Carlo AG, Mineralmahlwerk Jurasit, Switzerland, 7 April 2000.

Tab. 8.12 Derivation of the inventories for mineral fertilisers of agricultural relevance in Switzerland included in ecoinvent data. (Abbr. = common abbreviation)

No.	Inventory name	Location	Unit	Nutrient content	Basic reference for the unit-process inventory	Supplemented or modified inputs and/or emission factors with respect to the basic reference.
1	ammonium nitrate, as N, at regional storehouse	RER	kgN	35% N	Davis & Haglund (1999), Appendix 5.	No process emissions specified in Davis & Haglund (1999). Emission factors for NH ₃ and particulate matter into air and N _{tot} into water were supplemented with data given in EFMA (2000a, p. 23). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
2	calcium ammonium nitrate, as N, at regional storehouse	RER	kgN	27% N	Davis & Haglund (1999), Appendix 9.	Emission factors given in Davis & Haglund (1999) for particulates into air. Emission factors for N _{tot} into water are derived from the mean value of three plants given in Davis & Haglund (1999), Appendix 8. The emission factor for NH ₃ into air was approximated from the average of values given in Patyk & Reinhardt (1997), p. 92. Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
3	urea, as N, at regional storehouse	RER	kgN	46% N	Davis & Haglund (1999), Appendix 9.	Emission factors given in Davis & Haglund (1999) for N _{tot} into water, NH ₃ and particulates into air. Emission factors of CH ₄ and CO into air were supplemented with data from Patyk & Reinhardt (1997), p. 92. CO ₂ used as input (733 kg CO ₂ /t urea) in urea synthesis arises as a by-product during the production of ammonia (Althaus <i>et al.</i> 2007) and is therefore omitted. Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
4	urea ammonium nitrate, as N, at regional storehouse	RER	kgN	32% N	Davis & Haglund (1999), Appendix 9 *).	Emission factors for the production of liquid urea given in Davis & Haglund (1999) for N _{tot} into water and particulate emissions into air were supplemented with specifications for CH ₄ and CO emissions into air from Patyk & Reinhardt (1997), p. 92. Emissions associated with the production of liquid ammonium nitrate, i.e. NH ₃ and particulate emissions into air and N _{tot} into water, were taken in full from EFMA (2000a, p. 23). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.

No.	Inventory name	Location	Unit	Nutrient content	Basic reference for the unit-process inventory	Supplemented or modified inputs and/or emission factors with respect to the basic reference.
5	calcium nitrate, as N, at regional storehouse	RER	kgN	16% N	Approximated inventory based on the process energy in Kongshaug (1998), Table 7.	Process emissions were taken from EFMA (1995). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
6	ammonium sulphate, as N, at regional storehouse	RER	kgN	21% N	Approximated inventory based on the process energy in Kongshaug (1998), Table 7.	Process emissions were taken from EPA (1995). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
7	single superphosphate, as P ₂ O ₅ , at regional storehouse	RER	kg	21% P ₂ O ₅	Davis & Haglund (1999), Appendix 10.	Emission factors given in Davis & Haglund (1999) for P _{tot} into water, fluorides and particulates into air. Process emissions of heavy metals into water during fertiliser production were supplemented with data from Audsley <i>et al.</i> (1997) (see also Tab. 8.7). Particle size distribution was assumed to be PM _{2.5} /TPM= 20%, PM ₁₀ /TPM= 60%.
8	triple superphosphate, as P ₂ O ₅ , at regional storehouse	RER	kg	48% P ₂ O ₅	Davis & Haglund (1999), Appendix 10.	Emission factors given in Davis & Haglund (1999) for P _{tot} into water, fluorides and particulates into air. Process emissions of heavy metals into water during fertiliser production were supplemented with data from Audsley <i>et al.</i> (1997) (see also Tab. 8.7). Particle size distribution was assumed to be PM _{2.5} /TPM= 20%, PM ₁₀ /TPM= 60%.
9	Thomas meal, as P ₂ O ₅ , at regional storehouse	RER	kg	17% P ₂ O ₅ , 32% Ca ^{**})	Audsley <i>et al.</i> (1997), p. 31.	No process emissions specified in Audsley <i>et al.</i> (1997), no process emissions supplemented. See comments in chapter 8.2.4.
10	potassium sulphate, as K ₂ O, at regional storehouse	RER	kg	50% K ₂ O	Davis & Haglund (1999), Appendix 6.	No process emissions specified in Davis & Haglund (1999). Particulate emissions and particle size distribution supplemented with data from EPA (1995).
11	potassium chloride, as K ₂ O, at regional storehouse	RER	kg	60% K ₂ O	K+S Aktiengesellschaft (1999 & 2001).	Process emissions for HCl and particulate matter in air and emissions to water were taken from K+S Aktiengesellschaft (2001). Particle size distribution was taken as PM _{2.5} /TPM = 5%, PM ₁₀ /TPM= 9% from EPA (1995). Demand for the resource 'sylvite, 25% in sylvinitite, in ground' was included in the inventory.

No.	Inventory name	Location	Unit	Nutrient content	Basic reference for the unit-process inventory	Supplemented or modified inputs and/or emission factors with respect to the basic reference.
12	ammonium nitrate phosphate, at regional storehouse	RER	kg	8% N, 52% P ₂ O ₅	Davis & Haglund (1999), Appendix 16.	Emission factors given in Davis & Haglund (1999) for P _{tot} into water, NH ₃ and particulates into air; process emissions of heavy metals into water during fertiliser production were supplemented with data from Audsley <i>et al.</i> (1997) (see also Tab. 8.7). Process emissions of NO ₂ into air were completed from Patyk & Reinhardt (1997, p.92). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
13	ammonium nitrate phosphate, as N, at regional storehouse	RER	kg	8% N, 52% P ₂ O ₅	Allocated inventory from inventory 'ammonium nitrate phosphate, at regional storehouse', referring to Davis & Haglund (1999), Appendix 16.	Allocation of emissions from inventory 'ammonium nitrate phosphate, at regional storehouse': NH ₃ and NO ₂ into air, 100%; PM _{2.5} , 100%, PM _{2.5-10} , 60% and TPM-PM ₁₀ , 40%. Emissions of toxic metals into water were fully allocated to the nutrient P ₂ O ₅ . See also comments in section allocations.
14	ammonium nitrate phosphate, as P ₂ O ₅ , at regional storehouse	RER	kg	8% N, 52% P ₂ O ₅	Allocated inventory from inventory 'ammonium nitrate phosphate, at regional storehouse', referring to Davis & Haglund (1999), Appendix 16.	Allocation of emissions from inventory 'ammonium nitrate phosphate, at regional storehouse': Emissions of toxic metals into water allocated 100%; P _{tot} into water, 100%; PM _{2.5-10} , 40% and TPM-PM ₁₀ , 60%. See also comments in section allocations.
15	potassium nitrate, at regional storehouse	RER	kg	14% N, 44% K ₂ O	Davis & Haglund (1999), Appendix 7.	No process emissions specified in Davis & Haglund (1999), no process emissions supplemented.
16	potassium nitrate, as N, at regional storehouse	RER	kg	14% N, 44% K ₂ O	Allocated inventory from inventory 'potassium nitrate, as N, at regional storehouse', referring to Davis & Haglund (1999), Appendix 7.	Nitric acid allocated 100% (see also comments in section allocations).
17	potassium nitrate, as K ₂ O, at regional storehouse	RER	kg	14% N, 44% K ₂ O	Allocated inventory from inventory 'potassium nitrate, as N, at regional storehouse', referring to Davis & Haglund (1999), Appendix 7.	Potassium chloride allocated 100% (see also comments in section allocations).
18	monoammonium phosphate, at regional storehouse	RER	kg	11% N, 52% P ₂ O ₅	Davis & Haglund (1999), Appendix 11.	Emission factors given in Davis & Haglund (1999) for P _{tot} into water, NH ₃ and particulates into air; process emissions of heavy metals into water during fertiliser production were supplemented with data from Audsley <i>et al.</i> (1997) (see also Tab. 8.7). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.

No.	Inventory name	Location	Unit	Nutrient content	Basic reference for the unit-process inventory	Supplemented or modified inputs and/or emission factors with respect to the basic reference.
19	monoammonium phosphate, as N, at regional storehouse	RER	kg	11% N, 52% P ₂ O ₅	Allocated inventory from inventory 'monoammonium phosphate, at regional storehouse', referring to Davis & Haglund (1999), Appendix 11.	Allocation of emissions from inventory 'monoammonium phosphate, at regional storehouse': NH ₃ into air, 100%; PM _{2.5} , 100%, PM _{2.5-10} , 60% and TPM-PM ₁₀ , 40%. Emissions of toxic metals into water were fully allocated to the nutrient P ₂ O ₅ . See also comments in section allocations.
20	monoammonium phosphate, as P ₂ O ₅ , at regional storehouse	RER	kg	11% N, 52% P ₂ O ₅	Allocated inventory from inventory 'monoammonium phosphate, at regional storehouse', referring to Davis & Haglund (1999), Appendix 11.	Allocation of emissions from inventory 'monoammonium phosphate, at regional storehouse': Emissions of toxic metals into water allocated 100%; P _{tot} into water, 100%; PM _{2.5-10} , 40% and TPM-PM ₁₀ , 60%. See also comments in section allocations.
21	diammonium phosphate, at regional storehouse	RER	kg	18% N, 46% P ₂ O ₅	Davis & Haglund (1999), Appendix 11.	Emission factors given in Davis & Haglund (1999) for P _{tot} into water, NH ₃ and particulates into air; process emissions of heavy metals into water during fertiliser production were supplemented with data from Audsley <i>et al.</i> (1997) (see also Tab. 8.7). Particle size distribution was assumed to be PM _{2.5} /TPM= 40%, PM ₁₀ /TPM= 60%.
22	diammonium phosphate, as N, at regional storehouse	RER	kg	18% N, 46% P ₂ O ₅	Allocated inventory from inventory 'diammonium phosphate, at regional storehouse', referring to Davis & Haglund (1999), Appendix 11.	Allocation of emissions from inventory 'diammonium phosphate, at regional storehouse': NH ₃ into air, 100%; PM _{2.5} , 100%, PM _{2.5-10} , 60% and TPM-PM ₁₀ , 40%. Emissions of toxic metals into water were fully allocated to the nutrient P ₂ O ₅ and not to N. See also comments in section allocations.
23	diammonium phosphate, as P ₂ O ₅ , at regional storehouse	RER	kg	18% N, 46% P ₂ O ₅	Allocated inventory from inventory 'diammonium phosphate, at regional storehouse', referring to Davis & Haglund (1999), Appendix 11.	Allocation of emissions from inventory 'diammonium phosphate, at regional storehouse': Emissions of toxic metals into water were allocated 100%; P _{tot} into water, 100%; PM _{2.5-10} , 40% and TPM-PM ₁₀ , 60%. See also comments in section allocations.
24	lime, algae, at regional storehouse	CH	kg	36% Ca ^{**})	Survey by Garcia & Nemecek (2000).	See comments in chapter 8.2.4.
25	lime, from carbonation, at regional storehouse	CH	kg	18% Ca ^{***})	Survey by Garcia & Nemecek (2000).	See comments in chapter 8.2.4.

No.	Inventory name	Location	Unit	Nutrient content	Basic reference for the unit-process inventory	Supplemented or modified inputs and/or emission factors with respect to the basic reference.
26	stone meal, at regional storehouse	CH	kg		Survey by Garcia & Nemecek (2000).	See comments in chapter 8.2.4.

*) Process unit inventory for urea ammonium nitrate in Davis & Haglund (1999) specifies the electricity demand, in addition to the amounts of liquid urea and liquid ammonium nitrate required for production. For calculation, the process unit inventories given in Davis & Haglund (1999) for liquid urea (Appendix 10) and liquid ammonium nitrate (Appendix 9) were applied.

**) According to LBL (2001), p. 493.

***) According to BGBI (1991), p. 1450.

8.3 Data Quality Considerations

Mineral Fertilisers

When using mineral-fertiliser inventories, the following limitations associated therewith should be borne in mind:

Pedigree Judgement

The uncertainty assessment has been carried out according to the methodology described in Frischknecht *et al.* (2007).

i) Data based on the unit-process inventories specified in Davis & Haglund (1999), with emission factors partially supplemented by other sources:

- Reliability and temporal correlation of the input data originating from the Davis & Haglund (1999) study, which was judged to be detailed and scientifically sound, were scored as 1. Completeness and further technological correlation were scored as 2, because the data used to compile the inventories in Davis & Haglund (1999) cover a large portion but not the entirety of the European market defined as a geographical system boundary for mineral fertilisers in ecoinvent data. Sample size was scored as 3, as the figures were based on aggregate measurements.
- Emission factors originating from the Davis & Haglund (1999) study were all judged to have a lower reliability than the input data of the same study, as noted in Davis & Haglund (1999). Completeness, geographical and further technological correlation was thus scored as 3, and temporal correlation as 2. The same scoring was also applied to emission factors derived from EFMA (2000a) and Patyk & Reinhardt (1997). Sample size was unknown (score 5).

ii) Data unit-process inventory for potassium chloride from K+S Aktiengesellschaft (2001):

- Reliability was scored as 2, since the figures come from a published environmental report. Completeness scored a 3, since Germany has 27% of the European market. Temporal correlation scored a 1 (data for the year 2000). Geographical correlation received a 3, since the data are from a smaller area than the one under study. Technological correlation was scored as 1 (the intended technology). Sample size scored a 3, since the figures come from an environmental report.

iii) Input data approximated solely from process-energy specifications in Kongshaug (1998):

- The energy inputs for ammonium sulphate and calcium nitrate, derived from the process energy and split between different energy carriers under the assumption of a mean N fertiliser, were considered to have a very low completeness (score of 5), and a low technological correlation (score of 4). Sample size was scored as 3, as the figures were based on aggregate measurements. The other uncertainty indicators were scored as 2.

iv) Input data obtained from individual manufacturers based on the survey done by Garcia & Nemecek (2000), and input data derived from Audsley *et al.* (1997):

- The input data for the inventories 'lime from algae', 'lime from carbonation' and 'stone meal' were provided in individual manufacturers' personal communications. The information was considered to represent non-verified data based on qualified estimates. The plants included in the survey were relevant for the market, although they did not cover it entirely, i.e. reliability, completeness and geographical correlation were thus scored as 3. Temporal and technological correlation were scored as 1, while sample size was scored as 5 (unknown).

- For Thomas meal, only a rough parameter for the process energy and an assumed energy source is given in Audsley *et al.* (1997). Except for geographical and further technological correlation, which were scored as 1, all other uncertainty indicators were scored as 2 (with sample size being scored as unknown). Transport was scored similarly to the inventories based on Davis & Haglund (1999).

v) Transports for all mineral fertilisers were scored with the same uncertainty:

- The distances were judged as non-verified data partly based on qualified estimates. Reliability was thus scored as 3. Due to the uncertainty in defining the transport carrier, further technological correlation was scored as 3. All other uncertainty indicators were not scored.

9 Organic Fertilisers

9.1 Characteristics

Organic fertilisers have two basic functions in agriculture:

- To provide nutrients. This function is especially important in organic agriculture, where most mineral fertilisers cannot be used. Only products listed in FIBL (2002) are permitted in organic agriculture.
- To improve soil properties. Organic fertilisers usually have a high organic-matter content, which is beneficial for soil biology and structure.

The most important type of organic fertiliser is farmyard manure. However, the on-farm usage of farmyard manure is not the subject of this chapter.

Tab. 9.1 Organic-matter and N-, P- and K-nutrient content of the organic fertilisers dealt with in this chapter. Percentages refer to the fresh weight of the product.

Fertiliser	% organic matter	% N	% P ₂ O ₅	% K ₂ O	Source
poultry manure, dried	85	4.6	3.3	2.5	LANDI (2003)
horn meal	80	12	0	0	FIBL (2002)
compost	78.4	0.7	0.4	0.6	Candinas <i>et al.</i> (1999), data for 1998

Switzerland produced 263,000 tonnes of compost in 1997 (Candinas *et al.* 1999), and used 2,300 tonnes of horn meal as organic fertiliser in 1998 (Herter & Külling 2001). No data on use were found for the other two fertilisers.

The dataset “vinasse, at regional storehouse, CH”, described in this report for ecoinvent data V1.x is now described in Jungbluth *et al.* (2007).

9.2 Life Cycle Inventories of Organic Fertilisers from Biogenic Wastes

Tab. 9.2 Selected organic fertilisers of agricultural importance in Switzerland.

Fertiliser	Unit	Location	Basic reference for the unit-process inventory
poultry manure, dried, at regional storehouse	kg	CH	Garcia & Nemecek (2000) ^{*)}
horn meal, at regional storehouse	kg	CH	Survey by Garcia & Nemecek (2000), supplemented by energy specifications from Commission of the European Communities (1993)
compost, at plant	kg	CH	Edelmann & Schleiss (1999, p. 51)

^{*)} Personal communication from A. Grub, Optigal SA, Courtepin, Switzerland, 26 September, 2000, with updates 26 April, 2002.

Organic fertilisers included in ecoinvent data are listed in Tab. 9.2. These fertilisers represent by-products of manufacturing or waste-recycling processes, and as such their preliminary production process was not taken into account. Extraction and transport of raw materials for the primary production processes and the processes themselves were fully allocated to the primary products. Only

further treatment of the by-products, waste-recycling processes and transport to the regional storehouse were allocated to the organic fertilisers.

The inventories 'horn meal' and 'poultry manure' are based on a survey done by Garcia & Nemecek (2000) at several plants in Switzerland. Garcia & Nemecek (2000) considered the energy requirements connected with the by-product processing steps required to produce a finished fertiliser. Process emissions were also considered, to the extent that data were available, as well as transport from the processing site to the regional storehouse.

To compile the production inventories, Garcia & Nemecek (2000) conducted a survey of several organic-fertiliser manufacturers and distributors in Switzerland. Because the data contain confidential information, they cannot be described in detail in this report. If noted in Tab. 9.2, the data obtained from the survey were also supplemented by specifications from literature. Since it was assumed that the fertiliser production sites were located in an urban/industrial area, all process emissions were deemed to emanate from an area of high population density.

9.2.1 Dried Poultry Manure

The unit-process raw data were obtained from a commercial poultry-manure producer in Switzerland³¹. The production process was described as follows:

The fresh poultry manure is transported 15 km by lorry from the poultry farms to the fertiliser plant, where it is dried to a water content of 12 to 15% by weight, using fuel oil as an energy source (13 g fuel oil per kg dried poultry manure). The drying procedure is followed by a granulation step effected by an electrically driven press (0.11 kWh per kg final product).

During production, ammonia emerges into the air as a process emission. The emission factor comes to 106 mg NH₃ per kg dried poultry manure. Heat production from electricity use was quantified as an emission into the air. Significant progress in emission control has been achieved over the last few years.

2 g waste is produced per kg dried poultry manure, mainly from packaging. The commercial dried poultry manure is then transported by lorry to a regional storehouse in Switzerland. The transport distance was assumed to be 150 km.

No data were found on the use of infrastructure for dried poultry manure and horn meal. Since the most important processes for these inventories are drying processes, the building infrastructure was approximated by a grass- and maize-drying plant (see chapter 13.2.3). The following values were used: area of building, 800 m²; total built-up area (including building), 3000 m²; lifetime of building, 50 years; duration of construction phase, 2 years; annual production, 2170 tonnes. The infrastructure of the facilities is already included in the modules for thermal energy delivery.

9.2.2 Horn Meal

A survey of several horn-meal distributors in Switzerland revealed that this product is imported from overseas, mainly from India. To produce horn meal, the horns must be sterilised in an autoclave and milled.

No information was available on the energy required for processing horns into horn meal. Consequently, the energy requirement for animal-meal production specified in the Commission of the European Communities (1993), i.e. an electricity requirement of 0.136 MJ and a natural-gas consumption of 1.175 MJ for processing 1 kg horn meal, was used as an approximation. To compile the inventory, these energy inputs were related to Europe.

³¹ Personal communication from A. Grub, Optigal SA, Courtepin, Switzerland, 26 September 2000, with updates on 26 April 2002.

Heat production due to electricity use was quantified as an emission into the air (high population density area).

Mean transport distances from India to a regional storehouse in Switzerland were determined as follows: 11,800 km by transoceanic freighter, 830 km by barge, 400 km by train and 20 km by lorry. These transport carriers were related to RER.

9.2.3 Compost

Compost is produced from biogenic waste. Due to its high content of heavy metals and other toxic compounds, household waste is now incinerated rather than composted. Compost usually comes from a separate waste collection from gardens, kitchens, etc., and is widely used in agriculture as it helps improve both soil structure and nutrient content.

There are different composting technologies, such as open windrow and enclosed tunnel composting. In ecoinvent data, only open windrow composting is included. This inventory is based on a study by Edelmann & Schleiss (1999) comprising an analysis of different biogenic-waste treatment technologies in full-size Swiss waste-treatment plants.

Edelmann & Schleiss (1999) inventoried plant infrastructure, considering the construction, operation and dismantling of the plant. Lifetimes were assumed as follows: mobile machines, 5 years; stationary machines, 10 years; structural parts of the building, 25 years.

The treatment plant was standardised in Edelmann & Schleiss (1999) to a treatment capacity of 10,000 tonnes of biogenic waste per year, which approximates the typical size of a professional biogenic waste treatment plant in Switzerland.

Edelmann & Schleiss (1999) also inventoried energy requirements for treatment of the waste, as well as the emissions produced during operation of the plant. All inputs and outputs were related to the functional unit 'treatment capacity of 10,000 tonnes of biogenic waste per year' in the original reference, in which an average water content for the biogenic waste of 50% by weight was used for calculation.

The two inventories listed in Tab. 9.3 were created in ecoinvent data on the basis of the data given in Edelmann & Schleiss (1999).

Tab. 9.3 Compost-related inventories included in ecoinvent data.

Name	Unit	Location	Basic reference for the unit-process inventory
compost plant, open	unit	CH	Edelmann & Schleiss (1999)
compost, at plant	kg	CH	Edelmann & Schleiss (1999)

The following approach was used to convert the data given in Edelmann & Schleiss (1999) into inventories in ecoinvent data:

Compilation of the Inventory 'compost plant, open' in ecoinvent data

- Infrastructure requirements of a compost plant with a treatment capacity of 10,000 tonnes of biogenic waste per year were related to one year in Edelmann & Schleiss (1999), p.50. These values were multiplied by an assumed lifetime of 25 years to obtain the total per compost plant unit over the entire lifetime of the compost plant. This means that all plant components (mobile and stationary machines, structural parts) were calculated for a period of 25 years. The construction phase was assumed to be one year.

- The items specified in Edelmann & Schleiss (1999) were correlated with ecoinvent inventories as presented in Tab. 9.4.
- Transport of the building materials to the construction site was supplemented according to the transport specifications defined in the of ecoinvent 2000 quality guidelines for consumption in Switzerland (Frischknecht *et al.* 2007).
- The disposal inventories corresponding to the infrastructure components of the compost plant are also listed in Tab. 9.4. Steel and iron are described in Edelmann & Schleiss (1999) as machine components. Disposal of these materials was not inventoried, because they are recycled. Disposal of copper, sand and bitumen was not considered either, since these construction materials are recycled at the end of the compost plant's lifetime.

Tab. 9.4 Items required for a compost plant as specified in Edelmann & Schleiss (1999), assignment to inventories in ecoinvent data and disposal modules used.

Position defined in Edelmann & Schleiss (1999)	Inventories used for calculation in ecoinvent data	Location/ Category	Inventories used for calculation of the disposal in ecoinvent	Location/ Category of disposal inventories	Unit
Beton (ohne Armierungseisen)	concrete, B 35/25 with CEM I 42.5 u. BZS, at plant	CH	Disposal, building, concrete, not reinforced, to sorting plant	CH	m ³
Schotter	gravel, crushed, at mine	CH	Disposal, inert waste, to inert material landfill	CH	kg
Gusseisen	cast iron, at plant	RER	To recycling		kg
Holzbaustoff Brettschichtholz	glued laminated timber, outdoor use, at plant	RER	Disposal, building, waste wood, treated, to final disposal	CH	m ³
Kupfer	copper, at regional storage	RER	To recycling		kg
Sand für Bau	sand, at plant	CH	To recycling		kg
Stahl niedriglegiert	steel, low-alloyed, at plant	RER	To recycling		kg
Armierungsstahl	reinforcing steel, at plant	CH	Disposal, building, reinforcement steel, to recycling	CH	kg
Bitumen ab Raffinerie CH	bitumen, at refinery	CH	To recycling		kg
Fläche III-IV	Occupation, industrial area, built up	resource			m ² a
	Occupation, construction site	resource			m ² a
	Transformation, from unknown	resource			m ²
	Transformation, to industrial area, built up	resource			m ²

Compilation of the Inventory 'compost, at plant' in ecoinvent data

- Energy requirements, emissions and waste production relating to 10,000 tonnes of biogenic waste treated per year in Edelmann & Schleiss (1999) were related to 1 kg compost produced, with a final-compost-product water content of 50% by weight being used for calculation.
- Infrastructure of the 'compost plant, open' was calculated for 1 kg final compost.
- The inputs and outputs associated with compost production specified in Edelmann & Schleiss (1999) were linked to ecoinvent inventories as presented in Tab. 9.5.
- Compost production was assumed to take place in an urban area; consequently, all emissions were considered to emanate from an area of high population density.

- Heat production due to electricity use was added to heat emanating from biological activity during compost production. It was assumed that heat was discharged into the air.
- The input “biogenic waste” was not included in the inventory, since it was not considered to be a waste.

Tab. 9.5 Inputs and outputs related to compost production as specified in Edelmann & Schleiss (1999), and assignment to inventories in ecoinvent data.

Position defined in Edelmann & Schleiss (1999)	Inventories used for calculation in ecoinvent data	Location/Category	Unit
Diesel ab Regionallager CH	diesel, at regional storage	CH	kg
Strom-Mix UCPTE	electricity, low voltage, at grid	CH	kWh
Infrastructure	compost plant, open	CH	unit
Fremdstoffe, Abfall CH95: in KVA	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg
Abwasser: m ³ in CH-ARA Grössenklasse 2	treatment, sewage, to wastewater treatment, Class 2	CH	m ³
Kommunale Abfallsammlung	transport, municipal waste collection, lorry 21t	CH	tkm
Abwärme in Luft p (Kompost)	Heat, waste	air\high population density	MJ
CH ₄ Methan p	Methane, biogenic	air\high population density	kg
CO Kohlenmonoxid s	Carbon monoxide, fossil	air\high population density	kg
CO ₂ Kohlendioxid p	Carbon dioxide, biogenic	air\high population density	kg
CO ₂ Kohlendioxid s	Carbon dioxide, fossil	air\high population density	kg
NO _x Stickoxide als NO ₂ s	Nitrogen oxides	air\high population density	kg
NH ₃ Ammoniak p	Ammonia	air\high population density	kg
N ₂ O Lachgas p	Dinitrogen monoxide	air\high population density	kg
H ₂ S Schwefelwasserstoff	Hydrogen sulfide	air\high population density	kg

The inventory does not include transport from compost plant to user. For calculation, transport of 5 – 20 km in Switzerland with a lorry may be used³².

9.3 Data Quality Considerations

Poultry Manure

Uncertainties associated with the poultry manure unit-process inventory data were judged as follows:

- The data provided by the manufacturer represent both verified data based in part on assumptions, and non-verified data based on measurements. Reliability was thus scored as 2.
- Although no data were available on the total volume of poultry manure produced in Switzerland, there were numerous indications that the plant providing the data (production volume: 3,000 tonnes per year) is a major producer in Switzerland. Completeness was therefore scored as 3.
- Temporal, geographical and further technological correlation were deemed to comply fully with the inventory standards, and were scored as 1.
- Sample size was scored as 5 (unknown).

³² Personal communication from W. Edelmann und K. Schleiss (Arbeitsgemeinschaft Bioenergie), August 1999.

Horn Meal

Uncertainties associated with these unit-process inventory data were judged as follows:

- The energy data used were approximated from energy parameters of related processes. The data sources for these energy parameters were based on verified data resulting from assumptions as well as from measurements.
- Based on a survey, the production site considered in order to compile the inventory reflects the market situation of this product in Switzerland. However, quantitative data on the share of the selected process on the total consumption in Switzerland are missing. Completeness was thus scored as 3.
- Since the energy requirements were approximated from related processes and geographically extrapolated in part, these uncertainty indicators were scored as 3.
- Temporal correlation of the data from 1993 used for horn meal was scored as 3.
- Sample size was scored as 5 (unknown).

Compost, at Plant and Compost Plant

Uncertainties associated with the unit-process inventory data in Edelmann & Schleiss (1999) were judged as follows:

- The data provided by Edelmann & Schleiss (1999) represent both verified data based in part on assumptions, and non-verified data based on measurements. Reliability was scored as 2.
- Data were gathered from only one compost plant, which nonetheless may be considered to be typical of conditions in Switzerland. Completeness was thus scored as 3.
- Temporal, geographical and further technological correlation was scored as 1.
- Sample size was scored as 5 (unknown).

10 Pesticides

10.1 Characteristics

Categories of Pesticides

A pesticide is "any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest" (FIFRA 1947). Pesticides are used in various economic sectors, such as agriculture, forestry, transport (railroads), industrial zones, parks and households (gardens). Among these sectors, however, agriculture is by far the main user (approx. 80-90% of all pesticides sold) (Brouwer *et al.* 1994).

Based on the target-organism group, pesticides of agricultural importance can be broadly categorised as

- Herbicides (for weed control),
- Insecticides (for insect control),
- Fungicides (for fungal pathogen control),
- Others (such as nematicides, bactericides, rodenticides).

Other compounds such as plant growth regulators do not kill any organisms, but are used in agriculture in the same way as pesticides. These compounds are also dealt with in this chapter.

Pesticides may be organic or inorganic compounds. Organic pesticides belong to a variety of substance classes, including N- and P-containing substances. Based on their chemical family, the different molecules of organic pesticides may be grouped into classes, such as phenoxy- or thiocarbamate compounds. By contrast, inorganic pesticides are copper-containing products, or, to a lesser extent, zinc- and mercury-based compounds.

Importance of Pesticide Use and Production

Roughly 2.6 million tonnes of active pesticide ingredients with a market value of \$US 38 billion were used worldwide in 1995. Europe, North America and Japan together account for three quarters of the world's pesticide consumption, with herbicides predominating (WRI 1998). The remaining one quarter is consumed in developing countries, where a large variety of pesticides no longer used in the developed nations is still being applied.

Between 1950 and the end of the 1980s, pesticide use increased more than thirtyfold. In the 1990s, pesticide use showed a decreasing trend in developed countries (on the basis of the weight), partly as a result of the introduction of new, more-powerful chemicals that are used in much smaller quantities (Harrison & Pearce 2001).

By way of comparison, annual sales of agricultural pesticides in the EU came to about 300,000 tonnes of active ingredients in 1996. Fungicides accounted for 41% of the total weight of active ingredients, followed by herbicides (39%) and insecticides (12%) (Lucas & Vall 1999). In the European Union alone, more than 700 different active ingredients are licensed (Stenzel 2001). Around 350 pesticides (active ingredients) are permitted in Switzerland, and about 6,000 different pesticides are used worldwide³³.

In Switzerland, total sales of pesticides approximately corresponding to domestic use amounted to 1,527 tonnes of active ingredients in 1999, equivalent to a turnover of around \$US 72 million. Fungicides contributed roughly 46% of this total, followed by herbicides with 40% (SBV 2000b)

³³ Personal communication from R. Büchi, FAL, 8 January 2003.

(Tab. 10.1). Switzerland produces approx. one third of the pesticides it consumes; the remainder is imported³⁴.

Tab. 10.1 Turnover of pesticides in Switzerland in 2005 (SBV 2006).

Switzerland 2005	Turnover (active ingredients in t)	Turnover in %
Fungicides, bactericides	621.6	44.7
Herbicides	627.8	45.1
Insecticides	122.4	8.8
Rodenticides, growth regulators and others	19.9	1.4
Pesticides (sum)	1391.7	100

The Swiss pesticide inventory (*Pflanzenschutzmittelverzeichnis*), published annually (e.g. FAW & BLW 2006), lists the active-ingredient content of different pesticide products.

Characteristics of the Manufacturing Process

Most modern synthetic organic pesticides are manufactured entirely from intermediates derived from fossil fuels. Primary pesticide manufacturing conventionally entails several process steps involving a variety of unit operations such as heating, stirring, distilling, filtering, drying and similar processes to build up a biologically active chemical entity from raw materials and/or specific chemical intermediates (Bhat *et al.* 1994).

Secondary processing involves the formulation of the pesticide in a marketable form, such as wettable powders, dusts, emulsifiable concentrates, granules, etc. This normally involves purely physical operations such as vessel charging, mixing, milling, warming, cooling, product transfer, granulation, drying, sieving and packaging. No chemical reactions take place during secondary processing.

Both the production and formulation processes require direct energy inputs for processing and transport, in addition to the intrinsic energy inputs needed.

10.2 Life Cycle Inventories of Organic Pesticides

Data Sources

It is very difficult to obtain current, accurate and specific data on pesticide production. The reasons for this are various:

- Detailed information on the production processes is available to the pesticide industry, but not to the public. A Company sharing information on its patent-protected pesticides would expose itself to the risk of competitors using a similar pesticide-manufacturing method.
- The number of chemical compounds used as pesticides is very large – over 6,000 worldwide. The active substances belong to very different chemical categories and are synthesised by various, sometimes highly complex, chemical pathways.

A detailed description of how all the active substances are produced would therefore greatly exceed the scope of the project and the database. For the use of the inventories in LCAs of agricultural production, a simplified approach was chosen. LCA studies of agricultural production have shown the impact of pesticide production to be fairly small, usually below 5% (e.g. Frick *et al.* 2001, Gaillard &

³⁴ Personal communication from C. Müller, Syngenta Basel, 1 March 2002.

Nemecek 2002, Nemecek *et al.* 2002). The toxic impact of the substance applied in the field, however, can be very significant at times. This impact must be estimated by the impact assessment method.

The inventories for pesticide production in ecoinvent data are intended for the use of these modules in LCAs of agricultural production only. They are not suited to other purposes, such as comparisons with other inventories of chemicals.

A review of the literature has shown that even recent studies (e.g. Bhat *et al.* 1994, Schorb 2000, Wang 1999, Heuer & Flake 2000, Reganold *et al.* 2001, Risoud 2002) are based on Green (1987) and even older data. Kaltschmitt & Reinhardt (1997) compared different data sources on the use of energy in pesticide production and concluded that Green (1987) provided the best basis for calculating LCAs of agricultural production. Recent attempts to collect newer data from the pesticide industry were only partly successful (Schorb 2000, Geisler *et al.* 2001). Because of the great effort required to obtain more precise data on one or two active substances, it was decided to use the data from Green (1987) to compile the ecoinvent inventories.

Data on Energy Usage

The inventories of 34 different substances and classes of organic pesticides included in ecoinvent data are based on the production energy figures given in Green (1987) and Bhat *et al.* (1994), the latter of which is also based on Green (1987). Since manufacturer's data – generally for internal use only – were not available, Green (1987) approximated the energy inputs required for the manufacture of selected pesticides. To do this, he modelled material flow sheets and line diagrams of the manufacturing process based on information given in patents on the manufacturing methods, or in the case of pesticides, on detailed literature on production processes no longer protected by patent. The approximation relates primarily to conditions in the US. However, since the data are the results of simulations and theoretical models, and are not measured data, they are not particularly specific to US conditions. Because the technologies used for the production processes in the US and Europe are likely to be similar, the data may also be used for the life cycle inventories of pesticides used in Switzerland and Europe.

To compose the unit-process inventory, the energy inputs specified in Green (1987) were interpreted as follows:

- Naphtha, natural gas and coke, which the author refers to as 'indirect energies', were considered to be feedstock energies (Tab. 10.2), i.e. energy carriers which are not combusted, but used as materials.
- Fuel oil, electricity and steam, noted as 'direct energies', were considered to be process energies, i.e. energies used to produce heat, mechanical power, etc. It was assumed that steam was produced by the combustion of heavy fuel oil (Tab. 10.2).
- From the descriptions in the article, it was concluded that the values given in Green (1987) corresponded to the cumulative energy demand (CED).

To convert these values into the useful energy (in the case of fuel oil, electricity and steam), the ratio of useful energy to the CED, derived from Frischknecht *et al.* (1996), was used to estimate the quantity of useful energy for each energy carrier. The methodology is described in detail by Gaillard *et al.* (1997).

$$UE = CED_{Gr}/CED_{ESU96}$$

UE = useful energy from energy carriers that are used as process energies, used in ecoinvent data (MJ)

CED_{Gr} = energy values given by Green (1987) (MJ)

CED_{ESU96} = sum of the products of all raw energy resources multiplied by their respective net calorific values, required to produce 1 MJ of useful energy (MJ/MJ). Values were taken from Frischknecht *et al.* (1996).

- For energy carriers used as materials or feedstock energies (naphtha, natural gas and coke), the quantity required was calculated as follows:

$$EC = CED_{Gr}/(CED'_{ESU96} * NCV_{EI00})$$

EC = quantity of energy carrier required (kg or m³)

CED'_{ESU96} = sum of the products of all raw energy resources multiplied by their respective net calorific values per kg or m³, taken from Frischknecht *et al.* (1996) (MJ/MJ)

NCV_{EI00} = net calorific value used in ecoinvent data (MJ/kg resp. MJ/m³)

Tab. 10.2 Inventories in ecoinvent data assigned to the energy-carrier type required for pesticide production specified in Green (1987). See text for the location to which the ecoinvent inventories refer.

Energy carrier specified in Green (1987)	Energy category specified in Green (1987)	Inventory in ecoinvent data
Naphtha	Indirect	naphtha, at refinery
Natural gas	Indirect	natural gas, at long-distance transport
Coke	Indirect	hard coal, imported, at regional storage
Fuel oil	Direct	heat, heavy fuel oil, at industrial furnace, 1MW
Electricity	Direct	electricity, low voltage, production UCTE, at grid
Steam	Direct	heat, heavy fuel oil, at industrial furnace, 1MW (assumption: steam is produced by the combustion of fuel oil)

In ecoinvent data, both individual pesticides and classes of pesticides were inventoried.

- The unit-process inventory of an individual pesticide was obtained from the energy-input specifications for this selected pesticide listed in Green (1987).
- The energy input for 'pesticide unspecified' represents the arithmetic mean of the energy inputs of all substances specified in Green (1987).
- The unit-process inventories of classes of pesticides were derived by grouping the substances specified in Green (1987) into classes of pesticides according to their chemical family, as per the specifications in Hartley & Kidd (1987) and Tomlin (1997) (see 'classes of pesticides' column in Tab. 10.3)³⁵. The arithmetic mean of the input energies of all pesticides belonging to a class was

³⁵ This approach was based on the fact that the variability of energy inputs specified in Green (1987) was lower when substances were classified according their chemical family, as opposed to their use (i.e. classification into insecticides, fungicides, herbicides).

then calculated. Where only one pesticide was included in Green (1987) for a class, the energy input required to produce this single pesticide was considered to be representative for the class.

- For the classes ‘nitrile compounds’, ‘pyridazine compounds’ and ‘cyclic N compounds’, no pesticide(s) of the same class were specified in Green (1987). These were derived from specifications in Bhat *et al.* (1994)³⁶ for the CED requirements for a pesticide belonging to these classes, i.e. Bromoxynil, Norflurazon and Methazol. The CED specifications in Bhat *et al.* (1994) were divided among the energy carriers by applying the averaged ratio of energy carriers for ‘pesticide unspecified’ for the cyclic N compounds, and the ratio for all pesticides designated as herbicides for the other two classes.

To take into account pesticides produced in Europe and Switzerland, which are both consumed in Switzerland, the pesticide inventories were created for two locations: a) regional storehouse RER and b) regional storehouse CH. For the pesticide inventories relating to the situation in Europe (RER), only RER and UCTE energy inputs were included in the production inventory. The inventories relating to regional storehouse CH take into account that roughly one third of the pesticides sold in Switzerland are of domestic production³⁷. Here, energy inputs for electricity, natural gas and heat from heavy fuel oil were split such that one third refers to CH inventories and the remainder to RER and UCTE inventories.

Emissions, Waste and Use of Infrastructure

Green (1987) considered energy inputs only. Other inputs are not included. The World Bank (1998) gives the quantity of solid waste produced as 200 kg per tonne of active ingredient. This output was inventoried as “disposal, hazardous waste, 25% water, to hazardous waste incineration, CH”. The same data source gives maximum air- and water emission levels. Unfortunately, these emissions could not be related to the product, and hence could not be included.

The emissions of pesticide active ingredients to the environment during manufacture were not included in the ecoinvent inventories. According to the World Bank (1998), these amounted to only 0.03-14 mg/kg active matter maximum. Compared with the quantity released on the field, this is a negligible amount.

Waste heat production stemming from the use of electricity was quantified as emission into the air (high population density). As the inventories only contain energy carriers as inputs, the mass balances are not always balanced.

³⁶ Bhat *et al.* (1994) derived the CED of these pesticides by assigning the energy inputs of the substances or classes of pesticide specified in Green (1987) which most closely match that of the searched pesticide.

³⁷ Personal communication from C. Müller, Syngenta Basel, 1 March 2001.

Tab. 10.3 Classification of substances specified in Green (1987) and Bhat *et al.* (1994) according to Hartley & Kidd (1987) and Tomlin (1997). *Italics: Substances or classes of substances included in ecoinvent data.* (h= herbicide, f= fungicide and i= insecticide). Substances marked with * were not specified in Green (1987); specifications refer to Bhat *et al.* (1994).

No	Pesticide Class	Pesticide specified in Green (1987)	Pesticides of agricultural relevance of the same class, categorised according to Margni <i>et al.</i> (2002)
1	[sulfonyl]urea-compounds	Chlorsulfuron(h), <i>Diuron(h)</i> , Fluometuron(h), <i>Linuron(h)</i>	Amidosulfuron, Chlortoluron, Dimefuron, Halosulfuron, Isoproturon, Mesosulfuron, Metha-benzthiazuron, Metsulfuron-Methyl, Monolinuron, Nicosulfuron, Prosulfuron, Rimsulfuron, SulfosulfuronTeflubenzu-ron, Thifensulfuron-Methyl, Triasulfuron, Triflusulfuron
2	phenoxy-compounds	<i>MCPA(h)</i> , 2,4-D(h), 2,4,5-T(h)	Acifluorphen, Esfenvalerate, Fenoxaprop, Fenpropathrin, Fluoroxypyr, Fluoroxypyr-als Ester, MCPB, Mecoprop, Mecoprop-P, Propargite, Pyriproxyfen, Triclopyr,
3	[thio]carbamate-compounds	<i>Carbofuran(i)</i> , Carbaryl(i), EPTC(h), Butylate(h)	Asulam, Carbendazim, Carbetamid, Cymoxanil, Desmedipham, Indoxacarb, Metham, Methiocarb, Methomyl, Molinate, Orbencarb, Pirimicarb, Propamocarb-Hydrochlorid, Phenmedipham, Prosulfocarb, Thiobencarb, Triallate
4	acetamide-anillide-compounds	Propanil(h), <i>Alachlor(h)</i> , <i>Propachlor(h)</i> , <i>Metolachlor(h)</i>	Acetamide, Clodinafop-propargyl, Diclofop-methyl, Dimethenamid, Metalaxyl, Napropamid, Tebutam
5	benzoic-compounds	<i>Dicamba(h)</i> , Chloramben(h)	Clopyralid
6	triazine-compounds	<i>Atrazine(h)</i> , <i>Cyanazine(h)</i> , Chlorsulfuron(h)	Metamitron, Metribuzin, Prometryn, Pymetrozine, Simazin, Terbutylazin, Tribenuron-Methyl
7	nitro-compounds	Dinoseb(h)	DNOC
8	dithiocarbamate-compounds	<i>Maneb(f)</i> , Ferbam(f)	Mancozeb, Metiram
9	dinitroaniline-compounds	Trifluralin(h)	Aclonifen, Ethalfluralin, Fluazinam, Fluroglycofen-Ethyl, Pendimethalin
10	pyretroid-compounds	Cypermethrin(i)	Bifentrin, Deltamethrin, Lambda-Cyhalothrin
11	benzimidazole-compounds	Benomyl(f)	Carbendazim, Chlorothalonil, Cloquintozet-Mexyl, Ethofumesat
12	organophosphorus-compounds	<i>Glyphosat(h)</i> , Phorat(i), Malathion(i), <i>Parathion(i)</i> , Methylparathion(i)	Azinphos, Chlorpyrifos, Diazinon, Dicrotophos, Disulfoton, EPTC, Ethephon, Ethoprop, Glufosinat, Methamidophos, Naled, Phosmet, Profenofos, Terbufos, Tribufos, Trichlorfon, Vamidothion
13	benzo[thia]diazole-compounds	Bentazon(h)	Benazolin, Thidiazuron
14	nitrile-compounds	Bromoxynil*(h)	Cyprodinil, Dichlobenil, Fenpiclonil, Ioxynil
15	diphenylether-compounds	Fluazifop-butyl(h)	Bifenox, Clodinafop-Propargyl, Diflufenican, Fluazifop-P-butyl, Propaquizafop
16	pyridazine-compounds	Norflurazon*(h)	Chloridazon, Maleic, hydrazide, Pyridate
17	cyclic N-compounds	Methazol*(h)	Clomazon, Cyproconazole, Cyprodinil, Difenconazol, Dimethomorph, Epoxiconazole, Fenpropidin, Fenpropimorph, Fluorochloridone, Flusilazole, Hexaconazole, Metconazol, Oxadixyl, Prochloraz, Propiconazol, Tebuconazole, Tridemorph
18	phthalamide-compounds	Captan(f)	Chlorothalonil

No	Pesticide Class	Pesticide specified in Green (1987)	Pesticides of agricultural relevance of the same class, categorised according to Margni <i>et al.</i> (2002)
19	bipyridylum-compounds	Diquat(h)	Paraquat
20	pesticide unspecified	Mean of all pesticides given in Green (1987)	

Infrastructure and Land Use

The infrastructure was inventoried as “chemical plant, organics, RER” as an approximation (assuming a production of 50,000 tonnes/year and a plant lifetime of 50 years (Althaus *et al.* 2007). This module also accounts for land use by the plant. The use of infrastructure was corrected for the average active-substance content of 50% in the final product (the average active-substance content of pesticides sold in 2000, FAW & BLW 2000), yielding a value of $8 \cdot 10^{-8}$ units/kg pesticide active substance.

Transports

Transport from the pesticide factory to the regional storehouse was inventoried by taking the standard transport distances for organic chemicals, namely 600 km by train and 100 km by lorry for Europe, or 600 km by train and 50 km by lorry for Switzerland. The transported mass was calculated by assuming an average active-matter content of 50% in the final product.

System Boundaries and Reference Product

The figures given in Green (1987) refer to pesticides with a 100% active-ingredient content, not packaged, at the regional storehouse. According to Green (1987), formulation and packaging contribute very little to the total energy consumption of pesticide manufacture. Furthermore, Green (1987) provides no detailed or specific data. Energy use during formulation and packaging was therefore not considered.

Recommended Application of the Pesticide Inventories

We recommend that organic pesticides, for which no specific inventories exist in ecoinvent, are treated as follows:

- If the active ingredient is listed in Tab. 10.3, use the inventory of the corresponding pesticide class.
- If the active ingredient is not listed in Tab. 10.3, use the inventory “pesticide unspecified”.

10.3 Data Quality Considerations

When using organic-pesticide inventories, the following limitations associated therewith should be borne in mind:

- The data given in Green (1987) refer to the situation in the USA, and their applicability to conditions in the European Union and Switzerland could not be confirmed. It is most likely, however, that the main manufacturing processes, which are patent-based, would not differ greatly in Europe or Switzerland.
- The energy inputs used for the inventories stem from data given in a reference from the year 1987. These still appear to be the most recent and reliable data, however, and are thus still commonly used for life cycle analysis (e.g. Bhat *et al.* 1994, Schorb 2000, Wang 1999). Wang’s comparison of different studies (1999) showed no significant deviation from the data of Green (1987). These data can therefore still be considered valid.

- The data are intended for the use in LCAs of agricultural production. They are not suited for any other use.

Since no quantitative information on the uncertainty concerning the energy inputs is included in Green (1987), pedigree judgement of the inputs used to compute the pesticide was made for a rough assessment.

Pedigree Judgement

- With reference to Green's peer-reviewed article (1987) used as a source, reliability of the data used to compile the pesticide inventories was consistently scored as 2.
- Completeness was scored as 2 for pesticides, based on an inventory for the same substance in Green (1987), as well as for the inventory 'pesticide unspecified', which was averaged from 41 inventories investigated by Green (1987). In all other cases, completeness of inventories of classes of pesticides derived from pesticides investigated by Green (1987) was awarded a 3.
- Temporal correlation was rated a 4, bearing in mind that Green's article was published in 1987.
- Quantification of energy requirements by Green (1987) was based largely on patents. Given the universality of patents for pesticide manufacture, geographical correlation was scored as 2.
- Pesticide classes derived from only one or two inventories of a pesticide belonging to the class were judged to have a lower 'further technological correlation' than the classes derived from three to five inventories of pesticides belonging to that class.
- All energy inputs of an individual inventory were scored identically, except for the energy input steam, which obtained a lower rating for 'further technological correlation' than all other energy inputs, as it was approximated by the combustion of fuel oil.
- Sample size was not scored, since the data were derived from theoretical models.

11 Seed

11.1 Characteristics

Plants for agricultural production are propagated either generatively, i.e. by botanic seed, or vegetatively, using tubers, cuttings, etc. The inventories in this chapter include both reproduction paths, namely propagation by seed (for most agricultural crops) and by tubers (vegetative propagation of potatoes).

The most important categories of seed in quantitative terms are cereal seed and potato seed (Tab. 11.1), both of which are produced for the most part in Switzerland. A sizeable quantity of rye seed is imported, but this is mainly used for fodder production and not for cereal grain crops³⁸. In addition, Switzerland produces some of its own maize seed. About 20% of the grain legume seed (peas and soy beans) used in Switzerland is produced domestically³⁹. Seed for other crops is wholly or mostly imported. The main provenance of seed imports to Switzerland is Western Europe (Tab. 11.1).

Tab. 11.1 Production, import and total consumption of seed in Switzerland. Source: SBV (2001) and Eidgenössische Oberzolldirektion (1999) for Swiss data, FAO (2002) for worldwide production. Figures are expressed in tonnes of seed per year, and refer to the average of the years 1999 and 2000 (except for sugar beets which is 1999 only). Exports of 1866 t/year seed potatoes and 100 t/year seed maize^{*)} were subtracted from the total consumption. WEU = Western Europe, DE = Germany, HR = Croatia, HU = Hungary, FR = France, NL = Netherlands, n.a. = not available.

Crop	Production CH t/year	Import CH t/year	Consumption CH t/year	% Import	Provenances	World production t/year
Wheat	17,890	65	17,955	0%	WEU	32,961,089
Barley	7,070	63	7,133	1%	WEU	9,090,282
Rye	442	708	1,149	62%	DE,HR,HU	2,064,672
Triticale	1,929	2	1,931	0%	WEU	427,795
Oats	1,313	22	1,335	2%	WEU	3,458,231
Maize	507	1,557	1,963	79%	FR,DE	5,713,719
Potato	58,514	4,802	61,450	8%	NL,FR	34,729,772
Grass/Clover	120	4,980	5,099	98%	WEU	n.a.
Faba beans	0	51	51	100%	WEU	n.a.
Rape seed	0	600	600	100%	WEU	474,559
Beets	0	136	136	100%	WEU	13,000
Other	0	387	387	100%	WEU	n.a.

^{*)} Personal communication from M. Keller, Saatzuchtgenossenschaft Düringen, 5 April 2002.

Characteristics of the Production Process

Seed production comprises different phases:

Agricultural *seed production* in the field usually requires additional processes over and above those of commercial crop production. For many crops, however, these processes mainly require only additional manpower (for inspection, weeding, removal of foreign plants, etc.) with little or no use of machines or other inputs that would be relevant in the life cycle inventories. For some crops, such as sugar beet, seed production may be quite different from normal crop production. For reasons of consistency, the process of drying is included in the inventories of agricultural seed production (“at farm”).

³⁸ Personal communication from M. Keller, Saatzuchtgenossenschaft Düringen, 5 April 2002.

³⁹ Estimated from the seed production for 1998-99, the average seeding rate, and crop surface areas for 1999-2000.

Transport to the seed-processing centre: This transport can be by tractor, lorry or rail.

1. *Processing the seed* comprises the steps of pre-cleaning, cleaning, chemical seed dressing (if necessary) and bag filling (Narain & Singh 1988). Drying is already included in step 1.
2. *Seed storage:* seed can be stored in storage bins, or bagged and stored in warehouses for several years (McDonald & Copeland 1997). Potato seed must be cooled during storage.
3. *Transport to regional storehouse* or final user. Imported seed is mainly transported by road⁴⁰.

11.2 Life Cycle Inventories of Seed

The system was modelled according to the five steps mentioned above. In the following descriptions, only the points relevant to the life cycle inventories are noted. For general information on seed production, the reader is referred to Desai *et al.* (1997), McDonald & Copeland (1997) and Kelly & George (1998).

11.2.1 Agricultural Seed Production

The agricultural seed production of most crops is similar to the commercial crop production, but has higher requirements in terms of selection of the region and field, optimal fertilisation, plant protection and weed control, control of pollination, etc. Seed-crop yields are often lower than normal crop yields (e.g. seed-maize yield is 3-4 times lower than that of a commercial grain-maize crop). This leads to higher seed-production costs. Seed crops place a higher demand on manpower (e.g. for manual removal of weeds and foreign plants), but manpower is not included in the life cycle inventories.

As no inventories were defined for agricultural production outside Switzerland, the modules with location “CH” are used for both Swiss and foreign production.

Cereal seed production requires no special measures to be included in the inventories, so the same inventories as for food and fodder production are used. According to LBL *et al.* (2000, p. 3), on average only 80% of seed-crop yield meets quality requirements and can be used as seed; the rest can be used as food for humans or animals. As the total yield is usually not lower than that of a normal cereal crop, the same inventory per kg of grains can be used, which corresponds to a mass allocation. For the agricultural seed production of cereals, the modules “wheat grains IP/organic, at farm, CH”, “rye grains IP/organic, at farm, CH”, or “barley grains IP/organic, at farm, CH” are used.

Potato seed production differs mainly in two aspects from commercial potato production: to prevent viral infection of the tubers, seed-potato haulms are usually destroyed at the beginning of July. This leads to a reduced yield (25,000 kg/ha for IP and 19,500 kg/ha for organic production; data from LBL *et al.* 2000, p. 29). For the compilation of seed-potato inventories (“potato seed IP/organic, at farm”), the same procedure as for arable crops was used (see description in chapter 14).

*Maize seed production*⁴¹ differs from commercial maize crop production. Modern maize varieties are hybrids, which means that lines of male and female parents must be combined in the same field. Due to the much lower yield of hybrids, maize seed crops yield about 3000 kg/ha, which is about three times lower than a commercial crop. Fertilisation is similar to with a commercial crop. Since the male and female parents must be sown separately and different sowing dates are required to ensure synchronisation of flowering, sowing is done in three to four passes. Optimal plant density is about 50,000 plants/ha (Kelly & George 1998), about half that of a commercial crop. A seed quantity of 15 kg/ha was assumed. An additional harrowing is carried out. Chemical and mechanical weed control is more intensive, since the plants are not as vigorous. The tassels of the female parents must be removed to avoid self-pollination. This is done mechanically in about six passes by cutting, which

⁴⁰ Personal communication from A. Gysin, fenaco Wintherthur, 20 March 2002.

⁴¹ Personal communication from P. Bassetti, Swissmaïs, 20 February 2002 and 18 April 2002.

requires a medium-sized tractor and about 1 hour per ha and per pass. To ensure complete detasseling, this procedure must be completed by manual detasseling, performed at the same time or afterwards. As no inventory has been defined for mechanical detasseling in ecoinvent data, “mowing by rotary mower, CH” was used as an approximation. Harvesting is carried out by a picker or a pick-sheller. In ecoinvent data, “harvesting by combine harvester, CH” was used as an approximation. For organic maize seed, a lower yield of 2600 kg/ha was assumed, since commercial organic crops have 87% the yield of integrated crops. Fertilisation was assumed to be the same as for a commercial crop (see chapter 14).

Since the seed production of *rape* does not differ fundamentally from the production of commercial crops, basically the same inventories can be used (as is the case with cereals). Yields are lower (1500-2100 kg/ha according to Desai *et al.* 1997), especially when hybrids are grown. No special inventory was defined for the seed production of rape. The module “rape seed IP, at farm, CH” was used instead. To account for the lower yield of the seed production (1800 kg/ha compared with 3150 kg/ha for a commercial crop), a greater quantity of rape seed was taken, namely 1.75 kg per kg seed produced.

For *grain legume* seed (pea in ecoinvent data), the same inventories are used as for commercial crop production.

Seed production of *forage grasses and legumes* follows a different scheme than for the production of forage crops, although the two are often partly combined. Italian ryegrass and red clover are used as representative species for grass- or clover seed production. Forage grasses are sown in August, while forage legumes are usually sown in spring under a cover crop (often cereals)⁴². The seed rate was taken as 16 kg/ha for Italian ryegrass and 7.5 kg/ha for red clover from Kelly & George (1998). The crops can usually be harvested twice for fodder production in autumn and in spring, and then once for seed in July (yield about 1000 kg/ha for grass and 400 kg/ha for legumes). The P- and K-fertilisation corresponds to a medium-intensity meadow (80 kg P₂O₅/ha and 230 kg K₂O/ha for forage legume seed and 85 kg P₂O₅/ha and 225 kg K₂O/ha for grass seed, according to Walther *et al.* 2001). Forage-grass seed crops are given 100 kg N/ha for seed production and an additional 100 kg N/ha for fodder production. Because of the biological nitrogen fixation, forage legumes do not need N fertilisers. Since the same field yields seed and fodder, an allocation must be made between the two crops. Based on potential yield of the medium-intensity meadow (4 harvests/year), 50% of the burdens are allocated to seed and 50% to fodder production (assuming that one harvest for seed corresponds roughly to two harvests for fodder).

Sugar-beet seed production is substantially different from commercial sugar-beet production. Sugar beets produce seed only after vernalisation, which means that the crop must be overwintered. The plants are normally overwintered *in situ*, but can also be removed from the soil, overwintered as stecklings and replanted in the second year. Fertilisation requirements are similar to those of commercial crops (Kelly & George 1998). The seed may be harvested by combine harvester. Seed yield was estimated as 2700 kg/ha (US data for 1984-89 from McDonald & Copeland 1997). No special inventory was defined for sugar-beet seed production, owing to a lack of detailed, specific data. The module “sugar beet IP, at farm, CH” was used instead as a rough approximation. To account for the much lower yield of the seed crop (2700 kg/ha, compared with 71,000 kg/ha for a sugar beet crop), a quantity of 26.3 kg sugar beet was taken per kg seed produced.

For the agricultural production of integrated grass-, clover- and maize seed, it was assumed that only mineral fertilisers were applied. Tab. 11.2 shows how the seed modules are linked to the agricultural seed production modules.

The transport of inputs from the regional storehouse to the farm was calculated according to Tab. 14.4.

⁴² Personal communication from B. Boller, ART, 30 August 1999.

Tab. 11.2 Modules of processed seed at the regional storehouse, and modules from agricultural production, which are used.

Module of processed seed	Quantity kg/kg seed	Module of unprocessed seed at farm
barley seed IP, at regional storehouse, CH	1	barley grains IP, at farm, CH
barley seed organic, at regional storehouse, CH	1	barley grains organic, at farm, CH
clover seed IP, at regional storehouse, CH	1	clover seed IP, at farm, CH
grass seed IP, at regional storehouse, CH	1	grass seed IP, at farm, CH
maize seed IP, at regional storehouse, CH	1	maize seed IP, at farm, CH
maize seed organic, at regional storehouse, CH	1	maize seed organic, at farm, CH
pea seed IP, at regional storehouse, CH	1	protein peas, IP, at farm, CH
pea seed organic, at regional storehouse, CH	1	protein peas, organic, at farm, CH
potato seed IP, at regional storehouse, CH	1	potato seed IP, at farm, CH
potato seed organic, at regional storehouse, CH	1	potato seed organic, at farm, CH
rape seed IP, at regional storehouse, CH	1.75	rape seed IP, at farm, CH
rye seed IP, at regional storehouse, CH	1	rye grains IP, at farm, CH
rye seed organic, at regional storehouse, CH	1	rye grains organic, at farm, CH
sugar beet seed IP, at regional storehouse, CH	26.3	sugar beets IP, at farm, CH
wheat seed IP, at processing centre, CH	1	wheat grains IP, at farm, CH
wheat seed organic, at processing centre, CH	1	wheat grains organic, at farm, CH

Since the inventories for rape seed and sugar beet use approximations (more than 1 kg input needed, Tab. 11.2), corrections are needed to ensure a correct CO₂- and energy balance. The values were therefore corrected by subtracting the difference

Drying

In order to ensure the safe storage of grains, humidity must be kept low. If humidity exceeds a certain level, drying is required. Drying must be done at low temperatures to avoid degradation of seed, which would decrease the germination. Low-temperature drying requires more energy than high-temperature drying (see chapter 13). Fuel was assumed to be the energy source for drying⁴³. The following humidity values are used in the calculations:

- From 16% to 15% for cereals,
- From 35% to 12% for maize⁴⁴,
- From 16% to 13% for peas,
- From 30% to 10% for grass-, forage-legume and sugar-beet seed.

To ensure consistency throughout the modules, the drying process was included in the inventories “at farm” (see also chapter 14), although it is normally performed in the seed processing centre.

⁴³ According to a personal communication from M. Keller, Saatzeitgenossenschaft Düringen, dated 28 March 2002, energy for drying seed in the Düringen seed processing centre of can be completely covered by burning the dust from cleaning. This is not usually the case, however.

⁴⁴ Personal communication from P. Bassetti, Swissmaïs, 18 April 2002.

11.2.2 Transport to the Seed Processing Centre

Transport to the seed processing centre is by tractor for short distances and by lorry or rail for longer distances. Seed of the quantitatively most important crops in Switzerland (cereals and potatoes) is to a large extent produced domestically, which limits the need for transport.

Little is known about the distances over which seed is transported. About 50% of Swiss maize seed is produced in the Tessin and transported by rail over about 300 km to the processing centre in Moudon. The remaining maize seed is transported over 100 km by lorry (estimate).

In the life cycle inventories, the following values were used:

- Cereals and potatoes (primarily domestic production): 30 km by lorry,
- Maize (ca. 80% imported, 20% domestic production): 30 km by rail ($=50\%*20\%*300$ km), 10 km by lorry ($=50\%*20\%*100$ km for Switzerland) and 80 km by lorry ($80\%*100$ km for Europe),
- Other seed (primarily foreign production): 100 km by lorry.

11.2.3 Seed Processing

Seed processing comprises the steps of pre-cleaning, cleaning, drying (if necessary), chemical seed dressing (for conventional and integrated production)⁴⁵ and bag filling (Narain & Singh 1988). The drying process is already included in the inventories “at farm” (see chapter 11.2.1).

Narain & Singh (1988) give data on energy use for the processing of wheat seed (electricity, 32.8 kWh/t) and maize seed (electricity, 78.3 kWh/t without drying) in India. Swiss figures from the seed processing centre in Düringen are slightly lower (27.7 kWh/t if only the 8500 t of seed produced, mainly cereals, is considered, and 24.2 kWh/t if the 2500 t cereals for human and animal consumption are also considered, assuming that it uses 50% of the energy of seed)⁴⁶.

The following values for electricity consumption are used for the calculations:

- Cereals and grain legumes: 24 kWh/t seed⁴⁷
- Maize seed, rape seed and grass- and forage-legume seed: 58 kWh/t seed⁴⁸
- Sugar-beet seed: 100 kWh/t seed⁴⁹

For potato seed, grading alone was included.

The following active ingredients used for the chemical seed dressing are listed in Tab. 11.3⁵⁰. Only conventional and integrated production use chemically synthesised pesticides; these compounds are not permitted in organic production. Pesticides are not permitted for the treatment of rape seed (FAW & BLW 2000).

For life cycle assessments using seed from integrated production (IP), the active substances listed in Tab. 11.3 should be included in the life cycle inventories as emissions into agricultural soil.

⁴⁵ Not permitted for organic production. However, if no organic seed is available, seed from integrated or conventional production (without chemical seed dressing) may be used in organic production.

⁴⁶ Personal communication from M. Keller, Saatzüchtgenossenschaft Düringen, 28 March 2002.

⁴⁷ Due to the similarity of treatments, the processing of grain legume seed is considered to be similar to that of cereal seed.

⁴⁸ The value is based on the value for cereals, using the ratio between maize and cereal seed given by Narain & Singh (1988). Rape seed, grass and forage legumes were assumed to be similar to maize.

⁴⁹ Estimated value; sugar-beet seed treatment is more complex (e.g. including pelleting) than that of other crops.

⁵⁰ Personal communication from Th. Hebeisen, ART, August, 2001.

No data were available for aerial and water emissions or waste generation.

Tab. 11.3 Pesticide active ingredients and ecoinvent modules used in the seed modules.

seed type	active ingredient per kg seed	ecoinvent-module used
wheat/rye seed IP	0.1 g Difenoconazole	diphenylether-compounds, at regional storehouse, CH
barley seed IP	0.07 g Cyprodynil 0.018 g Cyproconazole	cyclic N-compounds, at regional storehouse, CH
maize seed IP	2.4 g Captan	phtalamide-compounds, at regional storehouse, CH
potato seed IP	0.05 g Fenpiclonil	nitrile-compounds, at regional storehouse, CH
sugar and fodder beet seed IP	54 g Thiram (TMTD)	pesticide unspecified, at regional storehouse, CH
pea seed IP	1.6 g Thiram (TMTD)	pesticide unspecified, at regional storehouse, CH

11.2.4 Seed Storage

Seed is usually stored from one season to the next, i.e. for 6-12 months depending on the crop (Kelly & George 1998), but also for shorter periods (2-3 months for winter cereals and 4-6 months for spring cereals⁵¹). For botanic seed, humidity control is very important to prevent fungal diseases; cooling is not usually necessary. The contrary applies to potatoes, where relative humidity should be kept over 95% and the temperature between 2-4°C. Potato stores must be cooled.

For the use of the warehouse, a storage density of 500 kg/m³ was assumed. The bulk density of cereals is 600-800 kg/m³, of potatoes, 650-700 kg/m³ (Rastovski & van Es 1987). Seed requires more storage space than grains or tubers for food or forage use, since it is mostly stored in bags and smaller quantities. In order to calculate the amount of building infrastructure used, the storage space was assumed to be used twice a year (based on average lengths of storage), except for potatoes, where it is used once and remains empty during the rest of the year.

The following values were retained for the inventories:

- use of the storage space for 12 months in the case of potatoes and 6 months in the case of other seed (used once a year for potatoes and twice a year for other seed),
- storage space requirement 2 m³/t seed,
- building height assumed to be 10 m, which corresponds to a land use of 0.2 m²/t seed; the total space occupied (including the surrounding area) was estimated to be twice as large (0.4 m²/t),
- electricity for cooling potatoes: 73 kWh/t for one season (Devres & Bishop 1995).

The lifetime of the buildings was assumed to be 50 years, the length of the construction phase, 2 years. For seed-processing and storage calculations, it was assumed that these steps are performed in Switzerland. In the case of imported seed, this would mean that unprocessed seed is imported into Switzerland and processed and stored there.

11.2.5 Transport to the Regional Storehouse

Transport distances to the regional storehouse depend on the crop and the region in which the seed is produced. In the case of Swiss production, a distance of 100 km from the processing centre to the regional storehouse was assumed. For imports, the distance was assumed to be 800 km. Most seed

⁵¹ Personal communication from M. Keller, Saatzüchtgenossenschaft Düringen, 2 April 2002.

imported into Switzerland comes from neighbouring countries. All transport was assumed to be by lorry⁵².

Combining the percentages of imports with these distances, the distances in Tab. 11.4 were used in the calculations.

Tab. 11.4 Transport distances from the processing centre to the regional storehouse.

Crops	transport, lorry 20-28t, fleet average, CH	transport, lorry >16t, fleet average, RER
Cereals	100	0
Maize	40	600
Potatoes	90	60
Grain legumes	20	640
Other	0	800

11.3 Data Quality Considerations

The seed of most crops is used in relatively small quantities. Life cycle assessment studies (e.g. Frick *et al.* 2001, Gaillard & Nemecek 2002, Nemecek *et al.* 2002) have shown that the environmental burdens of agricultural crops due to seed lie below 5% for most crops (potatoes being an exception, contributing about 10% to the total environmental impact). It is therefore possible to model seed production in a simplified manner.

For the seed modules “at farm”, the same uncertainty judgement as for arable crop production was applied. The scoring for seed transport, processing and storage is shown in Tab. 11.5.

Data on energy use in cereal-seed processing are quite reliable; data on other seed were extrapolated and therefore receive a worse rating. Sample size for the use of electricity was scored as 5 and not scored for the other figures, which are not based on measurements.

Tab. 11.5 Uncertainty judgement for the data on agricultural seed production.

Category	Reliability	Completeness	Temporal correlation	Geographical correlation	Further technological correlation
Quantity of seed: cereals and potatoes	1	1	1	1	1
Quantity of seed: maize, grass, clover and grain legumes	1	1	1	3	3
Quantity of seed: rape seed and sugar beet	4	1	1	3	3
Processing cereal seed	2	3	1	1	1
Processing maize seed	3	3	4	5	1
Processing other seed	5	3	4	5	1
Seed storage	5	3	1	2	1
Transport processes	2	1	1	1	1

⁵² According to a personal communication from A. Gysin, fenaco Wintherthur, dated 20 March 2002, all transport of seed from Europe is by lorry. Rail transport is used for seed of overseas origin, which is transported to Rotterdam by transoceanic liner, then by barge to Basel and finally by rail to Wintherthur. However, since Switzerland imports little seed from overseas, this method of transport is not taken into account.

12 Feedstuffs

12.1 Characteristics

The products used as feedstuffs for animal husbandry are numerous and of various origin (Tab. 12.1). While some raw materials for feedstuff are produced only for this purpose (main products), many others are by-products of human food production. Cereal crops might be used directly as feedstuffs, while other crops, such as oilseeds, usually undergo many processing steps before being used as feedstuffs. Recommendations on the feeding of ruminants and data on numerous feedstuffs can be found in RAP (1999). RAP (1995) provides the information for the feeding of pigs.

The most important category of feedstuffs is cereals crops, followed by oilseed cakes and cereal products (Tab. 12.1). Around 40% of livestock feedstuffs are imported into Switzerland, mainly from Europe. Cereals are largely produced in Switzerland, while oilseed cakes and cereal products are mainly imported from Europe.

Tab. 12.1 Production, import and total consumption of feedstuff in Switzerland. Source: SBV (2000a) and Eidg. Oberzolldirektion (1999). The figures are expressed in t/year and refer to the year 1999.

Feed	Production CH, t/year	Import CH, t/year	Consumption CH, t/year	% Import	Main provenance
Wheat and rye	178,100	13,500	191,600	7	EU
Oats	26,900	24,100	29,637	47	Australia (65%), EU
Barley and triticale	286,400	25,500	311,900	8	EU
Maize	191,300	20,000	211,300	10	Eastern Europe
Rice	0	10,200	10,200	100	Western Europe, 44% USA + Africa
Sorghum	0	7,126	7,126	100	FR
<i>Total Cereals</i>	<i>682,700</i>	<i>100,426</i>	<i>783,126</i>	<i>13</i>	
Legumes (not including soy beans)	10,000	19,726	29,726	66	FR
Soy beans	0	3,566	3,566	100	Western Europe
<i>Total Legumes</i>	<i>10,000</i>	<i>23,292</i>	<i>33,292</i>	<i>70</i>	
Rape-seed cake	31,300	2,910	34,210	9	DE
Soy-bean cake	0	77,058	77,058	100	Western Europe; 10% Brazil
Other cakes and oilseeds	0	84,492	84,492	100	
<i>Total oilseed cakes</i>	<i>31,300</i>	<i>164,460</i>	<i>195,760</i>	<i>84</i>	
Soy beans (beans for oil)	0	18,636	18,636	100	Brazil, USA
Milling products	50,900	87,000	137,900	63	
Maize gluten	0	33,250	33,250	100	USA (80%), EU
By-products of beer production and legume processing		16,800	16,800	100	
Sugar-beet molasses	31,200	3,297	34,497	10	DE
Glucose and fructose		15,803	15,803	100	Western Europe
Dried potatoes	5,900	0	5,900	0	
Potato proteins	0	20,616	20,616	100	DE, EU
Dextrin and other starch		30,344	30,344	100	DE, EU
Skimmed milk and whey powder	11,500	0	11,500	0	
Animal fats ^a	20,000	4,961	-	-	DE, Western Europe
Yeast	200	5,946	6,146	97	EU

Feed	Production CH, t/year	Import CH, t/year	Consumption CH, t/year	% Import	Main provenance
Animal proteins (fish meal and others)	0	10,295	10,295	100	Western Europe
Total	823,700	535,126	1,358,826	39	

^a The use of animal fat in animal feed has been prohibited since 2000 (see chapter 18).

Large quantities of these products are processed into compound feed. The proportions of the different ingredients in feed mixes vary widely from year to year and manufacturer to manufacturer, depending on price fluctuations and the current market situation. Nevertheless, the nutritive value of the feed mixes is kept constant. Animal-feed requirements and the properties of different feedstuffs are given in RAP (1995 & 1999). To calculate the quantities of a specific compound feedstuff the inventory user must calculate the required quantities of each feedstuff component. The processing of compound feed is already included in the inventories.

12.1.1 Characteristics of the Production Process

Feed-mix production comprises different processes, the importance of each of which depends on the proportions of the different feedstuff categories in each feed mix. The main processes are as follows:

1. *Production of the feed ingredients*: this production takes place on farm fields (agricultural production), as a main product, in factories (industrial production) or as a by-product, in food processing factories (by-products). Only feedstuffs stemming from agricultural production were considered in the ecoinvent database. The modules described in chapters 14, 15 and 17 provide information on the production of some of the raw materials of the feed mixes.
2. *Transport to the feed-processing centre and storage* of the raw materials: transport is by boat for overseas imports and mainly by lorry within Europe and Switzerland. The main factor in transport distance is the provenance of the raw materials: raw materials produced in Switzerland have the shortest transport distances, followed by raw materials from neighbouring countries (see Tab. 12.1). Transport distances are greatest for inputs imported from overseas.
3. *Processing the feedstuffs*: the main steps comprise rolling, crushing or milling, heat treatment, dosing, mixing, squeezing and pelleting. For some feed categories such as the oilseed cakes, the main processing steps are performed outside the feedstuff factory.
4. *Storage and packaging* of the feed mixes. The packaging materials are not considered in the inventories.
5. *Transport from the feed processing centre to the regional storehouse* or the final user. As all inventories included in ecoinvent refer to “at feed mill”, this final step was not included in the inventories. It must, however, be borne in mind by the user of these inventories. The relevant information is given below.

12.2 Life Cycle Inventories of Feedstuff

12.2.1 Agricultural Production of the Feed Ingredients

The agricultural crop production and drying inventories are documented in chapter 14. For the integrated production (IP) of wheat, barley and rye, a mix between the integrated intensive production (denoted by IP) and the integrated extensive production (called “Extenso”) was calculated for domestic production, since these products are not processed separately (unlike organic feedstuffs, which follow a different path). As shown in Tab. 14.11, 42% of wheat and rye and 63% of barley were produced according to the “Extenso”-rules in 2000. These percentages are used to calculate the mix for the respective cereal (Tab. 12.2).

No inventories for the production of raw materials outside Switzerland have been defined in ecoinvent data V1.0. Production abroad is therefore approximated by integrated production (denoted by “IP”) or organic production with location in Switzerland. As the “Extenso”-production exists only in Switzerland, it was assumed that imported cereals used in the IP feed stem from integrated intensive production.

The import statistics do not differentiate between production from conventional, integrated and organic farming. We therefore assumed the same proportion of imports for integrated and organic feedstuffs.

Tab. 12.2 Feedstuff-ingredient production processes, based on statistics from the year 1999 (see Tab. 12.1).

Feedstuff, at feed mill, CH	% import	% CH-production	Out of CH-production (%)			values used in ecoinvent data		
			IP	extensive	organic	IP	extensive	organic
wheat, IP	7	93	58	42	0	61	39	0
wheat, organic	7	93	0	0	100	0	0	100
rye, IP	7	93	58	42	0	61	39	0
rye, organic	7	93	0	0	100	0	0	100
barley, IP	8	92	37	63	0	42	58	0
barley, organic	8	92	0	0	100	0	0	100
grain maize, IP	10	90	100	-	0	100	0	0
grain maize, organic	10	90	0	-	100	0	0	100
protein peas, IP	66	34	100	-	0	100	0	0
fava beans, IP	66	34	0	-	100	0	0	100

12.2.2 Transport to the Feed Processing Centre

Transport of raw materials produced in Switzerland to the feed processing centre is by tractor for short distances and by lorry or rail for longer distances. For simplicity's sake, transport within Switzerland is assumed to be 100 km by lorry. Transport of raw materials from Europe is mainly by lorry, and was estimated to average 1,000 km. Overseas transport is by transoceanic liner, and was assumed to average 10,000 km by ship and 2,000 km by barge (transport from production location to pier and from pier to feed processing centre). Using the cereals as an example, the average transport distances are then calculated as follows: Provenance of the cereals is 87% domestic production, 10% EU import and 3% overseas import. Transport distance is then calculated as follows: $87\% \cdot 100 \text{ km} = 87 \text{ km}$ by lorry; $10\% \cdot 1,000 \text{ km} = 100 \text{ km}$ by lorry; $3\% \cdot 10,000 \text{ km} = 300 \text{ km}$ by transoceanic liner and $3\% \cdot 2,000 \text{ km} = 60 \text{ km}$ by barge.

Tab. 12.3 Values for cereals, maize and legumes used in the life cycle inventories for the average transport distance from the farm to the feed processing centre. The other values are not used in the inventories given in ecoinvent data. They are listed here as references for other applications.

Feed	Transport distance in km			
	lorry, CH	lorry, RER	Barge	Overseas ship
Cereals	87	100	60	300
Maize	90	100	0	0
Legumes	30	700	0	0
Oilseed cakes	16	800	80	400
Cereal products	30	550	300	1,500
Maize gluten	0	200	1,600	8,000

Feed	Transport distance in km			
	lorry, CH	lorry, RER	Barge	Overseas ship
Sugar and molasses	62	380	0	0
Plant fats	0	0	2,000	10,000
Potato protein	0	1,000	0	0
Dried potatoes	100	0	0	0
Yeast	3	970	0	0
Fish meal	0	1,000	0	0
Crude fibre products (straw)	50	500	0	0

12.2.3 Processing the Feedstuffs

Feed ingredients arise either as main products from production processes (e.g. yeast and minerals), or as by-products (e.g. cereal products, plant fats).

Given the great number of feedstuff production processes, considerable simplification is required to adequately describe these processes for the ecoinvent database. Only cereals and legumes are included as feedstuffs in the database, in addition to dried roughage, which is described in chapter 15. The processes considered are crushing or milling, heat treatment, dosing, mixing, squeezing and pelleting.

The feed-milling and treatment processes were described in detail by Rossel (2001). The actual processing and treatment may vary widely, as may the environmental impact. The processes can only be described for an average situation.

Salzgeber & Lörcher (1996) give the total energy and water consumption of a mill processing about 30,000 tonnes of cereals per year. This mill produces flour for human consumption. The total process energy consumption of this mill was approx. 350 MJ per tonne of cereal grain processed (98% as electricity and 2% as fuel) or 440 MJ per tonne of flour produced; the CED was calculated by the authors as 970 MJ/tonne cereals. The mill consumes 1800 m³ water and produces 1200 m³ wastewater per year.

For production of feed mixes in the feed processing centres, Cederberg (1998) gives an energy consumption (CED) of 374 MJ per tonne of expander-treated feed for dairy cows. In the process, the expander and pelleting treatments use about 80% of the total energy consumed (40% as electricity and 60% as gas). The grinding and mixing requires only 47 MJ per tonne of feed. For the production of non-expander-treated protein mix, Cederberg (1998) gives an energy consumption of 259 MJ per tonne.

Hilger *et al.* (1997) calculated that the production of feed mixes from raw materials on an industrial scale requires about 270 MJ per tonne feedstuff (CED).

InfoMil (1996) gives a total energy consumption for the production of feed mixes of 300 to 500 MJ (CED) per tonne of final product by units processing from a few thousand to more than 200,000 tonnes per year. The size of the production units corresponds to the conditions in Switzerland, where the largest feed processing centre produces 275,000 tonnes of feed per year. These data stem from a survey of the Dutch feed industry. The feed mills use about 25-45 kWh electricity and 2-6 m³ gas per tonne of feedstuff produced. The production step “dosing, milling and mixing” uses 18-32 MJ per tonne in production units processing 1 to 6 tons per charge and about 10 charges per hour. The 2-6 m³ of gas is used in the production step “pelleting and expanding”, which also uses 25-30 kWh electricity.

The data from the different sources seem to give a relatively consistent picture, despite the different conditions in the various countries. The CED reported levels of between 270 and 500 MJ/tonne of feedstuff. The values for the flour mill (Salzgeber and Lörcher 1996) are higher, but the quality requirements are stricter for the production of food than for feedstuffs, hence the processes are more complex. For ecoinvent the values from InfoMil (1996) were used, namely an average of 35 kWh electricity and 4 m³ natural gas per tonne of feedstuff. The natural gas was converted into final energy

(=145 MJ), using the lower heating value given in the quality guidelines of ecoinvent 2000. These values are valid for pelleted and expanded feedstuffs, and are applied to all cereals and maize.

Lower values were taken for the legumes “protein peas” and “broad beans”, since protein feedstuff is not expanded (Cederberg 1998, Rossel 2001). The value given by Cederberg (1998) for protein mixes is 70% that of expanded feedstuff; thus 70% of the above values were taken for legumes, i.e. 25 kWh electricity and 100 MJ final energy in the form of natural gas.

Water consumption and wastewater output was completed from Salzgeber and Lörcher (1996), namely 56 litres tap water and 38 litres wastewater per tonne of feedstuff produced.

No data on air emissions and waste generation were available.

12.2.4 Feedstuff Storage

The typical storage capacity of the cereal silo found in some feed processing centres in Switzerland is 10,000 tonnes, with an effective storage volume of about 80% of the building volume. With an average grain specific weight of 750 kg/m³, this gives a building volume of about 16,650 m³. Assuming that these feed processing centres process 25,000 tonnes of grain per year, the cereals will be stored for about three months before being processed.

The largest feed processing centre in Switzerland (UFA-Biblis) has a feedstuff storage compound of 29,440 m³ (80 m length * 23 m * 16 m height) with an effective storage capacity of 4,140 tonnes (5520 palettes of 750 kg), or about 150 kg/m³ (UFA 2002). This centre processes 275,000 tonnes of concentrate feeds yearly. After processing, the feedstuff must therefore be stored for approx. one week in the processing centre before being sent to the regional storehouse.

The average length of storage of the raw materials is set at 3 months in the calculations. The storage volume required was estimated as 1.7 m³/tonne of raw material (grain specific weight 750 kg/m³ and 80% of the building volume used for storage). The lifetime of the storage silo was assumed to be 50 years, the length of the construction phase 2 years. Land use was estimated from the height of Biblis's storage silo. Taking the height of 16 m, the area used is 0.1 m²/tonne of feedstuff for 3 months. The total area occupied as “industrial area, built up” was estimated as twice as high, namely 0.2 m²/tonne.

12.2.5 Transport to the Regional Storehouse and the Final User

Feed processing centres in Switzerland are widely scattered (e.g. Herzogenbuchsee, Lenzburg, Sursee, St. Margerethen and Puidoux for the UFA, the most important feedstuff supplier in Switzerland). On average, UFA lorries deliver feed to three regional storehouses and cover a distance of 213 km per load for delivery and return of the empty lorry⁵³. To calculate the transports in tkm, only half of this distance need be considered, since the return of the empty lorry is already included in the relevant transport modules. Taking into account that two to three farmers are supplied with feed, this results in an average distance of about 70 km per farmer.

This transport is not included in the modules, which refer to “at feed mill”. They must, however, be considered by the user of these modules.

12.3 Data Quality Considerations

Being based on statistics, the data on the provenance of the raw materials may be considered sufficiently accurate (Tab. 12.4). No information, however, was collected on the agricultural

⁵³ Personal communication from F. Friedli, UFA Herzogenbuchsee, 22 January 2001.

production of imported raw materials. These data were extrapolated from the Swiss production and technological correlation was therefore scored as 3.

Transport distances are based on statistics and assumptions, and their reliability was scored as 3.

Energy values for feedstuff processing were extrapolated from Dutch data from 1996. Since data for legumes were extrapolated from cereal processing, their technological correlation was scored as 3.

The data on water consumption and wastewater output were extrapolated from older data of a related technology. Their quality is thus lower.

Storage-space requirements are rough estimates from a few sites, but the underlying information comes from Swiss sites and is current.

Sample size was not scored for feedstuff.

Tab. 12.4 Uncertainty judgement for the feedstuff data.

Category	Reliability	Completeness	Temporal correlation	Geographical correlation	Further technological correlation
Provenances of raw materials (agricultural production)	1	1	1	1	3
Transport distances	3	1	1	1	1
Energy values for cereals and maize	2	3	2	3	1
Energy values for legumes	2	3	2	3	3
Water and wastewater quantities	3	3	3	3	4
Storage space requirement	3	3	1	1	1

13 Grass-, Maize- and Grain-Drying

13.1 Characteristics

Different ways exist of preserving food and feedstuffs for storage, such as drying, adding chemicals or silaging (Eichhorn 1999: 262). Drying has the disadvantage of a high energy consumption and a certain risk of nutrient loss, but new technologies and a series of advantages (see below) make drying a potentially useful method. Moreover, drying is the most effective way of preserving bread grains.

Forage-drying plants first and foremost provide livestock owners with an easy-to-handle product made for long-term storage and requiring little storage space. Drying animal fodder is also considered to improve on the raw material, yielding a product rich in nutrients and proteins. In Western Europe, especially after the BSE crisis, there is a major need for vegetable-protein-rich feedstuff, which in Switzerland is mainly satisfied by imports from the European Union (EU) and overseas. Since there are increasingly few suppliers able to provide the market with GMO-free soya bean, however, replacing a percentage of the soya-bean protein with clover grass protein becomes a more and more interesting proposition (Schmid 2001, Welter 2001).

There are three different ways to organise grass-, maize- and grain drying:

- drying plant on each individual farm,
- mobile drying plants,
- bigger, centralised drying plants producing dry forage for a whole region.

In Switzerland the third option has traditionally been the most common one. Swiss drying plants are organised into several associations. Two of them were contacted for this report: the *Verband Schweizerischer Trocknungsbetriebe* (VSTB; “Association of Swiss Drying Plants”), uniting most of the grass- and maize dryers (about 90 members), and the *Verband der Getreidesammelstellen der Schweiz* (VGS; “Swiss Association of Grain Collection Points”) with about 75 members.

Drying time depends on the initial water content (a function of grain ripeness, grain moisture, air humidity and temperature). Typical water content before and after drying is given in Tab. 13.1. For members of VSTB, pellets produced from the whole maize plant are by far the most important dried product (Tab. 13.1), followed by grass pellets.

Tab. 13.1 Quantity of dried products in VSTB-member plants (Schmid 2001, Christen 2002).

Product	Quantity [t/a] (Average 1998-2001)	% of dried quantity	typical water content before drying [%]	typical water content after drying [%]
Dried grass	30,404	15%	75	12-14
Maize plant pellets	103,581	53%	65-70	12-14
Grain maize	16,818	9%	35-40	12-14
Dried sugar beet cossettes	9,447	5%	78	11
Cereals	5,406	3%	16	13-14
Other (e.g. potatoes or straw)	31,625	15%		
Total	197,279	100%		

By far the highest percentage of grain is dried in specialised grain-drying plants. Tab.13.2 shows the average quantity of grain produced in Switzerland in the period from 1998 to 2001. In Switzerland,

depending on weather conditions, from 0 to 75 % of annual grain production is dried⁵⁴. Consequently, it is not useful to give an average quantity of dried grain for a single year. No statistical data are available on the quantity of dried cereals. As the amount of water to be evaporated is significantly higher for grain maize than for other grains, it is recommended to use the dataset “maize drying” in this case.

Tab.13.2 Surface area (average 1997-2000^a) and usable production (average 1997-99) of grain (including maize) and green crop (SBV 2000b, SBV 2001b).

		Surface area [ha]	Usable production [t]
Breadgrains	Wheat	94,580	552,144
	Rye	3,604	21,605
	Spelt	1,601	8,327
	Other breadgrains	94	643
Fodder cereals	Barley	47,954	296,104
	Oats	6,572	37,729
	Triticale	8,495	46,544
	Mixed cereals	406	2,375
	Grain maize	21,236	191,287
Total cereals		184,542	1,156,758
Temporary leys and green fodder		114,601	17,778,667

^a Data for the year 2000 are provisional.

Besides drying at regional drying plants, it is common practice to dry hay at farms. These data are described in chapter 5.

13.1.1 Classes of Grass-, Maize- and Grain-Drying Plants

The following list provides an overview of the most common classes of drying-plants:

- Feed-and-turn dryers,
- Belt dryers,
- Mixed-flow dryers / Batch dryers,
- Rotary dryers.

Within the first three classes, most of the dryers are designed for use on individual farms or for mobile drying plants. Most of the 70 centralised grass- and maize-drying plants in Switzerland use rotary dryers⁵⁵, whereas most of the grain-drying plants are equipped with mixed-flow or batch dryers.

13.1.2 Operational Characteristics of the Rotary Dryers

The technology of rotary dryers is described in Stela (2002) as follows: The drying drum is supplied with the wet product by means of a suitable feed device. Transporting scoops on the feed side of the drum take up the product and carry it to the drum internals. The choice of drum internals is determined

⁵⁴ Personal communication from Th. Häusermann, (VGS). Seengen, Switzerland, July 2002.

⁵⁵ Personal communication from J. Christen, (VSTB). Alberswil, Switzerland, July 2002.

by the characteristics of the product. Drying is usually in concurrent flow with the wet product coming into contact with the hot drying gas, the temperature of which possesses a wide adjustment range, allowing the most suitable drying temperature to be set for each product. The residence time of the product in the dryer depends on drum speed, drum declination, the rate of flow of the drying gas and the damming device at the end of the drum.

13.1.3 Operational Characteristics of Mixed-Flow / Batch Dryers

The technology of the mixed-flow / batch drying principle is described in Stela (2002) as follows: The product to be dried runs through the dryer from top to bottom due to the force of gravity. Horizontally arranged warm-air and exhaust-air ducts run through vertical product shafts. The optimally designed shape of the roof of these ducts guarantees the even flow of the product. The roof ducts are open downwards at full length. The warm air streams into the column at the front side of the warm air roofs and flows through the closely piled up product. The product warms up, delivers its water into the passing air and dries. As a countermovement, the air cools down and air humidity rises to saturation limit, depending on the product. The humid air escapes through the neighbouring exhaust-air roofs. This air is extracted from the dryer by the exhaust-air fan. The design of the roof ducts guarantees even product moisture and maximum protection of the product.

13.1.4 Operational Characteristics of Direct and Indirect Air Heaters

As is commonly known, drying is a far quicker process when the drying air is heated. The drying plants are thus equipped with air heaters that raise the air temperature to different levels, depending on the product to be dried. It is important not to overheat any dried product, but bread- and seed grains in particular must not be heated to over 40 °C. This is because the individual grain must preserve its chemical and physical properties, especially its germination capacity and suitability for baking for subsequent use as seed and in food production. Cereals used as feedstuffs are more tolerant of high temperatures, so the grains may be dried in the drying plants at an air temperature 5 to 10 °C higher warmer than the air temperature permitted for bread grains (80-90 °C). This allows for a more efficient drying process and a lower fuel consumption per kg of water extracted. The upper temperature limit to which grass- and maize-plant pellets may be heated is around 70-80 °C, which allows the air temperature to be increased to approx. 110–120 °C. Because of their protein content, fodder legumes are not resistant to high temperatures. The ecoinvent module "grain drying, low temperature" should therefore be borne in mind in order to evaluate the drying of beans, peas and other legumes.

Also depending on the product to be dried, direct or indirect air heaters can be fitted in both classes of dryers mentioned above. The operational characteristics of these heaters are described in the following sections.

The direct air heater consists of an outer casing, an inner sheath against radiation loss and a perforated combustion chamber of highly heat-resistant steel, placed centrally in the casing. The hot exhaust gases are mixed with the fresh air heated at the side of the combustion chamber, and warm air of a homogeneous temperature is generated. The combustion energy is consequently led directly into the dryer. This method is used primarily for drying feedstuffs and industrial raw materials (Stela 2002). Most grass- and maize-drying plants are equipped with this type of furnace⁵⁶.

The indirect air heater consists of an outer casing, an inner sheath against radiation loss, flue-gas funnels, and a closed combustion chamber of highly heat-resistant steel, placed centrally in the casing. In the indirect air heater, the heating energy is transferred exclusively via the heat-exchange surfaces of the closed combustion chamber and the fuel gas is funneled into the fresh air. The cooled combustion gases are drawn off through a chimney. The warm air is consequently completely free of

⁵⁶ Personal communication from J. Christen, VSTB, Alberswil, Switzerland, July 2002.

combustion gases. The indirect air heating method is used especially for drying products for food processing, particularly when the air is heated with heating oil (Stela 2002). Most members of the VGS in fact use indirect air heaters, while a minority who heat with gas use direct furnaces⁵⁷.

Typical air heaters used in Swiss drying plants have a nominal power of 4 to 5 MW.

13.2 Life Cycle Inventories of Grass-, Maize- and Grain-Drying

13.2.1 Scope of the Life Cycle Inventories

Most of the data describing the drying processes for grass, maize and grain are based on personal communications from experts of the two associations, VSTB and VGS. In addition, for the fuel consumption, the Swiss values were compared to values given in international references. Where no Swiss values were available, data from international literature were used.

The inventoried process for grass- and maize drying (Fig. 13.1) consists in the drying process itself, a dosage and a transport system, chopping the wet stuff, ventilation, milling, pressing of pellets, cooling, and the weighing and packing of the finished dry product⁵⁸. Besides drying, the inventory for grain includes cleaning (filtering and removing dust and foreign matter), a transport system, weighing, removing awns and cooling; not included is the milling of the dried grain⁵⁹.

The production of the wet educt entering the drying plant and the further use of the dry product leaving the plant were not included, as there are other ecoinvent modules for these processes. Nor was transport taken into account, since the user has better information on the distances which he can easily use with the appropriate ecoinvent module. The system boundaries are shown in Fig. 13.1.

⁵⁷ Personal communication from Th. Häusermann, VGS Seengen, Switzerland, July 2002.

⁵⁸ Personal communication from F. Scheidegger, Landi Landshut, Bätterkinden, Switzerland, November 2002.

⁵⁹ Personal communication from H. Müller, Trocknungs-Genossenschaft Strass, Frauenfeld, Switzerland, November 2002.

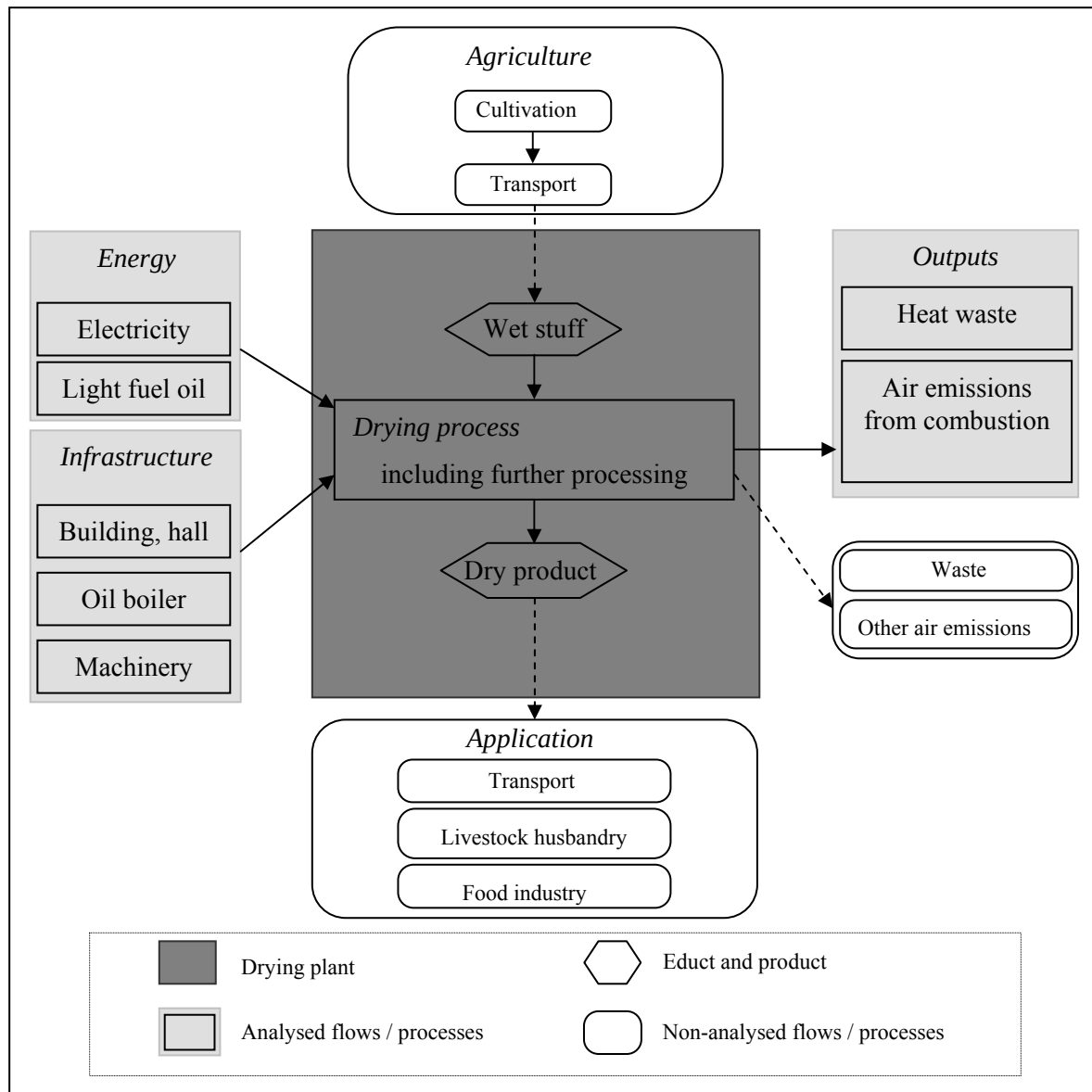


Fig. 13.1 System boundaries of the inventories for the drying of agricultural products.

13.2.2 Functional Unit and Application

Data are given for the functional unit "evaporation of 1 kg of water". This allows users to use their own data for water content before and after drying. For the calculation of typical amounts of water evaporated see chapter 13.2.4, Tab. 13.1 and Tab. 14.2.

To analyse e.g. the drying of 1200 kg of wheat from 18 to 14 % water content, a user should

1. calculate the amount of water evaporated per kg: $(0.14 - 0.18) / (0.18 - 1) = 0.05$ kg water evaporated per kg dried product (formula 13.2).
2. calculate the amount of water evaporated: $1200 \text{ kg} * 0.05 \text{ kg} = 60 \text{ kg}$ of water evaporated
3. link to the corresponding ecoinvent module: $60 * \text{module "grain drying, high temperature"}$.

13.2.3 Infrastructure and Land Use

A typical Swiss drying plant for grass and maize measures approx. 20 m * 40 m⁶⁰. This surface is included as "building, hall". Only a minority of the drying plants have a storage building. It was therefore assumed that a typical Swiss grass- and maize-drying plant does not have a storehouse. For "building, hall", a lifetime of 50 years was assumed. The land use for environs needed to deal with tractors and trailers and other vehicles or machines was estimated as 3,000 square metres (area only, material not included⁶¹), included the 800 m² of the plant itself⁶². It was assumed that there was a transformation from "pasture and meadow" occupancy to "industrial area" occupancy.

A typical grain-drying plant consists of the dryer itself, built on a surface area of 150 square metres (about 12 m * 12 m), and up to four storeys in height⁶³. Most plants do not have a storehouse. As with the drying plants for grass and maize, this surface is included as "building, hall", with a lifetime of 50 years as well. The total surface area of a plant, including building surface and surroundings, was also estimated to be 3000 square metres. Transformation was assumed to be from "pasture and meadow" occupancy to "industrial area" occupancy.

The total surface area of a plant was also included for the building and dismantling phases as "occupation, construction site".

The area of the building, and of the land transformation, calculated for one kilogram evaporated water, is obtained by dividing the total surface area of the building or of the environs by the annual yield of dried product, the water evaporated per kilogram of dried product and by the lifetime of the infrastructure. Land use by occupancy was calculated in almost the same way, the only difference being that the total surface area of the environs was not divided by the lifetime. The mean annual yield of a typical grass- and maize-drying plant is 2,170 t of dried products, including 330 t of dried grass and 1,160 t of dried maize (Christen 2002). A typical grain-drying plant produces 10,000 t of dried grain, 5,600 t from low-temperature and 4,400 from high-temperature drying (SBV 2000b, SBV 2001b and personal communication). Chapter 13.2.4 below describes how the amount of water evaporated was calculated.

For energy supply, infrastructure and outputs, ecoinvent modules (where available) were taken into account. For dryer and the other machinery employed in the further processing, no adequate module was available. An estimate of the total weight of the machines was made (65 tonnes for grass- and maize-drying plants, 40 tonnes for grain-drying plants) which was calculated as cast iron, the main component of this machinery⁶⁴. Solid waste was assumed to be irrelevant. The heater, the consumed fuel and the exhaust gases from combustion are included in the ecoinvent module "light fuel oil, burned in industrial furnace 1MW, non-modulating". The weight for the heater inventoried in this module (5 tonnes) was subtracted from the estimated weight of the machinery used in the drying process.

The weight of the machinery, balanced as cast iron, per kilogram of water evaporated, was calculated in the same way as the area of the building and the land transformation. The expected useful life of the machinery was 25 years. Construction processes for the machinery were not included. Maintenance of the building and heater are included in the respective ecoinvent module. Maintenance of other equipment is not included. The latter is assumed to be completely recycled at the end of its useful life; hence, no effects for disposal are allocated to the drying process.

⁶⁰ Personal communication from J. Christen, VSTB. Alberswil, Switzerland, July 2002.

⁶¹ Personal communication from H.J. Althaus, EMPA Dübendorf, 27 January 2003.

⁶² Personal communication from F. Scheidegger, Landi Landshut. Bätterkinden, Switzerland, November 2002.

⁶³ Personal communication from Th. Häusermann, VGS. Seengen, Switzerland, July 2002.

⁶⁴ Personal communication from W. Kunz, Kunz Drytec AG. Dintikon, Switzerland, November 2002 and personal communication from R. Walia, Wirtech AG. Uetendorf, Switzerland, November 2002.

13.2.4 Energy Carrier and Energy Demand

Air heaters in drying plants can be run on various energy carriers, such as electricity, heating oil, liquid gas, natural gas, solid fuels (coke, coal, wood, straw) or solar panels. In Switzerland, three quarters of the grass- and maize-drying plants use heating oil, and one quarter, natural gas⁶⁵. A clear majority of grain-drying plants also use heating oil to run the heater⁶⁶.

For electricity consumption, no data were published by the two associations, VSTB and VGS. The values inventoried in this study were therefore based on data from a single plant of each type.

Drying Maize and Grass

For drying maize from a 70 % water content at the outset of the drying process (W_i , [% water content]) to a 13 % water content at the end (W_e , [% water content]), the average heating oil consumption (OC, [kg oil/kg product]) of the VSTB-member plants was 13 kg oil per 100 kilograms dried product (Christen 2002 and Schmid 2001). According to formulas 13.1 and 13.2, this corresponds to an energy demand for heating (E , [MJ/kg $H_2O_{\text{evaporated}}$]) of 3 MJ per kilogram water evaporated (W_{evap} , [kg H_2O /kg product]).

$$E = OC * HV / W_{\text{evap}} \quad (13.1)$$

$$W_{\text{evap}} = (W_e - W_i) / (W_i - 100) \quad (13.2)$$

The heat value (HV, [MJ/kg oil]) for light heating oil is 42.6 MJ per kilogram oil.

In the same plants, drying grass from a water content of 75 % to 13 % uses an average of about 30 kg oil per 100 kg dried product (Christen 2002 and Schmid 2001). This corresponds to 5 MJ per kilogram water evaporated.

The drying and further processing of wet stuff in the one grass- and maize-drying plant for which data were available produces a total electricity consumption per year (C_y , [kWh/a]) of 615,000 kWh. The mean yield per year (Y , [kg product/a]) of this plant was 7,760 t of dried products, while the mean water evaporated from one kilogram dried product (W_{evap}) was 1.74 kg. According to formula 13.3, this means a consumption of 0.05 kWh electricity per kg water evaporated (C_w , [kWh/kg H_2O evaporated]) for the plant's entire production. As an approximation, this value was attributed to both grass and maize.

$$C_w = C_y / (Y * W_{\text{evap}}) \quad (13.3)$$

Drying Grain

For reasons described in chapter 13.1.4, grain drying must be separated into low- and high-temperature drying. Since no data were available for the heating-energy requirement in Swiss grain-drying plants, values had to be taken from international references, which provided widely varying and even sometimes contradictory data.

The theoretical energy requirement for evaporating one kg of water is 2.60 MJ according to Pimentel (1996), who also writes that real energy use is 2-6 times higher than that, or 5.2-15.6 MJ. Other data found in the literature suggest that real energy demand is 2-3 times the theoretical value (Carlsson & Faist 2000). Values found in the references are listed Tab. A. 17 in the Appendix. Based on this

⁶⁵ Personal communication from J. Christen, VSTB. Alberswil, Switzerland, July 2002.

⁶⁶ Personal communication from Th. Häusermann, VGS. Seengen, Switzerland, July 2002.

literature analysis, we estimated that low-temperature drying of bread grains required 7 MJ heating energy per kilogram water evaporated, whereas evaporating 1 kg of water at higher temperatures from the fodder cereals required 5 MJ.

Electricity consumption was analysed for one grain-processing plant with an annual yield of 10,000 t dried grain (the total for high- and low-temperature drying), and came to 240,000 kWh per year. Per kilogram dried grain, 0.024 kg water is evaporated (see Tab.13.3 and formula 13.2). This means that according to formula 13.3, 1 kWh electricity per kg water evaporated is needed to process grain.

Tab.13.3 summarises the energy requirements for drying maize, grass and grain as described above.

Tab.13.3 Energy demand for evaporating 1 kg of water during the drying process of different wet stuffs.

		Maize	Grass	Grain	
				Low temperature	High temperature
Water content ($W_i \Rightarrow W_e$)		70 $\Rightarrow$ 13 %	75 $\Rightarrow$ 13 %	16 $\Rightarrow$ 14 %	16 $\Rightarrow$ 14 %
Water evaporated [kg H ₂ O / kg dried product]		1.9	2.5	0.024	0.024
Energy demand per kg evaporated water	Oil [MJ]	3	5	7	5
	Electricity [kWh]	0.05	0.05	1	1
	Total [MJ]	3.2	5.2	10.6	8.6

13.2.5 Emissions

For indirect furnaces running on light fuel oil, the threshold values given in the Swiss Ordinance on Air Pollution Control, (LRV) (Schweizerische Eidgenossenschaft 1985) must be respected in Switzerland. According to expert estimates, bearing in mind modern technology and fuels, complying with these threshold values should not be a problem for either type of furnace ⁶⁷.

In the case of direct air heaters, one potential problem that may be the subject of further studies, especially if the dried stuff is destined for human consumption, is the sulphur dioxide. There were no data available, however, on the special flue gas that might develop owing to direct contact of the exhaust gas with the wet stuff⁶⁷. No data or indications were available on the organic compounds that might possibly form in the direct dryer⁶⁸.

Because of this lack of information, indirect and direct air heaters could not be treated separately, and emissions and waste heat from fuel consumption were inventoried as an approximation as "light fuel oil, low sulphur, burned in boiler 1MW".

Solid emissions were considered not to be relevant, and hence were not included. Waste heat from heating-oil combustion was included in the respective ecoinvent module. This left only the waste heat from electricity consumption to be calculated (3.6 MJ per kWh).

⁶⁷ Personal communication from A. Liechti, Swiss Agency for the Environment, Forests and Landscape (SAEFL), Bern, Switzerland, August 2002.

⁶⁸ Personal communication from J. Schleicher, Department for Waste, Water, Energy and Air (AWEL), Zurich, Switzerland, August 2002.

13.3 Data Quality Considerations

Data on heating-oil consumption for grass- and maize-drying plants were based on the mean values of measurements taken over the last two years in about 80 drying plants (representing the majority of plants in Switzerland). For the grain-drying plants, we made an assumption based on international literature published over the last six years.

Data on electricity consumption was based on measurements taken in 2001 in one drying plant per type, which represent typical plants in terms of Swiss conditions.

Expert estimates from the manufacturer were the basis for the amount of machinery balanced. According to the experts, this material can be calculated as cast iron.

Data on infrastructure, land occupation and transformation are expert estimates.

Emissions from heaters were inventoried as “light fuel oil, low sulphur, burned in boiler 1 MW”. There is no information on specific emissions from heaters used in drying processes.

Waste flows and air emissions other than from combustion were not analysed, because of a lack of data, and because of the assumption that their ecological influence was negligible compared to that of the analysed flows and processes.

Part IV: Agricultural Outputs

14 Arable Crop Production in Switzerland

This chapter describes the inventories for arable crop production (for human food and animal feed). All these inventories refer to Switzerland.

14.1 Characteristics

14.1.1 Farming systems

The three main farming systems practiced in Switzerland are the conventional, the integrated and the organic farming. They are characterised by increasingly restrictive rules and by increasing subsidies.

They are briefly characterised in the following:

- *Conventional production* in the Swiss context is agricultural production complying with legislation but not meeting the minimum requirements for integrated production.
- *Integrated production* refers to agriculture meeting the ecological requirements (*ökologischer Leistungsnachweis*, ÖLN) defined by the *Direktzahlungsverordnung* decree⁶⁹ (and of course also the requirements of the legislation).
- *Organic production* complies with the requirements for organic production. Application of synthetic pesticides and fast-acting mineral fertilisers is not permitted. By definition, organic production also meets the requirements of integrated production.

For the integrated production of cereals and rape seed there are two intensity levels in respect to plant protection: ordinary production (intensive) and *extensive production* (see the *Direktzahlungsverordnung* decree). In the latter the application of fungicides, insecticides and plant growth regulators is not permitted. Herbicides and mineral fertilisers are still allowed in this production system. Extensive production is also a form of integrated production. Organic farming automatically also complies with the rules of extensive production and integrated production.

14.1.2 Overview of the Life Cycle Inventories

An overview of the arable crop inventories is given in Tab. 14.2.

Three different farming systems or production methods were considered in the inventories:

- integrated intensive production (“IP”)
- integrated extensive production (“extensive”) and
- organic production (“organic”).

Tab. 14.1 summarises the requirements for the different production methods.

⁶⁹ Verordnung über die Direktzahlungen in der Landwirtschaft (Direktzahlungsverordnung, DZV), 7.12.1998.

Tab. 14.1 Categories of agricultural production inventories included in ecoinvent data and the standards with which they comply.

Category of module in ecoinvent	Meets requirements of		
	Integrated production ("ökologischer Leistungsnachweis", ÖLN)	Extensive production (<i>Extenso</i>), cereals and rape seed only	Organic farming (<i>Bio</i>)
IP	yes	no	no
extensive	yes	yes	no
organic	yes	yes	yes

Because conventional farming is only of minor and decreasing importance in Switzerland, no such inventories were included in ecoinvent data.

The inventories denoted by "**IP**" refer to crops from integrated production. For cereals (wheat, rye and barley) and rape seed, they refer to crops not fulfilling the requirements of the "extensive production of cereals and rape seed" (*Extenso*). IP therefore denotes an integrated production with intensive plant protection in the Swiss context for cereals and rape seed.

The inventories for "**extensive**" wheat, barley, rye and rape seed correspond to crops from integrated production without the application of fungicides, insecticides and plant growth regulators. For all other crops where there is no "extensive production" according to the *Direktzahlungsverordnung*, IP simply refers to integrated production meeting the requirements of the ÖLN. Only inventories for "IP" and "organic" have been included for these crops, according to their relevance.

The inventories for "**organic**" crops comply with the requirements for organic production.

All data were collected for the Swiss lowlands (*Talgebiet*: for definition, see BLW 2001), which account for by far the largest percentage of the arable surface. No inventories were included for the hill or mountain regions, where arable crop production is only of minor importance.

Inventories for the cereals are for their respective winter form, i.e. winter wheat, winter barley and winter rye⁷⁰.

Although the inventories for arable crops are based on data from past years, they were corrected so as to be representative for the current situation (for details see below).

The reference function of all arable crop inventories is **1 kg fresh matter**. Silage maize and fodder beets are often calculated with reference to dry matter. In ecoinvent, however, they were calculated as fresh matter for reasons of consistency. Tab. 14.2 summarises the inventories described in this chapter and their main characteristics.

⁷⁰ Studies have shown that the differences in the life cycle inventories between winter and spring cereals are not very great (Nemecek *et al.* 2005), so these inventories may also be used as an approximation for spring-sown cereal crops.

Tab. 14.2 Main characteristics of the inventories of arable crops in ecoinvent data. All modules refer to the location Switzerland (CH). The quantities of nutrients applied refer to the quantities of mineral fertilisers and farmyard manure applied, corrected for the crop residues (see chapter 14.2.3).

	yield main product (kg/ha)	humidity at harvest (%)	humidity at storage (%)	straw yield (kg/ha)	harvested straw	kg seed/ha	date of tillage	date of sowing	date of harvest	kg N/ha (available N)	kg P2O5/ha	kg K2O/ha	slurry and liquid manure (m3/ha)	solid manure t/ha	number of pesticide applications
wheat IP	6425	16%	15%	3915	49%	180	15.10	25.10	5.8	140	66.0	59	3.3	0.1	3.4
wheat extensive	5305	16%	15%	3232	49%	180	15.10	25.10	5.8	124	55	49	8.2	1	1
wheat organic	4069	16%	15%	3306	65%	200	15.10	25.10	5.8	66	65	196	46	9	0
rye IP	7540	16%	15%	5013	49%	140	25.9	5.10	5.8	90	75	79	6.8	1.7	2.7
rye extensive	5969	16%	15%	3968	49%	140	25.9	5.10	5.8	90	60	63	7.7	0.3	1.2
rye organic	4172	16%	15%	3698	65%	170	25.9	5.10	5.8	48	47	141	33.1	6.4	0
barley IP	6828	16%	15%	3606	49%	110	15.9	25.9	15.7	110	71	83	6.8	1.8	2.7
barley extensive	5403	16%	15%	2854	49%	110	15.9	25.9	15.7	99	56	68	7.7	0.3	1.2
barley organic	4153	16%	15%	2924	65%	140	15.9	25.9	15.7	53	52	157	36.9	7.3	0
sunflower IP	3151	17%	6%	6302	0%	3.75	25.4	25.4	20.9	13	40	69	0	0	0
rape seed IP	3113	12%	12%	5781	0%	5.5	15.8	25.8	15.7	125	50	31	8.5	5.5	3.5
rape seed extensive	2683	12%	6%	4983	0%	5.5	15.8	25.8	15.7	107	44	26	24.4	9.4	1.9
potatoes IP	37770	78%	78%	15108	0%	2520	25.3	15.4	15.9	71	67	226	13.1	13.7	6.8
potatoes organic	22908	78%	78%	9163	0%	2520	25.3	15.4	15.9	31	51	120	9.7	14	0
grain maize IP	9279	39%	14%	11019	0%	25	25.4	5.5	25.10	108	66	46	11.8	7.8	1.7
grain maize organic	7777	39%	14%	9235	0%	25	25.4	5.5	25.10	77	94	271	46.4	18.4	0
silage maize IP	61457	72%	72%	0	0%	27	15.5	15.5	25.9	98	124	269	23.7	11.9	1
silage maize organic	49166	72%	72%	0	0%	27	15.5	15.5	25.9	65	79	227	39	15.4	0
sugar beets IP	72310	77%	77%	55623	0%	2.1	5.3	25.3	5.10	81	58	132	13.1	9	4.4
fodder beets IP	97002	85%	85%	36376	0%	2.1	5.3	25.3	5.10	61	72	184	20	13.2	2.9
fava beans IP	3782	16%	13%	4255	0%	175	1.2	1.3	10.8	0	52	64	0	0	1
fava beans organic	3384	16%	13%	3807	0%	190	1.2	1.3	10.8	38	40	129	26.5	6.6	0
soy beans IP	2933	16%	11%	2933	0%	110	1.4	1.5	15.9	13	40	69	9.4	2.2	1.2
soy beans organic	2806	16%	11%	2806	0%	120	1.4	1.5	15.9	8	12	30	3.9	2.7	0
protein peas IP	3840	16%	13%	3840	0%	250	1.2	1.3	25.7	5	39	56	4.3	0.3	1.8
protein peas organic	3044	16%	13%	3044	0%	275	1.2	1.3	25.7	38	40	129	26.5	6.6	0

14.2 Life Cycle Inventories

A schematic description of the inventories of arable crops is presented in Fig. 4.1.

There is no single data source that could be used to model all aspects of crop production and that would be representative of Swiss production as a whole (see also Nemecek & Erzinger 2005 and Nemecek *et al.* 2005). Data were therefore gathered from different sources by a group of experts from the Swiss Federal Agricultural Research Stations ART and ACW. Tab. 14.3 summarises the data sources used to define the inventories. Detailed information is given in the following sections.

Tab. 14.3 Data sources used to compile the arable-crop production inventories.

Category of data	Data source(s)	Remarks	See Chapter
Yields main products	FADN FAT (years 1996-2003)	weighted means, data for sunflower and faba beans estimated by experts, silage maize taken from variety trials 1996-2003	14.2.1
Straw yields and crop residues	Walther <i>et al.</i> (2001)	assuming a constant grain/straw ratio	14.2.2
Humidity contents	LBL <i>et al.</i> (2000)		14.2.5
Quantity of seed	LBL <i>et al.</i> (2000)		14.2.7
Dates of sowing and harvest	Näf (1996)		
Quantity of fertilisers	Walther <i>et al.</i> (2001)	corrections for yield, green manure and hoeing according to Walther <i>et al.</i> (2001)	14.2.3
Types of fertilisers "IP" and "extensive" crop systems	Importation statistics (from Rossier 2000) for mineral fertilisers (years 1996-98) BLW <i>et al.</i> (1998) for farmyard manure (years 1994-96)		14.2.3
Types of fertilisers organic	BLW <i>et al.</i> (1998) for farmyard manure (years 1994-96)		14.2.3
Use of machinery	LBL <i>et al.</i> (2000)		14.2.5
Pesticide applications	BLW <i>et al.</i> (1998), (years 1994-96)		14.2.6
Chemical seed dressing	Information provided by seed suppliers and experts		

14.2.1 Yields

The two most important data sources for yields are *Schweizerischer Bauernverband* statistics (SBV 2000b) and farm accountancy data network (FADN) statistics of the ART (former FAT, FAT 2005). Although the first data source is based on larger samples, it is not detailed enough to provide the required information (e.g. no distinction is made between the different types of production for a given crop). The FADN data were therefore chosen. The ART's FADN statistics are based on an annual evaluation of the accountancies of 3,500 farms in Switzerland. The data were provided by FAT⁷¹. The 1996 – 2003 weighted averages for the Swiss lowlands were used for grain maize, potatoes, rape seed IP, sugar and fodder beets, soy beans IP, protein peas, wheat, rye and barley.

The yields for silage maize were calculated from the ART's (former FAL) variety-testing trials between 1996 and 2003⁷². Variety-trial yields are about 15% higher than average yields achieved by farmers. To take account of this, only 85% of the yield for "silage maize IP, at farm" was considered. No variety trials were available for organic production. For "silage maize organic, at farm", yields of 80% of the corresponding integrated crop (expert estimate) were assumed.

⁷¹ Personal communications from H. Eggimann, FAT, September 2001 and April 2005.

⁷² Personal communications from M. Menzi, FAL, October 2001 and May 2005.

IP sunflower and IP and organic faba bean yields were estimated by those responsible for these crops and the Swiss Federal Agricultural Research stations⁷³. Organic soya bean yield was assumed to be equal to IP soya bean yield (expert estimate: FADN data were not used, as the sample was too small).

Although the data source (FADN) did not differentiate between spring and winter forms of the grain for wheat, barley and rye yields, it is known from numerous other data sources that winter cereals have higher yields than spring cereals. In order to differentiate between winter and spring yields, the ratios of the spring and winter cereal areas from BLW *et al.* (1998) and the yields given by SBV (2006) were used to estimate winter wheat, winter barley and winter rye yields from FADN data (For details see Nemecek *et al.* 2005, Appendix 3.1.3). No inventories for spring cereals were included in ecoinvent data.

14.2.2 Co-Products and Crop Residues

The crops under investigation produce main products, co-products (straw, leaves) and crop residues. If the co-products are harvested, an allocation must be made. Only cereal straw, and to a minor extent sugar- and fodder-beet leaves, are generally harvested in Switzerland and can be considered co-products.

For straw from IP and extensive production, it was assumed that 75% of the (harvestable) straw is harvested and 25% is left in the field (expert estimate). For organic production, it was assumed that 100% of the (harvestable) straw is harvested. According to Walther *et al.* (2001, Tab. 38), only 65% of the straw is normally harvested (due to losses and to the fact that straw cannot be cut at the soil surface). The respective values are therefore 49% for IP and extensive cereals and 65% for organic cereals; the rest remains in the field and is ploughed back into the soil during subsequent cultivation. The contribution to gross profit for the cereals considered lies between 7 and 10% (Tab. 14.4). Straw can therefore be considered a co-product.

As beet leaves are only rarely harvested for animal fodder in Switzerland, it was assumed that they remain to 100% on the field. This means that beet leaves are not treated as co-products, but as crop residues.

The quantities of straw and crop residues were calculated from the yields of the main products by taking the ratio of main product to by-product from Walther *et al.* (2001, Tab. 2). This ratio was assumed to be constant, which is an acceptable simplification for the purpose of establishing the inventories.

14.2.3 Allocations

For the cereals wheat, rye and barley, allocations were made between the production of grains and of straw. For all other crops, no co-product was harvested according to the inventories.

For cereals, all inputs and outputs were allocated on the basis of an economic factor (Tab. 14.4), except for the inputs listed in Tab. 14.5, for which allocation to grains or straw was clear-cut. The economic factor was derived from the share of the returns from the grains and the straw (quantity harvested * market price), respectively in the total returns of the crop. The prices of the main products were taken from LBL *et al.* (2000). The straw price was taken as 0.10 CHF/kg straw from FAT (2000b, p. 5).

The resources “carbon dioxide, in air” and “energy, gross calorific value, in biomass” were allocated proportionally to the carbon and the heating energy exported into the grains and straw. Heavy metals were allocated proportionally to the quantity of each element exported into the grains or the straw.

⁷³ Personal communication from the responsible parties, Th. Hebeisen (FAL) and R. Charles (RAC), October 2001.

Tab. 14.4 Names of the modules for cereal grains and straw and standard allocation factors used for all inputs and outputs of each crop and cultivation type, except for the inputs listed in Tab. 14.5.

Name of the unallocated module	Name of the allocated module	Allocation factor for grains	Name of the allocated module	Allocation factor for straw
wheat extensive, at farm	wheat grains extensive, at farm	92.5%	wheat straw extensive, at farm	7.5%
wheat IP, at farm	wheat grains IP, at farm	92.5%	wheat straw IP, at farm	7.5%
wheat organic, at farm	wheat grains organic, at farm	93.1%	wheat straw organic, at farm	6.9%
rye extensive, at farm	rye grains extensive, at farm	90.3%	rye straw extensive, at farm	9.7%
rye IP, at farm	rye grains IP, at farm	90.3%	rye straw IP, at farm	9.7%
rye organic, at farm	rye grains organic, at farm	91.9%	rye straw organic, at farm	8.1%
barley extensive, at farm	barley grains extensive, at farm	89.9%	barley straw extensive, at farm	10.1%
barley IP, at farm	barley grains IP, at farm	89.9%	barley straw IP, at farm	10.1%
barley organic, at farm	barley grains organic, at farm	91.3%	barley straw organic, at farm	8.7%

Tab. 14.5 Inputs of the different crops listed in Tab. 14.4 for which the standard allocation factors were not used.

Name	Location	Unit	Allocation factor for grains	Allocation factor for straw
transport, tractor and trailer	CH	tkm	100%	0%
baling	CH	unit	0%	100%
loading bales	CH	unit	0%	100%
grain drying	CH	kg	100%	0%

14.2.4 Fertilisers

The amounts of the principal nutrients N, P and K were calculated from the recommended doses given by Walther *et al.* (2001, Tab. 2). For integrated production (IP and extensive), the following rules were observed:

1. First, the amounts of N, P or K recommended by Walther *et al.* (2001, Tab. 2) for an average crop were taken. All inventories refer to a production meeting the ecological requirements of the “Direktzahlungsverordnung”. One element of these requirements is an equilibrated nutrient balance. The nutrient balance in turn was calculated based on Walther *et al.* (2001). Therefore it seems reasonable to use this source to base the fertiliser data on it.
2. The recommended amounts of N, P or K were multiplied by the ratio EY/SY, where EY is the effective yield (adopted in the inventory, Tab. 14.2) and SY is the standard yield given by Walther *et al.* (2001, Tab. 2). For nitrogen, the correction was only applied if EY/SY ≤ 1, according to the recommendations of Walther *et al.* (2001). This means that the amount of nitrogen cannot exceed the recommended dose.
3. If hoeing was carried out more than once, the required nitrogen fertilisation was reduced by 10 kg/ha, according to Walther *et al.* (2001, Tab. 26).
4. For spring-sown crops, the amount of nitrogen required was reduced by 20 kg N/ha, to take into account the effect of the green manure (according to Walther *et al.* 2001, Tab. 23; see also chapter 14.2.9).
5. A further correction was carried out for the crop residues: a part of the nutrients P and K spread on the field in the form of fertilisers will remain in the crop residues on the field and can be used by the subsequent crop. It is recommended that this quantity be deducted from the fertilisation of the

following crop (Walther *et al.*, 2001, p. 46). This percentage of the nutrients must therefore be allocated to the following crop, and can be deducted from the fertilisation. To do this, the percentage of the nutrient remaining on the field after harvest was calculated from Walther *et al.* (2001, Tab. 2), then subtracted from the amount of P and K that must be allocated to the crop.

6. Apportionment of the quantities of N, P and K to the different types of mineral fertilisers in integrated production was performed as follows: the quantities of farmyard manure were taken from BLW *et al.* (1998). The quantities of nutrients spread with farmyard manure were calculated and subtracted from the total dose to be applied. The remaining dose was distributed among the different types of mineral fertilisers in proportion to their total use in Switzerland between 1996-98 (from Rossier 2000, Appendix 1). The proportions of the different fertilisers are shown in Tab. 14.6.
7. The production of mineral fertilisers was calculated by inventories referring to Europe (RER). Transports to the consumer in CH were added (see chapter 14.2.8). "Patentkali" was approximated by potassium chloride (KCl), and ammonium phosphate by diammonium phosphate (which accounts for over 90% of the ammonium phosphate used in Switzerland).
8. The production and storage of farmyard manure was fully allocated to animal production, and was therefore not included in the inventories (see chapter 4.4.2 and Fig. 4.1). Transport to the fields, machine use and emissions from spreading farmyard manure were, however, included (see respective inventories in Tab. 14.7).

A different procedure was followed for organic production, which is limited by the availability of farmyard manure, with the result that the quantities of nutrients are often lower than the recommended dose. The following rules were applied:

After applying steps 1 to 3 above, the amount of available nitrogen was estimated as 70% of the recommended dose for cereals, and 100% of the recommended dose for potatoes and maize⁷⁴. For grain legumes, the recommended N fertilisation is nil, since these crops produce nitrogen in symbiosis with N-fixing bacteria. For these crops, the average amounts of manure for the years 1994-96 were taken from BLW *et al.* (1998), a study in which 159 pilot farms were monitored over several years. The proportions of the different types of fertilisers (cattle slurry and manure, pig slurry and manure, and poultry manure) were taken from BLW *et al.* (1998). The average contributions of these fertilisers to total N fertilisation were calculated for cereals, maize and potatoes and were used to calculate the quantity of each fertiliser type based on the available nitrogen.

⁷⁴ Personal communication from S. Hartnagel, FiBL, 6 November 2001.

Tab. 14.6 Proportion of different fertilisers for the nutrients N, P and K (from Rossier 2000, Appendix 1). The proportion of N for ammonium phosphates (AP_N) was calculated from the quantity of P_2O_5 applied in ammonium phosphates (AP_P). The other N fertilisers were calculated after subtracting the AP_N.

	Share of different fertilisers for nutrient		
	N	P_2O_5	K_2O
Urea	$(100-AP_N)*18\%$		
Ammonium nitrate	$(100-AP_N)*52\%$		
Ammonium sulphate	$(100-AP_N)*4\%$		
Calcium ammonium nitrate	$(100-AP_N)*26\%$		
Ammonium phosphates (inventoried as DAP)	$18/46*AP_P$	AP_P: 28%	
Triple superphosphate		41%	
Single superphosphate		2%	
Thomas meal		5%	
Rock phosphate		24%	
Potassium chloride			86%
Potassium sulphate			6%
Patentkali (raw potassium)			8%
Sum	100%	100%	100%

14.2.5 Machine Usage

Machine use for crop production was derived from specifications in LBL *et al.* (2000). In this document, standard procedures for the production of crops and various other agricultural products are described with a view to calculating gross margins for these products. The production schemes are typical of a specific type of production and are defined by numerous experts, mainly from the extension services. Alternatively, the data from Näf (1996) could be used. The latter source, however, is older and less detailed (e.g. no distinction is made between integrated intensive, extensive and organic production). LBL *et al.* (2000) was therefore chosen as the basis for the inventories.

The different work processes from LBL *et al.* (2000) were assigned to ecoinvent-modules as shown in Tab. 14.7. These modules for field-work processes are described in chapter 7.

The following adjustments were applied to the machine data from LBL *et al.* (2000) to render them consistent with the assumptions made in the inventories:

- The number of passes by fertiliser spreader was adjusted to the assumptions applied in each inventory.
- The number of passes by field sprayer was taken from BLW *et al.* (1998) to be consistent with the pesticide applications.
- The number of passes by vacuum tanker and manure spreader were calculated from the quantities of slurry and manure respectively.
- The number of straw bales to be baled and loaded was calculated from the amount of straw to be harvested (average fresh weight of 1 bale = 160 kg). A correction of 0.23 was applied to estimate the effort needed to produce a straw bale compared to a silage bale (see footnote for Tab. 14.7).
- Use of the potato-grading machines was adjusted to the potato yield.

Harvested grains must be dried if their moisture content exceeds a maximum level; otherwise they cannot be stored. The processes “grain drying, high-temperature (CH)” and “grain drying, low-temperature (CH)” were included in the inventories. Low-temperature drying is used for food (wheat, rye), oilseed and legumes; high-temperature drying for feedstuffs except legumes (barley). The

process “maize drying” is used for grain maize. The difference in water content of the grains at harvest and at storage (i.e. the amount of water lost in between), taken from LBL *et al.* (2000), was used to estimate the requirements for grain drying. The values are given in Tab. 14.2. Formula 13.2 was applied to calculate the amount of water to be evaporated.

For the transport of the harvested good from the field the following distances were used:

- All grains (cereals, grain maize, oil seeds, grain legumes): 10 km to the regional processing centre, where the grain drying takes place,
- Potatoes, sugar and fodder beets: 1 km to the farm.

For silage maize the transport to the farm is included in the process “fodder loading, by self-loading trailer”.

Tab. 14.7 Machine-use inventories used to calculate the arable-crop production inventories.

Name of the module in ecoinvent data	Location	Unit	Machine specification according to LBL <i>et al.</i> (2000)
tillage, ploughing	CH	ha	Pflug, 2-scharig
tillage, rotary cultivator	CH	ha	Bodenfräse mit Stabkrümmer, 2,1 m
tillage, harrowing, by spring tine harrow	CH	ha	Federzinkenegge mit Krümmer, 3 m
tillage, harrowing, by rotary harrow	CH	ha	Kreiselegge mit Stabkrümmer, 3 m
fertilising, by broadcaster	CH	ha	Schleuderstreuer, bis 500 l
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	Misten (unit conversion t → kg necessary)
sowing	CH	ha	Sämaschine mit Fahrgassenschaltung, 2,5 m
potato planting	CH	ha	Kartoffellegemaschine, 2-reihig
sowing	CH	ha	Einzelkornsämaschine für Mais, 4-reihig
sowing	CH	ha	Einzelkornsämaschine für Rüben, 6-reihig
application of plant protection products, by field sprayer	CH	ha	Feldspritze, 12m
tillage, currying, by weeder	CH	ha	Hackstriegel, 6 m
hoeing	CH	ha	Scharhackgerät für Mais, 4-reihig
hoeing	CH	ha	Rübenhackgerät, 6-reihig
hoeing	CH	ha	Sternhackgerät für Mais, 4-reihig
hoeing	CH	ha	Sternhackgerät mit Vorschar für Rüben, 6-reihig
tillage, hoeing and earthing-up, potatoes	CH	ha	Kartoffelhack- und häufelgerät, 4-reihig
fertilising, by broadcaster	CH	ha	Schleuderstreuer, bis 500 l
slurry spreading, by vacuum tanker	CH	m ³	Gülldüngung
combine harvesting	CH	ha	Mähdrescher, 95 kW (129 PS), ohne
combine harvesting	CH	ha	Mähdrescher, 125 kW (170 PS), ohne
chopping, maize	CH	ha	Anbaumaishäcksler, 1-reihig
harvesting, by complete harvester, beets	CH	ha	Zuckerrübensvollernter, mittel, mit Bunker,
potato haulm cutting	CH	ha	Kartoffelkrautschläger, 1,8 m, 2-reihig
harvesting, by complete harvester, potatoes	CH	ha	Kartoffelvollernter, mittel, Rollbodenbunker, 1-reihig
mulching	CH	ha	Schlegelmulchgerät ohne Schwenkarm, 2-2,5 m
fodder loading, by self-loading trailer ^{*)}	CH	m ³	Ladewagen mit Schneidvorrichtung, 13-20 m ³
transport, tractor and trailer	CH	tkm	Pneuwagen, 2-achsig, 8 t, hydraulisch kippbar
baling	CH	unit	Hochdruckpresse, fresh weight 160 kg/bale, correction factor of 0.23 applied ^{**)}
loading bales	CH	unit	Ballenlader
tillage, cultivating, chisseling	CH	ha	Grubber mit Nachläufer, 2,2 m
potato grading	CH	kg	Kartoffelsortiermaschine mit Verleseband

^{*)} Conversion from hours into m³ using the factor 60m³/h given by Amman (1999): 3 cartloads per hour, 20m³ per cartload.

^{**)} The process “baling” refers to silage production, which takes much more time (0.13 h/bale) than straw-bale production (0.03 h/bale). The use of polyethylene is also lower for straw bales. A factor of 0.23 (=0.03/0.13) was therefore applied to the number of bales for the process “baling”, but not for “loading bales”.

14.2.6 Pesticides and Biological Control

There are no statistics on the use of pesticides in different crops, production methods and regions. Statistics on pesticide sales are not detailed enough to create the inventories. All available data stem from case studies, which are not representative for the whole of Swiss production.

The quantities of the different pesticides were derived from average values for the years 1994-96 from BLW *et al.* (1998). The experts judged this to be the best available data source specifying pesticide application in sufficient detail. The three most important active substances were selected from the category of herbicides, and the two most important from the category of fungicides. Only the most important active ingredient was taken from each of the pesticide categories “insecticides”, “molluscicides” and “growth regulators”. To take account of the pesticides that were not considered, the quantities of the active ingredient in each category were adjusted so that the total quantity remained constant. A check was carried out to determine whether all of these active ingredients were still in use in the year 2005. If not, the ingredients in question were removed and the quantities of the remaining pesticides adjusted to keep the total quantity of active ingredients constant.

The pesticide inventories listed in Tab. 10.3 were used to calculate pesticide production. Where necessary, organic pesticides were approximated with inventories of the same chemical class or group, according to the procedure described in 10.2 on page 103.

Copper-containing inorganic pesticides were approximated by the inventory for “copper oxide, at plant, RER”.

Trichogramma are used in maize (mainly in grain maize) to control the corn borer. *Trichogramma* are reared on the eggs of the flour moth. The electricity consumption for rearing the flour moths and the *Trichogramma* was estimated from data given by a *Trichogramma* provider. The details are confidential and cannot be given in this report. The applications of *Trichogramma* were taken from LBL *et al.* (2000).

14.2.7 Seed

For seeds, the corresponding seed inventories of each crop were used, where available. Organically grown seeds (where available) were used for crops from organic production, whilst seeds from IP production were taken for crops from both extensive and IP production. In the event that no seed inventory was available for a crop in ecoinvent data, the following approximations were made:

- green manure IP and organic⁷⁵ → rape seed IP, at regional storehouse CH,
- sunflower IP → rape seed IP, at regional storehouse CH,
- faba beans, soy beans IP/organic → pea seed IP/organic, at regional storehouse CH,
- fodder beet IP → sugar-beet seed IP, at regional storehouse CH.

The quantity of seed was taken from LBL *et al.* (2000).

14.2.8 Transports of Inputs to the Farm

The transport of inputs from the regional storehouse to the farm in Switzerland were included. The different inputs for which transport was added are listed in Tab. 14.8, along with the estimated transport distances and carriers. The sum of the transport for each transport carrier was arrived at by

⁷⁵ It was assumed that untreated IP seed was used for organic production. This exception is permitted in organic farming, if there is not enough organic seed available on the market.

multiplying the masses of the inputs by the corresponding transport distances for each transport carrier.

Pesticides and fertilisers were converted into the product weight in order to calculate the requirements for transport in tkm. For pesticides, a mean active-ingredient content of 50% was used. This mean value represents the average active-ingredient content of the pesticides authorised in 2000 (FAW & BLW 2000). For phosphate rock, the product weight was calculated based on a P_2O_5 content of 32%. For N-, P- and K fertilisers, the average nutrient contents from Tab. 8.2 were used.

The transport distances for N-, P-, and K fertilisers were provided by Landor GmbH⁷⁶ (see Tab. 8.6). For the transport distances of phosphate rock, see chapter 8.2.1 ("Transport" section). For the transport of pesticides and seeds, distance from the regional storehouse to the farm was estimated at 15 km. Most of the auxiliaries are sold by the agricultural cooperatives (<http://www.landi.ch>). As there are 450 sale points distributed throughout Switzerland (<http://www.landischweiz.ch/>), the distance from regional storehouse to farm is unlikely to be very great.

Tab. 14.8 Supplementary transport considered for crop production.

Inventories used as input for crop production	Transport carrier	transport, transoceanic freight ship	transport, barge	transport, lorry	transport, freight, rail	transport, lorry	transport, van <3.5t
	Location	OCE	RER	RER	CH	CH	CH
	Transport distance added						
Seed, at regional storehouse CH	to user CH						15
N-fertilisers, at regional storehouse RER	to user CH		900		100	100	
P-fertilisers, at regional storehouse RER	to user CH		400		100	100	
K-fertilisers, at regional storehouse RER	to user CH		100		100	100	
Phosphate rock/phosphoric acid, at plant MA	to regional storehouse RER	2500		400			
Pesticides, at regional storehouse CH	to user CH						15

14.2.9 Green Manure

With integrated and organic production, a soil cover during winter is generally required to reduce the risk of soil erosion and nitrate leaching. For autumn-sown crops this soil cover is to a certain extent provided by the crop itself. For spring-sown crops (all arable crops in ecoinvent except the cereals and rape seed), a green manure was included in the inventories to take account of the winter period, thereby ensuring that the period considered is approximately the same for all crops. This means that machine usage, any nitrogen applications (for integrated production) and field emissions during the winter are included in the inventories for the spring-sown crops. The end date for green manure was chosen on the basis of the date of soil cultivation of the spring-sown crop in question. As these dates differ between the various spring-sown crops, several inventories for green manure with different end dates were included in ecoinvent. Tab. 14.9 shows which green manure inventory was used for the

⁷⁶ Personal communication from C. Kopp, Landor GmbH, 6 February 2001.

different spring sown-crops. As the green manure has no harvested product, its reference function is 1 ha cultivated land. Each module for a spring-sown crop uses the fraction of 1 ha of green manure corresponding to the area required to produce 1 kg of product.

Tab. 14.9 Green manure inventories used for spring-sown crops.

Name	Location	Unit	Spring sown crops for which green manure inventory was used
green manure IP, until January	CH	ha	protein peas, IP, at farm fava beans IP, at farm
green manure organic, until January	CH	ha	protein peas, organic, at farm fava beans organic, at farm
green manure IP, until February	CH	ha	sugar beets IP, at farm fodder beets IP, at farm
green manure IP, until March	CH	ha	potatoes IP, at farm soy beans IP, at farm
green manure organic, until March	CH	ha	potatoes organic, at farm soy beans organic, at farm
green manure IP, until April	CH	ha	grain maize IP, at farm silage maize IP, at farm sunflower IP, at farm
green manure organic, until April	CH	ha	grain maize organic, at farm silage maize organic, at farm

The inventories in ecoinvent are for overwintering green manure established by mid-August, with no biological nitrogen fixation capability (e.g. *Cruciferae*). Inventories for green manure were compiled following the same approach as for crop production. The only difference between integrated and organic production is the application of 30 kg N/ha in the form of mineral fertiliser in the integrated variant (according to Walther *et al.*, 2001, Tab. 2), which is not done in the case of the organic variant (LBL *et al.* 2000, p. 37).

14.2.10 Land Use

Land occupation was calculated from the duration of land use (taking account of the time from soil cultivation until harvest) and the yield per area unit (see chapter 4.2.1). Land occupation by green manure was derived using the period from the time of seeding until the end of the month specified in each green manure inventory. The land occupied was always considered as “Occupation, arable, non-irrigated”, since the land was assumed not to be irrigated (see chapter 4.3).

Land transformation was calculated on the basis of the area required to produce 1 kg of product. The type of use before establishment of the crop was assumed to be 71% arable land and 29% meadow (sown on arable land) for all winter crops. These percentages correspond to the proportions of arable crops and leys out of the total arable surface in Switzerland (293,000 ha arable crops (71%), 118,000 ha leys (29%), 411,000 ha total (100%) arable surface in 2000), taken from BLW (2001, p. A4). Green manure is not established after meadow or pasture (as this would cause the meadow to assume the function of a green manure), but is always established between two arable crops. Land transformation to green manure was therefore calculated 100% as “Transformation, from arable, non-irrigated”. The spring-sown crops were assumed to follow a green manure. In these cases too, land transformation was calculated 100% as “Transformation, from arable, non-irrigated”.

The categories of land resources included are presented in Tab. 14.10.

Tab. 14.10 Consideration of land requirements affiliated with crop production.

Name	Category	Unit	Method of Compilation
Occupation, arable, non-irrigated	resource	m ² a	Time period from soil cultivation for the crop until harvesting and for the area required to produce 1 kg of product
Transformation, from meadow and pasture, intensive	resource	m ²	Derived from the area required to produce 1 kg of product
Transformation, from arable, non-irrigated	resource	m ²	
Transformation, to arable, non-irrigated	resource	m ²	

14.2.11 Direct Field Emissions

Direct field emissions were calculated using emission models (described in the chapter 4.4), the results of which were included in the inventories.

In addition, all pesticides applied for crop production were assumed to end up as emissions to the soil. The amounts of pesticides used as inputs were thus simultaneously calculated as outputs (emissions to agricultural soil). The substances specified in the inventories were used as references to correlate the corresponding emissions. Only for the inputs “pesticides, unspecified”, “fungicides, unspecified” and “insecticides, unspecified”, could no corresponding flow be assigned. Field emissions resulting from these admittedly small quantities of substances were thus not considered.

14.2.12 Straw Inventories

ecoinvent data include detailed straw inventories for the different production methods of wheat, rye and barley (see Tab. 14.1). The inventories “straw IP, at farm” and “straw organic, at farm” calculate a production mix for straw in Swiss agriculture. They are intended for those wishing to use average inventories for straw in their studies, without having detailed information about the exact production method of the cereals, which is the normal situation. Integrated and organic straw are differentiated, since they are marketed through separate channels.

General straw inventories were compiled by combining the straw inventories for wheat, barley and rye based on their relative importance over the total agricultural surface area designated for these crops in Switzerland in the year 2000 (BLW 2001, p. A4 & A40).

Since not all cereal crops were included in ecoinvent, the following assignments were applied:

- wheat/rye mix for bread production was calculated as wheat,
- spelt and triticale were calculated as rye,
- oats and mixes for fodder production were calculated as barley.

Tab. 14.11 Agricultural areas for wheat, barley and rye production in Switzerland in the year 2000 (BLW 2001, p. A4 & A40). Derivation of allocation factors for the general straw inventories.

Crop	Agricultural area in Switzerland in 2000 (ha)	Percentage on total area for cereals	Percentage of extensive production
Wheat (including wheat/rye mix for bread production)	94,150	59%	42%
Rye (including spelt and triticale)	5,110	3%	
Barley (including oats and wheat/rye mix for fodder production)	61,405	38%	63%
Total	178,194	100%	-

Two general straw inventories were created:

- straw from organic production (straw organic, at farm CH)
- straw from integrated production (straw IP, at farm CH). This inventory summarises the straw inventories from both IP and extensive production (the latter of which is in fact also a type of integrated production). The percentage of extensive production in 2000 was 42% of total production for cereals intended for human nutrition, and 63% of total production for cereals intended for feedstuff (BLW 2001, see Tab. 14.11). These percentages were used to calculate the production mix for cereal straw.

14.3 Data Quality Considerations

Fertiliser data come from recommendation and import statistics. As these data are derived from the nutrient content of the exported products and numerous field trials, their reliability is scored as 2 (Tab. 14.12).

The pesticide data were derived from a pilot farm network, and are of limited representativeness. As the data were collected about ten years before the reference year of 2005, temporal correlation is scored as 3.

Data on machine usage, seed quantity, grain drying and land use are scored as 2, since they stem from the calculation bases of the extension services.

Data on machine usage, seed quantity and grain drying and land use are scored as 2, since they stem from the calculation bases of the extension services.

Tab. 14.12 Uncertainty judgement for the data on arable crops.

Category	Reliability	Completeness	Temporal correlation	Geographical correlation	Further technological correlation
Fertilisers and fertiliser spreading	2	1	1	1	1
Pesticide usage, pesticide applications and pesticide emissions	2	2	3	1	1
Other machine usage, seed quantity, grain drying, land use	2	1	1	1	1
Transport processes	4	1	1	1	1
Field emissions	2	2	1	1	1

15 Arable Crop Production in the EU

In this chapter, 12 datasets for selected agricultural products at farm are described. The data were collected in a concerted action of the EU (GL-Pro, European extension network for the development of grain legume production in the EU, 5th framework programme EU, QLK5-CT-2002-02418). The goal of the project was to study the economic and environmental effects of introducing grain legumes (peas, beans) into European crop rotations (von Richthofen *et al.* 2006, Nemecek & Baumgartner 2006)

The methodology described in chapter 4 (System Characterisation) is also applicable to this chapter, except when stated otherwise.

Tab. 15.1 gives an overview of the agricultural land usage in the three considered countries France, Germany and Spain as well as for the whole EU-25.

Tab. 15.1 Agricultural land usage in the year 2003 (FAOSTAT 2006) in the three investigated countries and in EU-25. EU-25 refers to the 25 member countries in 2004.

Area in 1000 ha 2003	France	Germany	Spain	EU-25
Total permanent crops	1122	213	4977	
Total arable land	18451	11827	13738	
Total permanent pastures	10117	4968	11470	
Total agricultural land	29690	17008	30185	164367
Wheat	4876	2967	2152	19239
Barley	1758	2087	3170	12935
Total cereals (without maize)	7184	6394	5966	46068
Maize	1685	473	480	6486
Rape (and mustard) seeds	1083	1272	4	4499
Sunflower	694	38	787	2196
Peas	399	145	113	1934
Beans	124	24	71	
Potatoes	157	284	101	2174
Sugar beet	400	445	100	2204

15.1 Characteristics of the Investigated Production Regions

The data were collected for the regions Barrois (FR), Castilla y Leon (SP) and Saxony Anhalt (DE). In the GL-Pro project data were collected for five other regions as well, but these are not included in ecoinvent data and therefore not treated here. Following information about Barrois, Saxony-Anhalt and Castilla-y-Leon is extracted from von Richthofen *et al.* (2006). Tab. 15.2 gives a summary of climate and soil data in these three regions.

Tab. 15.2 Characteristics of the study regions and share of grain legumes in arable land.

	Annual average temperature (°C)	Annual average rainfall (mm)	Soil
Castilla y Leon (ES)	11	400-500	Calcareous clay
Barrois (FR)	10	730	Calcareous clay
Saxony-Anhalt (DE)	9	850	Loam

Crops normally are planted in specific crop rotations. According to the goal of the project GL-Pro two crop rotations were defined in each region: a typical crop rotation without grain legumes (crop rotation 1) and an alternative crop rotation with grain legumes (crop rotation 2). Depending on the position in the crop rotation, respectively the preceding crops, the yield and the inputs in terms of fertilisers, pesticides and machinery work vary and therefore the life cycle inventory results will also be different. Tab. 15.3 shows the considered crop rotations out of the GL-Pro concerted action and the crops that were included in this database. As crop rotation 1 is considered as typical for the region, all crops – except the grain legumes – were taken from crop rotation 1. In cases where the same crop occurred several times in the crop rotation an average of these crops was calculated (wheat in Saxony-Anhalt and Barrois, barley in Castilla y León). In the case of wheat in Saxony-Anhalt, only the average of the first two wheat crops was calculated, as a sequence of three times wheat was not considered to be common.

Tab. 15.3 Overview of the crop rotations used in this database (Nemecek & Baumgartner 2006). OSR = oilseed rape, W = winter wheat, wB = winter barley, sB = spring barley, P = spring peas, wP = winter peas, SF = sunflower. The boxes show the crops considered here; where two crops are in a box, the average of those was calculated.

Region	Crop rotation 1	Crop rotation 2
Saxony-Anhalt (D)	OSR-W-W-W-wB	OSR-W-P-W-wB
Barrois (F)	OSR-W-W-wB	OSR-W-wP-W-wB
Castilla y León (E)	SF-W-wB-sB	P-W-wB-sB

15.1.1 Barrois (France)

Barrois is a calcareous plateau in the Northeast of France with continental climate tendency. The annual average rainfall is around 800 to 950 mm. Soils have low depth (15 to 60 cm) and high permeability, i.e. they are susceptible to drought. 61% of farms in Barrois are specialised in arable crops, among which 26% have additionally herbivore livestock. The most important crops in arable land are wheat (33%), barley (22%) and rapeseed (22%). Grain legumes have only a share of 1.4% in arable land; the biggest parts of them are spring peas.

In farmers' fields the three years crop sequence winter oilseed rape – winter wheat – winter barley is dominating. This rotation with 33% rapeseed and 67% cereals was therefore chosen as standard rotation for the comparative analysis of crop rotations.

In addition the four-year rotation winter oilseed rape – winter wheat – winter wheat – winter barley (25% rapeseed, 75% cereals) was part of the analysis.

These rotations were compared with the five -year grain legume rotation winter oilseed rape – winter wheat – winter peas – winter wheat – winter barley (20% rapeseed, 60% cereals). Winter peas are selected, since the advice for farmers in Barrois is to choose winter peas as well adapted grain legume species.

15.1.2 Saxony-Anhalt (Germany)

In Saxony-Anhalt, situated in the East of Germany, grain legumes have the highest share in arable land in Germany. About 3.3 % (32,700 ha) of the arable land were covered in 2005 with grain legumes, compared to 1.4% (168,7000 ha) on the German average. The most important grain legume crops in Saxony-Anhalt are field peas, mainly cultivated in the fertile blackearth soils of the Magdeburger Börde.

The production, however, is limited by the continental climate with less than 500 mm of rainfall on the annual average and the risk of early summer drought, leading to higher yield fluctuations. In 2000 and 2003 grain yields were affected by very dry periods in spring. The low yields in 2002 were by contrast due to wet harvest conditions and the flood catastrophe at the river Elbe.

15.1.3 Castilla y Leon (Spain)

Castilla y Leon is situated in the northern centre of Spain. The climate is semiarid with 400 to 500 mm annual average rainfall and 11°C annual average temperature. About 4% of the arable land is cropped with grain legumes. In 2004 about 150,000 ha of grain legumes were cultivated in this region of Spain, of those about 60,000 ha of common vetch and nearly 60,000 ha of field peas. The pea acreage strongly increased since 2000, starting from approximately 15,000 ha.

15.2 Characteristics

15.2.1 Overview of the Life Cycle Inventories

Inventories for the cereals are for their respective winter form except for barley in Castilla y Leon (SP), where the average of the summer and winter form was calculated. The production inventories, i.e. agronomical-technical description of the cropping systems, were taken from the common data collection of the economic and environmental analyses in the GL-Pro project. The data represent the common practices in each region according to local experts:

- Castilla y Leon (Spain):
Pierre Casta, Instituto Tecnológico Agrario de Castilla y Leon (ITA), Valladolid
- Barrois and Picardie (France):
Gaëtan Dubois, UNIP, Paris
- Saxony-Anhalt (Germany):
J.-S. v. Richthofen, proPlant GmbH, Muenster
H. Pahl, Technical University of Munich (TUM), Freising-Weihenstephan

The datasets are described in von Richthofen *et al.* (2006).

Although the inventories for arable crops are based on data from past years, they were corrected so as to be representative for the current situation (for details see below).

The reference function of all arable crop inventories is 1 kg fresh matter. If grain drying was necessary, the datasets refer to the product after drying. Tab. 15.4 summarises the main characteristics of the inventories described in this chapter.

Tab. 15.4 Main characteristics of the inventories of rape seed, wheat, barley, peas and sunflowers in the three regions.

	Yield main product (kg/ha)	Moisture at harvest (%)	Moisture at storage (%)	kg seed/ha	Date of tillage	Date of sowing	Date of harvest	kg N/ha (available N)	kg P ₂ O ₅ /ha	kg K ₂ O/ha	Number of pesticide applications
rape seed conventional, Barrois, at farm, FR	3020	10%	10%	3.0	10.08	25.08	15.07	177	75	74	7.7
wheat grains conventional, Barrois, at farm, FR	6753	14.5%	14.5%	158	15.09	10.10	25.07	154	49	45	6.9
barley grains conventional, Barrois, at farm, FR	6760	14%	14%	141	15.09	25.09	05.07	164	79	22	5.5
protein peas conventional, Barrois, at farm, FR	4040	14%	14%	190	05.10	25.10	15.07	0	43	52	8.0
sunflower conventional, Castilla-y-Leon, at farm, ES	1032	6%	6%	3.3	15.12	15.04	01.10	0	0	1	2.0
wheat grains conventional, Castilla-y-Leon, at farm, ES	3049	15%	15%	175	01.11	15.10	15.07	80	72	111	1.0
barley grains conventional, Castilla-y-Leon, at farm, ES	2774	15%	15%	206	01.11	15.11	15.07	75	72	26	2.0
protein peas conventional, Castilla-y-Leon, at farm, ES	1202	13%	13%	220	15.12	15.02	30.06	0	0	1	3.0
rape seed conventional, Saxony-Anhalt, at farm, DE	3500	10%	9%	3.5	15.08	15.08	15.07	83	60	64	5.0
wheat grains conventional, Saxony-Anhalt, at farm, DE	7567	16%	14.5%	180	01.10	15.10	15.08	187	51	46	5.0
barley grains conventional, Saxony-Anhalt, at farm, DE	7500	14%	14%	160	01.09	15.09	31.07	135	45	29	4.0
protein peas conventional, Saxony-Anhalt, at farm, DE	3800	16%	14%	235	31.10	01.03	31.07	0	60	1	2.0

15.3 Life Cycle Inventories

Life cycle inventories of infrastructure, inputs and processes were taken from the ecoinvent database (Frischknecht *et al.* 2007).

15.3.1 Calculation of weighted averages for certain crops

The yield of a crop depends on the preceding crop and therefore on its position in the crop rotation. E.g. wheat following rape seed or pea has a higher yield than when it follows wheat (see sources summarised in von Richthofen *et al.* 2006). As there were several wheat crops in Saxony-Anhalt and

Barrois (see Tab. 15.3), the average of two wheat crops was used for the calculation of the LCIs. For Saxony-Anhalt we used the first two wheat crops, since a sequence of three wheat crops as assumed in crop rotation 1 is not very common. For Castilla-Leon we calculated the average of the two barley crops.

In these cases, where the average of two crops is calculated (see Tab. 15.3), we used the averages weighted by the yields. The respective weights w_1 and w_2 were calculated as follows:

$$w_1 = Y_1/(Y_1 + Y_2)$$

$$w_2 = Y_2/(Y_1 + Y_2)$$

where Y_1 and Y_2 are the yields of the first and second crop.

15.3.2 Yields

The data sources for yield are taken from the GL-Pro project (von Richthofen *et al.* 2006, Nemecek & Baumgartner 2006). Tab. 15.5 shows the average yields from 2000 to 2004 in the three areas. The yields used in this inventory sometimes differ from the average because the weighted averages of the yields corrected by their position in the crop rotation were used. The following sources were used: Statistisches Landesamt Sachsen-Anhalt (2004) for Saxony-Anhalt, M.A.P.A. (2005) for Castilla-y-Leon and OCERA (2004) for Barrois.

Tab. 15.5 Average yields and yields used in the inventories (kg/ha fresh weight, with moisture at storage).

	2000	2001	2002	2003	2004	ave- rage	Used in inventory
rape seed conventional, Barrois, at farm, FR	3400	2350	2900	2900	3550	3020	3020
wheat grains conventional, Barrois, at farm, FR	7600	6900	7100	5600	7800	7000	6753
barley grains conventional, Barrois, at farm, FR	7400	6500	7000	5700	7200	6760	6760
protein peas conventional, Barrois, at farm, FR	4600	4100	4300	2500	4700	4040	4040
sunflower conventional, Castilla-y-Leon, at farm, ES	1140	1100	910	1000	1010	1032	1032
wheat grains conventional, Castilla-y-Leon, at farm, ES	3751	2091	2918	3067	3420	3049	3049
barley grains conventional, Castilla-y-Leon, at farm, ES	3758	1634	2407	2773	3300	2774	2774
protein peas conventional, Castilla-y-Leon, at farm, ES	1331	654	856	1671	1501	1202	1202
rape seed conventional, Saxony-Anhalt, at farm, DE	3250	3730	2790	3330	4250	3470	3500
wheat grains conventional, Saxony-Anhalt, at farm, DE	7470	7900	6440	6830	8430	7414	7567
barley grains conventional, Saxony-Anhalt, at farm, DE	7700	8490	7170	6170	7750	7456	7500
protein peas conventional, Saxony-Anhalt, at farm, DE	3970	4000	3170	3930	4110	3836	3800

15.3.3 Co-products and Crop Residues

Part of the nutrients in the fertilisers is exported by the products and another part remains in the crop residues and is restored to the soil. These nutrients are fully or partly available to the following crop and therefore the fertiliser rate applied to the following crop can be reduced. In the same way the crop receives nutrients from the preceding crop, which reduces its fertiliser requirements (see Tab. 15.6, fertiliser rate applied by the farmer). Since we are dealing with individual crops and not the whole crop rotation, the fertiliser effect of crop residues was attributed to the crop causing it for the

calculation of the LCI (see Tab. 15.6, fertiliser rate used in the inventory). These two fertiliser rates differ for individual crops, but are equal, when the sum over the whole crop rotation is calculated. The quantities of nutrients in the crop residues (CR_i) were calculated from Walther *et al.* (2001, Tab. 2).

Tab. 15.6 Procedure for calculating the nutrients in the crop residues (example for the crop rotation with four crops).
NR = nutrient requirement, CR = nutrients in crop residues, FR = fertiliser rate.

	Crop1	Crop2	Crop3	Crop4
Nutrient requirement	NR1	NR2	NR3	NR4
Nutrients in crop residues	CR1	CR2	CR3	CR4
Fertiliser rate applied by the farmer	$FR1=NR1-CR4$	$FR2=NR2-CR1$	$FR3=NR3-CR2$	$FR4=NR4-CR3$
Fertiliser rate used in the inventory	$FR1=NR1-CR1$	$FR2=NR2-CR2$	$FR3=NR3-CR3$	$FR4=NR4-CR4$

15.3.4 Allocations

Since no co-products result from the systems investigated, no allocation for co-products is required. Straw is not harvested, in line with the assumption that the farm has no livestock. The restitution of nutrients from incorporated crop residues is considered according to Tab. 15.6.

15.3.5 Fertilisers

Only mineral fertilisers were considered in the GL-Pro project, assuming a stockless farm. The amounts of the principal nutrients N, P and K (see Tab. 15.7) were calculated according to the GL-Pro data (von Richthofen *et al.* 2006, Nemecek & Baumgartner 2006).

Tab. 15.7 Quantity and type of fertilisers used in the inventories (in kg/ha of the respective nutrient, as specified by the name of the fertiliser inventories).

	ammonium nitrate, as N, at regional storehouse	urea, as N, at regional storehouse	calcium ammonium nitrate, as N, at regional storehouse	ammonium sulphate, as N, at regional storehouse	triple superphosphate, as P ₂ O ₅ , at regional storehouse	potassium chloride, as K ₂ O, at regional storehouse	limestone, milled, loose, at plant
rape seed conventional, Barrois, at farm, FR	104.5	60.5	0.0	11.4	77.1	97.3	27.7
wheat grains conventional, Barrois, at farm, FR	101.5	81.4	0.0	15.1	71.9	42.2	12.6
barley grains conventional, Barrois, at farm, FR	74.6	73.5	0.0	15.7	74.1	0.0	12.6
protein peas conventional, Barrois, at farm, FR	0.0	0.0	0.0	0.0	46.8	137.4	24.1
sunflower conventional, Castilla-y-Leon, at farm, ES	0.0	0.0	0.0	0.0	0.0	0.0	0.0
wheat grains conventional, Castilla-y-Leon, at farm, ES	79.3	0.0	0.0	0.0	72.0	108.5	0.0
barley grains conventional, Castilla-y-Leon, at farm, ES	75.0	0.0	0.0	0.0	72.0	24.6	0.0
protein peas conventional, Castilla-y-Leon, at farm, ES	0.0	0.0	0.0	0.0	0.0	0.0	0.0
rape seed conventional, Saxony-Anhalt, at farm, DE	54.6	0.0	45.5	0.0	49.4	87.4	0.0
wheat grains conventional, Saxony-Anhalt, at farm, DE	0.0	0.0	173.9	0.0	56.2	35.8	0.0
barley grains conventional, Saxony-Anhalt, at farm, DE	0.0	0.0	142.6	0.0	44.6	6.5	0.0
protein peas conventional, Saxony-Anhalt, at farm, DE	0.0	0.0	0.0	0.0	44.3	0.0	0.0

15.3.6 Machine usage

The machine usage for crop production was derived from the GL-Pro project (von Richthofen *et al.* 2006, Nemecek & Baumgartner 2006). Tab. 15.8 shows the machine usage for the 12 crops.

Tab. 15.8 Machine-use inventories used to calculate the arable-crop production inventories (unit = number of passes)

	tillage, cultivating, chiselling	tillage, harrowing, by rotary harrow	tillage, harrowing, by spring tine harrow	tillage, ploughing	tillage, rolling	fertilising, by broadcaster	combine harvesting	application of plant protection products, by field sprayer	sowing
rape seed conventional, Barrois, at farm, FR		0.50	2.00	0.50		2.50	1.00	7.65	1.00
wheat grains conventional, Barrois, at farm, FR		0.48	1.74	0.52		3.50	1.00	6.86	1.00
barley grains conventional, Barrois, at farm, FR		0.20	1.60	0.80		2.50	1.00	5.50	1.00
protein peas conventional, Barrois, at farm, FR		0.65	1.82	0.35			1.00	7.95	1.00
sunflower conventional, Castilla-y-Leon, at farm, ES	1.00			1.00				2.00	1.00
wheat grains conventional, Castilla-y-Leon, at farm, ES	1.00			1.00	1.00	2.00		1.00	1.00
barley grains conventional, Castilla-y-Leon, at farm, ES	1.00			1.00	1.00	2.00		2.00	1.00
protein peas conventional, Castilla-y-Leon, at farm, ES	1.00			1.00				3.00	1.00
rape seed conventional, Saxony-Anhalt, at farm, DE		4.00		1.00		2.33	1.00	5.00	1.00
wheat grains conventional, Saxony-Anhalt, at farm, DE	0.52	2.96		0.48		3.83	1.00	4.96	1.00
barley grains conventional, Saxony-Anhalt, at farm, DE		4.00		1.00		3.75	1.00	4.00	1.00
protein peas conventional, Saxony-Anhalt, at farm, DE		4.00		1.00		0.67	1.00	2.00	1.00

15.3.7 Pesticides

The quantities of the different pesticides were derived from the GL-Pro project (von Richthofen *et al.* 2006, Nemecek & Baumgartner 2006). The pesticide inventories listed in Tab. 15.9 were used to calculate pesticide production.

Tab. 15.9 Pesticides used for the 12 crops (kg/ha).

	rape seed conventional, Barrois, at farm, FR	wheat grains conventional, Barrois, at farm, FR	barley grains conventional, Barrois, at farm, FR	protein peas conventional, Barrois, at farm, FR	sunflower conventional, Castilla-y-Leon, at farm, ES	wheat grains conventional, Castilla-y-Leon, at farm, ES	barley grains conventional, Castilla-y-Leon, at farm, ES	protein peas conventional, Castilla-y-Leon, at farm, ES	rape seed conventional, Saxony-Anhalt, at farm, DE	wheat grains conventional, Saxony-Anhalt, at farm, DE	barley grains conventional, Saxony-Anhalt, at farm, DE	protein peas conventional, Saxony-Anhalt, at farm, DE
2,4-MCPA (kg)		0.282	0.016									
2-4 D (kg)							0.206					
Aclonifen (kg)				0.600								
Alacloro (kg)					0.900							
Anthraquinone (kg)		0.053	0.070									
Azoxystrobine (kg)		0.042	0.076							0.125		
Bentazon (kg)				0.870								0.480
Bifenox (kg)		0.036	0.018									
Bitertanol (kg)		0.001	0.008									
Bromoxynil (kg)		0.101	0.045									
Bromuconazole (kg)		0.002										
Carbendazim (kg)	0.241											
Carfentrazone-ethyle (kg)		0.000	0.001									
Chloridazon (kg)										0.375		
Chlormequat (CCC, kg)	0.120	0.658	0.002							0.789		
Chlorothalonil (kg)				1.000								
Chlortoluron (kg)			0.108									
Choline chloride (kg)		0.458										
Cicloxidim (kg)								0.120				
Clodinafop-Propargyl (kg)		0.047										
Clomazon (kg)	0.054											
Clomazone (kg)								0.083				
Clopyralid (kg)		0.029	0.002									
Cloquintocet-mexyl (kg)		0.012										
Cypermethrin (kg)	0.028	0.008	0.005						0.010			
Cyproconazol (kg)		0.005	0.003							0.026		
Cyprodinil (kg)		0.179	0.214							0.192		
Deltamethrin (kg)	0.006	0.002	0.001									
Diclofop 24% (kg)						0.600	0.204					

	rape seed conventional, Barrois, at farm, FR	wheat grains conventional, Barrois, at farm, FR	barley grains conventional, Barrois, at farm, FR	protein peas conventional, Barrois, at farm, FR	sunflower conventional, Castilla-y-Leon, at farm, ES	wheat grains conventional, Castilla-y-Leon, at farm, ES	barley grains conventional, Castilla-y-Leon, at farm, ES	protein peas conventional, Castilla-y-Leon, at farm, ES	rape seed conventional, Saxony-Anhalt, at farm, DE	wheat grains conventional, Saxony-Anhalt, at farm, DE	barley grains conventional, Saxony-Anhalt, at farm, DE	protein peas conventional, Saxony-Anhalt, at farm, DE
Diclofop-méthyl (kg)		0.075	0.112									
Difenoconazol (kg)	0.013											
Diffufenican (kg)		0.042	0.023							0.063	0.063	
Dimetazachlore (kg)	0.352											
Epoxiconazole (kg)		0.028	0.015									
Ethephon (kg)		0.002	0.193									
Fenoxapron etil 2% (kg)					0.050	0.017						
Fenoxaprop-P-ethyl (kg)			0.005									
Fenpropidin (kg)		0.012								0.252	0.225	
Fenpropimorph (kg)		0.029	0.036									
Florasulam (kg)		0.000									0.004	
Fluazifop-P-Butyl (kg)									0.080			
Fludioxonil (kg)		0.005	0.004									
Flufenacet (kg)										0.150	0.150	
Flupyrsulphuron-méthyle (kg)		0.000										
Fluquinconazol (kg)										0.065		
Fluroxypyr (kg)		0.135	0.003									
Flurtamone (kg)										0.156	0.156	
Flusilazole (kg)			0.009									
Fuberidazol (kg)											0.006	
Glyphosate (kg)		0.360		0.216								
Imazalil (kg)											0.006	
Imidaclopride (kg)		0.004	0.035									
Iodosulfuron-methyl-sodium (kg)		0.000										
Ioxynil (kg)		0.134	0.027									
Iprodione (kg)	0.158											
Isoproturon (kg)		0.174	0.470									
kresoxim-methyl (kg)		0.008	0.001									
Lambda-cyhalothrin (kg)	0.006	0.002	0.001	0.006					0.008	0.008	0.008	0.008
Linuron (kg)					0.450			0.300				
MCPA (kg)							0.413					
Mecoprop (kg)		0.190										
Mecoprop-P (kg)		0.047										
Mefenpir 4% (kg)						0.100	0.034					
mefenpyr-diethyl (kg)			0.010									
mepiquat-chlorure (kg)		0.004	0.073									

	rape seed conventional, Barrois, at farm, FR	wheat grains conventional, Barrois, at farm, FR	barley grains conventional, Barrois, at farm, FR	protein peas conventional, Barrois, at farm, FR	sunflower conventional, Castilla-y-Leon, at farm, ES	wheat grains conventional, Castilla-y-Leon, at farm, ES	barley grains conventional, Castilla-y-Leon, at farm, ES	protein peas conventional, Castilla-y-Leon, at farm, ES	rape seed conventional, Saxony-Anhalt, at farm, DE	wheat grains conventional, Saxony-Anhalt, at farm, DE	barley grains conventional, Saxony-Anhalt, at farm, DE	protein peas conventional, Saxony-Anhalt, at farm, DE
mesosulfuron-methyl-sodium (kg)		0.001										
Metalaxyl-M (kg)									0.002			
Metaldehyd (kg)	0.255	0.150	0.240	0.066								
Metazachlore (kg)	0.156								0.625			
Metconazol (kg)	0.002	0.003							0.030			
Metosulam (kg)		0.001										
metsulfuron methyle (kg)		0.003										
Metsulfuron-Methyl (kg)		0.003	0.001									
Napropamid (kg)	0.404											
Natrium-Salz (kg)												0.524
Oxifluorfen (kg)					0.180							
Pendimethalin (kg)												0.600
Picoxystrobine (kg)		0.003	0.025							0.036		
Prochloraz (kg)		0.010								0.012		
Procymidone (kg)	0.056											
prohexadione-calcium (kg)		0.000										
Propiconazol (kg)		0.012	0.022							0.080	0.063	
propoxycarbazone sodium (kg)		0.001										
Propyzamide (kg)				0.200								
Prothioconazol (kg)									0.125			
Pyraclostrobine (kg)		0.027	0.009									
Quinoxifen (kg)		0.012										
quizalofop-ethyl (kg)	0.007											
Silthiofam (kg)		0.019										
Spiroxamine (kg)		0.007								0.130		
Tebuconazole (kg)	0.021	0.020	0.016						0.226	0.102		
Tefluthrine (kg)		0.001										
Thiram (kg)												0.423
Tralkoxidim (kg)							0.163					
Triadimenol (kg)		0.003									0.048	
Tribenuron (kg)						0.018	0.008					
Trifloxystrobine (kg)		0.015	0.010									
Trifluralin (kg)	0.922							0.600				
Trinexapac-ethyl (kg)		0.009	0.020							0.022	0.122	
Vinchlozoline (kg)	0.019											

15.3.8 Seed

For seeds, the corresponding ecoinvent seed inventories of each crop were used. Sunflower seeds were approximated with rape seed IP, at regional storehouse CH. The quantities of seeds base on personal communication (GL-Pro partners, see 15.2.1).

15.3.9 Transports

The different inputs for which transport was added are listed in Tab. 6, along with the estimated transport distances and carriers. The sum of the transport for each transport carrier was arrived at by multiplying the mass of the inputs by the corresponding transport distances for each transport carrier.

The transport of fertilisers from regional storehouse to the farm is already included in the fertiliser module. Pesticides and seeds were converted into the product weight according to chapter 14.2.8 in order to calculate the requirements for transport from regional storehouse to the farm in tkm. Transport from farm to field is also already included in the seed, fertiliser and pesticide modules.

Tab. 15.10 Supplementary transport considered for crop production

	Delivery van
	RER, km
Pesticides, at regional storehouse	15
Seeds, at regional storehouse	15

15.3.10 Land use

Land occupation and transformation was calculated according to the procedure described in chapter 14.2.10 with one exception: As the considered crop rotations do not include temporary meadows, the transformation was always assumed to be “from arable, non-irrigated” → “to arable, non-irrigated”.

15.3.11 Direct Field Emissions

Direct field emissions were calculated using emission models (described in the chapter 4.4), the results of which were included in the inventories. The methodology applied is the same as applied in chapter 14.

In addition, all pesticides applied for crop production were assumed to end up as emissions to the soil. The amounts of pesticides used as inputs were thus simultaneously calculated as outputs (emissions to agricultural soil). The substances specified in the inventories were used as references to correlate the corresponding emissions. Only for the inputs “pesticides, unspecified”, “fungicides, unspecified” and “insecticides, unspecified”, could no corresponding flow be assigned. Field emissions resulting from these admittedly small quantities of substances were thus not considered.

For the calculation of nitrate emissions (NO_3^-), the differences in climate in the regions investigated was taken into account. The transformation factor described in chapter 4.4.2 was corrected by the precipitation rate during winter, since leaching and therefore nitrate losses strongly depend on winter precipitations. The transformation factor was therefore multiplied by the rate of precipitations during the winter months (October to March) from the considered region and the site Reckenholz, where the SALCA nitrate model has been calibrated (Richner et al. 2006).

Tab. 15.11 Winter precipitations and nitrate leaching transformation factors for the different regions

	Winter precipitations (October-March) in mm	Nitrate leaching transformation factor
Swiss lowlands (site Reckenholz)	433	0.80
Barrois	381	0.70
Castilla-y-Leon	266	0.49
Saxony-Anhalt	183	0.34

15.3.12 Data Quality Considerations

The LCI for the 12 crops described in this chapter base on a survey done in the concerted action GL-Pro. Since the data collection was carried out by national partners for France, Germany and Spain and furthermore the data have been validated by local experts, they can be considered quite reliable and representative for the actual situation.

Yield data were taken from national and regional statistics. They are representative for an average situation, but cannot represent the deviations in individual cases.

Fertiliser requirements are derived from the nutrient needs of the crops, by procedures recommended by the local agricultural extension services. They can therefore be considered as quite reliable.

The machinery usage is derived from recommendation, expert knowledge, databases and surveys of farmers. The data were linked to the dataset for the machinery work processes for Switzerland in the ecoinvent database.

The scoring of the different aspects of uncertainty was done according to the arable crops in Switzerland (see chapter 14). The only exception is the temporal correlation for pesticide usage, which is set to 1, since recent data are used.

Tab. 15.12 Uncertainty judgement for the data on arable crops

Category	Reliability	Completeness	Temporal correlation	Geographical correlation	Further technol. correlation
Fertiliser and fertiliser spreading	2	1	1	1	1
Pesticide usage, pesticide applications and pesticide emissions	2	2	1	1	1
Other machine usage, seed quantity, grain drying, land use	2	1	1	1	1
Transport processes	4	1	1	1	1
Field emissions	2	2	1	1	1

16 Hay

16.1 Characteristics

Grassland is the most common type of agricultural land in Switzerland (Tab. 3.1), with permanent grassland accounting for the largest share (about 630,000 ha). The hay inventories included in ecoinvent data refer to hay harvested from permanent grassland in the Swiss lowlands.

Soil cultivation and sowing need not be considered, since they are not required for permanent grassland. Nemecek & Huguenin (2002), anyway, have shown that the difference between permanent and temporary grassland (on arable land) is relatively small except for nitrate leaching, where the temporary meadow has a higher risk of leaching during and after its establishment.

16.2 Life Cycle Inventories

Hay inventories for three different production types (Tab. 16.1) were included in ecoinvent.

Tab. 16.1 Hay inventories included in ecoinvent data and their major characteristics. The net yield is the yield after storage (including losses in the field and during conservation and storage).

Name	Unit	Number of harvests	Net yield kg DM/ha	Method of drying	Fertilisation
hay intensive IP, at farm, CH	kg	5	11603	ventilation with cold air	78 m ³ liquid manure (dilution 1:1), 29 kg N (as ammonium nitrate) 27 kg P ₂ O ₅ triple superphosphate
hay intensive organic, at farm, CH	kg	5	9863	ventilation with cold air	68 m ³ cattle slurry (dilution 1:1), 6.8 kg P ₂ O ₅ raw phosphate (hyperphosphate)
hay extensive, at farm, CH	kg	1	2702	field drying	no fertilisation

A schematic description of the system can be found in Fig. 4.2.

The system includes fertilisation for the intensive meadow (liquid manure and mineral fertiliser for the integrated hay, and cattle slurry and raw phosphate for the organic hay). The extensive hay is not fertilised. A chemical treatment against *Rumex* is included in the inventory for the integrated intensive production. No machine use was calculated for the pesticide application, as the treatment is applied manually. In addition, the harvest and preparation of the fodder in the meadow, as well as its transport to the farm and storage in the barn for 12 months (with cold-air ventilation for the intensive hay, and without ventilation for the extensive hay) are included in the inventories. Even if the hay is not stored for the entire period, the barn cannot usually be used for other purposes, and will remain partly empty for several months. This period is also included. The **reference function is 1 kg dry matter** hay after storage in all cases (net yield). Note that this reference function differs from that of the arable crop production products (chapter 14), where all inventories refer to the fresh weight of the products. No allocations were necessary for the compilation of the dried roughage inventories.

In the following sections, the procedure for compiling the hay inventories is described. A detailed description can also be found in Nemecek *et al.* (2005, Appendix 3.1.4).

16.2.1 Yields

The gross yields were taken from Dietl (1989) for “hay, intensive IP” (11603 kg/ha) and “hay extensive” (2702 kg/ha). These figures are based on a long series of experiments and modelling. Since the yields of meadows are not regularly recorded, there are no representative statistics available. This

means that data sources such as FAT (2001) or SBV (2000a) cannot be used. The gross yield of "hay, intensive organic" (9863 kg/ha) was estimated 15% lower than for "hay, intensive IP" according to SBV (2000b).

To calculate the net yield (after losses during conservation and storage, see Tab. 16.1), 10% field losses and 4.5% losses during conservation and storage were deducted for ventilated hay (intensive production), and 20% field losses and 3.5% losses during conservation and storage for field-dried hay (extensive production) (Nemecek *et al.* 2005; Appendix 3.1.4).

16.2.2 Fertilisers

The fertilisation of "hay, intensive IP" was calculated so that the total amount of nutrients corresponded to the dose recommended by Walther *et al.* (2001, Tab. 3). For "hay, intensive organic" the lower availability of fertilisers on the organic farm was taken into account. The extensive variant is not fertilised at all.

Application of mineral fertilisers was only taken into account for "hay, intensive IP, at farm". Farmyard manure application was considered for "hay, intensive IP" and "hay, intensive organic".

16.2.3 Machine Usage

Machine use for the hay inventories was derived from LBL *et al.* (2000, pp. 34-35). The different work processes from LBL *et al.* (2000, pp. 34-35) were assigned to ecoinvent modules as shown in Tab. 16.2.

As in the arable-crop inventories, the number of passes by vacuum tanker, manure spreader and fodder loader as well as the number of bales, were calculated from the corresponding figures in the inventory.

Tab. 16.2 Machine-use inventories used to calculate the hay inventories.

Name of the module in ecoinvent data	Location	Unit	Machine specification according to LBL <i>et al.</i> (2000, p. 34-35)	Comments
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	Misten, Hydrauliklader, 3t-Zetter, pro t	
fertilising, by broadcaster	CH	ha	Schleuderstreuer, bis 500 l	
slurry spreading, by vacuum tanker	CH	m ³	Güllen, 4m ³ -Vakuumfass, pro m ³	
mowing, by rotary mower	CH	ha	Kreiselmäher, 1,6-2 m	
haying, by rotary tedder	CH	ha	Kreiselheuer, 4,6-6 m	
swath, by rotary windrower	CH	ha	Kreiselschwader, 2,8-3,3 m	
fodder loading, by self-loading trailer	CH	m ³	Futter einführen Ladewagen, 20m ³	0.0208 m ³ /kg DM
baling	CH	unit	Hochdruckpresse klein	Fresh weight of 1 bale = 200 kg, correction factor of 0.23 applied ^{*)}
loading bales	CH	unit	Stroh, Heu laden und einführen, Ballenlader	
dried roughage store, cold-air dried, conventional, operation	CH	kg		Referring to the net yield
dried roughage store, non ventilated, operation	CH	kg		1.12 m ³ /kg DM lifetime of the store=50 years
dried roughage store, cold-air dried, conventional	CH	m ³		0.95 m ³ /kg DM lifetime of the store=50 years

^{*)} The process “baling” refers to silage production, which takes much more time (0.13 h/bale) than hay-bale production (0.03 h/bale). The use of polyethylene is also lower for hay bales. A factor of 0.23 (=0.03/0.13) was therefore applied to the number of bales for the process “baling”, but not for “loading bales”.

16.2.4 Pesticides

Pesticide application was taken into account solely for “hay, intensive IP”, and only against *Rumex*. Only the herbicide Asulam was used (0.5 l/ha, 400 g active ingredient/litre i.e. 200 g active matter/ha). *Rumex* is the only weed that is regularly treated in meadows. Other pesticides are usually not applied. It was assumed that Asulam is applied manually to treat individual plants.

16.2.5 Transports

Transport was supplemented for fertilisers and pesticides according to the procedure described in chapter 14.2.8.

16.2.6 Land Use

Land occupation and transformation categories specified in Tab. 16.3 were used for calculating land requirements associated with hay production. As the inventories refer to a period of 12 months, land occupation was related to one year.

It was assumed that type of land occupation does not change, i.e. that intensive meadow and pasture was the same type beforehand, as was extensive meadow and pasture. Permanent grassland has rarely been converted to arable land over the past few years. Although the contrary occurred in some cases, the surface areas are very small compared with the total area of permanent grassland. Intensive grassland has sometimes been converted to extensive grassland over the last few years, owing to the subsidies paid for extensive grassland. Some extensive meadows were also installed on set-aside

arable land. Extensive meadows must be installed for at least six years. No quantitative data are available on the history of extensive meadows, except for in case studies that are not representative of the entire agricultural area of Switzerland. To simplify matters, it was assumed that extensive meadows were extensive meadows before installation. According to ecoinvent quality guidelines, the lifetime of the intensive meadow is fixed at to 20 years, that of the extensive meadow at 50 years.

Tab. 16.3 Land occupation and transformation taken into consideration for hay inventories.

Name	Category	Unit	Land use category used for
Occupation, pasture and meadow, extensive	resource	m ² a	hay extensive
Occupation, pasture and meadow, intensive	resource	m ² a	hay intensive IP/organic
Transformation, from pasture and meadow, extensive Transformation, to pasture and meadow, extensive	resource	m ²	hay extensive
Transformation, from pasture and meadow, intensive Transformation, to pasture and meadow, intensive	resource	m ²	hay intensive IP/organic

16.2.7 Emissions

The calculation models for direct emissions are described in chapter 4.4.

16.3 Data Quality Considerations

The same uncertainty scoring as for arable crops is applied (see Tab. 14.12).

17 Starch

17.1 Characteristics of Starch Production

Starch is an important raw material for the food industry and for other non-food industrial sectors such as paper production. In Germany, the main resource for starch production is potato (67 % by weight of total raw-material processing for starch production), followed by wheat (18 %) and maize (15 %). Total starch production in Germany amounts to roughly 1.5 million tonnes, to which potato-, maize- and wheat starch contribute 40 %, 31 % and 29 %, respectively (Würdinger *et al.* 2003).

17.2 Life Cycle Inventories of Maize and Potato Starch

The unit process data for maize- and potato starch were taken from the study by Würdinger *et al.* (2003), based on a detailed literature survey of the production processes for these products. For a detailed description of the methodology used and the data sources, we refer to this study.

To compile the unit process, Würdinger *et al.* (2003) inventoried maize- and potato-starch production processes common in Germany. This study was chosen as a basis for the inventories since it was the most recent one found and compares and discusses different data sources.

a) Maize starch

- transport of the maize grains (14% water content by weight) to the starch plant (distance 100 km);
- mechanical separation of impurities;
- soaking of the maize grains for about 40 hours in 50 °C warm process water;
- milling of the soaked maize and separation of the germ. The germ is dried and pressed into corn oil;
- mechanical separation of starch from other components;
- mechanical desiccation and thermal drying of the extracted starch to obtain a final product with a maximum water content of 14% by weight;
- evaporation of the process water to obtain the by-products of maize gluten and maize-gluten feed.

b) Potato starch

- The potato-starch inventory includes the following manufacturing steps:
- transport of the potatoes (25% dry-matter content by weight⁷⁷) to the starch plant (distance 25 km);
- washing of the potatoes;
- chopping of the potatoes;
- separation of the potato juice, which is subsequently used to produce potato protein. The remaining juice is either evaporated or subjected to waste-water treatment;

⁷⁷ The dry-matter content of the module “potatoes IP, at farm” is 22%, which corresponds to potatoes of medium starch content as used for direct human consumption. Special potato varieties with a high starch content (25% dry matter content) are used for starch production. No such inventories were defined in ecoinvent data. As the agricultural production of these different varieties does not vary greatly, the module “potatoes IP, at farm” is used as an approximation. No adjustment is performed for the dry-matter content.

- washing, refining, and finally, drying of the starch. At the end of the transformation process, the potato starch is left with a water content of 20% by weight.

The data source contains details on the use of energy (electricity and heat), the quantity of tap water used, and the quantity and quality of wastewater produced. For the treatment of the wastewater, two modules were defined, namely “treatment, maize-starch production effluent, to wastewater treatment, class 2” and “treatment, potato-starch production effluent, to wastewater treatment, class 2”. No data were found on emissions to air and waste generation. It is unlikely that significant aerial emissions occur during starch production.

No data were found on the use of infrastructure. “chemical plant, organics, RER” was therefore used as a proxy module.

Würdinger *et al.* (2003) allocated the product “maize starch” and the by-products “corn oil”, “maize gluten” and “maize-gluten feed” on an economic basis, using the market values of these products. Maize starch accounted for 83 % of total earnings. Potato-starch production is allocated 90% to potato starch on the basis of total earnings. The inventories refer to Germany, since the original data were collected for that country.

17.3 Data Quality Considerations

The following limitations associated with starch-production inventories should be borne in mind when using them.

Pedigree Judgement

Reliability was scored as 2, since the verified data were combined with assumptions.

Completeness was scored as 2, since the study covers several important starch producers in Germany.

Temporal, geographical and technological correlations were all scored as 1, since the data come from a recent study, and cover the target area and the technology to be investigated.

Sample size was scored as 3, since the figures are aggregated from various measurements taken in several plants.

18 Tallow

18.1 Characteristics

Only tallow as a by-product of animal production was included in ecoinvent data.

Until the year 2000, meat meal, bone meal and tallow as slaughterhouse waste were used mainly as feedstuffs and as fertilisers in organic farming (rich in N and P). The process for producing meat meal and bonemeal from slaughterhouse waste is called rendering (COWI 2000). Since January 2001, this use has been completely banned in Switzerland due to the potential risk of BSE transmission. The European Union also banned the use of meat- and bonemeal as feedstuffs in January 2001. The same applies to tallow from slaughterhouse waste.

Since 2001, all meat meal, bone meal and tallow from slaughterhouse waste are combusted in cement ovens. Tallow has a net calorific value of 39 MJ/kg and a water content of 0.1-0.4% (Nottrodt 2001). Bone meal cannot be burnt in cement ovens, because this would deteriorate the quality of the cement. Instead, it is burnt together with sewage sludge waste in special ovens.

Switzerland produces 45,000 tonnes of meat meal, 20,000 tonnes of bone meal and 20,000 tonnes of tallow per year (Nottrodt 2001).

18.2 Life Cycle Inventory

Even if these products are no longer used in animal feed, they must still be sterilised to prevent the risk of BSE transmission. The sterilisation process remains the same, i.e. the waste materials are first sterilized, then combusted⁷⁸.

The tallow-production process is described as follows: The animal carcasses and abattoir wastes are crushed in a grinder, then sterilised by steam. They are then dried using steam as a heat source, and finally defatted, a process in which the fat is separated from the greaves. The fat is clarified, while the greaves are milled into meat- and bone meal. The processing paths of tallow and of meat- and bone meal are very similar, diverging only in the final step (milling). The minimum requirements for sterilisation are a temperature of 133 °C and a pressure of 3 bar for 20 minutes (Nottrodt 2001).

The data for Switzerland were taken from the Lyss slaughterhouse waste treatment plant⁷⁸. The process uses natural gas and electricity as energy sources. The data were compared to a study from Denmark (COWI 2000) and were found to be quite similar. The use of heating energy given by COWI (2000) is 6.3 MJ per kg tallow, which is slightly lower than the value for the Lyss plant (8.4 MJ/kg). According to COWI (2000), fuel oil is used as the heat source, while the Lyss plant uses natural gas. Electricity consumption is the same in both cases (0.18 kWh/kg tallow). The data were therefore judged to be sufficiently reliable. In addition, data on freshwater use and wastewater production were available for the Lyss plant.

Transport from the slaughterhouse to the rendering plant is included as 150 km by lorry, taking into account that the Lyss plant transforms approx. 50% of slaughterhouse wastes in Switzerland.

The rendering process produces large quantities of wastewater with high nitrogen content and high BOD values. The wastewater is treated in the plant (sterilisation). Since no detailed data were available on the quality of the wastewater produced, the module "treatment, sewage, to wastewater treatment, class 2, CH" was used as a proxy for the wastewater treatment.

An important emission from rendering is foul odour. Because this category of emission is not considered in ecoinvent data, it is omitted. No indications of other relevant air emissions were found,

⁷⁸ Personal communication from H. Soltermann, GZM Lyss, 17 April 2002.

so no air emissions (except those already included in the combustion of fuels) were added. There were no data available on waste generation.

The use of building infrastructure was estimated from aerial photographs: 3000 m² is occupied by the buildings (estimated height of 10 m, inventoried as “building, multi-storey, RER”), and the total area is 11,000 m², which was inventoried as “industrial area, built up”. The estimated lifetime of the plant was 50 years, and the occupation as construction site was assumed to be 2 years.

The inventory includes only the processing of tallow from slaughterhouse wastes. Animal husbandry, slaughtering and transport to the rendering plant are not included, because rendering must be considered as a waste disposal process with negative economic value (Kleinhanss *et al.* 2000).

The collected data refer to the total production of meat meal, bone meal and tallow in the rendering plant. Physical allocation to the products was carried out. The total production figures were divided by the total output of the plant, since the processing of meat- and bone meal on the one hand and tallow on the other are to a large extent the same. The inventory for tallow can therefore also be used for meat- and bone meal.

18.3 Data Quality Considerations

Reliability was scored as 3, since the values stem from a personal communication. The data cover approximately 50% of the Swiss production of tallow (9,560 tonnes out of about 20,000 tonnes/year). Completeness was scored as 2. Temporal, geographical and technical correlation were all scored as 1, since the data from 1999 are quite recent, come from Switzerland and refer to the technology currently in use. Sample size was scored as 3, as the values refer to statistics covering the whole year.

Part V: Publications

19 Selected Publications

In the following the reader is referred to a few selected publications using ecoinvent data.

Nemecek & Erzinger (2005) have presented the datasets and results of ecoinvent data V1.1. They put special emphasis to the method modelling data of arable crops by using different data sources.

Nemecek *et al.* (2005) have studied Swiss arable cropping and forage production systems by comparing the effects of the farming system (conventional, integrated, organic), the production intensity, grassland management, form and amount of fertilization, choice of arable crop and production region on the environmental impacts. Two new SALCA methods were applied in this study, namely SALCA biodiversity (Jeanneret *et al.* 2006) and SALCA soil quality (Oberholzer *et al.* 2006).

Kägi *et al.* (2007) present an evaluation of the environmental impacts of different biomass productions in Switzerland, by comparing different farming systems (integrated and organic), intensity levels and different crops.

Nemecek *et al.* (2007) evaluated the effect of the introduction of grain legumes into crop rotations in four European regions.

Baumgartner *et al.* (2007) studied the influence of replacement of soya bean meal by European grain legumes in animal feed.

Hölscher *et al.* (2007) compared the environmental impacts of a food with a energy crop rotation as well as with willow, *Miscanthus* and permanent grassland.

Part VI: Appendices and Literature

In the appendices additional information on the inventories as well as a complete listing of all unit-process inventories is given.

In the unit-process inventories, the following EcoSpold-fields are shown:

- Exchange: field 401 (Name)
- Location/Category: fields 662 (Location), 495 (Category) and 496 (SubCategory)
- Unit: field 403 (Unit)
- Value: field 3707 (meanValue)
- UncertType: field 3708 (uncertaintyType) (1 = lognormal)
- SD95%: field 3709 (standardDeviation95)
- UncertScores: field 3792 (generalComment)

The uncertainty scores have the following meaning in the order of the listing:

- Reliability
- Completeness
- Temporal correlation
- Geographical correlation
- Further technological correlation
- Sample size

na = not considered

(I) = infrastructure module

Further information on EcoSpold and the uncertainty assessment can be found in Frischknecht *et al.* (2007).

Appendix A4 to Chapter 4 (System Characterisation)

Tab. A. 1 Heavy-metal contents of plant material (mg/kg dry matter, from Freiermuth 2006).

	Cd	Cu	Zn	Pb	Ni	Cr	Hg
	[mg/kg DM]	[mg/kg DM]	[mg/kg DM]	[mg/kg DM]	[mg/kg DM]	[mg/kg DM]	[mg/kg DM]
<i>Generic mean</i>	0.10	6.6	32.0	0.54	1.04	0.55	0.04
Grass / Hay	0.13	8.6	40	1.2	1.68	1.09	0.15
Grain maize	0.03	2.5	21.5	0.3	1.16	0.32	0
Silage maize	0.1	5	34.5	1.61	0.48	0.7	0.01
Wheat grains	0.1	3.3	21.1	0.2	0.2	0.2	0.01
Wheat straw	0.2	2.5	9.6	0.6	0.6	0.7	
Barley grains	0.03	4.3	26.6	0.2	0.1	0.1	
Barley straw	0.1	4.8	11.1	0.6	0.8	1.2	
Rye straw	0.1	3.2	13	0.4	0.7	0.5	
Potatoes	0.04	6.45	15	0.55	0.33	0.57	0.09
Rape seed	1.6	3.3	48	5.25	2.6	0.5	0.1
Faba beans	0.04	6	30.1	0.87	1.3	0.69	0
Soya beans	0.06	15.1	47.7	0.08	5.32	0.52	0
Protein peas	0.09	10	73	0.16	0.83	0.32	0.01
Sugar beets	0.4	12	36.4	1.16	1.08	1.775	0.095

Tab. A. 2 Heavy-metal contents of mineral fertilisers (mg/kg nutrient) according to Desaulles & Studer (1993). No data available on Hg. Source: Freiermuth (2006).

Mineral fertilisers (%N/%P2O5/%K2O/%Mg)	Cd mg/kg nutrient	Cu mg/kg nutrient	Zn mg/kg nutrient	Pb mg/kg nutrient	Ni mg/kg nutrient	Cr mg/kg nutrient
Urea (46/0/0) kg N	0.11	13.04	95.65	2.39	4.35	4.35
Calcium ammonium nitrate (20/0/0) kg N	0.25	60.00	155.00	5.50	90.00	10.00
Ammonium nitrate (27.5/0/0) kg N	0.18	25.45	181.82	6.91	47.27	14.55
Ammonium sulphate (21/0/0) kg N	0.24	19.05	142.86	5.24	8.57	9.52
Calcium ammonium nitrate (27/0/0) kg N	0.19	8.52	100.00	5.93	12.59	2.96
Magnesium ammonium nitrate (23/0/0/5) kg N	0.43	56.52	4.35	4.35	21.74	6.09
<i>Generic mean N</i>	<i>0.21</i>	<i>22.25</i>	<i>121.43</i>	<i>5.37</i>	<i>17.17</i>	<i>7.81</i>
Triple superphosphate (0/46/0) kg P2O5	113.04	97.83	650.00	7.61	95.65	567.39
Superphosphate (0/19/0) kg P2O5	52.63	121.05	852.63	578.95	105.26	342.11
Thomas meal (0/16/0) kg P2O5	1.56	250.00	425.00	75.00	125.00	12212.50
Hyperphosphate/raw phosphate (0/26/0) kg P2O5	50.00	115.38	915.38	23.85	76.92	611.54
<i>Generic mean P</i>	<i>51.32</i>	<i>118.22</i>	<i>751.32</i>	<i>49.42</i>	<i>100.46</i>	<i>589.46</i>
Potassium chloride (KCl) (0/0/60) kg K2O	0.10	8.33	76.67	9.17	3.50	3.33
Potassium sulphate (0/0/50) kg K2O	0.10	4.00	64.00	6.60	1.60	4.00
Raw potassium (0/0/26/5) kg K2O	0.19	173.08	153.85	11.54	11.54	173.08
Lime kg CaO	0.12	4.00	8.00	3.60	12.20	314.00
<i>Generic mean K</i>	<i>0.11</i>	<i>6.17</i>	<i>70.33</i>	<i>7.88</i>	<i>7.52</i>	<i>88.54</i>

Tab. A. 3 Heavy-metal contents of farmyard manure and organic fertilisers (g/unit, compiled by Freiermuth 2006 from from Menzi & Kessler (1998) and Desaulles & Studer (1993, p. 152)). Dry matter (DM) contents from Walther *et al.* (2001, Tab. 44).

	Cd [mg/kg DM]	Cu [mg/kg DM]	Zn [mg/kg DM]	Pb [mg/kg DM]	Ni [mg/kg DM]	Cr [mg/kg DM]	Hg [mg/kg DM]	DM- content
Farmyard manure								
Cattle liquid manure	0.178	37.1	162.2	3.77	4.3	3.9	0.4	9.0%
Cattle slurry	0.16	19.1	123.3	2.92	3.1	2.1	0.6	7.5%
Cattle staple manure	0.172	23.9	117.7	3.77	4.3	3.9	0.4	19.0%
Cattle manure form loose housing	0.151	22.0	91.1	2.81	4.3	3.9	0.4	21.0%
Pig liquid manure	0.21	115.3	746.5	1.76	8.6	6.7	0.8	5.0%
Pig solid manure	0.21	115.3	746.5	1.76	8.6	6.7	0.8	27.0%
Litter from broilers	0.292	43.8	349.2	2.92	40	10	0.2	65.0%
Litter from belts from laying hens	0.2525	39.6	468.4	2.235	7.9	5.5	0.2	30.0%
Litter from deep pits from laying hens	0.2525	39.6	468.4	2.235	7.9	5.5	0.2	45.0%

Appendix A5 to Chapter 5 (Agricultural Buildings)

Description of Modules

Tab. A. 4 Building infrastructure modules in ecoinvent data.

Name	Description ^a , size	Service life	Functional Unit (FU)	FU explanation	Location
Housing system with fully-slatted floor, pig	Comprises: pen, fittings, slurry store, concentrate silo, straw storage area. construction plan: see Appendix Fig. A. 1 Feeding system: wet mix feed Dung removal: dung removal by flushing incl. pump and agitator. Ventilation: forced ventilation, porous ceiling. Size: 300 pig places	50 years	pig place	One fattening pig place during entire service life	CH
Label housing system, pig	Comprises: pen, fittings, slurry store, concentrate silo, straw storage area. Animal-friendly housing system with multispace pens and exercise area Compliant to typical restrictions from animal friendly housing labels. Floor: littered lying area; feeding area with slatted floor; exercise area with partially slatted floor. Construction plan: see Appendix Fig. A. 2 Feeding system: wet mix feed Dung removal: dung removal by flushing incl. pump and agitator. Ventilation: free ventilation (open front housing) Size: 300 pig places	50 years	pig place	One fattening pig place during entire service life	CH
Tied housing system, cattle	Comprises: Housing for dairy cattle incl. rearing to 6 months, fittings, slurry store, dried roughage store, exercise area, milking system. construction plan: see Appendix Fig. A. 3 Feeding system: hay and grass (no silage), hay store above stable, cold air dried. Milking: pipeline milking machine (2 MU) ^c and cooling vat. Dung removal: hydraulic manure removal system, slurry store, agitator, dung slab Ventilation: gravity flues (without fans). Size: 22 LU dairy cattle (not including rearing)	50 years	LU	One livestock unit (LU) place during entire service life	CH

Name	Description ^a , size	Service life	Functional Unit (FU)	FU explanation	Location
Loose housing system, cattle	Comprises: cubicle housing system for dairy cattle incl. rearing to 6 months, fittings, slurry store, dried roughage store, exercise yard, milking parlour. construction plan: see Appendix Fig. A. 4 Feeding system: hay and grass (no silage), hay store on ground floor, cold air dried. Milking: Herringbone milking parlour 1x4 (4 MU) and cooling vat. Dung removal: flat scraper system, slurry store, agitator. Ventilation: free ventilation (open front housing). Size: 22 LU dairy cattle (not including rearing)	50 years	LU	One livestock unit (LU) place during entire service life	CH
Milking parlour	Herringbone milking parlour 1x4 with 4 MU, pipeline milking machine Comprises: floor, walls and roof of milk room and milking parlour, fittings, boiler, pipeline milking machine. construction plan: see Appendix Fig. A. 5	50 years ^b	unit	One complete milking parlour during entire service life	CH
Dried roughage store, cold-air dried, conventional	Comprises: dried roughage store on ground floor, incl. floor, walls, roof. Construction plan: see Appendix Fig. A. 5 Ventilation: cold air with fan. Storage: impeller blower with telescopic spreader Removal: manual DM ^d content: on storage 60 %, on removal 88 %. Size: 830 m ³ storage space	50 years	m ³	One m ³ of dried roughage store during entire service life	CH
Dried roughage store, air dried, solar	Comprises: dried roughage store on ground floor, incl. floor, walls, roof Ventilation: fan-assisted: air heated by double roof structure („solar collector“). Storage: impeller blower with telescopic spreader Removal: manual DM content: on storage 55 %, on removal 88 %. Size: 830 m ³ storage space.	50 years	m ³	One m ³ of dried roughage store during entire service life	CH
Dried roughage store, non ventilated	Comprises: dried roughage store on ground floor, incl. floor, walls, roof Ventilation: none Storage: impeller blower with telescopic spreader. Removal: manual DM content : on storage 84 %, on removal 84 %. Size: 830 m ³ storage space	50 years	m ³	One m ³ of dried roughage store during entire service life	CH
Slurry store and processing	Concrete tank: underfloor, rectangular, covered. Agitator: 6 kW marine screw agitator Size: 300 m ³ storage space.	40 years	unit	One complete slurry store during entire service life	CH

Name	Description ^a , size	Service life	Functional Unit (FU)	FU explanation	Location
Tower silo, plastic	Material: glass fibre reinforced polyester Foundation: included (concrete) Size: 105 m ³ storage volume.	25 years	m ³	One m ³ of tower silo storage during entire service life	CH
Dung slab	Material: concrete Dung liquor: drainage to slurry store included. Size: 64 m ² storage area.	40 years	m ²	One m ³ of tower silo storage during entire service life	CH
Shed	Shed: closed on three sides, timber construction; of which garage: fireproofed, with doors. Construction plan: see Appendix Fig. A. 6 Size: 264 m ² useful area, of which garage 102 m ² .	50 years	m ²	One m ³ of shed during entire service life	CH

a For additional information on system boundaries see chapter 5.2.3

b Service life of the principal components of the milking parlour is 12 years. In order to have the same service life as the cattle housing, the milking parlour is maintained and replaced during 50 years.

c MU = Milking unit

d DM = Dry matter

Tab. A. 5 Basic operation modules for the building infrastructure in ecoinvent data.

Name	Description ^a	Functional Unit (FU)	FU explanation	Location
Housing system with fully-slatted floor, pig, operation	Building infrastructure: see Tab. A. 4, "housing system with fully-slatted floor, pig" Energy and auxiliary materials used by: lighting, ventilation, heating, feeding system, slurry agitator and pump, cleaning, drinking water.	pig place	1 fattening pig place during one year	CH
Label housing system, pig, operation	Building infrastructure: see Tab. A. 4, "label housing system, pig" Energy and auxiliary materials used by: lighting, feeding system, slurry agitator and pump, cleaning, drinking water.	pig place	1 fattening pig place during one year	CH
Tied housing system, cattle, operation	Building infrastructure: see Tab. A. 4, "tied housing system, cattle" Energy and auxiliary materials used by: lighting, dried roughage storage and drying, slurry agitator, milking machine, milk cooling, cleaning, drinking water.	LU	1 livestock unit (LU) during one year	CH
Loose housing system, cattle, operation	Building infrastructure: see Tab. A. 4, "loose housing system, cattle" Energy and auxiliary materials used by: lighting, dried roughage storage and drying, dung scraper, slurry agitator, milking machine, milk cooling, cleaning, drinking water.	LU	1 livestock unit (LU) during one year	CH
Milking	Building infrastructure: see Tab. A. 4, "milking parlour" Energy and auxiliary materials used by: lighting, milking machine, milking machine cleaning, milking parlour and milk room.	kg	Milking of 1 kg milk	CH
Dried roughage store, cold-air dried, conventional, operation	Building infrastructure: see Tab. A. 4, "dried roughage store, cold-air dried, conventional" Energy and auxiliary materials used by: lighting, impeller blower, fan.	kg	kg DM hay (DM: dry matter) Conversion to kg feed via DM content (Tab. A. 4)	CH
Dried roughage store, air dried, solar, operation	Building infrastructure: see Tab. A. 4, "dried roughage store, air dried, solar" Energy and auxiliary materials used by: lighting, impeller blower, fan.	kg	kg DM hay (DM: dry matter) conversion to kg feed via DM content (Tab. A. 4)	CH
Dried roughage store, non ventilated, operation	Building infrastructure: see Tab. A. 4, "dried roughage store, non ventilated" Energy and auxiliary materials used by: lighting, impeller blower.	kg	kg DM hay (DM: dry matter) Conversion to kg feed via DM content (Tab. A. 4)	CH
Slurry store and processing, operation	Building infrastructure: see Tab. A. 4, "slurry store and processing, operation" Energy and auxiliary materials used by: 6 kW marine screw agitator.	m ³	m ³ slurry ^b	CH

^a For additional information on system boundaries see chapter 5.2.3

^b Given as m³ slurry removed from the slurry store. Calculation based on "Assessment of water conservation on a farm" (FOAG & SAEFL 1994), assuming an average storage period of five months. Does not include dilution additional to waste water normally introduced (e.g. roof water or additional dilution at slurry application).

Construction Plans of Selected Buildings

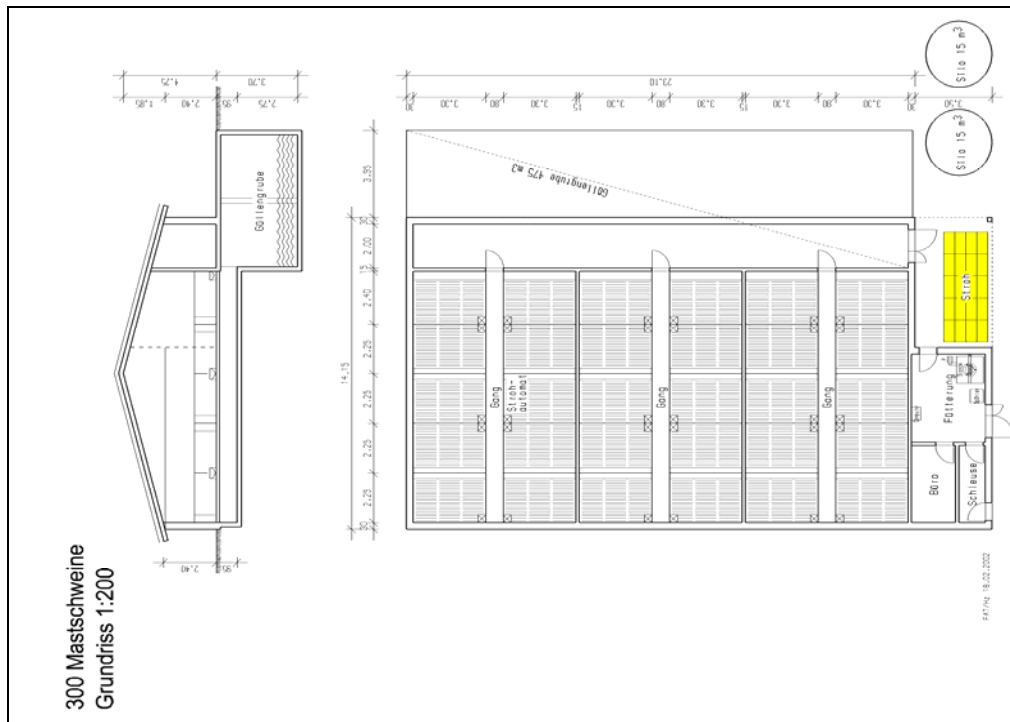


Fig. A. 1 Construction plan for fattening pig housing with fully slatted floor, 300 pig places.

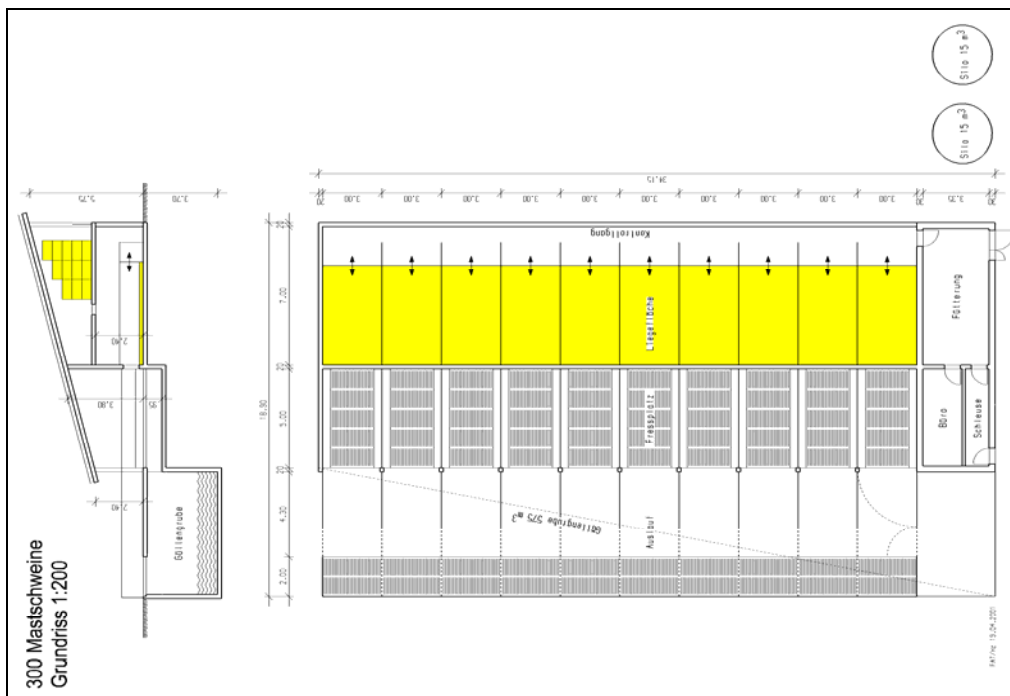
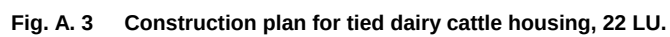


Fig. A. 2 Construction plan for label housing for fattening pigs, 300 pig places.





a)

b)

Fig. A.5 Construction plans for two important building parts. a) Herringbone milking parlour 1x4 incl. milk room; b) dried roughage store 830 m³.

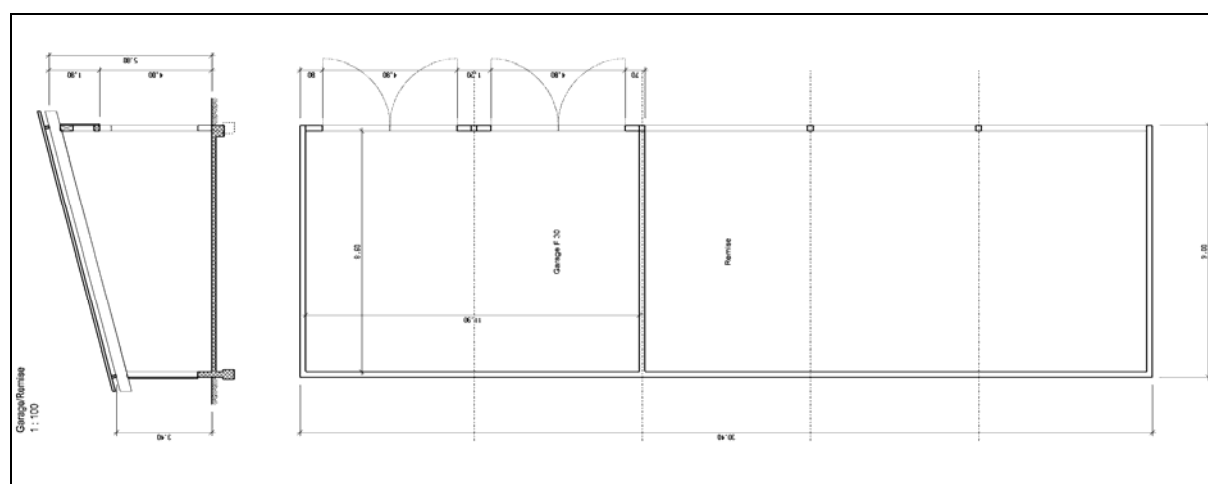


Fig. A.6 Construction plan for a shed with garage section.

Utilised Modules for Waste Disposal

Tab. A. 6 Assignment of the construction material to the ecoinvent modules of waste disposal.

Construction material	Disposal module in ecoinvent database	Remarks
aluminium construction		No disposal
bitumen sealing	disposal, building, bitumen sheet, to final disposal	
brick	disposal, building, brick, to final disposal	
cast iron	disposal, building, bulk iron (excluding reinforcement), to sorting plant	
cast iron, galvanised	disposal, building, bulk iron (excluding reinforcement), to sorting plant	
cement mortar	disposal, building, cement (in concrete) and mortar, to final disposal	
cement PC-CH	disposal, building, cement (in concrete) and mortar, to final disposal	
ceramics	disposal, building, glass sheet, to final disposal	
chromium steel	disposal, building, bulk iron (excluding reinforcement), to sorting plant	
concrete pc150	disposal, building, concrete, not reinforced, to final disposal	
concrete pc300	disposal, building, reinforced concrete, to final disposal	
concrete with expanded clay (Leca)	disposal, building, concrete, not reinforced, to final disposal disposal, building, brick, to final disposal	64% (weight) 36% (weight)
copper	disposal, building, bulk iron (excluding reinforcement), to sorting plant	
excavation hydraulic digger	-	no disposal
excavation skid-steer loader	-	no disposal
fibre cement shingle	disposal, building, brick, to final disposal	
fibre cement slab	disposal, building, brick, to final disposal	
Frinorm insulating wall panel 80 mm	disposal, building, polystyrene isolation, flame-retardant, to final disposal	
glass	disposal, building, glass sheet, to final disposal	
glued laminated timber	disposal, building, waste wood, treated, to final disposal	
gravel round	-	no disposal
iron enamel	disposal, building, bulk iron (excluding reinforcement), to sorting plant disposal, building, glass sheet, to final disposal	90% (weight) 10% (weight)
particle board	disposal, building, fibre board, to final disposal	
particle board with cement	disposal, building, cement (in concrete) and mortar, to final disposal	
polyester	disposal, building, polyethylene/polypropylene products, to final disposal	
polyester GF30	disposal, glass, 0% water, to municipal incineration disposal, plastics, mixture, 15.3% water, to municipal incineration	30% (weight) 70% (weight)
polyester GF50	disposal, glass, 0% water, to municipal incineration disposal, plastics, mixture, 15.3% water, to municipal incineration	50% (weight) 50% (weight)
polyester GF60	disposal, glass, 0% water, to municipal incineration disposal, plastics, mixture, 15.3% water, to municipal incineration	60% (weight) 40% (weight)
polyester reinforced concrete	disposal, building, concrete, not reinforced, to final disposal	
polyethylene HDPE	disposal, building, polyethylene/polypropylene products, to final disposal	
polypropylene (PP)	disposal, building, polyethylene/polypropylene products, to final disposal	
polystyrene foam slab (EPS)	disposal, building, polystyrene isolation, flame-retardant, to final disposal	
polyurethane foam slab (PUR)	disposal, building, polyurethane foam, to final disposal	
PVC	disposal, building, polyvinylchloride products, to final disposal	
PVC sealing	disposal, building, PVC sealing sheet, to final disposal	

Construction material	Disposal module in ecoinvent database	Remarks
reinforcing steel	disposal, building, reinforced concrete, to final disposal	
rock wool	disposal, building, mineral wool, to final disposal	
sawn timber, softwood, planed	disposal, building, waste wood, treated, to final disposal and disposal, building, waste wood, treated, to final disposal	
sheet steel, galvanised	disposal, building, bulk iron (excluding reinforcement), to sorting plant	
synthetic resin slab	disposal, building, polyethylene/polypropylene products, to final disposal	
synthetic rubber	disposal, building, polyethylene/polypropylene products, to final disposal	
fleece (PE)	disposal, building, polyethylene/polypropylene products, to final disposal	
Zinc	disposal, building, bulk iron (excluding reinforcement), to sorting plant	

Unit-Process Inventories from Chapter 5 (Agricultural Buildings)

(Last Changes 2004)

Tab. A. 7 Unit-process inventories for agricultural buildings.

Unit process inventory for: housing system with fully-slatted floor, pig, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	1.85E+00	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	9.52E+01	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	3.47E+00	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	6.16E+00	1	1.05	(1,2,1,1,1,na)
brick, at plant	RER	kg	3.28E+02	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	1.73E-01	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	1.64E+00	1	1.05	(1,2,1,1,1,na)
bitumen, at refinery	CH	kg	2.64E-01	1	1.05	(1,2,1,1,1,na)
flat glass, uncoated, at plant	RER	kg	1.98E+00	1	1.05	(1,2,1,1,1,na)
glass fibre, at plant	RER	kg	2.42E-01	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	4.21E+01	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	1.93E+00	1	1.05	(1,2,1,1,1,na)
sanitary ceramics, at plant	CH	kg	2.26E+01	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	5.04E+02	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	3.07E+00	1	1.05	(1,2,1,1,1,na)
cement mortar, at plant	CH	kg	8.14E+01	1	1.05	(1,2,1,1,1,na)
polypropylene, granulate, at plant	RER	kg	1.36E-02	1	1.05	(1,2,1,1,1,na)
polystyrene foam slab, at plant	RER	kg	9.53E-01	1	1.05	(1,2,1,1,1,na)
polyurethane, rigid foam, at plant	RER	kg	4.48E+00	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	7.89E+00	1	1.05	(1,2,1,1,1,na)
chromium steel 18/8, at plant	RER	kg	1.64E+01	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	8.18E+00	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	8.18E+00	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	1.74E+00	1	1.05	(1,2,1,1,1,na)
rock wool, at plant	CH	kg	3.19E+00	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	4.49E+01	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	1.91E+02	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	2.32E+00	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	9.58E+00	1	1.05	(1,2,1,1,1,na)
polyvinylchloride, bulk polymerised, at plant	RER	kg	7.64E+00	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	1.27E-01	1	1.05	(1,2,1,1,1,na)
glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	RER	kg	2.42E-01	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	5.32E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	6.40E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	3.81E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	3.70E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, bitumen sheet, to final disposal	CH	kg	2.64E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	2.46E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, mineral wool, to final disposal	CH	kg	3.19E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	2.02E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, polyurethane foam, to final disposal	CH	kg	4.48E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, polyvinylchloride products, to final disposal	CH	kg	7.64E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	3.81E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	2.66E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polystyrene isolation, flame-retardant, to final disposal	CH	kg	7.25E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	7.19E+01	1	1.05	(1,2,1,1,1,na)
disposal, glass, 0% water, to municipal incineration	CH	kg	2.41E-01	1	1.05	(1,2,1,1,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	1.61E-01	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	2.71E+02	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.62E+02	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.25E+00	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	3.25E+00	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	6.49E+00	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	1.91E+01	1	1.05	(1,2,1,1,1,na)
housing system with fully-slatted floor, pig (l)	CH	pig place	1.00E+00			

Unit process inventory for: label housing system, pig, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	1.28E+00	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	7.84E+01	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	3.44E+00	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	6.26E+00	1	1.05	(1,2,1,1,1,na)
brick, at plant	RER	kg	2.30E+02	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	1.30E-01	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	1.42E+00	1	1.05	(1,2,1,1,1,na)
bitumen, at refinery	CH	kg	4.78E-01	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	9.00E-03	1	1.05	(1,2,1,1,1,na)
flat glass, uncoated, at plant	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	3.29E+01	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	5.07E-01	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	4.93E+02	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	2.79E+00	1	1.05	(1,2,1,1,1,na)
epoxy resin, liquid, at plant	RER	kg	1.87E+01	1	1.05	(1,2,1,1,1,na)
cement mortar, at plant	CH	kg	5.78E+01	1	1.05	(1,2,1,1,1,na)
kraft paper, unbleached, at plant	RER	kg	2.58E+01	1	1.05	(1,2,1,1,1,na)
polypropylene, granulate, at plant	RER	kg	3.22E-02	1	1.05	(1,2,1,1,1,na)
polystyrene foam slab, at plant	RER	kg	4.45E-01	1	1.05	(1,2,1,1,1,na)
polyurethane, rigid foam, at plant	RER	kg	8.80E-02	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	9.35E+00	1	1.05	(1,2,1,1,1,na)
chromium steel 18/8, at plant	RER	kg	5.78E+01	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	7.87E+00	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	7.87E+00	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	1.67E+00	1	1.05	(1,2,1,1,1,na)
rock wool, at plant	CH	kg	1.03E+00	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	5.33E+01	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	1.48E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	1.74E+00	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	1.19E+00	1	1.05	(1,2,1,1,1,na)
polyvinylchloride, bulk polymerised, at plant	RER	kg	6.47E-01	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	2.93E-01	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	6.25E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	1.51E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	2.92E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	3.21E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, bitumen sheet, to final disposal	CH	kg	4.78E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, mineral wool, to final disposal	CH	kg	1.03E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	4.51E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, polyurethane foam, to final disposal	CH	kg	8.80E-02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyvinylchloride products, to final disposal	CH	kg	6.47E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	2.86E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	6.98E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, polystyrene isolation, flame-retardant, to final disposal	CH	kg	3.35E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	1.03E+02	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	2.33E+02	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.89E+02	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.78E+00	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	3.78E+00	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	7.56E+00	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	2.25E+01	1	1.05	(1,2,1,1,1,na)
label housing system, pig (l)	CH	pig place	1.00E+00			

Unit process inventory for: tied housing system, cattle, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	2.73E+01	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	6.93E+02	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	2.36E+01	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	5.15E+01	1	1.05	(1,2,1,1,1,na)
brick, at plant	RER	kg	3.14E+03	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	2.24E+00	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	1.31E+01	1	1.05	(1,2,1,1,1,na)
bitumen, at refinery	CH	kg	3.31E+00	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	5.27E-01	1	1.05	(1,2,1,1,1,na)
flat glass, uncoated, at plant	RER	kg	2.44E+01	1	1.05	(1,2,1,1,1,na)
glass fibre, at plant	RER	kg	3.83E+00	1	1.05	(1,2,1,1,1,na)
synthetic rubber, at plant	RER	kg	1.25E+02	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	6.02E+02	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	3.38E+01	1	1.05	(1,2,1,1,1,na)
sanitary ceramics, at plant	CH	kg	4.63E+01	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	1.20E+04	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	7.02E+01	1	1.05	(1,2,1,1,1,na)
cement mortar, at plant	CH	kg	7.73E+02	1	1.05	(1,2,1,1,1,na)
polypropylene, granulate, at plant	RER	kg	1.10E+00	1	1.05	(1,2,1,1,1,na)
polystyrene foam slab, at plant	RER	kg	2.43E+01	1	1.05	(1,2,1,1,1,na)
polyurethane, rigid foam, at plant	RER	kg	2.41E+01	1	1.05	(1,2,1,1,1,na)
calendering, rigid sheets	RER	kg	1.49E+00	1	1.05	(1,2,1,1,1,na)
sand, at mine	CH	kg	5.05E+02	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	9.36E+01	1	1.05	(1,2,1,1,1,na)
particle board, indoor use, at plant	RER	m3	9.22E-01	1	1.05	(1,2,1,1,1,na)
chromium steel 18/8, at plant	RER	kg	1.16E+02	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	5.92E+01	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	5.92E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	1.26E+01	1	1.05	(1,2,1,1,1,na)
rock wool, at plant	CH	kg	4.00E+01	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	5.33E+02	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	1.92E+03	1	1.05	(1,2,1,1,1,na)
roof tile, at plant	RER	kg	4.13E+02	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	2.45E+01	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	3.49E+01	1	1.05	(1,2,1,1,1,na)
polyvinylchloride, bulk polymerised, at plant	RER	kg	1.49E+00	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	5.65E+00	1	1.05	(1,2,1,1,1,na)
glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	RER	kg	3.31E+01	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	9.67E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	5.60E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	2.53E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	4.18E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	2.94E+04	1	1.05	(1,2,1,1,1,na)
disposal, building, bitumen sheet, to final disposal	CH	kg	3.31E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	7.07E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, mineral wool, to final disposal	CH	kg	4.00E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	1.77E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyurethane foam, to final disposal	CH	kg	2.41E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, PVC sealing sheet, to final disposal	CH	kg	1.49E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, fibre board, to final disposal	CH	kg	6.00E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	5.76E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	2.21E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, polystyrene isolation, flame-retardant, to final disposal	CH	kg	1.83E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	8.69E+02	1	1.05	(1,2,1,1,1,na)
disposal, glass, 0% water, to municipal incineration	CH	kg	3.81E+00	1	1.05	(1,2,1,1,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	2.54E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	2.93E+03	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	2.56E+03	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	5.12E+01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	5.12E+01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	1.02E+02	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	3.48E+02	1	1.05	(1,2,1,1,1,na)
tied housing system, cattle (I)	CH	LU	1.00E+00			

Unit process inventory for: loose housing system, cattle, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	2.41E+01	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	6.63E+02	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	2.49E+01	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	5.37E+01	1	1.05	(1,2,1,1,1,na)
brick, at plant	RER	kg	1.17E+03	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	2.04E+00	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	1.44E+01	1	1.05	(1,2,1,1,1,na)
bitumen, at refinery	CH	kg	1.08E+03	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	8.62E-01	1	1.05	(1,2,1,1,1,na)
flat glass, uncoated, at plant	RER	kg	1.24E+01	1	1.05	(1,2,1,1,1,na)
glass fibre, at plant	RER	kg	8.80E+00	1	1.05	(1,2,1,1,1,na)
synthetic rubber, at plant	RER	kg	3.71E+01	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	8.00E+02	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	1.06E+01	1	1.05	(1,2,1,1,1,na)
sanitary ceramics, at plant	CH	kg	2.26E+02	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	1.27E+04	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	7.55E+01	1	1.05	(1,2,1,1,1,na)
epoxy resin, liquid, at plant	RER	kg	4.83E+00	1	1.05	(1,2,1,1,1,na)
cement mortar, at plant	CH	kg	2.90E+02	1	1.05	(1,2,1,1,1,na)
kraft paper, unbleached, at plant	RER	kg	6.67E+00	1	1.05	(1,2,1,1,1,na)
polypropylene, granulate, at plant	RER	kg	1.75E+00	1	1.05	(1,2,1,1,1,na)
polystyrene foam slab, at plant	RER	kg	6.12E+00	1	1.05	(1,2,1,1,1,na)
polyurethane, rigid foam, at plant	RER	kg	5.43E+00	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	1.52E+02	1	1.05	(1,2,1,1,1,na)
particle board, indoor use, at plant	RER	m3	1.10E+00	1	1.05	(1,2,1,1,1,na)
chromium steel 18/8, at plant	RER	kg	1.17E+02	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	5.65E+01	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	5.65E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	1.20E+01	1	1.05	(1,2,1,1,1,na)
rock wool, at plant	CH	kg	1.43E+01	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	8.77E+02	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	3.77E+02	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	2.59E+01	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	1.24E+01	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	6.25E+00	1	1.05	(1,2,1,1,1,na)
glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	RER	kg	4.88E+01	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	1.20E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	4.98E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	3.06E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	2.20E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	3.23E+04	1	1.05	(1,2,1,1,1,na)
disposal, building, bitumen sheet, to final disposal	CH	kg	1.18E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	2.38E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, mineral wool, to final disposal	CH	kg	1.43E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	9.19E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, polyurethane foam, to final disposal	CH	kg	5.43E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, fibre board, to final disposal	CH	kg	7.18E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	4.59E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	5.39E+02	1	1.05	(1,2,1,1,1,na)
disposal, bitumen, 1.4% water, to sanitary landfill	CH	kg	1.08E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, polystyrene isolation, flame-retardant, to final disposal	CH	kg	4.61E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	1.07E+03	1	1.05	(1,2,1,1,1,na)
disposal, glass, 0% water, to municipal incineration	CH	kg	8.79E+00	1	1.05	(1,2,1,1,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	5.86E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	2.97E+03	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	3.05E+03	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	6.10E+01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	6.10E+01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	1.22E+02	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	4.30E+02	1	1.05	(1,2,1,1,1,na)
loose housing system, cattle (l)	CH	LU	1.00E+00			

Unit process inventory for: milking parlour, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	1.95E+02	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	1.63E+03	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	1.09E+01	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	6.61E+01	1	1.05	(1,2,1,1,1,na)
brick, at plant	RER	kg	2.49E+04	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	8.31E+00	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	3.88E+01	1	1.05	(1,2,1,1,1,na)
bitumen, at refinery	CH	kg	9.51E+03	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	3.79E+00	1	1.05	(1,2,1,1,1,na)
flat glass, uncoated, at plant	RER	kg	1.21E+02	1	1.05	(1,2,1,1,1,na)
synthetic rubber, at plant	RER	kg	8.16E+02	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	1.95E+03	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	1.92E+02	1	1.05	(1,2,1,1,1,na)
sanitary ceramics, at plant	CH	kg	4.97E+03	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	2.06E+04	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	5.37E+01	1	1.05	(1,2,1,1,1,na)
cement mortar, at plant	CH	kg	6.14E+03	1	1.05	(1,2,1,1,1,na)
polystyrene foam slab, at plant	RER	kg	3.30E+00	1	1.05	(1,2,1,1,1,na)
polyurethane, rigid foam, at plant	RER	kg	3.30E+01	1	1.05	(1,2,1,1,1,na)
calendering, rigid sheets	RER	kg	2.29E+01	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	2.84E+02	1	1.05	(1,2,1,1,1,na)
chromium steel 18/8, at plant	RER	kg	2.75E+02	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	1.44E+02	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	1.44E+02	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	3.06E+01	1	1.05	(1,2,1,1,1,na)
rock wool, at plant	CH	kg	7.95E+02	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	1.62E+03	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	2.34E+03	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	1.04E+02	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	1.92E+02	1	1.05	(1,2,1,1,1,na)
polyvinylchloride, bulk polymerised, at plant	RER	kg	2.29E+01	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	1.48E+01	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.62E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	9.73E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	8.33E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	2.68E+04	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	8.71E+04	1	1.05	(1,2,1,1,1,na)
disposal, building, bitumen sheet, to final disposal	CH	kg	2.51E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	5.09E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, mineral wool, to final disposal	CH	kg	7.95E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	1.01E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, polyurethane foam, to final disposal	CH	kg	3.30E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, PVC sealing sheet, to final disposal	CH	kg	2.29E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	1.83E+04	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	8.48E+03	1	1.05	(1,2,1,1,1,na)
disposal, bitumen, 1.4% water, to sanitary landfill	CH	kg	9.49E+03	1	1.05	(1,2,1,1,1,na)
disposal, building, polystyrene isolation, flame-retardant, to final disposal	CH	kg	3.30E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	2.52E+03	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	9.58E+03	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	2.70E+03	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	5.40E+01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	5.40E+01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	5.40E+01	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	9.44E+02	1	1.05	(1,2,1,1,1,na)
milking parlour (I)	CH	unit	1.00E+00			

Unit process inventory for: dried roughage store, cold-air dried, conventional, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	2.33E-01	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	1.86E+00	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	1.78E-03	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	9.80E-02	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	1.19E-03	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	6.78E-02	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	9.14E-03	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	2.79E+00	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	8.56E+01	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	1.40E+00	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	1.82E+00	1	1.05	(1,2,1,1,1,na)
particle board, indoor use, at plant	RER	m3	2.93E-02	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	2.81E-01	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	1.04E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	4.29E-03	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	7.81E-02	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	8.30E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	6.30E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	3.70E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	1.22E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	1.51E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
disposal, building, fibre board, to final disposal	CH	kg	1.90E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	2.62E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	5.52E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	1.60E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.58E+01	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.16E-01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	3.16E-01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	3.16E-01	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	2.99E+00	1	1.05	(1,2,1,1,1,na)
dried roughage store, cold-air dried, conventional (I)	CH	m3	1.00E+00			

Unit process inventory for: dried roughage store, air dried, solar, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	2.33E-01	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	1.86E+00	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	1.78E-03	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	9.80E-02	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	1.19E-03	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	6.78E-02	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	9.14E-03	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	2.79E+00	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	8.56E+01	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	1.40E+00	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	1.82E+00	1	1.05	(1,2,1,1,1,na)
particle board, indoor use, at plant	RER	m3	5.56E-02	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	2.81E-01	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	1.04E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	4.29E-03	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	7.81E-02	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	8.57E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	6.30E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	3.70E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	1.22E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	1.51E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
disposal, building, fibre board, to final disposal	CH	kg	3.61E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	2.62E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	5.52E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	1.68E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.58E+01	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.16E-01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	3.16E-01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	3.16E-01	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	3.09E+00	1	1.05	(1,2,1,1,1,na)
dried roughage store, air dried, solar (I)	CH	m3	1.00E+00			

Unit process inventory for: dried roughage store, non ventilated, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
aluminium, production mix, at plant	RER	kg	1.21E-01	1	1.05	(1,2,1,1,1,na)
reinforcing steel, at plant	RER	kg	1.86E+00	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	2.14E-03	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	1.15E-01	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	1.19E-03	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	6.78E-02	1	1.05	(1,2,1,1,1,na)
glued laminated timber, indoor use, at plant	RER	m3	9.14E-03	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	1.33E+00	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	1.03E+02	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	7.30E-01	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	1.82E+00	1	1.05	(1,2,1,1,1,na)
particle board, indoor use, at plant	RER	m3	4.00E-03	1	1.05	(1,2,1,1,1,na)
steel, low-alloyed, at plant	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
sheet rolling, steel	RER	kg	1.32E+00	1	1.05	(1,2,1,1,1,na)
zinc coating, coils	RER	m2	2.81E-01	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	1.04E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	4.29E-03	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	4.33E-02	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	7.54E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	6.30E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	2.00E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	1.22E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	1.51E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	6.31E-03	1	1.05	(1,2,1,1,1,na)
disposal, building, fibre board, to final disposal	CH	kg	2.60E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	2.62E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	3.39E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	1.50E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.58E+01	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.16E-01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	3.16E-01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	3.16E-01	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	2.71E+00	1	1.05	(1,2,1,1,1,na)
dried roughage store, non ventilated (I)	CH	m3	1.00E+00			

Unit process inventory for: slurry store and processing, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
reinforcing steel, at plant	RER	kg	2.26E+01	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	1.49E+00	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	2.53E+00	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	2.44E-02	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	3.67E-01	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	1.91E+00	1	1.05	(1,2,1,1,1,na)
chromium steel 18/8, at plant	RER	kg	1.74E+00	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	1.09E-01	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	2.72E-03	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	4.78E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	1.36E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	8.29E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	5.36E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	3.75E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	4.44E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.50E+01	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.76E-01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	3.76E-01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	3.76E-01	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	1.72E+00	1	1.05	(1,2,1,1,1,na)
slurry store and processing (I)	CH	m3	1.00E+00			

Unit process inventory for: tower silo, plastic, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
reinforcing steel, at plant	RER	kg	2.61E+00	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	1.18E-01	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	5.97E-03	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	7.43E-02	1	1.05	(1,2,1,1,1,na)
glass fibre, at plant	RER	kg	1.37E+01	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	2.58E+00	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	4.63E+01	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	1.12E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	1.48E-01	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	7.62E-05	1	1.05	(1,2,1,1,1,na)
glass fibre reinforced plastic, polyester resin, hand lay-up, at plant	RER	kg	9.12E+00	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	4.61E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	3.81E-02	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	1.66E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	1.31E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	1.12E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	2.72E+00	1	1.05	(1,2,1,1,1,na)
disposal, glass, 0% water, to municipal incineration	CH	kg	1.37E+01	1	1.05	(1,2,1,1,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	9.12E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	1.31E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.69E+01	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	6.75E-01	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	6.75E-01	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	3.38E-01	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	1.66E+00	1	1.05	(1,2,1,1,1,na)
tower silo, plastic (I)	CH	m3	1.00E+00			

Unit process inventory for: dung slab, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
reinforcing steel, at plant	RER	kg	1.08E+01	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	7.69E-01	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	5.37E-02	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	3.11E-01	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	1.71E-01	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	3.52E+02	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	1.71E-01	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	4.73E-04	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	1.19E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	2.37E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	6.95E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	1.71E-01	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	1.18E+02	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	5.83E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	4.00E+01	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	1.00E+00	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	1.00E+00	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	1.00E+00	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	4.30E-01	1	1.05	(1,2,1,1,1,na)
dung slab (I)	CH	m2	1.00E+00			

Unit process inventory for: shed, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
reinforcing steel, at plant	RER	kg	8.28E+00	1	1.05	(1,2,1,1,1,na)
excavation, skid-steer loader	RER	m3	1.28E-02	1	1.05	(1,2,1,1,1,na)
excavation, hydraulic digger	RER	m3	3.52E-01	1	1.05	(1,2,1,1,1,na)
poor concrete, at plant	CH	m3	8.33E-03	1	1.05	(1,2,1,1,1,na)
concrete, normal, at plant	CH	m3	2.83E-01	1	1.05	(1,2,1,1,1,na)
cast iron, at plant	RER	kg	5.33E-01	1	1.05	(1,2,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	2.68E-02	1	1.05	(1,2,1,1,1,na)
gravel, round, at mine	CH	kg	3.09E+02	1	1.05	(1,2,1,1,1,na)
copper, at regional storage	RER	kg	6.21E-01	1	1.05	(1,2,1,1,1,na)
fibre cement corrugated slab, at plant	CH	kg	6.73E+00	1	1.05	(1,2,1,1,1,na)
fibre cement facing tile, at plant	CH	kg	3.83E+01	1	1.05	(1,2,1,1,1,na)
portland cement, strength class Z 42.5, at plant	CH	kg	2.19E+01	1	1.05	(1,2,1,1,1,na)
zinc coating, pieces	RER	m2	3.05E-02	1	1.05	(1,2,1,1,1,na)
extrusion, plastic film	RER	kg	2.68E-02	1	1.05	(1,2,1,1,1,na)
sawn timber, softwood, planed, air dried, at plant	RER	m3	2.09E-01	1	1.05	(1,2,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.41E+00	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, treated, to final disposal	CH	kg	1.10E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, waste wood, untreated, to final disposal	CH	kg	7.72E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, brick, to final disposal	CH	kg	4.51E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, reinforced concrete, to final disposal	CH	kg	6.31E+02	1	1.05	(1,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	2.68E-02	1	1.05	(1,2,1,1,1,na)
disposal, building, concrete, not reinforced, to final disposal	CH	kg	1.83E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, cement (in concrete) and mortar, to final disposal	CH	kg	3.85E+01	1	1.05	(1,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	1.18E+00	1	1.05	(1,2,1,1,1,na)
transport, lorry 28t	CH	tkm	5.66E+01	1	2.00	(1,2,1,1,1,na)
Occupation, urban, discontinuously built	resource/land	m2a	1.24E+02	1	1.50	(1,2,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	2.47E+00	1	2.00	(1,2,1,1,1,na)
Transformation, to urban, discontinuously built	resource/land	m2	2.47E+00	1	2.00	(1,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.47E+00	1	1.50	(1,2,1,1,1,na)
Heat, waste	air/low population density	MJ	8.67E+00	1	1.05	(1,2,1,1,1,na)
shed (l)	CH	m2	1.00E+00			

Unit process inventory for: housing system with fully-slatted floor, pig, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	2.18E+02	1	1.07	(2,2,1,1,1,na)
tap water, at user	RER	kg	2.86E+03	1	1.07	(2,2,1,1,1,na)
lubricating oil, at plant	RER	kg	1.47E-03	1	1.07	(2,2,1,1,1,na)
hydrogen peroxide, 50% in H2O, at plant	RER	kg	1.17E+00	1	1.07	(2,2,1,1,1,na)
transport, lorry 28t	CH	tkm	5.85E-02	1	2.00	(2,2,1,1,1,na)
housing system with fully-slatted floor, pig (l)	CH	pig place	2.00E-02	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	7.86E+02	1	1.07	(2,2,1,1,1,na)
housing system with fully-slatted floor, pig, operation	CH	pig place	1.00E+00			

Unit process inventory for: label housing system, pig, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	2.32E+01	1	1.07	(2,2,1,1,1,na)
tap water, at user	RER	kg	2.86E+03	1	1.07	(2,2,1,1,1,na)
lubricating oil, at plant	RER	kg	1.47E-03	1	1.07	(2,2,1,1,1,na)
hydrogen peroxide, 50% in H2O, at plant	RER	kg	1.17E+00	1	1.07	(2,2,1,1,1,na)
transport, lorry 28t	CH	tkm	5.85E-02	1	2.00	(2,2,1,1,1,na)
label housing system, pig (l)	CH	pig place	2.00E-02	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	8.37E+01	1	1.07	(2,2,1,1,1,na)
label housing system, pig, operation	CH	pig place	1.00E+00			

Unit process inventory for: tied housing system, cattle, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	9.21E+02	1	1.07	(2,2,1,1,1,na)
tap water, at user	RER	kg	4.08E+04	1	1.07	(2,2,1,1,1,na)
lubricating oil, at plant	RER	kg	2.60E-01	1	1.07	(2,2,1,1,1,na)
sodium metasilicate pentahydrate, 58%, powder, at plant	RER	kg	2.49E+00	1	1.07	(2,2,1,1,1,na)
soda, powder, at plant	RER	kg	4.24E+00	1	1.07	(2,2,1,1,1,na)
water, deionised, at plant	CH	kg	2.10E+00	1	1.07	(2,2,1,1,1,na)
chemicals organic, at plant	GLO	kg	4.33E+00	1	1.07	(2,2,1,1,1,na)
transport, lorry 28t	CH	tkm	6.71E-01	1	2.00	(2,2,1,1,1,na)
tied housing system, cattle (l)	CH	LU	2.00E-02	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	3.31E+03	1	1.07	(2,2,1,1,1,na)
tied housing system, cattle, operation	CH	LU	1.00E+00			

Unit process inventory for: loose housing system, cattle, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	9.12E+02	1	1.07	(2,2,1,1,1,na)
tap water, at user	RER	kg	4.02E+04	1	1.07	(2,2,1,1,1,na)
lubricating oil, at plant	RER	kg	2.60E-01	1	1.07	(2,2,1,1,1,na)
sodium metasilicate pentahydrate, 58%, powder, at plant	RER	kg	2.36E+00	1	1.07	(2,2,1,1,1,na)
soda, powder, at plant	RER	kg	4.02E+00	1	1.07	(2,2,1,1,1,na)
water, deionised, at plant	CH	kg	1.99E+00	1	1.07	(2,2,1,1,1,na)
chemicals organic, at plant	GLO	kg	4.11E+00	1	1.07	(2,2,1,1,1,na)
transport, lorry 28t	CH	tkm	6.37E-01	1	2.00	(2,2,1,1,1,na)
loose housing system, cattle (l)	CH	LU	2.00E-02	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	3.28E+03	1	1.07	(2,2,1,1,1,na)
loose housing system, cattle, operation	CH	LU	1.00E+00			

Unit process inventory for: milking, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	4.57E-02	1	1.07	(2,2,1,1,1,na)
tap water, at user	RER	kg	6.29E-01	1	1.07	(2,2,1,1,1,na)
lubricating oil, at plant	RER	kg	2.92E-05	1	1.07	(2,2,1,1,1,na)
sodium metasilicate pentahydrate, 58%, powder, at plant	RER	kg	1.61E-04	1	1.07	(2,2,1,1,1,na)
soda, powder, at plant	RER	kg	2.77E-04	1	1.07	(2,2,1,1,1,na)
water, deionised, at plant	CH	kg	1.37E-04	1	1.07	(2,2,1,1,1,na)
chemicals organic, at plant	GLO	kg	2.84E-04	1	1.07	(2,2,1,1,1,na)
transport, lorry 28t	CH	tkm	4.44E-05	1	2.00	(2,2,1,1,1,na)
milking parlour (l)	CH	unit	1.33E-07	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	1.65E-01	1	1.07	(2,2,1,1,1,na)
milking	CH	kg	1.00E+00			

Unit process inventory for: dried roughage store, cold-air dried, conventional, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	1.31E-01	1	1.07	(2,2,1,1,1,na)
dried roughage store, cold-air dried, conventional (l)	CH	m3	2.53E-04	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	4.70E-01	1	1.07	(2,2,1,1,1,na)
dried roughage store, cold-air dried, conventional, operation	CH	kg	1.00E+00			

Unit process inventory for: dried roughage store, air dried, solar, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	8.91E-02	1	1.07	(2,2,1,1,1,na)
dried roughage store, air dried, solar (l)	CH	m3	2.53E-04	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	3.21E-01	1	1.07	(2,2,1,1,1,na)
Energy, solar	resource/in air	MJ	1.60E+01	1	1.07	(2,2,1,1,1,na)
dried roughage store, air dried, solar, operation	CH	kg	1.00E+00			

Unit process inventory for: dried roughage store, non ventilated, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	5.09E-03	1	1.07	(2,2,1,1,1,na)
dried roughage store, non ventilated (l)	CH	m3	2.65E-04	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	1.83E-02	1	1.07	(2,2,1,1,1,na)
dried roughage store, non ventilated, operation	CH	kg	1.00E+00			

Unit process inventory for: slurry store and processing, operation, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	3.75E-01	1	1.07	(2,2,1,1,1,na)
slurry store and processing (l)	CH	m3	3.47E-05	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	1.35E+00	1	1.07	(2,2,1,1,1,na)
slurry store and processing, operation	CH	m3	1.00E+00			

Appendix A6 to Chapter 6 (Agricultural Machinery)

Unit-Process Inventories from Chapter 6 (Agricultural Machinery)

(Last Changes 2004)

Tab. A. 8 Unit-process inventories for the six classes of agricultural machinery.

Unit process inventory for: tractor, production, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
steel, converter, unalloyed, at plant	RER	kg	8.06E-01	1	1.21	(4,2,1,2,1,na)
steel, low-alloyed, at plant	RER	kg	8.40E-02	1	1.21	(4,2,1,2,1,na)
chromium steel 18/8, at plant	RER	kg	3.60E-02	1	1.21	(4,2,1,2,1,na)
aluminium, production mix, at plant	RER	kg	5.40E-02	1	1.21	(4,2,1,2,1,na)
copper, at regional storage	RER	kg	2.40E-02	1	1.21	(4,2,1,2,1,na)
zinc for coating, at regional storage	RER	kg	1.20E-02	1	1.21	(4,2,1,2,1,na)
lead, at regional storage	RER	kg	4.00E-02	1	1.21	(4,2,1,2,1,na)
alkyd paint, white, 60% in solvent, at plant	RER	kg	6.00E-03	1	1.21	(4,2,1,2,1,na)
flat glass, uncoated, at plant	RER	kg	1.20E-02	1	1.21	(4,2,1,2,1,na)
synthetic rubber, at plant	RER	kg	2.96E-01	1	1.21	(4,2,1,2,1,na)
polypropylene, granulate, at plant	RER	kg	3.90E-02	1	1.21	(4,2,1,2,1,na)
lubricating oil, at plant	RER	kg	2.06E-01	1	1.21	(4,2,1,2,1,na)
paper, woodfree, coated, at integrated mill	RER	kg	6.00E-03	1	1.21	(4,2,1,2,1,na)
electricity, medium voltage, at grid	CH	kWh	2.66E+00	1	1.29	(3,4,3,2,3,na)
natural gas, burned in industrial furnace >100kW	RER	MJ	4.92E+00	1	1.29	(3,4,3,2,3,na)
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	1.39E+01	1	1.29	(3,4,3,2,3,na)
hard coal, burned in industrial furnace 1-10MW	RER	MJ	8.40E-01	1	1.29	(3,4,3,2,3,na)
transport, lorry 40t	CH	tkm	4.40E-01	1	2.06	(4,4,1,2,1,na)
transport, freight, rail	RER	tkm	1.00E-01	1	2.06	(4,4,1,2,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	1.20E-02	1	1.32	(4,4,1,2,3,na)
disposal, paper, 11.2% water, to municipal incineration	CH	kg	6.00E-03	1	1.32	(4,4,1,2,3,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	4.80E-02	1	1.32	(4,4,1,2,3,na)
disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	kg	2.06E-01	1	1.32	(4,4,1,2,3,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	1.15E-02	1	1.61	(3,4,3,2,3,na)
Heat, waste	air/high population density	MJ	9.35E+00	1	1.29	(3,4,3,2,3,na)
Carbon dioxide, fossil	air/low population density	kg	6.54E-03	1	1.29	(3,4,3,2,3,na)
tractor, production (I)	CH	kg	1.00E+00			

Unit process inventory for: harvester, production, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
steel, converter, unalloyed, at plant	RER	kg	9.25E-01	1	1.21	(4,2,1,2,1,na)
steel, low-alloyed, at plant	RER	kg	9.30E-02	1	1.21	(4,2,1,2,1,na)
chromium steel 18/8, at plant	RER	kg	4.00E-02	1	1.21	(4,2,1,2,1,na)
aluminium, production mix, at plant	RER	kg	5.90E-02	1	1.21	(4,2,1,2,1,na)
copper, at regional storage	RER	kg	2.60E-02	1	1.21	(4,2,1,2,1,na)
zinc for coating, at regional storage	RER	kg	1.30E-02	1	1.21	(4,2,1,2,1,na)
lead, at regional storage	RER	kg	1.80E-02	1	1.21	(4,2,1,2,1,na)
alkyd paint, white, 60% in solvent, at plant	RER	kg	7.00E-03	1	1.21	(4,2,1,2,1,na)
flat glass, uncoated, at plant	RER	kg	1.30E-02	1	1.21	(4,2,1,2,1,na)
synthetic rubber, at plant	RER	kg	1.06E-01	1	1.21	(4,2,1,2,1,na)
polypropylene, granulate, at plant	RER	kg	4.00E-02	1	1.21	(4,2,1,2,1,na)
lubricating oil, at plant	RER	kg	1.30E-02	1	1.21	(4,2,1,2,1,na)
paper, woodfree, coated, at integrated mill	RER	kg	1.00E-03	1	1.21	(4,2,1,2,1,na)
electricity, medium voltage, at grid	CH	kWh	2.23E+00	1	1.29	(3,4,3,2,3,na)
natural gas, burned in industrial furnace >100kW	RER	MJ	4.92E+00	1	1.29	(3,4,3,2,3,na)
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	8.33E+00	1	1.29	(3,4,3,2,3,na)
hard coal, burned in industrial furnace 1-10MW	RER	MJ	8.40E-01	1	1.29	(3,4,3,2,3,na)
transport, lorry 40t	CH	tkm	4.40E-01	1	2.06	(4,4,1,2,1,na)
transport, freight, rail	RER	tkm	1.00E-01	1	2.06	(4,4,1,2,1,na)
disposal, building, glass sheet, to final disposal	CH	kg	1.30E-02	1	1.32	(4,4,1,2,3,na)
disposal, paper, 11.2% water, to municipal incineration	CH	kg	1.00E-03	1	1.32	(4,4,1,2,3,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	5.03E-02	1	1.32	(4,4,1,2,3,na)
disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	kg	1.30E-02	1	1.32	(4,4,1,2,3,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	8.64E-03	1	1.61	(3,4,3,2,3,na)
Heat, waste	air/high population density	MJ	7.46E+00	1	1.29	(3,4,3,2,3,na)
Carbon dioxide, fossil	air/low population density	kg	7.00E-06	1	1.29	(3,4,3,2,3,na)
harvester, production (I)	CH	kg	1.00E+00			

Unit process inventory for: trailer, production, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
steel, converter, unalloyed, at plant	RER	kg	7.75E-01	1	1.21	(4,2,1,2,1,na)
steel, low-alloyed, at plant	RER	kg	3.30E-02	1	1.21	(4,2,1,2,1,na)
chromium steel 18/8, at plant	RER	kg	2.20E-02	1	1.21	(4,2,1,2,1,na)
aluminium, production mix, at plant	RER	kg	2.10E-01	1	1.21	(4,2,1,2,1,na)
brass, at plant	CH	kg	6.00E-03	1	1.21	(4,2,1,2,1,na)
alkyd paint, white, 60% in solvent, at plant	RER	kg	6.00E-03	1	1.21	(4,2,1,2,1,na)
synthetic rubber, at plant	RER	kg	8.40E-02	1	1.21	(4,2,1,2,1,na)
electricity, medium voltage, at grid	CH	kWh	1.62E+00	1	1.29	(3,4,3,2,3,na)
natural gas, burned in industrial furnace >100kW	RER	MJ	4.10E+00	1	1.29	(3,4,3,2,3,na)
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	3.59E+00	1	1.29	(3,4,3,2,3,na)
hard coal, burned in industrial furnace 1-10MW	RER	MJ	7.00E-01	1	1.29	(3,4,3,2,3,na)
transport, lorry 40t	CH	tkm	4.40E-01	1	2.06	(4,4,1,2,1,na)
transport, freight, rail	RER	tkm	1.00E-01	1	2.06	(4,4,1,2,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	8.77E-03	1	1.32	(4,4,1,2,3,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	5.48E-03	1	1.61	(3,4,3,2,3,na)
Heat, waste	air/high population density	MJ	5.88E+00	1	1.29	(3,4,3,2,3,na)
Carbon dioxide, fossil	air/low population density	kg	6.00E-06	1	1.29	(3,4,3,2,3,na)
trailer, production (I)	CH	kg	1.00E+00			

Unit process inventory for: agricultural machinery, general, production, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
steel, converter, unalloyed, at plant	RER	kg	1.14E+00	1	1.21	(4,2,1,2,1,na)
steel, low-alloyed, at plant	RER	kg	8.00E-02	1	1.21	(4,2,1,2,1,na)
chromium steel 18/8, at plant	RER	kg	6.70E-02	1	1.21	(4,2,1,2,1,na)
brass, at plant	CH	kg	7.00E-03	1	1.21	(4,2,1,2,1,na)
alkyd paint, white, 60% in solvent, at plant	RER	kg	7.00E-03	1	1.21	(4,2,1,2,1,na)
synthetic rubber, at plant	RER	kg	4.00E-02	1	1.21	(4,2,1,2,1,na)
electricity, medium voltage, at grid	CH	kWh	1.95E+00	1	1.29	(3,4,3,2,3,na)
natural gas, burned in industrial furnace >100kW	RER	MJ	4.10E+00	1	1.29	(3,4,3,2,3,na)
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	7.88E+00	1	1.29	(3,4,3,2,3,na)
hard coal, burned in industrial furnace 1-10MW	RER	MJ	7.00E-01	1	1.29	(3,4,3,2,3,na)
transport, lorry 40t	CH	tkm	4.40E-01	1	2.06	(4,4,1,2,1,na)
transport, freight, rail	RER	tkm	1.00E-01	1	2.06	(4,4,1,2,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	1.03E-02	1	1.32	(4,4,1,2,3,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	7.68E-03	1	1.61	(3,4,3,2,3,na)
Heat, waste	air/high population density	MJ	6.10E+00	1	1.29	(3,4,3,2,3,na)
Carbon dioxide, fossil	air/low population density	kg	7.00E-06	1	1.29	(3,4,3,2,3,na)
agricultural machinery, general, production (I)	CH	kg	1.00E+00			

Unit process inventory for: agricultural machinery, tillage, production, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
steel, converter, unalloyed, at plant	RER	kg	1.22E+00	1	1.21	(4,2,1,2,1,na)
steel, low-alloyed, at plant	RER	kg	7.30E-02	1	1.21	(4,2,1,2,1,na)
chromium steel 18/8, at plant	RER	kg	1.45E-01	1	1.21	(4,2,1,2,1,na)
brass, at plant	CH	kg	7.00E-03	1	1.21	(4,2,1,2,1,na)
alkyd paint, white, 60% in solvent, at plant	RER	kg	3.50E-03	1	1.21	(4,2,1,2,1,na)
synthetic rubber, at plant	RER	kg	3.50E-03	1	1.21	(4,2,1,2,1,na)
electricity, medium voltage, at grid	CH	kWh	2.14E+00	1	1.29	(3,4,3,2,3,na)
natural gas, burned in industrial furnace >100kW	RER	MJ	4.10E+00	1	1.29	(3,4,3,2,3,na)
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	1.03E+01	1	1.29	(3,4,3,2,3,na)
hard coal, burned in industrial furnace 1-10MW	RER	MJ	7.00E-01	1	1.29	(3,4,3,2,3,na)
transport, lorry 40t	CH	tkm	4.40E-01	1	2.06	(4,4,1,2,1,na)
transport, freight, rail	RER	tkm	1.00E-01	1	2.06	(4,4,1,2,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	8.00E-03	1	1.32	(4,4,1,2,3,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	8.92E-03	1	1.61	(3,4,3,2,3,na)
Heat, waste	air/high population density	MJ	6.52E+00	1	1.29	(3,4,3,2,3,na)
Carbon dioxide, fossil	air/low population density	kg	4.00E-06	1	1.29	(3,4,3,2,3,na)
agricultural machinery, tillage, production (I)	CH	kg	1.00E+00			

Unit process inventory for: slurry tanker, production, CH (Infrastructure)						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
steel, converter, unalloyed, at plant	RER	kg	9.90E-01	1	1.21	(4,2,1,2,1,na)
steel, low-alloyed, at plant	RER	kg	6.00E-02	1	1.21	(4,2,1,2,1,na)
chromium steel 18/8, at plant	RER	kg	6.00E-02	1	1.21	(4,2,1,2,1,na)
brass, at plant	CH	kg	6.00E-03	1	1.21	(4,2,1,2,1,na)
zinc for coating, at regional storage	RER	kg	2.40E-02	1	1.21	(4,2,1,2,1,na)
alkyd paint, white, 60% in solvent, at plant	RER	kg	6.00E-03	1	1.21	(4,2,1,2,1,na)
synthetic rubber, at plant	RER	kg	6.00E-02	1	1.21	(4,2,1,2,1,na)
electricity, medium voltage, at grid	CH	kWh	1.74E+00	1	1.29	(3,4,3,2,3,na)
natural gas, burned in industrial furnace >100kW	RER	MJ	4.10E+00	1	1.29	(3,4,3,2,3,na)
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	5.15E+00	1	1.29	(3,4,3,2,3,na)
hard coal, burned in industrial furnace 1-10MW	RER	MJ	7.00E-01	1	1.29	(3,4,3,2,3,na)
transport, lorry 40t	CH	tkm	4.40E-01	1	2.06	(4,4,1,2,1,na)
transport, freight, rail	RER	tkm	1.00E-01	1	2.06	(4,4,1,2,1,na)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	9.00E-03	1	1.32	(4,4,1,2,3,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	6.28E-03	1	1.61	(3,4,3,2,3,na)
Heat, waste	air/high population density	MJ	5.73E+00	1	1.29	(3,4,3,2,3,na)
Carbon dioxide, fossil	air/low population density	kg	6.00E-06	1	1.29	(3,4,3,2,3,na)
slurry tanker, production (l)	CH	kg	1.00E+00			

Appendix A7 to Chapter 7 (Agricultural Field Work Processes)

Name, Description and Fuel Consumption of the Work Processes

Tab. A. 9 ecoinvent module name, description, functional unit and operation time of the different inventoried work processes.

Name	Description	Function al Unit (FU)	Operation time ^a [h/FU]	Loca-tion
application of plant protection products, by field sprayer	Field sprayer, 15 m working width, 800 l carrying capacity, plant protection product not included	ha	0.70	CH
baling	Round baler for round bales of 1.4 m ³ , silage with wrapping foil, 700 kg. Time need for baling and wrapping, without loading and transport. Wrapping foil (PE-film) included. Fodder production and cutting not included. The module is calculated for silage baling. For baling of straw or hay, a rough estimation is possible: baling hay or straw = 0.23 * baling silage.	Ba ^c	0.13	CH
irrigating	Overhead watering of one ha during one year (4 times 300 m ³ water). Mobile sprinkler system, with fix installed pump (30 m ³ /h, 7-8 bar, 22 kW), water pipe and hydrant, turbine propulsion, 300 m water hose, exterior diameter 75 mm. Water amount of 1200 m ³ per ha and year included.	ha or m3 water use	4 × 0.25 tractor 4 × 10 automaton	CH
combine harvesting	Combine harvesting, working width 4.5 m. Grain production not included.	ha	1.30	CH
fertilising, by broadcaster	Fertiliser broadcaster, 500 l carrying capacity. Fertiliser not included.	ha	1.50	CH
hoeing	Inter-row hoeing roller, 4-row, working width 3 m	ha	1.00	CH
harvesting, by complete harvester, potatoes	Complete harvester, medium, moving apron bunker, 1-row. Potato production not included.	ha	13.40	CH
slurry spreading, by vacuum tanker	Slurry spreading with vacuum slurry tank 5000 l carrying capacity. Incl. pumping from slurry container at farm. Slurry and emissions from slurry not included.	m ³	0.06	CH
potato planting	Four-row potato planter. Driver plus 4 persons as crew. Planting material not included.	ha	5.30	CH
mowing, by motor mower	Motor mower, working width 1.9 m. Petrol engine 8 kW.	ha	2.00	CH
chopping, maize	Includes chopping maize and transport to the farm. Tractor-mounted maize chopper, 2-rows, working width 1.5m, 3 Th ^b plus 2 tractors with tyre trailer, 2 Th either, maize production not included	ha	7.00	CH

Name	Description	Function al Unit (FU)	Operation time ^a [h/FU]	Loca- tion
solid manure loading and spreading, by hydraulic loader and spreader	Solid manure loading with hydraulic loader (power take-off driven) from dung slab on manure spreader. Solid manure spreading with manure spreader. Working load 5 t. Solid manure production and emissions from slurry and manure not included.	kg	0.00019	CH
mulching	Flail mulcher, working width 2.5 m. Labour needs given for pasture. In special crops clearly higher (2-4 Th/ha).	ha	1.10	CH
sowing	Seeder, working width 3m. Seed not included.	ha	1.30	CH
planting	Two-row planter. Driver plus 2 persons as crew. Operation time very variable, depending on crop, stocking density; typical values are between 5 and 30 Th/ha. Planting material not included.	ha	10.00	CH
tillage, harrowing, by spring tine harrow	Spring tine harrow, working width 3 m	ha	0.80	CH
tillage, rotary cultivator	Rotary cultivator, working width 2.5 m	ha	1.50	CH
tillage, cultivating, chiselling	Working width 2.5 m	ha	1.20	CH
tillage, ploughing	Four-furrow plough	ha	2.10	CH
tillage, rolling	Rolling, working width 3 m	ha	0.90	CH
transport, tractor and trailer	Heavy road transport with tractor and 2 tyre-trailers of max. 8t loading capacity each. Mean velocity when loaded = 15 km/h. Mean velocity when empty = 25 km/h. Empty return over the same distance included.	tkm	0.0071	CH
harvesting, by complete harvester, beets	Complete harvester, big, with bunker, 1-row, without leaf salvage. "Medium" module for sugar beets (6.8 Th/ha) and fodder beets (7.7 Th/ha). For sugar beets 1-row complete harvester is rare, more frequent is the 6-row self driver. Beet production not included.	ha	8.00	CH
potato haulm cutting	Potato haulm cutter, working width 1.8 m, 2-rows	ha	1.50	CH
mowing, by rotary mower	Rotary mower, working width 3 m	ha	0.90	CH
haymaking, by rotary tedder	Rotary tedder, working width 5.5 m	ha	0.60	CH
swath, by rotary windrower	Rotary windrower, working width 3 m	ha	1.00	CH
fodder loading, by self-loading trailer	Self-loading trailer with cutter, 22 m ³ loading volume. Includes loading, transport and fast discharge. Time need variable depending on loaded good. Assumption: loading trailer is filled up completely. Fodder production, cutting, swath, etc. not included.	m ³	0.02	CH
loading bales	Loading of straw bales with bale gripper onto trailer. Without transport to farm and discharging. Straw production not included.	Ba	0.023	CH

Name	Description	Functional Unit (FU)	Operation time ^a [h/FU]	Location
tillage, hoeing and earthing-up, potatoes	Four-row potato lister cultivator	ha	1.10	CH
tillage, currying, by weeder	Weeder, working width 6 m	ha	0.50	CH
tillage, harrowing, by rotary harrow	Rotary harrow, working width 3 m	ha	1.20	CH
potato grading	Potato grader (2t potatoes/h), palox handler and weigh. FU is 1 kg ungraded potatoes.	kg	0.50	CH

a Personal communication in July 2002 from R. Stark. FAT, Ettenhausen, Switzerland.

b Th = tractor hour

c Ba = bale

Tab. A. 10 Characteristics of the employed agricultural machinery, fuel consumption and emission for the work processes.

Module-name ^a	Vehicles						Machines				Fuel consumption ^c		Emission per hour ^{c, d}		
	Class	Weight ^b [kg]	Space requirement ^b [m ²]	Annual employment ^b [h/a]	Nominal power ^c [kW]	Mean power ^c [kW]	Class	Weight ^b [kg]	Space requirement ^b [m ²]	Annual employment ^b [h/a]	per hour [l/h]	per FU ^a [kg/FU]	HC [g/h]	NO _x [g/h]	CO [g/h]
application of plant protection products, by field sprayer	Tractor	3300	15.2	600	50	4.1	AM - general	477	6.1	75	3	1.76	10	113	14
baling	Tractor	3700	15.8	600	62	17.9	AM - general	2286	13.4	60	6.8	0.74	14	237	30
irrigating (per ha)	Tractor	3300	15.2	600	50	9.7	AM - general	1300	14.6	200	4.5	3.78	17	193	36
combine harvesting	Combine harvester	9445	53.6	125	150	105.0	–	–	–	114	30.5	33.31	4.5	51	9.6
fertilising, by broadcaster	Tractor	3300	15.2	600	50	8.6	AM - general	193	3.8	80	4.2	5.29	10	154	14
hoeing	Tractor	3300	15.2	600	50	7.3	AM - tillage	559	12.2	29	3.9	3.28	12	144	17
harvesting, by complete harvester, potatoes	Tractor	3300	15.2	600	50	2.6	AM - general	3198	24.4	143	2.5	28.14	13	121	24
slurry spreading, by vacuum tanker	Tractor	3300	15.2	600	50	8.9	Slurry tankers	1690	19.9	167	4.3	0.22	12	156	26
potato planting	Tractor	3300	15.2	600	50	0.6	AM - general	447	7.9	60	2	8.90	7	76	10
mowing, by motor mower	Tractor	370	7.8	70	8	0.8	–	–	–	70	2	3.40	35	30	950
chopping, maize	Tractor	11100	47.4	600	3 × 62	34 + 2 × 20	AM - general	9515	102.6	53 + 2 × 80	–	52.75	46	836	139
solid manure loading and spreading, by hydraulic loader and spreader	Tractor	7000	31	600	50 + 62	2 + 18	AM - general	3848	29.7	115 + 100	–	5.31E-04	28	367	53
mulching	Tractor	3300	15.2	600	50	7.3	AM - tillage	664	8.1	63	3.8	3.51	11	143	15
sowing	Tractor	3300	15.2	600	50	6.1	AM - general	507	8.4	29	3.5	3.82	10	131	11
planting	Tractor	3300	15.2	600	50	0.6	AM - general	175	5	143	2	16.80	7	76	10
tillage, harrowing, by spring tine harrow	Tractor	3700	15.8	600	62	17.2	AM - tillage	538	12.2	28	6.6	4.44	15	247	33

Module-name ^a	Vehicles						Machines				Fuel consumption ^c		Emission per hour ^{c, d}		
	Class	Weight ^b [kg]	Space requirement ^b [m ²]	Annual employment ^b [h/a]	Nominal power ^c [kW]	Mean power ^c [kW]	Class	Weight ^b [kg]	Space requirement ^b [m ²]	Annual employment ^b [h/a]	per hour [l/h]	per FU ^a [kg/FU]	HC [g/h]	NO _x [g/h]	CO [g/h]
tillage, rotary cultivator	Tractor	3300	15.2	600	50	34.4	AM - tillage	902	6.8	31	11.2	14.11	25	387	126
tillage, cultivating, chiselling	Tractor	5300	19.4	600	78	48.9	AM - tillage	709	9.9	44	15.4	15.52	27	507	108
tillage, ploughing	Tractor	5300	19.4	600	78	46.5	AM - tillage	1039	13.3	73	14.8	26.11	26	490	99
tillage, rolling	Tractor	3300	15.2	600	50	8.6	AM - tillage	910	6.1	21	4.2	3.18	10	154	14
transport, tractor and trailer	Tractor	3700	15.8	600	62	19.8	Trailers	4084	43.6	80	7.3	0.04	14	239	36
harvesting, by complete harvester, beets	Tractor	5300	19.4	600	78	48.9	AM - general	2660	22	200	15.4	103.49	27	507	108
potato haulm cutting	Tractor	3300	15.2	600	50	7.3	AM - general	540	7	24	3.8	4.79	11	143	15
mowing, by rotary mower	Tractor	3300	15.2	600	50	14.1	AM - general	643	6.1	28	5.7	4.31	12	186	19
haymaking, by rotary tedder	Tractor	3300	15.2	600	50	7.3	AM - general	552	12.6	60	3.8	1.92	11	143	15
swath, by rotary windrower	Tractor	3300	15.2	600	50	6.1	AM - general	315	12.2	54	3.5	2.94	11	133	16
fodder loading, by self-loading trailer	Tractor	3700	15.8	600	62	16.3	AM - general	3035	27.3	100	6.3	0.11	13	214	23
loading bales	Tractor	3300	15.2	600	50	8.6	AM - general	450	13.4	112	4.2	0.08	12	142	18
tillage, hoeing and earthing-up, potatoes	Tractor	3300	15.2	600	50	7.3	AM - tillage	495	10.2	44	3.9	3.60	12	144	17
tillage, currying, by weeder	Tractor	3300	15.2	600	50	7.3	AM - tillage	475	8.1	25	3.8	1.60	11	143	15
tillage, harrowing, by rotary harrow	Tractor	3700	15.8	600	62	34.4	AM - tillage	1183	7	32	11.4	11.49	17	389	77
potato grading	—	-	-	-	-	-	AM – general	962	20.2	200	- ^e	- ^e	-	-	-

a See Tab. A. 9 for corresponding module description

b Ammann (2001), Ammann (2002) and personal communication in September 2002 from H. Ammann. FAT, Ettenhausen, Switzerland.

c Personal communication in July 2002 from M. Rinaldi. FAT, Ettenhausen, Switzerland.

d For calculation of other air emissions see chapter 7.2.6.

e Electricity usage: 2 kWh/hour, 1 kWh/ton of ungraded potatoes.

Unit-Process Inventories from Chapter 7 (Agricultural Field Work Processes)

(Last Changes 2004)

Tab. A. 11 Unit-process inventories for agricultural field work processes.

Unit process inventory for: application of plant protection products, by field sprayer, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	3.21E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	5.30E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.76E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.98E-03	1	3.01	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	6.75E-03	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	7.91E-02	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	9.80E-03	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	5.49E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.78E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	2.28E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.29E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	4.63E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.76E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	8.82E-08	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	3.00E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	2.12E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.23E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.76E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	5.29E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	5.80E-06	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	8.01E+01	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	4.60E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	7.80E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.75E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	3.53E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.76E-08	1	1.56	(1,2,1,1,3,na)
application of plant protection products, by field sprayer	CH	ha	1.00E+00			

Unit process inventory for: baling, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	6.68E-02	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.52E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	7.43E-01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	2.47E-04	1	3.01	(1,4,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	1.00E+00	1	1.11	(1,4,1,1,1,na)
extrusion, plastic film	RER	kg	1.00E+00	1	1.11	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.72E-03	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.08E-02	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.90E-03	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	2.31E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	7.49E-04	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	9.58E-05	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	5.42E-06	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	3.53E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	7.43E-09	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	3.71E-08	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	1.26E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	8.91E-05	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	5.20E-08	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	7.43E-07	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	2.23E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	2.44E-06	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	3.37E+01	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	9.98E-05	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.71E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	3.82E-08	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.49E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	7.43E-09	1	1.56	(1,2,1,1,3,na)
baling	CH	unit	1.00E+00			

Unit process inventory for: irrigating, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	4.58E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	2.17E+01	1	1.13	(2,4,2,1,1,na)
diesel, at regional storage	CH	kg	3.78E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	5.89E-02	1	3.01	(1,4,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	2.31E+01	1	1.11	(1,4,1,1,1,na)
extrusion, plastic film	RER	kg	2.42E+01	1	1.11	(1,4,1,1,1,na)
Water, river	resource/in water	m3	1.20E+03	1	1.11	(1,4,1,1,1,na)
excavation, hydraulic digger	RER	m3	4.00E+00	1	1.11	(1,4,1,1,1,na)
cast iron, at plant	RER	kg	4.27E+00	1	1.11	(1,4,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	8.76E+02	1	1.11	(1,4,1,1,1,na)
polyvinylchloride, bulk polymerised, at plant	RER	kg	1.13E+00	1	1.11	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.65E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.93E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.63E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.17E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.81E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.88E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.76E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.56E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.78E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.89E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	6.43E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	4.54E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.65E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.78E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.13E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.24E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	3.33E+03	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	2.71E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	5.30E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.09E-06	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	7.56E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.78E-08	1	1.56	(1,2,1,1,3,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	4.27E-03	1	1.11	(1,4,1,1,1,na)
disposal, building, polyvinylchloride products, to final disposal	CH	kg	1.13E-03	1	1.11	(1,4,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	6.67E-03	1	1.11	(1,4,1,1,1,na)
Occupation, construction site	resource/land	m2a	6.67E+00	1	1.52	(1,4,1,1,1,na)
irrigating	CH	ha	1.00E+00			

Unit process inventory for: irrigating, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
irrigating	CH	ha	8.33E-04	1	1.00	
irrigating	CH	m3	1.00E+00			

Unit process inventory for: combine harvesting, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
harvester, production (I)	CH	kg	6.30E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	3.33E+01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	8.58E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.45E-01	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.70E+00	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.20E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.03E+02	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.36E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.30E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.43E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.49E-01	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.33E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.67E-06	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	5.66E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	4.00E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.33E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.33E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	9.99E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.10E-04	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.51E+03	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	2.38E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	3.78E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	8.82E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	6.66E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.33E-07	1	1.56	(1,2,1,1,3,na)
combine harvesting	CH	ha	1.00E+00			

Unit process inventory for: fertilising, by broadcaster, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	6.88E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	2.41E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	5.29E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.71E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.43E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.31E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	2.10E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.65E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	5.33E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	6.83E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.86E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	2.08E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	5.29E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.65E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	9.00E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	6.35E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.70E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	5.29E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.59E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.74E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	2.40E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	8.97E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.49E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	3.40E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.06E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	5.29E-08	1	1.56	(1,2,1,1,3,na)
fertilising, by broadcaster	CH	ha	1.00E+00			

Unit process inventory for: hoeing, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	4.58E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (I)	CH	kg	9.32E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	3.28E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	6.61E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.15E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.44E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.70E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.02E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.30E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.23E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.39E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.19E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.28E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.64E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	5.57E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	3.93E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.29E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.28E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	9.83E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.08E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.49E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	5.83E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	9.63E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.20E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	6.55E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.28E-08	1	1.56	(1,2,1,1,3,na)
hoeing	CH	ha	1.00E+00			

Unit process inventory for: harvesting, by complete harvester, potatoes, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	6.14E+00	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	2.67E+01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	2.81E+01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	5.56E-02	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.70E-01	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.62E+00	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.22E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	8.74E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	2.84E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	3.63E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.05E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	5.67E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	2.81E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.41E-06	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	4.78E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	3.38E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.97E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	2.81E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	8.44E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	9.26E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.28E+03	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.04E-02	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.82E-05	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	4.01E-06	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	5.63E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	2.81E-07	1	1.56	(1,2,1,1,3,na)
harvesting, by complete harvester, potatoes	CH	ha	1.00E+00			

Unit process inventory for: slurry spreading, by vacuum tanker, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	2.75E-02	1	1.11	(1,4,1,1,1,na)
slurry tanker, production (I)	CH	kg	5.63E-02	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	2.17E-01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.90E-04	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	6.90E-04	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	9.36E-03	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.56E-03	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	6.74E-01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	2.18E-04	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	2.80E-05	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.58E-06	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	8.62E-04	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	2.17E-09	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.08E-08	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	3.68E-07	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	2.60E-05	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.52E-08	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	2.17E-07	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	6.50E-09	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	7.13E-07	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	9.84E+00	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	5.59E-05	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	9.16E-08	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.11E-08	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	4.33E-06	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	2.17E-09	1	1.56	(1,2,1,1,3,na)
slurry spreading, by vacuum tanker	CH	m3	1.00E+00			

Unit process inventory for: potato planting, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	2.43E+00	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.99E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	8.90E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.32E-02	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	3.59E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.03E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	5.30E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	2.77E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	8.98E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	1.15E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	6.50E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	4.82E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	8.90E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	4.45E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	1.51E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.07E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	6.23E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	8.90E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	2.67E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	2.93E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	4.04E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	3.28E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	5.50E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.25E-06	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.78E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	8.90E-08	1	1.56	(1,2,1,1,3,na)
potato planting	CH	ha	1.00E+00			

Unit process inventory for: mowing, by motor mower, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	8.81E-01	1	1.11	(1,4,1,1,1,na)
petrol, two-stroke blend, at regional storage	CH	kg	3.00E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	4.46E-03	1	3.01	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	3.28E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.00E-02	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.90E+00	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	7.10E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	2.16E-04	1	1.21	(1,2,1,1,3,na)
Lead	air/low population density	kg	4.38E-04	1	5.05	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	8.76E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.84E-02	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	7.97E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.00E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.50E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	5.10E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	3.90E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.10E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.00E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.20E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	5.78E-06	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.35E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.12E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.85E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	4.23E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.20E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.00E-08	1	1.56	(1,2,1,1,3,na)
mowing, by motor mower	CH	ha	1.00E+00			

Unit process inventory for: chopping, maize, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	3.60E+00	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	8.42E+00	1	1.11	(1,4,1,1,1,na)
trailer, production (I)	CH	kg	1.36E+01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	5.28E+01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	6.65E-02	1	3.01	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.03E-01	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.03E+00	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.45E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.64E+02	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	5.32E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	6.81E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.85E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	2.74E-01	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	5.28E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.64E-06	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	8.97E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	6.33E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.69E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	5.28E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.58E-06	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.74E-04	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	2.39E+03	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.46E-02	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	2.42E-05	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	5.47E-06	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.06E-03	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	5.28E-07	1	1.56	(1,2,1,1,3,na)
chopping, maize	CH	ha	1.00E+00			

Unit process inventory for: solid manure loading and spreading, by hydraulic loader and spreader, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	5.88E-05	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.89E-04	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	5.31E-04	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	4.40E-07	1	3.01	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.59E-06	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.41E-05	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.17E-06	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.65E-03	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	5.35E-07	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	6.85E-08	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.88E-09	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	2.25E-06	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	5.31E-12	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.65E-11	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	9.03E-10	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	6.37E-08	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.72E-11	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	5.31E-10	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.59E-11	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.75E-09	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	2.41E-02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	9.33E-08	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.61E-10	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	3.58E-11	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.06E-08	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	5.31E-12	1	1.56	(1,2,1,1,3,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.00E+00			

Unit process inventory for: mulching, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	5.04E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.11E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	3.51E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	3.80E-03	1	3.01	(1,4,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.16E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.57E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.65E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.09E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.54E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.53E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.56E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.29E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.51E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.76E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	5.97E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	4.21E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.46E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.51E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.05E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.16E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.59E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	7.49E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.28E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.86E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	7.02E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.51E-08	1	1.56	(1,2,1,1,3,na)
mulching	CH	ha	1.00E+00			

Unit process inventory for: sowing, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	5.96E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	9.66E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	3.82E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	5.46E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.25E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.70E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.43E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.19E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.85E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.93E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.79E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.28E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.82E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.91E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	6.50E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	4.59E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.68E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.82E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.15E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.26E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.74E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	8.52E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.44E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	3.25E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	7.64E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.82E-08	1	1.56	(1,2,1,1,3,na)
sowing	CH	ha	1.00E+00			

Unit process inventory for: planting, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	4.58E+00	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.17E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.68E+01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.51E-02	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	6.77E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	7.60E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.00E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	5.23E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.69E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	2.17E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.23E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	9.10E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.68E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	8.40E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	2.86E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	2.02E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.18E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.68E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	5.04E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	5.53E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	7.63E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	5.94E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	9.86E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.25E-06	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	3.36E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.68E-07	1	1.56	(1,2,1,1,3,na)
planting	CH	ha	1.00E+00			

Unit process inventory for: tillage, harrowing, by spring tine harrow, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (l)	CH	kg	4.11E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (l)	CH	kg	9.96E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	4.44E+00	1	1.11	(1,4,1,1,1,na)
shed (l)	CH	m2	5.84E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.14E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.98E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	2.64E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.38E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	4.47E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	5.72E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.24E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	2.09E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	4.44E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.22E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	7.54E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	5.32E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.10E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	4.44E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.33E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.46E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	2.01E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	5.23E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	8.63E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.97E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	8.87E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	4.44E-08	1	1.56	(1,2,1,1,3,na)
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00			

Unit process inventory for: tillage, rotary cultivator, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (l)	CH	kg	6.88E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (l)	CH	kg	3.76E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.41E+01	1	1.11	(1,4,1,1,1,na)
shed (l)	CH	m2	7.56E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	3.55E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	5.81E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.89E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.38E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.42E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	1.82E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.03E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	8.35E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.41E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	7.06E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	2.40E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.69E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	9.88E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.41E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	4.23E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	4.64E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	6.41E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	8.74E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.44E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	3.30E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	2.82E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.41E-07	1	1.56	(1,2,1,1,3,na)
tillage, rotary cultivator	CH	ha	1.00E+00			

Unit process inventory for: tillage, cultivating, chiselling, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (l)	CH	kg	8.83E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (l)	CH	kg	1.48E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.55E+01	1	1.11	(1,4,1,1,1,na)
shed (l)	CH	m2	5.73E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	3.02E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.08E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.30E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.83E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.56E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	2.00E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.13E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	8.28E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.55E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	7.76E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	2.64E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.86E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.09E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.55E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	4.66E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	5.11E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	7.05E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.12E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.86E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	4.24E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	3.10E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.55E-07	1	1.56	(1,2,1,1,3,na)
tillage, cultivating, chiselling	CH	ha	1.00E+00			

Unit process inventory for: tillage, ploughing, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (l)	CH	kg	1.55E+00	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (l)	CH	kg	2.16E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	2.61E+01	1	1.11	(1,4,1,1,1,na)
shed (l)	CH	m2	8.01E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	5.10E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.03E+00	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	2.08E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	8.12E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	2.63E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	3.37E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.91E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.38E-01	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	2.61E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.31E-06	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	4.44E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	3.13E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.83E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	2.61E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	7.83E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	8.59E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.19E+03	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.96E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	3.25E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	7.42E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	5.22E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	2.61E-07	1	1.56	(1,2,1,1,3,na)
tillage, ploughing	CH	ha	1.00E+00			

Unit process inventory for: tillage, rolling, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (l)	CH	kg	4.13E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (l)	CH	kg	1.82E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	3.18E+00	1	1.11	(1,4,1,1,1,na)
shed (l)	CH	m2	5.34E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	8.56E-03	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.39E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.26E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	9.88E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.20E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.10E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.32E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.25E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.18E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.59E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	5.40E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	3.81E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.22E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.18E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	9.53E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.04E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.44E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	5.24E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	8.66E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.98E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	6.35E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.18E-08	1	1.56	(1,2,1,1,3,na)
tillage, rolling	CH	ha	1.00E+00			

Unit process inventory for: transport, tractor and trailer, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (l)	CH	kg	3.65E-03	1	1.11	(1,4,1,1,1,na)
trailer, production (l)	CH	kg	2.42E-02	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	4.36E-02	1	1.11	(1,4,1,1,1,na)
shed (l)	CH	m2	8.13E-05	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	9.35E-05	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.70E-03	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	2.56E-04	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.36E-01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	4.40E-05	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	5.63E-06	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.18E-07	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	2.14E-04	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	4.36E-10	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.18E-09	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	7.41E-08	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	5.23E-06	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.05E-09	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	4.36E-08	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.31E-09	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.43E-07	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.98E+00	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	2.10E-05	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	3.43E-08	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	7.80E-09	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	8.72E-07	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	4.36E-10	1	1.56	(1,2,1,1,3,na)
transport, tractor and trailer	CH	tkm	1.00E+00			

Unit process inventory for: harvesting, by complete harvester, beets, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	5.89E+00	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	7.39E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.03E+02	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.98E-02	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	2.02E-01	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.06E+00	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	8.64E-01	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	3.22E+02	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.04E-01	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	1.33E-02	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	7.55E-04	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	5.52E-01	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.03E-06	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	5.17E-06	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	1.76E-04	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.24E-02	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	7.24E-06	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.03E-04	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	3.10E-06	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	3.40E-04	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	4.70E+03	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	8.21E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.38E-05	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	3.12E-06	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	2.07E-03	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.03E-06	1	1.56	(1,2,1,1,3,na)
harvesting, by complete harvester, beets	CH	ha	1.00E+00			

Unit process inventory for: potato haulm cutting, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	6.88E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	3.00E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	4.79E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	1.24E-02	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.58E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.15E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	2.25E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.49E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	4.83E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	6.18E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.50E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.76E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	4.79E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.39E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	8.14E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	5.75E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.35E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	4.79E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.44E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.58E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	2.17E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.17E-03	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	2.04E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	4.50E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	9.58E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	4.79E-08	1	1.56	(1,2,1,1,3,na)
potato haulm cutting	CH	ha	1.00E+00			

Unit process inventory for: mowing, by rotary mower, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	4.13E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.07E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	4.31E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	2.90E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.02E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.67E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.71E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.34E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	4.34E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	5.56E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	3.15E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	2.06E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	4.31E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	2.15E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	7.33E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	5.17E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	3.02E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	4.31E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.29E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.42E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.96E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	6.29E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.08E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.41E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	8.62E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	4.31E-08	1	1.56	(1,2,1,1,3,na)
mowing, by rotary mower	CH	ha	1.00E+00			

Unit process inventory for: haying, by rotary tedder, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	2.75E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	3.83E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.92E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	2.40E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	6.33E-03	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	8.58E-02	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	9.00E-03	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	5.96E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.93E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	2.47E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.40E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	7.05E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.92E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	9.58E-08	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	3.26E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	2.30E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.34E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.92E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	5.75E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	6.30E-06	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	8.70E+01	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	3.87E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	6.54E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.47E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	3.83E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.92E-08	1	1.56	(1,2,1,1,3,na)
haying, by rotary tedder	CH	ha	1.00E+00			

Unit process inventory for: swath, by rotary windrower, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	4.58E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	4.04E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	2.94E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	4.26E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.06E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.33E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.60E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	9.15E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	2.96E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	3.79E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.15E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	9.94E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	2.94E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.47E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	5.00E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	3.53E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.06E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	2.94E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	8.82E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	9.67E-06	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.33E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	6.22E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.04E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.36E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	5.88E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	2.94E-08	1	1.56	(1,2,1,1,3,na)
swath, by rotary windrower	CH	ha	1.00E+00			

Unit process inventory for: fodder loading, by self-loading trailer, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	1.03E-02	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	3.83E-02	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.06E-01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	9.33E-05	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	2.45E-04	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.28E-03	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	4.60E-04	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	3.29E-01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.07E-04	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	1.37E-05	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	7.73E-07	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	4.95E-04	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.06E-09	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	5.29E-09	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	1.80E-07	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.27E-05	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	7.41E-09	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.06E-07	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	3.18E-09	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	3.48E-07	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	4.81E+00	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.68E-05	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	2.92E-08	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	6.47E-09	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	2.12E-06	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.06E-09	1	1.56	(1,2,1,1,3,na)
fodder loading, by self-loading trailer	CH	m3	1.00E+00			

Unit process inventory for: loading bales, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	1.05E-02	1	1.11	(1,4,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	7.70E-03	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	8.11E-02	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	6.67E-05	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	2.65E-04	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.27E-03	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	4.14E-04	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	2.53E-01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	8.18E-05	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	1.05E-05	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	5.92E-07	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	3.21E-04	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	8.11E-10	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	4.06E-09	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	1.38E-07	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	9.74E-06	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	5.68E-09	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	8.11E-08	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	2.43E-09	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	2.67E-07	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	3.68E+00	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	1.42E-05	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	2.37E-08	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	5.37E-09	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	1.62E-06	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	8.11E-10	1	1.56	(1,2,1,1,3,na)
loading bales	CH	unit	1.00E+00			

Unit process inventory for: tillage, hoeing and earthing-up, potatoes, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	5.04E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (I)	CH	kg	9.43E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	3.60E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	6.39E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.27E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.58E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	1.87E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.12E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	3.63E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	4.65E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	2.63E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	1.30E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	3.60E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	1.80E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	6.13E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	4.32E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	2.52E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	3.60E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	1.08E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.19E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	1.64E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	6.41E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.06E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.42E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	7.21E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	3.60E-08	1	1.56	(1,2,1,1,3,na)
tillage, hoeing and earthing-up, potatoes	CH	ha	1.00E+00			

Unit process inventory for: tillage, currying, by weeder, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	2.29E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (I)	CH	kg	6.33E-01	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.60E+00	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	3.49E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	5.28E-03	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	7.15E-02	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	7.50E-03	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.97E+00	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.61E-03	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	2.06E-04	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	1.17E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	5.87E-03	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.60E-08	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	7.98E-08	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	2.71E-06	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.92E-04	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	1.12E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.60E-06	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	4.79E-08	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	5.25E-06	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	7.25E+01	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	2.91E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	4.81E-07	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	1.10E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	3.19E-05	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.60E-08	1	1.56	(1,2,1,1,3,na)
tillage, currying, by weeder	CH	ha	1.00E+00			

Unit process inventory for: tillage, harrowing, by rotary harrow, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	6.17E-01	1	1.11	(1,4,1,1,1,na)
agricultural machinery, tillage, production (I)	CH	kg	3.29E+00	1	1.11	(1,4,1,1,1,na)
diesel, at regional storage	CH	kg	1.15E+01	1	1.11	(1,4,1,1,1,na)
shed (I)	CH	m2	5.30E-03	1	3.01	(1,4,1,1,1,na)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.88E-02	1	1.52	(1,4,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.67E-01	1	1.52	(1,4,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	9.24E-02	1	5.01	(1,4,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	3.57E+01	1	1.21	(1,2,1,1,3,na)
Sulfur dioxide	air/low population density	kg	1.16E-02	1	1.21	(1,2,1,1,3,na)
Methane, fossil	air/low population density	kg	1.48E-03	1	1.56	(1,2,1,1,3,na)
Benzene	air/low population density	kg	8.39E-05	1	1.56	(1,2,1,1,3,na)
Particulates, < 2.5 um	air/low population density	kg	6.27E-02	1	3.05	(1,2,1,1,3,na)
Cadmium	air/low population density	kg	1.15E-07	1	5.05	(1,2,1,1,3,na)
Chromium	air/low population density	kg	5.75E-07	1	5.05	(1,2,1,1,3,na)
Copper	air/low population density	kg	1.95E-05	1	5.05	(1,2,1,1,3,na)
Dinitrogen monoxide	air/low population density	kg	1.38E-03	1	1.56	(1,2,1,1,3,na)
Nickel	air/low population density	kg	8.04E-07	1	5.05	(1,2,1,1,3,na)
Zinc	air/low population density	kg	1.15E-05	1	5.05	(1,2,1,1,3,na)
Benzo(a)pyrene	air/low population density	kg	3.45E-07	1	5.05	(1,2,1,1,3,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	3.78E-05	1	3.05	(1,2,1,1,3,na)
Heat, waste	air/low population density	MJ	5.22E+02	1	1.11	(1,4,1,1,1,na)
Zinc	soil/agricultural	kg	7.84E-04	1	1.52	(1,4,1,1,1,na)
Lead	soil/agricultural	kg	1.30E-06	1	1.52	(1,4,1,1,1,na)
Cadmium	soil/agricultural	kg	2.96E-07	1	1.52	(1,4,1,1,1,na)
Ammonia	air/low population density	kg	2.30E-04	1	1.56	(1,2,1,1,3,na)
Selenium	air/low population density	kg	1.15E-07	1	1.56	(1,2,1,1,3,na)
tillage, harrowing, by rotary harrow	CH	ha	1.00E+00			

Unit process inventory for: potato grading, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
agricultural machinery, general, production (I)	CH	kg	2.16E-04	1	1.07	(1,3,1,1,1,na)
shed (I)	CH	m2	1.09E-06	1	3.00	(1,3,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	1.02E-03	1	1.07	(1,3,1,1,1,na)
Heat, waste	air/low population density	MJ	3.67E-03	1	1.07	(1,3,1,1,1,na)
potato grading	CH	kg	1.00E+00			

Appendix A8 to Chapter 8 (Mineral Fertilisers)

Unit-process inventories from Chapter 8 (Mineral Fertilisers)

(Last Changes 2004)

Tab. A. 12 Unit-process inventories for mineral fertilisers.

Unit process inventory for: ammonium nitrate, as N, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
transport, freight, rail	RER	tkm	1.71E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.86E-01	1	2.09	(4,5,na,na,na,na)
ammonia, steam reforming, liquid, at plant	RER	kg	6.09E-01	1	1.09	(2,1,1,2,1,3)
nitric acid, 50% in H ₂ O, at plant	RER	kg	2.25E+00	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	1.14E-09	1	3.01	(2,1,1,2,1,3)
Ammonium, ion	water/river	kg	7.36E-04	1	1.63	(2,3,2,3,3,5)
Ammonia	air/high population density	kg	5.71E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 µm	air/high population density	kg	5.71E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	2.86E-04	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 µm	air/high population density	kg	5.71E-04	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	-2.40E+00	1	1.31	(2,3,2,3,3,5)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: calcium ammonium nitrate, as N, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
transport, barge	RER	tkm	6.46E-02	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	RER	tkm	2.40E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	4.14E-01	1	2.09	(4,5,na,na,na,na)
ammonia, steam reforming, liquid, at plant	RER	kg	6.08E-01	1	1.09	(2,1,1,2,1,3)
nitric acid, 50% in H ₂ O, at plant	RER	kg	2.25E+00	1	1.09	(2,1,1,2,1,3)
limestone, milled, loose, at plant	CH	kg	9.17E-01	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	1.51E-09	1	3.01	(2,1,1,2,1,3)
Ammonium, ion	water/river	kg	9.63E-04	1	1.63	(2,3,2,3,3,5)
Ammonia	air/high population density	kg	3.21E-03	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 µm	air/high population density	kg	7.46E-05	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	3.73E-05	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 µm	air/high population density	kg	7.46E-05	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	-2.40E+00	1	1.31	(2,3,2,3,3,5)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: urea, as N, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, natural gas, at industrial furnace >100kW	RER	MJ	8.05E+00	1	1.09	(2,1,1,2,1,3)
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	3.20E-01	1	1.09	(2,1,1,2,1,3)
transport, freight, rail	RER	tkm	1.30E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.17E-01	1	2.09	(4,5,na,na,na,na)
ammonia, steam reforming, liquid, at plant	RER	kg	1.23E+00	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	8.70E-10	1	3.01	(2,1,1,2,1,3)
Carbon monoxide, fossil	air/high population density	kg	2.87E-03	1	5.11	(2,3,2,3,3,5)
Methane, fossil	air/high population density	kg	7.83E-04	1	1.63	(2,3,2,3,3,5)
Ammonium, ion	water/river	kg	3.64E-04	1	1.63	(2,3,2,3,3,5)
Ammonia	air/high population density	kg	3.48E-03	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 µm	air/high population density	kg	8.09E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	4.04E-04	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 µm	air/high population density	kg	8.09E-04	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	1.15E+00	1	1.31	(2,3,2,3,3,5)
urea, as N, at regional storehouse	RER	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX8X to Chapter X8X (XMineral FertilisersX)

Unit process inventory for: urea ammonium nitrate, as N, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, natural gas, at industrial furnace >100kW	RER	MJ	1.79E+00	1	1.09	(2,1,1,2,1,3)
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.57E-01	1	1.09	(2,1,1,2,1,3)
transport, freight, rail	RER	tkm	1.88E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	3.13E-01	1	2.09	(4,5,na,na,na,na)
ammonia, steam reforming, liquid, at plant	RER	kg	9.21E-01	1	1.09	(2,1,1,2,1,3)
nitric acid, 50% in H ₂ O, at plant	RER	kg	1.13E+00	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	1.25E-09	1	3.01	(2,1,1,2,1,3)
Carbon monoxide, fossil	air/high population density	kg	1.44E-03	1	5.11	(2,3,2,3,3,5)
Methane, fossil	air/high population density	kg	3.92E-04	1	1.63	(2,3,2,3,3,5)
Ammonium, ion	water/river	kg	5.50E-04	1	1.63	(2,3,2,3,3,5)
Ammonia	air/high population density	kg	2.86E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 µm	air/high population density	kg	6.90E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	3.45E-04	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 µm	air/high population density	kg	6.90E-04	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	-2.58E+00	1	1.31	(2,3,2,3,3,5)
urea ammonium nitrate, as N, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: calcium nitrate, as N, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, at hard coal industrial furnace 1-10MW	RER	MJ	2.06E+00	1	1.57	(2,5,2,2,4,3)
heat, natural gas, at industrial furnace >100kW	RER	MJ	3.22E+01	1	1.57	(2,5,2,2,4,3)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.35E+00	1	1.57	(2,5,2,2,4,3)
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	2.12E-01	1	1.57	(2,5,2,2,4,3)
diesel, burned in building machine	GLO	MJ	1.66E-02	1	1.57	(2,5,2,2,4,3)
transport, freight, rail	RER	tkm	3.87E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	6.45E-01	1	2.09	(4,5,na,na,na,na)
chemical plant, organics (I)	RER	unit	2.58E-09	1	3.32	(2,5,2,2,4,5)
Particulates, > 10 µm	air/high population density	kg	1.08E-03	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	5.40E-04	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 µm	air/high population density	kg	1.08E-03	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	7.63E-01	1	1.31	(2,3,2,3,3,5)
Nitrate	water/river	kg	8.60E-03	1	1.63	(2,3,2,3,3,5)
Nitrate	air/high population density	kg	1.20E-04	1	1.63	(2,3,2,3,3,5)
calcium nitrate, as N, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: ammonium sulphate, as N, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, at hard coal industrial furnace 1-10MW	RER	MJ	1.43E+00	1	1.57	(2,5,2,2,4,3)
heat, natural gas, at industrial furnace >100kW	RER	MJ	2.23E+01	1	1.57	(2,5,2,2,4,3)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.40E+00	1	1.57	(2,5,2,2,4,3)
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.47E-01	1	1.57	(2,5,2,2,4,3)
diesel, burned in building machine	GLO	MJ	1.51E-02	1	1.57	(2,5,2,2,4,3)
transport, freight, rail	RER	tkm	2.86E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	4.76E-01	1	2.09	(4,5,na,na,na,na)
chemical plant, organics (I)	RER	unit	1.90E-09	1	3.32	(2,5,2,2,4,5)
NM VOC, non-methane volatile organic compounds, unspecified origin	air/high population density	kg	5.20E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 µm	air/high population density	kg	1.52E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	7.60E-05	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 µm	air/high population density	kg	1.52E-04	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	5.29E-01	1	1.31	(2,3,2,3,3,5)
ammonium sulphate, as N, at regional storehouse	RER	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX8X to Chapter X8X (XMineral FertilisersX)

Unit process inventory for: single superphosphate, as P2O5, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.85E+00	1	1.09	(2,1,1,2,1,3)
transport, transoceanic freight ship	OCE	tkm	2.44E+01	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	RER	tkm	2.86E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	4.38E+00	1	2.09	(4,5,na,na,na,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.00E+00	1	1.09	(2,1,1,2,1,3)
sulphuric acid, liquid, at plant	RER	kg	1.75E+00	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	1.90E-09	1	3.01	(2,1,1,2,1,3)
Zinc, ion	water/river	kg	2.60E-05	1	5.11	(2,3,2,3,3,5)
Lead	water/river	kg	1.90E-05	1	5.11	(2,3,2,3,3,5)
Nickel, ion	water/river	kg	1.70E-05	1	5.11	(2,3,2,3,3,5)
Mercury	water/river	kg	4.20E-06	1	5.11	(2,3,2,3,3,5)
Copper, ion	water/river	kg	2.20E-05	1	5.11	(2,3,2,3,3,5)
Chromium, ion	water/river	kg	2.20E-05	1	5.11	(2,3,2,3,3,5)
Cadmium, ion	water/river	kg	4.40E-06	1	5.11	(2,3,2,3,3,5)
Arsenic, ion	water/river	kg	4.40E-06	1	5.11	(2,3,2,3,3,5)
Phosphate	water/river	kg	4.42E-03	1	1.63	(2,3,2,3,3,5)
Hydrogen fluoride	air/high population density	kg	1.05E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 um	air/high population density	kg	1.24E-03	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 um, and < 10um	air/high population density	kg	1.24E-03	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 um	air/high population density	kg	6.19E-04	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	6.67E+00	1	1.31	(2,3,2,3,3,5)
single superphosphate, as P2O5, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: triple superphosphate, as P2O5, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, natural gas, at industrial furnace >100kW	RER	MJ	1.46E+00	1	1.09	(2,1,1,2,1,3)
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	7.52E-01	1	1.09	(2,1,1,2,1,3)
transport, transoceanic freight ship	OCE	tkm	1.08E+01	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	RER	tkm	1.25E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	1.93E+00	1	2.09	(4,5,na,na,na,na)
phosphoric acid, fertiliser grade, 70% in H2O, at plant	MA	kg	9.66E-01	1	1.09	(2,1,1,2,1,3)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	3.00E-01	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	8.33E-10	1	3.01	(2,1,1,2,1,3)
Zinc, ion	water/river	kg	2.60E-05	1	5.11	(2,3,2,3,3,5)
Lead	water/river	kg	1.90E-05	1	5.11	(2,3,2,3,3,5)
Nickel, ion	water/river	kg	1.70E-05	1	5.11	(2,3,2,3,3,5)
Mercury	water/river	kg	4.20E-06	1	5.11	(2,3,2,3,3,5)
Copper, ion	water/river	kg	2.20E-05	1	5.11	(2,3,2,3,3,5)
Chromium, ion	water/river	kg	2.20E-05	1	5.11	(2,3,2,3,3,5)
Cadmium, ion	water/river	kg	4.40E-06	1	5.11	(2,3,2,3,3,5)
Arsenic, ion	water/river	kg	4.40E-06	1	5.11	(2,3,2,3,3,5)
Phosphate	water/river	kg	4.42E-03	1	1.63	(2,3,2,3,3,5)
Hydrogen fluoride	air/high population density	kg	1.05E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 um	air/high population density	kg	5.42E-04	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 um, and < 10um	air/high population density	kg	5.42E-04	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 um	air/high population density	kg	2.71E-04	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	2.71E+00	1	1.31	(2,3,2,3,3,5)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: thomas meal, as P2O5, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.19E+00	1	1.22	(2,1,1,2,1,5)
chemical plant, organics (I)	RER	unit	4.00E-10	1	2.09	(4,5,na,na,na,na)
transport, lorry 16t	RER	tkm	5.88E-01	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	5.88E-01	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	RER	tkm	2.94E-01	1	2.09	(4,5,na,na,na,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: potassium sulphate, as K2O, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	5.76E+00	1	1.09	(2,1,1,2,1,3)
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	1.22E-01	1	1.09	(2,1,1,2,1,3)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	2.09	(4,5,na,na,na,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	1.03E+00	1	1.09	(2,1,1,2,1,3)
sulphuric acid, liquid, at plant	RER	kg	1.13E+00	1	1.09	(2,1,1,2,1,3)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.01	(2,1,1,2,1,3)
Particulates, > 10 um	air/high population density	kg	3.76E-08	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 um, and < 10um	air/high population density	kg	1.65E-08	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 um	air/high population density	kg	1.19E-08	1	3.10	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	4.40E-01	1	1.31	(2,3,2,3,3,5)
potassium sulphate, as K2O, at regional storehouse	RER	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX8X to Chapter X8X (XMineral FertilisersX)

Unit process inventory for: potassium chloride, as K2O, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
natural gas, burned in industrial furnace >100kW	RER	MJ	5.25E-01	1	1.10	(2,3,1,3,1,3)
natural gas, burned in cogen 1MWe lean burn	RER	MJ	3.23E+00	1	1.10	(2,3,1,3,1,3)
diesel, burned in building machine	GLO	MJ	1.39E-01	1	1.10	(2,3,1,3,1,3)
transport, freight, rail	RER	tkm	1.00E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	1.67E-01	1	2.09	(4,5,na,na,na,na)
chemicals inorganic, at plant	GLO	kg	1.01E-02	1	1.10	(2,3,1,3,1,3)
disposal, salt tailings potash mining, 0% water, to residual material landfill	CH	kg	4.58E+00	1	1.10	(2,3,1,3,1,3)
disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	7.17E-04	1	1.10	(2,3,1,3,1,3)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	3.16E-04	1	1.10	(2,3,1,3,1,3)
chemical plant, organics (I)	RER	unit	6.67E-10	1	3.01	(2,3,1,3,1,3)
sylvite, 25 % in sylvite, in ground	resource/in ground	kg	1.89E+00	1	1.10	(2,3,1,3,1,3)
Water, cooling, unspecified natural origin	resource/in water	m3	1.30E-02	1	1.10	(2,3,1,3,1,3)
Water, river	resource/in water	m3	3.50E-03	1	1.10	(2,3,1,3,1,3)
Sodium, ion	water/river	kg	7.36E-02	1	1.63	(2,3,2,3,3,5)
Potassium, ion	water/river	kg	1.61E-03	1	1.63	(2,3,2,3,3,5)
Chloride	water/river	kg	1.26E-01	1	1.63	(2,3,2,3,3,5)
Calcium, ion	water/river	kg	2.09E-03	1	1.63	(2,3,2,3,3,5)
Magnesium	water/river	kg	1.21E-02	1	1.63	(2,3,2,3,3,5)
Sulfur	water/river	kg	1.25E-02	1	1.63	(2,3,2,3,3,5)
Hydrogen chloride	air/high population density	kg	9.28E-06	1	1.63	(2,3,2,3,3,5)
Particulates, > 10 um	air/high population density	kg	1.40E-05	1	1.63	(2,3,2,3,3,5)
Particulates, > 2.5 um, and < 10um	air/high population density	kg	6.16E-07	1	2.10	(2,3,2,3,3,5)
Particulates, < 2.5 um	air/high population density	kg	7.70E-07	1	3.10	(2,3,2,3,3,5)
potassium chloride, as K2O, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: ammonium nitrate phosphate, at regional storehouse, RER								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	ammonium nitrate phosphate, as N, at regional storehouse RER (kg)	ammonium nitrate phosphate, as P2O5, at regional storehouse RER (kg)
heat, natural gas, at industrial furnace >100kW	RER	MJ	2.29E+00	1	1.09	(2,1,1,2,1,3)	52%	48%
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	2.73E-01	1	1.09	(2,1,1,2,1,3)	52%	48%
transport, transoceanic freight ship	OCE	tkm	1.27E+01	1	2.09	(4,5,na,na,na,na)	14%	86%
transport, freight, rail	RER	tkm	6.00E-01	1	2.09	(4,5,na,na,na,na)	14%	86%
transport, lorry 32t	RER	tkm	2.12E+00	1	2.09	(4,5,na,na,na,na)	14%	86%
ammonia, steam reforming, liquid, at plant	RER	kg	1.02E-01	1	1.09	(2,1,1,2,1,3)	100%	
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	5.20E-01	1	1.09	(2,1,1,2,1,3)		100%
chemical plant, organics (I)	RER	unit	4.00E-10	1	3.01	(2,1,1,2,1,3)	52%	48%
Zinc, ion	water/river	kg	1.35E-05	1	5.11	(2,3,2,3,3,5)		100%
Lead	water/river	kg	9.88E-06	1	5.11	(2,3,2,3,3,5)		100%
Nickel, ion	water/river	kg	8.84E-06	1	5.11	(2,3,2,3,3,5)		100%
Mercury	water/river	kg	2.18E-06	1	5.11	(2,3,2,3,3,5)		100%
Copper, ion	water/river	kg	1.14E-05	1	5.11	(2,3,2,3,3,5)		100%
Chromium, ion	water/river	kg	1.14E-05	1	5.11	(2,3,2,3,3,5)		100%
Cadmium, ion	water/river	kg	2.29E-06	1	5.11	(2,3,2,3,3,5)		100%
Arsenic, ion	water/river	kg	2.29E-06	1	5.11	(2,3,2,3,3,5)		100%
Phosphate	water/river	kg	5.09E-04	1	1.63	(2,3,2,3,3,5)		100%
Ammonia	air/high population density	kg	1.02E-04	1	1.63	(2,3,2,3,3,5)	100%	
Nitrogen oxides	air/high population density	kg	2.80E-03	1	1.78	(4,5,4,3,3,5)	100%	
Particulates, > 10 um	air/high population density	kg	2.00E-04	1	1.63	(2,3,2,3,3,5)	40%	60%
Particulates, > 2.5 um, and < 10um	air/high population density	kg	1.00E-04	1	2.10	(2,3,2,3,3,5)	60%	40%
Particulates, < 2.5 um	air/high population density	kg	2.00E-04	1	3.10	(2,3,2,3,3,5)	100%	
Heat, waste	air/high population density	MJ	9.83E-01	1	1.31	(2,3,2,3,3,5)	52%	48%
ammonium nitrate phosphate, as N, at regional storehouse	RER	kg	8.40E-02				100%	
ammonium nitrate phosphate, as P2O5, at regional storehouse	RER	kg	5.20E-01					100%

Unit process inventory for: potassium nitrate, at regional storehouse, RER								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	potassium nitrate, as N, at regional storehouse RER (kg)	potassium nitrate, as K2O, at regional storehouse RER (kg)
heat, natural gas, at industrial furnace >100kW	RER	MJ	4.56E+00	1	1.09	(2,1,1,2,1,3)	61%	39%
transport, freight, rail	RER	tkm	6.00E-01	1	2.09	(4,5,na,na,na,na)	24%	76%
nitric acid, 50% in H2O, at plant	RER	kg	6.30E-01	1	1.09	(2,1,1,2,1,3)	100%	
potassium chloride, as K2O, at regional storehouse	RER	kg	4.40E-01	1	1.09	(2,1,1,2,1,3)		100%
chemical plant, organics (I)	RER	unit	4.00E-10	1	3.01	(2,1,1,2,1,3)	61%	39%
Heat, waste	air/high population density	MJ	-1.44E+00	1	1.31	(2,3,2,3,3,5)	61%	39%
potassium nitrate, as N, at regional storehouse	RER	kg	1.40E-01				100%	
potassium nitrate, as K2O, at regional storehouse	RER	kg	4.40E-01					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX8X to Chapter X8X (XMineral FertilisersX)

Unit process inventory for: monoammonium phosphate, at regional storehouse, RER								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	monoammonium phosphate, as N, at regional storehouse RER (kg)	monoammonium phosphate, as P2O5, at regional storehouse RER (kg)
heat, natural gas, at industrial furnace >100kW	RER	MJ	3.30E-01	1	1.09	(2,1,1,2,1,3)	45%	55%
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	3.89E-02	1	1.09	(2,1,1,2,1,3)	45%	55%
transport, transoceanic freight ship	OCE	tkm	2.56E+00	1	2.09	(4,5,na,na,na,na)	17%	83%
transport, freight, rail	RER	tkm	6.00E-01	1	2.09	(4,5,na,na,na,na)	17%	83%
transport, lorry 32t	RER	tkm	4.91E-01	1	2.09	(4,5,na,na,na,na)	17%	83%
ammonia, steam reforming, liquid, at plant	RER	kg	1.30E-01	1	1.09	(2,1,1,2,1,3)	100%	
phosphoric acid, fertiliser grade, 70% in H2O, at plant	MA	kg	7.16E-01	1	1.09	(2,1,1,2,1,3)		100%
chemical plant, organics (I)	RER	unit	4.00E-10	1	3.01	(2,1,1,2,1,3)	45%	55%
Zinc, ion	water/river	kg	1.35E-05	1	5.11	(2,3,2,3,3,5)		100%
Lead	water/river	kg	9.88E-06	1	5.11	(2,3,2,3,3,5)		100%
Nickel, ion	water/river	kg	8.84E-06	1	5.11	(2,3,2,3,3,5)		100%
Mercury	water/river	kg	2.18E-06	1	5.11	(2,3,2,3,3,5)		100%
Copper, ion	water/river	kg	1.14E-05	1	5.11	(2,3,2,3,3,5)		100%
Chromium, ion	water/river	kg	1.14E-05	1	5.11	(2,3,2,3,3,5)		100%
Cadmium, ion	water/river	kg	2.29E-06	1	5.11	(2,3,2,3,3,5)		100%
Arsenic, ion	water/river	kg	2.29E-06	1	5.11	(2,3,2,3,3,5)		100%
Phosphate	water/river	kg	2.76E-05	1	1.63	(2,3,2,3,3,5)		100%
Ammonia	air/high population density	kg	1.32E-04	1	1.63	(2,3,2,3,3,5)	100%	
Particulates, > 10 µm	air/high population density	kg	2.00E-04	1	1.63	(2,3,2,3,3,5)	40%	60%
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	1.00E-04	1	2.10	(2,3,2,3,3,5)	60%	40%
Particulates, < 2.5 µm	air/high population density	kg	2.00E-04	1	3.10	(2,3,2,3,3,5)	100%	
Heat, waste	air/high population density	MJ	1.40E-01	1	1.31	(2,3,2,3,3,5)	45%	55%
monoammonium phosphate, as N, at regional storehouse	RER	kg	1.10E-01				100%	
monoammonium phosphate, as P2O5, at regional storehouse	RER	kg	5.20E-01					100%

Unit process inventory for: diammonium phosphate, at regional storehouse, RER								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	diammonium phosphate, as N, at regional storehouse RER (kg)	diammonium phosphate, as P2O5, at regional storehouse RER (kg)
heat, natural gas, at industrial furnace >100kW	RER	MJ	2.90E-01	1	1.09	(2,1,1,2,1,3)	60%	40%
electricity, medium voltage, production UCTE, at grid	UCTE	kWh	3.33E-02	1	1.09	(2,1,1,2,1,3)	60%	40%
transport, transoceanic freight ship	OCE	tkm	2.27E+00	1	2.09	(4,5,na,na,na,na)	28%	72%
transport, freight, rail	RER	tkm	6.00E-01	1	2.09	(4,5,na,na,na,na)	28%	72%
transport, lorry 32t	RER	tkm	4.63E-01	1	2.09	(4,5,na,na,na,na)	28%	72%
ammonia, steam reforming, liquid, at plant	RER	kg	2.20E-01	1	1.09	(2,1,1,2,1,3)	100%	
phosphoric acid, fertiliser grade, 70% in H2O, at plant	MA	kg	6.36E-01	1	1.09	(2,1,1,2,1,3)		100%
chemical plant, organics (I)	RER	unit	4.00E-10	1	3.01	(2,1,1,2,1,3)	60%	40%
Zinc, ion	water/river	kg	1.20E-05	1	5.11	(2,3,2,3,3,5)		100%
Lead	water/river	kg	8.74E-06	1	5.11	(2,3,2,3,3,5)		100%
Nickel, ion	water/river	kg	7.82E-06	1	5.11	(2,3,2,3,3,5)		100%
Mercury	water/river	kg	1.93E-06	1	5.11	(2,3,2,3,3,5)		100%
Copper, ion	water/river	kg	1.01E-05	1	5.11	(2,3,2,3,3,5)		100%
Chromium, ion	water/river	kg	1.01E-05	1	5.11	(2,3,2,3,3,5)		100%
Cadmium, ion	water/river	kg	2.02E-06	1	5.11	(2,3,2,3,3,5)		100%
Arsenic, ion	water/river	kg	2.02E-06	1	5.11	(2,3,2,3,3,5)		100%
Phosphate	water/river	kg	2.45E-05	1	1.63	(2,3,2,3,3,5)		100%
Ammonia	air/high population density	kg	2.16E-04	1	1.63	(2,3,2,3,3,5)	100%	
Particulates, > 10 µm	air/high population density	kg	2.00E-04	1	1.63	(2,3,2,3,3,5)	40%	60%
Particulates, > 2.5 µm, and < 10µm	air/high population density	kg	1.00E-04	1	2.10	(2,3,2,3,3,5)	60%	40%
Particulates, < 2.5 µm	air/high population density	kg	2.00E-04	1	3.10	(2,3,2,3,3,5)	100%	
Heat, waste	air/high population density	MJ	1.20E-01	1	1.31	(2,3,2,3,3,5)	60%	40%
diammonium phosphate, as N, at regional storehouse	RER	kg	1.80E-01				100%	
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	4.60E-01					100%

Unit process inventory for: lime, algae, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, natural gas, at industrial furnace >100kW	RER	MJ	1.50E+00	1	1.24	(3,3,1,3,1,5)
electricity, low voltage, at grid	FR	kWh	2.00E-02	1	1.24	(3,3,1,3,1,5)
chemical plant, organics (I)	RER	unit	4.00E-10	1	2.06	(3,3,1,3,1,5)
transport, barge	RER	tkm	7.80E-02	1	2.06	(3,3,1,3,1,5)
transport, lorry 32t	RER	tkm	1.95E-02	1	2.06	(3,3,1,3,1,5)
transport, freight, rail	RER	tkm	1.00E+00	1	2.06	(3,3,1,3,1,5)
Calcite, in ground	resource/in ground	kg	8.92E-01	1	1.24	(3,3,1,3,1,5)
Heat, waste	air/high population density	MJ	7.20E-02	1	1.24	(3,3,1,3,1,5)
lime, algae, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: lime, from carbonation, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
transport, lorry 28t	CH	tkm	6.00E-02	1	2.06	(3,3,1,3,1,5)
lime, from carbonation, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: stone meal, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
electricity, low voltage, at grid	CH	kWh	5.00E-03	1	1.24	(3,3,1,3,1,5)
chemical plant, organics (I)	RER	unit	4.00E-10	1	2.06	(3,3,1,3,1,5)
transport, lorry 28t	CH	tkm	5.00E-02	1	2.06	(3,3,1,3,1,5)
transport, freight, rail	CH	tkm	5.00E-02	1	2.06	(3,3,1,3,1,5)
Heat, waste	air/high population density	MJ	1.80E-02	1	1.24	(3,3,1,3,1,5)
stone meal, at regional storehouse	CH	kg	1.00E+00			

Appendix A9 to Chapter 9 (Organic Fertilisers)

Unit-Process Inventories from Chapter 9 (Organic Fertilisers)

(Last Changes 2004)

Tab. A. 13 Unit-process inventories for organic fertilisers.

Unit process inventory for: poultry manure, dried, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
light fuel oil, burned in boiler 100kW, non-modulating	CH	MJ	5.50E-01	1	1.22	(2,3,1,1,1,5)
electricity, low voltage, at grid	CH	kWh	1.10E-01	1	1.22	(2,3,1,1,1,5)
disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	kg	2.00E-03	1	1.22	(2,3,1,1,1,5)
building, hall (l)	CH	m2	7.37E-06	1	3.05	(2,3,1,1,1,5)
Occupation, industrial area	resource/land	m2a	1.38E-03	1	1.57	(2,3,1,1,1,5)
Occupation, construction site	resource/land	m2a	5.53E-05	1	1.57	(2,3,1,1,1,5)
Transformation, from pasture and meadow	resource/land	m2	2.76E-05	1	2.05	(2,3,1,1,1,5)
Transformation, to industrial area	resource/land	m2	2.76E-05	1	2.05	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.84E-01	1	2.05	(2,3,1,1,1,5)
Ammonia	air/high population density	kg	1.06E-04	1	1.57	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	3.96E-01	1	1.22	(2,3,1,1,1,5)
poultry manure, dried, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: horn meal, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, natural gas, at industrial furnace >100kW	RER	MJ	1.18E+00	1	1.33	(2,3,3,3,3,5)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.79E-02	1	1.33	(2,3,3,3,3,5)
building, hall (l)	CH	m2	7.37E-06	1	3.11	(2,3,3,3,3,5)
Occupation, industrial area	resource/land	m2a	1.38E-03	1	1.64	(2,3,3,3,3,5)
Occupation, construction site	resource/land	m2a	5.53E-05	1	1.64	(2,3,3,3,3,5)
Transformation, from pasture and meadow	resource/land	m2	2.76E-05	1	2.12	(2,3,3,3,3,5)
Transformation, to industrial area	resource/land	m2	2.76E-05	1	2.12	(2,3,3,3,3,5)
transport, transoceanic freight ship	OCE	tkm	1.18E+01	1	2.12	(2,3,3,3,3,5)
transport, barge	RER	tkm	8.30E-01	1	2.12	(2,3,3,3,3,5)
transport, freight, rail	RER	tkm	4.00E-01	1	2.12	(2,3,3,3,3,5)
transport, lorry 32t	RER	tkm	2.00E-02	1	2.12	(2,3,3,3,3,5)
Heat, waste	air/high population density	MJ	1.36E-01	1	1.33	(2,3,3,3,3,5)
horn meal, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: compost, at plant, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
diesel, at regional storage	CH	kg	2.68E-03	1	1.22	(2,3,1,1,1,5)
electricity, low voltage, at grid	CH	kWh	1.18E-02	1	1.22	(2,3,1,1,1,5)
compost plant, open (l)	CH	unit	7.41E-09	1	3.05	(2,3,1,1,1,5)
disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	kg	1.85E-05	1	1.22	(2,3,1,1,1,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	8.33E-04	1	1.22	(2,3,1,1,1,5)
transport, municipal waste collection, lorry 21t	CH	tkm	6.22E-03	1	2.05	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	6.60E+00	1	1.22	(2,3,1,1,1,5)
Methane, biogenic	air/high population density	kg	1.01E-02	1	1.31	(2,3,1,1,1,5)
Carbon monoxide, fossil	air/high population density	kg	1.28E-04	1	5.06	(2,3,1,1,1,5)
Carbon dioxide, biogenic	air/high population density	kg	5.20E-01	1	1.22	(2,3,1,1,1,5)
Carbon dioxide, fossil	air/high population density	kg	8.43E-03	1	1.22	(2,3,1,1,1,5)
Nitrogen oxides	air/high population density	kg	4.53E-04	1	1.57	(2,3,1,1,1,5)
Ammonia	air/high population density	kg	9.78E-04	1	1.31	(2,3,1,1,1,5)
Dinitrogen monoxide	air/high population density	kg	2.81E-04	1	1.48	(2,3,1,1,1,5)
Hydrogen sulfide	air/high population density	kg	5.28E-04	1	1.57	(2,3,1,1,1,5)
compost, at plant	CH	kg	1.00E+00			

Appendix A10 to Chapter 10 (Pesticides)

Unit-Process Inventories from Chapter 10 (Pesticides)

(Last Changes 2004)

Tab. A. 14 Unit-process inventories for pesticides.

Unit process inventory for: [sulfonyl]urea-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.58E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	1.62E+00	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.01E+00	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.18E+00	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.29E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.07E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.09E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.26E+01	1	1.22	(2,2,4,2,1,na)
[sulfonyl]urea-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: diuron, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.32E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.50E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.07E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.73E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	6.59E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.36E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.87E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.01E+01	1	1.22	(2,2,4,2,1,na)
diuron, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: linuron, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.44E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.57E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.16E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.85E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	7.22E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.78E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.92E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.08E+01	1	1.22	(2,2,4,2,1,na)
linuron, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: phenoxy-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.14E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	7.34E-01	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.98E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.42E+00	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	5.70E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	9.90E-02	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	7.12E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	7.69E+00	1	1.22	(2,2,4,2,1,na)
phenoxy-compounds, at regional storehouse	CH	kg	1.00E+00			

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Unit process inventory for: MCPA, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.37E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	8.67E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	2.04E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.28E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	6.87E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	1.02E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	6.42E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	6.94E+00	1	1.22	(2,2,4,2,1,na)
MCPA, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: 2,4-D, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	9.84E+00	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	6.34E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.09E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	4.92E+00	1	1.31	(2,2,4,2,3,na)
electricity, low voltage, at grid	CH	kWh	5.47E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	5.91E+00	1	1.22	(2,2,4,2,1,na)
2,4-D, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: [thio]carbamate-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	3.10E-01	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.97E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	8.40E-01	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.80E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.06E+00	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	9.83E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	3.90E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.53E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.65E+01	1	1.22	(2,2,4,2,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: carbofuran, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	3.22E-02	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.96E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	2.23E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.07E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.47E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	2.48E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.35E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.74E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.96E+01	1	1.22	(2,2,4,2,1,na)
carbofuran, at regional storehouse	CH	kg	1.00E+00			

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Unit process inventory for: acetamide-anilide-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.53E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	1.50E+00	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	5.28E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.42E+00	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.27E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.64E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.71E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.85E+01	1	1.22	(2,2,4,2,1,na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: alachlor, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.55E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.60E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	4.72E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.76E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.27E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.36E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.88E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.03E+01	1	1.22	(2,2,4,2,1,na)
alachlor, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: propachlor, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.75E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.74E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	4.92E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.66E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.38E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.46E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.83E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.98E+01	1	1.22	(2,2,4,2,1,na)
propachlor, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: metolachlor, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.71E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.65E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	4.69E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.42E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.35E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.34E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.71E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.85E+01	1	1.22	(2,2,4,2,1,na)
metolachlor, at regional storehouse	CH	kg	1.00E+00			

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Unit process inventory for: benzoic-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.22E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.31E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	8.66E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.07E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	6.10E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	4.33E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.54E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.66E+01	1	1.22	(2,2,4,2,1,na)
benzoic-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: dicamba, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.24E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.12E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.24E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.17E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.12E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	6.20E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.08E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.25E+01	1	1.22	(2,2,4,2,1,na)
dicamba, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: triazine-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.72E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	1.03E+00	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	9.63E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.77E+00	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.36E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	4.82E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.38E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.49E+01	1	1.22	(2,2,4,2,1,na)
triazine-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: atrazine, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.54E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	7.03E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.17E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.69E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	7.69E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.84E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	8.46E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	9.14E+00	1	1.22	(2,2,4,2,1,na)
atrazine, at regional storehouse	CH	kg	1.00E+00			

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Unit process inventory for: cyanazine, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.65E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	8.88E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.12E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.75E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	8.26E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.59E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	8.76E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	9.46E+00	1	1.22	(2,2,4,2,1,na)
cyanazine, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: nitro-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	7.48E+00	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	7.97E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.53E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.53E-01	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	3.74E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	7.64E-02	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.26E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.36E+00	1	1.22	(2,2,4,2,1,na)
nitro-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: dithiocarbamate-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	1.77E-01	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	7.67E+00	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	2.20E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	5.52E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	9.26E-01	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	3.84E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.76E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	4.63E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	5.00E+00	1	1.22	(2,2,4,2,1,na)
dithiocarbamate-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: maneb, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	2.57E-01	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.30E+00	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	4.39E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	3.90E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.18E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	3.15E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	1.95E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	5.90E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	6.37E+00	1	1.22	(2,2,4,2,1,na)
maneb, at regional storehouse	CH	kg	1.00E+00			

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Unit process inventory for: dinitroaniline-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	9.44E+00	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	9.17E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	2.17E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.42E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	4.72E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	1.09E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.21E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.31E+01	1	1.22	(2,2,4,2,1,na)
dinitroaniline-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: pyrethroid-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	8.67E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.45E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.21E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	8.53E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	4.33E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	6.04E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	4.26E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	4.60E+01	1	1.22	(2,2,4,2,1,na)
pyrethroid-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: benzimidazole-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.64E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.41E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.21E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.23E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	2.32E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	6.04E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.62E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.82E+01	1	1.22	(2,2,4,2,1,na)
benzimidazole-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: organophosphorus-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	7.20E-02	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.58E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	7.26E-01	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.32E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.66E+00	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	7.92E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	3.66E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.33E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.52E+01	1	1.22	(2,2,4,2,1,na)
organophosphorus-compounds, at regional storehouse	CH	kg	1.00E+00			

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Unit process inventory for: glyphosate, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.97E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	5.37E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.58E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	9.69E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.99E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	7.89E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	4.84E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	5.23E+01	1	1.22	(2,2,4,2,1,na)
glyphosate, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: parathion, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	1.67E-01	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.93E+00	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	5.69E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	3.92E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.53E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	3.46E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	1.96E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.27E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.37E+01	1	1.22	(2,2,4,2,1,na)
parathion, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: benzo[thia]diazole-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.74E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	2.09E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.12E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.12E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	2.37E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.61E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.56E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.76E+01	1	1.22	(2,2,4,2,1,na)
benzo[thia]diazole-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: nitrile-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	8.72E-03	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.46E+01	1	2.06	(2,3,4,2,5,na)
naphtha, at refinery	RER	kg	6.55E-01	1	1.57	(2,3,4,2,4,na)
natural gas, at long-distance pipeline	RER	Nm3	4.34E-01	1	1.57	(2,3,4,2,4,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.97E+00	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	7.29E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.17E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	9.87E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.07E+01	1	1.22	(2,2,4,2,1,na)
nitrile-compounds, at regional storehouse	CH	kg	1.00E+00			

(XPesticidesX)

Unit process inventory for: diphenylether-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.84E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.45E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.22E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	7.85E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	3.42E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	6.08E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	3.92E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	4.24E+01	1	1.22	(2,2,4,2,1,na)
diphenylether-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: pyridazine-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	8.72E-03	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.46E+01	1	2.06	(2,3,4,2,5,na)
naphtha, at refinery	RER	kg	6.55E-01	1	1.57	(2,3,4,2,4,na)
natural gas, at long-distance pipeline	RER	Nm3	4.34E-01	1	1.57	(2,3,4,2,4,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.97E+00	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	7.29E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.17E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	9.87E-01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.07E+01	1	1.22	(2,2,4,2,1,na)
pyridazine-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: cyclic N-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	3.08E-02	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.42E+01	1	2.06	(2,3,4,2,5,na)
naphtha, at refinery	RER	kg	6.11E-01	1	1.57	(2,3,4,2,4,na)
natural gas, at long-distance pipeline	RER	Nm3	4.41E-01	1	1.57	(2,3,4,2,4,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.08E+00	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	7.11E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	2.21E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.04E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.12E+01	1	1.22	(2,2,4,2,1,na)
cyclic N-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: phtalamide-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.33E+00	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	6.18E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	2.38E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.32E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	2.16E+00	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	1.19E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.16E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.25E+01	1	1.22	(2,2,4,2,1,na)
phtalamide-compounds, at regional storehouse	CH	kg	1.00E+00			

(XPesticidesX)

Unit process inventory for: bipyridylum-compounds, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.49E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.14E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.10E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.34E+00	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	3.25E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	5.52E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	2.17E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.34E+01	1	1.22	(2,2,4,2,1,na)
bipyridylum-compounds, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: pesticide unspecified, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	5.01E-02	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.31E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	9.94E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.18E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.38E+00	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	CH	MJ	1.16E+01	1	1.31	(2,2,4,2,3,na)
natural gas, at long-distance pipeline	CH	Nm3	3.59E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.69E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	2.09	(4,5,na,na,na,na)
transport, lorry 28t	CH	tkm	1.00E-01	1	2.09	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.83E+01	1	1.22	(2,2,4,2,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: [sulfonyl]urea-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.86E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	1.62E+00	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.52E+00	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	6.26E+00	1	1.22	(2,3,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,3,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,3,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.26E+01	1	1.22	(2,3,4,2,1,na)
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: diuron, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.98E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.50E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.61E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.60E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.01E+01	1	1.31	(2,3,4,2,3,na)
diuron, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: linuron, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.17E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.57E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.73E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.77E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.08E+01	1	1.31	(2,3,4,2,3,na)
linuron, at regional storehouse	RER	kg	1.00E+00			

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Unit process inventory for: phenoxy-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.71E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	7.34E-01	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	2.97E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.14E+00	1	1.22	(2,3,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,3,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,3,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	7.69E+00	1	1.22	(2,3,4,2,1,na)
phenoxy-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: MCPA, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.06E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	8.67E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	3.06E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.93E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	6.94E+00	1	1.22	(2,2,4,2,1,na)
MCPA, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: 2,4-D, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.48E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	6.34E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.64E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	5.91E+00	1	1.22	(2,2,4,2,1,na)
2,4-D, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: [thio]carbamate-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	3.10E-01	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.95E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	8.40E-01	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.17E+00	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.59E+00	1	1.22	(2,3,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,3,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,3,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.65E+01	1	1.22	(2,3,4,2,1,na)
[thio]carbamate-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: carbofuran, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	3.22E-02	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	7.44E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	2.23E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.60E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	8.21E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.96E+01	1	1.22	(2,2,4,2,1,na)
carbofuran, at regional storehouse	RER	kg	1.00E+00			

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Unit process inventory for: acetamide-anilide-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.80E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	1.50E+00	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.92E-01	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.13E+00	1	1.22	(2,3,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,3,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,3,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.85E+01	1	1.22	(2,3,4,2,1,na)
acetamide-anilide-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: alachlor, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.82E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.60E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.08E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.65E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.03E+01	1	1.22	(2,2,4,2,1,na)
alachlor, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: propachlor, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.13E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.74E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.38E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.50E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.98E+01	1	1.22	(2,2,4,2,1,na)
propachlor, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: metolachlor, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.06E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.65E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	7.03E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.13E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.85E+01	1	1.22	(2,2,4,2,1,na)
metolachlor, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: benzoic-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.83E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.31E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.30E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.61E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.66E+01	1	1.31	(2,3,4,2,3,na)
benzoic-compounds, at regional storehouse	RER	kg	1.00E+00			

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Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.36E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	1.12E+00	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.86E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	6.25E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.25E+01	1	1.22	(2,2,4,2,1,na)
dicamba, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: triazine-compounds, at regional storehouse, RER

Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	4.08E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	1.03E+00	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.44E+00	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	4.15E+00	1	1.22	(2,3,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,3,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,3,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.49E+01	1	1.22	(2,3,4,2,1,na)
triazine-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: atrazine, at regional storehouse, RER

Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.31E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	7.03E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.75E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.54E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	9.14E+00	1	1.22	(2,2,4,2,1,na)
atrazine, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: cyanazine, at regional storehouse, RER

Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.48E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	8.88E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.68E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.63E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	9.46E+00	1	1.22	(2,2,4,2,1,na)
cyanazine, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: nitro-compounds, at regional storehouse, RER

Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.12E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	7.97E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	2.29E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.79E-01	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.36E+00	1	1.31	(2,3,4,2,3,na)
nitro-compounds, at regional storehouse	RER	kg	1.00E+00			

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Unit process inventory for: dithiocarbamate-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	1.77E-01	1	1.31	(2,3,4,2,3,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.15E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	2.20E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	8.28E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.39E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	5.00E+00	1	1.31	(2,3,4,2,3,na)
dithiocarbamate-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: maneb, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	2.57E-01	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	9.44E+00	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	4.39E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	5.86E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.77E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	6.37E+00	1	1.22	(2,2,4,2,1,na)
maneb, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: dinitroaniline-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.42E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	9.17E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	3.26E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.63E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.31E+01	1	1.31	(2,3,4,2,3,na)
dinitroaniline-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: pyrethroid-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.30E+02	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.45E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.81E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.28E+01	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	4.60E+01	1	1.31	(2,3,4,2,3,na)
pyrethroid-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: benzimidazole-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.96E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.41E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.81E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	7.85E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.82E+01	1	1.31	(2,3,4,2,3,na)
benzimidazole-compounds, at regional storehouse	RER	kg	1.00E+00			

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Unit process inventory for: organophosphorus-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	7.20E-02	1	1.22	(2,3,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.38E+01	1	1.31	(2,3,4,2,3,na)
naphtha, at refinery	RER	kg	7.26E-01	1	1.22	(2,3,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.10E+00	1	1.22	(2,3,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	7.00E+00	1	1.22	(2,3,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,3,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,3,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.52E+01	1	1.22	(2,3,4,2,1,na)
organophosphorus-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: glyphosate, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	5.96E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	5.37E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	2.37E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.45E+01	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	5.23E+01	1	1.22	(2,2,4,2,1,na)
glyphosate, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: parathion, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	1.67E-01	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.04E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	5.69E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	5.88E-01	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.80E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.37E+01	1	1.22	(2,2,4,2,1,na)
parathion, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: benzo[thia]diazole-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	7.11E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	2.09E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.68E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	7.67E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.76E+01	1	1.31	(2,3,4,2,3,na)
benzo[thia]diazole-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: nitrile-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	8.72E-03	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.19E+01	1	2.06	(2,3,4,2,5,na)
naphtha, at refinery	RER	kg	6.55E-01	1	1.57	(2,3,4,2,4,na)
natural gas, at long-distance pipeline	RER	Nm3	6.51E-01	1	1.57	(2,3,4,2,4,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.96E+00	1	1.57	(2,3,4,2,4,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.57	(2,3,4,2,4,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.28	(2,3,4,2,4,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.07E+01	1	1.57	(2,3,4,2,4,na)
nitrile-compounds, at regional storehouse	RER	kg	1.00E+00			

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Unit process inventory for: diphenylether-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	1.03E+02	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.45E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.82E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	1.18E+01	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	4.24E+01	1	1.31	(2,3,4,2,3,na)
diphenylether-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: pyridazine-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	8.72E-03	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.19E+01	1	2.06	(2,3,4,2,5,na)
naphtha, at refinery	RER	kg	6.55E-01	1	1.57	(2,3,4,2,4,na)
natural gas, at long-distance pipeline	RER	Nm3	6.51E-01	1	1.57	(2,3,4,2,4,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	2.96E+00	1	1.57	(2,3,4,2,4,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.57	(2,3,4,2,4,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.28	(2,3,4,2,4,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.07E+01	1	1.57	(2,3,4,2,4,na)
pyridazine-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: cyclic N-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	3.08E-02	1	1.57	(2,3,4,2,4,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	2.13E+01	1	2.06	(2,3,4,2,5,na)
naphtha, at refinery	RER	kg	6.11E-01	1	1.57	(2,3,4,2,4,na)
natural gas, at long-distance pipeline	RER	Nm3	6.62E-01	1	1.57	(2,3,4,2,4,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.12E+00	1	1.57	(2,3,4,2,4,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.57	(2,3,4,2,4,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.28	(2,3,4,2,4,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.12E+01	1	1.57	(2,3,4,2,4,na)
cyclic N-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: phtalamide-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	6.49E+00	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	6.18E-01	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	3.56E-01	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	3.47E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.25E+01	1	1.31	(2,3,4,2,3,na)
phtalamide-compounds, at regional storehouse	RER	kg	1.00E+00			

Unit process inventory for: bipyridylium-compounds, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	9.74E+01	1	1.57	(2,3,4,2,4,na)
naphtha, at refinery	RER	kg	1.14E+00	1	1.31	(2,3,4,2,3,na)
natural gas, at long-distance pipeline	RER	Nm3	1.66E+00	1	1.31	(2,3,4,2,3,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	6.51E+00	1	1.31	(2,3,4,2,3,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.31	(2,3,4,2,3,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.10	(2,3,4,2,3,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	2.34E+01	1	1.31	(2,3,4,2,3,na)
bipyridylium-compounds, at regional storehouse	RER	kg	1.00E+00			

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Unit process inventory for: pesticide unspecified, at regional storehouse, RER						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
hard coal mix, at regional storage	UCTE	kg	5.01E-02	1	1.22	(2,2,4,2,1,na)
heat, heavy fuel oil, at industrial furnace 1MW	RER	MJ	3.47E+01	1	1.31	(2,2,4,2,3,na)
naphtha, at refinery	RER	kg	9.94E-01	1	1.22	(2,2,4,2,1,na)
natural gas, at long-distance pipeline	RER	Nm3	1.08E+00	1	1.22	(2,2,4,2,1,na)
electricity, low voltage, production UCTE, at grid	UCTE	kWh	5.07E+00	1	1.22	(2,2,4,2,1,na)
disposal, hazardous waste, 25% water, to hazardous waste incineration	CH	kg	2.00E-01	1	1.22	(2,2,4,2,1,na)
chemical plant, organics (I)	RER	unit	8.00E-10	1	3.05	(2,2,4,2,1,na)
transport, freight, rail	RER	tkm	1.20E+00	1	1.30	(4,5,na,na,na,na)
transport, lorry 32t	RER	tkm	2.00E-01	1	1.30	(4,5,na,na,na,na)
Heat, waste	air/high population density	MJ	1.83E+01	1	1.22	(2,2,4,2,1,na)
pesticide unspecified, at regional storehouse	RER	kg	1.00E+00			

Appendix A11 to Chapter 11 (Seed)

Unit-Process Inventories from Chapter 11 (Seed)

Tab. A. 15 Unit-process inventories for seed.

Unit process inventory for: barley seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
barley grains IP, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.40E-02	1	1.22	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	8.64E-02	1	1.22	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.30E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	2.00	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	2.00	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	3.23	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	3.23	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	1.78	(5,3,1,2,1,na)
cyclic N-compounds, at regional storehouse	CH	kg	8.80E-05	1	2.24	(2,2,1,1,1,5)
barley seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: barley seed organic, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
barley grains organic, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.40E-02	1	1.22	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	8.64E-02	1	1.22	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.30E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
barley seed organic, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: clover seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
clover seed IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,3,3,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 32t	RER	tkm	9.00E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
clover seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: grass seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
grass seed IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,3,3,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 32t	RER	tkm	9.00E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
grass seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: maize seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
maize seed IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,3,3,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 28t	CH	tkm	5.00E-02	1	2.00	(2,1,1,1,1,na)
transport, lorry 32t	RER	tkm	6.80E-01	1	2.00	(2,1,1,1,1,na)
transport, freight, rail	RER	tkm	3.00E-02	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
phthalamide-compounds, at regional storehouse	CH	kg	2.40E-03	1	1.22	(2,2,1,1,1,5)
maize seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: maize seed organic, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
maize seed organic, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,3,3,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.35	(3,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.35	(3,3,4,5,1,5)
transport, lorry 28t	CH	tkm	5.00E-02	1	2.00	(2,1,1,1,1,na)
transport, lorry 32t	RER	tkm	6.80E-01	1	2.00	(2,1,1,1,1,na)
transport, freight, rail	RER	tkm	3.00E-02	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
maize seed organic, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: pea seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
protein peas, IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,3,3,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 28t	CH	tkm	2.00E-02	1	2.00	(2,1,1,1,1,na)
transport, lorry 32t	RER	tkm	7.40E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.60E-03	1	1.22	(2,2,1,1,1,5)
pea seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: pea seed organic, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
protein peas, organic, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,3,3,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 28t	CH	tkm	2.00E-02	1	2.00	(2,1,1,1,1,na)
transport, lorry 32t	RER	tkm	7.40E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
pea seed organic, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: potato seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
potato seed IP, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	7.30E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.63E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 28t	CH	tkm	1.20E-01	1	2.00	(2,1,1,1,1,na)
transport, lorry 32t	RER	tkm	6.00E-02	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	4.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	4.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	1.60E-05	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	8.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	8.00E-06	1	2.24	(5,3,1,2,1,na)
nitrile-compounds, at regional storehouse	CH	kg	5.00E-05	1	1.22	(2,2,1,1,1,5)
potato seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: potato seed organic, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
potato seed organic, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	7.30E-02	1	1.64	(5,3,4,5,1,5)
Heat, waste	air/high population density	MJ	2.63E-01	1	1.64	(5,3,4,5,1,5)
transport, lorry 28t	CH	tkm	1.20E-01	1	2.00	(2,1,1,1,1,na)
transport, lorry 32t	RER	tkm	6.00E-02	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	4.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	4.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	1.60E-05	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	8.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	8.00E-06	1	2.24	(5,3,1,2,1,na)
potato seed organic, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: rape seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
rape seed IP, at farm	CH	kg	1.75E+00	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	5.80E-02	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	9.00E-01	1	1.07	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	1.07	(2,1,1,1,1,na)
Heat, waste	air/high population density	MJ	2.09E-01	1	1.07	(2,1,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.50	(2,1,1,1,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.50	(2,1,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	-1.87E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	-2.02E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: sugar beet seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
sugar beets IP, at farm	CH	kg	2.63E+01	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	1.00E-01	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	9.00E-01	1	1.07	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	1.07	(2,1,1,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	5.40E-02	1	1.07	(2,1,1,1,1,na)
Heat, waste	air/high population density	MJ	3.60E-01	1	1.07	(2,1,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.50	(2,1,1,1,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.50	(2,1,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	-9.87E-01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	-1.16E+01	1	1.07	(2,1,1,1,1,na)
sugar beet seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: rye seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
rye grains IP, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.40E-02	1	1.22	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	8.64E-02	1	1.22	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.30E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
diphenylether-compounds, at regional storehouse	CH	kg	1.00E-04	1	1.22	(2,2,1,1,1,5)
rye seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: rye seed organic, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
rye grains organic, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.40E-02	1	1.22	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	8.64E-02	1	1.22	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.30E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
rye seed organic, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: wheat seed IP, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
wheat grains IP, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.40E-02	1	1.22	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	8.64E-02	1	1.22	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.30E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
diphenylether-compounds, at regional storehouse	CH	kg	1.00E-04	1	1.22	(2,2,1,1,1,5)
wheat seed IP, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: wheat seed organic, at regional storehouse, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
wheat grains organic, at farm	CH	kg	1.00E+00	1	1.05	(1,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	2.40E-02	1	1.22	(2,3,1,1,1,5)
Heat, waste	air/high population density	MJ	8.64E-02	1	1.22	(2,3,1,1,1,5)
transport, lorry 28t	CH	tkm	1.30E-01	1	2.00	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	3.23	(5,3,1,2,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.78	(5,3,1,2,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.78	(5,3,1,2,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.24	(5,3,1,2,1,na)
wheat seed organic, at regional storehouse	CH	kg	1.00E+00			

Unit process inventory for: clover seed IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
clover seed IP, at regional storehouse	CH	kg	1.88E-02	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.50E-03	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.25E-03	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.50E-03	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.50E-03	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	2.86E-01	1	1.07	(2,1,1,1,1,na)
[thio]carbamate-compounds, at regional storehouse	RER	kg	5.00E-04	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.00E-01	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	2.88E-01	1	1.07	(2,1,1,1,1,na)
transport, freight, rail	CH	tkm	6.88E-02	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	6.88E-02	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	6.19E-01	1	2.09	(4,5,na,na,na,na)
transport, van <3.5t	CH	tkm	2.96E-04	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.25E+01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	8.88E+00	1	1.21	(2,1,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	3.63E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.25E+01	1	1.21	(2,1,1,1,1,na)
Asulam	soil/agricultural	kg	5.00E-04	1	1.24	(2,2,3,1,1,na)
Phosphorus	water/river	kg	2.46E-04	1	1.51	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	7.15E-03	1	1.41	(2,2,1,1,1,na)
Nitrate	water/ground-	kg	9.34E-03	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	2.89E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	2.89E-04	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.50E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	4.81E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-5.39E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-3.92E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	7.12E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-3.56E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-1.96E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-5.12E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	6.94E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	5.37E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.20E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	4.37E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	8.88E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.53E-06	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	4.71E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	5.41E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.29E-05	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.04E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	5.14E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	5.50E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.16E-05	1	1.80	(2,2,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.30E+00	1	1.80	(2,2,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.63E+01	1	1.80	(2,2,1,1,1,na)
clover seed IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: grass seed IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
grass seed IP, at regional storehouse	CH	kg	1.60E-02	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E-03	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	5.00E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	5.00E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.50E-03	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.00E-03	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	1.00E-03	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	2.86E-01	1	1.07	(2,1,1,1,1,na)
benzimidazole-compounds, at regional storehouse	RER	kg	2.00E-03	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.00E-01	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P ₂ O ₅ , at regional storehouse	RER	kg	4.25E-02	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K ₂ O, at regional storehouse	RER	kg	1.13E-01	1	1.07	(2,1,1,1,1,na)
transport, freight, rail	CH	tkm	5.62E-02	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	5.62E-02	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	5.06E-01	1	2.09	(4,5,na,na,na,na)
transport, van <3.5t	CH	tkm	3.00E-04	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	5.00E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.55E+00	1	1.21	(2,1,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	1.45E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	5.00E+00	1	1.21	(2,1,1,1,1,na)
Ethofumesate	soil/agricultural	kg	2.00E-03	1	1.24	(2,2,3,1,1,na)
Phosphorus	water/river	kg	9.85E-05	1	1.51	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.01E-03	1	1.41	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.21E-03	1	1.21	(2,2,1,1,1,na)
Nitrate	water/ground-	kg	3.74E-03	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.17E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.17E-04	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.12E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	2.05E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-9.68E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-1.64E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	4.37E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	4.58E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-2.25E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	2.80E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	2.18E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	1.44E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.52E-07	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	1.45E-06	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	2.52E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	2.18E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	9.29E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.37E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.96E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	6.20E-06	1	1.80	(2,2,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.37E+00	1	1.80	(2,2,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.60E+01	1	1.80	(2,2,1,1,1,na)
grass seed IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: maize seed IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
maize seed IP, at regional storehouse	CH	kg	5.00E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.17E-03	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.33E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	1.00E-03	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.00E-03	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	5.67E-04	1	1.07	(2,1,1,1,1,na)
mowing, by rotary mower	CH	ha	2.00E-03	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.33E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	3.54E-01	1	1.07	(2,1,1,1,1,na)
triazine-compounds, at regional storehouse	RER	kg	2.20E-04	1	1.07	(2,1,1,1,1,na)
acetamide-anillide-compounds, at regional storehouse	RER	kg	1.97E-04	1	1.07	(2,1,1,1,1,na)
organophosphorus-compounds, at regional storehouse	RER	kg	8.33E-05	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	1.30E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	3.54E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.18E-02	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	1.50E-02	1	1.07	(2,1,1,1,1,na)
transport, freight, rail	CH	tkm	1.71E-02	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	1.71E-02	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	1.54E-01	1	2.09	(4,5,na,na,na,na)
transport, van <3.5t	CH	tkm	9.00E-05	1	2.09	(4,5,na,na,na,na)
green manure IP, until April	CH	ha	3.33E-04	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.67E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.37E+00	1	1.21	(2,1,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	9.67E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.33E+00	1	1.21	(2,1,1,1,1,na)
Heat, waste	air/high population density	MJ	4.68E-02	1	1.07	(2,1,1,1,1,na)
Atrazine	soil/agricultural	kg	2.20E-04	1	1.24	(2,2,3,1,1,na)
Metolachlor	soil/agricultural	kg	1.97E-04	1	1.24	(2,2,3,1,1,na)
Glyphosate	soil/agricultural	kg	8.33E-05	1	1.24	(2,2,3,1,1,na)
Phosphorus	water/river	kg	4.78E-05	1	1.51	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.99E-03	1	1.41	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	8.59E-04	1	1.21	(2,2,1,1,1,na)
Nitrate	water/ground-	kg	9.77E-02	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	4.29E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	4.29E-05	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.18E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	2.42E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	5.10E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-3.16E-07	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	4.32E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	2.35E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	8.39E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	1.10E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.11E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	8.08E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	7.95E-08	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	7.83E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.05E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.52E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.46E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	9.58E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.62E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	4.62E-06	1	1.80	(2,2,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.40E+00	1	1.80	(2,2,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.62E+01	1	1.80	(2,2,1,1,1,na)
maize seed IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: maize seed organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
maize seed organic, at regional storehouse	CH	kg	5.77E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.35E-03	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.85E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	1.15E-03	1	1.07	(2,1,1,1,1,na)
mowing, by rotary mower	CH	ha	2.31E-03	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.85E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-02	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	1.78E-02	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and	CH	kg	7.07E+00	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	3.54E-01	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	1.50E-02	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	8.65E-05	1	2.09	(4,5,na,na,na,na)
green manure organic, until April	CH	ha	3.85E-04	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.92E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.73E+00	1	1.21	(2,1,1,1,1,na)
Transformation, from pasture and meadow	resource/land	m2	1.12E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.85E+00	1	1.21	(2,1,1,1,1,na)
Heat, waste	air/high population density	MJ	5.40E-02	1	1.07	(2,1,1,1,1,na)
Phosphorus	water/river	kg	5.52E-05	1	1.51	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.68E-03	1	1.41	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.36E-02	1	1.21	(2,2,1,1,1,na)
Nitrate	water/ground-	kg	8.15E-02	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	4.97E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	4.97E-05	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.52E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	3.17E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	1.32E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	5.38E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	6.03E-06	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	8.71E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	7.55E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	2.18E-04	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	7.88E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.20E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	1.15E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	5.51E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	4.15E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	1.09E-06	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	2.53E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.09E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	7.01E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.36E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.12E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	4.25E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.11E-05	1	1.80	(2,2,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.40E+00	1	1.80	(2,2,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.62E+01	1	1.80	(2,2,1,1,1,na)
maize seed organic, at farm	CH	kg	1.00E+00			

Unit process inventory for: potato seed IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	4.00E-05	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by rotary harrow	CH	ha	8.00E-05	1	1.07	(2,1,1,1,1,na)
tillage, hoeing and earthing-up, potatoes	CH	ha	8.00E-05	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	4.00E-05	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	8.00E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	5.23E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	5.48E-01	1	1.07	(2,1,1,1,1,na)
harvesting, by complete harvester, potatoes	CH	ha	4.00E-05	1	1.07	(2,1,1,1,1,na)
potato grading	CH	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
potato haulm cutting	CH	ha	4.00E-05	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.72E-04	1	1.07	(2,1,1,1,1,na)
potato planting	CH	ha	4.00E-05	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-03	1	1.07	(2,1,1,1,1,na)
potato seed IP, at regional storehouse	CH	kg	1.25E-01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	6.83E-04	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	2.40E-04	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	3.42E-04	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	5.17E-05	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	6.22E-04	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	4.08E-05	1	1.07	(2,1,1,1,1,na)
phthalamide-compounds, at regional storehouse	CH	kg	1.31E-04	1	1.13	(2,2,3,1,1,na)
dithiocarbamate-compounds, at regional storehouse	CH	kg	1.70E-04	1	1.13	(2,2,3,1,1,na)
nitrile-compounds, at regional storehouse	CH	kg	6.24E-06	1	1.13	(2,2,3,1,1,na)
metolachlor, at regional storehouse	CH	kg	6.00E-06	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	3.24E-05	1	1.13	(2,2,3,1,1,na)
[sulfonyl]urea-compounds, at regional storehouse	CH	kg	4.00E-07	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	1.88E-03	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	5.28E-04	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	5.28E-04	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	3.72E-03	1	2.09	(4,5,na,na,na,na)
green manure IP, until march	CH	ha	4.00E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	3.08E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.71E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.39E-01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	4.00E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	4.00E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	6.39E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	4.19E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.71E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.68E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	7.31E-04	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.88E-04	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.89E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.33E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.87E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.81E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	3.19E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	3.93E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	4.01E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	5.26E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	2.47E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	4.86E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	8.46E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	8.69E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	3.36E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	3.11E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	7.59E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	9.17E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.75E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.41E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	9.37E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	4.37E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.17E-06	1	1.80	(2,2,1,1,1,na)
Chlorothalonil	soil/agricultural	kg	1.31E-04	1	1.24	(2,2,3,1,1,na)
Fenpiclonil	soil/agricultural	kg	6.24E-06	1	1.24	(2,2,3,1,1,na)
Mancozeb	soil/agricultural	kg	1.70E-04	1	1.24	(2,2,3,1,1,na)
Metribuzin	soil/agricultural	kg	6.00E-06	1	1.24	(2,2,3,1,1,na)
Orbencarb	soil/agricultural	kg	3.24E-05	1	1.24	(2,2,3,1,1,na)
Teflubenzuron	soil/agricultural	kg	4.00E-07	1	1.24	(2,2,3,1,1,na)
potato seed IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: potato seed organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	1.03E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by rotary harrow	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
tillage, hoeing and earthing-up, potatoes	CH	ha	1.03E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	8.28E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.05E+00	1	1.07	(2,1,1,1,1,na)
harvesting, by complete harvester, potatoes	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
potato grading	CH	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
potato haulm cutting	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.26E-04	1	1.07	(2,1,1,1,1,na)
potato planting	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-03	1	1.07	(2,1,1,1,1,na)
potato seed organic, at regional storehouse	CH	kg	1.60E-01	1	1.07	(2,1,1,1,1,na)
copper oxide, at plant	RER	kg	1.09E-04	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	2.40E-03	1	2.09	(4,5,na,na,na,na)
green manure organic, until march	CH	ha	5.13E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.96E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.60E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.78E-01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	5.13E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	5.13E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	8.08E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	5.37E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.65E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	4.73E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.09E-03	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.07E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.94E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	3.63E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.04E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	8.75E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	7.50E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	7.34E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	8.42E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	2.50E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	7.38E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.13E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	1.14E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	5.24E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	4.02E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	1.01E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	2.42E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.39E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	9.01E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.84E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.46E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	5.64E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.46E-06	1	1.80	(2,2,1,1,1,na)
potato seed organic, at farm	CH	kg	1.00E+00			

Appendix A12 to Chapter 12 (Feedstuffs)

Unit-Process Inventories from Chapter 12 (Feedstuffs)

(Last Changes 2004)

Tab. A. 16 Unit-process inventories for feedstuffs.

Unit process inventory for: wheat IP, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
wheat grains IP, at farm	CH	kg	6.10E-01	1	1.21	(1,1,1,1,3,na)
wheat grains extensive, at farm	CH	kg	3.90E-01	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	8.70E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
transport, barge	RER	tkm	6.00E-02	1	2.01	(3,1,1,1,1,na)
transport, transoceanic freight ship	OCE	tkm	3.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
wheat IP, at feed mill	CH	kg	1.00E+00			

Unit process inventory for: wheat organic, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
wheat grains organic, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	8.70E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
transport, barge	RER	tkm	6.00E-02	1	2.01	(3,1,1,1,1,na)
transport, transoceanic freight ship	OCE	tkm	3.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
wheat organic, at feed mill	CH	kg	1.00E+00			

(XFeedstuffsX)

Unit process inventory for: rye IP, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
rye grains IP, at farm	CH	kg	6.10E-01	1	1.21	(1,1,1,1,3,na)
rye grains extensive, at farm	CH	kg	3.90E-01	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	8.70E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
transport, barge	RER	tkm	6.00E-02	1	2.01	(3,1,1,1,1,na)
transport, transoceanic freight ship	OCE	tkm	3.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
rye IP, at feed mill	CH	kg	1.00E+00			

Unit process inventory for: rye organic, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
rye grains organic, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	8.70E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
transport, barge	RER	tkm	6.00E-02	1	2.01	(3,1,1,1,1,na)
transport, transoceanic freight ship	OCE	tkm	3.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
rye organic, at feed mill	CH	kg	1.00E+00			

Unit process inventory for: barley IP, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
barley grains IP, at farm	CH	kg	4.20E-01	1	1.21	(1,1,1,1,3,na)
barley grains extensive, at farm	CH	kg	5.80E-01	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	8.70E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
transport, barge	RER	tkm	6.00E-02	1	2.01	(3,1,1,1,1,na)
transport, transoceanic freight ship	OCE	tkm	3.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
barley IP, at feed mill	CH	kg	1.00E+00			

(XFeedstuffsX)

Unit process inventory for: barley organic, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
barley grains organic, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	8.70E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
transport, barge	RER	tkm	6.00E-02	1	2.01	(3,1,1,1,1,na)
transport, transoceanic freight ship	OCE	tkm	3.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
barley organic, at feed mill	CH	kg	1.00E+00			

Unit process inventory for: grain maize IP, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
grain maize IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	9.00E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
grain maize IP, at feed mill	CH	kg	1.00E+00			

Unit process inventory for: grain maize organic, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
grain maize organic, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	3.50E-02	1	1.23	(2,3,2,3,1,5)
Heat, waste	air/high population density	MJ	1.26E-01	1	1.23	(2,3,2,3,1,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.45E-01	1	1.23	(2,3,2,3,1,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	9.00E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	1.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
grain maize organic, at feed mill	CH	kg	1.00E+00			

(XFeedstuffsX)

Unit process inventory for: protein peas IP, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
protein peas, IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	2.50E-02	1	1.31	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	9.00E-02	1	1.31	(2,3,2,3,3,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.00E-01	1	1.31	(2,3,2,3,3,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	3.00E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	7.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
protein peas IP, at feed mill	CH	kg	1.00E+00			

Unit process inventory for: fava beans IP, at feed mill, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
fava beans IP, at farm	CH	kg	1.00E+00	1	1.21	(1,1,1,1,3,na)
electricity, low voltage, at grid	CH	kWh	2.50E-02	1	1.31	(2,3,2,3,3,5)
Heat, waste	air/high population density	MJ	9.00E-02	1	1.31	(2,3,2,3,3,5)
natural gas, burned in industrial furnace >100kW	RER	MJ	1.00E-01	1	1.31	(2,3,2,3,3,5)
tap water, at user	RER	kg	5.60E-02	1	1.60	(3,3,3,3,4,5)
treatment, sewage, to wastewater treatment, class 2	CH	m3	3.80E-05	1	1.60	(3,3,3,3,4,5)
transport, lorry 28t	CH	tkm	3.00E-02	1	2.01	(3,1,1,1,1,na)
transport, lorry 32t	RER	tkm	7.00E-01	1	2.01	(3,1,1,1,1,na)
building, multi-storey (I)	RER	m3	8.50E-06	1	3.02	(3,3,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	5.00E-05	1	1.52	(3,3,1,1,1,na)
Occupation, construction site	resource/land	m2a	2.00E-06	1	1.52	(3,3,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
Transformation, from unknown	resource/land	m2	1.00E-06	1	2.02	(3,3,1,1,1,na)
fava beans IP, at feed mill	CH	kg	1.00E+00			

Appendix A13 to Chapter 13 (Grass-, Maize- and Grain-Drying)

Literature Data on Energy Demand of Drying Processes

Tab. A. 17 Energy demand in drying processes.

Dried stuff	Reference	Fuel type	Energy demand [MJ / kg water evaporated]	From X% to Y% of water content	Drying temperature [°C]	Observations
Grass	Baumgartner 1991	Heating oil and electricity	3.9	82 ⇔ 12		
	Schmid 2001	Heating oil	4.9			
		Gas	4			
	Christen 2002	Heating oil	5.6			
		Gas	4.3			
	This study	Heating oil Electricity Total	5 + 0.2 = 5.2	75 ⇔ 13	110 - 120	
Maize (whole plant)	Schmid 2001	Heating oil	2.9			
		Gas	2.6			
	Christen 2002	Heating oil	3			
		Gas	2.8			
	This study	Heating oil Electricity Total	3 + 0.2 = 5.2	70 ⇔ 13	110 - 120	
Grain and grain maize	Audsley 2000	Diesel, gas or straw	6	20 ⇔ 15	90	High-temperature drying
		Diesel, gas or straw	10	20 ⇔ 15	40	Low-temperature drying
		Electricity	2.3		ambient	Ventilation only
	Strehler 1985	Heating oil, wood or straw	5			
	Böckelmann 2000	Heating oil, wood or straw	5.1			
Grain and grain maize	Hellevang 1994	Heating oil, wood or straw	3.2			Low-temperature drying
		Heating oil, wood or straw	5.7			High-temperature drying
	Jakob <i>et al.</i> 1993	Heating oil	3.3			Grain maize
	Zimmer 1992	Heating oil	5.1			
	Quade 1993	Heating oil	2.9	~ 80 ⇔ 12		
			5.3	20 ⇔ 14	70 - 100	Cereals human consumption
			6.3	20 ⇔ 14	50	Seed malting barley
			5.1	14 ⇔ 8	70 - 80	Lay rape
			4.4	40 ⇔ 14	70 - 100	Grain maize (feeding)
Grain	Sonesson 1993	Heating oil	6.6			
	Thompsson	Heating oil	5.7			

Dried stuff	Reference	Fuel type	Energy demand [MJ / kg water evaporated]	From X% to Y% of water content	Drying temperature [°C]	Observations
and grain maize	1999	and electricity	4.6			
	Johansson 1998	Heating oil and electricity	6.4			
	This study	Heating oil Electricity Total	7 + 3.6 = 10.6	16 ⇔ 14	80 - 90	Low-temperature drying
		Heating oil Electricity Total	5 + 3.6 = 8.6	16 ⇔ 14	95 - 100	High-temperature drying

Unit-Process Inventories from Chapter 13 (Grass-, Maize- and Grain-Drying)

(Last Changes 2004)

Tab. A. 18 Unit-process inventories for drying processes.

Unit process inventory for: grass drying, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
light fuel oil, burned in industrial furnace 1MW, non-modulating	CH	MJ	5.00E+00	1	1.09	(2,3,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	5.00E-02	1	1.22	(2,5,1,1,1,na)
cast iron, at plant	RER	kg	7.05E-04	1	1.21	(4,na,1,1,1,na)
building, hall (I)	CH	m2	4.70E-06	1	3.01	(3,2,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	4.41E-02	1	1.52	(3,2,1,1,1,3)
Occupation, construction site	resource/land	m2a	2.65E-03	1	1.52	(3,2,1,1,1,3)
Transformation, from pasture and meadow	resource/land	m2	1.76E-05	1	2.02	(3,2,1,1,1,3)
Transformation, to industrial area, built up	resource/land	m2	1.76E-05	1	2.02	(3,2,1,1,1,3)
Heat, waste	air/low population density	MJ	1.80E-01	1	1.14	(2,4,1,1,1,3)
grass drying	CH	kg	1.00E+00			

Unit process inventory for: maize drying, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
light fuel oil, burned in industrial furnace 1MW, non-modulating	CH	MJ	3.00E+00	1	1.09	(2,3,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	5.00E-02	1	1.22	(2,5,1,1,1,na)
cast iron, at plant	RER	kg	7.05E-04	1	1.21	(4,na,1,1,1,na)
building, hall (I)	CH	m2	4.70E-06	1	3.01	(3,2,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	4.41E-02	1	1.52	(3,2,1,1,1,3)
Occupation, construction site	resource/land	m2a	2.65E-03	1	1.52	(3,2,1,1,1,3)
Transformation, from pasture and meadow	resource/land	m2	1.76E-05	1	2.02	(3,2,1,1,1,3)
Transformation, to industrial area, built up	resource/land	m2	1.76E-05	1	2.02	(3,2,1,1,1,3)
Heat, waste	air/low population density	MJ	1.80E-01	1	1.14	(2,4,1,1,1,3)
maize drying	CH	kg	1.00E+00			

Unit process inventory for: grain drying, low temperature, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
light fuel oil, burned in industrial furnace 1MW, non-modulating	CH	MJ	7.00E+00	1	1.08	(2,na,2,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.00E+00	1	1.22	(2,5,2,2,1,na)
cast iron, at plant	RER	kg	5.83E-03	1	1.21	(4,na,1,1,1,na)
building, hall (I)	CH	m2	1.25E-05	1	3.01	(3,2,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	6.25E-01	1	1.52	(3,2,1,1,1,3)
Occupation, construction site	resource/land	m2a	3.75E-02	1	1.52	(3,2,1,1,1,3)
Transformation, from pasture and meadow	resource/land	m2	2.50E-04	1	2.02	(3,2,1,1,1,3)
Transformation, to industrial area, built up	resource/land	m2	2.50E-04	1	2.02	(3,2,1,1,1,3)
Heat, waste	air/low population density	MJ	3.60E+00	1	1.13	(2,4,2,2,1,na)
grain drying, low temperature	CH	kg	1.00E+00			

Unit process inventory for: grain drying, high temperature, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
light fuel oil, burned in industrial furnace 1MW, non-modulating	CH	MJ	5.00E+00	1	1.08	(2,na,2,2,1,na)
electricity, low voltage, at grid	CH	kWh	1.00E+00	1	1.22	(2,5,2,2,1,na)
cast iron, at plant	RER	kg	5.83E-03	1	1.21	(4,na,1,1,1,na)
building, hall (I)	CH	m2	1.25E-05	1	3.01	(3,2,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	6.25E-01	1	1.52	(3,2,1,1,1,3)
Occupation, construction site	resource/land	m2a	3.75E-02	1	1.52	(3,2,1,1,1,3)
Transformation, from pasture and meadow	resource/land	m2	2.50E-04	1	2.02	(3,2,1,1,1,3)
Transformation, to industrial area, built up	resource/land	m2	2.50E-04	1	2.02	(3,2,1,1,1,3)
Heat, waste	air/low population density	MJ	3.60E+00	1	1.13	(2,4,2,2,1,na)
grain drying, high temperature	CH	kg	1.00E+00			

Appendix A14 to Chapter 14 (Arable Crop Production in Switzerland)

Unit-Process Inventories from Chapter 14 (Arable Crop Production in Switzerland)

Tab. A. 19 Unit-process inventories for arable crop production.

Unit process inventory for: wheat extensive, CH								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	wheat grains extensive, at farm CH (kg)	wheat straw extensive, at farm CH (kg)
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
tillage, currying, by weeder	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
fertilising, by broadcaster	CH	ha	4.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
slurry spreading, by vacuum tanker	CH	m3	8.25E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	9.62E+02	1	1.07	(2,1,1,1,1,na)	92%	8%
baling	CH	unit	4.65E+00	1	1.07	(2,1,1,1,1,na)		100%
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
loading bales	CH	unit	2.02E+01	1	1.07	(2,1,1,1,1,na)		100%
application of plant protection products, by field sprayer	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
transport, tractor and trailer	CH	tkm	5.37E+01	1	1.07	(2,1,1,1,1,na)	100%	
wheat seed IP, at regional storehouse	CH	kg	1.80E+02	1	1.07	(2,1,1,1,1,na)	92%	8%
grain drying, low temperature	CH	kg	6.31E+01	1	1.07	(2,1,1,1,1,na)	100%	
ammonium nitrate, as N, at regional storehouse	RER	kg	5.64E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
urea, as N, at regional storehouse	RER	kg	1.98E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	5.02E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.83E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
ammonium sulphate, as N, at regional storehouse	RER	kg	4.27E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.89E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
single superphosphate, as P2O5, at regional storehouse	RER	kg	7.60E-01	1	1.07	(2,1,1,1,1,na)	92%	8%
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.28E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.10E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
thomas meal, as P2O5, at regional storehouse	RER	kg	2.38E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
potassium chloride, as K2O, at regional storehouse	RER	kg	1.51E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
potassium sulphate, as K2O, at regional storehouse	RER	kg	9.89E-01	1	1.07	(2,1,1,1,1,na)	92%	8%
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	7.20E-01	1	1.13	(2,2,3,1,1,na)	92%	8%
cyclic N-compounds, at regional storehouse	RER	kg	1.80E-02	1	1.13	(2,2,3,1,1,na)	92%	8%
nitrile-compounds, at regional storehouse	RER	kg	2.00E-01	1	1.13	(2,2,3,1,1,na)	92%	8%
phenoxy-compounds, at regional storehouse	RER	kg	3.50E-01	1	1.13	(2,2,3,1,1,na)	92%	8%
transport, van <3.5t	CH	tkm	2.74E+00	1	2.09	(4,5,na,na,na,na)	92%	8%
transport, lorry 20-28t, fleet average	CH	tkm	4.80E+01	1	2.09	(4,5,na,na,na,na)	92%	8%
transport, freight, rail	CH	tkm	4.80E+01	1	2.09	(4,5,na,na,na,na)	92%	8%
transport, barge	RER	tkm	3.53E+02	1	2.09	(4,5,na,na,na,na)	92%	8%
transport, transoceanic freight ship	OCE	tkm	8.62E+01	1	2.09	(4,5,na,na,na,na)	92%	8%
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.35E+05	1	1.07	(2,1,1,1,1,na)	59%	41%
Carbon dioxide, in air	resource/in air	kg	1.12E+04	1	1.07	(2,1,1,1,1,na)	61%	39%
Occupation, arable	resource/land	m2a	7.94E+03	1	1.11	(2,1,1,1,1,na)	92%	8%
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	92%	8%
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	92%	8%
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	92%	8%
Nitrate	water/ground-	kg	3.88E+02	1	1.51	(2,2,1,1,1,na)	92%	8%
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	92%	8%
Phosphate	water/river	kg	7.89E-01	1	1.51	(2,2,1,1,1,na)	92%	8%
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	92%	8%
Ammonia	air/low population density	kg	1.05E-01	1	1.21	(2,2,1,1,1,na)	92%	8%
Carbon dioxide, fossil	air/low population density	kg	1.55E+01	1	1.07	(2,2,1,1,1,na)	92%	8%
Nitrogen oxides	air/low population density	kg	1.53E+00	1	1.41	(2,2,1,1,1,na)	92%	8%
Dinitrogen monoxide	air/low population density	kg	7.27E+00	1	1.41	(2,2,1,1,1,na)	92%	8%
Cadmium	soil/agricultural	kg	2.84E-03	1	1.51	(2,2,1,1,1,na)	55%	45%
Chromium	soil/agricultural	kg	3.50E-02	1	1.51	(2,2,1,1,1,na)	41%	59%
Copper	soil/agricultural	kg	-1.54E-03	1	1.51	(2,2,1,1,1,na)	77%	23%
Lead	soil/agricultural	kg	3.42E-03	1	1.51	(2,2,1,1,1,na)	30%	70%
Mercury	soil/agricultural	kg	1.40E-04	1	1.51	(2,2,1,1,1,na)	21%	79%
Nickel	soil/agricultural	kg	6.54E-03	1	1.51	(2,2,1,1,1,na)	45%	55%
Zinc	soil/agricultural	kg	3.19E-02	1	1.51	(2,2,1,1,1,na)	88%	12%
Cadmium, ion	water/river	kg	2.60E-05	1	1.80	(2,2,1,1,1,na)	55%	45%
Chromium, ion	water/river	kg	2.92E-03	1	1.80	(2,2,1,1,1,na)	41%	59%
Copper, ion	water/river	kg	2.35E-03	1	1.80	(2,2,1,1,1,na)	77%	23%
Lead	water/river	kg	4.87E-04	1	1.80	(2,2,1,1,1,na)	30%	70%
Mercury	water/river	kg	7.44E-06	1	1.80	(2,2,1,1,1,na)	21%	79%
Nickel, ion	water/river	kg	1.92E-03	1	1.80	(2,2,1,1,1,na)	45%	55%
Zinc, ion	water/river	kg	3.72E-03	1	1.80	(2,2,1,1,1,na)	88%	12%
Cadmium, ion	water/ground-	kg	4.22E-05	1	1.80	(2,2,1,1,1,na)	55%	45%
Chromium, ion	water/ground-	kg	2.00E-02	1	1.80	(2,2,1,1,1,na)	41%	59%
Copper, ion	water/ground-	kg	3.27E-03	1	1.80	(2,2,1,1,1,na)	77%	23%
Lead	water/ground-	kg	1.17E-04	1	1.80	(2,2,1,1,1,na)	30%	70%
Mercury	water/ground-	kg	8.98E-06	1	1.80	(2,2,1,1,1,na)	21%	79%
Zinc, ion	water/ground-	kg	1.93E-02	1	1.80	(2,2,1,1,1,na)	88%	12%
Difenoconazole	soil/agricultural	kg	1.80E-02	1	1.24	(2,2,3,1,1,na)	92%	8%
Ioxynil	soil/agricultural	kg	2.00E-01	1	1.24	(2,2,3,1,1,na)	92%	8%
Isoproturon	soil/agricultural	kg	7.20E-01	1	1.24	(2,2,3,1,1,na)	92%	8%
Mecoprop-P	soil/agricultural	kg	3.50E-01	1	1.24	(2,2,3,1,1,na)	92%	8%
wheat grains extensive, at farm	CH	kg	5.30E+03				100%	
wheat straw extensive, at farm	CH	kg	3.23E+03					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: wheat IP, CH									
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	wheat grains IP, at farm CH (kg)	wheat straw IP, at farm CH (kg)	
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
tillage, currying, by weeder	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
fertilising, by broadcaster	CH	ha	4.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
slurry spreading, by vacuum tanker	CH	m3	3.28E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.11E+02	1	1.07	(2,1,1,1,1,na)	92%	8%	
baling	CH	unit	5.63E+00	1	1.07	(2,1,1,1,1,na)		100%	
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
loading bales	CH	unit	2.45E+01	1	1.07	(2,1,1,1,1,na)		100%	
application of plant protection products, by field sprayer	CH	ha	3.40E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
transport, tractor and trailer	CH	tkm	6.50E+01	1	1.07	(2,1,1,1,1,na)	100%		
wheat seed IP, at regional storehouse	CH	kg	1.80E+02	1	1.07	(2,1,1,1,1,na)	92%	8%	
grain drying, low temperature	CH	kg	7.65E+01	1	1.07	(2,1,1,1,1,na)	100%		
ammonium nitrate, as N, at regional storehouse	RER	kg	6.71E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
urea, as N, at regional storehouse	RER	kg	2.36E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
diammonium phosphate, as N, at regional storehouse	RER	kg	6.98E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	3.37E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
ammonium sulphate, as N, at regional storehouse	RER	kg	5.08E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.62E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
single superphosphate, as P2O5, at regional storehouse	RER	kg	1.06E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.78E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.53E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
thomas meal, as P2O5, at regional storehouse	RER	kg	3.31E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
potassium chloride, as K2O, at regional storehouse	RER	kg	4.49E+01	1	1.07	(2,1,1,1,1,na)	92%	8%	
potassium sulphate, as K2O, at regional storehouse	RER	kg	2.95E+00	1	1.07	(2,1,1,1,1,na)	92%	8%	
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	1.27E+00	1	1.13	(2,2,3,1,1,na)	92%	8%	
cyclic N-compounds, at regional storehouse	RER	kg	6.28E-01	1	1.13	(2,2,3,1,1,na)	92%	8%	
nitrile-compounds, at regional storehouse	RER	kg	1.90E-01	1	1.13	(2,2,3,1,1,na)	92%	8%	
pesticide unspecified, at regional storehouse	RER	kg	2.60E-01	1	1.13	(2,2,3,1,1,na)	92%	8%	
phenoxy-compounds, at regional storehouse	RER	kg	3.10E-01	1	1.13	(2,2,3,1,1,na)	92%	8%	
transport, van <3.5t	CH	tkm	2.78E+00	1	2.09	(4,5,na,na,na,na)	92%	8%	
transport, lorry 20-28t, fleet average	CH	tkm	6.51E+01	1	2.09	(4,5,na,na,na,na)	92%	8%	
transport, freight, rail	CH	tkm	6.51E+01	1	2.09	(4,5,na,na,na,na)	92%	8%	
transport, barge	RER	tkm	4.36E+02	1	2.09	(4,5,na,na,na,na)	92%	8%	
transport, transoceanic freight ship	OCE	tkm	1.20E+02	1	2.09	(4,5,na,na,na,na)	92%	8%	
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.64E+05	1	1.07	(2,1,1,1,1,na)	59%	41%	
Carbon dioxide, in air	resource/in air	kg	1.36E+04	1	1.07	(2,1,1,1,1,na)	61%	39%	
Occupation, arable	resource/land	m2a	7.94E+03	1	1.11	(2,1,1,1,1,na)	92%	8%	
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	92%	8%	
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	92%	8%	
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	92%	8%	
Nitrate	water/ground-	kg	3.32E+02	1	1.51	(2,2,1,1,1,na)	92%	8%	
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	92%	8%	
Phosphate	water/river	kg	8.13E-01	1	1.51	(2,2,1,1,1,na)	92%	8%	
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	92%	8%	
Ammonia	air/low population density	kg	9.06E+00	1	1.21	(2,2,1,1,1,na)	92%	8%	
Carbon dioxide, fossil	air/low population density	kg	1.85E+01	1	1.07	(2,2,1,1,1,na)	92%	8%	
Nitrogen oxides	air/low population density	kg	1.53E+00	1	1.41	(2,2,1,1,1,na)	92%	8%	
Dinitrogen monoxide	air/low population density	kg	7.29E+00	1	1.41	(2,2,1,1,1,na)	92%	8%	
Cadmium	soil/agricultural	kg	3.98E-03	1	1.51	(2,2,1,1,1,na)	55%	45%	
Chromium	soil/agricultural	kg	5.58E-02	1	1.51	(2,2,1,1,1,na)	41%	59%	
Copper	soil/agricultural	kg	-1.18E-02	1	1.51	(2,2,1,1,1,na)	77%	23%	
Lead	soil/agricultural	kg	3.21E-03	1	1.51	(2,2,1,1,1,na)	30%	70%	
Mercury	soil/agricultural	kg	1.64E-05	1	1.51	(2,2,1,1,1,na)	21%	79%	
Nickel	soil/agricultural	kg	7.19E-03	1	1.51	(2,2,1,1,1,na)	45%	55%	
Zinc	soil/agricultural	kg	3.30E-04	1	1.51	(2,2,1,1,1,na)	88%	12%	
Cadmium, ion	water/river	kg	2.71E-05	1	1.80	(2,2,1,1,1,na)	55%	45%	
Chromium, ion	water/river	kg	2.96E-03	1	1.80	(2,2,1,1,1,na)	41%	59%	
Copper, ion	water/river	kg	2.26E-03	1	1.80	(2,2,1,1,1,na)	77%	23%	
Lead	water/river	kg	4.76E-04	1	1.80	(2,2,1,1,1,na)	30%	70%	
Mercury	water/river	kg	5.00E-06	1	1.80	(2,2,1,1,1,na)	21%	79%	
Nickel, ion	water/river	kg	1.99E-03	1	1.80	(2,2,1,1,1,na)	45%	55%	
Zinc, ion	water/river	kg	3.33E-03	1	1.80	(2,2,1,1,1,na)	88%	12%	
Cadmium, ion	water/ground-	kg	4.40E-05	1	1.80	(2,2,1,1,1,na)	55%	45%	
Chromium, ion	water/ground-	kg	2.03E-02	1	1.80	(2,2,1,1,1,na)	41%	59%	
Copper, ion	water/ground-	kg	3.16E-03	1	1.80	(2,2,1,1,1,na)	77%	23%	
Lead	water/ground-	kg	1.14E-04	1	1.80	(2,2,1,1,1,na)	30%	70%	
Mercury	water/ground-	kg	6.03E-06	1	1.80	(2,2,1,1,1,na)	21%	79%	
Zinc, ion	water/ground-	kg	1.73E-02	1	1.80	(2,2,1,1,1,na)	88%	12%	
Difenoconazole	soil/agricultural	kg	1.80E-02	1	1.24	(2,2,3,1,1,na)	92%	8%	
Fenpropimorph	soil/agricultural	kg	4.50E-01	1	1.24	(2,2,3,1,1,na)	92%	8%	
Tebuconazole	soil/agricultural	kg	1.60E-01	1	1.24	(2,2,3,1,1,na)	92%	8%	
Chlormequat	soil/agricultural	kg	2.30E-01	1	1.24	(2,2,3,1,1,na)	92%	8%	
Metaldehyde	soil/agricultural	kg	3.00E-02	1	1.24	(2,2,3,1,1,na)	92%	8%	
Ioxynil	soil/agricultural	kg	1.90E-01	1	1.24	(2,2,3,1,1,na)	92%	8%	
Isoproturon	soil/agricultural	kg	1.27E+00	1	1.24	(2,2,3,1,1,na)	92%	8%	
Mecoprop-P	soil/agricultural	kg	3.10E-01	1	1.24	(2,2,3,1,1,na)	92%	8%	
wheat grains IP, at farm	CH	kg	6.43E+03				100%		
wheat straw IP, at farm	CH	kg	3.92E+03					100%	

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: wheat organic, CH								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	wheat grains organic, at farm CH (kg)	wheat straw organic, at farm CH (kg)
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	93%	7%
tillage, currying, by weeder	CH	ha	3.00E+00	1	1.07	(2,1,1,1,1,na)	93%	7%
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	93%	7%
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	93%	7%
slurry spreading, by vacuum tanker	CH	m3	4.90E+01	1	1.07	(2,1,1,1,1,na)	93%	7%
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	6.04E+03	1	1.07	(2,1,1,1,1,na)	93%	7%
balancing	CH	unit	4.75E+00	1	1.07	(2,1,1,1,1,na)		100%
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	93%	7%
loading bales	CH	unit	2.07E+01	1	1.07	(2,1,1,1,1,na)		100%
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	93%	7%
transport, tractor and trailer	CH	tkm	4.12E+01	1	1.07	(2,1,1,1,1,na)	100%	
wheat seed organic, at regional storehouse	CH	kg	2.00E+02	1	1.07	(2,1,1,1,1,na)	93%	7%
grain drying, low temperature	CH	kg	4.84E+01	1	1.07	(2,1,1,1,1,na)	100%	
transport, van <3.5t	CH	tkm	3.00E+00	1	2.09	(4,5,na,na,na,na)	93%	7%
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.17E+05	1	1.07	(2,1,1,1,1,na)	52%	48%
Carbon dioxide, in air	resource/in air	kg	9.62E+03	1	1.07	(2,1,1,1,1,na)	54%	46%
Occupation, arable	resource/land	m2a	7.94E+03	1	1.11	(2,1,1,1,1,na)	93%	7%
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	93%	7%
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	93%	7%
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	93%	7%
Nitrate	water/ground-	kg	4.00E+02	1	1.51	(2,2,1,1,1,na)	93%	7%
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	93%	7%
Phosphate	water/river	kg	1.05E+00	1	1.51	(2,2,1,1,1,na)	93%	7%
Phosphate	water/ground-	kg	2.23E-01	1	1.51	(2,2,1,1,1,na)	93%	7%
Ammonia	air/low population density	kg	3.39E+01	1	1.21	(2,2,1,1,1,na)	93%	7%
Nitrogen oxides	air/low population density	kg	1.33E+00	1	1.41	(2,2,1,1,1,na)	93%	7%
Dinitrogen monoxide	air/low population density	kg	6.34E+00	1	1.41	(2,2,1,1,1,na)	93%	7%
Cadmium	soil/agricultural	kg	1.26E-04	1	1.51	(2,2,1,1,1,na)	48%	52%
Chromium	soil/agricultural	kg	-7.58E-03	1	1.51	(2,2,1,1,1,na)	35%	65%
Copper	soil/agricultural	kg	1.29E-01	1	1.51	(2,2,1,1,1,na)	71%	29%
Lead	soil/agricultural	kg	8.27E-03	1	1.51	(2,2,1,1,1,na)	25%	75%
Mercury	soil/agricultural	kg	1.42E-03	1	1.51	(2,2,1,1,1,na)	17%	83%
Nickel	soil/agricultural	kg	1.21E-02	1	1.51	(2,2,1,1,1,na)	38%	62%
Zinc	soil/agricultural	kg	7.09E-01	1	1.51	(2,2,1,1,1,na)	85%	15%
Cadmium, ion	water/river	kg	1.39E-05	1	1.80	(2,2,1,1,1,na)	48%	52%
Chromium, ion	water/river	kg	2.44E-03	1	1.80	(2,2,1,1,1,na)	35%	65%
Copper, ion	water/river	kg	2.54E-03	1	1.80	(2,2,1,1,1,na)	71%	29%
Lead	water/river	kg	8.84E-04	1	1.80	(2,2,1,1,1,na)	25%	75%
Mercury	water/river	kg	9.06E-06	1	1.80	(2,2,1,1,1,na)	17%	83%
Nickel, ion	water/river	kg	2.20E-03	1	1.80	(2,2,1,1,1,na)	38%	62%
Zinc, ion	water/river	kg	5.74E-03	1	1.80	(2,2,1,1,1,na)	85%	15%
Cadmium, ion	water/ground-	kg	2.26E-05	1	1.80	(2,2,1,1,1,na)	48%	52%
Chromium, ion	water/ground-	kg	1.67E-02	1	1.80	(2,2,1,1,1,na)	35%	65%
Copper, ion	water/ground-	kg	3.54E-03	1	1.80	(2,2,1,1,1,na)	71%	29%
Lead	water/ground-	kg	2.12E-04	1	1.80	(2,2,1,1,1,na)	25%	75%
Mercury	water/ground-	kg	1.09E-05	1	1.80	(2,2,1,1,1,na)	17%	83%
Zinc, ion	water/ground-	kg	2.98E-02	1	1.80	(2,2,1,1,1,na)	85%	15%
wheat grains organic, at farm	CH	kg	4.07E+03				100%	
wheat straw organic, at farm	CH	kg	3.31E+03					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: rye extensive, CH								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	rye grains extensive, at farm CH (kg)	rye straw extensive, at farm CH (kg)
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, currying, by weeder	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
fertilising, by broadcaster	CH	ha	3.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
slurry spreading, by vacuum tanker	CH	m3	7.68E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	2.78E+02	1	1.07	(2,1,1,1,1,na)	90%	10%
baling	CH	unit	5.70E+00	1	1.07	(2,1,1,1,1,na)		100%
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
loading bales	CH	unit	2.48E+01	1	1.07	(2,1,1,1,1,na)		100%
application of plant protection products, by field sprayer	CH	ha	1.20E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
transport, tractor and trailer	CH	tkm	6.04E+01	1	1.07	(2,1,1,1,1,na)	100%	
rye seed IP, at regional storehouse	CH	kg	1.40E+02	1	1.07	(2,1,1,1,1,na)	90%	10%
grain drying, low temperature	CH	kg	7.11E+01	1	1.07	(2,1,1,1,1,na)	100%	
ammonium nitrate, as N, at regional storehouse	RER	kg	3.93E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
urea, as N, at regional storehouse	RER	kg	1.38E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
diammonium phosphate, as N, at regional storehouse	RER	kg	5.83E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	1.97E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
ammonium sulphate, as N, at regional storehouse	RER	kg	2.97E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.19E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
single superphosphate, as P2O5, at regional storehouse	RER	kg	8.82E-01	1	1.07	(2,1,1,1,1,na)	90%	10%
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.49E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.28E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
thomas meal, as P2O5, at regional storehouse	RER	kg	2.77E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
potassium chloride, as K2O, at regional storehouse	RER	kg	3.42E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
potassium sulphate, as K2O, at regional storehouse	RER	kg	2.24E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	1.16E+00	1	1.13	(2,2,3,1,1,na)	90%	10%
cyclic N-compounds, at regional storehouse	RER	kg	1.40E-02	1	1.13	(2,2,3,1,1,na)	90%	10%
dinitroaniline-compounds, at regional storehouse	RER	kg	4.20E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
pesticide unspecified, at regional storehouse	RER	kg	3.00E-02	1	1.13	(2,2,3,1,1,na)	90%	10%
transport, van <3.5t	CH	tkm	2.15E+00	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, lorry 20-28t, fleet average	CH	tkm	4.39E+01	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, freight, rail	CH	tkm	4.39E+01	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, barge	RER	tkm	2.75E+02	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, transoceanic freight ship	OCE	tkm	1.00E+02	1	2.09	(4,5,na,na,na,na)	90%	10%
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.58E+05	1	1.07	(2,1,1,1,1,na)	57%	43%
Carbon dioxide, in air	resource/in air	kg	1.31E+04	1	1.07	(2,1,1,1,1,na)	59%	41%
Occupation, arable	resource/land	m2a	8.50E+03	1	1.11	(2,1,1,1,1,na)	90%	10%
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	90%	10%
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	90%	10%
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	90%	10%
Nitrate	water/ground-	kg	3.40E+02	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphate	water/river	kg	7.99E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Ammonia	air/low population density	kg	8.02E+00	1	1.21	(2,2,1,1,1,na)	90%	10%
Carbon dioxide, fossil	air/low population density	kg	1.48E+01	1	1.07	(2,2,1,1,1,na)	90%	10%
Nitrogen oxides	air/low population density	kg	9.89E-01	1	1.41	(2,2,1,1,1,na)	90%	10%
Dinitrogen monoxide	air/low population density	kg	4.71E+00	1	1.41	(2,2,1,1,1,na)	90%	10%
Cadmium	soil/agricultural	kg	3.51E-03	1	1.51	(2,2,1,1,1,na)	69%	31%
Chromium	soil/agricultural	kg	4.35E-02	1	1.51	(2,2,1,1,1,na)	47%	53%
Copper	soil/agricultural	kg	-9.71E-03	1	1.51	(2,2,1,1,1,na)	70%	30%
Lead	soil/agricultural	kg	3.22E-03	1	1.51	(2,2,1,1,1,na)	38%	62%
Mercury	soil/agricultural	kg	7.96E-05	1	1.51	(2,2,1,1,1,na)	100%	
Nickel	soil/agricultural	kg	5.36E-03	1	1.51	(2,2,1,1,1,na)	39%	61%
Zinc	soil/agricultural	kg	4.80E-03	1	1.51	(2,2,1,1,1,na)	84%	16%
Cadmium, ion	water/river	kg	2.65E-05	1	1.80	(2,2,1,1,1,na)	69%	31%
Chromium, ion	water/river	kg	2.94E-03	1	1.80	(2,2,1,1,1,na)	47%	53%
Copper, ion	water/river	kg	2.31E-03	1	1.80	(2,2,1,1,1,na)	70%	30%
Lead	water/river	kg	4.62E-04	1	1.80	(2,2,1,1,1,na)	38%	62%
Mercury	water/river	kg	6.80E-06	1	1.80	(2,2,1,1,1,na)	100%	
Nickel, ion	water/river	kg	1.86E-03	1	1.80	(2,2,1,1,1,na)	39%	61%
Zinc, ion	water/river	kg	3.51E-03	1	1.80	(2,2,1,1,1,na)	84%	16%
Cadmium, ion	water/ground-	kg	4.30E-05	1	1.80	(2,2,1,1,1,na)	69%	31%
Chromium, ion	water/ground-	kg	2.01E-02	1	1.80	(2,2,1,1,1,na)	47%	53%
Copper, ion	water/ground-	kg	3.23E-03	1	1.80	(2,2,1,1,1,na)	70%	30%
Lead	water/ground-	kg	1.11E-04	1	1.80	(2,2,1,1,1,na)	38%	62%
Mercury	water/ground-	kg	8.20E-06	1	1.80	(2,2,1,1,1,na)	100%	
Zinc, ion	water/ground-	kg	1.82E-02	1	1.80	(2,2,1,1,1,na)	84%	16%
Difenoconazole	soil/agricultural	kg	1.40E-02	1	1.24	(2,2,3,1,1,na)	90%	10%
Metalddehyde	soil/agricultural	kg	3.00E-02	1	1.24	(2,2,3,1,1,na)	90%	10%
Chlorotoluron	soil/agricultural	kg	6.30E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Isoproturon	soil/agricultural	kg	5.30E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Pendimethalin	soil/agricultural	kg	4.20E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
rye grains extensive, at farm	CH	kg	5.97E+03				100%	
rye straw extensive, at farm	CH	kg	3.97E+03					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: rye IP, CH								
	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	rye grains IP, at farm CH (kg)	rye straw IP, at farm CH (kg)
Exchanges								
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, currying, by weeder	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
fertilising, by broadcaster	CH	ha	3.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
slurry spreading, by vacuum tanker	CH	m3	6.79E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.84E+03	1	1.07	(2,1,1,1,1,na)	90%	10%
baling	CH	unit	7.21E+00	1	1.07	(2,1,1,1,1,na)		100%
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
loading bales	CH	unit	3.13E+01	1	1.07	(2,1,1,1,1,na)		100%
application of plant protection products, by field sprayer	CH	ha	2.70E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
transport, tractor and trailer	CH	tkm	7.63E+01	1	1.07	(2,1,1,1,1,na)	100%	
rye seed IP, at regional storehouse	CH	kg	1.40E+02	1	1.07	(2,1,1,1,1,na)	90%	10%
grain drying, low temperature	CH	kg	8.98E+01	1	1.07	(2,1,1,1,1,na)	100%	
ammonium nitrate, as N, at regional storehouse	RER	kg	3.79E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
urea, as N, at regional storehouse	RER	kg	1.33E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
diammonium phosphate, as N, at regional storehouse	RER	kg	7.07E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	1.90E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
ammonium sulphate, as N, at regional storehouse	RER	kg	2.87E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.65E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
single superphosphate, as P2O5, at regional storehouse	RER	kg	1.07E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.81E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.55E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
thomas meal, as P2O5, at regional storehouse	RER	kg	3.36E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
potassium chloride, as K2O, at regional storehouse	RER	kg	4.27E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
potassium sulphate, as K2O, at regional storehouse	RER	kg	2.80E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	1.34E+00	1	1.13	(2,2,3,1,1,na)	90%	10%
benzimidazole-compounds, at regional storehouse	RER	kg	3.50E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
cyclic N-compounds, at regional storehouse	RER	kg	3.84E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
dinitroaniline-compounds, at regional storehouse	RER	kg	4.20E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
organophosphorus-compounds, at regional storehouse	RER	kg	3.20E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
pesticide unspecified, at regional storehouse	RER	kg	4.00E-02	1	1.13	(2,2,3,1,1,na)	90%	10%
transport, van <3.5t	CH	tkm	2.19E+00	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, lorry 20-28t, fleet average	CH	tkm	4.78E+01	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, freight, rail	CH	tkm	4.78E+01	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, barge	RER	tkm	2.83E+02	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, transoceanic freight ship	OCE	tkm	1.21E+02	1	2.09	(4,5,na,na,na,na)	90%	10%
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.00E+05	1	1.07	(2,1,1,1,1,na)	57%	43%
Carbon dioxide, in air	resource/in air	kg	1.66E+04	1	1.07	(2,1,1,1,1,na)	59%	41%
Occupation, arable	resource/land	m2a	8.50E+03	1	1.11	(2,1,1,1,1,na)	90%	10%
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	90%	10%
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	90%	10%
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	90%	10%
Nitrate	water/ground-	kg	3.30E+02	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphate	water/river	kg	8.14E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Ammonia	air/low population density	kg	8.50E+00	1	1.21	(2,2,1,1,1,na)	90%	10%
Carbon dioxide, fossil	air/low population density	kg	1.05E+01	1	1.07	(2,2,1,1,1,na)	90%	10%
Nitrogen oxides	air/low population density	kg	9.93E-01	1	1.41	(2,2,1,1,1,na)	90%	10%
Dinitrogen monoxide	air/low population density	kg	4.73E+00	1	1.41	(2,2,1,1,1,na)	90%	10%
Cadmium	soil/agricultural	kg	4.23E-03	1	1.51	(2,2,1,1,1,na)	69%	31%
Chromium	soil/agricultural	kg	5.80E-02	1	1.51	(2,2,1,1,1,na)	47%	53%
Copper	soil/agricultural	kg	9.84E-03	1	1.51	(2,2,1,1,1,na)	70%	30%
Lead	soil/agricultural	kg	4.33E-03	1	1.51	(2,2,1,1,1,na)	38%	62%
Mercury	soil/agricultural	kg	1.67E-04	1	1.51	(2,2,1,1,1,na)	100%	
Nickel	soil/agricultural	kg	6.73E-03	1	1.51	(2,2,1,1,1,na)	39%	61%
Zinc	soil/agricultural	kg	4.76E-03	1	1.51	(2,2,1,1,1,na)	84%	16%
Cadmium, ion	water/river	kg	2.72E-05	1	1.80	(2,2,1,1,1,na)	69%	31%
Chromium, ion	water/river	kg	2.96E-03	1	1.80	(2,2,1,1,1,na)	47%	53%
Copper, ion	water/river	kg	2.38E-03	1	1.80	(2,2,1,1,1,na)	70%	30%
Lead	water/river	kg	5.90E-04	1	1.80	(2,2,1,1,1,na)	38%	62%
Mercury	water/river	kg	7.74E-06	1	1.80	(2,2,1,1,1,na)	100%	
Nickel, ion	water/river	kg	2.01E-03	1	1.80	(2,2,1,1,1,na)	39%	61%
Zinc, ion	water/river	kg	3.97E-03	1	1.80	(2,2,1,1,1,na)	84%	16%
Cadmium, ion	water/ground-	kg	4.41E-05	1	1.80	(2,2,1,1,1,na)	69%	31%
Chromium, ion	water/ground-	kg	2.03E-02	1	1.80	(2,2,1,1,1,na)	47%	53%
Copper, ion	water/ground-	kg	3.32E-03	1	1.80	(2,2,1,1,1,na)	70%	30%
Lead	water/ground-	kg	1.42E-04	1	1.80	(2,2,1,1,1,na)	38%	62%
Mercury	water/ground-	kg	9.34E-06	1	1.80	(2,2,1,1,1,na)	100%	
Zinc, ion	water/ground-	kg	2.06E-02	1	1.80	(2,2,1,1,1,na)	84%	16%
Chlorothalonil	soil/agricultural	kg	3.50E-01	1	1.22	(2,3,1,1,1,na)	90%	10%
Difenoconazole	soil/agricultural	kg	1.40E-02	1	1.24	(2,2,3,1,1,na)	90%	10%
Fenpropimorph	soil/agricultural	kg	3.70E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Ethephon	soil/agricultural	kg	3.20E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Metalddehyde	soil/agricultural	kg	4.00E-02	1	1.24	(2,2,3,1,1,na)	90%	10%
Chlorotoluron	soil/agricultural	kg	3.90E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Isoproturon	soil/agricultural	kg	9.50E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Pendimethalin	soil/agricultural	kg	4.20E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
rye grains IP, at farm	CH	kg	7.54E+03				100%	
rye straw IP, at farm	CH	kg	5.01E+03					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: rye organic, CH								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	rye grains organic, at farm CH (kg)	rye straw organic, at farm CH (kg)
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
tillage, currying, by weeder	CH	ha	3.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
slurry spreading, by vacuum tanker	CH	m3	3.52E+01	1	1.07	(2,1,1,1,1,na)	92%	8%
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	4.34E+03	1	1.07	(2,1,1,1,1,na)	92%	8%
baling	CH	unit	5.32E+00	1	1.07	(2,1,1,1,1,na)	92%	100%
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
loading bales	CH	unit	2.31E+01	1	1.07	(2,1,1,1,1,na)	92%	100%
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	92%	8%
transport, tractor and trailer	CH	tkm	4.22E+01	1	1.07	(2,1,1,1,1,na)	100%	
rye seed organic, at regional storehouse	CH	kg	1.70E+02	1	1.07	(2,1,1,1,1,na)	92%	8%
grain drying, low temperature	CH	kg	4.97E+01	1	1.07	(2,1,1,1,1,na)	100%	
transport, van <3.5t	CH	tkm	2.55E+00	1	2.09	(4,5,na,na,na,na)	92%	8%
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.25E+05	1	1.07	(2,1,1,1,1,na)	50%	50%
Carbon dioxide, in air	resource/in air	kg	1.03E+04	1	1.07	(2,1,1,1,1,na)	52%	48%
Occupation, arable	resource/land	m2a	8.50E+03	1	1.11	(2,1,1,1,1,na)	92%	8%
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	92%	8%
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	92%	8%
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	92%	8%
Nitrate	water/ground-	kg	3.66E+02	1	1.51	(2,2,1,1,1,na)	92%	8%
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	92%	8%
Phosphate	water/river	kg	1.03E+00	1	1.51	(2,2,1,1,1,na)	92%	8%
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	92%	8%
Ammonia	air/low population density	kg	2.62E+01	1	1.21	(2,2,1,1,1,na)	92%	8%
Nitrogen oxides	air/low population density	kg	1.24E+00	1	1.41	(2,2,1,1,1,na)	92%	8%
Dinitrogen monoxide	air/low population density	kg	5.90E+00	1	1.41	(2,2,1,1,1,na)	92%	8%
Cadmium	soil/agricultural	kg	1.31E-04	1	1.51	(2,2,1,1,1,na)	63%	37%
Chromium	soil/agricultural	kg	-9.54E-03	1	1.51	(2,2,1,1,1,na)	40%	60%
Copper	soil/agricultural	kg	8.26E-02	1	1.51	(2,2,1,1,1,na)	64%	36%
Lead	soil/agricultural	kg	5.90E-03	1	1.51	(2,2,1,1,1,na)	31%	69%
Mercury	soil/agricultural	kg	1.00E-03	1	1.51	(2,2,1,1,1,na)	100%	
Nickel	soil/agricultural	kg	7.60E-03	1	1.51	(2,2,1,1,1,na)	33%	67%
Zinc	soil/agricultural	kg	4.66E-01	1	1.51	(2,2,1,1,1,na)	80%	20%
Cadmium, ion	water/river	kg	1.14E-05	1	1.80	(2,2,1,1,1,na)	63%	37%
Chromium, ion	water/river	kg	2.26E-03	1	1.80	(2,2,1,1,1,na)	40%	60%
Copper, ion	water/river	kg	2.52E-03	1	1.80	(2,2,1,1,1,na)	64%	36%
Lead	water/river	kg	7.05E-04	1	1.80	(2,2,1,1,1,na)	31%	69%
Mercury	water/river	kg	8.94E-06	1	1.80	(2,2,1,1,1,na)	100%	
Nickel, ion	water/river	kg	2.00E-03	1	1.80	(2,2,1,1,1,na)	33%	67%
Zinc, ion	water/river	kg	5.53E-03	1	1.80	(2,2,1,1,1,na)	80%	20%
Cadmium, ion	water/ground-	kg	1.85E-05	1	1.80	(2,2,1,1,1,na)	63%	37%
Chromium, ion	water/ground-	kg	1.55E-02	1	1.80	(2,2,1,1,1,na)	40%	60%
Copper, ion	water/ground-	kg	3.52E-03	1	1.80	(2,2,1,1,1,na)	64%	36%
Lead	water/ground-	kg	1.69E-04	1	1.80	(2,2,1,1,1,na)	31%	69%
Mercury	water/ground-	kg	1.08E-05	1	1.80	(2,2,1,1,1,na)	100%	
Zinc, ion	water/ground-	kg	2.87E-02	1	1.80	(2,2,1,1,1,na)	80%	20%
rye grains organic, at farm	CH	kg	4.17E+03				100%	
rye straw organic, at farm	CH	kg	3.70E+03					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop Production in SwitzerlandX)

Unit process inventory for: barley extensive, CH									
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	barley grains extensive, at farm CH (kg)	barley straw extensive, at farm CH (kg)	
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
tillage, currying, by weeder	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
fertilising, by broadcaster	CH	ha	4.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
slurry spreading, by vacuum tanker	CH	m3	7.68E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	2.78E+02	1	1.07	(2,1,1,1,1,na)	90%	10%	
baling	CH	unit	4.10E+00	1	1.07	(2,1,1,1,1,na)		100%	
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
loading bales	CH	unit	1.78E+01	1	1.07	(2,1,1,1,1,na)		100%	
application of plant protection products, by field sprayer	CH	ha	1.20E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
transport, tractor and trailer	CH	tkm	5.47E+01	1	1.07	(2,1,1,1,1,na)	100%		
barley seed IP, at regional storehouse	CH	kg	1.10E+02	1	1.07	(2,1,1,1,1,na)	90%	10%	
grain drying, high temperature	CH	kg	6.43E+01	1	1.07	(2,1,1,1,1,na)	100%		
ammonium nitrate, as N, at regional storehouse	RER	kg	4.42E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
urea, as N, at regional storehouse	RER	kg	1.55E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
diammonium phosphate, as N, at regional storehouse	RER	kg	5.43E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.22E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
ammonium sulphate, as N, at regional storehouse	RER	kg	3.34E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.04E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
single superphosphate, as P2O5, at regional storehouse	RER	kg	8.22E-01	1	1.07	(2,1,1,1,1,na)	90%	10%	
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.39E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.19E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
thomas meal, as P2O5, at regional storehouse	RER	kg	2.58E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
potassium chloride, as K2O, at regional storehouse	RER	kg	3.69E+01	1	1.07	(2,1,1,1,1,na)	90%	10%	
potassium sulphate, as K2O, at regional storehouse	RER	kg	2.42E+00	1	1.07	(2,1,1,1,1,na)	90%	10%	
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	1.16E+00	1	1.13	(2,2,3,1,1,na)	90%	10%	
cyclic N-compounds, at regional storehouse	RER	kg	1.94E-03	1	1.13	(2,2,3,1,1,na)	90%	10%	
dinitroaniline-compounds, at regional storehouse	RER	kg	4.20E-01	1	1.13	(2,2,3,1,1,na)	90%	10%	
nitrile-compounds, at regional storehouse	RER	kg	7.70E-03	1	1.13	(2,2,3,1,1,na)	90%	10%	
pesticide unspecified, at regional storehouse	RER	kg	3.00E-02	1	1.13	(2,2,3,1,1,na)	90%	10%	
transport, van <3.5t	CH	tkm	1.70E+00	1	2.09	(4,5,na,na,na,na)	90%	10%	
transport, lorry 20-28t, fleet average	CH	tkm	4.64E+01	1	2.09	(4,5,na,na,na,na)	90%	10%	
transport, freight, rail	CH	tkm	4.64E+01	1	2.09	(4,5,na,na,na,na)	90%	10%	
transport, barge	RER	tkm	2.98E+02	1	2.09	(4,5,na,na,na,na)	90%	10%	
transport, transoceanic freight ship	OCE	tkm	9.32E+01	1	2.09	(4,5,na,na,na,na)	90%	10%	
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.31E+05	1	1.07	(2,1,1,1,1,na)	63%	37%	
Carbon dioxide, in air	resource/in air	kg	1.09E+04	1	1.07	(2,1,1,1,1,na)	65%	35%	
Occupation, arable	resource/land	m2a	8.19E+03	1	1.11	(2,1,1,1,1,na)	90%	10%	
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	90%	10%	
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	90%	10%	
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	90%	10%	
Nitrate	water/ground-	kg	4.67E+02	1	1.51	(2,2,1,1,1,na)	90%	10%	
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	90%	10%	
Phosphate	water/river	kg	7.94E-01	1	1.51	(2,2,1,1,1,na)	90%	10%	
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	90%	10%	
Ammonia	air/low population density	kg	8.52E+00	1	1.21	(2,2,1,1,1,na)	90%	10%	
Carbon dioxide, fossil	air/low population density	kg	1.22E+01	1	1.07	(2,2,1,1,1,na)	90%	10%	
Nitrogen oxides	air/low population density	kg	1.19E+00	1	1.41	(2,2,1,1,1,na)	90%	10%	
Dinitrogen monoxide	air/low population density	kg	5.68E+00	1	1.41	(2,2,1,1,1,na)	90%	10%	
Cadmium	soil/agricultural	kg	3.63E-03	1	1.51	(2,2,1,1,1,na)	60%	40%	
Chromium	soil/agricultural	kg	4.02E-02	1	1.51	(2,2,1,1,1,na)	37%	63%	
Copper	soil/agricultural	kg	-9.66E-03	1	1.51	(2,2,1,1,1,na)	70%	30%	
Lead	soil/agricultural	kg	3.26E-03	1	1.51	(2,2,1,1,1,na)	56%	44%	
Mercury	soil/agricultural	kg	1.17E-04	1	1.51	(2,2,1,1,1,na)	53%	47%	
Nickel	soil/agricultural	kg	6.05E-03	1	1.51	(2,2,1,1,1,na)	48%	52%	
Zinc	soil/agricultural	kg	2.94E-03	1	1.51	(2,2,1,1,1,na)	88%	12%	
Cadmium, ion	water/river	kg	2.62E-05	1	1.80	(2,2,1,1,1,na)	60%	40%	
Chromium, ion	water/river	kg	2.93E-03	1	1.80	(2,2,1,1,1,na)	37%	63%	
Copper, ion	water/river	kg	2.31E-03	1	1.80	(2,2,1,1,1,na)	70%	30%	
Lead	water/river	kg	4.57E-04	1	1.80	(2,2,1,1,1,na)	56%	44%	
Mercury	water/river	kg	6.79E-06	1	1.80	(2,2,1,1,1,na)	53%	47%	
Nickel, ion	water/river	kg	1.85E-03	1	1.80	(2,2,1,1,1,na)	48%	52%	
Zinc, ion	water/river	kg	3.49E-03	1	1.80	(2,2,1,1,1,na)	88%	12%	
Cadmium, ion	water/ground-	kg	4.26E-05	1	1.80	(2,2,1,1,1,na)	60%	40%	
Chromium, ion	water/ground-	kg	2.01E-02	1	1.80	(2,2,1,1,1,na)	37%	63%	
Copper, ion	water/ground-	kg	3.23E-03	1	1.80	(2,2,1,1,1,na)	70%	30%	
Lead	water/ground-	kg	1.10E-04	1	1.80	(2,2,1,1,1,na)	56%	44%	
Mercury	water/ground-	kg	8.19E-06	1	1.80	(2,2,1,1,1,na)	53%	47%	
Zinc, ion	water/ground-	kg	1.81E-02	1	1.80	(2,2,1,1,1,na)	88%	12%	
Cyproconazole	soil/agricultural	kg	1.94E-03	1	1.24	(2,2,3,1,1,na)	90%	10%	
Cyprodinil	soil/agricultural	kg	7.70E-03	1	1.24	(2,2,3,1,1,na)	90%	10%	
Metaldehyde	soil/agricultural	kg	3.00E-02	1	1.24	(2,2,3,1,1,na)	90%	10%	
Chlorotoluron	soil/agricultural	kg	6.30E-01	1	1.24	(2,2,3,1,1,na)	90%	10%	
Isoproturon	soil/agricultural	kg	5.30E-01	1	1.24	(2,2,3,1,1,na)	90%	10%	
Pendimethalin	soil/agricultural	kg	4.20E-01	1	1.24	(2,2,3,1,1,na)	90%	10%	
barley grains extensive, at farm	CH	kg	5.40E+03				100%		
barley straw extensive, at farm	CH	kg	2.85E+03					100%	

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: barley IP, CH								
	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	barley grains IP, at farm CH (kg)	barley straw IP, at farm CH (kg)
Exchanges								
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, currying, by weeder	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
fertilising, by broadcaster	CH	ha	4.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
slurry spreading, by vacuum tanker	CH	m3	6.79E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.84E+03	1	1.07	(2,1,1,1,1,na)	90%	10%
baling	CH	unit	5.18E+00	1	1.07	(2,1,1,1,1,na)	100%	100%
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
loading bales	CH	unit	2.25E+01	1	1.07	(2,1,1,1,1,na)	100%	100%
application of plant protection products, by field sprayer	CH	ha	2.70E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
transport, tractor and trailer	CH	tkm	6.91E+01	1	1.07	(2,1,1,1,1,na)	100%	100%
barley seed IP, at regional storehouse	CH	kg	1.10E+02	1	1.07	(2,1,1,1,1,na)	90%	10%
grain drying, high temperature	CH	kg	8.13E+01	1	1.07	(2,1,1,1,1,na)	100%	100%
ammonium nitrate, as N, at regional storehouse	RER	kg	4.85E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
urea, as N, at regional storehouse	RER	kg	1.70E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
diammonium phosphate, as N, at regional storehouse	RER	kg	6.57E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.43E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
ammonium sulphate, as N, at regional storehouse	RER	kg	3.67E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.47E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
single superphosphate, as P2O5, at regional storehouse	RER	kg	9.94E-01	1	1.07	(2,1,1,1,1,na)	90%	10%
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.68E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.44E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
thomas meal, as P2O5, at regional storehouse	RER	kg	3.12E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
potassium chloride, as K2O, at regional storehouse	RER	kg	4.62E+01	1	1.07	(2,1,1,1,1,na)	90%	10%
potassium sulphate, as K2O, at regional storehouse	RER	kg	3.03E+00	1	1.07	(2,1,1,1,1,na)	90%	10%
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	1.34E+00	1	1.13	(2,2,3,1,1,na)	90%	10%
benzimidazole-compounds, at regional storehouse	RER	kg	3.50E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
cyclic N-compounds, at regional storehouse	RER	kg	3.72E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
dinitroaniline-compounds, at regional storehouse	RER	kg	4.20E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
nitrile-compounds, at regional storehouse	RER	kg	7.70E-03	1	1.13	(2,2,3,1,1,na)	90%	10%
organophosphorus-compounds, at regional storehouse	RER	kg	3.20E-01	1	1.13	(2,2,3,1,1,na)	90%	10%
pesticide unspecified, at regional storehouse	RER	kg	4.00E-02	1	1.13	(2,2,3,1,1,na)	90%	10%
transport, van <3.5t	CH	tkm	1.74E+00	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, lorry 20-28t, fleet average	CH	tkm	5.36E+01	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, freight, rail	CH	tkm	5.36E+01	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, barge	RER	tkm	3.34E+02	1	2.09	(4,5,na,na,na,na)	90%	10%
transport, transoceanic freight ship	OCE	tkm	1.13E+02	1	2.09	(4,5,na,na,na,na)	90%	10%
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.66E+05	1	1.07	(2,1,1,1,1,na)	63%	37%
Carbon dioxide, in air	resource/in air	kg	1.38E+04	1	1.07	(2,1,1,1,1,na)	65%	35%
Occupation, arable	resource/land	m2a	8.19E+03	1	1.11	(2,1,1,1,1,na)	90%	10%
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	90%	10%
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	90%	10%
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	90%	10%
Nitrate	water/ground-	kg	4.26E+02	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphate	water/river	kg	8.08E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	90%	10%
Ammonia	air/low population density	kg	9.63E+00	1	1.21	(2,2,1,1,1,na)	90%	10%
Carbon dioxide, fossil	air/low population density	kg	1.34E+01	1	1.07	(2,2,1,1,1,na)	90%	10%
Nitrogen oxides	air/low population density	kg	1.13E+00	1	1.41	(2,2,1,1,1,na)	90%	10%
Dinitrogen monoxide	air/low population density	kg	5.39E+00	1	1.41	(2,2,1,1,1,na)	90%	10%
Cadmium	soil/agricultural	kg	4.40E-03	1	1.51	(2,2,1,1,1,na)	60%	40%
Chromium	soil/agricultural	kg	5.39E-02	1	1.51	(2,2,1,1,1,na)	37%	63%
Copper	soil/agricultural	kg	-9.66E-03	1	1.51	(2,2,1,1,1,na)	70%	30%
Lead	soil/agricultural	kg	4.45E-03	1	1.51	(2,2,1,1,1,na)	56%	44%
Mercury	soil/agricultural	kg	2.20E-04	1	1.51	(2,2,1,1,1,na)	53%	47%
Nickel	soil/agricultural	kg	7.90E-03	1	1.51	(2,2,1,1,1,na)	48%	52%
Zinc	soil/agricultural	kg	2.91E-03	1	1.51	(2,2,1,1,1,na)	88%	12%
Cadmium, ion	water/river	kg	2.69E-05	1	1.80	(2,2,1,1,1,na)	60%	40%
Chromium, ion	water/river	kg	2.96E-03	1	1.80	(2,2,1,1,1,na)	37%	63%
Copper, ion	water/river	kg	2.38E-03	1	1.80	(2,2,1,1,1,na)	70%	30%
Lead	water/river	kg	5.88E-04	1	1.80	(2,2,1,1,1,na)	56%	44%
Mercury	water/river	kg	7.74E-06	1	1.80	(2,2,1,1,1,na)	53%	47%
Nickel, ion	water/river	kg	2.01E-03	1	1.80	(2,2,1,1,1,na)	48%	52%
Zinc, ion	water/river	kg	3.97E-03	1	1.80	(2,2,1,1,1,na)	88%	12%
Cadmium, ion	water/ground-	kg	4.37E-05	1	1.80	(2,2,1,1,1,na)	60%	40%
Chromium, ion	water/ground-	kg	2.03E-02	1	1.80	(2,2,1,1,1,na)	37%	63%
Copper, ion	water/ground-	kg	3.32E-03	1	1.80	(2,2,1,1,1,na)	70%	30%
Lead	water/ground-	kg	1.41E-04	1	1.80	(2,2,1,1,1,na)	56%	44%
Mercury	water/ground-	kg	9.33E-06	1	1.80	(2,2,1,1,1,na)	53%	47%
Zinc, ion	water/ground-	kg	2.06E-02	1	1.80	(2,2,1,1,1,na)	88%	12%
Chlorothalonil	soil/agricultural	kg	3.50E-01	1	1.22	(2,3,1,1,1,na)	90%	10%
Cyproconazole	soil/agricultural	kg	1.94E-03	1	1.24	(2,2,3,1,1,na)	90%	10%
Cyprodinil	soil/agricultural	kg	7.70E-03	1	1.24	(2,2,3,1,1,na)	90%	10%
Fenpropimorph	soil/agricultural	kg	3.70E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Ethephon	soil/agricultural	kg	3.20E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Metaldehyde	soil/agricultural	kg	4.00E-02	1	1.24	(2,2,3,1,1,na)	90%	10%
Chlorotoluron	soil/agricultural	kg	3.90E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Isoproturon	soil/agricultural	kg	9.50E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
Pendimethalin	soil/agricultural	kg	4.20E-01	1	1.24	(2,2,3,1,1,na)	90%	10%
barley grains IP, at farm	CH	kg	6.83E+03				100%	
barley straw IP, at farm	CH	kg	3.61E+03					100%

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: barley organic, CH									
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	barley grains organic, at farm CH (kg)	barley straw organic, at farm CH (kg)	
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	91%	9%	
tillage, currying, by weeder	CH	ha	3.00E+00	1	1.07	(2,1,1,1,1,na)	91%	9%	
tillage, harrowing, by spring tine harrow	CH	ha	2.00E+00	1	1.07	(2,1,1,1,1,na)	91%	9%	
tillage, ploughing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	91%	9%	
slurry spreading, by vacuum tanker	CH	m3	3.93E+01	1	1.07	(2,1,1,1,1,na)	91%	9%	
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	4.84E+03	1	1.07	(2,1,1,1,1,na)	91%	9%	
baling	CH	unit	4.20E+00	1	1.07	(2,1,1,1,1,na)		100%	
combine harvesting	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	91%	9%	
loading bales	CH	unit	1.83E+01	1	1.07	(2,1,1,1,1,na)		100%	
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)	91%	9%	
transport, tractor and trailer	CH	tkm	4.20E+01	1	1.07	(2,1,1,1,1,na)	100%		
barley seed organic, at regional storehouse	CH	kg	1.40E+02	1	1.07	(2,1,1,1,1,na)	91%	9%	
grain drying, high temperature	CH	kg	4.94E+01	1	1.07	(2,1,1,1,1,na)	100%		
transport, van <3.5t	CH	tkm	2.10E+00	1	2.09	(4,5,na,na,na,na)	91%	9%	
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.13E+05	1	1.07	(2,1,1,1,1,na)	56%	44%	
Carbon dioxide, in air	resource/in air	kg	9.32E+03	1	1.07	(2,1,1,1,1,na)	58%	42%	
Occupation, arable	resource/land	m2a	8.19E+03	1	1.11	(2,1,1,1,1,na)	91%	9%	
Transformation, from pasture and meadow, intensive	resource/land	m2	2.90E+03	1	1.21	(2,1,1,1,1,na)	91%	9%	
Transformation, from arable, non-irrigated	resource/land	m2	7.10E+03	1	1.21	(2,1,1,1,1,na)	91%	9%	
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)	91%	9%	
Nitrate	water/ground-	kg	4.06E+02	1	1.51	(2,2,1,1,1,na)	91%	9%	
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)	91%	9%	
Phosphate	water/river	kg	1.03E+00	1	1.51	(2,2,1,1,1,na)	91%	9%	
Phosphate	water/ground-	kg	2.21E-01	1	1.51	(2,2,1,1,1,na)	91%	9%	
Ammonia	air/low population density	kg	2.92E+01	1	1.21	(2,2,1,1,1,na)	91%	9%	
Nitrogen oxides	air/low population density	kg	1.30E+00	1	1.41	(2,2,1,1,1,na)	91%	9%	
Dinitrogen monoxide	air/low population density	kg	6.18E+00	1	1.41	(2,2,1,1,1,na)	91%	9%	
Cadmium	soil/agricultural	kg	2.80E-04	1	1.51	(2,2,1,1,1,na)	53%	47%	
Chromium	soil/agricultural	kg	-8.45E-03	1	1.51	(2,2,1,1,1,na)	30%	70%	
Copper	soil/agricultural	kg	9.41E-02	1	1.51	(2,2,1,1,1,na)	64%	36%	
Lead	soil/agricultural	kg	6.72E-03	1	1.51	(2,2,1,1,1,na)	49%	51%	
Mercury	soil/agricultural	kg	1.16E-03	1	1.51	(2,2,1,1,1,na)	46%	54%	
Nickel	soil/agricultural	kg	9.37E-03	1	1.51	(2,2,1,1,1,na)	41%	59%	
Zinc	soil/agricultural	kg	5.24E-01	1	1.51	(2,2,1,1,1,na)	85%	15%	
Cadmium, ion	water/river	kg	1.21E-05	1	1.80	(2,2,1,1,1,na)	53%	47%	
Chromium, ion	water/river	kg	2.32E-03	1	1.80	(2,2,1,1,1,na)	30%	70%	
Copper, ion	water/river	kg	2.53E-03	1	1.80	(2,2,1,1,1,na)	64%	36%	
Lead	water/river	kg	7.62E-04	1	1.80	(2,2,1,1,1,na)	49%	51%	
Mercury	water/river	kg	8.98E-06	1	1.80	(2,2,1,1,1,na)	46%	54%	
Nickel, ion	water/river	kg	2.07E-03	1	1.80	(2,2,1,1,1,na)	41%	59%	
Zinc, ion	water/river	kg	5.61E-03	1	1.80	(2,2,1,1,1,na)	85%	15%	
Cadmium, ion	water/ground-	kg	1.96E-05	1	1.80	(2,2,1,1,1,na)	53%	47%	
Chromium, ion	water/ground-	kg	1.59E-02	1	1.80	(2,2,1,1,1,na)	30%	70%	
Copper, ion	water/ground-	kg	3.53E-03	1	1.80	(2,2,1,1,1,na)	64%	36%	
Lead	water/ground-	kg	1.83E-04	1	1.80	(2,2,1,1,1,na)	49%	51%	
Mercury	water/ground-	kg	1.08E-05	1	1.80	(2,2,1,1,1,na)	46%	54%	
Zinc, ion	water/ground-	kg	2.91E-02	1	1.80	(2,2,1,1,1,na)	85%	15%	
barley grains organic, at farm	CH	kg	4.15E+03				100%		
barley straw organic, at farm	CH	kg	2.92E+03					100%	

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: grain maize IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	2.16E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.08E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	3.23E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	1.27E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	8.38E-01	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	1.08E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.83E-04	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	1.08E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.08E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.41E-02	1	1.07	(2,1,1,1,1,na)
maize seed IP, at regional storehouse	CH	kg	2.69E-03	1	1.07	(2,1,1,1,1,na)
maize drying	CH	kg	4.10E-01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	4.32E-03	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.52E-03	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	3.41E-04	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.16E-03	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	3.27E-04	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.28E-03	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	5.16E-05	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	8.72E-04	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	7.50E-04	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	1.62E-04	1	1.07	(2,1,1,1,1,na)
atrazine, at regional storehouse	CH	kg	7.11E-05	1	1.13	(2,2,3,1,1,na)
glyphosate, at regional storehouse	CH	kg	2.69E-05	1	1.13	(2,2,3,1,1,na)
metolachlor, at regional storehouse	CH	kg	6.36E-05	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	6.52E-06	1	1.13	(2,2,3,1,1,na)
electricity, low voltage, at grid	CH	kWh	4.20E-03	1	1.07	(2,1,1,1,1,na)
Heat, waste	air/high population density	MJ	1.51E-02	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	4.55E-05	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	3.33E-03	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	3.33E-03	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	2.63E-02	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	5.86E-03	1	2.09	(4,5,na,na,na,na)
green manure IP, until April	CH	ha	1.08E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.59E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.37E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	5.39E-01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.08E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.08E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	3.16E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.55E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	4.92E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.39E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.81E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.19E-03	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.54E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	7.32E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	2.61E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	2.19E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	3.93E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	7.72E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	8.11E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	1.04E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	1.74E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.10E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	3.63E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.15E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.04E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.12E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.93E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	6.33E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	4.27E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.12E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.74E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.12E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.14E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.79E-06	1	1.80	(2,2,1,1,1,na)
Atrazine	soil/agricultural	kg	7.11E-05	1	1.24	(2,2,3,1,1,na)
Glyphosate	soil/agricultural	kg	2.69E-05	1	1.24	(2,2,3,1,1,na)
Metolachlor	soil/agricultural	kg	6.36E-05	1	1.24	(2,2,3,1,1,na)
grain maize IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: grain maize organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	1.29E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	2.57E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.29E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	5.96E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	2.36E+00	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	1.29E-04	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	2.57E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.29E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.41E-02	1	1.07	(2,1,1,1,1,na)
maize seed organic, at regional storehouse	CH	kg	3.21E-03	1	1.07	(2,1,1,1,1,na)
maize drying	CH	kg	4.10E-01	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	CH	kWh	5.02E-03	1	1.07	(2,1,1,1,1,na)
Heat, waste	air/high population density	MJ	1.81E-02	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	4.82E-05	1	2.09	(4,5,na,na,na,na)
green manure organic, until April	CH	ha	1.29E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.59E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.37E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	6.43E-01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.29E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.29E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.72E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.85E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	5.71E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.66E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.54E-03	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.18E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	5.61E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	9.64E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.96E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	1.60E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.15E-06	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	2.63E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	1.96E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	6.90E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	2.64E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	3.98E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.84E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.94E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.38E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.65E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	8.53E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	3.64E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.32E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.56E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	3.94E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.42E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.76E-06	1	1.80	(2,2,1,1,1,na)
grain maize organic, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: silage maize IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	3.25E-05	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.63E-05	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	4.88E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	3.86E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.94E-01	1	1.07	(2,1,1,1,1,na)
chopping, maize	CH	ha	1.63E-05	1	1.07	(2,1,1,1,1,na)
fodder loading, by self-loading trailer	CH	m3	4.00E-03	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.63E-05	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	1.63E-05	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.63E-05	1	1.07	(2,1,1,1,1,na)
maize seed IP, at regional storehouse	CH	kg	4.39E-04	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	4.07E-04	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.43E-04	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	1.22E-04	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.04E-04	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	3.08E-05	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	4.58E-04	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	1.85E-05	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	3.12E-04	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	2.68E-04	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	5.79E-05	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	1.75E-03	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	1.15E-04	1	1.07	(2,1,1,1,1,na)
atrazine, at regional storehouse	CH	kg	9.11E-06	1	1.13	(2,2,3,1,1,na)
glyphosate, at regional storehouse	CH	kg	3.25E-06	1	1.13	(2,2,3,1,1,na)
metolachlor, at regional storehouse	CH	kg	3.91E-06	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.06E-06	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	7.11E-06	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	8.91E-04	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	8.91E-04	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	3.80E-03	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	2.09E-03	1	2.09	(4,5,na,na,na,na)
green manure IP, until April	CH	ha	1.63E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	5.31E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	4.81E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	5.88E-02	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.63E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.63E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.66E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	2.34E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	5.56E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.50E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	3.74E-04	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.12E-04	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.10E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	5.23E-05	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	6.87E-09	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	5.66E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-3.39E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	-5.08E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	1.03E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-6.56E-08	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-2.18E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	5.28E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	5.72E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	4.82E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.21E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.73E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	4.55E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.03E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	7.29E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	3.33E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	5.72E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	4.49E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.77E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	4.51E-07	1	1.80	(2,2,1,1,1,na)
Atrazine	soil/agricultural	kg	9.11E-06	1	1.24	(2,2,3,1,1,na)
Glyphosate	soil/agricultural	kg	3.25E-06	1	1.24	(2,2,3,1,1,na)
Metolachlor	soil/agricultural	kg	3.91E-06	1	1.24	(2,2,3,1,1,na)
silage maize IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: silage maize organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	2.03E-05	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	4.07E-05	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.03E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	7.92E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	3.14E-01	1	1.07	(2,1,1,1,1,na)
chopping, maize	CH	ha	2.03E-05	1	1.07	(2,1,1,1,1,na)
fodder loading, by self-loading trailer	CH	m3	4.00E-03	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	4.07E-05	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.03E-05	1	1.07	(2,1,1,1,1,na)
maize seed organic, at regional storehouse	CH	kg	5.49E-04	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	8.24E-06	1	2.09	(4,5,na,na,na,na)
green manure organic, until April	CH	ha	2.03E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	5.31E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	4.81E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	7.34E-02	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.03E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.03E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.17E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	2.92E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	6.21E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.87E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	7.68E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	7.26E-06	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.46E-05	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-3.76E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-6.43E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-2.52E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	-4.53E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	2.52E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-1.77E-08	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.87E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.85E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	6.12E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	6.06E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.80E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	2.18E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	5.58E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.32E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	5.31E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	3.56E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	7.18E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	5.71E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	2.23E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	5.82E-07	1	1.80	(2,2,1,1,1,na)
silage maize organic, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: potatoes IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	2.65E-05	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by rotary harrow	CH	ha	5.30E-05	1	1.07	(2,1,1,1,1,na)
tillage, hoeing and earthing-up, potatoes	CH	ha	5.30E-05	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.65E-05	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	7.94E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	3.46E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	3.63E-01	1	1.07	(2,1,1,1,1,na)
harvesting, by complete harvester, potatoes	CH	ha	2.65E-05	1	1.07	(2,1,1,1,1,na)
potato grading	CH	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
potato haulm cutting	CH	ha	2.65E-05	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.80E-04	1	1.07	(2,1,1,1,1,na)
potato planting	CH	ha	2.65E-05	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-03	1	1.07	(2,1,1,1,1,na)
potato seed IP, at regional storehouse	CH	kg	6.67E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	4.49E-04	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.58E-04	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	2.56E-05	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.25E-04	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	3.39E-05	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	9.60E-05	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	3.87E-06	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	6.54E-05	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	5.62E-05	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	1.21E-05	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	2.29E-03	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	1.50E-04	1	1.07	(2,1,1,1,1,na)
phtalamide-compounds, at regional storehouse	CH	kg	8.68E-05	1	1.13	(2,2,3,1,1,na)
dithiocarbamate-compounds, at regional storehouse	CH	kg	1.13E-04	1	1.13	(2,2,3,1,1,na)
nitrile-compounds, at regional storehouse	CH	kg	3.34E-06	1	1.13	(2,2,3,1,1,na)
metolachlor, at regional storehouse	CH	kg	3.97E-06	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	2.14E-05	1	1.13	(2,2,3,1,1,na)
[sulfonyl]urea-compounds, at regional storehouse	CH	kg	2.65E-07	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	1.01E-03	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	7.98E-04	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	7.98E-04	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	3.10E-03	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	4.39E-04	1	2.09	(4,5,na,na,na,na)
green manure IP, until march	CH	ha	2.65E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	3.64E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	3.20E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.25E-01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.65E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.65E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	6.93E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	2.77E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.61E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.41E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.85E-04	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.24E-04	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.25E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.55E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	2.78E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	4.65E-08	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	1.74E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.69E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	2.04E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	3.61E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	1.57E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	4.55E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	6.19E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	5.75E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.33E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	2.06E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	5.11E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.29E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	8.60E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	4.93E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	9.34E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	6.51E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	2.89E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	7.78E-07	1	1.80	(2,2,1,1,1,na)
Chlorothalonil	soil/agricultural	kg	8.68E-05	1	1.24	(2,2,3,1,1,na)
Fenpiclonil	soil/agricultural	kg	3.34E-06	1	1.24	(2,2,3,1,1,na)
Mancozeb	soil/agricultural	kg	1.13E-04	1	1.24	(2,2,3,1,1,na)
Metribuzin	soil/agricultural	kg	3.97E-06	1	1.24	(2,2,3,1,1,na)
Orbencarb	soil/agricultural	kg	2.14E-05	1	1.24	(2,2,3,1,1,na)
Teflubenzuron	soil/agricultural	kg	2.65E-07	1	1.24	(2,2,3,1,1,na)
potatoes IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: rape seed extensive, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	7.45E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.73E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.12E-03	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	9.09E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	3.51E+00	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.73E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	7.08E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.73E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.07E-02	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	2.05E-03	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	6.82E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.31E-02	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	4.61E-03	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	6.58E-03	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	9.92E-04	1	1.07	(2,1,1,1,1,na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	7.01E-04	1	1.13	(2,2,3,1,1,na)
dinitroaniline-compounds, at regional storehouse	CH	kg	1.71E-04	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	7.83E-05	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	8.94E-05	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	6.19E-05	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	7.66E-03	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	7.66E-03	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	6.89E-02	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.48E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.68E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	3.36E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from pasture and meadow, intensive	resource/land	m2	1.08E+00	1	1.21	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.65E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.73E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	7.08E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	4.54E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	2.71E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	8.23E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	8.27E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	3.61E-03	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.46E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.12E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-4.07E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-3.02E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	2.35E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	1.83E-06	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	3.06E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	2.60E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	8.97E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	4.69E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	8.66E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	9.32E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	3.35E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	3.33E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	7.81E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.90E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	7.61E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	5.94E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.30E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	8.03E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	4.02E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	9.85E-06	1	1.80	(2,2,1,1,1,na)
Metaldehyde	soil/agricultural	kg	7.83E-05	1	1.24	(2,2,3,1,1,na)
Carbetamide	soil/agricultural	kg	8.94E-05	1	1.24	(2,2,3,1,1,na)
Tebutam	soil/agricultural	kg	7.01E-04	1	1.24	(2,2,3,1,1,na)
Trifluralin	soil/agricultural	kg	1.71E-04	1	1.24	(2,2,3,1,1,na)
rape seed extensive, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: rape seed IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	6.43E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.21E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	9.64E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	2.74E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.78E+00	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.21E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.12E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.21E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.07E-02	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.77E-03	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	6.82E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.70E-02	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	5.98E-03	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	7.89E-04	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	8.54E-03	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	1.29E-03	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.96E-03	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	1.19E-04	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	2.02E-03	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	1.73E-03	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	3.74E-04	1	1.07	(2,1,1,1,1,na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	4.98E-04	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	8.35E-05	1	1.13	(2,2,3,1,1,na)
pyrethroid-compounds, at regional storehouse	CH	kg	9.64E-06	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	1.90E-04	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	4.99E-05	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	1.18E-02	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	1.18E-02	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	9.79E-02	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	1.35E-02	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.48E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.68E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	2.90E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from pasture and meadow, intensive	resource/land	m2	9.32E-01	1	1.21	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.28E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.21E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.08E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	3.92E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	3.40E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	7.09E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	3.88E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.70E-03	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.91E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.86E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-4.07E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	3.92E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	1.79E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	6.25E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	1.36E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	1.85E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	8.48E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	7.39E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	9.01E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	8.01E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.05E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	2.81E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	6.64E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.67E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.20E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.18E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.12E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	4.92E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	3.39E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	8.66E-06	1	1.80	(2,2,1,1,1,na)
Metaldehyde	soil/agricultural	kg	8.35E-05	1	1.24	(2,2,3,1,1,na)
Carbetamide	soil/agricultural	kg	1.90E-04	1	1.24	(2,2,3,1,1,na)
Napropamide	soil/agricultural	kg	1.48E-04	1	1.24	(2,2,3,1,1,na)
Tebutam	soil/agricultural	kg	3.50E-04	1	1.24	(2,2,3,1,1,na)
Cypermethrin	soil/agricultural	kg	9.64E-06	1	1.24	(2,2,3,1,1,na)
rape seed IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: sugar beets IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	2.77E-05	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.38E-05	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	5.53E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	1.82E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.24E-01	1	1.07	(2,1,1,1,1,na)
harvesting, by complete harvester, beets	CH	ha	1.38E-05	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	6.08E-05	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	2.77E-05	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.38E-05	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-03	1	1.07	(2,1,1,1,1,na)
sugar beet seed IP, at regional storehouse	CH	kg	2.90E-05	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	3.49E-04	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.23E-04	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	2.30E-05	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	1.75E-04	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	2.64E-05	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	8.63E-05	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	3.48E-06	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	5.88E-05	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	5.05E-05	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	1.09E-05	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	3.56E-04	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	2.34E-05	1	1.07	(2,1,1,1,1,na)
phtalamide-compounds, at regional storehouse	CH	kg	9.68E-07	1	1.13	(2,2,3,1,1,na)
cyclic N-compounds, at regional storehouse	CH	kg	2.07E-06	1	1.13	(2,2,3,1,1,na)
benzimidazole-compounds, at regional storehouse	CH	kg	7.33E-06	1	1.13	(2,2,3,1,1,na)
triazine-compounds, at regional storehouse	CH	kg	2.79E-05	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	2.83E-06	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	5.12E-06	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	1.82E-06	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	3.32E-04	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	3.32E-04	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	2.15E-03	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	3.95E-04	1	2.09	(4,5,na,na,na,na)
green manure IP, until February	CH	ha	1.38E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	3.78E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	3.22E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	8.07E-02	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.38E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.38E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	5.88E-04	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.98E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	6.19E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.78E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	2.14E-04	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	9.62E-05	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.73E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	8.26E-05	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-1.94E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-1.11E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-8.07E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	-1.88E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-5.73E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-4.17E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.79E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.51E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	4.49E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	4.06E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.39E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.44E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.75E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	8.27E-08	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	4.85E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.61E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.81E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.84E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.48E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.64E-07	1	1.80	(2,2,1,1,1,na)
Chlorothalonil	soil/agricultural	kg	9.68E-07	1	1.24	(2,2,3,1,1,na)
Fenpropimorph	soil/agricultural	kg	2.07E-06	1	1.24	(2,2,3,1,1,na)
Metaldehyde	soil/agricultural	kg	1.11E-06	1	1.24	(2,2,3,1,1,na)
Ethofumesate	soil/agricultural	kg	7.33E-06	1	1.24	(2,2,3,1,1,na)
Metamitron	soil/agricultural	kg	2.79E-05	1	1.24	(2,2,3,1,1,na)
Phenmedipham	soil/agricultural	kg	5.12E-06	1	1.24	(2,2,3,1,1,na)
sugar beets IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: fodder beets IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	2.06E-05	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.03E-05	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	4.12E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	2.07E-04	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.36E-01	1	1.07	(2,1,1,1,1,na)
harvesting, by complete harvester, beets	CH	ha	1.03E-05	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.99E-05	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	2.06E-05	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.03E-05	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.00E-03	1	1.07	(2,1,1,1,1,na)
sugar beet seed IP, at regional storehouse	CH	kg	2.16E-05	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	9.98E-05	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	3.51E-05	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	1.67E-05	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	5.01E-05	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	7.55E-06	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	6.26E-05	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	2.53E-06	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	4.27E-05	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	3.67E-05	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	7.92E-06	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	3.21E-04	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	2.11E-05	1	1.07	(2,1,1,1,1,na)
benzimidazole-compounds, at regional storehouse	CH	kg	3.92E-06	1	1.13	(2,2,3,1,1,na)
triazine-compounds, at regional storehouse	CH	kg	1.43E-05	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.28E-06	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	3.51E-06	1	1.13	(2,2,3,1,1,na)
organophosphorus-compounds, at regional storehouse	CH	kg	6.19E-07	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	1.03E-06	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	1.65E-04	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	1.65E-04	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	7.68E-04	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	2.86E-04	1	2.09	(4,5,na,na,na,na)
green manure IP, until February	CH	ha	1.03E-05	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.48E+00	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.10E-01	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	6.30E-02	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.03E-01	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.03E-01	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.53E-04	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.48E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	5.14E-06	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.52E-06	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.85E-04	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	2.75E-05	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	9.78E-06	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	4.66E-05	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-2.02E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-1.13E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-8.14E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	-2.49E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-5.73E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-4.81E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.91E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	2.67E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	3.37E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.05E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.26E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.09E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.67E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	6.38E-08	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	3.68E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.96E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.61E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.56E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.12E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.81E-07	1	1.80	(2,2,1,1,1,na)
Ethofumesate	soil/agricultural	kg	3.92E-06	1	1.24	(2,2,3,1,1,na)
Metamitron	soil/agricultural	kg	1.43E-05	1	1.24	(2,2,3,1,1,na)
Phenmedipham	soil/agricultural	kg	3.51E-06	1	1.24	(2,2,3,1,1,na)
Terbufos	soil/agricultural	kg	6.19E-07	1	1.24	(2,2,3,1,1,na)
fodder beets IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: sunflower IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	6.35E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.17E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	6.35E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.17E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	3.17E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.17E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.13E-02	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.19E-03	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	1.33E-01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	6.44E-03	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	2.26E-03	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as N, at regional storehouse	RER	kg	1.23E-03	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	3.23E-03	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	4.88E-04	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	4.61E-03	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	1.86E-04	1	1.07	(2,1,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	3.14E-03	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	2.70E-03	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	5.83E-04	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	7.94E-03	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	5.21E-04	1	1.07	(2,1,1,1,1,na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	2.38E-06	1	1.13	(2,2,3,1,1,na)
dinitroaniline-compounds, at regional storehouse	CH	kg	9.52E-04	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	7.93E-05	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	4.89E-05	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	8.33E-03	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	8.33E-03	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	4.85E-02	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	2.11E-02	1	2.09	(4,5,na,na,na,na)
green manure IP, until April	CH	ha	3.17E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.84E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.20E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.28E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.17E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.17E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	6.07E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	4.55E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	9.26E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.40E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	7.54E-04	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.78E-03	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.43E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.16E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-1.25E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	6.65E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-1.52E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.23E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-2.73E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	9.95E-08	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-3.58E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	9.52E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.07E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	7.16E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.54E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.39E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	5.68E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	8.74E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.31E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.23E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	8.49E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	3.13E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.43E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.85E-06	1	1.80	(2,2,1,1,1,na)
Metalaxil	soil/agricultural	kg	2.38E-06	1	1.24	(2,2,3,1,1,na)
Metaldehyde	soil/agricultural	kg	7.93E-05	1	1.24	(2,2,3,1,1,na)
Acetonifon	soil/agricultural	kg	9.52E-04	1	1.24	(2,2,3,1,1,na)
sunflower IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: fava beans IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	5.29E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.04E-02	1	1.07	(2,1,1,1,1,na)
pea seed IP, at regional storehouse	CH	kg	4.63E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	3.57E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	7.77E-03	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	3.13E-04	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	4.55E-03	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	9.82E-04	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	1.57E-02	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	1.03E-03	1	1.07	(2,1,1,1,1,na)
benzo[thia]diazole-compounds, at regional storehouse	CH	kg	4.76E-05	1	1.13	(2,2,3,1,1,na)
diphenylether-compounds, at regional storehouse	CH	kg	2.12E-05	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	6.96E-04	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	7.04E-03	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	7.04E-03	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	1.83E-02	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	3.55E-02	1	2.09	(4,5,na,na,na,na)
green manure IP, until January	CH	ha	2.64E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.54E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.21E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.39E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.64E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.64E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.22E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	2.77E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	8.00E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	2.84E-05	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.44E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.64E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.08E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	1.30E-05	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-3.11E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	4.23E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	4.65E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-2.82E-07	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	6.01E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	6.70E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	4.41E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	5.94E-08	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.20E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	4.64E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.14E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	5.34E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	7.16E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.66E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.80E-06	1	1.80	(2,2,1,1,1,na)
Bentazone	soil/agricultural	kg	4.76E-05	1	1.24	(2,2,3,1,1,na)
Fluazifop-P-butyl	soil/agricultural	kg	2.12E-05	1	1.24	(2,2,3,1,1,na)
fava beans IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: fava beans organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	2.96E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	5.91E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.96E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	7.84E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	1.96E+00	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.96E-04	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	2.96E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.96E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.04E-02	1	1.07	(2,1,1,1,1,na)
pea seed organic, at regional storehouse	CH	kg	5.61E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	3.57E-02	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	8.42E-04	1	2.09	(4,5,na,na,na,na)
green manure organic, until January	CH	ha	2.96E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.52E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.20E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.55E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.96E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.96E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.66E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	3.10E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.08E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.26E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.19E-03	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.48E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.14E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	9.57E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.81E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	1.30E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.02E-06	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	2.56E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	1.68E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	6.22E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	2.81E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	5.54E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	6.32E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.98E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	2.25E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	4.80E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.25E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	5.31E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	4.42E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.03E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	5.54E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	3.16E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	7.54E-06	1	1.80	(2,2,1,1,1,na)
fava beans organic, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: soy beans IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	6.82E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.41E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	3.41E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	3.21E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	7.64E-01	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.41E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	4.09E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.41E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.06E-02	1	1.07	(2,1,1,1,1,na)
pea seed IP, at regional storehouse	CH	kg	3.75E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	5.95E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	5.15E-03	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	2.08E-04	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	3.02E-03	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	6.52E-04	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	7.59E-03	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	4.98E-04	1	1.07	(2,1,1,1,1,na)
linuron, at regional storehouse	CH	kg	1.17E-04	1	1.13	(2,2,3,1,1,na)
metolachlor, at regional storehouse	CH	kg	8.45E-04	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	5.91E-04	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	4.08E-03	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	4.08E-03	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	1.16E-02	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	2.36E-02	1	2.09	(4,5,na,na,na,na)
green manure IP, until march	CH	ha	3.41E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.97E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.37E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.55E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.41E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.41E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	6.83E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	4.89E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.30E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.76E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.94E-03	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	5.99E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.85E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	7.32E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	5.78E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-4.72E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	1.12E-06	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	9.81E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-9.77E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	1.28E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	9.51E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.14E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	9.52E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.75E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	3.23E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	6.14E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.52E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.31E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.64E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.13E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	3.56E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	3.31E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	6.70E-06	1	1.80	(2,2,1,1,1,na)
Linuron	soil/agricultural	kg	1.17E-04	1	1.24	(2,2,3,1,1,na)
Metolachlor	soil/agricultural	kg	8.45E-04	1	1.24	(2,2,3,1,1,na)
soy beans IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: soy beans organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	3.56E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	7.13E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.56E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	1.38E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	9.69E-01	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.56E-04	1	1.07	(2,1,1,1,1,na)
hoeing	CH	ha	3.56E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.56E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.06E-02	1	1.07	(2,1,1,1,1,na)
pea seed organic, at regional storehouse	CH	kg	4.28E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	5.95E-02	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	6.41E-04	1	2.09	(4,5,na,na,na,na)
green manure organic, until march	CH	ha	3.56E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.96E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.36E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.62E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.56E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.56E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	6.93E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	5.11E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.30E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.93E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.23E-03	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.05E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.88E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	3.15E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.94E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-6.98E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	7.31E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	8.72E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-9.80E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	4.61E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	1.93E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	5.39E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	9.58E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.23E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	3.30E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	4.29E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.34E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	2.66E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	3.14E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.14E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.51E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	3.38E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	5.90E-06	1	1.80	(2,2,1,1,1,na)
soy beans organic, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: protein peas, IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	2.60E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	5.21E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.60E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	2.60E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	1.12E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	7.24E-02	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.60E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	4.69E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.60E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.04E-02	1	1.07	(2,1,1,1,1,na)
pea seed IP, at regional storehouse	CH	kg	6.51E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	3.57E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	5.25E-03	1	1.07	(2,1,1,1,1,na)
single superphosphate, as P2O5, at regional storehouse	RER	kg	2.12E-04	1	1.07	(2,1,1,1,1,na)
phosphate rock, as P2O5, beneficiated, dry, at plant	MA	kg	3.07E-03	1	1.07	(2,1,1,1,1,na)
thomas meal, as P2O5, at regional storehouse	RER	kg	6.63E-04	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	9.88E-03	1	1.07	(2,1,1,1,1,na)
potassium sulphate, as K2O, at regional storehouse	RER	kg	6.48E-04	1	1.07	(2,1,1,1,1,na)
benzo[thia]diazole-compounds, at regional storehouse	CH	kg	1.93E-04	1	1.13	(2,2,3,1,1,na)
nitrile-compounds, at regional storehouse	CH	kg	1.30E-05	1	1.13	(2,2,3,1,1,na)
dinitroaniline-compounds, at regional storehouse	CH	kg	3.78E-04	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.04E-04	1	1.13	(2,2,3,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	5.73E-05	1	1.13	(2,2,3,1,1,na)
transport, van <3.5t	CH	tkm	9.99E-04	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	4.60E-03	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	4.60E-03	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	1.22E-02	1	2.09	(4,5,na,na,na,na)
transport, transoceanic freight ship	OCE	tkm	2.40E-02	1	2.09	(4,5,na,na,na,na)
green manure IP, until January	CH	ha	2.60E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.50E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.20E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.26E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.60E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.60E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.35E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	2.73E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.01E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	2.87E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.69E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.14E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.49E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	6.96E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	7.38E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-4.98E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	5.02E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	1.35E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	5.46E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.35E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	5.58E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	6.47E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	4.93E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	6.09E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.32E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.97E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	6.50E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.05E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	5.16E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	8.01E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.70E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.85E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.92E-06	1	1.80	(2,2,1,1,1,na)
Fenpiclonil	soil/agricultural	kg	1.30E-05	1	1.24	(2,2,3,1,1,na)
Acclonifen	soil/agricultural	kg	3.78E-04	1	1.24	(2,2,3,1,1,na)
Bentazone	soil/agricultural	kg	1.93E-04	1	1.24	(2,2,3,1,1,na)
Carbetamide	soil/agricultural	kg	3.91E-05	1	1.24	(2,2,3,1,1,na)
Pirimicarb	soil/agricultural	kg	1.82E-05	1	1.24	(2,2,3,1,1,na)
protein peas, IP, at farm	CH	kg	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: protein peas, organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, currying, by weeder	CH	ha	6.57E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	6.57E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	8.71E-03	1	1.07	(2,1,1,1,1,na)
solid manure loading and spreading, by hydraulic loader and spreader	CH	kg	2.18E+00	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	1.04E-02	1	1.07	(2,1,1,1,1,na)
pea seed organic, at regional storehouse	CH	kg	9.03E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	3.57E-02	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.35E-03	1	2.09	(4,5,na,na,na,na)
green manure organic, until January	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.46E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.16E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable, non-irrigated	resource/land	m2a	1.59E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.28E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.28E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.95E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	3.44E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.20E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.62E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.66E-03	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.59E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.71E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	9.62E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.86E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	1.21E-05	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.43E-06	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	2.78E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	2.20E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	4.64E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.20E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	6.15E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	7.03E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.19E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	2.50E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	5.33E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.40E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	6.05E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	4.91E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.14E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	6.10E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	3.51E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	8.45E-06	1	1.80	(2,2,1,1,1,na)
protein peas, organic, at farm	CH	kg	1.00E+00			

Unit process inventory for: green manure organic, until January, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	4.61E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.44E+01	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.44E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	2.68E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.07E-01	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.92E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.87E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	9.86E-07	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.55E-05	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.30E-05	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.48E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.53E-04	1	1.80	(2,2,1,1,1,na)
green manure organic, until January	CH	ha	1.00E+00			

Unit process inventory for: green manure IP, until January, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.56E+01	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	5.46E+00	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	7.80E+00	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	1.18E+00	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	8.17E+01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	4.61E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.38E+02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.44E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	2.68E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.07E-01	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.68E+00	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.29E+00	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.89E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.28E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.34E-06	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.55E-03	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	7.06E-04	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	6.95E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.64E-03	1	1.80	(2,2,1,1,1,na)
green manure IP, until January	CH	ha	1.00E+00			

Unit process inventory for: green manure IP, until February, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.56E+01	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	5.46E+00	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	7.80E+00	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	1.18E+00	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	8.17E+01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	5.50E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.38E+02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	3.13E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.25E-01	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.68E+00	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.29E+00	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.89E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.28E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.34E-06	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.55E-03	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	7.06E-04	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	6.95E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.64E-03	1	1.80	(2,2,1,1,1,na)
green manure IP, until February	CH	ha	1.00E+00			

Life cycle inventories of Swiss and European agricultural production systems - Appendix AX14X to Chapter X14X (XArable Crop
Production in SwitzerlandX)

Unit process inventory for: green manure organic, until February, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	5.50E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.44E+01	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	3.13E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.25E-01	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.92E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.87E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	9.86E-07	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.55E-05	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.30E-05	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.48E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.53E-04	1	1.80	(2,2,1,1,1,na)
green manure organic, until February	CH	ha	1.00E+00			

Unit process inventory for: green manure organic, until march, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	6.28E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.44E+01	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	3.58E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.43E-01	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.92E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.87E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	9.86E-07	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.55E-05	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.30E-05	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.48E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.53E-04	1	1.80	(2,2,1,1,1,na)
green manure organic, until march	CH	ha	1.00E+00			

Unit process inventory for: green manure IP, until march, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.56E+01	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	5.46E+00	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	7.80E+00	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	1.18E+00	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	8.17E+01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	6.28E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.38E+02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	3.58E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.43E-01	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.68E+00	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.29E+00	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.89E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.28E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.34E-06	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.55E-03	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	7.06E-04	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	6.95E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.64E-03	1	1.80	(2,2,1,1,1,na)
green manure IP, until march	CH	ha	1.00E+00			

Unit process inventory for: green manure organic, until April, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	7.11E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.44E+01	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	4.02E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.61E-01	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.92E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.87E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	9.86E-07	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.55E-05	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.30E-05	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.48E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.53E-04	1	1.80	(2,2,1,1,1,na)
green manure organic, until April	CH	ha	1.00E+00			

Unit process inventory for: green manure IP, until April, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by spring tine harrow	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
mulching	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.00E+00	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E+01	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.56E+01	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	5.46E+00	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	7.80E+00	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	1.18E+00	1	1.07	(2,1,1,1,1,na)
transport, van <3.5t	CH	tkm	1.50E-01	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, freight, rail	CH	tkm	9.08E+00	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	8.17E+01	1	2.09	(4,5,na,na,na,na)
Occupation, arable, non-irrigated	resource/land	m2a	7.11E+03	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.00E+04	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.38E+02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.22E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	4.02E-01	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.61E-01	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	1.68E+00	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	4.29E+00	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.89E-01	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.28E+00	1	1.41	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.34E-06	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.55E-03	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	7.06E-04	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	6.95E-06	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.64E-03	1	1.80	(2,2,1,1,1,na)
green manure IP, until April	CH	ha	1.00E+00			

Appendix A15 to Chapter 15 (Arable Crop Production in the EU)

Unit-Process Inventories from Chapter 15 (Arable Crop Production in the EU)

Tab. A. 20 Unit-process inventories for arable crop production inventories in the EU

Unit process inventory for: protein peas conventional, Barrois, at farm, FR						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	1.61E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	4.51E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	8.66E-05	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.48E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.97E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.48E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	2.00E-03	1	1.07	(2,1,1,1,1,na)
pea seed IP, at regional storehouse	CH	kg	4.70E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	8.31E-03	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	5.19E-02	1	1.07	(2,1,1,1,1,na)
limestone, milled, loose, at plant	CH	kg	5.97E-03	1	1.07	(2,1,1,1,1,na)
benzimidazole-compounds, at regional storehouse	RER	kg	2.48E-04	1	1.07	(2,2,1,1,1,na)
benzothiazole-compounds, at regional storehouse	RER	kg	2.15E-04	1	1.07	(2,2,1,1,1,na)
dinitroaniline-compounds, at regional storehouse	RER	kg	1.49E-04	1	1.07	(2,2,1,1,1,na)
glyphosate, at regional storehouse	RER	kg	5.35E-05	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	7.28E-05	1	1.07	(2,2,1,1,1,na)
pyrethroid-compounds, at regional storehouse	RER	kg	1.55E-06	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	7.28E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.51E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.20E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	2.48E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.48E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.48E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	6.95E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	3.13E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	1.53E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	5.31E-05	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.48E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.66E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.23E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	1.13E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-5.83E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	3.31E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-3.23E-11	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	5.51E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.37E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	6.99E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	7.00E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	4.99E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	5.18E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	7.61E-11	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.65E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	6.03E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.09E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	4.62E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	6.72E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.20E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	8.85E-11	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.01E-06	1	1.80	(2,2,1,1,1,na)
Chlorothalonil	soil/agricultural	kg	2.48E-04	1	1.21	(2,2,1,1,1,na)
Bentazone	soil/agricultural	kg	2.15E-04	1	1.21	(2,2,1,1,1,na)
Acifluorfen	soil/agricultural	kg	1.49E-04	1	1.21	(2,2,1,1,1,na)
Glyphosate	soil/agricultural	kg	5.35E-05	1	1.21	(2,2,1,1,1,na)
Metaldehyde	soil/agricultural	kg	1.63E-05	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	1.55E-06	1	1.21	(2,2,1,1,1,na)
Pronamide	soil/agricultural	kg	4.95E-05	1	1.21	(2,2,1,1,1,na)
protein peas conventional, Barrois, at farm	FR	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: rape seed conventional, Barrois, at farm, FR						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	1.66E-04	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	6.62E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.66E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	8.28E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.31E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.53E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.31E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	2.00E-03	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	9.93E-04	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	3.28E-02	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.90E-02	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	3.56E-03	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.46E-02	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	5.69E-02	1	1.07	(2,1,1,1,1,na)
limestone, milled, loose, at plant	CH	kg	9.18E-03	1	1.07	(2,1,1,1,1,na)
[thio]carbamate-compounds, at regional storehouse	RER	kg	7.98E-05	1	1.07	(2,2,1,1,1,na)
acetamide-anilide-compounds, at regional storehouse	RER	kg	1.34E-04	1	1.07	(2,2,1,1,1,na)
dinitroaniline-compounds, at regional storehouse	RER	kg	3.05E-04	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	3.80E-04	1	1.07	(2,2,1,1,1,na)
pyretroid-compounds, at regional storehouse	RER	kg	1.32E-05	1	1.07	(2,2,1,1,1,na)
cyclic N-compounds, at regional storehouse	RER	kg	2.95E-05	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	4.32E-05	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.38E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.57E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	3.59E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.31E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.31E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	9.30E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	4.19E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	2.13E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	7.10E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.60E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	2.98E-02	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	4.56E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.17E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.57E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	7.13E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-3.74E-07	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	1.26E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-4.11E-10	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	1.29E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	3.29E-07	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	9.79E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	9.83E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	7.30E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	8.84E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.69E-11	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	6.65E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.02E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.53E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.50E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	9.83E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.04E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.97E-11	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	5.09E-06	1	1.80	(2,2,1,1,1,na)
Carbendazim	soil/agricultural	kg	7.98E-05	1	1.21	(2,2,1,1,1,na)
Napropamide	soil/agricultural	kg	1.34E-04	1	1.21	(2,2,1,1,1,na)
Trifluralin	soil/agricultural	kg	3.05E-04	1	1.21	(2,2,1,1,1,na)
Chlormequat	soil/agricultural	kg	3.97E-05	1	1.21	(2,2,1,1,1,na)
Metaldehyde	soil/agricultural	kg	8.44E-05	1	1.21	(2,2,1,1,1,na)
Cypermethrin	soil/agricultural	kg	9.11E-06	1	1.21	(2,2,1,1,1,na)
Deltamethrin	soil/agricultural	kg	2.05E-06	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	2.05E-06	1	1.21	(2,2,1,1,1,na)
Clomazone	soil/agricultural	kg	1.80E-05	1	1.21	(2,2,1,1,1,na)
Difenoconazole	soil/agricultural	kg	4.14E-06	1	1.21	(2,2,1,1,1,na)
Metconazole	soil/agricultural	kg	5.22E-07	1	1.21	(2,2,1,1,1,na)
Tebuconazole	soil/agricultural	kg	6.83E-06	1	1.21	(2,2,1,1,1,na)
Metazachlor	soil/agricultural	kg	5.15E-05	1	1.21	(2,2,1,1,1,na)
Dimethachlor	soil/agricultural	kg	1.17E-04	1	1.21	(2,2,1,1,1,na)
Iprodion	soil/agricultural	kg	5.22E-05	1	1.21	(2,2,1,1,1,na)
Vinclozolin	soil/agricultural	kg	6.21E-06	1	1.21	(2,2,1,1,1,na)
Procymidone	soil/agricultural	kg	1.86E-05	1	1.21	(2,2,1,1,1,na)
Quizalofop ethyl ester	soil/agricultural	kg	2.38E-06	1	1.21	(2,2,1,1,1,na)
rape seed conventional, Barrois, at farm	FR	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: wheat grains conventional, Barrois, at farm, FR						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	7.05E-05	1	1.07	(2.1.1.1.1.na)
tillage, harrowing, by spring tine harrow	CH	ha	2.58E-04	1	1.07	(2.1.1.1.1.na)
tillage, ploughing	CH	ha	7.76E-05	1	1.07	(2.1.1.1.1.na)
fertilising, by broadcaster	CH	ha	5.18E-04	1	1.07	(2.1.1.1.1.na)
combine harvesting	CH	ha	1.48E-04	1	1.07	(2.1.1.1.1.na)
application of plant protection products, by field sprayer	CH	ha	1.02E-03	1	1.07	(2.1.1.1.1.na)
sowing	CH	ha	1.48E-04	1	1.07	(2.1.1.1.1.na)
transport, tractor and trailer	CH	tkm	1.99E-03	1	1.07	(2.1.1.1.1.na)
wheat seed IP, at regional storehouse	CH	kg	2.33E-02	1	1.07	(2.1.1.1.1.na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.51E-02	1	1.07	(2.1.1.1.1.na)
urea, as N, at regional storehouse	RER	kg	1.21E-02	1	1.07	(2.1.1.1.1.na)
ammonium sulphate, as N, at regional storehouse	RER	kg	2.24E-03	1	1.07	(2.1.1.1.1.na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.09E-02	1	1.07	(2.1.1.1.1.na)
potassium chloride, as K2O, at regional storehouse	RER	kg	6.59E-03	1	1.07	(2.1.1.1.1.na)
limestone, milled, loose, at plant	CH	kg	1.87E-03	1	1.07	(2.1.1.1.1.na)
(sulfonyl)urea-compounds, at regional storehouse	RER	kg	2.62E-05	1	1.07	(2.2.1.1.1.na)
diphenylether-compounds, at regional storehouse	RER	kg	1.85E-05	1	1.07	(2.2.1.1.1.na)
glyphosate, at regional storehouse	RER	kg	5.33E-05	1	1.07	(2.2.1.1.1.na)
nitrile-compounds, at regional storehouse	RER	kg	6.13E-05	1	1.07	(2.2.1.1.1.na)
organophosphorus-compounds, at regional storehouse	RER	kg	3.41E-07	1	1.07	(2.2.1.1.1.na)
pesticide unspecified, at regional storehouse	RER	kg	2.82E-04	1	1.07	(2.2.1.1.1.na)
phenoxy-compounds, at regional storehouse	RER	kg	5.50E-05	1	1.07	(2.2.1.1.1.na)
pyretroid-compounds, at regional storehouse	RER	kg	1.61E-06	1	1.07	(2.2.1.1.1.na)
cyclic N-compounds, at regional storehouse	RER	kg	1.76E-05	1	1.07	(2.2.1.1.1.na)
transport, lorry 3.5-16t, fleet average	RER	tkm	3.65E-04	1	2.09	(4.5.na.na.na.na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.51E+01	1	1.07	(2.1.1.1.1.na)
Carbon dioxide, in air	resource/in air	kg	1.29E+00	1	1.07	(2.1.1.1.1.na)
Occupation, arable	resource/land	m2a	1.48E+00	1	1.11	(2.1.1.1.1.na)
Transformation, from arable, non-irrigated	resource/land	m2	1.48E+00	1	1.21	(2.1.1.1.1.na)
Transformation, to arable, non-irrigated	resource/land	m2	1.48E+00	1	1.21	(2.1.1.1.1.na)
Nitrate	water/ground-	kg	4.17E-02	1	1.51	(2.2.1.1.1.na)
Phosphorus	water/river	kg	1.87E-05	1	1.51	(2.2.1.1.1.na)
Phosphate	water/river	kg	9.33E-05	1	1.51	(2.2.1.1.1.na)
Phosphate	water/ground-	kg	3.18E-05	1	1.51	(2.2.1.1.1.na)
Ammonia	airflow, population density	kg	2.79E-03	1	1.21	(2.2.1.1.1.na)
Carbon dioxide, fossil	airflow, population density	kg	1.90E-02	1	1.07	(2.2.1.1.1.na)
Nitrogen oxides	airflow, population density	kg	2.23E-04	1	1.41	(2.2.1.1.1.na)
Dinitrogen monoxide	airflow, population density	kg	1.06E-03	1	1.41	(2.2.1.1.1.na)
Cadmium	soil/agricultural	kg	1.12E-06	1	1.51	(2.2.1.1.1.na)
Chromium	soil/agricultural	kg	2.75E-06	1	1.51	(2.2.1.1.1.na)
Copper	soil/agricultural	kg	1.50E-06	1	1.51	(2.2.1.1.1.na)
Lead	soil/agricultural	kg	1.82E-07	1	1.51	(2.2.1.1.1.na)
Mercury	soil/agricultural	kg	1.11E-10	1	1.51	(2.2.1.1.1.na)
Nickel	soil/agricultural	kg	9.22E-07	1	1.51	(2.2.1.1.1.na)
Zinc	soil/agricultural	kg	1.23E-07	1	1.51	(2.2.1.1.1.na)
Cadmium, ion	water/river	kg	4.36E-09	1	1.80	(2.2.1.1.1.na)
Chromium, ion	water/river	kg	4.37E-07	1	1.80	(2.2.1.1.1.na)
Copper, ion	water/river	kg	3.24E-07	1	1.80	(2.2.1.1.1.na)
Lead	water/river	kg	2.97E-08	1	1.80	(2.2.1.1.1.na)
Mercury	water/river	kg	3.84E-11	1	1.80	(2.2.1.1.1.na)
Nickel, ion	water/river	kg	2.80E-07	1	1.80	(2.2.1.1.1.na)
Zinc, ion	water/river	kg	4.41E-07	1	1.80	(2.2.1.1.1.na)
Cadmium, ion	water/ground-	kg	6.82E-09	1	1.80	(2.2.1.1.1.na)
Chromium, ion	water/ground-	kg	2.89E-06	1	1.80	(2.2.1.1.1.na)
Copper, ion	water/ground-	kg	4.36E-07	1	1.80	(2.2.1.1.1.na)
Lead	water/ground-	kg	6.87E-09	1	1.80	(2.2.1.1.1.na)
Mercury	water/ground-	kg	4.46E-11	1	1.80	(2.2.1.1.1.na)
Zinc, ion	water/ground-	kg	2.20E-06	1	1.80	(2.2.1.1.1.na)
Isoproturon	soil/agricultural	kg	2.58E-05	1	1.21	(2.2.1.1.1.na)
Metsulfuron-methyl	soil/agricultural	kg	3.87E-07	1	1.21	(2.2.1.1.1.na)
Bifenox	soil/agricultural	kg	5.28E-06	1	1.21	(2.2.1.1.1.na)
Clodinafop-propargyl	soil/agricultural	kg	7.03E-06	1	1.21	(2.2.1.1.1.na)
Diffenican	soil/agricultural	kg	6.19E-06	1	1.21	(2.2.1.1.1.na)
Glyphosate	soil/agricultural	kg	5.33E-05	1	1.21	(2.2.1.1.1.na)
Bromoxynil	soil/agricultural	kg	1.50E-05	1	1.21	(2.2.1.1.1.na)
Cyprodinil	soil/agricultural	kg	2.64E-05	1	1.21	(2.2.1.1.1.na)
Ioxynil	soil/agricultural	kg	1.98E-05	1	1.21	(2.2.1.1.1.na)
Ethephon	soil/agricultural	kg	3.41E-07	1	1.21	(2.2.1.1.1.na)
Chlormequat	soil/agricultural	kg	9.74E-05	1	1.21	(2.2.1.1.1.na)
Metaldehyde	soil/agricultural	kg	2.22E-05	1	1.21	(2.2.1.1.1.na)
Trinexapac-ethyl	soil/agricultural	kg	1.39E-06	1	1.21	(2.2.1.1.1.na)
Fluroxypyr	soil/agricultural	kg	2.01E-05	1	1.21	(2.2.1.1.1.na)
Mecoprop	soil/agricultural	kg	2.81E-05	1	1.21	(2.2.1.1.1.na)
Mecoprop-P	soil/agricultural	kg	6.91E-06	1	1.21	(2.2.1.1.1.na)
Cypermethrin	soil/agricultural	kg	1.11E-06	1	1.21	(2.2.1.1.1.na)
Deltamethrin	soil/agricultural	kg	2.50E-07	1	1.21	(2.2.1.1.1.na)
Lambda-Cyhalothrin	soil/agricultural	kg	2.50E-07	1	1.21	(2.2.1.1.1.na)
Cyproconazole	soil/agricultural	kg	7.13E-07	1	1.21	(2.2.1.1.1.na)
Epoxiconazole	soil/agricultural	kg	4.08E-06	1	1.21	(2.2.1.1.1.na)
Fenpropidin	soil/agricultural	kg	1.72E-06	1	1.21	(2.2.1.1.1.na)
Fenpropimorph	soil/agricultural	kg	4.32E-06	1	1.21	(2.2.1.1.1.na)
Metconazole	soil/agricultural	kg	5.04E-07	1	1.21	(2.2.1.1.1.na)
Prochloraz	soil/agricultural	kg	1.49E-06	1	1.21	(2.2.1.1.1.na)
Propiconazole	soil/agricultural	kg	1.84E-06	1	1.21	(2.2.1.1.1.na)
Tebuconazole	soil/agricultural	kg	2.97E-06	1	1.21	(2.2.1.1.1.na)
Cloquintocet-mexyl	soil/agricultural	kg	1.76E-06	1	1.21	(2.2.1.1.1.na)
MCPA	soil/agricultural	kg	4.18E-05	1	1.21	(2.2.1.1.1.na)
Mepiquat chloride	soil/agricultural	kg	6.22E-07	1	1.21	(2.2.1.1.1.na)
Azoxystrobin	soil/agricultural	kg	6.22E-06	1	1.21	(2.2.1.1.1.na)
Spiroxamine	soil/agricultural	kg	1.02E-06	1	1.21	(2.2.1.1.1.na)
Tefluthrin	soil/agricultural	kg	8.45E-08	1	1.21	(2.2.1.1.1.na)
Triadimenol	soil/agricultural	kg	4.20E-07	1	1.21	(2.2.1.1.1.na)
Anthraquinone	soil/agricultural	kg	7.91E-06	1	1.21	(2.2.1.1.1.na)
Bitertanol	soil/agricultural	kg	1.18E-07	1	1.21	(2.2.1.1.1.na)
Bromuconazole	soil/agricultural	kg	3.57E-07	1	1.21	(2.2.1.1.1.na)
Carfentrazone ethyl ester	soil/agricultural	kg	6.22E-08	1	1.21	(2.2.1.1.1.na)
Choline chloride	soil/agricultural	kg	6.78E-05	1	1.21	(2.2.1.1.1.na)
Clopyralid	soil/agricultural	kg	4.27E-06	1	1.21	(2.2.1.1.1.na)
Diclofop-methyl	soil/agricultural	kg	1.11E-05	1	1.21	(2.2.1.1.1.na)
Florasulam	soil/agricultural	kg	3.33E-08	1	1.21	(2.2.1.1.1.na)
Fludioxonil	soil/agricultural	kg	7.52E-07	1	1.21	(2.2.1.1.1.na)
Flupyr-sulfuron-methyl	soil/agricultural	kg	4.77E-08	1	1.21	(2.2.1.1.1.na)
Imidacloprid	soil/agricultural	kg	5.49E-07	1	1.21	(2.2.1.1.1.na)
Iodosulfuron-methyl-sodium	soil/agricultural	kg	2.98E-08	1	1.21	(2.2.1.1.1.na)
Kresoxim-methyl	soil/agricultural	kg	1.14E-06	1	1.21	(2.2.1.1.1.na)
Mesosulfuron-methyl (prop)	soil/agricultural	kg	1.49E-07	1	1.21	(2.2.1.1.1.na)
Metosulam	soil/agricultural	kg	9.40E-08	1	1.21	(2.2.1.1.1.na)
Picoxystrobin	soil/agricultural	kg	4.28E-07	1	1.21	(2.2.1.1.1.na)
Prohexadione-calcium	soil/agricultural	kg	3.70E-08	1	1.21	(2.2.1.1.1.na)
Propoxycarbazon-sodium (prop)	soil/agricultural	kg	2.05E-07	1	1.21	(2.2.1.1.1.na)
Pyraclostrobin (prop)	soil/agricultural	kg	3.93E-06	1	1.21	(2.2.1.1.1.na)
Quinoxifen	soil/agricultural	kg	1.79E-06	1	1.21	(2.2.1.1.1.na)
Silthiofam	soil/agricultural	kg	2.76E-06	1	1.21	(2.2.1.1.1.na)
Trifloxystrobin	soil/agricultural	kg	2.16E-06	1	1.21	(2.2.1.1.1.na)
wheat grains conventional, Barrois, at farm	FR	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: barley grains conventional, Barrois, at farm, FR						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	2.96E-05	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	2.37E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.18E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	3.70E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	1.48E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	8.14E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.48E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	2.00E-03	1	1.07	(2,1,1,1,1,na)
barley seed IP, at regional storehouse	CH	kg	2.08E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.16E-02	1	1.07	(2,1,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.14E-02	1	1.07	(2,1,1,1,1,na)
ammonium sulphate, as N, at regional storehouse	RER	kg	2.45E-03	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.08E-02	1	1.07	(2,1,1,1,1,na)
limestone, milled, loose, at plant	CH	kg	1.87E-03	1	1.07	(2,1,1,1,1,na)
[sulfonyl]urea-compounds, at regional storehouse	RER	kg	8.56E-05	1	1.07	(2,2,1,1,1,na)
diphenylether-compounds, at regional storehouse	RER	kg	6.08E-06	1	1.07	(2,2,1,1,1,na)
nitrile-compounds, at regional storehouse	RER	kg	4.23E-05	1	1.07	(2,2,1,1,1,na)
organophosphorus-compounds, at regional storehouse	RER	kg	2.85E-05	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	1.06E-04	1	1.07	(2,2,1,1,1,na)
phenoxy-compounds, at regional storehouse	RER	kg	4.73E-07	1	1.07	(2,2,1,1,1,na)
pyrethroid-compounds, at regional storehouse	RER	kg	1.07E-06	1	1.07	(2,2,1,1,1,na)
cyclic N-compounds, at regional storehouse	RER	kg	1.49E-05	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	3.21E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.53E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.31E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	1.48E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.48E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.48E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	4.60E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.87E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	9.42E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	3.17E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	2.60E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	1.79E-02	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.09E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	9.95E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.21E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	2.93E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-2.25E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	1.19E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	8.53E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-2.28E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	4.36E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	4.37E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.19E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.00E-08	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.68E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	4.19E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	6.83E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.89E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	4.30E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	4.63E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.09E-06	1	1.80	(2,2,1,1,1,na)
Chlorotoluron	soil/agricultural	kg	1.60E-05	1	1.21	(2,2,1,1,1,na)
Isoproturon	soil/agricultural	kg	6.95E-05	1	1.21	(2,2,1,1,1,na)
Metsulfuron-methyl	soil/agricultural	kg	1.55E-07	1	1.21	(2,2,1,1,1,na)
Bifenox	soil/agricultural	kg	2.65E-06	1	1.21	(2,2,1,1,1,na)
Diffenican	soil/agricultural	kg	3.42E-06	1	1.21	(2,2,1,1,1,na)
Bromoxynil	soil/agricultural	kg	6.66E-06	1	1.21	(2,2,1,1,1,na)
Cyprodinil	soil/agricultural	kg	3.17E-05	1	1.21	(2,2,1,1,1,na)
Ioxynil	soil/agricultural	kg	3.99E-06	1	1.21	(2,2,1,1,1,na)
Ethephon	soil/agricultural	kg	2.85E-05	1	1.21	(2,2,1,1,1,na)
Chlomequat	soil/agricultural	kg	2.66E-07	1	1.21	(2,2,1,1,1,na)
Metaldehyde	soil/agricultural	kg	3.55E-05	1	1.21	(2,2,1,1,1,na)
Trinexapac-ethyl	soil/agricultural	kg	2.88E-06	1	1.21	(2,2,1,1,1,na)
Fluroxypyr	soil/agricultural	kg	4.73E-07	1	1.21	(2,2,1,1,1,na)
Cypermethrin	soil/agricultural	kg	7.40E-07	1	1.21	(2,2,1,1,1,na)
Deltamethrin	soil/agricultural	kg	1.66E-07	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	1.66E-07	1	1.21	(2,2,1,1,1,na)
Cyproconazole	soil/agricultural	kg	4.05E-07	1	1.21	(2,2,1,1,1,na)
Epoxiconazole	soil/agricultural	kg	2.29E-06	1	1.21	(2,2,1,1,1,na)
Fenpropimorph	soil/agricultural	kg	5.30E-06	1	1.21	(2,2,1,1,1,na)
Flusilazole	soil/agricultural	kg	1.28E-06	1	1.21	(2,2,1,1,1,na)
Propiconazole	soil/agricultural	kg	3.30E-06	1	1.21	(2,2,1,1,1,na)
Tebuconazole	soil/agricultural	kg	2.34E-06	1	1.21	(2,2,1,1,1,na)
MCPA	soil/agricultural	kg	2.37E-06	1	1.21	(2,2,1,1,1,na)
Mepiquat chloride	soil/agricultural	kg	1.09E-05	1	1.21	(2,2,1,1,1,na)
Azoxystrobin	soil/agricultural	kg	1.12E-05	1	1.21	(2,2,1,1,1,na)
Antraquinone	soil/agricultural	kg	1.04E-05	1	1.21	(2,2,1,1,1,na)
Bitteranol	soil/agricultural	kg	1.12E-06	1	1.21	(2,2,1,1,1,na)
Carfentrazone ethyl ester	soil/agricultural	kg	1.04E-07	1	1.21	(2,2,1,1,1,na)
Clpyralid	soil/agricultural	kg	2.37E-07	1	1.21	(2,2,1,1,1,na)
Diclofop-methyl	soil/agricultural	kg	1.66E-05	1	1.21	(2,2,1,1,1,na)
Fenoxaprop-P ethyl ester	soil/agricultural	kg	7.69E-07	1	1.21	(2,2,1,1,1,na)
Fludioxonil	soil/agricultural	kg	6.63E-07	1	1.21	(2,2,1,1,1,na)
Imidacloprid	soil/agricultural	kg	5.22E-06	1	1.21	(2,2,1,1,1,na)
Kresoxim-methyl	soil/agricultural	kg	1.02E-07	1	1.21	(2,2,1,1,1,na)
Mefenpyr-diethyl	soil/agricultural	kg	1.54E-06	1	1.21	(2,2,1,1,1,na)
Picoxystrobin	soil/agricultural	kg	3.68E-06	1	1.21	(2,2,1,1,1,na)
Pyraclostrobin (prop)	soil/agricultural	kg	1.29E-06	1	1.21	(2,2,1,1,1,na)
Trifloxystrobin	soil/agricultural	kg	1.52E-06	1	1.21	(2,2,1,1,1,na)
barley grains conventional, Barrois, at farm	FR	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: sunflower conventional, Castilla-y-Leon, at farm, ES						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, cultivating, chiselling	CH	ha	9.69E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	9.69E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	9.69E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.94E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	9.69E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	4.70E-03	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	3.15E-03	1	1.07	(2,1,1,1,1,na)
linuron, at regional storehouse	RER	kg	4.36E-04	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	1.05E-03	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	9.17E-05	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.83E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.19E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	1.21E+01	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	9.69E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	9.69E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.78E-01	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	2.02E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	2.60E-04	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	5.93E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.82E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	3.17E-09	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-8.87E-09	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-4.57E-08	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	1.03E-08	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-9.02E-11	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-7.59E-10	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	2.29E-08	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.31E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	2.01E-09	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	1.67E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	3.33E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	8.83E-11	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	6.63E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.60E-08	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	3.15E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	8.05E-09	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.37E-08	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	4.66E-10	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	6.23E-11	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	4.85E-08	1	1.80	(2,2,1,1,1,na)
Linuron	soil/agricultural	kg	4.36E-04	1	1.21	(2,2,1,1,1,na)
Alachlor	soil/agricultural	kg	4.36E-04	1	1.21	(2,2,1,1,1,na)
Oxyfluorfen	soil/agricultural	kg	1.74E-04	1	1.21	(2,2,1,1,1,na)
sunflower conventional, Castilla-y-Leon, at farm	ES	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: wheat grains conventional, Castilla-y-Leon, at farm, ES						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, cultivating, chiselling	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
tillage, rolling	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	6.56E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.28E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	5.00E-03	1	1.07	(2,1,1,1,1,na)
wheat seed IP, at regional storehouse	CH	kg	5.74E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	2.58E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.22E-02	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	1.61E-01	1	1.07	(2,1,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	2.52E-04	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	8.68E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.45E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.24E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	2.73E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.28E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.28E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.25E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	5.57E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	5.28E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	6.26E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.55E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	7.40E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	2.50E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	5.66E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-8.73E-07	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	5.37E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	6.31E-12	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	2.54E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	9.25E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	1.29E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.30E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	9.54E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.16E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.26E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	8.95E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.28E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.51E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.39E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	9.57E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.99E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.10E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	4.77E-06	1	1.80	(2,2,1,1,1,na)
Mefenpyr	soil/agricultural	kg	1.64E-05	1	1.21	(2,2,1,1,1,na)
Diclofop	soil/agricultural	kg	9.84E-05	1	1.21	(2,2,1,1,1,na)
Fenoxaprop ethyl ester	soil/agricultural	kg	8.20E-06	1	1.21	(2,2,1,1,1,na)
Tribenuron-methyl	soil/agricultural	kg	2.95E-06	1	1.21	(2,2,1,1,1,na)
wheat grains conventional, Castilla-y-Leon, at farm	ES	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: barley grains conventional, Castilla-y-Leon, at farm, ES						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, cultivating, chiselling	CH	ha	3.60E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	3.60E-04	1	1.07	(2,1,1,1,1,na)
tillage, rolling	CH	ha	3.60E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	7.21E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	3.60E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	5.41E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	3.60E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	5.00E-03	1	1.07	(2,1,1,1,1,na)
barley seed IP, at regional storehouse	CH	kg	6.70E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	2.78E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	2.14E-02	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	4.39E-03	1	1.07	(2,1,1,1,1,na)
MCPA, at regional storehouse	RER	kg	7.43E-05	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	2.28E-04	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	1.01E-03	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.44E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.24E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	3.60E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	3.60E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	3.60E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	9.63E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	5.56E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	7.73E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	6.74E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	3.15E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.50E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	2.36E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	4.24E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-1.72E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	3.36E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	2.44E-06	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	6.20E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	1.27E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	1.27E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	9.18E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	6.84E-08	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	8.47E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.14E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.62E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	6.88E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.01E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.29E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	4.61E-06	1	1.80	(2,2,1,1,1,na)
MCPA	soil/agricultural	kg	7.43E-05	1	1.21	(2,2,1,1,1,na)
2,4-D	soil/agricultural	kg	7.43E-05	1	1.21	(2,2,1,1,1,na)
Mefenpyr	soil/agricultural	kg	6.13E-06	1	1.21	(2,2,1,1,1,na)
Diclofop	soil/agricultural	kg	3.68E-05	1	1.21	(2,2,1,1,1,na)
Fenoxaprop ethyl ester	soil/agricultural	kg	3.06E-06	1	1.21	(2,2,1,1,1,na)
Tralkoxydim	soil/agricultural	kg	5.86E-05	1	1.21	(2,2,1,1,1,na)
Tribenuron-methyl	soil/agricultural	kg	1.35E-06	1	1.21	(2,2,1,1,1,na)
barley grains conventional, Castilla-y-Leon, at farm	ES	kg	1.00E+00			

Unit process inventory for: protein peas conventional, Castilla-y-Leon, at farm, ES						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, cultivating, chiselling	CH	ha	8.32E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	8.32E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	8.32E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	2.49E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	8.32E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	5.00E-03	1	1.07	(2,1,1,1,1,na)
pea seed IP, at regional storehouse	CH	kg	1.83E-01	1	1.07	(2,1,1,1,1,na)
dinitroaniline-compounds, at regional storehouse	RER	kg	4.99E-04	1	1.07	(2,2,1,1,1,na)
linuron, at regional storehouse	RER	kg	2.49E-04	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	1.16E-04	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	2.77E-03	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.31E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.04E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	7.62E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	8.32E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	8.32E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.57E-01	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.16E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	1.64E-04	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	5.12E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	2.44E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.06E-08	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-2.90E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-4.62E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.03E-08	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	5.93E-10	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	3.27E-08	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.27E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	6.93E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	4.78E-08	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	1.08E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	3.83E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	3.24E-10	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	7.80E-08	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	8.00E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	9.87E-10	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.88E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	1.32E-06	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	8.06E-10	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	3.43E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.64E-06	1	1.80	(2,2,1,1,1,na)
Trifluralin	soil/agricultural	kg	4.99E-04	1	1.21	(2,2,1,1,1,na)
Linuron	soil/agricultural	kg	2.49E-04	1	1.21	(2,2,1,1,1,na)
Cycloxydim	soil/agricultural	kg	9.98E-05	1	1.21	(2,2,1,1,1,na)
protein peas conventional, Castilla-y-Leon, at farm	ES	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: rape seed conventional, Saxony-Anhalt, at farm, DE						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	1.14E-03	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.86E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	6.66E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.86E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	1.43E-03	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.86E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	4.00E-03	1	1.07	(2,1,1,1,1,na)
rape seed IP, at regional storehouse	CH	kg	1.00E-03	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	1.01E-02	1	1.07	(2,1,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.38E-02	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	1.15E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	1.29E-02	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	4.83E-02	1	1.07	(2,1,1,1,1,na)
diphenylether-compounds, at regional storehouse	RER	kg	2.29E-05	1	1.07	(2,1,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	2.39E-04	1	1.07	(2,2,1,1,1,na)
pyrethroid-compounds, at regional storehouse	RER	kg	5.00E-06	1	1.07	(2,2,1,1,1,na)
cyclic N-compounds, at regional storehouse	RER	kg	2.39E-02	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	7.39E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.41E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.60E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	2.86E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.86E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.86E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.81E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	3.61E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	6.13E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	6.16E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.14E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.02E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	3.33E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	3.14E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-3.97E-07	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	3.31E-08	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-4.88E-10	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	4.94E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-4.50E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	8.12E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	8.29E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	6.40E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	6.88E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.70E-11	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	5.31E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	7.20E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.27E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	5.48E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	8.62E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	1.59E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.98E-11	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	3.60E-06	1	1.80	(2,2,1,1,1,na)
Fluazifop-P-butyl	soil/agricultural	kg	2.29E-05	1	1.21	(2,2,1,1,1,na)
Cypermethrin	soil/agricultural	kg	2.86E-06	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	2.14E-06	1	1.21	(2,2,1,1,1,na)
Clomazone	soil/agricultural	kg	2.38E-02	1	1.21	(2,2,1,1,1,na)
Metconazole	soil/agricultural	kg	8.57E-06	1	1.21	(2,2,1,1,1,na)
Tebuconazole	soil/agricultural	kg	6.46E-05	1	1.21	(2,2,1,1,1,na)
Metazachlor	soil/agricultural	kg	1.79E-04	1	1.21	(2,2,1,1,1,na)
Prothioconazol	soil/agricultural	kg	1.18E-05	1	1.21	(2,2,1,1,1,na)
rape seed conventional, Saxony-Anhalt, at farm	DE	kg	1.00E+00			

Unit process inventory for: wheat grains conventional, Saxony-Anhalt, at farm, DE						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, cultivating, chiselling	CH	ha	6.88E-05	1	1.07	(2,1,1,1,1,na)
tillage, harrowing, by rotary harrow	CH	ha	3.91E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	6.34E-05	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	5.06E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	1.32E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	6.55E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.32E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	3.98E-03	1	1.07	(2,1,1,1,1,na)
wheat seed IP, at regional storehouse	CH	kg	2.38E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	1.53E-02	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.32E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	7.75E-03	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	6.56E-03	1	1.07	(2,1,1,1,1,na)
diphenylether-compounds, at regional storehouse	RER	kg	8.26E-06	1	1.07	(2,1,1,1,1,na)
nitrile-compounds, at regional storehouse	RER	kg	2.53E-05	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	1.96E-04	1	1.07	(2,2,1,1,1,na)
pyrethroid-compounds, at regional storehouse	RER	kg	9.91E-07	1	1.07	(2,2,1,1,1,na)
pyridazine-compounds, at regional storehouse	RER	kg	4.95E-05	1	1.07	(2,2,1,1,1,na)
cyclic N-compounds, at regional storehouse	RER	kg	6.24E-05	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	3.67E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.51E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.29E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	1.43E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.32E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.32E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.78E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.67E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	2.96E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	5.64E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.54E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	7.35E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	7.59E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	1.59E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-1.60E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	2.03E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-1.40E-10	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	7.02E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.35E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.81E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	3.84E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	2.88E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	3.26E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	3.90E-11	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.39E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	3.55E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	5.96E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.54E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.88E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	7.53E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	4.54E-11	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.78E-06	1	1.80	(2,2,1,1,1,na)
Diflufenican	soil/agricultural	kg	8.26E-06	1	1.21	(2,2,1,1,1,na)
Cyprodinil	soil/agricultural	kg	2.53E-05	1	1.21	(2,2,1,1,1,na)
Chlormequat	soil/agricultural	kg	1.04E-04	1	1.21	(2,2,1,1,1,na)
Trinexapac-ethyl	soil/agricultural	kg	2.93E-06	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	9.91E-07	1	1.21	(2,2,1,1,1,na)
Chloridazon	soil/agricultural	kg	4.95E-05	1	1.21	(2,2,1,1,1,na)
Cyproconazole	soil/agricultural	kg	3.38E-06	1	1.21	(2,2,1,1,1,na)
Fenpropidin	soil/agricultural	kg	3.33E-05	1	1.21	(2,2,1,1,1,na)
Prochloraz	soil/agricultural	kg	1.60E-06	1	1.21	(2,2,1,1,1,na)
Propiconazole	soil/agricultural	kg	1.06E-05	1	1.21	(2,2,1,1,1,na)
Tebuconazole	soil/agricultural	kg	1.35E-05	1	1.21	(2,2,1,1,1,na)
Azoxystrobin	soil/agricultural	kg	6.80E-06	1	1.21	(2,2,1,1,1,na)
Spiroxamine	soil/agricultural	kg	1.72E-05	1	1.21	(2,2,1,1,1,na)
Flufenacet	soil/agricultural	kg	4.96E-06	1	1.21	(2,2,1,1,1,na)
Flurtamone	soil/agricultural	kg	5.17E-06	1	1.21	(2,2,1,1,1,na)
Picoxystrobin	soil/agricultural	kg	1.14E-06	1	1.21	(2,2,1,1,1,na)
wheat grains conventional, Saxony-Anhalt, at farm	DE	kg	1.00E+00			

Production in the EUX)

Unit process inventory for: barley grains conventional, Saxony-Anhalt, at farm, DE						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	5.33E-04	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	1.33E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	5.00E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	1.33E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	5.33E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	1.33E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	4.00E-03	1	1.07	(2,1,1,1,1,na)
barley seed IP, at regional storehouse	CH	kg	2.13E-02	1	1.07	(2,1,1,1,1,na)
grain drying, high temperature	CH	kg	1.49E-02	1	1.07	(2,1,1,1,1,na)
calcium ammonium nitrate, as N, at regional storehouse	RER	kg	2.01E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	5.75E-03	1	1.07	(2,1,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	9.81E-05	1	1.07	(2,1,1,1,1,na)
diphenylether-compounds, at regional storehouse	RER	kg	8.33E-06	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	8.23E-05	1	1.07	(2,2,1,1,1,na)
pyrethroid-compounds, at regional storehouse	RER	kg	1.00E-06	1	1.07	(2,2,1,1,1,na)
cyclic N-compounds, at regional storehouse	RER	kg	3.83E-05	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	3.24E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.53E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.31E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	1.22E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	1.33E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	1.33E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	1.67E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.69E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	2.62E-05	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	4.88E-04	1	1.21	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.33E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	6.36E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	6.43E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	5.06E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-2.44E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	1.31E-07	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	5.52E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-3.37E-06	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.74E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	3.75E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	2.57E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	2.20E-08	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	2.15E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	3.03E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	5.86E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	2.48E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.46E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	5.09E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	1.51E-06	1	1.80	(2,2,1,1,1,na)
Diflufenican	soil/agricultural	kg	8.33E-06	1	1.21	(2,2,1,1,1,na)
Trinexapac-ethyl	soil/agricultural	kg	1.63E-05	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	1.00E-06	1	1.21	(2,2,1,1,1,na)
Fenprovidin	soil/agricultural	kg	3.00E-05	1	1.21	(2,2,1,1,1,na)
Propiconazole	soil/agricultural	kg	8.33E-06	1	1.21	(2,2,1,1,1,na)
Florasulam	soil/agricultural	kg	2.50E-07	1	1.21	(2,2,1,1,1,na)
Flufenacet	soil/agricultural	kg	1.00E-05	1	1.21	(2,2,1,1,1,na)
Flurtamone	soil/agricultural	kg	1.04E-05	1	1.21	(2,2,1,1,1,na)
barley grains conventional, Saxony-Anhalt, at farm	DE	kg	1.00E+00			

Unit process inventory for: protein peas conventional, Saxony-Anhalt, at farm, DE						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tillage, harrowing, by rotary harrow	CH	ha	1.05E-03	1	1.07	(2,1,1,1,1,na)
tillage, ploughing	CH	ha	2.63E-04	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	1.76E-04	1	1.07	(2,1,1,1,1,na)
combine harvesting	CH	ha	2.63E-04	1	1.07	(2,1,1,1,1,na)
application of plant protection products, by field sprayer	CH	ha	5.26E-04	1	1.07	(2,1,1,1,1,na)
sowing	CH	ha	2.63E-04	1	1.07	(2,1,1,1,1,na)
transport, tractor and trailer	CH	tkm	4.00E-03	1	1.07	(2,1,1,1,1,na)
pea seed IP, at regional storehouse	CH	kg	6.18E-02	1	1.07	(2,1,1,1,1,na)
grain drying, low temperature	CH	kg	2.05E-02	1	1.07	(2,1,1,1,1,na)
triple superphosphate, as P2O5, at regional storehouse	RER	kg	9.21E-03	1	1.07	(2,1,1,1,1,na)
benzo[thia]diazole-compounds, at regional storehouse	RER	kg	1.26E-04	1	1.07	(2,2,1,1,1,na)
dinitroaniline-compounds, at regional storehouse	RER	kg	1.58E-04	1	1.07	(2,2,1,1,1,na)
pesticide unspecified, at regional storehouse	RER	kg	2.49E-04	1	1.07	(2,2,1,1,1,na)
pyrethroid-compounds, at regional storehouse	RER	kg	1.97E-06	1	1.07	(2,2,1,1,1,na)
transport, lorry 3.5-16t, fleet average	RER	tkm	9.44E-04	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.49E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.19E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	2.41E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from arable, non-irrigated	resource/land	m2	2.63E+00	1	1.21	(2,1,1,1,1,na)
Transformation, to arable, non-irrigated	resource/land	m2	2.63E+00	1	1.21	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.58E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	3.00E-05	1	1.51	(2,2,1,1,1,na)
Phosphate	water/ground-	kg	5.17E-05	1	1.51	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.78E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.33E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.24E-06	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	8.59E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	-5.72E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	7.96E-08	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-7.28E-12	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	5.18E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	-1.23E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	6.65E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	6.64E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	4.60E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.19E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	8.95E-11	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.24E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	5.09E-07	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	1.16E-08	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	4.87E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	6.88E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	3.06E-09	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.16E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.82E-06	1	1.80	(2,2,1,1,1,na)
Bentazone	soil/agricultural	kg	1.26E-04	1	1.21	(2,2,1,1,1,na)
Pendimethalin	soil/agricultural	kg	1.58E-04	1	1.21	(2,2,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	1.97E-06	1	1.21	(2,2,1,1,1,na)
protein peas conventional, Saxony-Anhalt, at farm	DE	kg	1.00E+00			

Appendix A16 to Chapter 16 (Hay)

Unit-Process Inventories from Chapter 16 (Hay)

Tab. A. 21 Unit-process inventories for hay .

Unit process inventory for: hay intensive IP, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
ammonium nitrate, as N, at regional storehouse	RER	kg	2.51E-03	1	1.07	(2,1,1,1,1,na)
fertilising, by broadcaster	CH	ha	8.62E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	6.70E-03	1	1.07	(2,1,1,1,1,na)
mowing, by rotary mower	CH	ha	4.31E-04	1	1.07	(2,1,1,1,1,na)
swath, by rotary windrower	CH	ha	4.31E-04	1	1.07	(2,1,1,1,1,na)
haying, by rotary tedder	CH	ha	1.08E-03	1	1.07	(2,1,1,1,1,na)
fodder loading, by self-loading trailer	CH	m3	2.18E-02	1	1.07	(2,1,1,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	4.31E-05	1	1.07	(2,1,1,1,1,na)
dried roughage store, cold-air dried, conventional, operation	CH	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
transport, freight, rail	CH	tkm	7.18E-04	1	2.09	(4,5,na,na,na,na)
transport, lorry 20-28t, fleet average	CH	tkm	7.18E-04	1	2.09	(4,5,na,na,na,na)
transport, van <3.5t	CH	tkm	1.29E-06	1	2.09	(4,5,na,na,na,na)
transport, barge	RER	tkm	6.46E-03	1	2.09	(4,5,na,na,na,na)
Occupation, pasture and meadow, intensive	resource/land	m2a	8.62E-01	1	1.11	(2,1,1,1,1,na)
Transformation, from pasture and meadow, intensive	resource/land	m2	4.31E-02	1	1.21	(2,1,1,1,1,na)
Transformation, to pasture and meadow, intensive	resource/land	m2	4.31E-02	1	1.21	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.65E+00	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.79E+01	1	1.07	(2,1,1,1,1,na)
Asulam	soil/agricultural	kg	4.31E-05	1	1.24	(2,2,3,1,1,na)
Nitrate	water/ground-	kg	6.44E-04	1	1.50	(2,1,1,1,1,na)
Phosphorus	water/river	kg	1.70E-05	1	1.50	(2,1,1,1,1,na)
Phosphate	water/river	kg	9.07E-05	1	1.50	(2,1,1,1,1,na)
Phosphate	water/ground-	kg	2.01E-05	1	1.50	(2,1,1,1,1,na)
Ammonia	air/low population density	kg	5.38E-03	1	1.21	(2,1,1,1,1,na)
Nitrogen oxides	air/low population density	kg	7.84E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.73E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	1.97E-07	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-4.80E-07	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	2.01E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	4.36E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-3.35E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-3.58E-08	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	1.11E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	4.65E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	3.82E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.21E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.85E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.52E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.10E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.00E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	3.62E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.63E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.05E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.18E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	9.40E-10	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.47E-06	1	1.80	(2,2,1,1,1,na)
hay intensive IP, at farm	CH	kg	1.00E+00			

Unit process inventory for: hay extensive, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
mowing, by rotary mower	CH	ha	3.70E-04	1	1.07	(2,1,1,1,1,na)
swath, by rotary windrower	CH	ha	7.40E-04	1	1.07	(2,1,1,1,1,na)
haying, by rotary tedder	CH	ha	1.30E-03	1	1.07	(2,1,1,1,1,na)
baling	CH	unit	1.44E-03	1	1.07	(2,1,1,1,1,na)
loading bales	CH	unit	6.25E-03	1	1.07	(2,1,1,1,1,na)
dried roughage store, non ventilated, operation	CH	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
Occupation, pasture and meadow, extensive	resource/land	m2a	3.70E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from pasture and meadow, extensive	resource/land	m2	7.40E-02	1	1.21	(2,1,1,1,1,na)
Transformation, to pasture and meadow, extensive	resource/land	m2	7.40E-02	1	1.21	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.65E+00	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.89E+01	1	1.07	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	2.36E-03	1	1.50	(2,1,1,1,1,na)
Phosphorus	water/river	kg	7.29E-05	1	1.50	(2,1,1,1,1,na)
Phosphate	water/river	kg	1.53E-04	1	1.50	(2,1,1,1,1,na)
Phosphate	water/ground-	kg	6.12E-05	1	1.50	(2,1,1,1,1,na)
Nitrogen oxides	air/low population density	kg	2.36E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	1.13E-04	1	1.41	(2,2,1,1,1,na)
hay extensive, at farm	CH	kg	1.00E+00			

Unit process inventory for: hay intensive organic, at farm, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
fertilising, by broadcaster	CH	ha	2.33E-05	1	1.07	(2,1,1,1,1,na)
slurry spreading, by vacuum tanker	CH	m3	6.90E-03	1	1.07	(2,1,1,1,1,na)
mowing, by rotary mower	CH	ha	5.07E-04	1	1.07	(2,1,1,1,1,na)
swath, by rotary windrower	CH	ha	5.07E-04	1	1.07	(2,1,1,1,1,na)
haying, by rotary tedder	CH	ha	1.27E-03	1	1.07	(2,1,1,1,1,na)
fodder loading, by self-loading trailer	CH	m3	2.18E-02	1	1.07	(2,1,1,1,1,na)
dried roughage store, cold-air dried, conventional, operation	CH	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
Occupation, pasture and meadow, intensive	resource/land	m2a	1.01E+00	1	1.11	(2,1,1,1,1,na)
Transformation, from pasture and meadow, intensive	resource/land	m2	5.07E-02	1	1.21	(2,1,1,1,1,na)
Transformation, to pasture and meadow, intensive	resource/land	m2	5.07E-02	1	1.21	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.65E+00	1	1.07	(2,1,1,1,1,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.79E+01	1	1.07	(2,1,1,1,1,na)
Phosphorus	water/river	kg	2.00E-05	1	1.50	(2,1,1,1,1,na)
Phosphate	water/river	kg	9.98E-05	1	1.50	(2,1,1,1,1,na)
Phosphate	water/ground-	kg	2.34E-05	1	1.50	(2,1,1,1,1,na)
Ammonia	air/low population density	kg	5.72E-03	1	1.21	(2,1,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.77E-05	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.22E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	7.28E-09	1	1.51	(2,2,1,1,1,na)
Chromium	soil/agricultural	kg	-1.46E-06	1	1.51	(2,2,1,1,1,na)
Copper	soil/agricultural	kg	2.04E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	4.80E-07	1	1.51	(2,2,1,1,1,na)
Mercury	soil/agricultural	kg	-2.96E-08	1	1.51	(2,2,1,1,1,na)
Nickel	soil/agricultural	kg	-2.04E-07	1	1.51	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	1.15E-05	1	1.51	(2,2,1,1,1,na)
Cadmium, ion	water/river	kg	3.63E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/river	kg	4.11E-07	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/river	kg	3.77E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/river	kg	1.99E-07	1	1.80	(2,2,1,1,1,na)
Mercury	water/river	kg	1.78E-09	1	1.80	(2,2,1,1,1,na)
Nickel, ion	water/river	kg	3.35E-07	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/river	kg	1.15E-06	1	1.80	(2,2,1,1,1,na)
Cadmium, ion	water/ground-	kg	2.83E-09	1	1.80	(2,2,1,1,1,na)
Chromium, ion	water/ground-	kg	1.75E-06	1	1.80	(2,2,1,1,1,na)
Copper, ion	water/ground-	kg	3.58E-07	1	1.80	(2,2,1,1,1,na)
Lead	water/ground-	kg	2.34E-08	1	1.80	(2,2,1,1,1,na)
Mercury	water/ground-	kg	1.10E-09	1	1.80	(2,2,1,1,1,na)
Zinc, ion	water/ground-	kg	2.84E-06	1	1.80	(2,2,1,1,1,na)
hay intensive organic, at farm	CH	kg	1.00E+00			

Appendix A17 to Chapter 17 (Starch)

Unit-Process Inventories from Chapter 17 (Starch)

Tab. A. 22 Unit-process inventories for starch (ecoinvent data v.1.01).

Unit process inventory for: potato starch, at plant, DE						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
potatoes IP, at farm	CH	kg	3.92E+00	1	1.09	(2,2,1,1,1,3)
tap water, at user	RER	kg	3.73E+00	1	1.09	(2,2,1,1,1,3)
electricity, medium voltage, at grid	DE	kWh	1.68E-01	1	1.09	(2,2,1,1,1,3)
heat, natural gas, at industrial furnace >100kW	RER	MJ	1.62E+00	1	1.09	(2,2,1,1,1,3)
transport, lorry >16t, fleet average	RER	tkm	9.80E-02	1	2.09	(4,5,na,na,na,na)
chemical plant, organics (l)	RER	unit	4.00E-10	1	3.01	(2,2,1,1,1,3)
treatment, potato starch production effluent, to wastewater treatment, class 2	CH	m3	5.81E-03	1	1.09	(2,2,1,1,1,3)
Heat, waste	air/high population density	MJ	6.03E-01	1	1.09	(2,2,1,1,1,3)
potato starch, at plant	DE	kg	1.00E+00			

Unit process inventory for: maize starch, at plant, DE						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
grain maize IP, at farm	CH	kg	1.26E+00	1	1.09	(2,2,1,1,1,3)
tap water, at user	RER	kg	1.96E+00	1	1.09	(2,2,1,1,1,3)
electricity, medium voltage, at grid	DE	kWh	2.54E-01	1	1.09	(2,2,1,1,1,3)
heat, natural gas, at industrial furnace >100kW	RER	MJ	3.99E+00	1	1.09	(2,2,1,1,1,3)
transport, lorry >16t, fleet average	RER	tkm	1.26E-01	1	2.09	(4,5,na,na,na,na)
chemical plant, organics (l)	RER	unit	4.00E-10	1	3.01	(2,2,1,1,1,3)
treatment, maize starch production effluent, to wastewater treatment, class 2	CH	m3	6.91E-04	1	1.09	(2,2,1,1,1,3)
Heat, waste	air/high population density	MJ	9.16E-01	1	1.09	(2,2,1,1,1,3)
maize starch, at plant	DE	kg	1.00E+00			

Appendix A18 to Chapter 18 (Tallow)

Unit-Process Inventory from Chapter 18 (Tallow)

(Last Changes 2004)

Tab. A. 23 Unit-process inventory for tallow (ecoinvent data v.1.01).

Unit process inventory for: tallow, at plant, CH						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
heat, natural gas, at industrial furnace >100kW	RER	MJ	8.39E+00	1	1.13	(3,2,1,1,1,3)
electricity, low voltage, at grid	CH	kWh	1.75E-01	1	1.13	(3,2,1,1,1,3)
treatment, sewage, to wastewater treatment, class 2	CH	m3	7.30E-03	1	1.13	(3,2,1,1,1,3)
Heat, waste	air/high population density	MJ	6.31E-01	1	1.13	(3,2,1,1,1,3)
tap water, at user	RER	kg	5.34E+00	1	1.13	(3,2,1,1,1,3)
transport, lorry 28t	CH	tkm	1.50E-01	1	2.02	(3,2,1,1,1,3)
building, multi-storey (l)	RER	m3	1.90E-05	1	1.13	(3,2,1,1,1,3)
Occupation, industrial area, built up	resource/land	m2a	3.49E-04	1	2.02	(3,2,1,1,1,3)
Occupation, construction site	resource/land	m2a	1.40E-05	1	1.52	(3,2,1,1,1,3)
Transformation, from unknown	resource/land	m2	6.98E-06	1	2.02	(3,2,1,1,1,3)
Transformation, to industrial area, built up	resource/land	m2	6.98E-06	1	2.02	(3,2,1,1,1,3)
tallow, at plant	CH	kg	1.00E+00			

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Life Cycle Inventories of U.S. Agricultural Production Systems

Data v2.0 (2007)



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Life Cycle Inventories of U.S. Agricultural Production Systems

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Summary

The relevant input data for five arable crops (namely wheat, rape seed, potato, rice, cotton) and for wool and sheep production are included. Products at farm level are included for arable crops and for wool and sheep.

The datasets refer to the USA. In case where the original sources given by NREL were used, these data were updated in all cases where more recent data were available. The datasets described in this report are thus not merely implementations of the original NREL datasets, but are new datasets that have been substantially remodelled, extended and updated, by using NREL as one source among others.

The relevant input data for modelling agricultural systems are included for arable crops and for sheep production.

Data for fertiliser types and amounts are partly based on assumptions. They are derived from national statistics and are representative for the USA. Pesticide usage is derived from detailed national statistics and is representative for the USA. Data for field operations and machine usage are derived from up to date sources from single states and extrapolated to the USA. Seed quantity and grain drying data are based on qualified estimates derived from single sources. In agriculture land use strongly depends on the data quality of the yield. The yields for the five crops are taken from the national database for agriculture. The average yield of the last six years is used in order to consider fluctuations of yields due to losses. Transport processes mostly base on estimates by reason of lack of data.

Direct field emissions are derived from national sources if possible. Direct field emissions of N_2O , heavy metals and the tractor exhaust gases NMVOC, NO_x and CO have been calculated using emission models.

ecoinvent data provides datasets for calculating LCAs in some of the most important production branches in U.S. conditions.

1 Crop Production in the USA

This report describes the inventories for arable crop production (for human food and animal feed). All these inventories refer to USA.

1.1 Characteristics

Of the total U.S. area excluding the outbound States Alaska and Hawaii (9.63 million km²) 4.09 million km² are used for agricultural production (42.2% of the total area, FAO 2006). Permanent pastures comprise 57.1% and arable land 42.9% of the area used for agricultural production. In 2005 only 8900 km² of agricultural land (0.22%) was cultivated according to the rules of organic farming. Tab. 1.1 lists the area of the most important crops.

Tab. 1.1 : Agricultural land usage in the year 2003 (FAO 2006)

Area 2003 (FAO 2006) in 1000 ha	USA
Total area	962909
Total agricultural land	409300
Total arable land (can potentially be used for arable crops)	175500
Total permanent pastures	233800
Total harvested area	99079
Soybean	29932
Maize	29798
Wheat	20233
Cotton	5286
Rice	1345
Potatoes	472
Rape and mustard seeds	366
Other crops	11648

There are about 2,101,000 farms in the USA. Their average size is 180 ha (USDA 2004a).

Agricultural workers held about 834,000 jobs in 2004. Of these, farmworkers were the most numerous, holding about 690,000 jobs. Graders and sorters held about 45,000 jobs, agricultural inspectors 14,000 jobs, agricultural equipment operators 60,000 jobs, and animal breeders 12,000 jobs. More than 66 % of all agricultural workers were engaged in crop and livestock production, while more than 5 % worked for agricultural service providers, mostly farm labor contractors (BLS 2006).

Tab. 1.2 lists the major field crops with the value of production in the year 1997. **Fehler! Verweisquelle konnte nicht gefunden werden.** gives an overview of the climate regions of the USA.

Tab. 1.2: Major Crops in the USA (NASS 2006)

Major Crops in the U.S.A. in 1997 (in US\$ billions)	
Corn	24.4
Soybean	17.7
Wheat	8.6
Alfalfa	8.3
Cotton	6.1
Hay, other than alfalfa	5.1
Tobacco	3.0
Rice	1.7
Sorghum	1.4
Barley	0.9

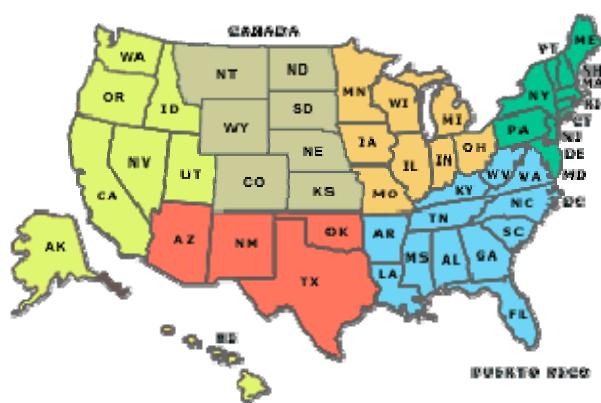


Fig. 1.1: Climate zones of the USA

USA can be divided into six climate regions, excluding Alaska, Hawaii and outlying territories. The climate varies considerably between different regions.

- Northwest Pacific
- Mid/South Pacific
- Midwest
- Northeast
- Southeast
- Southwest

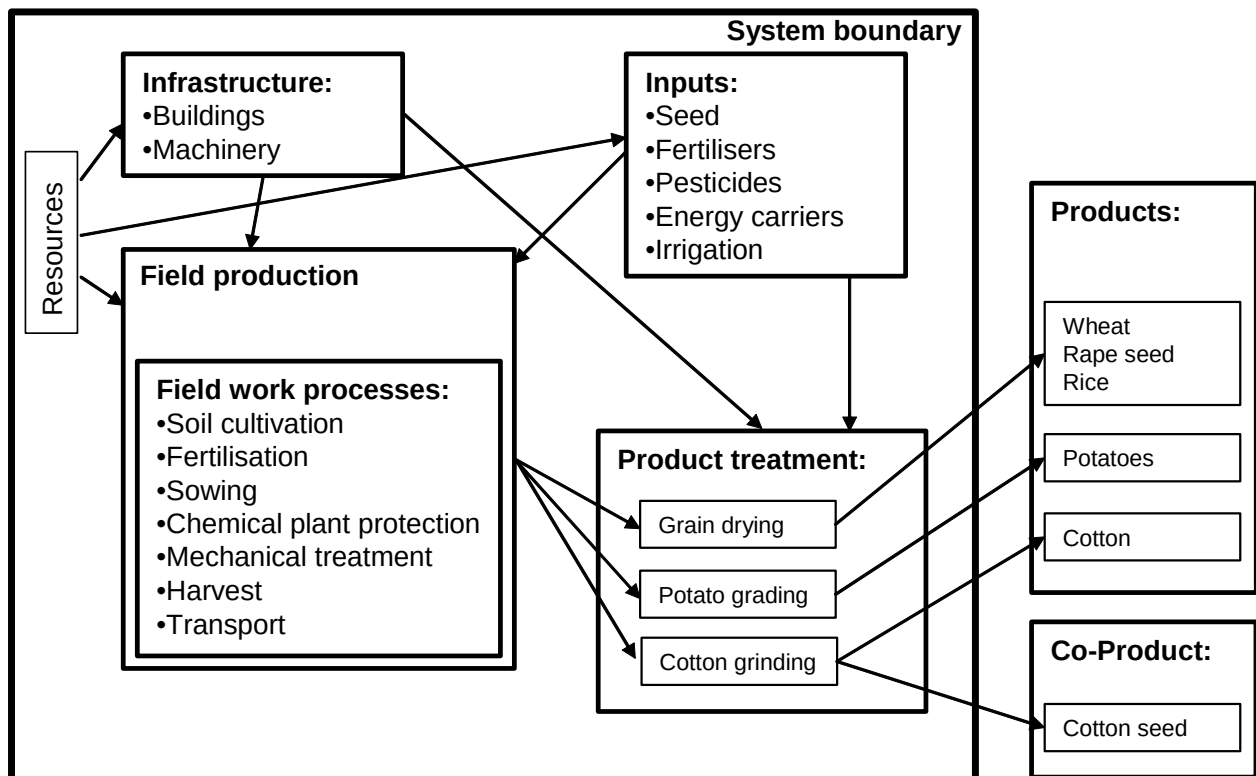
Due to its large size and wide range of geographic features, a wide variety of climates can be encountered in the United States. The climate is temperate in most areas, tropical in southern Florida, semi-arid in the Great Plains west of the 100th meridian, Mediterranean in coastal California and arid in the Great Basin. The climate is comparatively generous with infrequent severe drought in the major agricultural regions, a general lack of widespread flooding, and a mainly temperate climate that receives adequate precipitation.

1.2 Arable Crop Production in the USA

1.2.1 Overview of the Life Cycle Inventories

Fig. 1.2 gives an overview of the processes included in the agricultural system. The reference function of all arable crop inventories is 1 kg fresh matter (after drying, if this is necessary). Tab. 1.3 summarises the inventories described in this chapter and their main characteristics. The system includes the processes including consumption of raw materials, energy, infrastructure and land use as well as the emissions to air, water, and soil. It also comprises transportation of the raw materials, storage and transportation of the final product. The emissions into water are assumed to occur into ground water and rivers.

Fig. 1.2: Overview of the agricultural system



Tab. 1.3: Overview of the main characteristics of the inventories

	Yield main product (kg/ha)	Moisture at harvest (%)	Moisture at storage (%)	Co-product (kg/ha)	Crop residues (kg/ha)	Kg seed/ha	Kg N/ha (available N)	Kg P ₂ O ₅ /ha	Kg K ₂ O/ha	Number of pesticide applications
Cotton	775	6%	6%	1144	2338	14.6	104	56	89	13.9
Potatoes	41001	78%	78%	-	16427	2354	230	177	169	4.0
Rapeseeds	1182	12%	6%	-	6331	5.6	112	22	34	4.0
Rice	6856	21%	13%	-	9096	141.2	141	30	26	6.5
Wheat	2253	15%	15%	-	2927	94.2	76	42	53	1.0

There is no single source that could be used to model all aspects of crop and wool production that would be representative of USA production as a whole. The NREL database (NREL 2006) was used as a baseline to collect further references. Tab. 1.4 summarises the data sources used to define the inventories. Detailed information is given in the following sections.

Tab. 1.4: Data sources to compile the arable crop production inventories

Category of Data	Data source(s)	Chapter
Yields of main products and co-products	NASS 2006, NREL 2006, Anthony & McAlister (2005), Nemecek <i>et al.</i> (2004), Thompson (1998)	1.2.2
Allocations	NASS 2006	1.2.3
Fertilisers	ARMS 2006, NREL 2006, IFA 2006, Livezey & Foremann 2004	1.2.4
Irrigation	USDA 2001 & 2004a, ARMS 2006, Hutson <i>et al.</i> 2004, NASS 1998, NREL 2006, Nemecek <i>et al.</i> 2004, UofA 2007	1.2.5
Pesticides	USDA 2004b, NCFAP 1997	1.2.7
Machine usage	NREL 2006, Nemecek <i>et al.</i> 2004, Jungbluth <i>et al.</i> 2007, KTBL 2004, Hogan <i>et al.</i> 2005 & 2006, Watkins <i>et al.</i> 2006, Epplin <i>et al.</i> 2005, Patterson 2002	1.2.6
Seeds	Brooks 2001, Mir 2002, Berglund 2002, Mohinder 2002, Nemecek <i>et al.</i> 2004	1.2.8
Transports	NREL 2006, Nemecek <i>et al.</i> 2004, Spielmann <i>et al.</i> 2007	1.2.9
Drying	Nemecek <i>et al.</i> 2004, Anthony & McAlister 2005, Thompson 1998	1.2.11
Land use	Nemecek <i>et al.</i> 2004	1.2.12
CO ₂ -uptake and biomass energy	Nemecek <i>et al.</i> 2004, Qiuxia <i>et al.</i> 2005, TU-Wien 2007	1.2.13
Direct emissions	Nemecek <i>et al.</i> 2004, Goolsby <i>et al.</i> 1999, Bennett <i>et al.</i> 1999, Teherani 1987, Walther <i>et al.</i> 2001, Freiermuth <i>et al.</i> 2006, Swain <i>et al.</i> 2005, Boquet & Breitenbeck 2000, Prasuhn 2006, Wolfensberger & Dinkel 1997	1.2.14, 1.2.15, 1.2.16, 1.2.17

1.2.2 Yields and Co-products/Crop Residues

The crops under investigation produce main products, co-products (straw, leaves or seeds in the case of cotton) and crop residues. NASS (2006) reports data for the production of field crops (see Tab. 1.5). The area of rapeseed is significantly lower than the area reported in Tab. 1.1, since the latter figure contains also mustard, which is grown on a much larger area than rapeseed. The quantities of straw and crop residues (left on the field) were taken from NREL (2006). It is assumed, that none of the co-products are harvested except for cotton, where cottonseed is generated as a co-product. According to NASS (2006), 1144 kg cotton seed are harvested beside the 775 kg harvested cotton fibres. The data on moisture at harvest and at storage are taken from Anthony & McAlister (2005) for cotton, from Nemecek *et al.* (2004) for potato, rapeseed and wheat and from Thompson (1998) for rice. Not all the area planted is also harvested due to droughts, storms, pests or similar events. The variation of the average yield per ha and year is up to 15% by cotton, 5% by potato, 25% by rapeseed, 3% by rice, 14% by wheat.

Part of the nutrients in the fertilisers is exported in the products and another part remains in the crop residues and is restored to the soil. These nutrients are fully or partly available to the following crop and therefore the fertiliser rate applied to the following crop can be reduced. In this way the crop receives nutrients from the preceding crop, which reduces its fertiliser requirements (see Nemecek *et al.* 2004, fertiliser rate applied by the farmer). Since we are modelling individual crops and not the whole crop rotation, the fertiliser effect of crop residues would normally be attributed to the crop causing it for the calculation of the LCI. In these inventories we use national statistics of the total amount of used industrial fertilisers. Therefore, we need not to incorporate the fertilizer effects of crop residues.

Tab. 1.5: Field crop production in the USA (NASS 2006), averages for the years 2001-2006

	Cotton	Potato	Rapeseed	Rice	Wheat
area planted (ha)	5826373	489413	1410	1292633	24020653
area harvested (%)	90%	98%	87%	99%	83%
production (t)	4516365	20066292	1667	8862292	54128897
yield of products (kg/ha planted)	fibres: 775 seed: 1144	41001	1182	6856	2253
Moisture at harvest (%)	6%	78%	12%	21%	15%
Moisture at storage (%)	6%	78%	6%	13%	15%
Crop residues (kg/ha planted)	2338	16427	6331	9096	2927

1.2.3 Allocations

Allocation is required only for the co-products cotton fibres and cottonseed. According to NASS (2006), 87% of the total revenue per hectare is made by cotton fibres and 13% by cottonseed. Economic allocation is applied between cotton fibres and cottonseed for all inputs and outputs with the following exceptions: For transportation of cotton fibres and seed from field to farm and the cotton ginning process mass allocation is used. Mass allocation is also used for heavy metal contents in cotton because for rawcotton the heavy metal contents for average biomass provided by the SALCA-heavy metal tool are used due to missing data. Rawcotton contains 40% cotton fibres and 60% cotton seed by weight. For energy in biomass and CO₂ uptake the corresponding amount is calculated.

1.2.4 Fertilisers

Quantities for the crop-specific fertiliser use in the USA are given in Tab. 1.6. In this study values for the amounts of used fertilisers are taken from ARMS (2006) for cotton and wheat, NREL (2006) for potato and rapeseed and Livezey & Foremann (2004) for rice. It was not possible to build average fer-

tiliser use from considering the last 6 years as we did for production because of poor data availability. Yield losses occur mostly in the period after fertilising. Therefore we assumed, that the fertiliser amount per ha is every year the same. Data for the products that are used as fertilisers are taken from IFA (2006). Tab. 1.7 lists the share of fertiliser type in the total amount of nutrient application calculated on the basis of the N content. Tab. 1.8 shows the use of fertilisers per hectare in this study.

Tab. 1.6: Fertiliser use for field crops in the USA (ARMS 2006, Livezey & Foremann 2004, NREL 2006)

Fertiliser [kg/ha/y]	Cotton	Potato	Rapeseed	Rice	Wheat
N	104	230	112	141	76
P ₂ O ₅	56	177	22	30	42
K ₂ O	89	169	34	26	53

Tab. 1.7: Fertiliser use by product type in the USA (IFA 2006). The share is referring to the main nutrient content.

Nutrient	Product	%
N	Anhydrous ammonia	50
	Urea	21
	Ammonium nitrate	29
P ₂ O ₅	DAP	100
K ₂ O	Potassium chloride	100

Tab. 1.8: Calculated use of fertilisers in this study

Product	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha	kg/ha	kg/ha	kg/ha	kg/ha
ammonia, liquid, at plant as NH ₃	49.8	97.5	62.6	78.5	36.4
urea as N	17.2	33.7	21.7	27.1	12.6
ammonium nitrate as N	23.8	46.6	29.9	37.5	17.4
diammonium phosphate as N	21.9	69.4	8.8	11.7	16.3
diammonium phosphate as P ₂ O ₅	55.9	177.4	22.5	30.0	41.7
potassium chloride as K ₂ O	89.3	169.3	33.6	26.0	53.4

1.2.5 Irrigation

In some regions of the USA, rainfall alone does not provide agricultural surfaces with sufficient water for plant cultivation and crops must be supplied with additional water by irrigation. Irrigated agriculture is the economic sector with the highest consumption of freshwater (other sectors being thermoelectric, domestic&commercial, other industry). With a consumptive use of freshwater of 8.7E+06 acre-feet (=1.07E+11 m³), irrigated agriculture accounts for almost 80% of the national total (USDA 2001). About 13% of agricultural land is irrigated (see also Tab. 1.9). Of the 22 million irrigated hectares, 49% are served by sprinkler systems whereas 42% are irrigated by gravity flow systems. Half of the used water irrigating 58% of all irrigated hectares is taken from wells. Of all irrigation pumps, 71% are pumps for wells. The average pumping depth is 48 m (157 feet). According to USDA (2004a), 64% of

the pumps are powered by electricity and 23% by diesel fuel. Data for irrigation infrastructure is taken from Nemecek *et al.* (2004).

Tab. 1.9: Statistical data on irrigation in the USA (USDA 2004a)

Area	Unit		
total arable land	ha	175500000	100%
irrigated land	ha	21642586	12% of total arable land
Irrigation system			% of total ha irrigated
sprinkler systems	ha	10901354	50%
gravity flow systems	ha	9358003	43%
drip or trickle systems	ha	1209241	6%
subirrigation	ha	113119	1%
Water source			% of total ha irrigated
from wells	ha	13071345	60%
from on farm surface source	ha	2946111	14%
from off-farm water suppliers	ha	5625130	26%
from wells	Mm ³	53656	50%
from on farm surface source	Mm ³	14555	14%
from off-farm water suppliers	Mm ³	38978	36%
Wells			
wells used	number	401193	100%
pumped wells	number	375851	94%
average pumping depth	m	48	
Pumps			% of total powered pumps
pumps powered	number	529393	100%
pumps for wells	number	375851	71%
pumps for tailwater pits	number	21618	4%
pumps for ponds, lakes, rivers	number	97378	18%
relift pumps	number	34546	7%
Pumps powered by			
electricity	amount	319102	64.1%
natural gas	amount	41771	8.4%
LP gas	amount	17792	3.6%
Diesel fuel	amount	112600	22.6%
gasoline	amount	6176	1.2%
solar & renewable energy	amount	411	0.1%

Since almost 2/3 of all pumps are electric powered, only electric powered pumps were considered. The energy use of water pumped from 150 feet depth powered by electricity is 0.24 kWh/m³ (see Tab. 1.10) according to UofA (2007).

Tab. 1.10: Typical energy use of different pump systems for the average pumping depth of 48m (UofA 2007). 1 acre-inch = 102.79 m³. The energy values are valid for each of the different systems.

energy use of different pump systems for average pumping depth = 48 m	Unit	per acre-inch	per m ³
electric	kWh	24.6	0.239
diesel	l	6.1	0.059
LP gas	l	11.4	0.110
gasoline	l	9.1	0.088
natural gas	m ³	8.5	0.083

The water use for cotton and wheat is calculated with the irrigated area and the water use per irrigated area (ARMS 2006). Data for the total water use of potato and rice production per year and the total cultivated area stem from Hutson *et al.* (2004) and NASS (1998). NREL (2006) reports no irrigation for rapeseed and no other data could be found. Tab. 1.11 shows the average use of water per crop and hectare for the five crops considering the planted area.

Tab. 1.11: Average amount of water used per planted area for the irrigation of the different crops (m³/ha/year).

Crop	Cotton	Potato	Rapeseed	Rice	Wheat
Source	ARMS 2006	Hutson <i>et al.</i> 2004, NASS 1998	NREL 2006	Hutson <i>et al.</i> 2004, NASS 1998	ARMS 2006
Water use (m ³ /ha/year)	1306	4615	0	7393	163

1.2.6 Machine Usage

NREL (2006) reports data for the diesel use in the cultivation of the field crops in the USA. These data are based on statistics and are representative for the harvested area. Jungbluth *et al.* (2007) extrapolated the usage of machines from the total diesel used and the proportion of the diesel used during different field operations in the cultivation in Germany (KTBL 2004) for corn. These proportions of the diesel consumptions are taken to calculate the machine usage based on the diesel consumptions of the several field works, which are given in Nemecek *et al.* (2004). There is a certain uncertainty because farm management in the USA is not truly comparable with that of Germany.

Therefore another method is used here to calculate the machine usage: The division of agriculture of the University of Arkansas (UofA) provides detailed data for field operations of rice production (Clearfield grade on silt loam soils) in Arkansas (Watkins *et al.* 2006), where 39% of the total U.S. rice is produced. There are also data available by Hogan *et al.* (2005) for field operations of cotton production (Roundup ready grade with conventional till in the Southeast/Central Boll Weevil Eradication Zone) in Arkansas (10% of nationwide production) and by Hogan *et al.* (2006) for field operations of wheat production (wheat following crops other than rice on sand/silt loam soils) in Arkansas (about 10% of nationwide production). Data for field operations of potato production in Idaho (accounting for 30% of U.S. production) is provided by Patterson (2002). Epplin *et al.* (2005) list field operations of rape seed in Oklahoma. Although only 6% of the nationwide canola production is located in Oklahoma, no better data could be found. Tab. 1.12 lists the field operations of the field crops. The energy use and emissions of the machine usage was calculated according to Nemecek *et al.* (2004).

Most of the field operations are performed on the entire planted area, but not the whole planted area is harvested due to losses. Losses most often occur after sowing, tillaging, fertilising and application of pesticides. Harvesting and pesticides applications shortly before harvesting cotton were considered to be performed only on the harvested area.

The process of ginning in cotton was approximated by “grain drying, high temperature” by means of an equal energy demand.

Tab. 1.12: Field operations for the cultivation of one hectare (number of trips per season, where not stated otherwise)

Field operation	Cotton	Potato	Rapeseed	Rice	Wheat
mulching	1	0	0	0	0
sowing	1	0	1	1.1	1
potato planting	0	1	0	0	0
potato haulm cutting	0	1	0	0	0
tillage, cultivating, chiselling	1	1	0.8	1	0
tillage, harrowing, by rotary harrow	1	0	0	0	0
tillage, harrowing, by spring tine harrow	4	2	4	5	2
tillage, hoeing and earthing-up, potatoes	0	2	0	0	0
tillage, ploughing	0	1	0.2	0.2	0.1
tillage, rolling	0	0	0	1	1
application of plant protection products, by field sprayer	13.9	4	4	6.5	1
fertilising, by broadcaster	3	2	3	3	3
combine harvesting	0.90	0	0.87	0.99	0.83
harvesting, by complete harvester, potatoes	0	0.98	0	0	0
potato grading (kg/ha)	0	41001	0	0	0
baling (number of bales/ha)	12.0	0	0	0	0
grain drying, low temperature (kg water/kg yield)	0	0	0.0682	0.1299	0
ginning, approximated by grain drying, high temperature (kg water/kg cotton yield)	0.092	0	0	0	0

1.2.7 Pesticides

USDA (2004b) reports amounts of pesticide used in cotton, potato, winter wheat cultivation (60% of total wheat production) and rice cultivation in the USA. Rape seed (Canola) is mainly produced in North Dakota. Data were available for the year 1997 (NCFAP 1997), based on official statistics of the USA (Tab. 1.13). The very high use of pesticides in potatoes is explained by two substances: metam sodium is used as a soil fumigant before planting potatoes to free the soil of nematodes, fungi, weed, etc. Sulfuric acid is used before harvest to desiccate the vines of potatoes in order to make harvest easier.

In the wheat and rape seed cultivation, 0.26 kg/ha and 0.64 kg/ha of pesticides are used, respectively. The average pesticide use in cotton and rice cultivation is 3.7 kg/ha and 3.85 kg/ha, respectively. Potato cultivation in the USA has by far the higher pesticide use with about 74 kg/ha active ingredients, which is made up mainly of production to desiccate the potato vines.

Tab. 1.13: Pesticides use in kg per hectare and year (NCFAP 1997)

	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y
Fungicides					
Azoxystrobin	1.43E-03	8.03E-02		2.43E-02	1.56E-04
Carboxin	6.69E-05				
Chlorothalonil		2.06E+00			
Copper hydroxide		6.99E-02			
Copper sulfate		2.01E-03		1.51E-01	
Cymoxanil		7.49E-03			
Dimethomorph		1.61E-03			
Etridiazole	3.21E-03				
Flutolanil		3.03E-02			
Iprodione	3.57E-04	5.43E-02			
Mancozeb		2.42E+00			
Maneb		3.93E-03			
Mefenoxam	1.25E-03	6.50E-02			
Metalaxyl		2.17E-02			
Metiram		1.15E-01			
PCNB	1.53E-02	1.67E-01			
Propamocarb hydroch.		1.34E-03			
Propiconazole				5.62E-03	8.92E-03
Pyraclostrobin					8.92E-04
Sulfur		4.66E-01			
Tebuconazole					3.57E-04
Trifloxystrobin					1.78E-04
Triphenyltin hydrox.		2.67E-02			

Tab. 1.13: Pesticides use in kg per hectare and year (NCFAP 1997) (contd.)

	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y
Herbicides					
2,4-D	2.75E-02	2.45E-04		1.26E-01	8.82E-02
Acetamide					7.58E-04
Acifluorfen				6.42E-03	
Atrazine					6.51E-03
Bensulfuron, methyl				8.48E-03	
Bentazon				2.73E-02	
Bromoxynil	1.52E-03				5.00E-03
Bromoxynil octanoate					2.14E-03
Carfentrazone-ethyl	1.87E-03				6.69E-05
Chlorsulfuron					7.14E-04
Clodinafop-propargil					3.57E-04
Clomazone	4.91E-04			1.28E-01	
Clethodim	2.90E-04	2.01E-04			
Cyanazine	1.38E-03				
Clopyralid			8.30E-03		8.03E-04
Dicamba	4.46E-04				6.42E-03
Diclofop-methyl					1.49E-03
Diquat		1.28E-01			
Diuron	1.22E-01				2.45E-03
DSMA	3.30E-03				
EPTC		6.65E-01			
Fenoxaprop				5.35E-04	1.78E-04
Flucarbazone-sodium					4.46E-05
Fluometuron	5.50E-02				
Fluroxypyr					8.03E-04
Fluroxypyr 1-methylh					2.23E-04
Glufosinate, ammonium		1.32E-02			
Glyphosate	1.06E+00	9.99E-03		9.74E-02	9.05E-02
Halosulfuron				2.41E-03	
Imazamox					2.68E-04
Lactofen	3.35E-04				
Linuron	3.75E-03	3.59E-02			
Metolachlor	1.66E-02	3.91E-02			
MCPA				3.39E-03	1.44E-02
Mesosulfuron-Methyl					2.23E-05
Metribuzin		2.65E-01			4.10E-03
Metsulfuron-Methyl			1.52E-05		2.68E-04
Molinate				7.22E-01	
MSMA	8.06E-02				
Nicosulfuron					
Norflurazon	7.58E-04				
Paraquat	4.42E-02	8.56E-03		1.23E-03	
Pendimethalin	1.30E-01	1.70E-01		6.64E-02	

Tab. 1.13: Pesticides use in kg per hectare and year (NCFAP 1997) (contd.)

	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y
Herbicides					
Picloram					8.92E-05
Prometryn	8.05E-02				
Propanil				1.87E+00	
Prosulfuron					4.46E-05
Quinclorac				3.12E-02	
Rimsulfuron		4.10E-03			
Pyrithiobac, sodium	8.56E-03				
S, Metolachlor	1.82E-02	1.47E-01			
Sethoxydim		3.57E-03	9.01E-02		
Sulfosulfuron					1.07E-03
Thiobencarb				4.00E-01	
Thifensulfuron	4.46E-04				7.23E-04
Tralkoxydim					3.12E-04
Trifluralin	2.89E-01	2.01E-02	2.02E-01		
Triallate					2.41E-03
Triasulfuron					7.14E-04
Triclopyr				4.34E-02	
Tribenuron-Methyl					3.57E-04

Tab. 1.13: Pesticides use in kg per hectare and year (NCFAP 1997) (contd.)

	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y
Insecticides					
Abamectin	1.87E-04				
Acephate	1.79E-01				
Aldicarb	1.38E-01	1.70E-01			
Azinphos, methyl		1.62E-02			
Bifenthrin	1.56E-04				
Buprofezin	7.14E-04				
Carbaryl		2.50E-03	3.84E-03	9.01E-03	
Carbofuran	1.43E-03	2.78E-02	9.81E-03	8.21E-03	
Chlorpyrifos	1.71E-02				9.64E-03
Cyfluthrin	4.82E-03	1.11E-02			
Cypermethrin	5.71E-03				
Zeta-Cypermethrin	3.12E-03				2.68E-04
Deltamethrin	2.68E-04				
Diazinon		6.16E-02			
Dicofol	2.16E-03				
Dicrotophos	5.89E-02				
Dimethoate	7.58E-04	4.26E-02			6.91E-04
Disulfoton	1.34E-03				
Endosulfan	6.78E-03	5.57E-02			
Esfenvalerate	2.68E-04	9.10E-03			
Ethoprop		1.54E-01			
Fenpropathrin	7.14E-04				
Imidacloprid	1.34E-03	4.39E-02	1.78E-05		
Indoxacarb	2.14E-03				
Lambda-cyhalothrin	2.94E-03		6.51E-04	3.48E-03	1.78E-04
Malathion	4.17E-01	2.43E-03	8.92E-04	1.27E-03	
Methamidophos	2.68E-04				
Methomyl	8.03E-04				
Naled	1.96E-03				
Oxamyl	8.56E-03	1.29E-01			
Parathion (ethyl, methyl)	1.11E-02		5.09E-03	8.91E-02	5.62E-03
Permethrin	1.56E-04	5.35E-03			
Petroleum distillate	1.45E-03				
Phorate	2.32E-02	2.79E-01			
Phosmet		3.07E-02			
Piperonyl butoxide		9.14E-04			
Profenofos	1.23E-02				
Propargite	3.14E-03	8.94E-02			
Pymetrozine		8.56E-03			
Pyriproxyfen	1.12E-04				
Spinosad	1.16E-03	1.78E-04			
Tebufozide	3.35E-04				
Thiamethoxam	3.12E-03	7.85E-03			
Tralomethrin	2.68E-04				
Trichlorfon		1.34E-04			

Tab. 1.13: Pesticides use in kg per hectare and year (NCFAP 1997) (contd.)

	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y
Other Pesticides					
Arsenic acid	2.19E-03				
Busan 881		2.60E+00			
Cyclanilide	1.28E-02				
Dichloropropene		4.16E+00			
Dimethipin	8.92E-04				
Endothall	1.78E-04	5.13E-04	8.92E-03		
Ethalfuralin			2.90E-01		
Ethephon	4.91E-01				
Gossypure	1.34E-05				
Harpin protein		1.34E-05			
Hexadecadien (Z, Z)	1.56E-05				
Maleic hydrazide		1.69E-01			
Mepiquat chloride	2.73E-02				
Metam, sodium		2.75E+01			
Quizalofob			2.00E-02		
Sodium chlorate	9.37E-02			2.41E-02	
Sulfuric acid		3.15E+01			
Thidiazuron	2.28E-02				
Tribufos	1.69E-01				
	Cotton	Potato	Rapeseed	Rice	Wheat
	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y	kg/ha/y
Total pesticides	3.70	74.08	0.64	3.85	0.26

1.2.8 Seed Inputs

Brooks (2001) reports 14.6 kg seeds per hectare for cotton and Mir (2002) reports 94.2 kg for wheat. Berglund (2002) reports 5.6 kg seeds per hectare for rapeseed. According to Mohinder (1992), 141.2 kg rice seeds per hectare are needed. There is no information available on the amount of potato tubers. Therefore, the amount of tuber used per hectare is taken from Nemecek *et al.* (2004).

Tab. 1.14: Amount of seeds used in this inventory

	Cotton	Potato*	Rapeseed	Rice	Wheat
Amount of seeds (kg/ha)	14.6	2354	5.6	141.2	94.2

* tubers

1.2.9 Seed Production

The use of seeds is calculated for potato, rape seed and wheat with the datasets from Nemecek *et al.* (2004) assuming that the seed processing in Europe is the same than in the USA. For cotton and rice seed new inventories were calculated based on:

- the impacts of 1kg cotton or rice of the inventories "cotton seed, at farm" and "rice, at farm",
- the seed processing of wheat and rape seed inventory assuming that the seed processing of cotton is similar to rapeseed and the seed processing of rice is similar to wheat,
- the assumption of 50 km transportation distance from farm to seed processing and 150 km from

seed processing to regional storehouse.

1.2.10 Transports

According to NREL (2006) the transport distance of raw materials to the farm is 400 km on railway and 200 km on road but no source is reported. The ecoinvent inventories of pesticides, seeds and fertilisers already include the transports to the regional storehouse. Therefore an own estimation was made for the transport distance from the regional storehouse to the farm (50 km) assuming that the fertiliser modules RER are applicable for the USA. Seed potatoes are produced in specialised regions and then sold to the commercial producers. In ecoinvent data V1.x transport of tuber was treated like this (Nemecek *et al.* 2004). For U.S. we made the same assumption. Tab. 1.15 shows supplementary transports considered for the cultivation of 1 ha using the transport dataset for lorry 40t described by Spielmann *et al.* (2007).

Tab. 1.15: Supplementary transports considered for 1 ha crop cultivation.

Material	Transport type		Cotton	Potato	Rapeseed	Rice	Wheat
From storehouse to farm		Distance (km)	tkm/ha	tkm/ha	tkm/ha	tkm/ha	tkm/ha
Pesticides	Lorry 40t	50	4.15E-01	8.30E+00	7.16E-02	4.35E-01	2.89E-02
Seeds	Lorry 40t	50	7.29E-01	1.18E+02	2.80E-01	7.06E+00	4.71E+00
Fertiliser	Lorry 40t	50	28.2E+00	65.8E+00	21.1 E+00	25.4E+00	19.8E+00
Total tkm	Lorry 40t		29.3E+00	191.8E+00	21.5E+00	32.9E+00	24.5E+00

1.2.11 Drying

Whereas cotton (6% moisture according to Anthony & McAlister 2005), potato and wheat (78% and 15% moisture respectively according to Nemecek *et al.* 2004) do not need any further drying (see Tab. 1.5), it is necessary to dry rape seed from 12% moisture at harvest to 6% moisture required for storage according to Nemecek *et al.* (2004) and rice from 21% to 13% according to Thompson (1998). The drying is modelled with the dataset “grain drying, high temperature” for rice and “grain drying, low temperature” for rape seed according to Nemecek *et al.* (2004). For cotton, the process drying is used to approximate ginning of cotton, based on the energy demand of the process.

1.2.12 Land Use

In Nemecek *et al.* (2004) land occupation was calculated from the duration of land use (taking the time from soil cultivation until harvest into account) and the yield per area unit. The land occupied was always considered as “Occupation, arable”. Land transformation was calculated on the basis of the area required to produce 1 kg yield. The type of use before establishment of the crop was assumed to be arable land, as the crops are typically in an arable crop rotation. In Tab. 1.16 the accounted amounts of land use are given. They are calculated with the yields listed in Tab. 1.5. The occupation is calculated as 12 months per year for all crops.

Tab. 1.16: Amounts of land transformation and occupation for the crop cultivation in the USA.

Land use	Cotton fibre	Cotton-seed	Potato	Rape-seed	Rice	Wheat
Transformation from arable [m ² per kg yield]	11.3	1.12	0.24	8.46	1.46	4.44
Transformation to arable [m ² per kg yield]	11.3	1.12	0.24	8.46	1.46	4.44
Occupation, arable, [m ² a per kg yield]	11.3	1.12	0.24	8.46	1.46	4.44

1.2.13 CO₂-uptake and Energy in Biomass

The uptake of CO₂ is calculated from the carbon balance. The biomass energy is calculated from the energy content of the main product. The carbon and energy content of the seeds is calculated as carbon and energy input respectively. Values for wheat, potato and rapeseed are taken from Nemecek *et al.* (2004). Qiuxia *et al.* (2005) report the carbon content of rice. Energy content of rice and cotton is reported by TU-Wien (2007). Tab. 1.17 lists biogenic CO₂ uptake and energy in biomass per hectare.

The allocation for cotton fibres and cottonseed is calculated on the basis of the amount of carbon respectively energy exported through the harvested products.

Tab. 1.17 Uptake of CO₂ and biomass energy contained in the harvested product and co-products per hectare

	Cotton	Allocation to cotton fibres	Allocation to cotton seed	Potato	Rape-seed	Rice	Wheat
CO ₂ , biogenic [kg/ha]	4197	27.4%	72.6%	13179	3163	10012	2845
Energy, biomass [MJ/ha]	39834	33.1%	66.9%	149556	29169	112041	33239

1.2.14 Heavy Metal Output

The heavy metal emissions were calculated by the SALCA heavy metal model (Swiss Agricultural Life Cycle Assessment-Heavy Metal). Introduction into farm land and discharge into surface water and groundwater are determined on the basis of heavy metal input from seed, fertilizers, plant protection and growth products and deposition. An allocation factor is used to distinguish between diffuse and agriculture-related introduction (Freiermuth 2006).

The uptake of heavy metals is given in Tab. 1.18. Bennett *et al.* (2000) and Teherani (1987) report heavy metal contents of rice. For potato, rapeseed, and wheat existing data from Nemecek *et al.* (2004) were used. No data on heavy metal uptake of cotton was available. Therefore, average contents in biomass reported in Freiermuth (2006) were used for raw cotton using mass allocation for cotton fibres and seeds.

Tab. 1.18: Heavy metal uptake of harvested products per kg dry matter

	Cotton	Potato	Rapeseed	Rice grains	Wheat grains
Source	Freiermuth 2004	Nemecek <i>et al.</i> (2004)	Nemecek <i>et al.</i> (2004)	Bennett <i>et al.</i> (2000) & for Nickel Teherani (1987)	Nemecek <i>et al.</i> (2004)
Cadmium (Cd, mg)	1.02E-01	4.00E-02	1.60E+00	2.10E-02	1.50E-01
Chromium (Cr, mg)	5.45E-01	5.70E-01	5.00E-01	4.90E-01	3.00E-01
Copper (Cu, mg)	6.63E+00	6.45E+00	3.30E+00	5.27E+00	5.00E+00
Lead (Pb, mg)	5.41E-01	5.50E-01	5.25E+00	9.60E-01	1.60E-01
Mercury (Hg, mg)	4.06E-02	9.00E-02	1.00E-01	3.50E-02	1.00E-02
Nickel (Ni, mg)	1.04E+00	3.30E-01	2.60E+00	9.65E-01	3.00E-01
Zinc (Zn, mg)	3.20E+01	1.50E+01	4.80E+01	4.39E+01	4.50E+01

1.2.15 Emissions to the Air

The ammonia (NH₃) emissions are calculated with the average emission factors for the several mineral fertilisers which are described in Goolsby *et al.* (1999).

The N₂O emissions are calculated with a direct emission factor of 1.25% of the N-input fertiliser, crop residue and an indirect emission factor of 2.5% from the N that is leached as nitrate (Nemecek *et al.* 2004). The N contents of crop residue are reported in Walther *et al.* (2001) for wheat, rapeseed and potato, in Swain *et al.* (2005) for rice and in Boquet & Breitenbeck (2000) for cotton.

The NO_x emissions are estimated from the emission of N₂O (Wolfensberger & Dinkel 1997):

$$\text{kg NO}_x = 0.21 * \text{kg N}_2\text{O}.$$

Tab. 2.17 shows direct emissions to air for the five crops.

Tab. 1.19: Emissions to air from crop cultivation

Emission	Cotton	Potato	Rapeseed	Rice	Wheat	Source
NH ₃ [kg/ha]	6.40	12.52	8.04	10.08	4.68	Goolsby <i>et al.</i> 1999
N ₂ O [kg/ha]	2.72	5.61	3.78	3.90	1.98	Nemecek <i>et al.</i> 2004
NO _x [kg/ha]	0.57	1.18	0.79	0.82	0.41	Nemecek <i>et al.</i> 2004
CO ₂ [kg/ha] from urea application	27.04	52.96	34.01	42.62	19.79	Nemecek <i>et al.</i> 2004

1.2.16 Emissions to the Water

Goolsby *et al.* (1999) report the phosphorus amount washed into water. The emission factor for P-emissions to ground water is taken from Prasuhn (2006).

Nitrate leaching is calculated with an emission factor of 61 % of the total N-loss to water according to Goolsby *et al.* (1999).

Heavy metal emissions to water were calculated by the SALCA-h tool (Freiermuth 2006).

Tab. 2.18 lists all direct emissions to water for the five crops.

Tab. 1.20 Direct emissions to water per hectare and year.

Emission	Cotton	Potato	Rapeseed	Rice	Wheat	Source
Phosphorus as P [kg/ha]	1.94E+00	6.16E+00	7.80E-01	1.04E+00	1.45E+00	Goosby <i>et al.</i> 1999, Litke 1999
Phosphate as PO ₄ to surface water [kg/ha]	2.62E+00	8.32E+00	1.05E+00	1.41E+00	1.96E+00	Goosby <i>et al.</i> 1999, Litke 1999
Phosphate as PO ₄ to ground water (kg/ha)	7.00E-02	7.00E-02	7.00E-02	7.00E-02	7.00E-02	Prasuhn 2006
Nitrate leaching as NO ₃ [kg/ha]	2.79E+01	7.61E+01	5.93E+01	5.10E+01	2.25E+01	Goolsby <i>et al.</i> 1999
Heavy metals to surface water						
Cadmium (Cd, g/ha)	9.88E-01	1.10E+00	8.12E-01	8.78E-01	9.41E-01	Freiermuth 2006
Copper (Cu, g/ha)	8.07E+01	9.51E+01	6.87E+01	7.40E+01	7.63E+01	
Zink (Zn, g/ha)	1.11E+02	1.83E+02	7.24E+01	8.65E+01	9.37E+01	
Lead (Pb, g/ha)	2.03E+01	4.17E+01	1.05E+01	1.25E+01	1.55E+01	
Nickel (Ni, g/ha)	7.44E+01	9.45E+01	5.88E+01	6.49E+01	6.61E+01	

Emission	Cotton	Potato	Rapeseed	Rice	Wheat	Source
Chrome (Cr, g/ha)	1.08E+02	1.14E+02	9.82E+01	1.02E+02	1.06E+02	
Mercury (Hg, g/ha)	-	1.62E-01	3.45E-03	8.45E-03	5.68E-03	
Heavy metals to groundwater						
Cadmium (Cd, g/ha)	4.26E-02	4.74E-02	3.50E-02	3.78E-02	4.05E-02	Freiermuth 2006
Copper (Cu, g/ha)	2.99E+00	3.52E+00	2.54E+00	2.74E+00	2.83E+00	
Zink (Zn, g/ha)	1.53E+01	2.51E+01	9.96E+00	1.19E+01	1.29E+01	
Lead (Pb, g/ha)	1.29E-01	2.66E-01	6.67E-02	7.97E-02	9.87E-02	
Nickel (Ni, g/ha)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	
Chrome (Cr, g/ha)	1.97E+01	2.07E+01	1.79E+01	1.86E+01	1.92E+01	
Mercury (Hg, g/ha)	-	5.18E-03	1.10E-04	2.71E-04	1.82E-04	

1.2.17 Emissions to the Soil

The applied pesticides are calculated as emissions to the agricultural soil. The emissions of heavy metals were calculated by the SALCA heavy metal tool (Freiermuth 2006), where the differences between the inputs of heavy metals contained in the fertilisers and the seeds as well as the outputs through harvested products and emissions to water are assumed to be heavy metal emissions to soil (Freiermuth 2006). In cases, where the total outputs exceed the total inputs, the emission to the soil becomes negative (i.e. net flow to the plant). Tab. 1.21 lists the seven heavy metals considered for the five crops.

Tab. 1.21: Heavy metal emissions to soil

Emission	Cotton	Potato	Rapeseed	Rice	Wheat	Source
Cadmium (Cd, g/ha)	2.98E+00	1.13E+01	-3.81E-01	6.56E-01	1.81E+00	Freiermuth 2006
Copper (Cu, g/ha)	-7.19E+01	-4.52E+01	-6.79E+01	-8.90E+01	-7.70E+01	
Zink (Zn, g/ha)	-4.84E+01	-1.55E+01	-5.84E+01	-1.08E+02	-6.96E+01	
Lead (Pb, g/ha)	-1.53E+01	-2.91E+01	-8.82E+00	-9.95E+00	-1.20E+01	
Nickel (Ni, g/ha)	-6.33E+01	-6.59E+01	-5.41E+01	-5.82E+01	-5.84E+01	
Chrome (Cr, g/ha)	-8.07E+01	9.05E+00	-9.70E+01	-9.63E+01	-9.02E+01	
Mercury (Hg, g/ha)	-	-4.63E-01	-4.08E-03	-9.39E-03	-5.46E-03	

1.2.18 Data Quality Considerations

In a first step, the sources reported in NREL were analysed. The original data contained in NREL did not cover the following items: infrastructure, land occupation and transformation, the use of the resources biogenic CO₂ and energy in biomass, emissions from machinery operations, emissions of heavy metals and emissions of phosphate to groundwater. Furthermore, some processes were not modelled detailed enough to be consistent with the ecoinvent methodology (e.g. production of seed, tractor operations, emissions of nitrate, etc.). Therefore new data sources were searched to complete the lacking data and the datasets were remodelled. In case where the original sources given by NREL were used, these data were updated in all cases where more recent data were available (e.g. for yearly statistics the most recent data were used). The datasets described in this report are thus not merely implementations of the original NREL datasets, but are new datasets that have been substantially remodelled, extended and updated, by using NREL as one source among others.

Data for fertiliser types and amounts are partly based on assumptions. They are derived from national statistics and are representative for the USA. They are not older than 2004 beside the NREL data for potato and rapeseed which are from the year 1997.

Pesticide usage is derived from detailed national statistics and is representative for the USA. Only for rapeseed data from the year 1997 are used.

Data for field operations and machine usage are derived from up to date sources from single states and extrapolated to the USA. The described field operations are representative for 10% of nationwide cotton and wheat production, 30% of potato production, 6% of rapeseed production and 39% of rice production.

Seed quantity and grain drying data are based on qualified estimates derived from single sources.

In agriculture land use strongly depends on the data quality of the yield. The yields for the five crops are taken from the national database for agriculture. The average yield of the last six years is used in order to consider fluctuations of yields due to losses.

Transport processes mostly base on estimates by reason of lack of data.

Field emissions are derived from national sources if possible. Heavy metals are calculated by the SALCA-heavy metal method which is primarily suited for Swiss conditions.

Tab. 1.22: Uncertainty assessment for the data on U.S. arable crops

Category	Reliability	Completeness	Temporal correlation	Geographical correlation	Further technol. correlation
Fertiliser	2	2	1 (3 for potato and rapeseed)	1	1
Pesticide usage, pesticide applications and pesticide emissions	2	1	1 (3 for rapeseed)	1	1
Machine usage, field operations	3	1	1	3	1
Seed quantity, grain drying	3	2	1	1	1
Land use, yield	1	1	1	1	1
Transport processes	4	5	9	9	9
Field emissions	2	2	1	1 (5 for heavy metals and PO ₄)	1

1.2.19 Outlook

For further improvement of these datasets, more information about following topics should be included (sorted according to decreasing importance):

- Considering direct emissions, statistical data should be replaced by suitable modelling under U.S. conditions
- The emissions of phosphorus and heavy metals to water should be calculated with a model better suited to U.S. conditions
- New modelling of field operations for U.S. conditions, especially for rape seed
- The consumption of fertilizers especially for potato and rape seed
- Fertiliser, pesticide, fuel and machinery production inventories specific to U.S. conditions
- More detailed modelling of the irrigation systems in the U.S. Only the most common irrigation system (with electric powered pumps) was considered. But there are other systems widely used, which should be included.
- Transport distances from field to farm, from farm to storehouse, from plant to storehouse.
- Heavy metal contents of cotton.

2 Sheep Production in the USA

2.1 Characteristics

Of USA's total area (9.63 million km²) 4.09 million km² are used for agricultural production (42.2% of the total area, FAO 2006). Permanent pastures comprise 57.1%, arable land 42.9% of the area used for agricultural production (see Tab. 2.1).

Tab. 2.1: Agricultural land usage in the year 2003 (FAO 2006)

Area in 1000 ha 2003 (FAO 2006)	USA
Total area	962'909
Total arable land	175'500
Total permanent pastures	233'800
Total agricultural land	409'300

More than two-thirds of the 68'300 U.S. sheep farms are located in the Southern Plains, Pacific and Mountain Regions. Texas and California are the largest sheep producing states (ASI 2006). The USA produced 16'795 tons of sheep wool and 68'246 tons of sheep carcasses (136'938 tons of live weight) in 2006 (see Tab. 2.2). Texas, California and Wyoming are the states with the highest sheep numbers (see Tab. 2.3). About 40% of sheep produced in the U.S are raised on western public rangelands, where the sheep need to be under herder supervision. Another 40% of sheep are produced in fenced pastures in semi-arid rangelands without supervision. The remaining 20% are produced in pastures in the Midwest and East (Wells *et al.* 2000). China and Australia have the highest numbers of sheep worldwide, followed by India, New Zealand and the United Kingdom (see Tab. 1.4). Australia, China and New Zealand are the leading wool production countries (see Tab. 2.4).

Compared to other livestock and agricultural industries, the U.S. Sheep Industry is very small, accounting for less than 1 percent of total U.S. livestock industry receipts.

Tab. 2.2: U.S. production of sheep wool and carcass in 2006 (NASS 2007)

U.S. production in 2006 (NASS 2006)	tons/year
Wool	16795
Live weight	136938

Tab. 2.3: States with the highest sheep numbers according to ASI (2006)

State	Sheep
Texas	1100000
California	680000
Wyoming	430000
South Dakota	370000
Colorado	360000

Tab. 2.4: World sheep inventory according to ASI (2006)

2003 World sheep inventory (in millions heads)	
World	1024
China	143.8
Australia	98.2
India	59.0
New Zealand	39.2
United Kingdom	35.7
United States	6.4

Tab. 2.4: The top ten wool producing countries according to FAOSTAT (FAO 2006). The USA are number 24 in the list of the countries with the highest wool production.

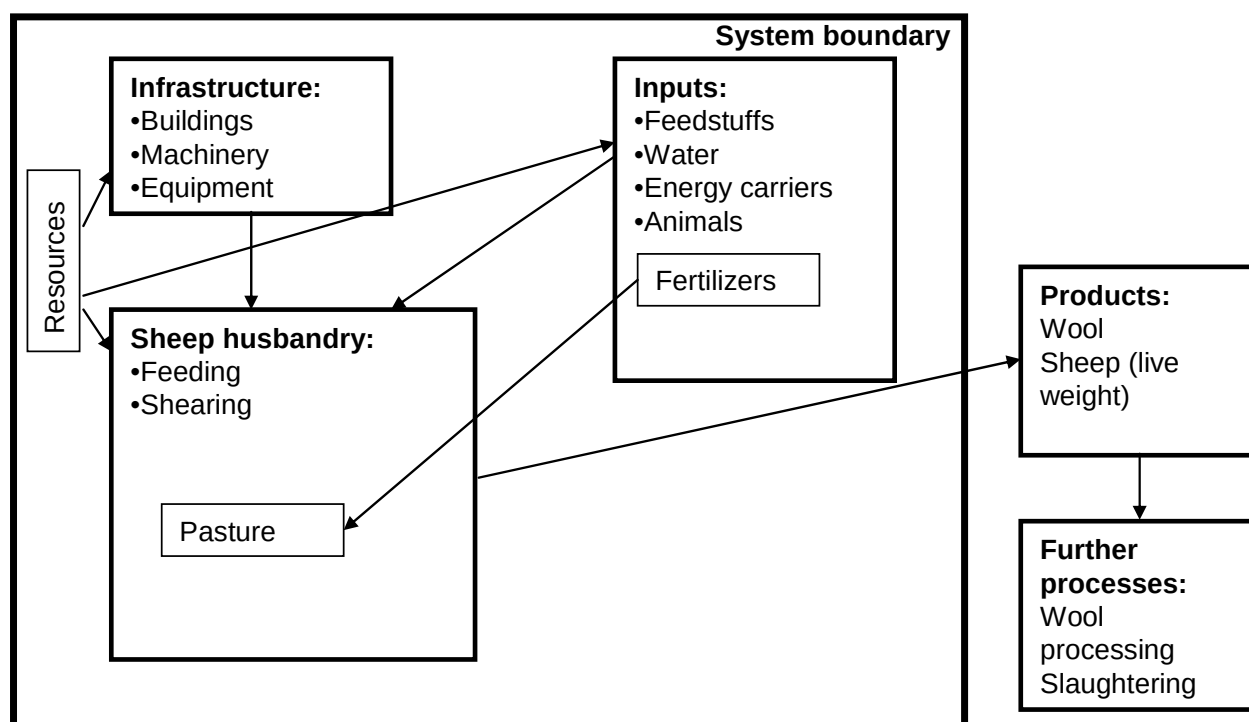
2006 Top 10 Wool Producing Countries (in t)	
Australia	519660
China	388777
New Zealand	209250
Iran, Islamic Rep of	75000
Argentina	60000
United Kingdom	60000
Russian Federation	48000
Sudan	46000
Turkey	46000
India	45200

2.2 Wool / Meat Production from Sheep in the USA

2.2.1 Overview of the Life Cycle Inventories

Sheep, in the United States, are raised for meat, wool, milk and breeding stock. In this study only wool and sheep for meat are considered as the two main products. We considered sheep husbandry, per head and year as a multi-output dataset with the two outputs kg “wool, at farm” for wool production and kg “sheep, live weight, at farm” for meat production. Other steps such as the processing of wool or of carcasses (in slaughterhouses) are not included in these inventories (see system boundary in Fig. 2.1). Tab. 2.5 summarises the inventories described in this chapter and their main characteristics. The system includes the process with the consumption of raw materials, energy, infrastructure and land use as well as the emissions to the air, water, and soil (Fig. 2.1). It also includes transportation of the raw materials. All further processes after the farm gate like transport to the slaughterhouse, slaughtering, cooling, etc. are excluded. The emissions into the water are assumed to be emitted into ground water and rivers.

Fig. 2.1: Sheep and wool production system



Tab. 2.5: Main characteristics of the inventories

	heads	Yield main product (kg/head/y)	Kg N/head/y (available N)	Kg P2O5/head/y	Kg K2O/head/y	Number of pesticide applications/head/y	Supplemental feeding (concentrates) (kg/head/y)
Sheep total	6230100		6.85	2.28	3.43	0.1	79.1
Meatsheep	2188500	62.8	6.85	2.28	3.43	0.1	79.1
Woolsheep	4041600	4.2	6.85	2.28	3.43	0.1	79.1

There is no single source that could be used to model all aspects related to wool and sheep production representative for the U.S. production. Data were therefore gathered from the NREL U.S. LCI Database (2006) and completed by other sources. Tab. 2.6 summarises the data sources used to define the inventories. Detailed information is given in the following sections.

Tab. 2.6: Data sources to compile the arable crop production inventories

Category of Data	Data source(s)	Chapter
Yields of main products and co-products	NASS 2007, USDA 2006	2.2.2
Allocations	NREL 2006, EPA 2006	2.2.3
Intensive & extensive pas-	AVNA 2007, Wells <i>et al.</i> 2000, NREL	2.2.4

Category of Data	Data source(s)	Chapter
tures	2006	
Fertilisers and Nutrition	NREL 2006, ARMS 2006	2.2.5
Water use	NREL 2006	2.2.6
Machine usage	Nemecek <i>et al.</i> 2004, Wells <i>et al.</i> 2000	2.2.7
Pesticides	NREL 2006, Wells <i>et al.</i> 2000	2.2.8
Feedstuff	NREL 2006	2.2.9
Transports	Nemecek <i>et al.</i> 2004, Spielmann <i>et al.</i> 2007, NREL 2006	2.2.10
Land use	NREL 2006, AVNA 2007	2.2.11
Energy in biomass & biogenic CO ₂	TVS 2007, Uni-Jena 2007	2.2.12
Direct emissions	Nemecek <i>et al.</i> 2004, Goolsby <i>et al.</i> 1999, Özyol 1990, Freiermuth 2006, Taipale 2006, EPA 2006, Litke 1999, Prasuhn 2006	2.2.13, 2.2.14, 2.2.15, 2.2.16

2.2.2 Yields

Sheep are shorn once or twice a year. The average fleece per shear weights 3.3 kg according to NASS (2007, see also Tab. 2.7). The average wool yield per sheep and year is about 4.2 kg. The average live weight of an U.S. sheep to be slaughtered is about 62.8 kg (USDA 2006).

Tab. 2.7: Wool and meat production in the USA in the year 2006

		Source
Wool Sheep (number)	4049565	Calculation
Fleece production (number/y)	5072000	NASS 2007
Fleece weight (kg/unit)	3.3	NASS 2007
Yield wool (kg/head/y)	4.2	own calculation
Meat Sheep (amount)	2180535	own calculation
Live weight (kg)	62.8	USDA 2006

2.2.3 Allocations

It is assumed that sheep for meat are slaughtered after one year, whereas sheep for wool live about eight years (NREL 2006). In 2006, only 43% of the 6,230,100 sheep were slaughtered (EPA 2006). Combining this information with the assumed average lifespan of sheep for meat production and sheep for wool production, sheep for meat production represent 35.1% and sheep for wool production 64.9% of the whole population (see Tab. 2.8). This physical allocation factor for the products wool and live weight was used for most inputs and outputs. For energy in sheep body and CO₂ uptake in sheep body the actual energy and CO₂ content in the products were estimated.

Tab. 2.8: Calculation of the wool and meat sheep contingent of the total amount of sheep.

	variable		comment	Source
Average lifespan meat sheep		1 year		NREL 2006
Average lifespan wool sheep		8 year		NREL 2006
Total amount of sheep	t	6,230,100	all meat sheep (m) and all wool sheep (w)	NASS 2007
Slaughtered sheep per year	s	2,693,700	all meat sheep (m) and 1/8 of wool sheep (1/8w)	USDA 2006
Amount of wool sheep	w	$= (t - s) * 8/7$	4,049,565	
Amount of meat sheep	m	$= (s - 1/8t) * 8/7$	2,180,535	
Allocation factor wool sheep		$= w / t$	64.9%	
Allocation factor meat sheep		$= m / t$	35.1%	

2.2.4 Intensive and Extensive Pastures

Wells *et al.* (2000) report that 80% of the sheep are held on extensive pastures with no additional fertilising and other field operations, whereas 20% of the sheep are held on intensive pastures similar to cattle. Data given by NREL (2006) refer to intensive pastures only, similar to cattle pastures, according to the cited sources. On these intensive pastures, application of fertilisers and pesticides occurs regularly and machinery operations are carried out. On the extensive pastures, these inputs and operations do not occur regularly. The respective figures reported by NREL (2006) were therefore multiplied by a factor 0.2.

On the other hand, extensive pastures have lower yields. Therefore the land occupation estimated by NREL (personal information O. Muller, Ecobilan, June 2006) was increased by a factor 1.96 using the estimated stocking rate of 12 sheep per ha (AVNA 2007).

2.2.5 Fertilisers

We assume that fertilisers are only used on 20% of the pastures used for sheep production (see chapter 2.2.3). Average quantities for the fertilizer use on pastures in the USA are given in Tab. 2.9. In this study values for the amounts of fertilisers used on intensive pastures are taken from NREL (2006) and multiplied by the factor 0.2 (share of intensive pastures).

Tab. 2.9: Average fertilizer use in the USA per sheep and year (ARMS 2006, NREL 2006)

Fertilizer	Unit kg/head/y
Ammoniumnitrate as N	1.37
Triple-Superphosphate as P ₂ O ₅	0.46
Potassium fertiliser as K ₂ O	0.69
Lime (CaO)	18.26

2.2.6 Water Use

Water used by sheep (see Tab. 2.6) is taken from NREL database 2006, in which water use per sheep is estimated.

Tab. 2.10: Water usage for sheep production

	l/head/y	
Water use	2071	NREL 2006

2.2.7 Machine Usage

Most of the area used by sheep is presumably permanent extensive pasture land. According to Wells *et al.* (2003) 80% of sheep produced are raised on vast rangelands with no specific field operations. 20% of pastures for sheep production are assumed to be intensive pasture land similar to pastures used for cattle production. Tab. 2.11 lists the average field operations. It is assumed that the amount and kind of field operations on intensive permanent pastureland is similar in the USA and CH. Besides the field operation rolling, four applications of fertilisers per year were assumed.

Tab. 2.11: Average field operations on permanent pastureland per sheep and year

Field operation	ha/head/y
tillage, rolling	8.24E-03
fertilising, by broadcaster	3.30E-02

2.2.8 Pesticides

According to NREL (2006) pesticides are used in pasture land. NREL does not distinguish between cattle and sheep pastures (personal information Muller, Ecobilan, June 2007). No data were found about pasture land explicitly used by sheep. Sheep are sturdier graziers than cattles and are able to eat plants otherwise poisonous to other grazing animals. Sheep have been used as an alternative to herbicides for suppressing weeds (Wells *et al.* 2003). Therefore, we assume that normally no pesticides are used on the pasture land for sheep.

2.2.9 Feedstuff

Sheep utilize roughage as their primary feed supply and usually do not require large amounts of purchased feed. NREL (2006) reports the quantities of concentrate feed supplied to sheep (see Tab. 2.12). No distinction between nutrition for wool and meat sheep and between sheep on extensive and intensive pasture was made. Even in extensive pasture systems it is likely that some amount of supplementary feedstuffs (concentrates) is given. The amount of grass grazed on pastures is not included as such, but is indirectly included through the management of the pastures (machinery, fertiliser and pesticide use, according to Fig. 2.1).

Tab. 2.12: Amount of concentrate feed used per sheep (NREL 2006)

	kg/head/y
Soy Meal	27.2
Corn (grain)	41.4
Limestone	6.62
Sodium chloride	3.88

2.2.10 Transports

According to NREL (2006) the transport distance of raw materials to the farm is 400 km on railway and 200 km on road but no source is reported. The ecoinvent inventories of fertilisers already include the transports to the regional storehouse. Therefore an own estimation was made for the transport distance from the regional storehouse to the farm (50 km) assuming that the fertiliser modules RER are applicable for the USA. Feedstuff (corn and soy) is transported from the cornbelt to the sheepfarms. The average transport distance to the five states with the highest sheep numbers is estimated to be approximately 1200 km. Tab. 2.13 shows the transports considered for sheep production using the transport datasets described by Spielmann *et al.* (2007). Transport from farm to slaughterhouse is not included in the system. Sheep transportation from farm to field is not considered in this report.

Tab. 2.13: Transports considered for raw materials used per sheep and year

Material	Transport type	Distance (km)	tkm/head
From corn belt to farm			
Corn and soy	Train	800	54.9
	Lorry 40t	400	27.4
From storehouse to farm			
Fertilizer and supplementary feedstuff (concentrates)	Lorry 40t	50	0.3
Total tkm	Train		54.9
Total tkm	Lorry 40t		27.7

2.2.11 Land Use

NREL (personal Information O. Muller, Ecobilan, June 2007) estimated 412 m² per year for the land use of a sheep (= 24 sheep / ha). This represents a rather intensive sheep production and may be true for 20% of the area used for sheep production. In this report a land use of 809 m² per year (=12.4 sheep / ha) is used based on Information given by AVNA (2007).

2.2.12 Energy in Biomass and Biogenic CO₂ Uptake

The total energy in biomass (sheep) is calculated from the energy content of the products. One kg wool contains about 23 MJ according to Textilverband Schweiz (TVS 2007). The biomass energy per live weight (without wool) approximated by the energy content of sheep meat is about 5.25 MJ according to Uni-Jena (2007). The difference between the energy input (energy content of the additional feedstuff) and the energy output (energy content in the sheep) is assessed as heat waste. The energy content of grass and manure is not assessed, since these processes occur within the system boundary. Tab. 2.14 lists the energy content of a sheep and the heat waste output as well as the CO₂ content of a sheep. The latter is roughly assumed by the C-content of carcass meal (40%).

Tab. 2.14: Biomass energy and CO₂ in product

Unit	Energy in biomass (sheep)	Heat waste	CO ₂ converted into sheep body	CO ₂ balance
MJ/sheep	425	790		
kg/ sheep			99.2	3.5 (net uptake)

2.2.13 Heavy Metals in the Products

Heavy metals are taken out of the system in products like wool and sheep (live weight). It is assumed that all manure remains on the pasture land. The uptake of heavy metals for wool (Özyol 1990) and sheep (live weight) is approximated by the heavy metal content of sheep meat) is given in Tab. 2.15. (Freiermuth 2006).

Tab. 2.15: Heavy metal uptake of wool and sheep (Freiermuth 2006, Özyol 1990)

	Wool	Sheep
Heavy metal	mg/kg	mg/kg
Cd	0.2232	0.0005

	Wool	Sheep
Heavy metal	mg/kg	mg/kg
Cu	35.32	0.9
Zn	475.6	23
Pb	2.778	0.065
Ni	*	0.02
Cr	*	0.04
Hg	0.1454	0.0005

*no data available

2.2.14 Emissions to the Air

There are two sources of direct emissions into air: Emissions from mineral fertilisers and emissions from the sheep and its manure. Taipale (2006) reports NH_3 production from manure per sheep and year in Colorado. NH_3 emissions from fertilisers are calculated according to Goolsby *et al.* (1999). NO_x and N_2O emissions from fertilisers are calculated with the emission factors given by Nemecek *et al.* (2004). The U.S. emission inventory 2006 (EPA 2006) reports the amount of N_2O from sheep manure as well as methane emissions from enteric fermentation and manure. Emissions to air are listed in Tab. 2.16.

Tab. 2.16: Emissions to air from sheep production

	Emission	Sheep	Source
		kg/head/y	
CH_4	Enteric fermentation	7.96E+00	EPA 2006
	Manure	7.11E-01	EPA 2006
	Total	8.67E+00	
N_2O	Fertilizer	2.22E-01	Nemecek <i>et al.</i> 2004
	Manure	3.55E-02	EPA 2006
	Total	2.57E-01	
NH_3	Fertilizer	3.33E-02	Goolsby <i>et al.</i> 1999
	Enteric Fermentation & Manure	3.40E+00	Taipale 2006
	Total	3.44E+00	
NO_x	Total	5.40E-02	Nemecek <i>et al.</i> 2004

2.2.15 Emissions to the Water

Litke (1999) reports data for the phosphorus emissions to water from sheep production. According to Goolsby *et al.* (1999), two third of the total phosphorus emission is particulate phosphorus from fertilizers.

The emission factor for phosphorus emissions to ground water is taken from Prasuhn (2006).

Goolsby *et al.* (1999) estimated the nitrate leaching in the Mississippi region with an emission factor of 61 % of the N-input (fertilisers and N fixation). We extrapolate this emission factor to the whole USA.

Heavy metal emissions to water were calculated by the SALCA-heavy metal tool (Freiermuth 2006). The SALCA-heavy metal tool is developed for Swiss conditions. Because of missing heavy metal data we used this method, which considers inputs through fertilizer, pesticides and nutrition and outputs through wool and sheep body. Because of different climate and soil conditions the method does not really represent the U.S. case and the uncertainty is rather high.

Tab. 2.17: Emissions to water from sheep production

Emission	kg/head/y
Particulate Phosphorus as P to surface water	4.75E-02
Phosphate as PO ₄ to surface water	6.42E-02
Phosphate as PO ₄ to ground water	6.00E-02
Nitrate leaching as NO ₃	4.96E+00
Heavy metals to surface water	
Cadmium (Cd)	5.62E-06
Copper (Cu)	1.14E-02
Zink (Zn)	3.36E-02
Lead (Pb)	3.43E-05
Nickel (Ni)	2.44E-03
Chrome (Cr)	1.20E-02
Mercury (Hg)	0.00E+00
Heavy metals to groundwater	
Cadmium (Cd)	2.42E-07
Copper (Cu)	4.21E-04
Zink (Zn)	4.62E-03
Lead (Pb)	2.18E-07
Nickel (Ni)	0.00E+00
Chrome (Cr)	2.19E-03
Mercury (Hg)	0.00E+00

2.2.16 Emissions to the Soil

The differences between outputs through products (wool and sheep) and inputs of heavy metals contained in the fertilisers and nutrition are assumed to be heavy metal emissions to soil (Freiermuth 2006). For some heavy metals, the outputs exceed the inputs and therefore a negative emission into soil results.

Tab. 2.18: Heavy metal input to soil based on own calculation with SALCA-heavy metal tool.

emission	kg/head/year
Cd	-2.48E-06
Cu	-1.39E-02
Zn	-1.15E-02
Pb	-2.77E-05
Ni	0.00E+00
Cr	-2.06E-03
Hg	-2.57E-02

2.2.17 Data Quality Considerations

In a first step, the sources reported in NREL were analysed. The original data contained in NREL did not cover the following items: infrastructure, land occupation and transformation, energy contained in the biomass, emissions from machinery operations, emissions of heavy metals and emissions of phos-

phate to groundwater. Furthermore, some processes were not detailed enough to be consistent with the ecoinvent methodology (e.g. tractor operations, emissions of nitrate, etc.). Therefore new data sources were searched to complete the lacking data and the datasets were remodelled. In case where the original sources given by NREL were used, these data were updated in all cases where more recent data were available. The datasets described in this report are thus not merely implementations of the original NREL datasets, but are new datasets that have been substantially remodelled, extended and updated, by using NREL as one source among others. Tab. 2.19 lists the uncertainty judgment of the used data.

Data for feedstuff use were taken from the estimates by NREL (2006) trying to represent the average U.S. conditions.

Fertiliser type and amount are reported by NREL (2006). Data are older than 1996 and are valid for intensive pastures only. For extensive pasture we assumed that no fertilisers are used.

The land use depends on the stock density of sheep. NREL (2006) assumes an average stock density based on a source from 1990. No current data on this topic could be found.

Data for field operations were extrapolated from Swiss data for pasture cultivation.

Transport processes mostly base on estimates due to missing data.

Field emissions are derived from national sources if possible. Heavy metals are calculated by the SALCA-heavy metal method which is primarily suited for Swiss conditions.

Tab. 2.19: Uncertainty judgement for the data on arable crops

Category	Reliability	Completeness	Temporal correlation	Geographical correlation	Further technol. correlation
Feedstuff use	2	3	4	3	1
Fertiliser	2	3	4	3	1
Land use	3	3	4	3	1
Field operations	3	3	3	5	1
Transport processes	4	5	9	9	9
Field emissions	2	2 (3 for Ammonia and Nitrate leaching)	2	3 (5 for heavy metals)	1

2.2.18 Outlook

For further improvement of these datasets, more information about following topics should be included:

- More precise average lifespan of a sheep for meat and wool production
- More detailed differentiation of input materials for wool and meat production
- Land use (stock density) of the most important sheep production states
- Transport distances from field to farm
- Estimation of direct emissions: statistical data should be replaced by suitable modelling
- The emissions of heavy metals to water should be calculated with a model better suited to U.S. conditions
- Heavy metal contents of sheep body parts
- Inclusion of transport to and processes at slaughterhouse
- To convert the U.S. sheep data sets, suited data sets like stock density (land use), type of sheep (wool production, live weight), feedstuffs and pasture cultivation have to be developed.

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Appendix A to Chapter 1 (Crop Production in the USA)

Unit-Process Inventories from Chapter 1 (Crop Production in the USA)

Tab. A. 1 Unit-process inventories for arable crop production.

Unit process inventory for: rice seed, at regional storehouse, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
rice, at farm	US	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	GR	kWh	2.40E-02	1	1.07	(2,1,1,1,1,na)
Heat, waste	soil/agricultural	MJ	8.64E-02	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	2.00E-01	1	1.07	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	1.07	(2,1,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.50	(2,1,1,1,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.50	(2,1,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
diphenylether-compounds, at regional storehouse	RER	kg	1.00E-04	1	1.21	(2,1,1,1,1,na)
rice seed, at regional storehouse	US	kg	1.00E+00			

Unit process inventory for: cotton seed, at regional storehouse, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
cotton seed, at farm	US	kg	1.00E+00	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	GR	kWh	5.80E-02	1	1.07	(2,1,1,1,1,na)
Heat, waste	soil/agricultural	MJ	2.09E-01	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	2.00E-01	1	1.07	(2,1,1,1,1,na)
building, multi-storey (I)	RER	m3	2.00E-05	1	1.07	(2,1,1,1,1,na)
Occupation, industrial area, built up	resource/land	m2a	2.00E-04	1	1.50	(2,1,1,1,1,na)
Occupation, construction site	resource/land	m2a	8.00E-06	1	1.50	(2,1,1,1,1,na)
Transformation, to industrial area, built up	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
Transformation, from unknown	resource/land	m2	4.00E-06	1	2.00	(2,1,1,1,1,na)
diphenylether-compounds, at regional storehouse	RER	kg	1.00E-04	1	1.21	(2,1,1,1,1,na)
cotton seed, at regional storehouse	US	kg	1.00E+00			

Life cycle inventories of U.S. agricultural production systems - Appendix A to Chapter 197H1 (198HCrop Production in the USA)

Unit process inventory for: cotton, US									
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	cotton fibres, US (kg)	cotton seed, at farm (kg)	cotton seed, at farm (kg)
mulching	CH	ha	1.00E+00	1	1.07	(2.1.1.3.1.na)	87%	13%	
sowing	CH	ha	1.00E+00	1	1.07	(2.1.1.3.1.na)	87%	13%	
tillage, cultivating, chiselling	CH	ha	1.00E+00	1	1.07	(2.1.1.3.1.na)	87%	13%	
tillage, harrowing, by rotary harrow	CH	ha	1.00E+00	1	1.07	(2.1.1.3.1.na)	87%	13%	
tillage, harrowing, by spring tine harrow	CH	ha	4.00E+00	1	1.07	(2.1.1.3.1.na)	87%	13%	
application of plant protection products, by field sprayer	CH	ha	1.39E+01	1	1.07	(2.1.1.3.1.na)	87%	13%	
fertilising, by broadcaster	CH	ha	3.00E+00	1	1.07	(2.1.1.3.1.na)	87%	13%	
combine harvesting	CH	ha	9.01E-01	1	1.07	(2.1.1.3.1.na)	87%	13%	
baling	CH	unit	1.20E+01	1	1.07	(2.1.1.3.1.na)	87%	13%	
grain drying, high temperature	CH	kg	9.17E-02	1	1.07	(2.2.1.1.1.na)	87%	13%	
cotton seed, at regional storehouse	US	kg	1.46E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
ammonia, liquid, at regional storehouse	RER	kg	4.98E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
urea, as N, at regional storehouse	RER	kg	1.72E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
ammonium nitrate, as N, at regional storehouse	RER	kg	2.38E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	5.58E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
potassium chloride, as K2O, at regional storehouse	RER	kg	8.93E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
irrigating	US	m3	1.31E+00	1	1.07	(2.1.1.1.1.na)	87%	13%	
pesticide unspecified, at regional storehouse	CH	kg	6.06E-01	1	1.07	(2.1.1.1.1.na)	87%	13%	
acetamide-anilide-compounds, at regional storehouse	CH	kg	3.48E-02	1	1.07	(2.1.1.1.1.na)	87%	13%	
cyclic N-compounds, at regional storehouse	CH	kg	4.91E-04	1	1.07	(2.1.1.1.1.na)	87%	13%	
phenoxo-compounds, at regional storehouse	CH	kg	3.17E-02	1	1.07	(2.1.1.1.1.na)	87%	13%	
triazine-compounds, at regional storehouse	CH	kg	8.19E-02	1	1.07	(2.1.1.1.1.na)	87%	13%	
benzothiadiazole-compounds, at regional storehouse	CH	kg	2.28E-02	1	1.07	(2.1.1.1.1.na)	87%	13%	
nitrile-compounds, at regional storehouse	CH	kg	1.52E-03	1	1.07	(2.1.1.1.1.na)	87%	13%	
sulfonylurea-compounds, at regional storehouse	CH	kg	5.54E-02	1	1.07	(2.1.1.1.1.na)	87%	13%	
dicamba, at regional storehouse	CH	kg	4.46E-04	1	1.07	(2.1.1.1.1.na)	87%	13%	
bipyridylum-compounds, at regional storehouse	CH	kg	4.42E-02	1	1.07	(2.1.1.1.1.na)	87%	13%	
diuron, at regional storehouse	CH	kg	1.22E-01	1	1.07	(2.1.1.1.1.na)	87%	13%	
organophosphorus-compounds, at regional storehouse	CH	kg	1.20E+00	1	1.07	(2.1.1.1.1.na)	87%	13%	
glyphosate, at regional storehouse	CH	kg	1.06E+00	1	1.07	(2.1.1.1.1.na)	87%	13%	
linuron, at regional storehouse	CH	kg	3.75E-03	1	1.07	(2.1.1.1.1.na)	87%	13%	
thiolcarbamate-compounds, at regional storehouse	CH	kg	4.37E-03	1	1.07	(2.1.1.1.1.na)	87%	13%	
dinitroaniline-compounds, at regional storehouse	CH	kg	4.19E-01	1	1.07	(2.1.1.1.1.na)	87%	13%	
pyretroid-compounds, at regional storehouse	CH	kg	9.10E-03	1	1.07	(2.1.1.1.1.na)	87%	13%	
transport, lorry >18, fleet average	RER	kg	1.00E+04	1	2.00	(4.5.na.na.na.na)	87%	13%	
Energy, gross calorific value, in biomass	resource/biotic	MJ	3.98E+04	1	1.07	(2.1.1.1.1.na)	33%	67%	
Carbon dioxide, in air	resource/in air	kg	4.20E-03	1	1.07	(2.1.1.1.1.na)	27%	73%	
Occupation, arable	resource/land	m2a	1.00E+04	1	2.00	(2.1.1.1.1.na)	87%	13%	
Transformation, from arable	resource/land	m2	1.00E+04	1	2.00	(2.1.1.1.1.na)	87%	13%	
Transformation, to arable	resource/land	m2	1.00E+04	1	2.00	(2.1.1.1.1.na)	87%	13%	
Nitrate	water/ground	kg	2.79E+01	1	1.51	(2.2.1.1.1.na)	87%	13%	
Phosphorus	water/river	kg	1.94E+00	1	1.51	(2.2.1.1.1.na)	87%	13%	
Phosphate	water/river	kg	2.62E+00	1	1.52	(2.2.1.5.1.na)	87%	13%	
Phosphate	water/ground	kg	7.00E-02	1	1.52	(2.2.1.5.1.na)	87%	13%	
Ammonia	air/low population density	kg	6.40E+00	1	1.21	(2.2.1.1.1.na)	87%	13%	
Carbon dioxide, fossil	air/low population density	kg	2.70E+01	1	1.07	(2.2.1.1.1.na)	87%	13%	
Nitrogen oxides	air/low population density	kg	5.72E-01	1	1.41	(2.2.1.1.1.na)	87%	13%	
Dinitrogen monoxide	air/low population density	kg	2.72E+00	1	1.41	(2.2.1.1.1.na)	87%	13%	
Cadmium	soil/agricultural	kg	2.98E-06	1	1.52	(2.2.1.5.1.na)	40%	60%	
Chromium	soil/agricultural	kg	-8.07E-05	1	1.52	(2.2.1.5.1.na)	40%	60%	
Copper	soil/agricultural	kg	-7.19E-05	1	1.52	(2.2.1.5.1.na)	40%	60%	
Lead	soil/agricultural	kg	-1.53E-05	1	1.52	(2.2.1.5.1.na)	40%	60%	
Nickel	soil/agricultural	kg	-6.33E-05	1	1.52	(2.2.1.5.1.na)	40%	60%	
Zinc	soil/agricultural	kg	-4.84E-05	1	1.52	(2.2.1.5.1.na)	40%	60%	
Cadmium, ion	water/river	kg	9.88E-07	1	1.82	(2.2.1.5.1.na)	40%	60%	
Chromium, ion	water/river	kg	1.08E-04	1	1.82	(2.2.1.5.1.na)	40%	60%	
Copper, ion	water/river	kg	8.07E-05	1	1.82	(2.2.1.5.1.na)	40%	60%	
Lead	water/river	kg	2.03E-05	1	1.82	(2.2.1.5.1.na)	40%	60%	
Nickel, ion	water/river	kg	7.44E-05	1	1.82	(2.2.1.5.1.na)	40%	60%	
Zinc, ion	water/river	kg	1.11E-04	1	1.82	(2.2.1.5.1.na)	40%	60%	
Cadmium, ion	water/ground	kg	4.26E-08	1	1.82	(2.2.1.5.1.na)	40%	60%	
Chromium, ion	water/ground	kg	1.97E-05	1	1.82	(2.2.1.5.1.na)	40%	60%	
Copper, ion	water/ground	kg	2.99E-06	1	1.82	(2.2.1.5.1.na)	40%	60%	
Lead	water/ground	kg	1.29E-07	1	1.82	(2.2.1.5.1.na)	40%	60%	
Zinc, ion	water/ground	kg	1.53E-05	1	1.82	(2.2.1.5.1.na)	40%	60%	
Azoxystrobin	soil/agricultural	kg	1.43E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Carbofen	soil/agricultural	kg	6.69E-05	1	1.21	(2.1.1.1.1.na)	87%	13%	
Enidiazole	soil/agricultural	kg	3.21E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Iprodion	soil/agricultural	kg	3.57E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Metazolin	soil/agricultural	kg	1.25E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Quintozene	soil/agricultural	kg	1.53E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
2,4-D	soil/agricultural	kg	2.75E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Bromoxynil	soil/agricultural	kg	1.52E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Carfentrazone ethyl ester	soil/agricultural	kg	1.87E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Clomazone	soil/agricultural	kg	4.91E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Clethodim	soil/agricultural	kg	2.30E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Cyanazine	soil/agricultural	kg	1.38E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Dicamba	soil/agricultural	kg	4.46E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Diuron	soil/agricultural	kg	1.22E-01	1	1.21	(2.1.1.1.1.na)	87%	13%	
DSMA	soil/agricultural	kg	3.30E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Flumetron	soil/agricultural	kg	5.50E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Glyphosate	soil/agricultural	kg	1.06E+00	1	1.21	(2.1.1.1.1.na)	87%	13%	
Lactofen	soil/agricultural	kg	3.35E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Linuron	soil/agricultural	kg	3.75E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Metolachlor	soil/agricultural	kg	3.48E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
MSMA	soil/agricultural	kg	8.06E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Norflurazon	soil/agricultural	kg	7.58E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Paraquat	soil/agricultural	kg	4.42E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Pendimethalin	soil/agricultural	kg	1.30E-01	1	1.21	(2.1.1.1.1.na)	87%	13%	
Prometryn	soil/agricultural	kg	8.05E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Pyriminobac sodium salt	soil/agricultural	kg	8.59E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Thifensulfuron-methyl	soil/agricultural	kg	4.46E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Trifluralin	soil/agricultural	kg	2.89E-01	1	1.21	(2.1.1.1.1.na)	87%	13%	
Abamectin	soil/agricultural	kg	1.87E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Acephate	soil/agricultural	kg	1.79E-01	1	1.21	(2.1.1.1.1.na)	87%	13%	
Aldicarb	soil/agricultural	kg	1.38E-01	1	1.21	(2.1.1.1.1.na)	87%	13%	
Bifenoxin	soil/agricultural	kg	1.85E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Buprofezin	soil/agricultural	kg	7.14E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Carbofuran	soil/agricultural	kg	1.43E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Chlorpyrifos	soil/agricultural	kg	1.71E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Cyfluthrin	soil/agricultural	kg	4.82E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Cypermethrin	soil/agricultural	kg	8.83E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Deltamethrin	soil/agricultural	kg	2.68E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Dicofol	soil/agricultural	kg	2.16E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Dicrotophos	soil/agricultural	kg	5.89E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Dimethoate	soil/agricultural	kg	7.58E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Disulfoton	soil/agricultural	kg	1.34E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Endosulfan	soil/agricultural	kg	6.78E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Esfenvalerate	soil/agricultural	kg	2.68E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Fenpropathrin	soil/agricultural	kg	7.14E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Imidacloprid	soil/agricultural	kg	1.34E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Indoxacarb	soil/agricultural	kg	2.14E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Lambda-Cyhalothrin	soil/agricultural	kg	2.94E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Malathion	soil/agricultural	kg	4.17E-01	1	1.21	(2.1.1.1.1.na)	87%	13%	
Methamidophos	soil/agricultural	kg	2.68E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Methomyl	soil/agricultural	kg	8.03E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Naled	soil/agricultural	kg	1.96E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Oxamyl	soil/agricultural	kg	8.66E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Parathion	soil/agricultural	kg	1.11E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Permethrin	soil/agricultural	kg	1.56E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Oils, unspecified	soil/agricultural	kg	1.45E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Phorate	soil/agricultural	kg	2.32E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Profluthrin	soil/agricultural	kg	1.23E-02	1	1.21	(2.1.1.1.1.na)	87%	13%	
Propargite	soil/agricultural	kg	3.14E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Pyriproxyfen	soil/agricultural	kg	1.12E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Spinosad	soil/agricultural	kg	1.16E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Tebuconazole	soil/agricultural	kg	3.35E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Thiamethoxam	soil/agricultural	kg	3.12E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Tralomep	soil/agricultural	kg	2.68E-04	1	1.21	(2.1.1.1.1.na)	87%	13%	
Arsenic	soil/agricultural	kg	2.19E-03	1	1.21	(2.1.1.1.1.na)	87%	13%	
Cyfluthrin	soil/agricultural	kg	1.28E-02	1	1.21	(2.1.1.1.1.na)</			

Life cycle inventories of U.S. agricultural production systems - Appendix A to Chapter 197H1 (198HCrop Production in the USA)

Unit process inventory for: potatoes, at farm, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
potato planting	CH	ha	2.44E-05	1	1.07	(2.1,1.3,1.na)
tillage, cultivating, chiselling	CH	ha	2.44E-05	1	1.07	(2.1,1.3,1.na)
potato haulm cutting	CH	ha	2.44E-05	1	1.07	(2.1,1.3,1.na)
tillage, harrowing, by spring tine harrow	CH	ha	4.88E-05	1	1.07	(2.1,1.3,1.na)
tillage, hoeing and earthing-up, potatoes	CH	ha	4.88E-05	1	1.07	(2.1,1.3,1.na)
tillage, ploughing	CH	ha	2.44E-05	1	1.07	(2.1,1.3,1.na)
application of plant protection products, by field sprayer	CH	ha	9.76E-05	1	1.07	(2.1,1.3,1.na)
fertilising, by broadcaster	CH	ha	4.88E-05	1	1.07	(2.1,1.3,1.na)
harvesting, by complete harvester, potatoes	CH	ha	2.39E-05	1	1.07	(2.1,1.3,1.na)
potato grading	CH	kg	1.00E+00	1	1.07	(2.1,1.3,1.na)
potato seed IP, at regional storehouse	CH	kg	5.74E-02	1	1.07	(2.2,1.1,1.na)
ammonia, liquid, at regional storehouse	RER	kg	2.38E-03	1	1.13	(2.2,3,1,1.na)
urea, as N, at regional storehouse	RER	kg	8.23E-04	1	1.13	(2.2,3,1,1.na)
ammonium nitrate, as N, at regional storehouse	RER	kg	1.14E-03	1	1.13	(2.2,3,1,1.na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	4.33E-03	1	1.13	(2.2,3,1,1.na)
potassium chloride, as K2O, at regional storehouse	RER	kg	4.13E-03	1	1.13	(2.2,3,1,1.na)
irrigating	US	m3	1.13E-04	1	1.07	(2.1,1,1,1.na)
pesticide unspecified, at regional storehouse	CH	kg	1.02E-03	1	1.07	(2.1,1,1,1.na)
dithiocarbamate-compounds, at regional storehouse	CH	kg	6.17E-05	1	1.07	(2.1,1,1,1.na)
maneb, at regional storehouse	CH	kg	9.57E-08	1	1.07	(2.1,1,1,1.na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	5.06E-06	1	1.07	(2.1,1,1,1.na)
phenoxy-compounds, at regional storehouse	CH	kg	2.41E-06	1	1.07	(2.1,1,1,1.na)
triazine-compounds, at regional storehouse	CH	kg	6.67E-06	1	1.07	(2.1,1,1,1.na)
[sulfonyl]urea-compounds, at regional storehouse	CH	kg	1.00E-07	1	1.07	(2.1,1,1,1.na)
bipyridylum-compounds, at regional storehouse	CH	kg	3.34E-06	1	1.07	(2.1,1,1,1.na)
organophosphorus-compounds, at regional storehouse	CH	kg	2.98E-05	1	1.07	(2.1,1,1,1.na)
glyphosate, at regional storehouse	CH	kg	2.44E-07	1	1.07	(2.1,1,1,1.na)
linuron, at regional storehouse	CH	kg	8.75E-07	1	1.07	(2.1,1,1,1.na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	6.70E-04	1	1.07	(2.1,1,1,1.na)
dinitroaniline-compounds, at regional storehouse	CH	kg	4.62E-06	1	1.07	(2.1,1,1,1.na)
pyridazine-compounds, at regional storehouse	CH	kg	4.11E-06	1	1.07	(2.1,1,1,1.na)
transport, lorry >16t, fleet average	RER	tkm	4.68E-03	1	2.09	(4.5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	3.65E+00	1	1.07	(2.1,1,1,1.na)
Carbon dioxide, in air	resource/in air	kg	3.21E-01	1	1.07	(2.1,1,1,1.na)
Occupation, arable	resource/land	m2a	2.44E-01	1	2.00	(2.1,1,1,1.na)
Transformation, from arable	resource/land	m2	2.44E-01	1	2.00	(2.1,1,1,1.na)
Transformation, to arable	resource/land	m2	2.44E-01	1	2.00	(2.1,1,1,1.na)
Nitrate	water/ground-	kg	1.86E-03	1	1.51	(2.2,1,1,1.na)
Phosphorus	water/river	kg	1.50E-04	1	1.51	(2.2,1,1,1.na)
Phosphate	water/river	kg	2.03E-04	1	1.52	(2.2,1,5,1.na)
Phosphate	water/ground-	kg	1.71E-06	1	1.52	(2.2,1,5,1.na)
Ammonia	airflow population density	kg	3.05E-04	1	1.21	(2.2,1,1,1.na)
Carbon dioxide, fossil	airflow population density	kg	1.29E-03	1	1.07	(2.2,1,1,1.na)
Nitrogen oxides	airflow population density	kg	2.87E-05	1	1.41	(2.1,1,1,1.na)
Dinitrogen monoxide	airflow population density	kg	1.37E-04	1	1.41	(2.1,1,1,1.na)
Cadmium	soil/agricultural	kg	2.75E-10	1	1.52	(2.2,1,5,1.na)
Chromium	soil/agricultural	kg	2.21E-10	1	1.52	(2.2,1,5,1.na)
Copper	soil/agricultural	kg	1.75E-09	1	1.52	(2.2,1,5,1.na)
Lead	soil/agricultural	kg	-7.11E-10	1	1.52	(2.2,1,5,1.na)
Mercury	soil/agricultural	kg	-1.13E-11	1	1.52	(2.2,1,5,1.na)
Nickel	soil/agricultural	kg	-1.61E-09	1	1.52	(2.2,1,5,1.na)
Zinc	soil/agricultural	kg	-3.79E-10	1	1.52	(2.2,1,5,1.na)
Cadmium, ion	water/river	kg	2.68E-11	1	1.82	(2.2,1,5,1.na)
Chromium, ion	water/river	kg	2.77E-09	1	1.82	(2.2,1,5,1.na)
Copper, ion	water/river	kg	2.32E-09	1	1.82	(2.2,1,5,1.na)
Lead	water/river	kg	1.02E-09	1	1.82	(2.2,1,5,1.na)
Mercury	water/river	kg	3.95E-12	1	1.82	(2.2,1,5,1.na)
Nickel, ion	water/river	kg	2.30E-09	1	1.82	(2.2,1,5,1.na)
Zinc, ion	water/river	kg	4.46E-09	1	1.82	(2.2,1,5,1.na)
Cadmium, ion	water/ground-	kg	1.16E-12	1	1.82	(2.2,1,5,1.na)
Chromium, ion	water/ground-	kg	5.05E-10	1	1.82	(2.2,1,5,1.na)
Copper, ion	water/ground-	kg	8.59E-11	1	1.82	(2.2,1,5,1.na)
Lead	water/ground-	kg	6.48E-12	1	1.82	(2.2,1,5,1.na)
Mercury	water/ground-	kg	1.26E-13	1	1.82	(2.2,1,5,1.na)
Zinc, ion	water/ground-	kg	6.13E-10	1	1.82	(2.2,1,5,1.na)
Azoxystrobin	soil/agricultural	kg	1.96E-06	1	1.21	(2.1,1,1,1.na)
Chlorothalonil	soil/agricultural	kg	5.03E-05	1	1.21	(2.1,1,1,1.na)
Cymoxanil	soil/agricultural	kg	1.83E-07	1	1.21	(2.1,1,1,1.na)
Dimethomorph	soil/agricultural	kg	3.92E-08	1	1.21	(2.1,1,1,1.na)
Flutolanil	soil/agricultural	kg	7.40E-07	1	1.21	(2.1,1,1,1.na)
Iprodion	soil/agricultural	kg	1.33E-06	1	1.21	(2.1,1,1,1.na)
Mancozeb	soil/agricultural	kg	5.89E-05	1	1.21	(2.1,1,1,1.na)
Maneb	soil/agricultural	kg	9.57E-08	1	1.21	(2.1,1,1,1.na)
Metolachlor	soil/agricultural	kg	2.12E-06	1	1.21	(2.1,1,1,1.na)
Metiram	soil/agricultural	kg	2.79E-06	1	1.21	(2.1,1,1,1.na)
Quintozene	soil/agricultural	kg	4.08E-06	1	1.21	(2.1,1,1,1.na)
Propamocarb HCl	soil/agricultural	kg	3.26E-08	1	1.21	(2.1,1,1,1.na)
Sulfur	soil/forestry	kg	1.14E-05	1	1.21	(2.1,1,1,1.na)
Fentin hydroxide	soil/agricultural	kg	6.51E-07	1	1.21	(2.1,1,1,1.na)
2,4-D	soil/agricultural	kg	5.98E-09	1	1.21	(2.1,1,1,1.na)
Clethodim	soil/agricultural	kg	4.90E-09	1	1.21	(2.1,1,1,1.na)
Diquat	soil/agricultural	kg	3.13E-06	1	1.21	(2.1,1,1,1.na)
EPTC	soil/agricultural	kg	1.62E-05	1	1.21	(2.1,1,1,1.na)
Glufosinate	soil/agricultural	kg	3.22E-07	1	1.21	(2.1,1,1,1.na)
Glyphosate	soil/agricultural	kg	2.44E-07	1	1.21	(2.1,1,1,1.na)
Linuron	soil/agricultural	kg	8.75E-07	1	1.21	(2.1,1,1,1.na)
Metolachlor	soil/agricultural	kg	4.53E-08	1	1.21	(2.1,1,1,1.na)
Metribuzin	soil/agricultural	kg	6.46E-06	1	1.21	(2.1,1,1,1.na)
Paraquat	soil/agricultural	kg	2.09E-07	1	1.21	(2.1,1,1,1.na)
Pendimethalin	soil/agricultural	kg	4.13E-06	1	1.21	(2.1,1,1,1.na)
Rimsulfuron	soil/agricultural	kg	1.00E-07	1	1.21	(2.1,1,1,1.na)
Sethoxydim	soil/agricultural	kg	8.70E-08	1	1.21	(2.1,1,1,1.na)
Trifluralin	soil/agricultural	kg	4.90E-07	1	1.21	(2.1,1,1,1.na)
Aldicarb	soil/agricultural	kg	4.14E-06	1	1.21	(2.1,1,1,1.na)
Azinphos-methyl	soil/agricultural	kg	3.96E-07	1	1.21	(2.1,1,1,1.na)
Carbaryl	soil/agricultural	kg	6.09E-08	1	1.21	(2.1,1,1,1.na)
Carbolfuran	soil/agricultural	kg	6.79E-07	1	1.21	(2.1,1,1,1.na)
Cyfluthrin	soil/agricultural	kg	2.70E-07	1	1.21	(2.1,1,1,1.na)
Diazinon	soil/agricultural	kg	1.50E-06	1	1.21	(2.1,1,1,1.na)
Dimethoate	soil/agricultural	kg	1.04E-06	1	1.21	(2.1,1,1,1.na)
Endosulfan	soil/agricultural	kg	1.36E-06	1	1.21	(2.1,1,1,1.na)
Esfenvalerate	soil/agricultural	kg	2.22E-07	1	1.21	(2.1,1,1,1.na)
Ethoprop	soil/agricultural	kg	3.76E-06	1	1.21	(2.1,1,1,1.na)
Imidacloprid	soil/agricultural	kg	1.07E-06	1	1.21	(2.1,1,1,1.na)
Malathion	soil/agricultural	kg	5.93E-08	1	1.21	(2.1,1,1,1.na)
Oxamyl	soil/agricultural	kg	3.15E-06	1	1.21	(2.1,1,1,1.na)
Permethrin	soil/agricultural	kg	1.31E-07	1	1.21	(2.1,1,1,1.na)
Phorate	soil/agricultural	kg	6.80E-06	1	1.21	(2.1,1,1,1.na)
Phosmet	soil/agricultural	kg	7.49E-07	1	1.21	(2.1,1,1,1.na)
Piperonyl butoxide	soil/agricultural	kg	2.23E-08	1	1.21	(2.1,1,1,1.na)
Propargite	soil/agricultural	kg	2.18E-06	1	1.21	(2.1,1,1,1.na)
Pymetrozine	soil/agricultural	kg	2.09E-07	1	1.21	(2.1,1,1,1.na)
Spinosad	soil/agricultural	kg	4.35E-09	1	1.21	(2.1,1,1,1.na)
Thiamethoxam	soil/agricultural	kg	1.91E-07	1	1.21	(2.1,1,1,1.na)
Trichlorfon	soil/agricultural	kg	3.26E-09	1	1.21	(2.1,1,1,1.na)
TCMTB	soil/agricultural	kg	6.34E-05	1	1.21	(2.1,1,1,1.na)
Dichlorprop-P	soil/agricultural	kg	1.01E-04	1	1.21	(2.1,1,1,1.na)
Endothal	soil/agricultural	kg	1.25E-08	1	1.21	(2.1,1,1,1.na)
Maleic hydrazide	soil/agricultural	kg	4.11E-06	1	1.21	(2.1,1,1,1.na)
Metam-sodium	soil/agricultural	kg	6.70E-04	1	1.21	(2.1,1,1,1.na)
Sulfuric acid	soil/agricultural	kg	7.67E-04	1	1.21	(2.1,1,1,1.na)
potatoes, at farm	US	kg	1.00E+00			

Unit process inventory for: rape seed, at farm, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
sowing	CH	ha	8.46E-04	1	1.07	(2,1,1,3,1,na)
tillage, cultivating, chiselling	CH	ha	6.77E-04	1	1.07	(2,1,1,3,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	3.38E-03	1	1.07	(2,1,1,3,1,na)
tillage, ploughing	CH	ha	1.69E-04	1	1.07	(2,1,1,3,1,na)
application of plant protection products, by field sprayer	CH	ha	3.38E-03	1	1.07	(2,1,1,3,1,na)
fertilising, by broadcaster	CH	ha	2.54E-03	1	1.07	(2,1,1,3,1,na)
combine harvesting	CH	ha	7.37E-04	1	1.07	(2,1,1,3,1,na)
grain drying, low temperature	CH	kg	6.82E-02	1	1.07	(2,1,1,3,1,na)
rape seed IP, at regional storehouse	CH	kg	4.74E-03	1	1.07	(2,2,1,1,1,na)
ammonia, liquid, at regional storehouse	RER	kg	5.30E-02	1	1.13	(2,2,3,1,1,na)
urea, as N, at regional storehouse	RER	kg	1.83E-02	1	1.13	(2,2,3,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	2.53E-02	1	1.13	(2,2,3,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.90E-02	1	1.13	(2,2,3,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	2.84E-02	1	1.13	(2,2,3,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.01E-04	1	1.07	(2,1,1,1,1,na)
[sulfonyl]urea-compounds, at regional storehouse	CH	kg	1.28E-08	1	1.07	(2,1,1,1,1,na)
organophosphorus-compounds, at regional storehouse	CH	kg	5.06E-06	1	1.07	(2,1,1,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	1.15E-05	1	1.07	(2,1,1,1,1,na)
dinitroaniline-compounds, at regional storehouse	CH	kg	4.16E-04	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	1.82E-02	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	2.47E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	2.68E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	8.46E+00	1	2.00	(2,1,1,1,1,na)
Transformation, from arable	resource/land	m2	8.46E+00	1	2.00	(2,1,1,1,1,na)
Transformation, to arable	resource/land	m2	8.46E+00	1	2.00	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	5.01E-02	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	6.60E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	8.92E-04	1	1.52	(2,2,1,5,1,na)
Phosphate	water/ground-	kg	5.92E-05	1	1.52	(2,2,1,5,1,na)
Ammonia	air/low population density	kg	6.80E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	2.88E-02	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	6.71E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.19E-03	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	-3.22E-10	1	1.52	(2,2,1,5,1,na)
Chromium	soil/agricultural	kg	-8.20E-08	1	1.52	(2,2,1,5,1,na)
Lead	soil/agricultural	kg	-7.46E-09	1	1.52	(2,2,1,5,1,na)
Mercury	soil/agricultural	kg	-3.45E-12	1	1.52	(2,2,1,5,1,na)
Nickel	soil/agricultural	kg	-4.58E-08	1	1.52	(2,2,1,5,1,na)
Zinc	soil/agricultural	kg	-4.94E-08	1	1.52	(2,2,1,5,1,na)
Cadmium, ion	water/river	kg	6.87E-10	1	1.82	(2,2,1,5,1,na)
Chromium, ion	water/river	kg	8.30E-08	1	1.82	(2,2,1,5,1,na)
Copper, ion	water/river	kg	5.81E-08	1	1.82	(2,2,1,5,1,na)
Lead	water/river	kg	8.87E-09	1	1.82	(2,2,1,5,1,na)
Mercury	water/river	kg	2.91E-12	1	1.82	(2,2,1,5,1,na)
Nickel, ion	water/river	kg	4.98E-08	1	1.82	(2,2,1,5,1,na)
Zinc, ion	water/river	kg	6.12E-08	1	1.82	(2,2,1,5,1,na)
Cadmium, ion	water/ground-	kg	2.96E-11	1	1.82	(2,2,1,5,1,na)
Chromium, ion	water/ground-	kg	1.51E-08	1	1.82	(2,2,1,5,1,na)
Copper, ion	water/ground-	kg	2.15E-09	1	1.82	(2,2,1,5,1,na)
Lead	water/ground-	kg	5.64E-11	1	1.82	(2,2,1,5,1,na)
Mercury	water/ground-	kg	9.33E-14	1	1.82	(2,2,1,5,1,na)
Zinc, ion	water/ground-	kg	8.43E-09	1	1.82	(2,2,1,5,1,na)
Clpyralid	soil/agricultural	kg	7.02E-06	1	1.21	(2,1,1,1,1,na)
Metsulfuron-methyl	soil/agricultural	kg	1.28E-08	1	1.21	(2,1,1,1,1,na)
Sethoxydim	soil/agricultural	kg	7.62E-05	1	1.21	(2,1,1,1,1,na)
Trifluralin	soil/agricultural	kg	1.71E-04	1	1.21	(2,1,1,1,1,na)
Carbaryl	soil/agricultural	kg	3.24E-06	1	1.21	(2,1,1,1,1,na)
Carbofuran	soil/agricultural	kg	8.30E-06	1	1.21	(2,1,1,1,1,na)
Imidacloprid	soil/agricultural	kg	1.51E-08	1	1.21	(2,1,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	5.51E-07	1	1.21	(2,1,1,1,1,na)
Malathion	soil/agricultural	kg	7.55E-07	1	1.21	(2,1,1,1,1,na)
Parathion	soil/agricultural	kg	4.31E-06	1	1.21	(2,1,1,1,1,na)
Endothall	soil/agricultural	kg	7.55E-06	1	1.21	(2,1,1,1,1,na)
Ethalfuralin	soil/agricultural	kg	2.45E-04	1	1.21	(2,1,1,1,1,na)
Quizalofop-P	soil/agricultural	kg	1.69E-05	1	1.21	(2,1,1,1,1,na)
rape seed, at farm	US	kg	1.00E+00			

Unit process inventory for: rice, at farm, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
sowing	CH	ha	1.63E-04	1	1.07	(2,1,1,3,1,na)
tillage, cultivating, chiselling	CH	ha	1.46E-04	1	1.07	(2,1,1,3,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	7.29E-04	1	1.07	(2,1,1,3,1,na)
tillage, ploughing	CH	ha	2.92E-05	1	1.07	(2,1,1,3,1,na)
tillage, rolling	CH	ha	1.46E-04	1	1.07	(2,1,1,3,1,na)
application of plant protection products, by field sprayer	CH	ha	9.48E-04	1	1.07	(2,1,1,3,1,na)
fertilising, by broadcaster	CH	ha	4.38E-04	1	1.07	(2,1,1,3,1,na)
combine harvesting	CH	ha	1.45E-04	1	1.07	(2,1,1,3,1,na)
grain drying, low temperature	CH	kg	1.30E-01	1	1.07	(2,2,1,1,1,na)
rice seed, at regional storehouse	US	kg	2.06E-02	1	1.07	(2,2,1,1,1,na)
ammonia, liquid, at regional storehouse	RER	kg	1.14E-02	1	1.07	(2,2,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	3.96E-03	1	1.07	(2,2,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	5.47E-03	1	1.07	(2,2,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	4.38E-03	1	1.07	(2,2,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	3.79E-03	1	1.07	(2,2,1,1,1,na)
irrigating	US	m3	1.08E-03	1	1.07	(2,1,1,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	3.53E-05	1	1.07	(2,1,1,1,1,na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	2.73E-04	1	1.07	(2,1,1,1,1,na)
cyclic N-compounds, at regional storehouse	CH	kg	1.96E-05	1	1.07	(2,1,1,1,1,na)
phenoxy-compounds, at regional storehouse	CH	kg	2.57E-05	1	1.07	(2,1,1,1,1,na)
benzo[thia]diazole-compounds, at regional storehouse	CH	kg	3.98E-06	1	1.07	(2,1,1,1,1,na)
[sulfonyl]urea-compounds, at regional storehouse	CH	kg	3.51E-07	1	1.07	(2,1,1,1,1,na)
bipyridylum-compounds, at regional storehouse	CH	kg	1.79E-07	1	1.07	(2,1,1,1,1,na)
organophosphorus-compounds, at regional storehouse	CH	kg	1.32E-05	1	1.07	(2,1,1,1,1,na)
glyphosate, at regional storehouse	CH	kg	1.42E-05	1	1.07	(2,1,1,1,1,na)
MCPA, at regional storehouse	CH	kg	4.95E-07	1	1.07	(2,1,1,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	1.66E-04	1	1.07	(2,1,1,1,1,na)
dinitroaniline-compounds, at regional storehouse	CH	kg	9.68E-06	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	4.80E-03	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.63E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.46E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	1.46E+00	1	2.00	(2,1,1,1,1,na)
Transformation, from arable	resource/land	m2	1.46E+00	1	2.00	(2,1,1,1,1,na)
Transformation, to arable	resource/land	m2	1.46E+00	1	2.00	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	7.44E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	1.52E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	2.05E-04	1	1.52	(2,2,1,5,1,na)
Phosphate	water/ground-	kg	1.02E-05	1	1.52	(2,2,1,5,1,na)
Ammonia	air/low population density	kg	1.47E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	6.22E-03	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.20E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	5.69E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	9.57E-11	1	1.52	(2,2,1,5,1,na)
Chromium	soil/agricultural	kg	-1.40E-08	1	1.52	(2,2,1,5,1,na)
Copper	soil/agricultural	kg	2.20E-08	1	1.52	(2,2,1,5,1,na)
Lead	soil/agricultural	kg	-1.45E-09	1	1.52	(2,2,1,5,1,na)
Mercury	soil/agricultural	kg	-1.37E-12	1	1.52	(2,2,1,5,1,na)
Nickel	soil/agricultural	kg	-8.48E-09	1	1.52	(2,2,1,5,1,na)
Zinc	soil/agricultural	kg	-1.57E-08	1	1.52	(2,2,1,5,1,na)
Cadmium, ion	water/river	kg	1.28E-10	1	1.82	(2,2,1,5,1,na)
Chromium, ion	water/river	kg	1.49E-08	1	1.82	(2,2,1,5,1,na)
Copper, ion	water/river	kg	1.08E-08	1	1.82	(2,2,1,5,1,na)
Lead	water/river	kg	1.83E-09	1	1.82	(2,2,1,5,1,na)
Mercury	water/river	kg	1.23E-12	1	1.82	(2,2,1,5,1,na)
Nickel, ion	water/river	kg	9.47E-09	1	1.82	(2,2,1,5,1,na)
Zinc, ion	water/river	kg	1.26E-08	1	1.82	(2,2,1,5,1,na)
Cadmium, ion	water/ground-	kg	5.51E-12	1	1.82	(2,2,1,5,1,na)
Chromium, ion	water/ground-	kg	2.71E-09	1	1.82	(2,2,1,5,1,na)
Copper, ion	water/ground-	kg	4.00E-10	1	1.82	(2,2,1,5,1,na)
Lead	water/ground-	kg	1.16E-11	1	1.82	(2,2,1,5,1,na)
Mercury	water/ground-	kg	3.95E-14	1	1.82	(2,2,1,5,1,na)
Zinc, ion	water/ground-	kg	1.74E-09	1	1.82	(2,2,1,5,1,na)
Azoxystrobin	soil/agricultural	kg	3.54E-06	1	1.21	(2,1,1,1,1,na)
Propiconazole	soil/agricultural	kg	8.20E-07	1	1.21	(2,1,1,1,1,na)
2,4-D	soil/agricultural	kg	1.84E-05	1	1.21	(2,1,1,1,1,na)
Mecoprop-P	soil/agricultural	kg	9.37E-07	1	1.21	(2,1,1,1,1,na)
Bensulfuron methyl ester	soil/agricultural	kg	1.24E-06	1	1.21	(2,1,1,1,1,na)
Bentazone	soil/agricultural	kg	3.98E-06	1	1.21	(2,1,1,1,1,na)
Clomazone	soil/agricultural	kg	1.87E-05	1	1.21	(2,1,1,1,1,na)
Fenoxaprop	soil/agricultural	kg	7.81E-08	1	1.21	(2,1,1,1,1,na)
Glyphosate	soil/agricultural	kg	1.42E-05	1	1.21	(2,1,1,1,1,na)
Halosulfuron-methyl	soil/agricultural	kg	3.51E-07	1	1.21	(2,1,1,1,1,na)
MCPA	soil/agricultural	kg	4.95E-07	1	1.21	(2,1,1,1,1,na)
Molinate	soil/agricultural	kg	1.05E-04	1	1.21	(2,1,1,1,1,na)
Paraquat	soil/agricultural	kg	1.79E-07	1	1.21	(2,1,1,1,1,na)
Pendimethalin	soil/agricultural	kg	9.68E-06	1	1.21	(2,1,1,1,1,na)
Propanil	soil/agricultural	kg	2.73E-04	1	1.21	(2,1,1,1,1,na)
Quinclorac	soil/agricultural	kg	4.55E-06	1	1.21	(2,1,1,1,1,na)
Thiobencarb	soil/agricultural	kg	5.84E-05	1	1.21	(2,1,1,1,1,na)
Triclopyr	soil/agricultural	kg	6.32E-06	1	1.21	(2,1,1,1,1,na)
Carbaryl	soil/agricultural	kg	1.31E-06	1	1.21	(2,1,1,1,1,na)
Carbofuran	soil/agricultural	kg	1.20E-06	1	1.21	(2,1,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	5.08E-07	1	1.21	(2,1,1,1,1,na)
Malathion	soil/agricultural	kg	1.85E-07	1	1.21	(2,1,1,1,1,na)
Parathion	soil/agricultural	kg	1.30E-05	1	1.21	(2,1,1,1,1,na)
Sodium chlorate	air/high population density	kg	3.51E-06	1	1.21	(2,1,1,1,1,na)
rice, at farm	US	kg	1.00E+00			

Unit process inventory for: wheat grains, at farm, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
sowing	CH	ha	4.44E-04	1	1.07	(2,1,1,3,1,na)
tillage, harrowing, by spring tine harrow	CH	ha	8.88E-04	1	1.07	(2,1,1,3,1,na)
tillage, ploughing	CH	ha	4.44E-05	1	1.07	(2,1,1,3,1,na)
tillage, rolling	CH	ha	4.44E-04	1	1.07	(2,1,1,3,1,na)
application of plant protection products, by field sprayer	CH	ha	4.44E-04	1	1.07	(2,1,1,3,1,na)
fertilising, by broadcaster	CH	ha	1.33E-03	1	1.07	(2,1,1,3,1,na)
combine harvesting	CH	ha	3.67E-04	1	1.07	(2,1,1,3,1,na)
wheat seed IP, at regional storehouse	CH	kg	4.18E-02	1	1.07	(2,2,1,1,1,na)
ammonia, liquid, at regional storehouse	RER	kg	1.62E-02	1	1.07	(2,2,1,1,1,na)
urea, as N, at regional storehouse	RER	kg	5.59E-03	1	1.07	(2,2,1,1,1,na)
ammonium nitrate, as N, at regional storehouse	RER	kg	7.72E-03	1	1.07	(2,2,1,1,1,na)
diammonium phosphate, as P2O5, at regional storehouse	RER	kg	1.85E-02	1	1.07	(2,2,1,1,1,na)
potassium chloride, as K2O, at regional storehouse	RER	kg	2.37E-02	1	1.07	(2,2,1,1,1,na)
irrigating	US	m3	7.25E-05	1	1.07	(2,1,1,1,1,na)
pesticide unspecified, at regional storehouse	CH	kg	1.28E-06	1	1.07	(2,1,1,1,1,na)
acetamide-anilide-compounds, at regional storehouse	CH	kg	3.37E-07	1	1.07	(2,1,1,1,1,na)
cyclic N-compounds, at regional storehouse	CH	kg	4.12E-06	1	1.07	(2,1,1,1,1,na)
phenoxy-compounds, at regional storehouse	CH	kg	3.97E-05	1	1.07	(2,1,1,1,1,na)
triazine-compounds, at regional storehouse	CH	kg	4.87E-06	1	1.07	(2,1,1,1,1,na)
nitrile-compounds, at regional storehouse	CH	kg	3.17E-06	1	1.07	(2,1,1,1,1,na)
(sulfonyl)urea-compounds, at regional storehouse	CH	kg	1.58E-06	1	1.07	(2,1,1,1,1,na)
diphenylether-compounds, at regional storehouse	CH	kg	8.22E-07	1	1.07	(2,1,1,1,1,na)
dicamba, at regional storehouse	CH	kg	2.85E-06	1	1.07	(2,1,1,1,1,na)
diuron, at regional storehouse	CH	kg	1.09E-06	1	1.07	(2,1,1,1,1,na)
organophosphorus-compounds, at regional storehouse	CH	kg	6.77E-06	1	1.07	(2,1,1,1,1,na)
glyphosate, at regional storehouse	CH	kg	4.01E-05	1	1.07	(2,1,1,1,1,na)
MCPA, at regional storehouse	CH	kg	6.38E-06	1	1.07	(2,1,1,1,1,na)
[thio]carbamate-compounds, at regional storehouse	CH	kg	1.07E-06	1	1.07	(2,1,1,1,1,na)
pyretroid-compounds, at regional storehouse	CH	kg	1.19E-07	1	1.07	(2,1,1,1,1,na)
transport, lorry >16t, fleet average	RER	tkm	1.09E-02	1	2.09	(4,5,na,na,na,na)
Energy, gross calorific value, in biomass	resource/biotic	MJ	1.48E+01	1	1.07	(2,1,1,1,1,na)
Carbon dioxide, in air	resource/in air	kg	1.26E+00	1	1.07	(2,1,1,1,1,na)
Occupation, arable	resource/land	m2a	4.44E+00	1	2.00	(2,1,1,1,1,na)
Transformation, from arable	resource/land	m2	4.44E+00	1	2.00	(2,1,1,1,1,na)
Transformation, to arable	resource/land	m2	4.44E+00	1	2.00	(2,1,1,1,1,na)
Nitrate	water/ground-	kg	9.98E-03	1	1.51	(2,2,1,1,1,na)
Phosphorus	water/river	kg	6.42E-04	1	1.51	(2,2,1,1,1,na)
Phosphate	water/river	kg	8.68E-04	1	1.52	(2,2,1,5,1,na)
Phosphate	water/ground-	kg	3.11E-05	1	1.52	(2,2,1,5,1,na)
Ammonia	air/low population density	kg	2.08E-03	1	1.21	(2,2,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	8.78E-03	1	1.07	(2,2,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.84E-04	1	1.41	(2,2,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	8.76E-04	1	1.41	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	8.03E-10	1	1.52	(2,2,1,5,1,na)
Chromium	soil/agricultural	kg	-4.00E-08	1	1.52	(2,2,1,5,1,na)
Lead	soil/agricultural	kg	-5.33E-09	1	1.52	(2,2,1,5,1,na)
Mercury	soil/agricultural	kg	-2.42E-12	1	1.52	(2,2,1,5,1,na)
Nickel	soil/agricultural	kg	-2.59E-08	1	1.52	(2,2,1,5,1,na)
Zinc	soil/agricultural	kg	-3.09E-08	1	1.52	(2,2,1,5,1,na)
Cadmium, ion	water/river	kg	4.18E-10	1	1.82	(2,2,1,5,1,na)
Chromium, ion	water/river	kg	4.68E-08	1	1.82	(2,2,1,5,1,na)
Copper, ion	water/river	kg	3.39E-08	1	1.82	(2,2,1,5,1,na)
Lead	water/river	kg	6.89E-09	1	1.82	(2,2,1,5,1,na)
Mercury	water/river	kg	2.52E-12	1	1.82	(2,2,1,5,1,na)
Nickel, ion	water/river	kg	2.93E-08	1	1.82	(2,2,1,5,1,na)
Zinc, ion	water/river	kg	4.16E-08	1	1.82	(2,2,1,5,1,na)
Cadmium, ion	water/ground-	kg	1.80E-11	1	1.82	(2,2,1,5,1,na)
Chromium, ion	water/ground-	kg	8.52E-09	1	1.82	(2,2,1,5,1,na)
Copper, ion	water/ground-	kg	1.25E-09	1	1.82	(2,2,1,5,1,na)
Lead	water/ground-	kg	4.38E-11	1	1.82	(2,2,1,5,1,na)
Mercury	water/ground-	kg	8.07E-14	1	1.82	(2,2,1,5,1,na)
Zinc, ion	water/ground-	kg	5.72E-09	1	1.82	(2,2,1,5,1,na)
Azoxystrobin	soil/agricultural	kg	6.93E-08	1	1.21	(2,1,1,1,1,na)
Propiconazole	soil/agricultural	kg	3.96E-06	1	1.21	(2,1,1,1,1,na)
Pyraclostrobin (prop)	soil/agricultural	kg	3.96E-07	1	1.21	(2,1,1,1,1,na)
Tebuconazole	soil/agricultural	kg	1.58E-07	1	1.21	(2,1,1,1,1,na)
Trifloxystrobin	soil/agricultural	kg	7.92E-08	1	1.21	(2,1,1,1,1,na)
2,4-D	soil/agricultural	kg	3.91E-05	1	1.21	(2,1,1,1,1,na)
Ioxynil	soil/agricultural	kg	3.37E-07	1	1.21	(2,1,1,1,1,na)
Atrazine	soil/agricultural	kg	2.89E-06	1	1.21	(2,1,1,1,1,na)
Bromoxynil	soil/agricultural	kg	3.17E-06	1	1.21	(2,1,1,1,1,na)
Carfentrazone ethyl ester	soil/agricultural	kg	2.97E-08	1	1.21	(2,1,1,1,1,na)
Chlorsulfuron	soil/agricultural	kg	3.17E-07	1	1.21	(2,1,1,1,1,na)
Clodinafop-propargyl	soil/agricultural	kg	1.58E-07	1	1.21	(2,1,1,1,1,na)
Clopyralid	soil/agricultural	kg	3.56E-07	1	1.21	(2,1,1,1,1,na)
Dicamba	soil/agricultural	kg	2.85E-06	1	1.21	(2,1,1,1,1,na)
Diclofop-methyl	soil/agricultural	kg	6.63E-07	1	1.21	(2,1,1,1,1,na)
Diuron	soil/agricultural	kg	1.09E-06	1	1.21	(2,1,1,1,1,na)
Fenoxaprop	soil/agricultural	kg	7.92E-08	1	1.21	(2,1,1,1,1,na)
Flucarbazone sodium salt	soil/agricultural	kg	1.98E-08	1	1.21	(2,1,1,1,1,na)
Fluroxypyr	soil/agricultural	kg	4.55E-07	1	1.21	(2,1,1,1,1,na)
Glyphosate	soil/agricultural	kg	4.01E-05	1	1.21	(2,1,1,1,1,na)
Imazamox	soil/agricultural	kg	1.19E-07	1	1.21	(2,1,1,1,1,na)
MCPA	soil/agricultural	kg	6.38E-06	1	1.21	(2,1,1,1,1,na)
Mesosulfuron-methyl (prop)	soil/agricultural	kg	9.90E-09	1	1.21	(2,1,1,1,1,na)
Metribuzin	soil/agricultural	kg	1.82E-06	1	1.21	(2,1,1,1,1,na)
Metsulfuron-methyl	soil/agricultural	kg	1.19E-07	1	1.21	(2,1,1,1,1,na)
Picloram	soil/agricultural	kg	3.96E-08	1	1.21	(2,1,1,1,1,na)
Prosulfuron	soil/agricultural	kg	1.98E-08	1	1.21	(2,1,1,1,1,na)
Sulfosulfuron	soil/agricultural	kg	4.75E-07	1	1.21	(2,1,1,1,1,na)
Thifensulfuron-methyl	soil/agricultural	kg	3.21E-07	1	1.21	(2,1,1,1,1,na)
Tralkoxydim	soil/agricultural	kg	1.39E-07	1	1.21	(2,1,1,1,1,na)
Tri-allate	soil/agricultural	kg	1.07E-06	1	1.21	(2,1,1,1,1,na)
Trasulfuron	soil/agricultural	kg	3.17E-07	1	1.21	(2,1,1,1,1,na)
Tribenuron-methyl	soil/agricultural	kg	1.58E-07	1	1.21	(2,1,1,1,1,na)
Chlorpyrifos	soil/agricultural	kg	4.28E-06	1	1.21	(2,1,1,1,1,na)
Cypermethrin	soil/agricultural	kg	1.19E-07	1	1.21	(2,1,1,1,1,na)
Dimethoate	soil/agricultural	kg	3.07E-07	1	1.21	(2,1,1,1,1,na)
Lambda-Cyhalothrin	soil/agricultural	kg	7.92E-08	1	1.21	(2,1,1,1,1,na)
Parathion	soil/agricultural	kg	2.49E-06	1	1.21	(2,1,1,1,1,na)
wheat grains, at farm	US	kg	1.00E+00			

Unit process inventory for: irrigating, US						
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores
tractor, production (I)	CH	kg	3.82E-04	1	1.07	(2,1,1,1,1,na)
agricultural machinery, general, production (I)	CH	kg	1.81E-02	1	1.07	(2,1,1,1,1,na)
diesel, at regional storage	CH	kg	3.15E-03	1	1.07	(2,1,1,1,1,na)
shed (I)	CH	m2	4.91E-05	1	1.07	(2,1,1,1,1,na)
polyethylene, HDPE, granulate, at plant	RER	kg	1.92E-02	1	1.07	(2,1,1,1,1,na)
extrusion, plastic film	RER	kg	2.02E-02	1	1.07	(2,1,1,1,1,na)
excavation, hydraulic digger	RER	m3	3.33E-03	1	1.07	(2,1,1,1,1,na)
cast iron, at plant	RER	kg	3.56E-03	1	1.07	(2,1,1,1,1,na)
electricity, low voltage, at grid	US	kWh	2.39E-01	1	1.07	(2,1,1,1,1,na)
polyvinylchloride, bulk polymerised, at plant	RER	kg	9.38E-04	1	1.07	(2,1,1,1,1,na)
NMVOC, non-methane volatile organic compounds, unspecified origin	air/low population density	kg	1.37E-05	1	1.50	(2,1,1,1,1,na)
Nitrogen oxides	air/low population density	kg	1.61E-04	1	1.50	(2,1,1,1,1,na)
Carbon monoxide, fossil	air/low population density	kg	3.02E-05	1	5.00	(2,1,1,1,1,na)
Carbon dioxide, fossil	air/low population density	kg	9.79E-03	1	1.07	(2,1,1,1,1,na)
Sulfur dioxide	air/low population density	kg	3.18E-06	1	1.07	(2,1,1,1,1,na)
Methane, fossil	air/low population density	kg	4.06E-07	1	1.50	(2,1,1,1,1,na)
Benzene	air/low population density	kg	2.30E-08	1	2.00	(2,1,1,1,1,na)
Particulates, < 2.5 um	air/low population density	kg	1.30E-05	1	3.00	(2,1,1,1,1,na)
Cadmium	air/low population density	kg	3.15E-11	1	5.00	(2,1,1,1,1,na)
Chromium	air/low population density	kg	1.58E-10	1	5.00	(2,1,1,1,1,na)
Copper	air/low population density	kg	5.36E-09	1	5.00	(2,1,1,1,1,na)
Dinitrogen monoxide	air/low population density	kg	3.78E-07	1	1.50	(2,1,1,1,1,na)
Nickel	air/low population density	kg	2.21E-10	1	5.00	(2,1,1,1,1,na)
Zinc	air/low population density	kg	3.15E-09	1	5.00	(2,1,1,1,1,na)
Benzo(a)pyrene	air/low population density	kg	9.45E-11	1	2.00	(2,1,1,1,1,na)
PAH, polycyclic aromatic hydrocarbons	air/low population density	kg	1.04E-08	1	3.00	(2,2,1,1,1,na)
Heat, waste	air/low population density	MJ	9.04E-01	1	1.07	(2,2,1,1,1,na)
Zinc	soil/agricultural	kg	2.25E-06	1	1.51	(2,2,1,1,1,na)
Lead	soil/agricultural	kg	4.41E-09	1	1.51	(2,2,1,1,1,na)
Cadmium	soil/agricultural	kg	9.06E-10	1	1.51	(2,2,1,1,1,na)
Ammonia	air/low population density	kg	6.30E-08	1	1.51	(2,2,1,1,1,na)
Selenium	air/low population density	kg	3.15E-11	1	5.00	(2,2,1,1,1,na)
disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	kg	3.56E-06	1	1.07	(2,2,1,1,1,na)
disposal, building, polyvinylchloride products, to final disposal	CH	kg	9.38E-07	1	1.07	(2,2,1,1,1,na)
disposal, building, polyethylene/polypropylene products, to final disposal	CH	kg	5.56E-06	1	1.07	(2,2,1,1,1,na)
Occupation, construction site	resource/land	m2a	5.56E-03	1	1.51	(2,2,1,1,1,na)
Water, well, in ground	resource/in water	m3	1.00E+00	1	1.07	(2,2,1,1,1,na)
irrigating	US	m3	1.00E+00			

Appendix B to Chapter 2 (Sheep Production in the USA)

Unit-Process Inventory from Chapter 2 (Sheep Production in the USA)

Tab. B. 2 Unit-process inventory for sheep production.

Unit process inventory for: sheep husbandry, per head and year, US								
Exchanges	Location/Category	Unit	Value	Uncert Type	SD 95%	Uncert Scores	wool, sheep, at farm US (kg)	sheep for slaughtering, live weight, at farm US (kg)
tillage, rolling	CH	ha	1.62E-02	1	1.20	(3,3,3,5,1,na)	65%	35%
fertilising, by broadcaster	CH	ha	6.47E-02	1	1.20	(3,3,3,5,1,na)	65%	35%
soybean meal, at oil mill	US	kg	2.72E+01	1	1.22	(2,3,4,3,1,na)	65%	35%
grain maize IP, at feed mill	CH	kg	4.14E+01	1	1.22	(2,3,4,3,1,na)	65%	35%
limestone, milled, packed, at plant	CH	kg	6.62E+00	1	1.22	(2,3,4,3,1,na)	65%	35%
sodium chloride, powder, at plant	RER	kg	3.88E+00	1	1.22	(2,3,4,3,1,na)	65%	35%
ammonium nitrate, as N, at regional storehouse	RER	kg	1.37E+00	1	1.22	(2,3,4,3,1,na)	65%	35%
triple superphosphate, as P2O5, at regional storehouse	RER	kg	4.56E-01	1	1.22	(2,3,4,3,1,na)	65%	35%
potassium chloride, as K2O, at regional storehouse	RER	kg	6.86E-01	1	1.22	(2,3,4,3,1,na)	65%	35%
limestone, milled, loose, at plant	CH	kg	1.83E+01	1	1.22	(2,3,4,3,1,na)	65%	35%
irrigating	US	m3	3.43E-03	1	1.20	(3,3,3,5,1,na)	65%	35%
shed (l)	CH	m2	2.00E+00	1	1.20	(3,3,3,5,1,na)	65%	35%
transport, freight, rail	RER	tkm	5.49E+01	1	2.09	(4,5,na,na,na,na)	65%	35%
transport, lorry >16t, fleet average	RER	tkm	2.77E+01	1	2.09	(4,5,na,na,na,na)	65%	35%
Energy, gross calorific value, in biomass	resource/biotic	MJ	4.25E+02	1	1.07	(2,1,1,1,1,na)	22%	78%
Heat, waste	air/low population density	MJ	7.90E+02	1	1.07	(2,1,1,1,1,na)	22%	78%
Carbon dioxide, in air	resource/in air	kg	9.57E+01	1	1.07	(2,1,1,1,1,na)	6%	94%
Occupation, pasture and meadow, intensive	resource/land	m2a	8.09E+02	1	2.06	(3,3,4,3,1,na)	65%	35%
Transformation, from pasture and meadow, intensive	resource/land	m2	8.09E+02	1	2.06	(3,3,4,3,1,na)	65%	35%
Transformation, to pasture and meadow, intensive	resource/land	m2	8.09E+02	1	2.06	(3,3,4,3,1,na)	65%	35%
Nitrate	water/ground-	kg	4.96E+00	1	1.51	(2,2,2,3,1,na)	65%	35%
Phosphorus	water/river	kg	4.75E-02	1	1.51	(2,2,2,3,1,na)	65%	35%
Phosphate	water/river	kg	6.42E-02	1	1.51	(2,2,2,3,1,na)	65%	35%
Phosphate	water/ground-	kg	6.00E-02	1	1.51	(2,2,2,3,1,na)	65%	35%
Ammonia	air/low population density	kg	3.44E+00	1	1.22	(2,3,2,3,1,na)	65%	35%
Nitrogen oxides	air/low population density	kg	5.40E-02	1	1.41	(2,2,2,3,1,na)	65%	35%
Dinitrogen monoxide	air/low population density	kg	2.57E-01	1	1.41	(2,2,2,3,1,na)	65%	35%
Methane, biogenic	air/low population density	kg	8.67E+00	1	1.41	(2,2,2,3,1,na)	65%	35%
Cadmium	soil/agricultural	kg	-2.00E-07	1	1.52	(2,2,2,5,1,na)	65%	35%
Chromium	soil/agricultural	kg	-1.12E-03	1	1.52	(2,2,2,5,1,na)	65%	35%
Copper	soil/agricultural	kg	-9.31E-04	1	1.52	(2,2,2,5,1,na)	65%	35%
Lead	soil/agricultural	kg	-2.24E-06	1	1.52	(2,2,2,5,1,na)	65%	35%
Nickel	soil/agricultural	kg	-1.66E-04	1	1.52	(2,2,2,5,1,na)	65%	35%
Zinc	soil/agricultural	kg	-2.08E-03	1	1.52	(2,2,2,5,1,na)	65%	35%
Cadmium, ion	water/river	kg	4.55E-07	1	1.82	(2,2,2,5,1,na)	65%	35%
Chromium, ion	water/river	kg	9.19E-04	1	1.82	(2,2,2,5,1,na)	65%	35%
Copper, ion	water/river	kg	2.72E-03	1	1.82	(2,2,2,5,1,na)	65%	35%
Lead	water/river	kg	2.78E-06	1	1.82	(2,2,2,5,1,na)	65%	35%
Mercury	water/river	kg	1.97E-04	1	1.82	(2,2,2,5,1,na)	65%	35%
Nickel, ion	water/river	kg	9.75E-04	1	1.82	(2,2,2,5,1,na)	65%	35%
Cadmium, ion	water/ground-	kg	1.96E-08	1	1.82	(2,2,2,5,1,na)	65%	35%
Chromium, ion	water/ground-	kg	3.40E-05	1	1.82	(2,2,2,5,1,na)	65%	35%
Copper, ion	water/ground-	kg	3.74E-04	1	1.82	(2,2,2,5,1,na)	65%	35%
Lead	water/ground-	kg	1.77E-08	1	1.82	(2,2,2,5,1,na)	65%	35%
Nickel, ion	water/ground-	kg	1.77E-04	1	1.82	(2,2,2,5,1,na)	65%	35%
wool, sheep, at farm	US	kg	4.16E+00				100%	
sheep for slaughtering, live weight, at farm	US	kg	6.28E+01					100%